CHAPTER 12
UTILITY SYSTEM

CHAPTER OVERVIEW

SECTION 12-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

12-1-1 FIRE DETECTION SYSTEM DESCRIPTION AND DATA.

12-1-1.1 Fire Detection System Description.

The fire detection system provides fire warning in the cockpit if there is a fire in either the main engine compartment or the APU compartment. The fire detection system warns the pilot and copilot when infrared radiation caused by a fire or extreme overheating is detected in either engine compartment or APU compartment (Figure 12-1-1-1 and Figure 12-1-1-2). The system consists of three control amplifiers, one for each engine and one for the APU, in the forward cabin ceiling; five sensors, two in each engine compartment, one on the firewall and the other on the engine deck, and one in the APU compartment; No. 1 and No. 2 engine T-handle fire warning lights, in the engine control quadrant; and an APU T-handle fire warning light and FIRE DET TEST/OPER/1/2 switch on the upper console. The fire detection system functionally interfaces with the caution/advisory warning system through the left relay panel. The caution/advisory warning system provides BRT/DIM/TEST control for the FIRE capsules, one each on the pilot's and copilot's master warning panel. It also provides T-handle light dimming control. The fire detection system is associated with the discharge-type fire extinguishing system described in paragraph 12-1-2.

12-1-1.2 Theory Of Operation.

12-1-1.2.1 Power Distribution.

Electrical power of 28 vdc for the fire detection system is supplied by the dc essential bus and the battery bus (Figure 12-1-1-2). The 28 vdc from the dc essential bus is routed through the FIRE DET NO. 1 ENG and the FIRE DET NO. 2 ENG circuit breakers on the upper console. The FIRE DET NO. 1 ENG circuit breaker supplies 28 vdc to both No. 1 engine fire detectors, the No. 1 engine fire detector control amplifier, and the left relay panel. The FIRE DET NO. 2 ENG circuit breaker supplies 28 vdc to both No. 2 engine fire detectors, the No. 2 engine fire detector control amplifier, and the left relay panel. The 28 vdc from the battery bus is routed through the APU FIRE DET circuit breaker on the lower console. The APU FIRE DET circuit breaker supplies 28 vdc to the APU fire detector, APU fire detector control amplifier, and the left relay panel. Lighting intensity of the three T-handle warning capsules and two FIRE capsules on the pilot's and copilot's master warning panels are controlled by the BRT/DIM-TEST switch on the caution/advisory panel. The BRT/DIM-TEST switch controls the dimming circuits in the left relay panel.

12-1-1.2.2 Normal Operation.

The system consists of five identical fire detector sensors. Each sensor is a dual-element photo resistive light sensor, whose electrical resistance decreases with the intensity and color of the light energy reaching the detector elements. One element is sensitive to red light, the other is sensitive to blue. In normal operation (selected by the OPER position of the FIRE DET TEST switch), the detector circuit configuration produces an output between 9 and 11 vdc when the blue component of ambient light reaches the detector and an output between 13 and 15 vdc when the red component of the fire's flame reaches the detector. The fire detector amplifiers control switching of 28 vdc to the FIRE capsules through a relay in the amplifiers. When detector output/amplifier input is between 9 and 11 vdc, amplifier output is 0 vdc; when detector output/amplifier input is between 13 and 15 vdc, amplifier output is 28 vdc.

12-1-1.2.3 Test Operation.

The FIRE DET TEST switch is used to simulate a fire condition. When the FIRE DET TEST switch is placed to 1, a simulated fire detected signal is applied to the APU fire detector, No. 1 engine firewall-mounted detector, and No. 2 engine firewall-mounted detector. This causes the two FIRE warning capsules on the master warning panels, #1 and #2 ENG EMER OFF, and APU T-handle capsules to go on. When the FIRE DET TEST switch is placed to 2, a simulated fire detected signal is applied to the No. 1 and No. 2 engine deck-mounted detectors. This causes the two FIRE warning capsules on the master warning panels as well as the #1 and #2 ENG EMER OFF T-handle capsules to go on. When the FIRE DET TEST switch is placed to OPER, all simulated fire detected signals are removed and the system operates in the normal mode.

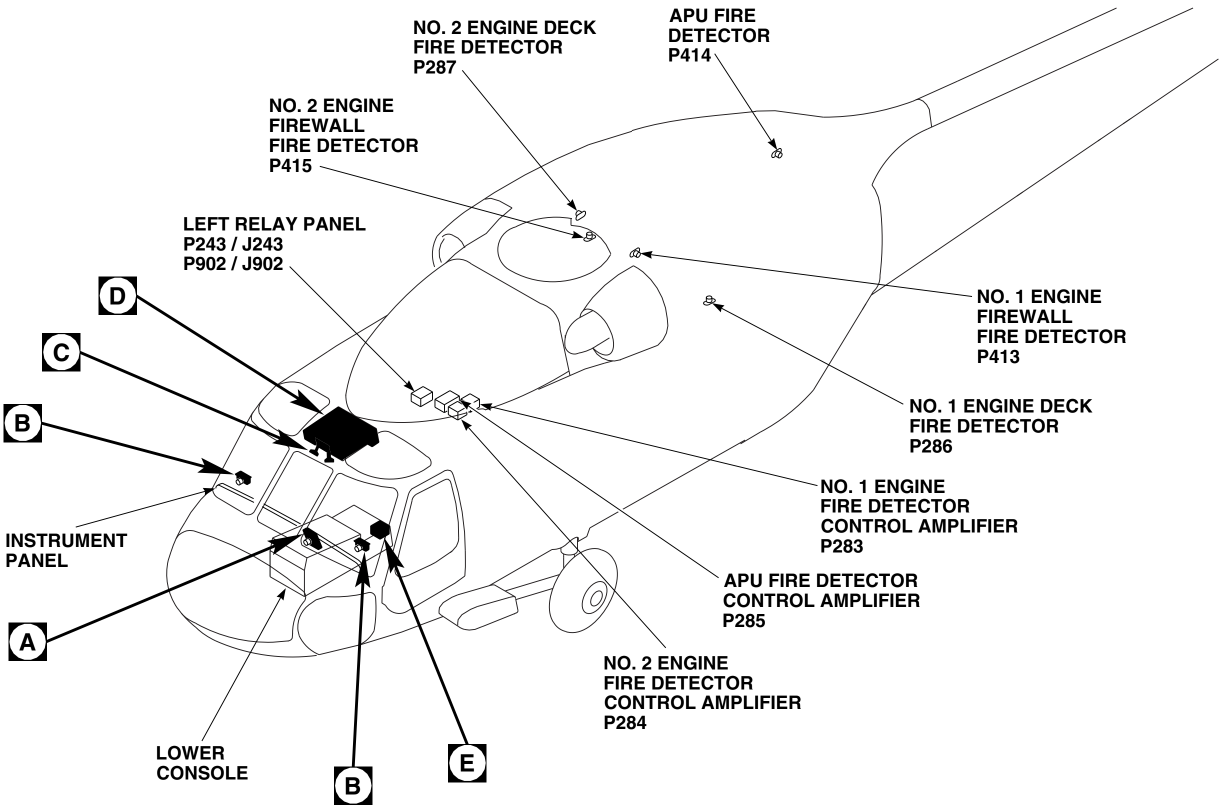
12-1-1.3 Component Description.

12-1-1.3.1 Fire Detector.

The fire detector is a nonrepairable unit incorporating a red-light-sensitive photo resistive element connected in series with a blue-light-sensitive photo resistive element. A fixed resistor is connected in parallel across the blue-sensitive element. Input power connects to the input of the red-sensitive element. The output side of the blue-sensitive element is connected to airframe ground. Two connections are made at the junction of the two elements. One connection provides a line from the output side of the red-sensitive element to the amplifier input. The other connection provides an input line for a 14 vdc test signal.

12-1-1.3.2 Detector Amplifier.

The fire detector amplifier is a nonrepairable module which switches 28 vdc to light the master warning panel FIRE capsules and the related T-handle. The amplifier is activated by a test signal or a fire detected signal between 13 and 15 vdc from the flame detector. When the flame detected signal falls below 11 vdc, the amplifier removes power from the fire warning lights.

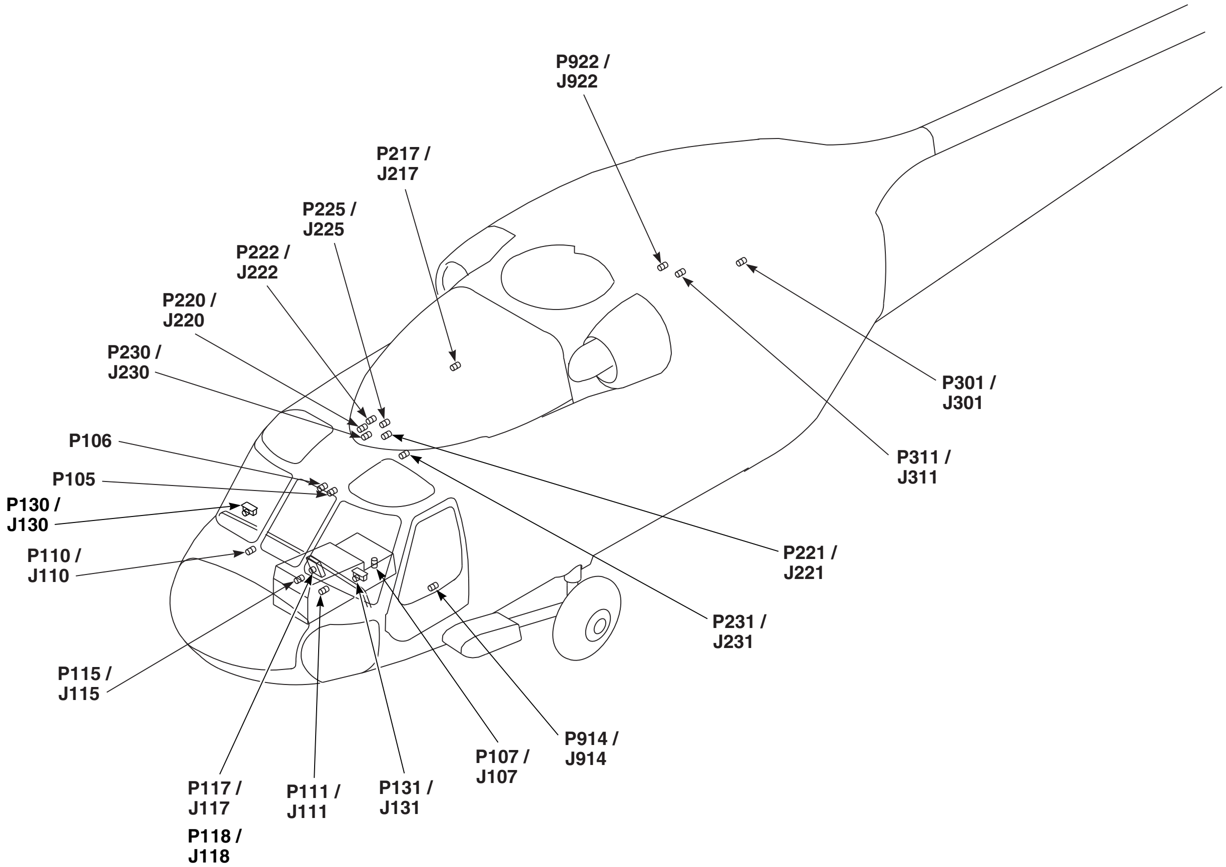


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P105	NO. 1 ENGINE QUADRANT
P106	NO. 2 ENGINE QUADRANT
P107 / J107	COCKPIT, BL 9 LH, STA 236
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P115 / J115	COCKPIT, BL 6 RH, STA 210

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P130 / J130	PILOT'S MASTER WARNING PANEL
P131 / J131	COPILOT'S MASTER WARNING PANEL
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P220 / J220	MAIN ROTOR PYLON DECK BL 3 RH, STA 243
P221 / J221	MAIN ROTOR PYLON DECK BL 3 LH, STA 245.5
P222 / J222	MAIN ROTOR PYLON DECK BL 3 RH, STA 244
P225 / J225	MAIN ROTOR PYLON DECK BL 3 LH, STA 244
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247

AA3410_1A
SA

FIGURE 12-1-1-1. FIRE DETECTION SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P243 / J243	LEFT RELAY PANEL
P283	NO. 1 ENGINE FIRE DETECTOR CONTROL AMPLIFIER
P284	NO. 2 ENGINE FIRE DETECTOR CONTROL AMPLIFIER
P285	APU FIRE DETECTOR CONTROL AMPLIFIER
P286	NO. 1 ENGINE DECK FIRE DETECTOR
P287	NO. 2 ENGINE DECK FIRE DETECTOR

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P301 / J301	TRANSITION DECK BL 12.88 LH, STA 415
P311 / J311	CABIN CEILING, BL 16.5 LH, STA 383
P413	NO. 1 ENGINE FIREWALL FIRE DETECTOR
P414	APU FIRE DETECTOR
P415	NO. 2 ENGINE FIREWALL FIRE DETECTOR
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380

AA3410_2A
SA

FIGURE 12-1-1-1. FIRE DETECTION SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)

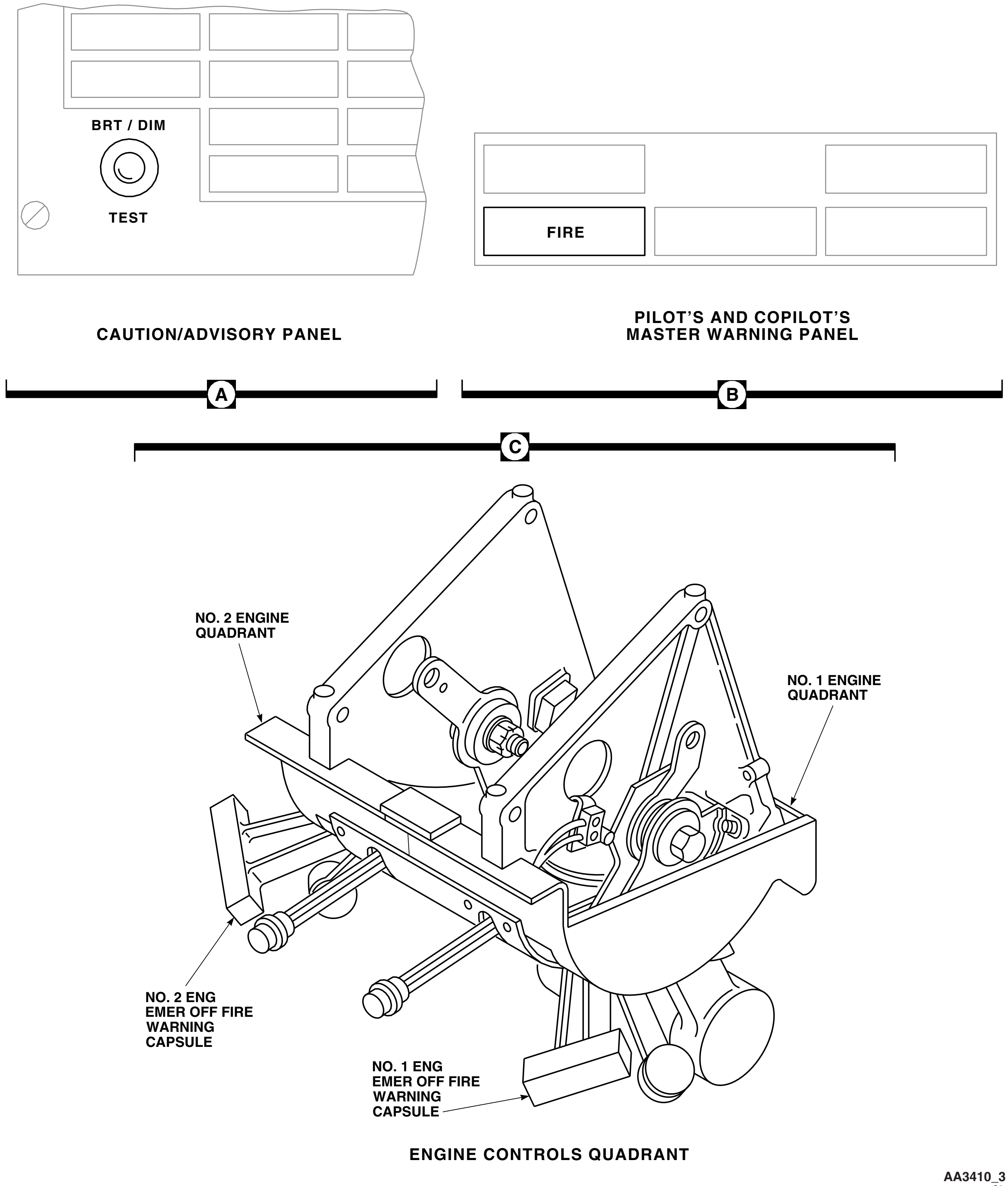


FIGURE 12-1-1-1. FIRE DETECTION SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)

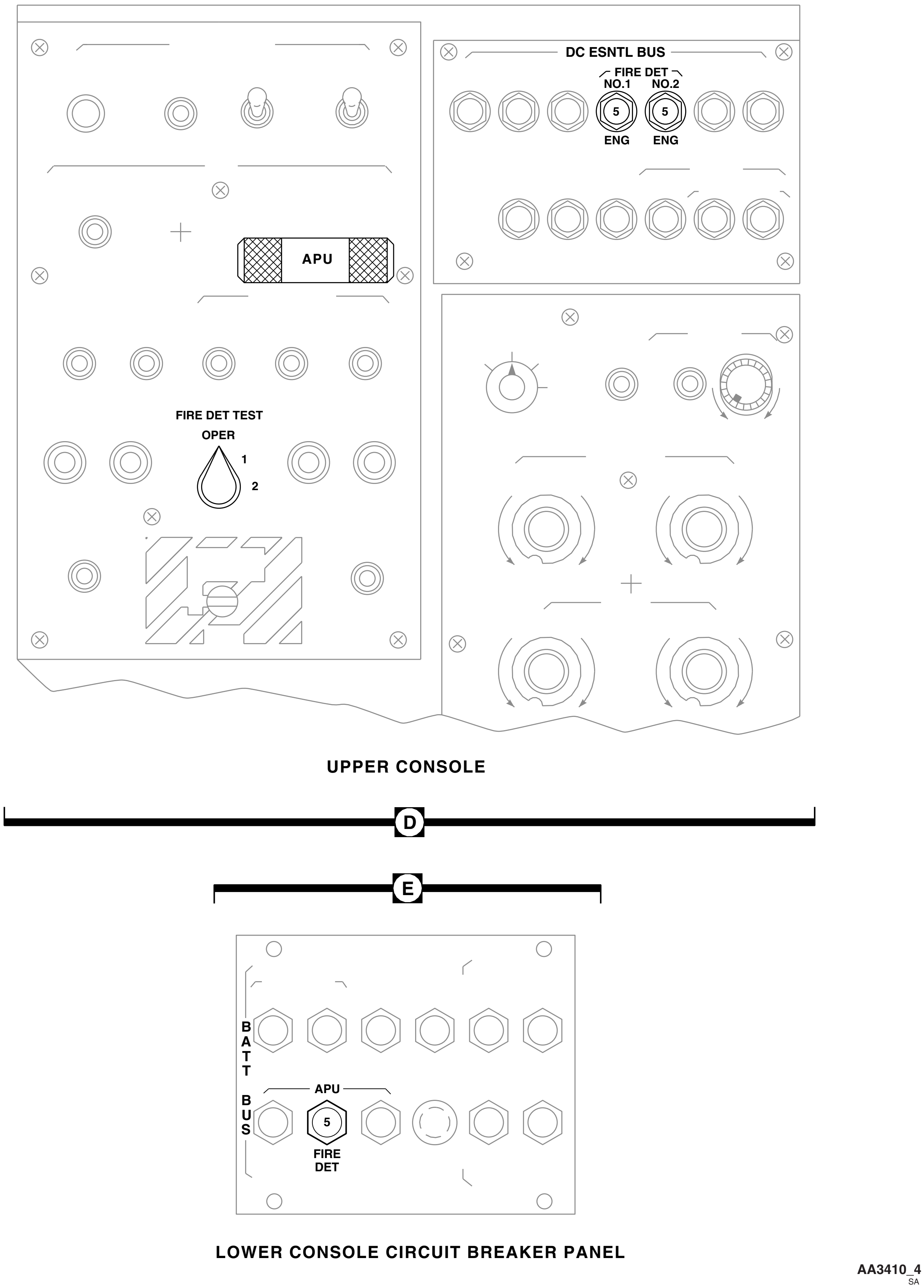


FIGURE 12-1-1-1. FIRE DETECTION SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)

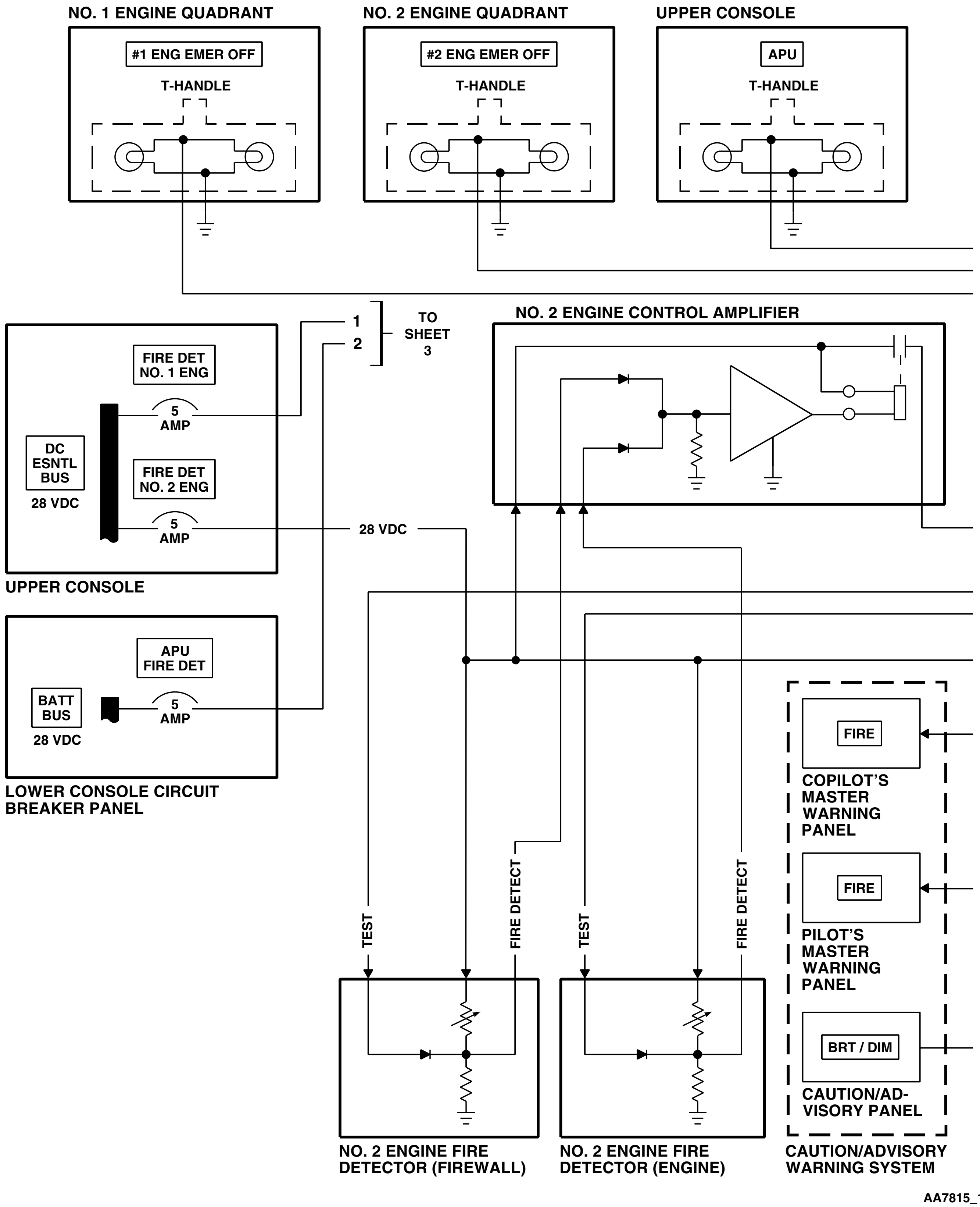


FIGURE 12-1-1-2. FIRE DETECTION SYSTEM BLOCK DIAGRAM (SHEET 1 OF 4)

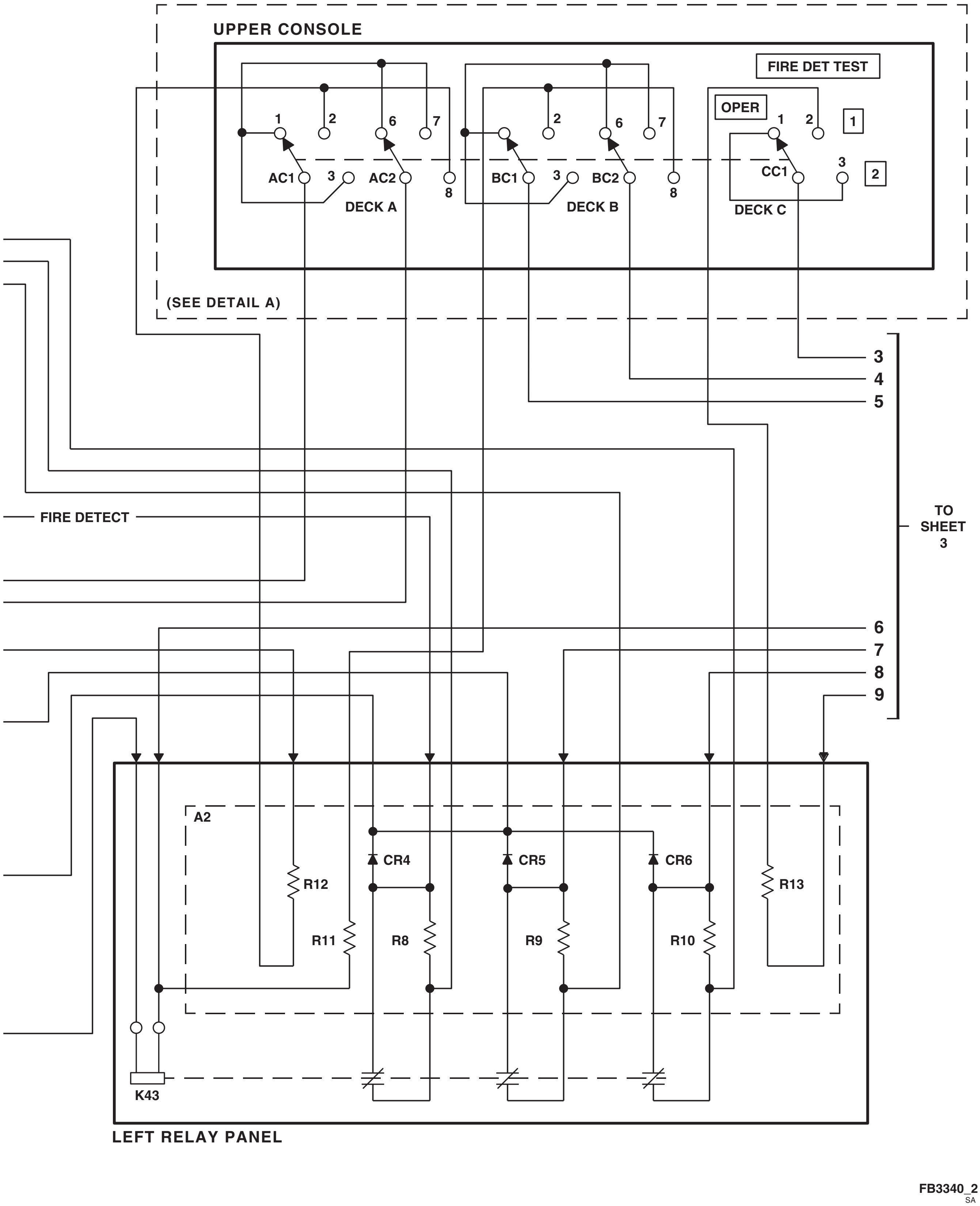


FIGURE 12-1-1-2. FIRE DETECTION SYSTEM BLOCK DIAGRAM (SHEET 2 OF 4)

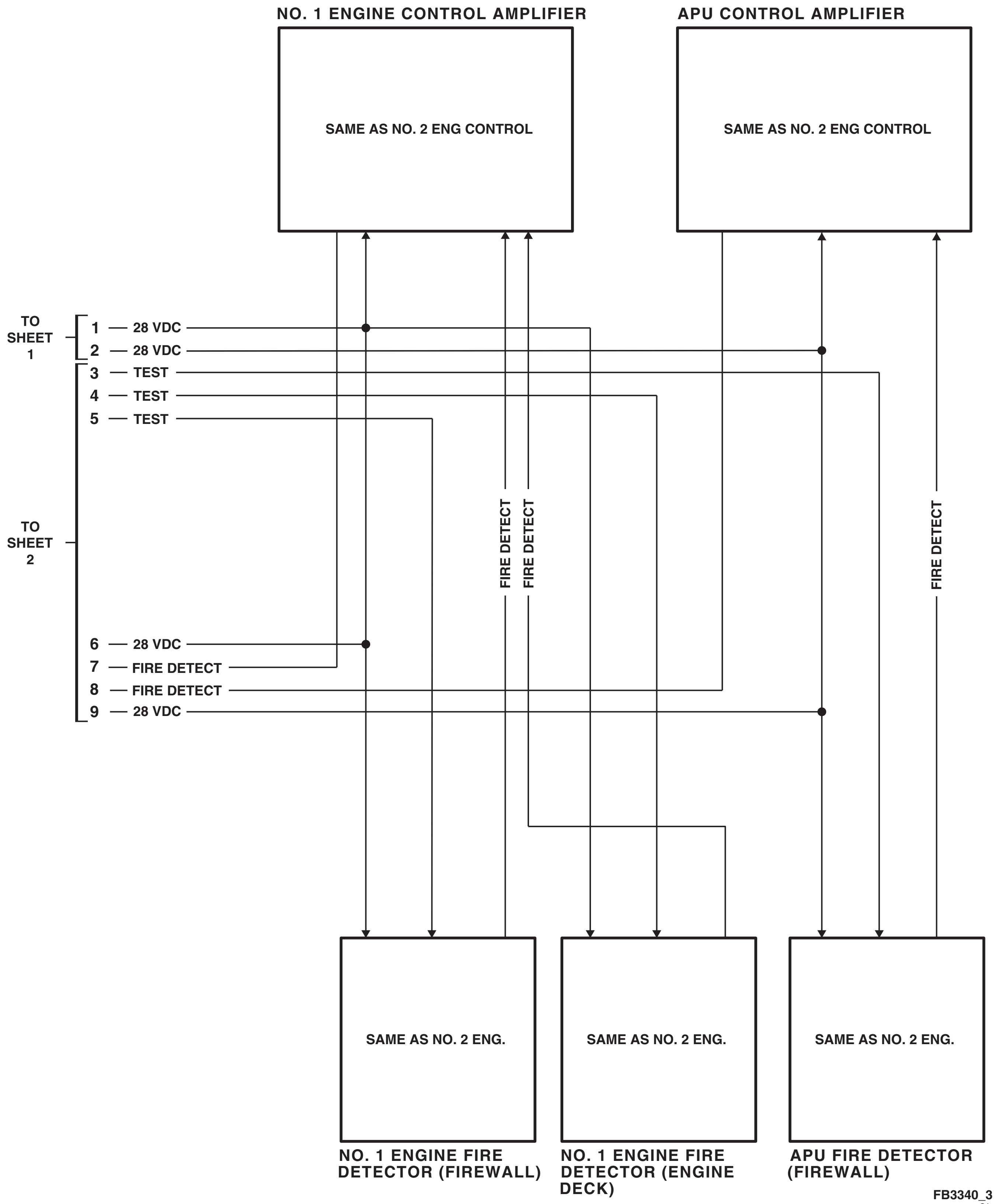
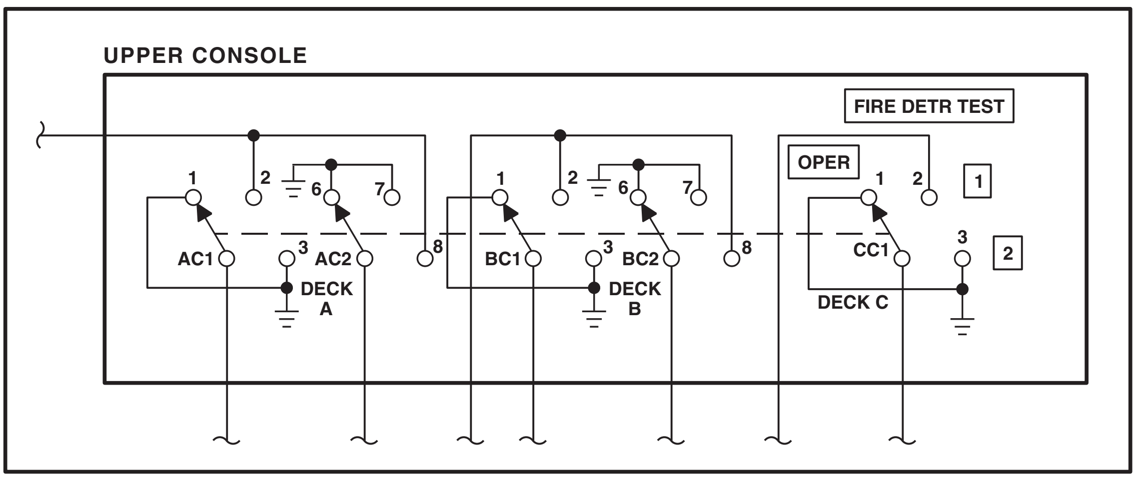


FIGURE 12-1-1-2. FIRE DETECTION SYSTEM BLOCK DIAGRAM (SHEET 3 OF 4)



NOTE
W/O EMEP

AA7815_4
SA

FIGURE 12-1-1-2. FIRE DETECTION SYSTEM BLOCK DIAGRAM (SHEET 4 OF 4)

12-1-2 FIRE EXTINGUISHING SYSTEM DESCRIPTION AND DATA.

12-1-2.1 Fire Extinguishing System Description.

The Fire Extinguishing System puts out fire in the APU or in either of the main engine compartments (Figure 12-1-2-1 and Figure 12-1-2-2).

The Fire Extinguishing System consists of:

a. Two Pressurized Containers (pressurized fire bottles)

b. Overboard Discharge Line

c. Discharge Indicator On The Right Side Of The Helicopter

d. Electrical Switches

e. Electrical Wiring

The main and reserve pressurized fire bottles are charged with 2.5 pounds of bromotrifluoromethane and both have a pressure gage. Each fire bottle serves as a backup for the other, thereby providing a two-shot capability to extinguish fires in either main engine compartment or in the APU compartment.

A red indicator disc, at the end of the overboard discharge line, provides visual aid to determine whether the fire bottles have been discharged. A broken-out red disc indicates that a fire bottle has discharged and the fire bottle must be replaced.

Upon impact of a crash of 10 Gs or more, an inertia switch automatically fires both containers into both main engine compartments.

12-1-2.2 Principles of Operation.

12-1-2.2.1 Power Distribution

Electrical power of 28 VDC is supplied by NO. 2 DC PRI BUS through the FIRE EXTGH circuit breaker, on pilot's circuit breaker panel, to the FIRE EXTGH/RESERVE/OFF/MAIN switch and to the # 2 ENG EMER OFF T-handle.

Electrical power of 28 VDC is supplied by BATT UTIL BUS through FIRE EXTGH circuit breaker, on lower console circuit breaker panel, to the FIRE EXTGH/RESERVE/OFF/MAIN switch and to the # 1 ENG EMER OFF T-handle.

12-1-2.2.2 Fire Extinguishing System Operation.

If a fire is detected in the No. 1 engine, No. 2 engine, or APU compartment, a light in the extinguishing agent T-handle on the upper console will go on (Figure 12-1-2-2).

Pulling out the lighted T-handle selects the compartment to which the extinguishing agent will be discharged (Figure 12-1-2-2).

The FIRE EXTGH switch on the upper console is spring-loaded OFF and controls the release of the extinguishing agent. Moving the switch to either MAIN or RESERVE selects the container from which the extinguishing agent will be discharged.

12-1-2.2.3 T-handle Controls.

Three T-shaped handles select the compartment into which the fire extinguishing agent is to be directed.

Two of the T-handles are on the engine control quadrant. The T-handles are marked #1 ENG EMER OFF and #2 ENG EMER OFF. The third T-handle, on the upper console, is marked APU.

Pulling a T-handle arms the fire extinguisher circuit for the engine compartment selected. It also shuts off fuel to the affected engine.

12-1-2.2.4 No. 1 Engine Main and Reserve Extinguisher Operation.

Pulling the #1 ENG EMER OFF T-handle closes the T-handle switch. This applies 28 VDC to the fire extinguisher logic module, arming the Fire Extinguishing System.

Setting the FIRE EXTGH switch (S20) to MAIN supplies control voltages directly into the logic module.

The control voltages are then routed to an explosive squib. This fires the squib, emptying the fire bottles contents into the piping leading to the No. 1 engine compartment.

To actuate the reserve fire bottle squib, the FIRE EXTGH switch (S20) must be set to RESERVE.

12-1-2.2.5 No. 2 Engine Main and Reserve Extinguisher Operation.

The extinguishing agent is applied to the No. 2 engine compartment in an identical manner as the No. 1 engine compartment.

When the #2 ENG EMER OFF T-handle for No. 2 engine is pulled, and the FIRE EXTGH switch (S20) set to MAIN, squib is fired.

To actuate the reserve fire bottle squib, the FIRE EXTGH switch (S20) must be set to RESERVE.

12-1-2.2.6 APU and Reserve Extinguisher Operation.

Pulling the APU T-handle supplies 28 VDC to the logic module. This supplies a control voltage to a directional valve.

The directional valve, when actuated, switches to an APU outlet from its normal No. 1 engine position. Setting the FIRE EXTGH switch (S20) to MAIN will apply control voltage to squib on the main fire bottle. This fires the extinguishing agent into the APU compartment.

To actuate the reserve fire bottle squib, the FIRE EXTGH switch (S20) must be set to RESERVE.

Two-way check valves placed in the pressure line ensure proper flow of fire extinguishing agent into the selected compartment.

In a crash, the impact switch S1, mounted in the left relay panel, closes. This actuates a control relay K24. Contacts of the relay apply 28 VDC to hold the relay coil in an energized state. Relay K24 also feed voltages to the squibs, discharging the contents of the bottle into both main engine compartments.

12-1-2.3 Component Description.

12-1-2.3.1 Pressurized Fire Bottles.

Both Pressurized Fire Bottles are mounted on the upper deck. Each fire bottle is charged with 600 psi of nitrogen pressure and 2.50 pounds of bromotrifluoromethane agent.

There are two outlets on each fire bottle, each with its own explosive squib. A pressure gage is supplied, as well as a thermal discharge relief port.

Thermal plugs permit discharge of the fire bottle contents, should ambient temperature go above a range of 214 - 226°F (101 - 108°C).

If the red plastic indicator disc is broken, it indicates a thermally discharged fire.

12-1-2.3.2 Directional Control Valve.

The directional control valve, mounted in the piping, is used for fire bottle discharge into the No. 1 engine compartment or into the APU compartment.

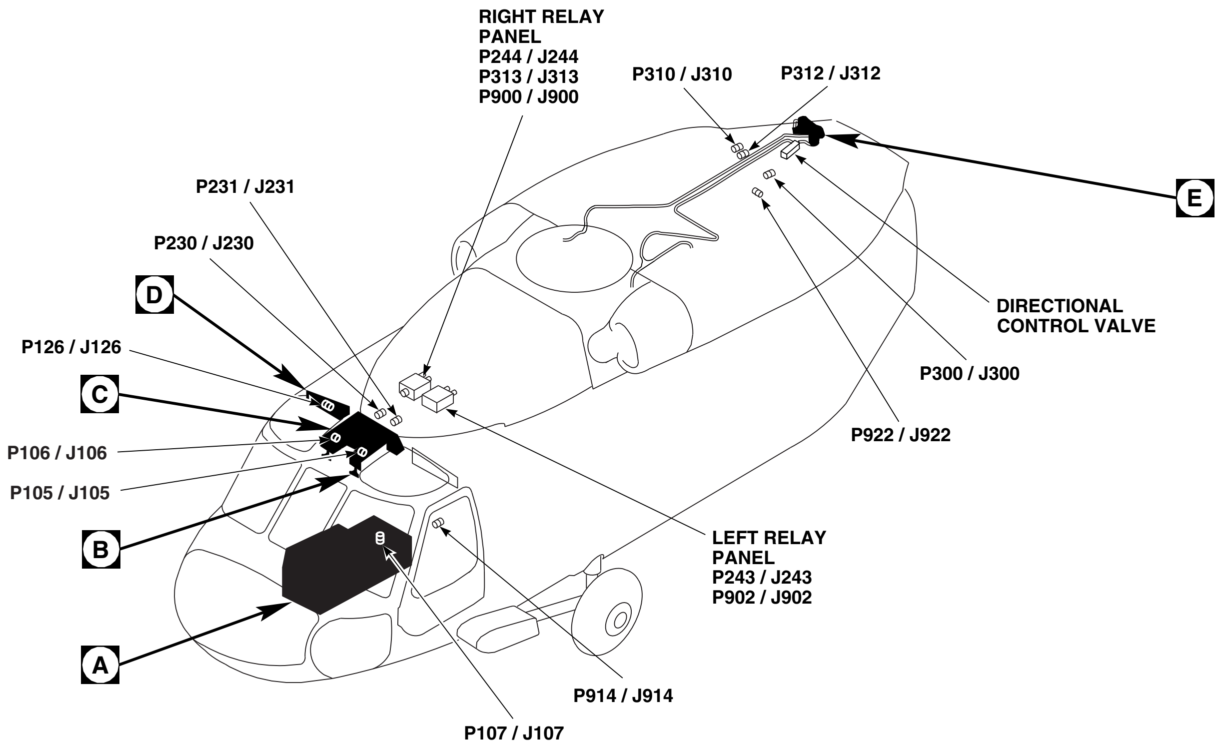
Pulling the APU T-handle applies a signal to the logic module, activating the directional control valve. This permits discharge of the agent into the APU compartment.

12-1-2.3.3 Impact Switch.

The impact switch (S1) is mounted in the left relay panel.

When the impact switch senses a 10 G force, it automatically discharges both fire bottles.

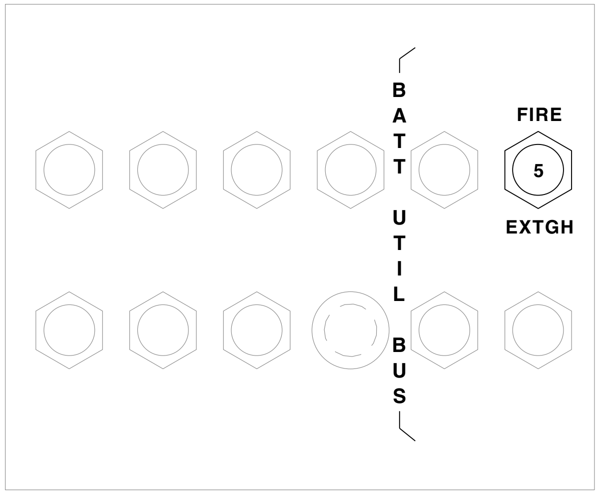
When the impact switch is closed, 28 VDC is applied to relay coil K24 in the right relay panel. The contacts of K24 place 28 VDC at the explosive squibs on the main fire bottle. This discharges the agent into both engine compartments.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P105 / J105	NO. 1 ENGINE CONTROL QUADRANT
P106 / J106	NO. 2 ENGINE CONTROL QUADRANT
P107 / J107	BATT BUS PANEL
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P243 / J243	LEFT RELAY PANEL ASSEMBLY
P244 / J244	RIGHT RELAY PANEL ASSEMBLY

FL0583_1
SA

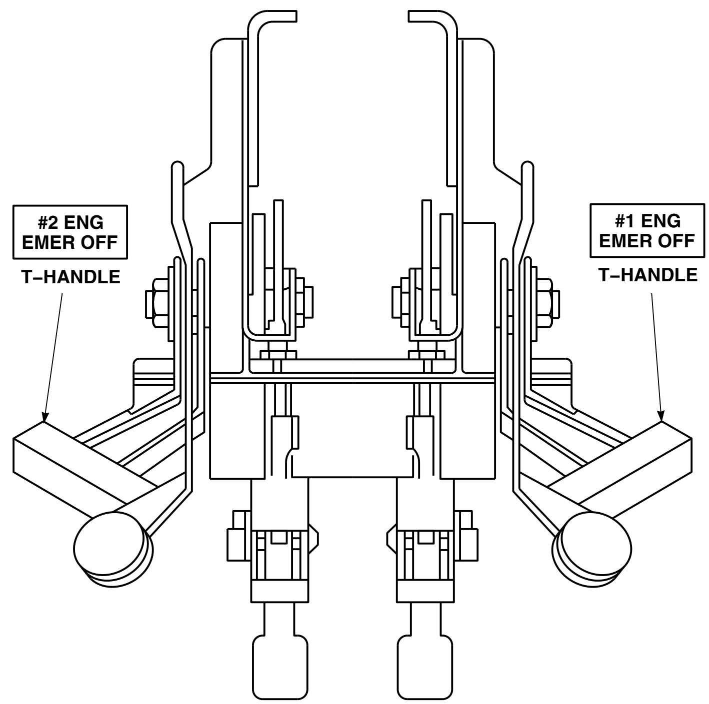
FIGURE 12-1-2-1. FIRE EXTINGUISHING SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)



LOWER CONSOLE
CIRCUIT BREAKER PANEL

A

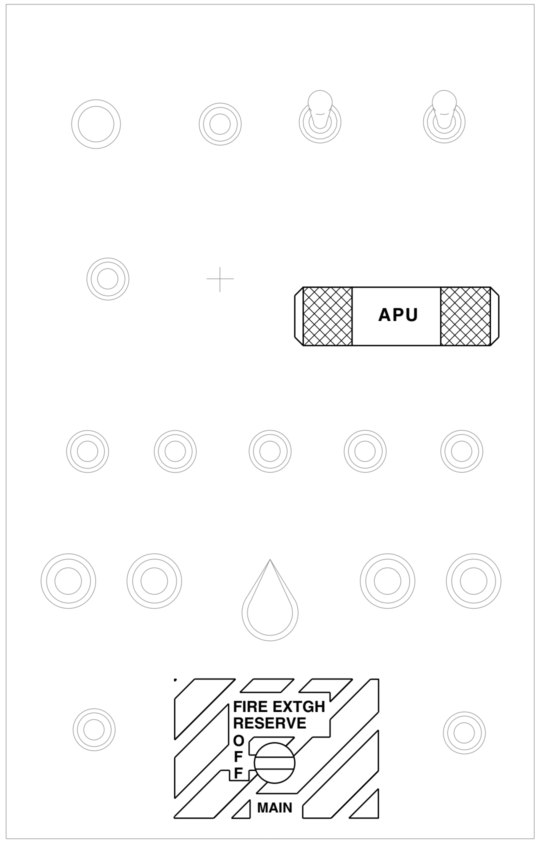
B



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P300 / J300	TRANSITION DECK BL 15 RH, STA 415
P310 / J310	CABIN CEILING BL 17 RH, STA 387
P312 / J312	CABIN CEILING BL 16.50 LH, STA 383
P313 / J313	RIGHT RELAY PANEL ASSEMBLY
P900 / J900	RIGHT RELAY PANEL ASSEMBLY
P902 / J902	LEFT RELAY PANEL ASSEMBLY
P914 / J914	COCKPIT BL 24 LH, STA 247
P922 / J922	CABIN CEILING BL 10.7 LH, STA 380

FB4464_2
SA

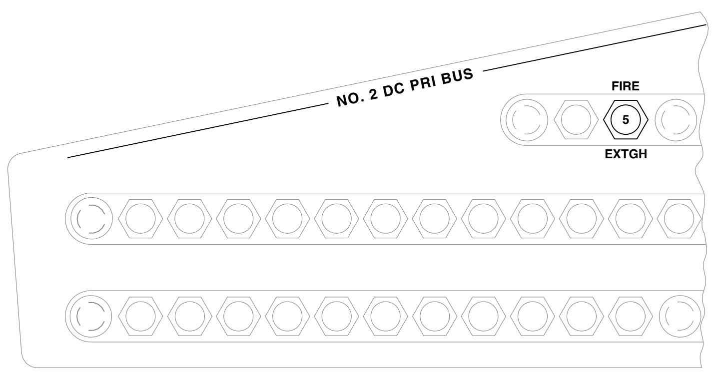
FIGURE 12-1-2-1. FIRE EXTINGUISHING SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)



UPPER CONSOLE

C

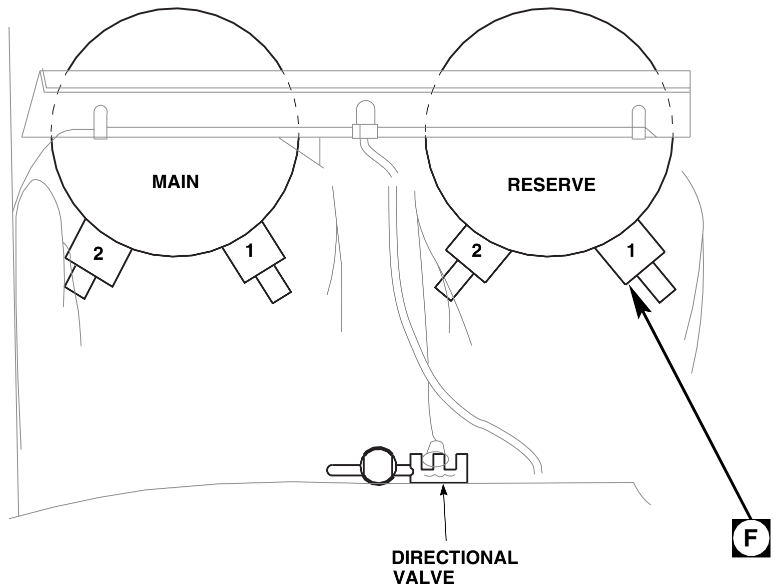
D



PILOT'S CIRCUIT BREAKER PANEL

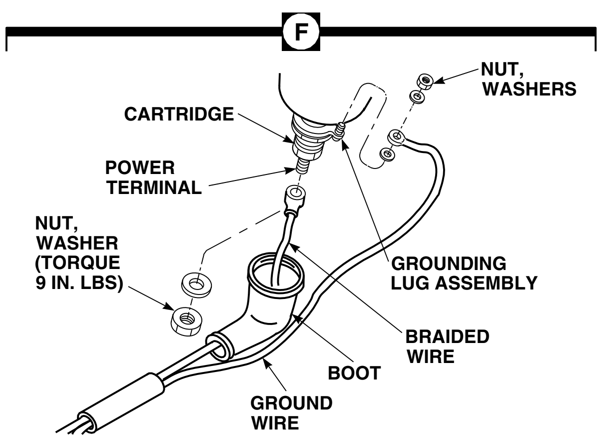
AA2794_3A
SA

FIGURE 12-1-2-1. FIRE EXTINGUISHING SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)

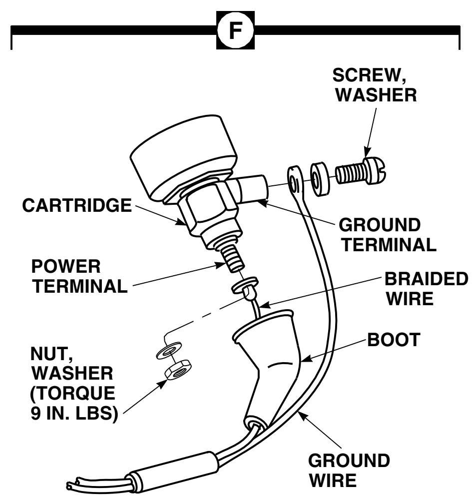


MAIN AND RESERVE FIRE BOTTLES
(LOOKING TO REAR)

E



EFFECTIVITY
WALTER KIDDE FIRE EXTINGUISHING CONTAINER, 898153.



EFFECTIVITY
HTL FIRE EXTINGUISHING CONTAINER, 30400004.

FO0069_4A
SA

FIGURE 12-1-2-1. FIRE EXTINGUISHING SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)

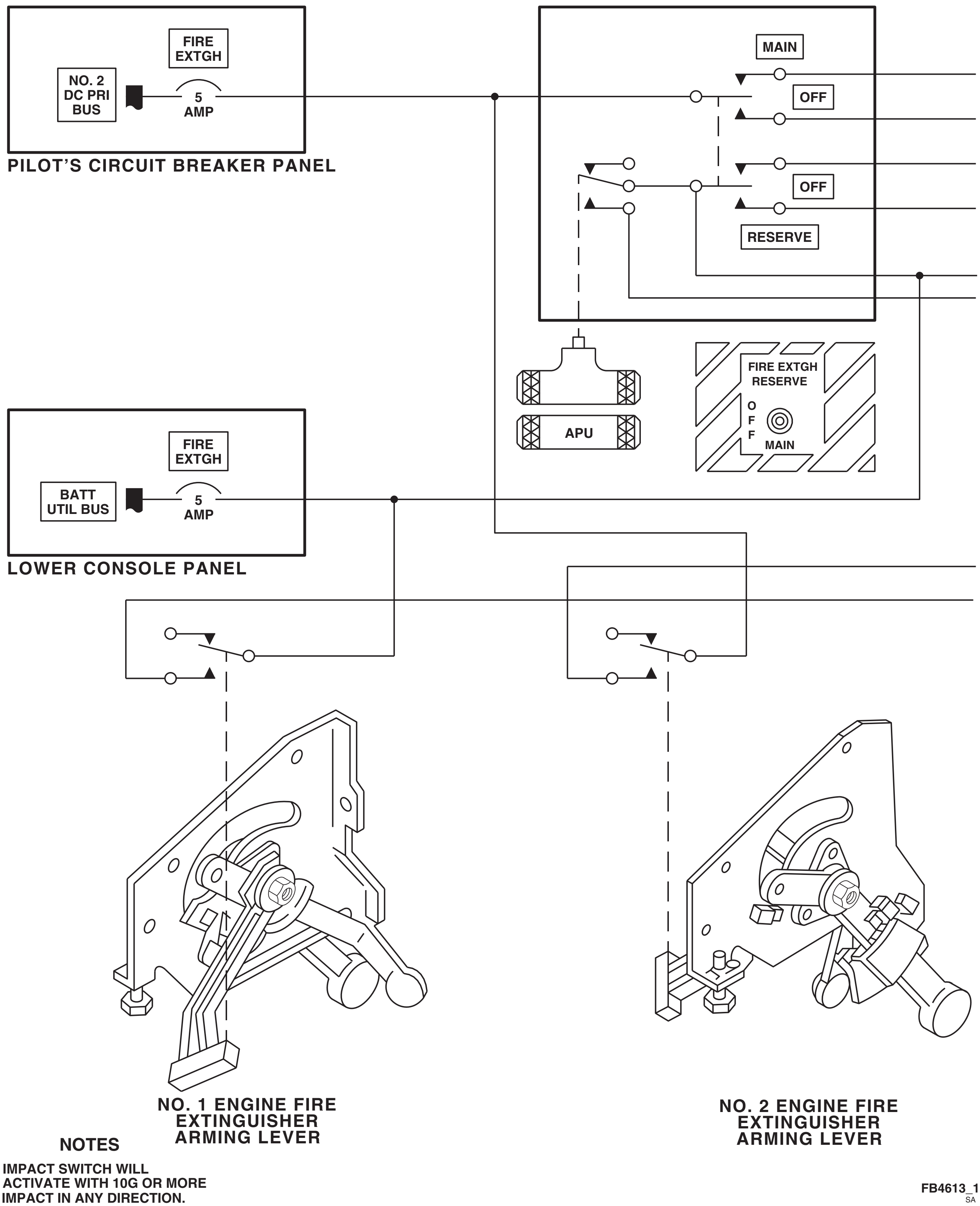


FIGURE 12-1-2-2. FIRE EXTINGUISHING SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

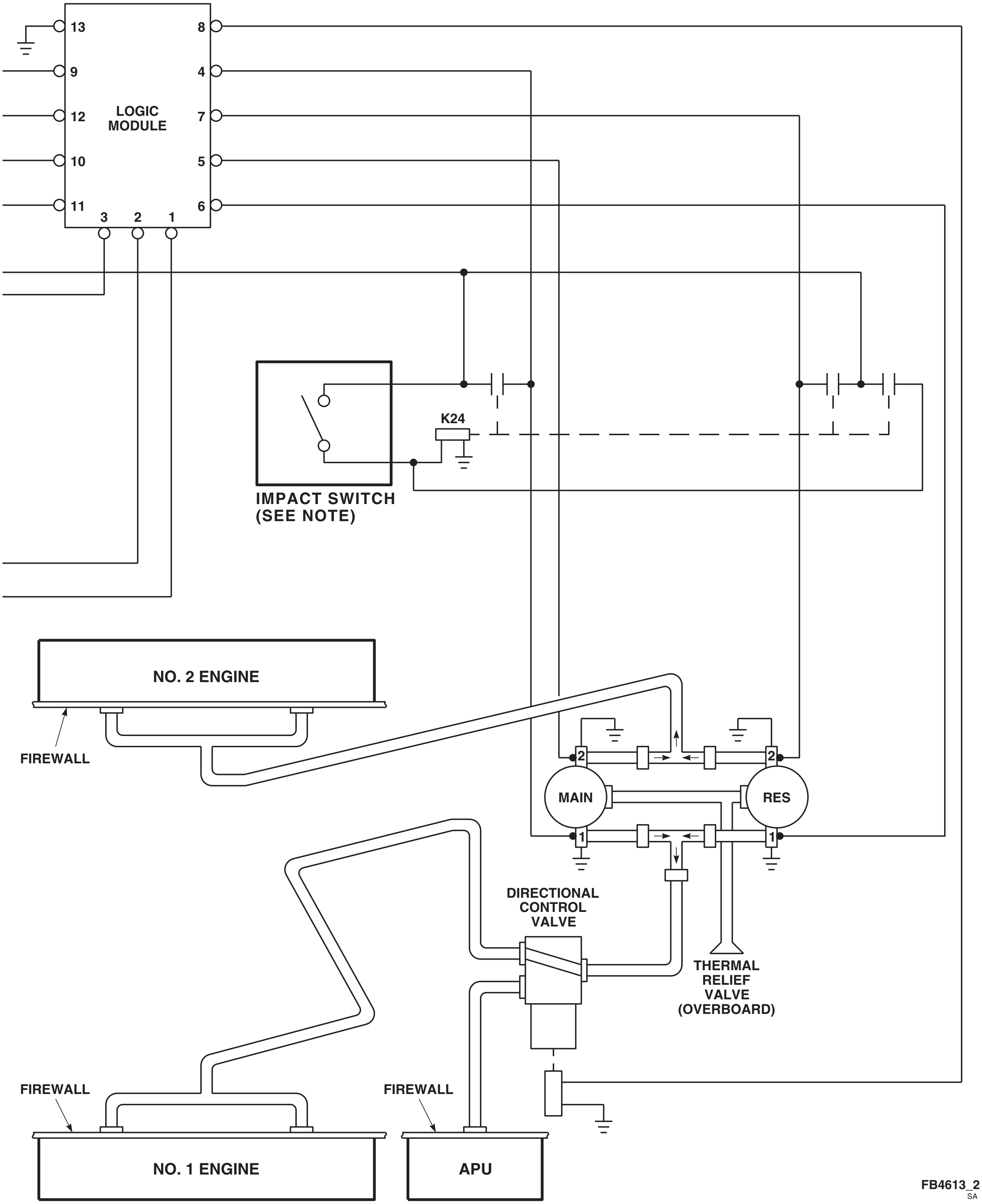


FIGURE 12-1-2-2. FIRE EXTINGUISHING SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

12-1-3 WINDSHIELD WIPER SYSTEM DESCRIPTION AND DATA.

12-1-3.1 Windshield Wiper System Description.

The electrically-operated windshield wiper system consists of a two-speed ac motor, two converters, two wipers, and a control switch (Figure 12-1-3-1). The HI and LOW positions of the WINDSHIELD WIPER control switch, on the upper console, control the wiper blade speed. The PARK position is used to return the wiper blade to the inboard edge of the windshields. Power for the wiper system is supplied by the No. 1 ac primary bus through the WSHLD WIPER circuit breaker on the copilot's circuit breaker panel.

12-1-3.2 Principles Of Operation.

12-1-3.2.1 Power Distribution.

Electrical power of 115 vac is supplied from the No. 1 ac primary bus through the WSHLD WIPER circuit breaker on the copilot's circuit breaker panel (Figure 12-1-3-2). The 115 vac is routed to the upper console panel and windshield wiper motor.

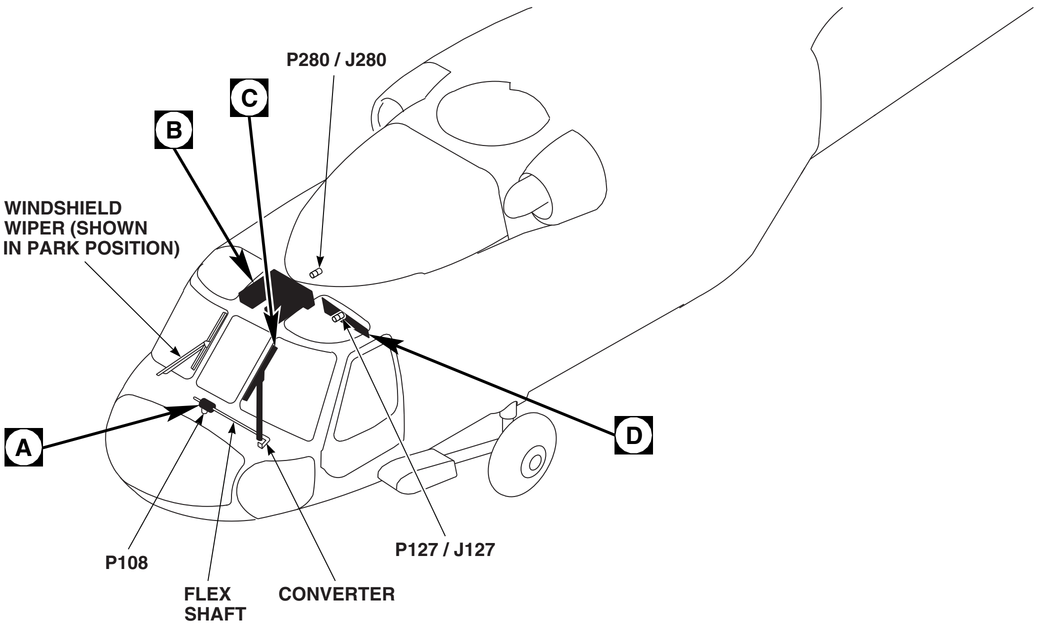
12-1-3.3 Component Description.

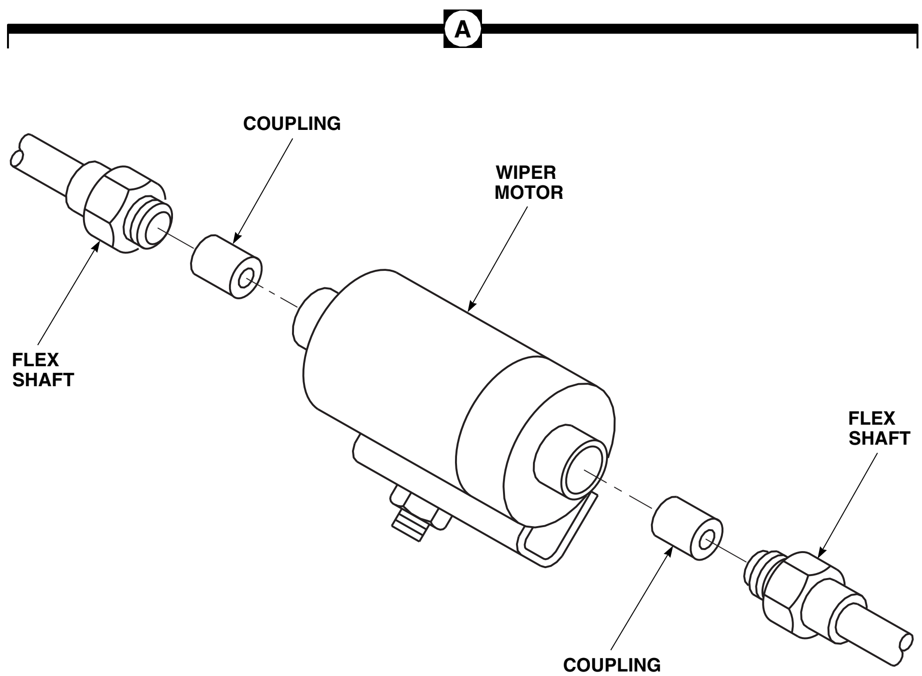
12-1-3.3.1 Wiper Motor.

The windshield wiper motor operates on 115 vac, three-phase 400 Hz. The motor contains a gear and cam-operated microswitch assembly and a diode rectifier. When the WINDSHIELD WIPER rotary switch is set to PARK and the cam-operated switch assembly closes, the diode rectifier converts the voltage and energizes the stator windings. The resulting magnetic field drags the rotor assembly to an almost instantaneous stop.

12-1-3.3.2 Converter.

Flexible drive arms transfer motor motion to the two converters. The converters contain gears that change motor rotary motion to a sweeping motion. Each wiper has one converter.





WIPER MOTOR

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P108	WINDSHIELD WIPER MOTOR
P127 / J127	COCKPIT CEILING BL 28 LH, STA 247
P280 / J280	CABIN CEILING, BL 6 LH, STA 247

AA3412_1
SA

FIGURE 12-1-3-1. WINDSHIELD WIPER SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

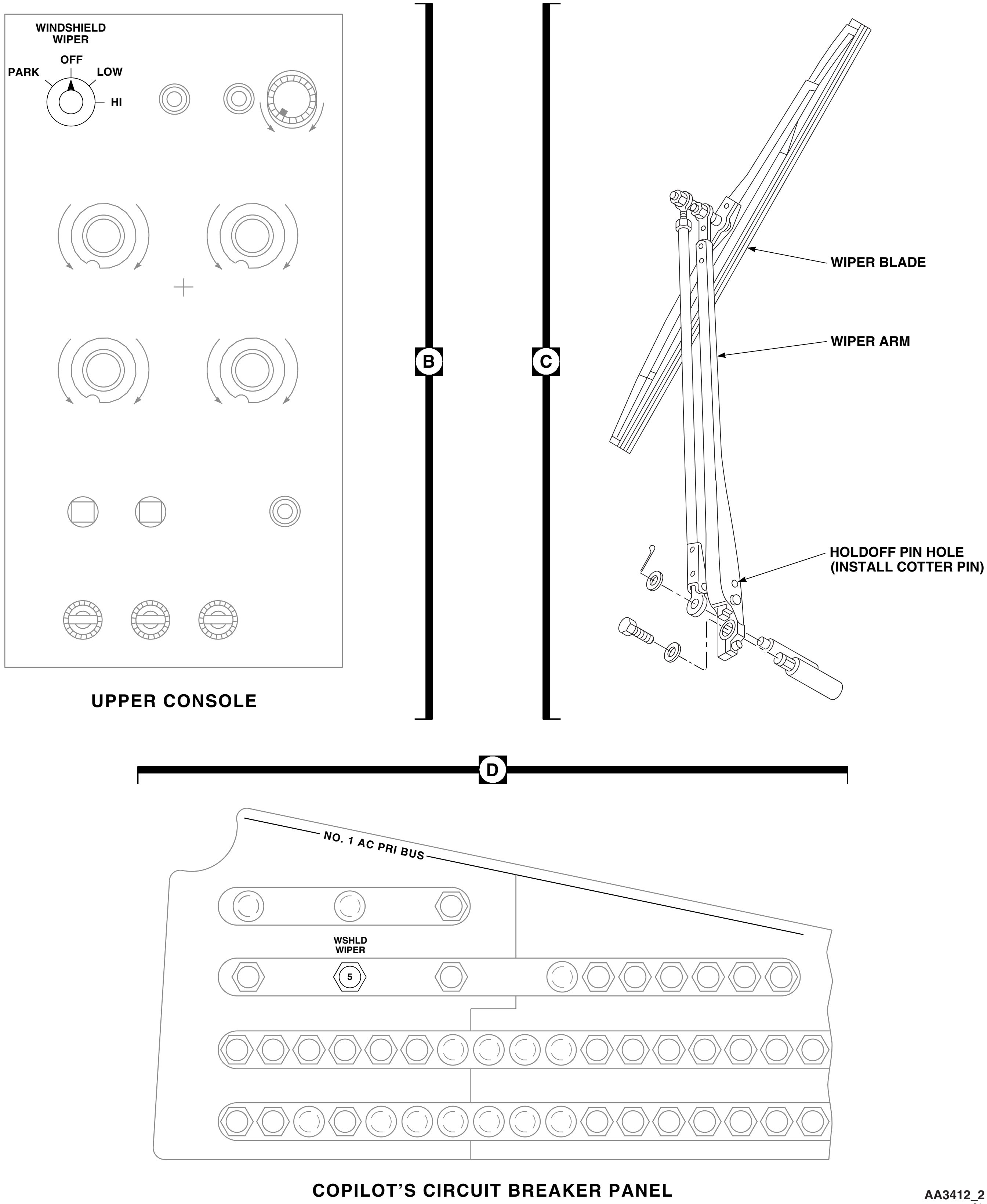


FIGURE 12-1-3-1. WINDSHIELD WIPER SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)

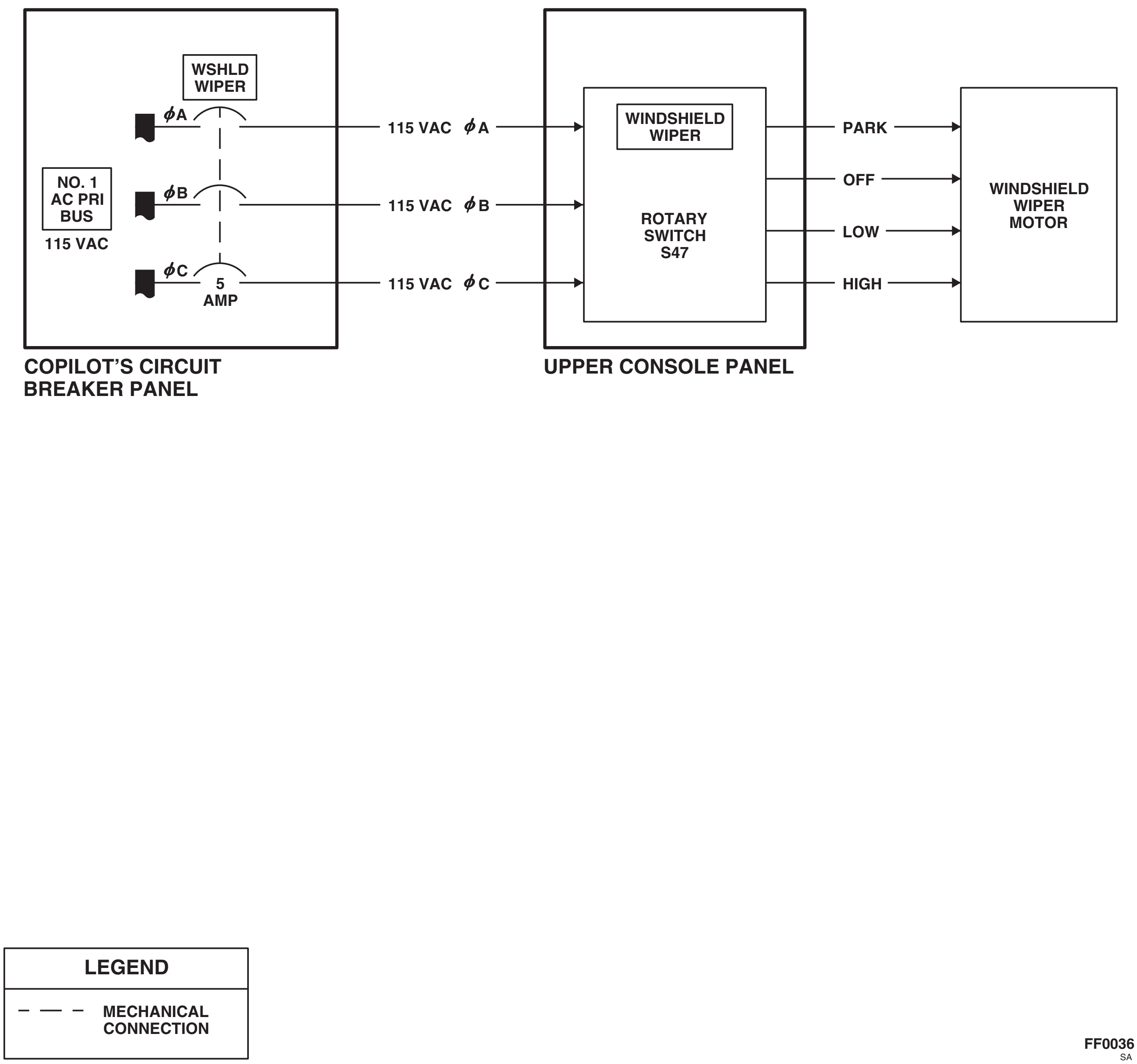


FIGURE 12-1-3-2. WINDSHIELD WIPER SYSTEM BLOCK DIAGRAM

12-1-4 WINDSHIELD ANTI-ICE SYSTEM DESCRIPTION AND DATA.

12-1-4.1 Windshield Anti-ice System.

The windshield anti-ice system prevents ice and fog from forming on the pilot's and copilot's windshields (Figure 12-1-4-1). The center windshield is also equipped with the anti-ice system. The anti-ice system applies three-phase, 115 vac power to heater elements within the windshield to keep a temperature of 21.1°C to 46.1°C (70°F to 115°F) on the windshield surface. The ac power is supplied to the windshield heater elements by anti-ice controllers that are activated by two temperature sensors within each windshield. The system consists of heater elements and temperature sensors in the pilot's and copilot's windshields, pilot's and copilot's anti-ice controllers on the right and left sides of the cabin overhead, and control switches on the upper console. The system also consists of heater elements and temperature sensors in the center windshield and center anti-ice controller on the left side of the cabin overhead, and a control switch on the upper console.

12-1-4.2 Power Distribution.

The windshield anti-ice system gets ac and dc electrical power from the pilot's and copilot's circuit breaker panels (Figure 12-1-4-2 and Figure 12-1-4-3). Electrical power of 28 vdc is supplied by the No. 1 and No. 2 dc primary buses and routed through the CPLT WSHLD ANTI-ICE, WINDSHIELD ANTI-ICE PILOT, and WINDSHIELD ANTI-ICE CTR circuit breakers. It is also routed through the WINDSHIELD ANTI-ICE COPILOT, WINDSHIELD ANTI-ICE PILOT, and WINDSHIELD ANTI-ICE CTR switches. Electrical power of 115 vac is supplied by the No. 1 and No. 2 ac primary buses and routed through the CPLT WSHLD ANTI-ICE, PILOT WSHLD ANTI-ICE, and CTR WSHLD ANTI-ICE circuit breakers to the copilot's, pilot's, and center anti-ice controllers.

12-1-4.3 Normal Operation.

Anti-ice operation for the pilot's, copilot's, and center windshields is the same. The copilot's anti-ice operation is described. When the WINDSHIELD ANTI-ICE COPILOT switch is placed ON, 28 vdc control voltage is applied through the switch and normally-closed contacts of automatic cutout relay K21, in the right relay panel, to the copilot's anti-ice controller. Windshield heater element power of 115 vac is also applied to the controller. The temperature of the copilot's windshield is monitored by two parallel connected temperature sensors within the windshield. The sensors form one leg of a resistance bridge circuit in the anti-ice controller. Changes in windshield temperature cause corresponding changes in sensor resistance, resulting in a bridge unbalance. The bridge unbalance produces a signal of specific phase that corresponds to a sensor resistance above or below the bridge balance point. This signal is phase-detected and compared with a 400 Hz reference signal within the controller. When windshield temperature decreases to a value giving a sensor resistance between 167 and 169 ohms, an in-phase bridge unbalance signal is produced. This signal turns on the controller that applies 115 vac power to the windshield heater elements to heat the windshield. When the temperature of the windshield increases to a value giving a sensor resistance between 4 and 5 ohms above the turn-on resistance value, the unbalanced bridge circuit produces an out-of-phase signal that turns off the controller.

The windshield anti-ice system contains protection circuitry to prevent damage to the windshield in case of a fault. If a windshield temperature sensor opens or shorts, or if there is a loss of ac or dc power, the anti-ice controller removes power from the windshield heater elements. The system is also automatically shut off if the auxiliary power unit (APU) generator is the only source of electrical power and the backup hydraulic pump is on. Under these conditions, 28 vdc from the APU is applied through the normally closed contacts of No. 2 and No. 1 ac generator contactors K2 and K1, through the energized contacts of APU/external power contactor K3, and the energized contacts of hydraulic emergency relay K19 to energize automatic cutout relay K21. With K21 energized, the 28 vdc is removed from the anti-ice controller and the system shuts off.

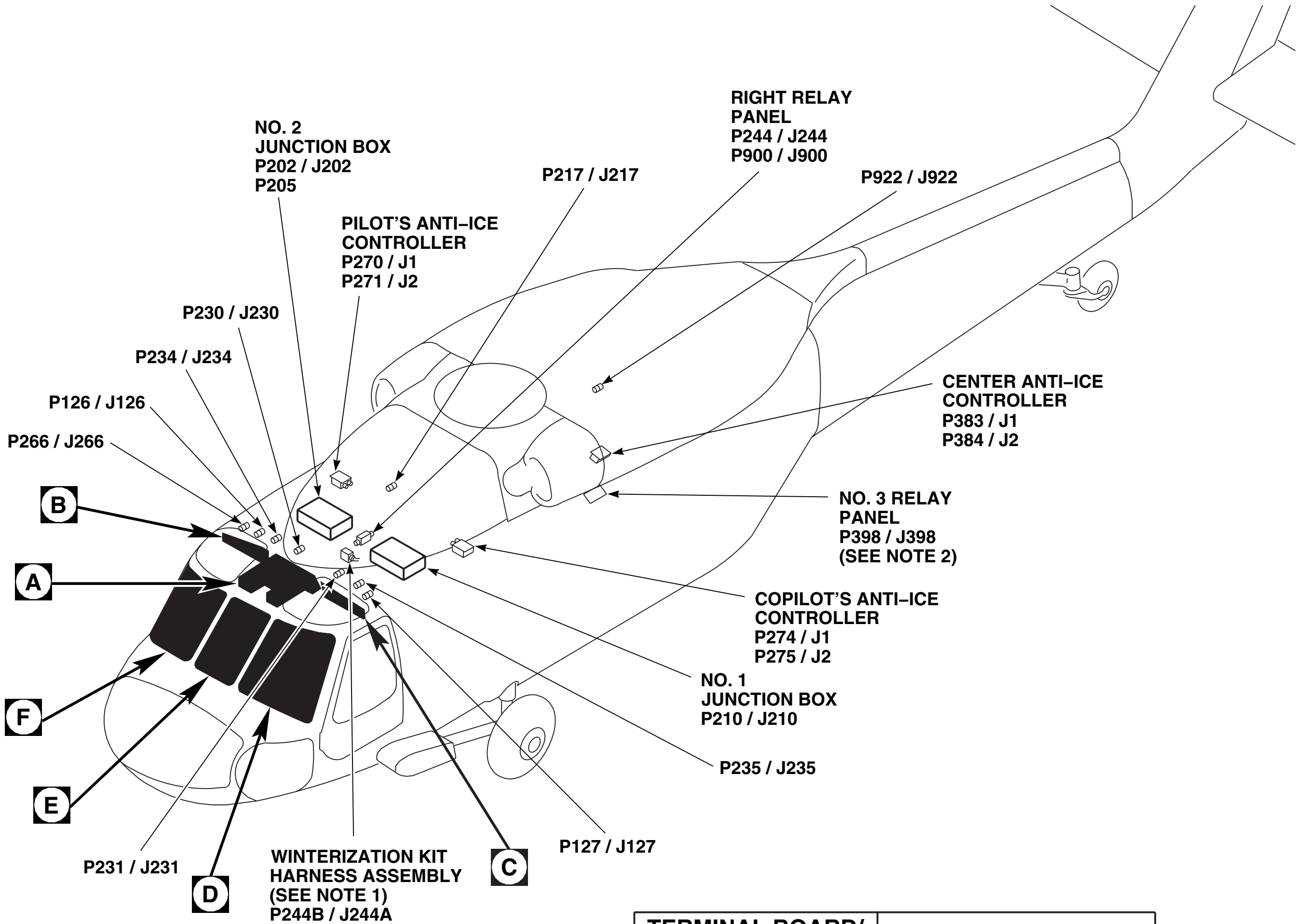
12-1-4.4 Component Description.

12-1-4.4.1 Temperature Sensors.

Two temperature sensors are embedded in each windshield. Each sensor is a variable resistor that changes with temperature changes. A decrease in temperature results in a larger sensor resistance. If there is an increase in temperature, it results in a smaller sensor resistance. Each pair of temperature sensors form a leg of a bridge circuit in its respective windshield anti-ice controller.

12-1-4.4.2 Windshield Anti-ice Controller.

The pilot's windshield anti-ice controller is in the right cabin overhead. The copilot's windshield anti-ice controller is in the left cabin overhead. The center windshield anti-ice controller is in the left cabin overhead. The windshield anti-ice controller applies or removes power to the windshield heating elements, maintaining the windshield temperatures at 21°C to 46°C (70°F to 115°F).



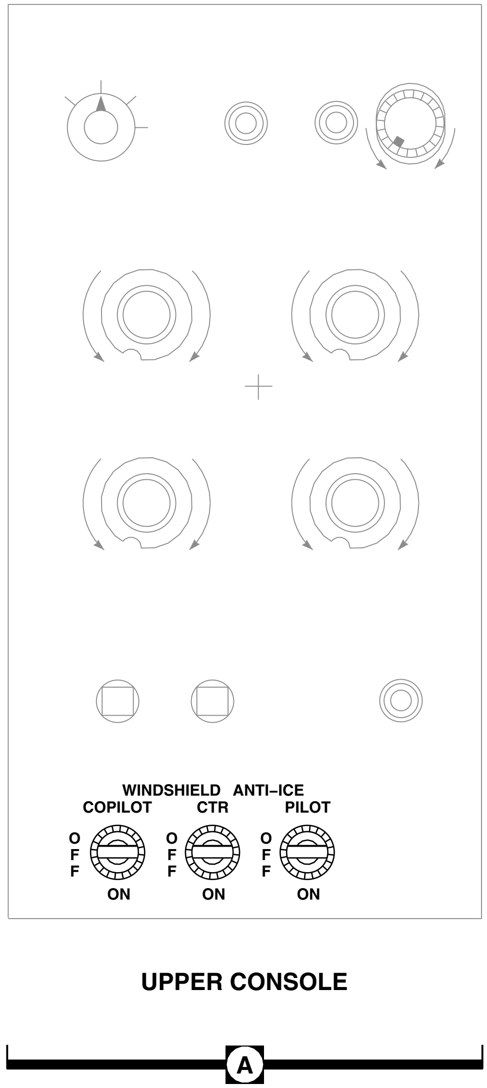
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P126 / J126	COCKPIT CEILING, BL 28 RH, STA 243
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243
P202 / J202	NO. 2 JUNCTION BOX
P205	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P217 / J217	MAIN ROTOR PYLON DECK, BL 0, STA 284
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P234 / J234	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 20 RH, STA 247
P235 / J235	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 20 LH, STA 247
P244 / J244	RIGHT RELAY PANEL
J244A / P244B (SEE NOTE 1)	WINTERIZATION KIT HARNESS ASSEMBLY

NOTES

1. WINTER
2. WITH ECS INSTALLED.

FB1603_1C
SA

FIGURE 12-1-4-1. WINDSHIELD ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 25 RH
P270 / J1	PILOT'S ANTI–ICE CONTROLLER
P271 / J2	PILOT'S ANTI–ICE CONTROLLER
P274 / J1	COPILOT'S ANTI–ICE CONTROLLER
P275 / J2	COPILOT'S ANTI–ICE CONTROLLER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P383 / J1	CENTER ANTI–ICE CONTROLLER
P384 / J2	CENTER ANTI–ICE CONTROLLER
P398 / J398	NO. 3 RELAY PANEL
P900 / J900	RIGHT RELAY PANEL
P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380

FB1603_2A
SA

FIGURE 12-1-4-1. WINDSHIELD ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)

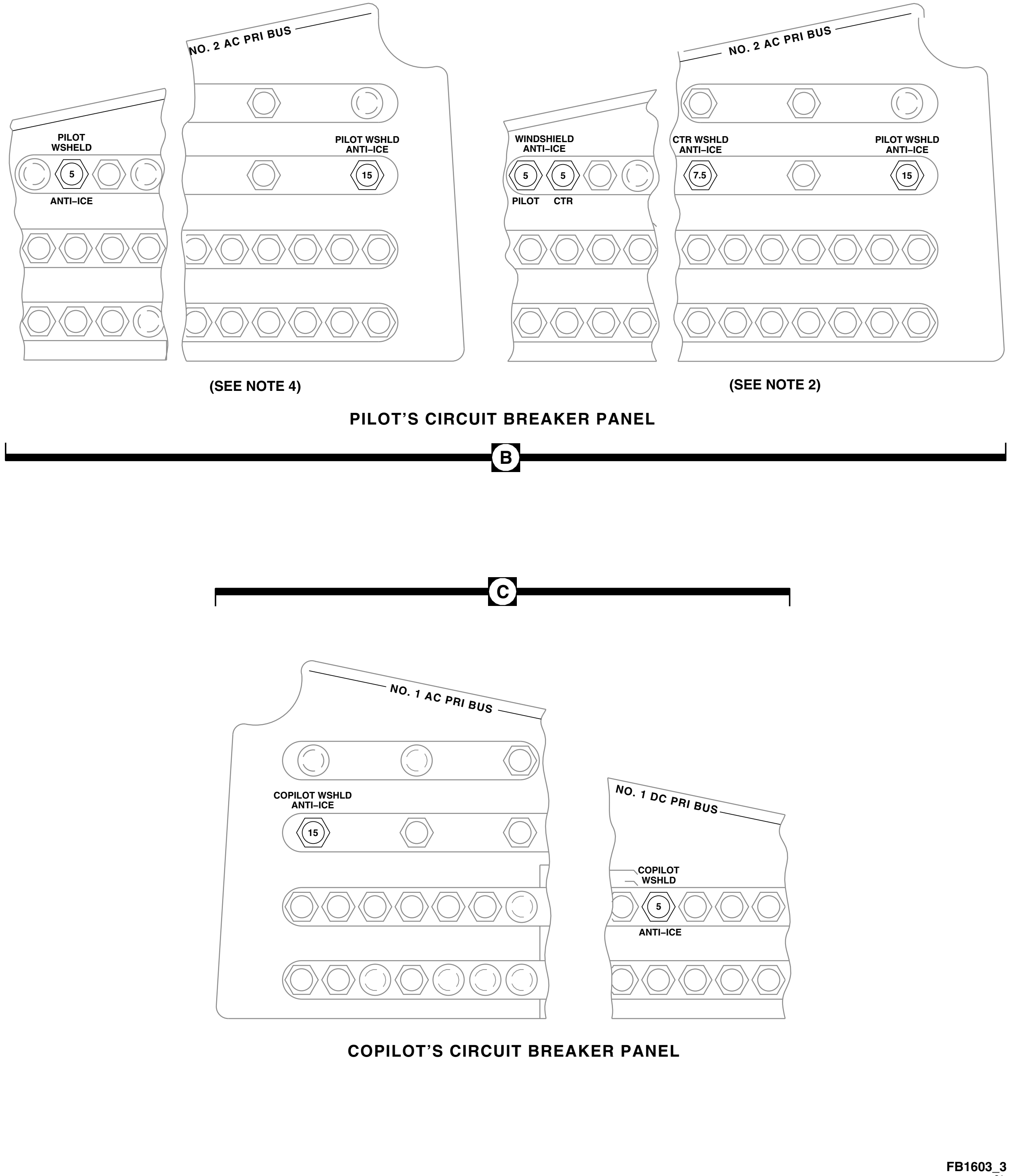


FIGURE 12-1-4-1. WINDSHIELD ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)

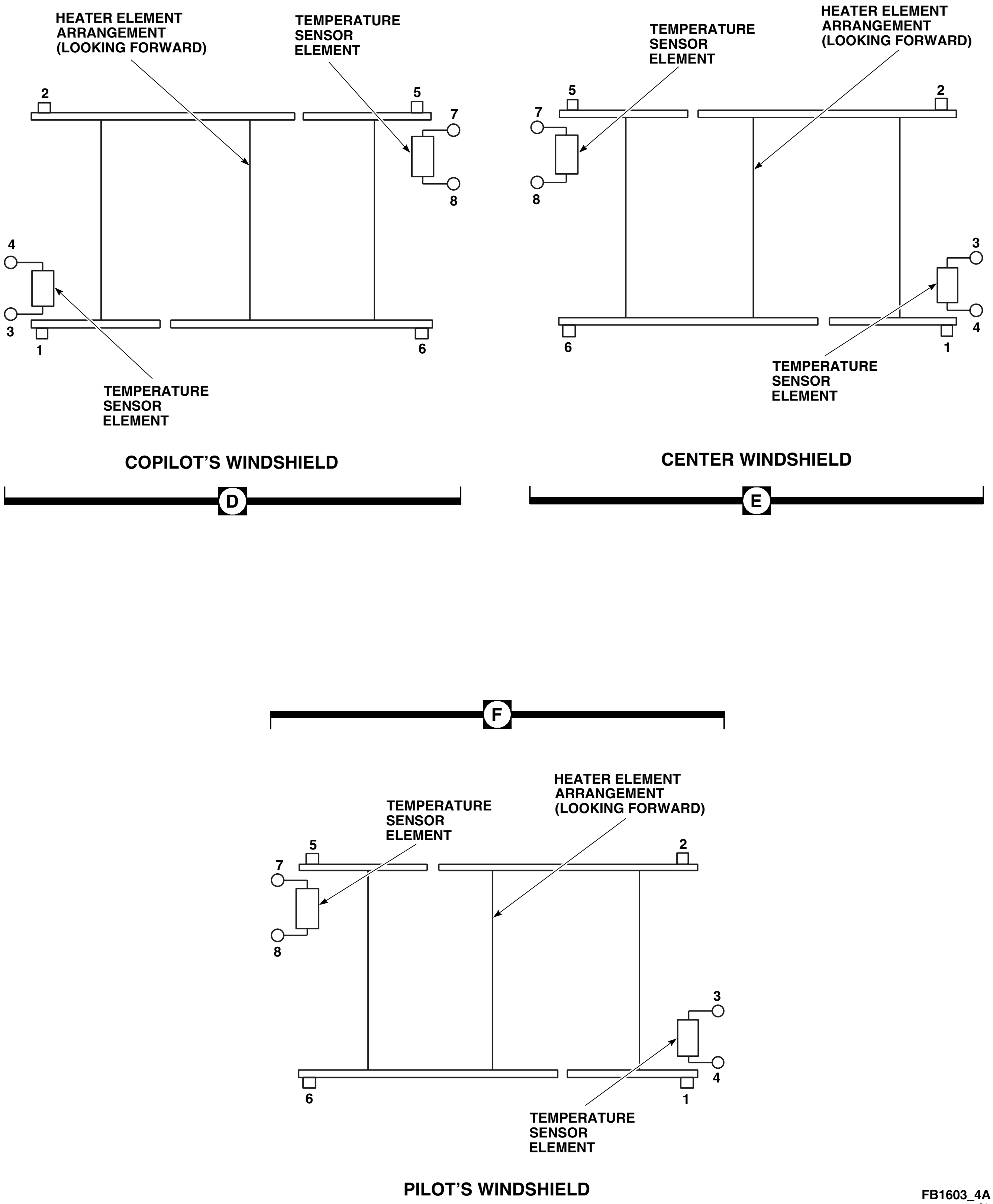
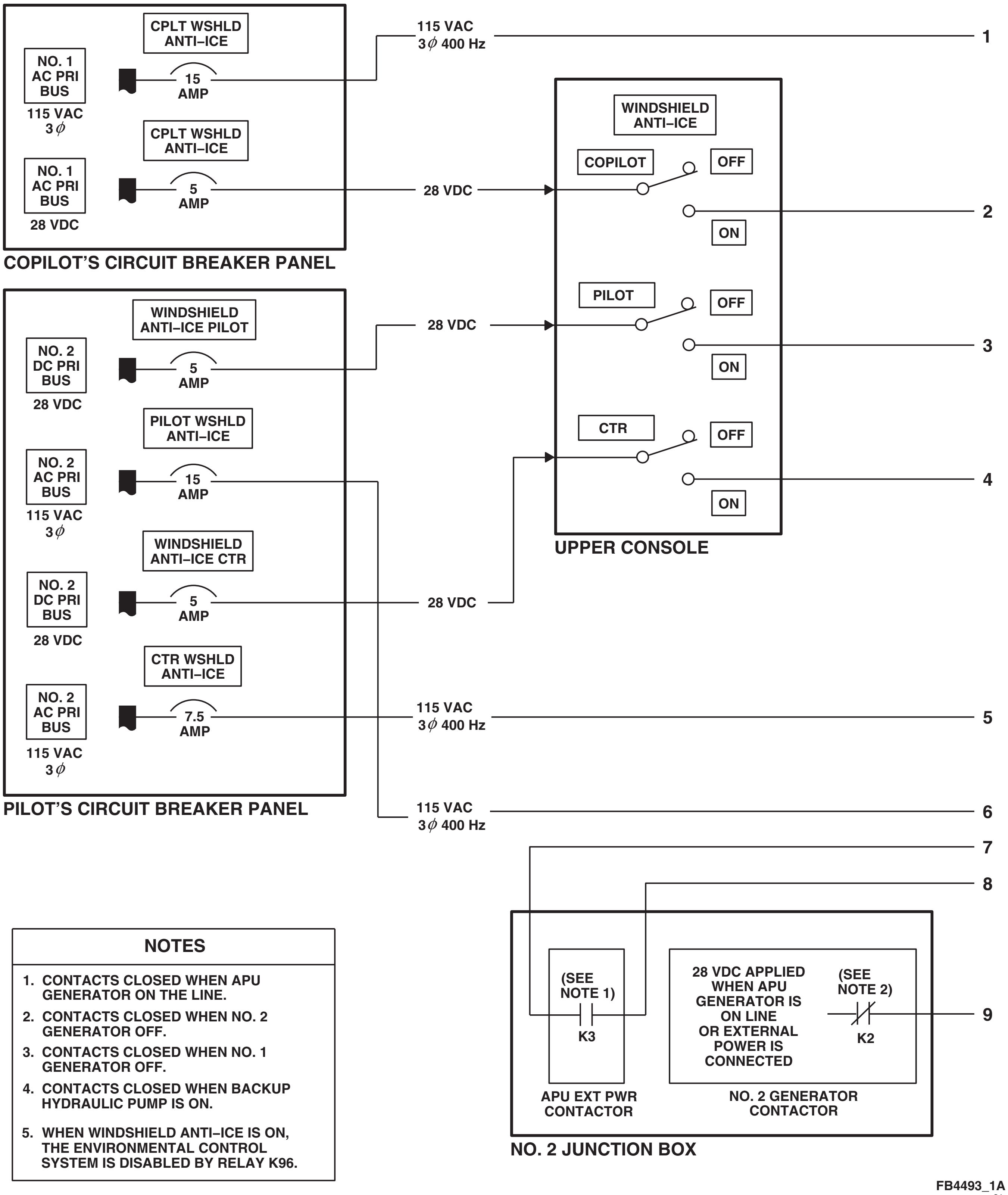


FIGURE 12-1-4-1. WINDSHIELD ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)



NOTES

1. CONTACTS CLOSED WHEN APU GENERATOR ON THE LINE.
2. CONTACTS CLOSED WHEN NO. 2 GENERATOR OFF.
3. CONTACTS CLOSED WHEN NO. 1 GENERATOR OFF.
4. CONTACTS CLOSED WHEN BACKUP HYDRAULIC PUMP IS ON.
5. WHEN WINDSHIELD ANTI–ICE IS ON, THE ENVIRONMENTAL CONTROL SYSTEM IS DISABLED BY RELAY K96.

FIGURE 12-1-4-2. WINDSHIELD ANTI-ICE SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

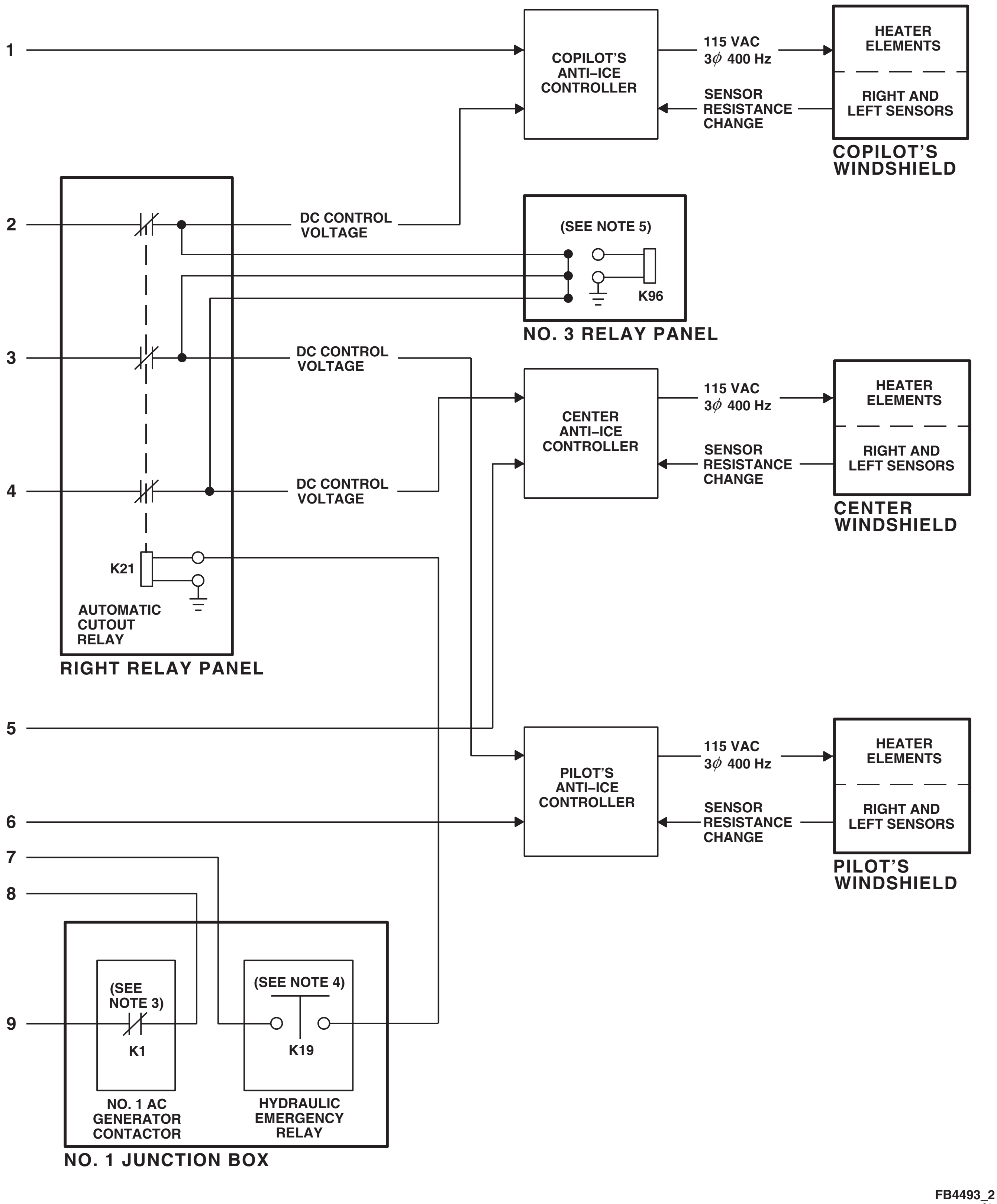


FIGURE 12-1-4-2. WINDSHIELD ANTI-ICE SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

12-1-5 BLADE DE-ICING SYSTEM DESCRIPTION AND DATA.

12-1-5.1 Blade De-icing System.

The blade de-icing system provides for de-icing of the main and tail rotor blades and droop stops. The de-icing system may be operated in either of two selectable modes, manual or automatic (Figure 12-1-5-1 and Figure 12-1-5-2). The automatic mode uses a signal from the outside air temperature (OAT) sensor to determine the amount of time the heating current is applied to the main and tail rotor blades. The manual mode provides selection of three predetermined de-icing conditions: trace, light, and moderate. The main and tail rotor channels of the de-ice system electrically heat mats bonded into the leading edges of each blade. Each main rotor blade contains four separate heating zones. De-icing current is sequentially supplied to identical zones of opposite blades to assure symmetrical ice shedding. The amount of time heating current is applied to each blade zone is determined by the OAT. This is sensed by an OAT sensor. The time between heating cycles is determined by an ice detector probe. The signals from both the OAT sensor and the ice detector probe are applied to the main and tail rotor channels in the de-ice controller. Each tail rotor blade contains only one heating zone, and all tail rotor blades are simultaneously pulsed by de-icing current. The amount of time heating current is supplied to the blades in the automatic mode is greater as the OAT decreases. In the manual mode, the time between heating cycles is determined by the mode selector switch setting. A blade de-ice test panel permits testing of the internal fail detection circuits in the de-ice controller. At NORM, the test panel permits normal system operation. At SYNC 1 and SYNC 2, the main rotor synchronization fail detection circuits are tested. At OAT, a short-circuited OAT sensor is simulated, and fail detection circuits in main and tail rotor channels are tested. At element on time (EOT), faulty element-on-timers are simulated in the main and tail channels, and their associated failure detection circuits are tested. The PWR MAIN RTR and PWR TAIL RTR lamps monitor power application to the main and tail rotor blades. The lamps will go on during system test, any time power remains applied to the blades with the MR DE-ICE FAIL or TR DE-ICE FAIL caution capsule on and the system POWER switch ON, or any time power remains applied to the blades with the system POWER switch OFF. The droop stops are continuously heated as long as the blade de-ice control panel POWER switch is in the TEST or ON mode. The blade de-icing system is made up of an OAT sensor mounted on the fuselage forward of the pilot's window; an ice detector mounted on the right engine air inlet duct, with a sensing probe exposed to outside air; an icing rate meter, blade de-ice control panel, and blade de-ice test panel, all of which are mounted on the instrument panel. The blade de-icing system also includes a slipring mounted on the main and tail gear boxes, auxiliary ac and main rotor blade de-ice junction boxes, a blade de-ice controller mounted in the cabin containing separate main and tail rotor channels, and droop stop heaters built into the rotor head droop stops. The main and tail rotor blades are not part of the system. All blades contain resistive heating mats built into the blade during its construction.

Panel lighting of the icing rate meter panel is controlled by the INST LT NON FLT control on the upper console. For a further description of panel lighting, refer to instrument panel lights (PARA 9-1-8). The left relay panel connects the TEST IN PROGRESS lamp on the blade de-ice control panel and the PWR MAIN RTR and PWR TAIL RTR lamps on the blade de-ice test panel to dimming voltage and permits indicator lamp dimming and disabling functions. For a further description of indicator lighting, refer to instrument panel and consoles indicator lights dimming (PARA 9-1-8).

12-1-5.2 Power Distribution.

Power of 28 vdc from the No. 2 dc primary bus is applied through the DE-ICE CNTRLR circuit breaker, on the mission readiness circuit breaker panel, as follows: to the blade de-ice controller as a power supply voltage; through de-energized contacts of ac bus tie contactor and energized contacts of APU/external power contactor, or through diode CR5 energizing No. 1 and No. 2 generator contactors, and diode CR11 to the blade de-ice controller power monitor circuits; to the blade de-ice test panel as press-to-test lamp voltage; and to the blade de-ice control panel POWER switch (Figure 12-1-5-2). If this switch is placed ON, 28 vdc will energize the icing rate meter, the blade de-ice controller, and relay K64 via diode CR16. If the POWER switch is placed to TEST, 28 vdc will energize relay K64 through diode CR15 and contacts of relay K65, if the backup pump is not operating. This test voltage is also applied to the blade de-ice controller. No. 2 dc primary voltage is applied through the ICE DET circuit breaker, on the mission readiness circuit breaker panel, to the icing rate meter. Power of 28 vdc from the battery bus is applied through the APU GEN CONTR circuit breaker, through normally-open or normally-closed contacts of No. 1 and No. 2 generator contactors, and energized contacts of relay K64, to energize contactors K62 and K63.

Three-phase, 115 vac tail rotor heater power may come from one of two sources: from the No. 1 ac primary bus through the DE-ICE PWR TAIL ROTOR circuit breaker and normally-open contacts of contactor K63, or from the APU generator through normally-closed contacts of contactor K63. From contactor K63, power is routed through current limiters CL13, CL14, and CL15, contacts of contactor K61, and fault detection circuitry within the blade de-ice controller to the tail rotor sliprings. Three-phase, 115 vac main rotor heater power may come from one of two sources: from the No. 2 ac primary bus through current limiters CL7, CL8, and CL9, and normally-open contacts of contactor K62, or from the APU generator through normally-closed contacts of contactor K62. From contactor K62, power is routed through current limiters CL10, CL11, and CL12, contacts of contactor K60, and MN RTR BLADE DE-ICE CT transformers T14 and T14A to the main rotor sliprings. Single-phase, 115 vac for the blade de-ice test panel and icing rate meter is supplied by the No. 2 ac primary bus via the ICE DET circuit breaker.

12-1-5.3 Ice Detector Operation.

The ice detector senses ice accumulation on a vibrating probe by a change in probe frequency. The probe excitation frequency is supplied by the icing rate meter (nulling voltage from icing rate meter). A sensing coil returns the probe's oscillation frequency to the icing rate meter (icing output to icing rate meter drive coil). The icing rate meter processes the signal from the ice detector and visually displays icing intensity. Also, the icing rate meter sends an ice-detected 28 vdc signal to the ICE DETECTED caution capsule when the blade de-ice control panel POWER switch is OFF or at TEST. When the POWER switch is placed ON, the ice detector aspirator heater is turned on and the ICE DETECTED caution capsule is turned off. If the blade de-ice control panel MODE switch is at AUTO, the icing rate meter sends an ice rate signal through the switch to the de-ice controller. After the probe has generated a signal proportional to the rate of icing, 28 vdc from the icing rate meter is applied to the probe strut heater. This melts the ice accumulated on the probe. The probe heater operates for about five to seven seconds and then cools down to allow another icing rate sensing cycle. The icing rate signal is held by circuitry within the icing rate meter panel. The hold circuits prevent the controller from receiving a false no-ice signal. An aspirator built into the detector uses engine bleed-air to create a vacuum near the probe sensing area. This draws air over the sensing probe when the helicopter is in a hover and there is no ram air flowing over the probe.

12-1-5.4 Main Rotor Blade System Automatic Mode Operation.

If the main rotor distributor is not in its proper reference position, 28 vdc power is applied to the de-ice controller from the ON position of the blade de-ice control panel POWER switch, and the main rotor channel automatically enters the fast synchronization mode. During this time, the distributor driver delivers a series of short pulses (375 milliseconds) to step the distributor to a reference position. During synchronization, the controller monitors the sync pulse input signal from the distributor to check for synchronization and proper sequencing. The distributor will provide a -3 to -7 volt reference signal to the controller when the stepping relay is in position for de-icing to start. If the reference signal is not present, the controller continues to generate pulses (to a maximum of eight) to advance a stepping relay in the distributor. If the -3 to -7 volt sync signal is not received within the first eight-pulse cycle, the controller generates a main blade de-ice fail signal, which turns on the MR DE-ICE FAIL capsule. If the reference signal is present, the output from the EOT pulse counter and sync control circuit is a synchronizer functional signal to the main blade fail/fault detect circuits. A no-fail signal from the main blade fail/fault detect circuit is applied to the main rotor power monitor. The power monitor is enabled when the POWER switch on the blade de-ice control panel is ON or at TEST and the synchronizer is functional.

The output of the main rotor power monitor enables the main rotor contactor driver, which energizes the main rotor blade de-ice contactor K60 through the contacts of relay K64. Contactor K60 supplies 115 vac, three-phase power to the distributor through current transformers T14 and T14A and the main rotor sliprings. The controller now begins a normal main rotor cycle. The eight EOT heating control pulses used to step the stepping relay in the distributor are generated by the main blade element-on-timer of the blade de-ice controller. The EOT distributor drive pulses are applied to the input control logic circuit in the distributor. The 28 vdc EOT signal simultaneously energizes stepping relay K2 and contactors K1 and K3. The gate circuits are also enabled. Stepping relay K2 cocks a ratchet mechanism to advance the rotary distributor. The gate circuits control the silicon controlled rectifiers (SCRs), which provide arc suppression for contactors K1 and K3. When the EOT signal goes low, contactors K1 and K3 open. The input control logic circuit and gate circuits keep the SCRs conducting until after the contacts of K1 and K3 have opened. Finally, the stepping relay solenoid is de-energized, allowing its contacts to advance. Sync information is derived from stepping relay K2 and a sync forming circuit. This provides a negative output pulse of five volts for position 1 and 30 volts for positions 3, 5, and 7. Positions 2, 4, 6, and 8 are sensed by zero volts on the sync pulse input line.

12-1-5.5 Tail Rotor Blade System Automatic Mode Operation.

The icing rate signal is also applied to the tail de-ice integrator circuit through the normally-closed contacts of a test relay. The output of the integrator is used to determine the off-time (OT) of the tail blade heating elements. With the POWER switch ON, 28 vdc energizes relay K64 through diode CR16. The 28 vdc is also supplied to the ON command input of the blade de-ice controller's tail rotor power monitor circuit.

The tail rotor power monitor is enabled because the ON input is present and a no-fail signal from the fail detect circuit enables the power monitor. The output of the tail rotor contactor driver energizes the TL RTR DE-ICE CNTOR K61 through the contacts of relay K64 when both inputs to the tail rotor contactor driver are present. The ON signal to OR-gate U6 provides one input to AND-gate U5. Then, a no-fail signal enables U5. The output of U5 enables the contactor driver and supplies one input to AND-gate U4. The other input to U4 is the tail rotor element on-timer signal. The pulse width of the timer signal is a function of the OAT. The lower the temperature, the greater the pulse width. The timer output pulse enables AND-gate U4, which energizes the heat control contactor in the controller. Heating power is connected through the tail rotor sliprings to the tail rotor blade heating elements. The time between EOT cycles is determined by the icing rate signal as sensed by the ice detector. The icing rate signal is applied to the tail de-ice integrator. The output of the integrator controls the operation of the tail rotor element-on-timer.

12-1-5.6 Main and Tail Blade De-ice System Test Mode.

Placing the blade de-ice control panel POWER switch to TEST applies 28 vdc to the test control circuits in the controller, which energizes the test relay and applies a command to all test circuits in the blade de-ice controller. The controller also feeds back 28 vdc to turn on the TEST IN PROGRESS lamp in the blade de-ice control panel for 105 to 135 seconds. During the test mode, the controller overrides existing EOT, OT, and any MANUAL or AUTO command to execute a preset test program. Test relay contacts remove the icing rate signal supplied by the AUTO or MANUAL position of the MODE switch on the blade de-ice control panel to the main and tail blade integrators in the controller. A fixed resistor is substituted, which provides a nominal off-time to 100 seconds for the main rotor channel and 100 seconds for the tail rotor channel. Another set of test relay contacts removes the OAT signal to the main and tail blade EOT circuits and substitutes a fixed resistor, which provides for eight 375 millisecond main blade EOT pulses and a single tail blade EOT pulse of one second. A test command input to the EOT pulse counter and sync check circuit causes the main rotor channel to enter the fast cycle mode. During this mode, the main blade element-on-timer generates eight EOT pulses. The EOT test program is generated by the same timing circuits that generate EOT and OT during normal operation. The main and tail blade systems fault detection circuits signal the existence of any malfunction in the same way as in normal operation. During the test cycle, the OT timers are checked for proper operation. If the tail or main blade OT is less than 90 seconds or more than 110 seconds, a failure indication will be generated. If the blade de-ice control panel POWER switch is placed OFF before the test cycle is complete, the controller terminates the test sequence and resets any existing warning outputs. When the test is allowed to terminate, all warning outputs are reset as if the POWER switch is placed OFF and then ON with no warning capsules on. The blade de-ice test panel provides a check of blade de-ice system for failures that are not detected during the normal TEST mode. The panel does this by inserting specific failure signals into the system, which should be detected by the built-in-test circuits in the controller. When the blade de-ice test panel switch is at SYNC 1 or SYNC 2, the test panel interrupts the distributor sync line and injects a false sync signal to the controller. The test panel provides the controller with a -30 vdc pulse when the mode switch is at SYNC 1 and presents an open circuit with switch at SYNC 2. This causes the MR DE-ICE FAIL capsule to go on. When the blade de-ice test panel switch is placed to OAT, the switch short-circuits the OAT sensor input to the controller. The main blade and tail blade fail detect circuits sense the simulated failure and cause the MR DE-ICE FAIL and TR DE-ICE FAIL caution capsules to go on.

When the blade de-ice test panel switch is placed to EOT, the switch connects grounds from relay K3 to the main blade and tail blade fault detect circuits in the controller to simulate a malfunction in the EOT timing circuits. Thus, the MR DE-ICE FAIL and TR DE-ICE FAIL caution capsules are turned on. When the test function switch on the blade de-ice test panel is at NORM, the test panel does not inject failure signals into the system and allows normal system operation. The blade de-ice test panel also functions to sense faults in the de-ice power circuits. If electrical power remains applied to either the main or tail rotor heating elements after a fail condition or when blade de-ice power is switched off, the blade de-ice test panel causes the corresponding PWR monitor indicator to go on. The tail rotor power monitor circuit monitors the three-phase ac voltage to the tail rotor heating elements. If voltage is present on at least two phases, a ground return is provided for fault detector relay K2. When the POWER switch on the blade de-ice control panel is OFF or the controller generates a tail rotor de-ice fail signal, relay K2 is energized. The PWR TAIL RTR lamp is turned on through the contacts of K2, warning that the tail rotor blade heater power has not been turned off as required. The main rotor power monitor circuit monitors the voltage from three main rotor blade current transformers. If power is being applied to any rotor blade heating element, the main rotor power monitor circuit provides a ground return for fault detector relay K1. If the POWER switch on the blade de-ice control panel is OFF or a main blade de-ice fail signal is generated by the controller, relay K1 is energized. The PWR MAIN RTR lamp on the blade de-ice test panel is turned on, warning that the main rotor blade heater power has not been turned off as required.

12-1-5.7 Icing Rate Meter Test.

If there is no ice on the ice detector probe, pressing and releasing the PRESS TO TEST push button on the icing rate meter will cause the meter's needle to move to center scale (1.0) and then to drop to zero or below. The ICE DETECTED caution capsule goes on. The probe heater and aspirator heater turn on at the same time. When the indicator needle goes below zero, the ICE DETECTED capsule, probe heater, and aspirator heater turn off.

12-1-5.8 Main and Tail Blade Manual Mode Operation.

The MANUAL mode of system operation provides the ability to maintain blade de-ice operation.

Should either the ice detector or icing rate meter fail, 28 vdc from the No. 2 dc primary bus DE-ICE CNTRLR circuit breaker supplies the blade de-ice controller power supply. The blade de-ice power supply powers the manual mode power supply, which produces three dc voltages for the three icing rate signals: T (trace) 2.0 vdc, L (light) 3.5 vdc, and M (moderate) 5.0 vdc. The dc voltages are supplied from the blade de-ice controller to the blade de-ice control panel. By turning the MODE select switch out of AUTO and to MANUAL, one of three discrete levels of off-time, T, L, or M, may be selected. The selected voltage is routed from the blade de-ice control panel to the main and tail de-ice integrators in the blade de-ice controller. In the MANUAL mode, the OT is constant for any selected position and no updating occurs as in AUTO mode.

12-1-5.9 Main Blade Fail/Fault Detection.

During EOT periods, three-phase pulsed current is supplied to the main blade heating elements through the current transformers of T14/T14A. Each of the current transformer outputs are applied to the blade de-ice controller's current sensing circuits in the main blade fail/fault circuits. The three-phase voltages are also supplied to the controller, which makes proportional adjustments to counteract the effect of power line voltage fluctuations. An increase of any single blade line current to between 58 and 64 amperes, a decrease to between 22 and 26 amperes, or a ground current of between one to ten amperes, will cause the current sensing circuits in the controller to produce a fail signal, causing the MR DE-ICE FAIL capsule on the caution panel to go on. The same fail signal is applied to the main blade power monitor, which removes 28 vdc from the main blade contactor driver, de-energizing MN RTR BLADE DE-ICE CNTOR K60, and removing 115 vac, three-phase power from the distributor. The fail signal applied to the main blade power monitor is also applied to the main blade control circuits to inhibit the element-on-timer. These malfunctions will cause an identical shutdown of the main blade de-icing system: EOT failure, OAT sensor failed open or shorted, synchronization pulse failure, or distributor advance failure. When a failure is detected, the controller's main blade de-ice channel is latched off by the action of the main rotor power monitor circuit. To attempt to restore system operation, the POWER switch on the blade de-ice control panel must be cycled to OFF and returned to ON. If the problem is transient, the system will return to operation. If the problem remains, the system will again be returned to the failed condition. If a single blade line is sensed to decrease between 0.39 and 0.43 amperes, the blade de-ice controller fail/fault circuit will produce a fault output, causing the MR DE-ICE FAULT capsule to go on. However, the blade de-ice system, though degraded, will continue to operate.

12-1-5.10 Tail Blade Fail Detection.

During EOT periods three-phase pulsed current is supplied to the tail rotor blades through three current transformers in the blade de-ice controller. Each of the three current transformer outputs is applied to the blade de-ice controllers current sensing circuit in the tail blade fail detect circuits. The three-phase voltages are also supplied to the controller, which makes proportional adjustments to counteract the effect of power line voltage fluctuations. An increase of any single blade line current to between 15 and 18 amperes, a decrease of between five and seven amperes, or a ground current between one to ten amperes, will cause the current sensing circuit to the controller to produce a fail signal that causes the TR DE-ICE FAIL capsule on the caution/advisory panel to go on. The same signal is applied to the tail rotor power monitor, which removes 28 vdc from the tail rotor contactor driver. This removes one enabling input from AND-gate U4, thereby de-energizing the tail rotor contactor and removing 115 vac, three-phase power from the tail rotor heating elements. The tail rotor power monitor disabling signal from the tail blade fail circuitry also inhibits the EOT timer in the tail blade de-ice control circuits. These malfunctions will cause an identical shutdown of the tail de-icing system: EOT failure or OAT sensor failure if a heating element is open or shorted. When a failure is detected, the controller's tail blade de-ice channel is latched off by the action of the tail rotor power monitor circuit. To attempt to restore system operation, the POWER switch on the blade de-ice control panel must be cycled to OFF and returned to ON. If the problem remains, the system will again be returned to the failed condition.

12-1-5.11 Blade De-ice Test Panel.

The blade de-ice test panel applies simulated malfunctions to the blade de-ice system and produces failure indications if system shutdown occurs (Figure 12-1-5-3 and Figure 12-1-5-4). The test panel contains a lighted information plate, a five-position rotary switch, and two push-button indicator lamps. The blade de-ice test panel also consists of a printed wiring board, a circuit board, and electrical connectors P2/J1. Connector P2 electrically interfaces the rotary switch and the indicator lamps with the printed wiring board. Electrical interface with the helicopter is accomplished through electrical connector J1.

12-1-5.12 Blade De-ice Test Panel Operation.

Internal circuits within the blade de-ice test panel are controlled by selecting specific modes on the five-position rotary switch. The five rotary switch positions are NORM-SYNC 1-SYNC 2-OAT-EOT. When the switch is at SYNC 1 or SYNC 2, the blade de-ice test panel interrupts the distributor sync pulse line and injects a false sync signal to the controller. The blade de-ice test panel sync circuitry provides the controller with a -30 vdc pulse when the switch is at SYNC 1 and provides an open circuit with the switch at SYNC 2. When at OAT, the switch short-circuits the OAT sensor input to the controller. When the switch is placed to EOT, the EOT test circuit grounds relay K3, which electrically grounds the circuit to the controller. When at NORM, the blade de-ice test panel does not inject failure signals into the system (outputs) and allows normal operation.

12-1-5.13 Blade De-ice Test Panel Fault Detection.

The main rotor power monitor circuit monitors the voltage from three main rotor blade current transformers. If a main rotor fault is detected and power remains applied to the main rotor circuitry, relay K1 is energized and the PWR MAIN RTR indicator lamp is turned on. The energized relay K1 also supplies 28 vdc output to the main rotor fail light system indicator. The tail rotor power monitor circuit monitors the three-phase ac voltage to the tail rotor heating elements. If a tail rotor fault is detected and power remains applied to the tail rotor circuitry, a ground return is provided to relay K2 and the PWR TAIL RTR indicator lamp is turned on. The energized relay K2 also supplies 28 vdc output to the tail rotor light system indicator.

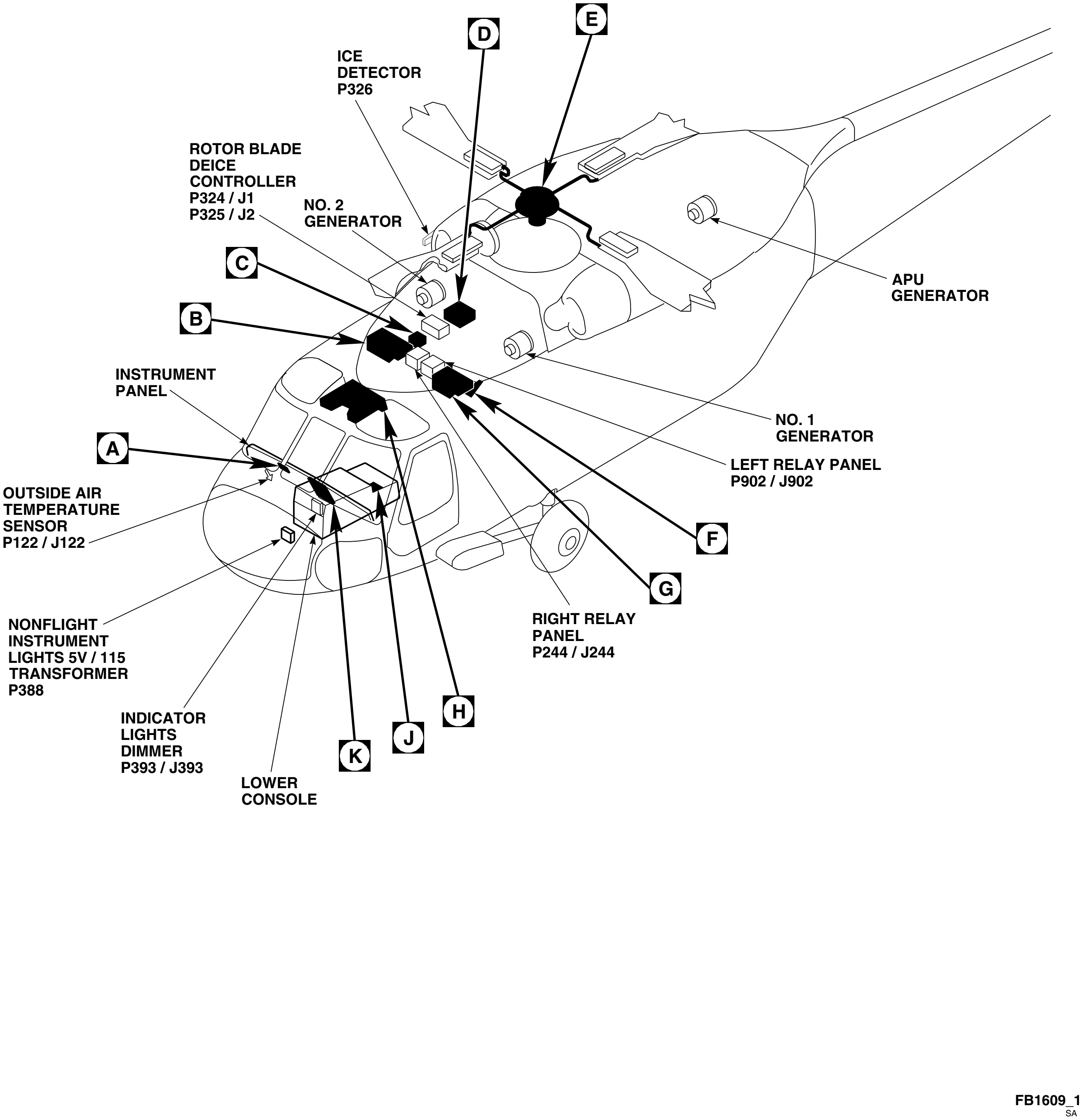
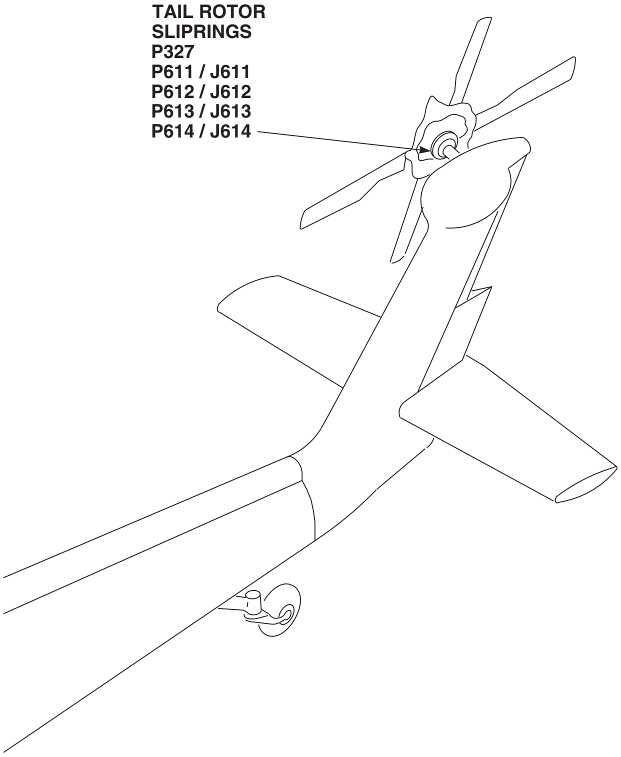


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 1 OF 10)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P107 / J107	COCKPIT, BL 9 LH, STA 236
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P114 / J114	COCKPIT, BL 8 RH, STA 200
P115 / J115	COCKPIT, BL 6 RH, STA 210
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P121 / J121	COCKPIT, BL 10 RH, STA 915
P122 / J122	(OAT) OUTSIDE AIR– TEMPERATURE SENSOR
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243
P132	ICING RATE METER
P133 / J133	BLADE DEICE CONTROL PANEL
P134 / J1	BLADE DEICE TEST PANEL
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P204 / J204	NO. 2 JUNCTION BOX
P205	NO. 2 JUNCTION BOX
P206	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P212 / J212	NO. 1 JUNCTION BOX
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P224 / J224	COCKPIT CEILING, BL 3 RH, STA 247
P230 / J230	CABIN CEILING, BL 8 RH, STA 245
P231 / J231	UPPER COCKPIT LH, AFT OUTBD BL 6.05, STA 247
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P244 / J244	RIGHT RELAY PANEL
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 25 RH
P280 / J280	CABIN CEILING, BL 6 LH, STA 247
P324 / J1	ROTOR BLADE DEICE CONTROLLER
P325 / J2	ROTOR BLADE DEICE CONTROLLER

FB1609_2
SA

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 2 OF 10)

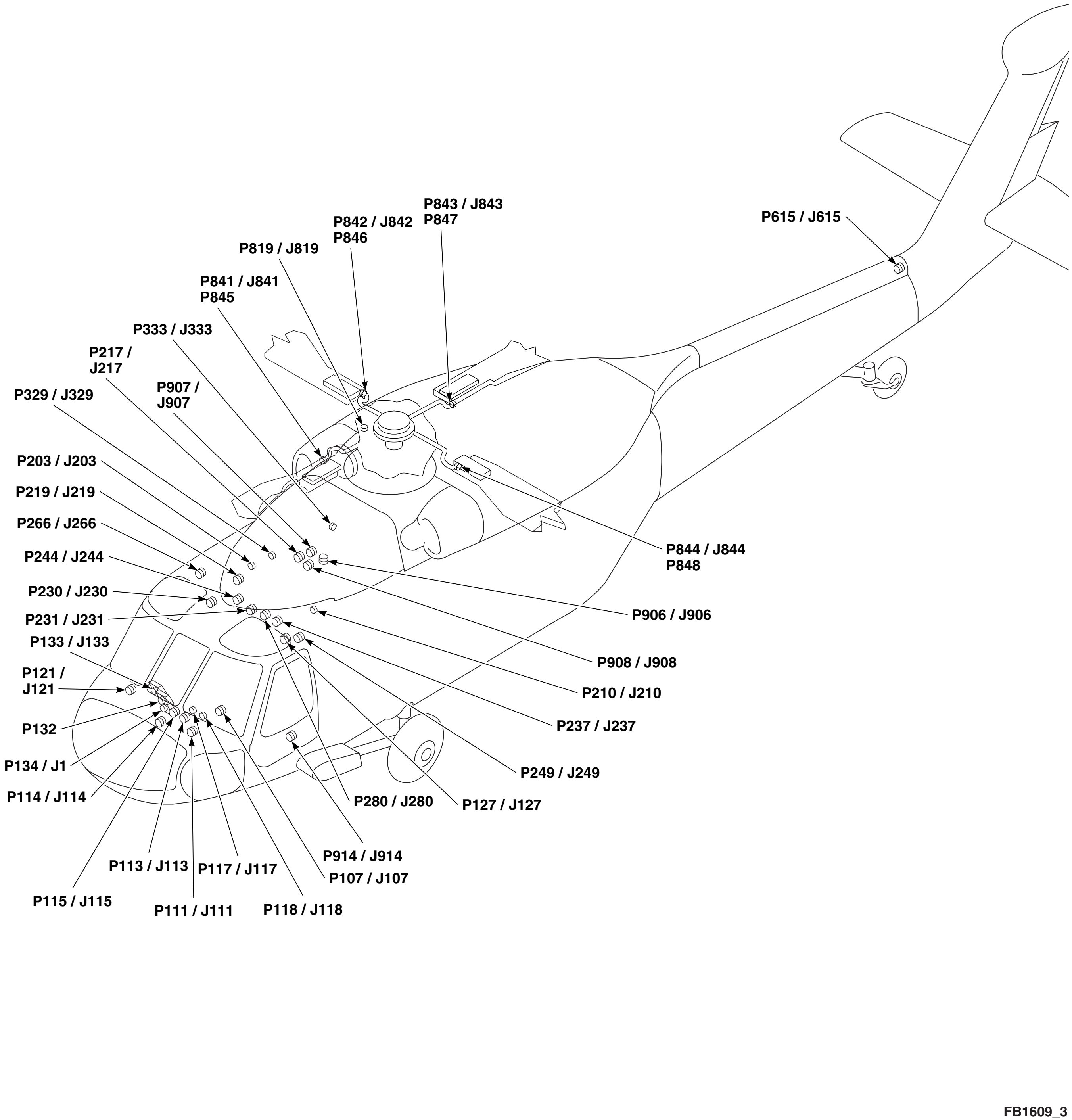
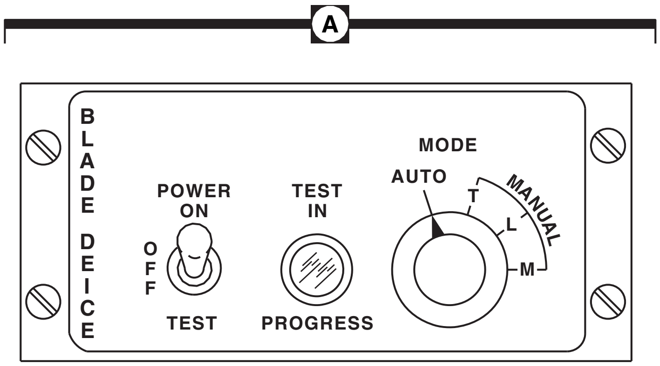
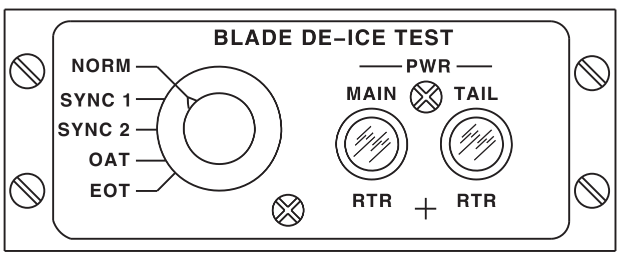


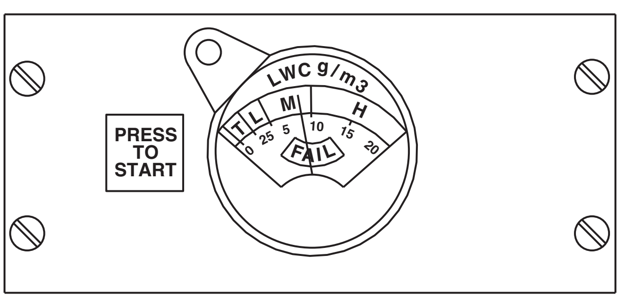
FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 3 OF 10)



BLADE DEICE CONTROL PANEL



BLADE DE–ICE TEST PANEL



ICING RATE METER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P326	ICE DETECTOR
P327	TAIL ROTOR SLIPRINGS
P329 / J329	MAIN ROTOR BLADE DEICE JUNCTION BOX
P332 / J332	AUXILIARY AC JUNCTION BOX
P333 / J333	AUXILIARY AC JUNCTION BOX
P388	NONFLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMERS
P393 / J393	INDICATOR LIGHTS DIMMER
P611 / J611	TAIL ROTOR BLADE
P612 / J612	TAIL ROTOR BLADE
P613 / J613	TAIL ROTOR BLADE
P614 / J614	TAIL ROTOR BLADE
P615 / J615	TAIL CONE, BL 3 RH, STA 651
P819 / J819	MAIN ROTOR PYLON DECK BL 23 RH, STA 352
P841 / J841	NO. 1 MAIN ROTOR BLADE
P842 / J842	NO. 2 MAIN ROTOR BLADE
P843 / J843	NO. 3 MAIN ROTOR BLADE
P844 / J844	NO. 4 MAIN ROTOR BLADE
P845	DROOP STOP HEATER NO. 1
P846	DROOP STOP HEATER NO. 2
P847	DROOP STOP HEATER NO. 3
P848	DROOP STOP HEATER NO. 4
P902 / J902	LEFT RELAY PANEL
P906 / J906	CABIN CEILING, BL 0, STA 284
P907 / J907	CABIN CEILING, BL 0, STA 288
P908 / J908	CABIN CEILING, BL 0, STA 279
P914 / J914	COCKPIT, BL 24 LH, STA 247
P1001 / J1001	MAIN ROTOR BLADE DEICE DISTRIBUTOR
P1002 / J1002	MAIN ROTOR BLADE DEICE DISTRIBUTOR
P1003 / J1003	MAIN ROTOR BLADE DEICE DISTRIBUTOR
P1004 / J1004	MAIN ROTOR BLADE DEICE DISTRIBUTOR

FB1609_4
SA

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 4 OF 10)

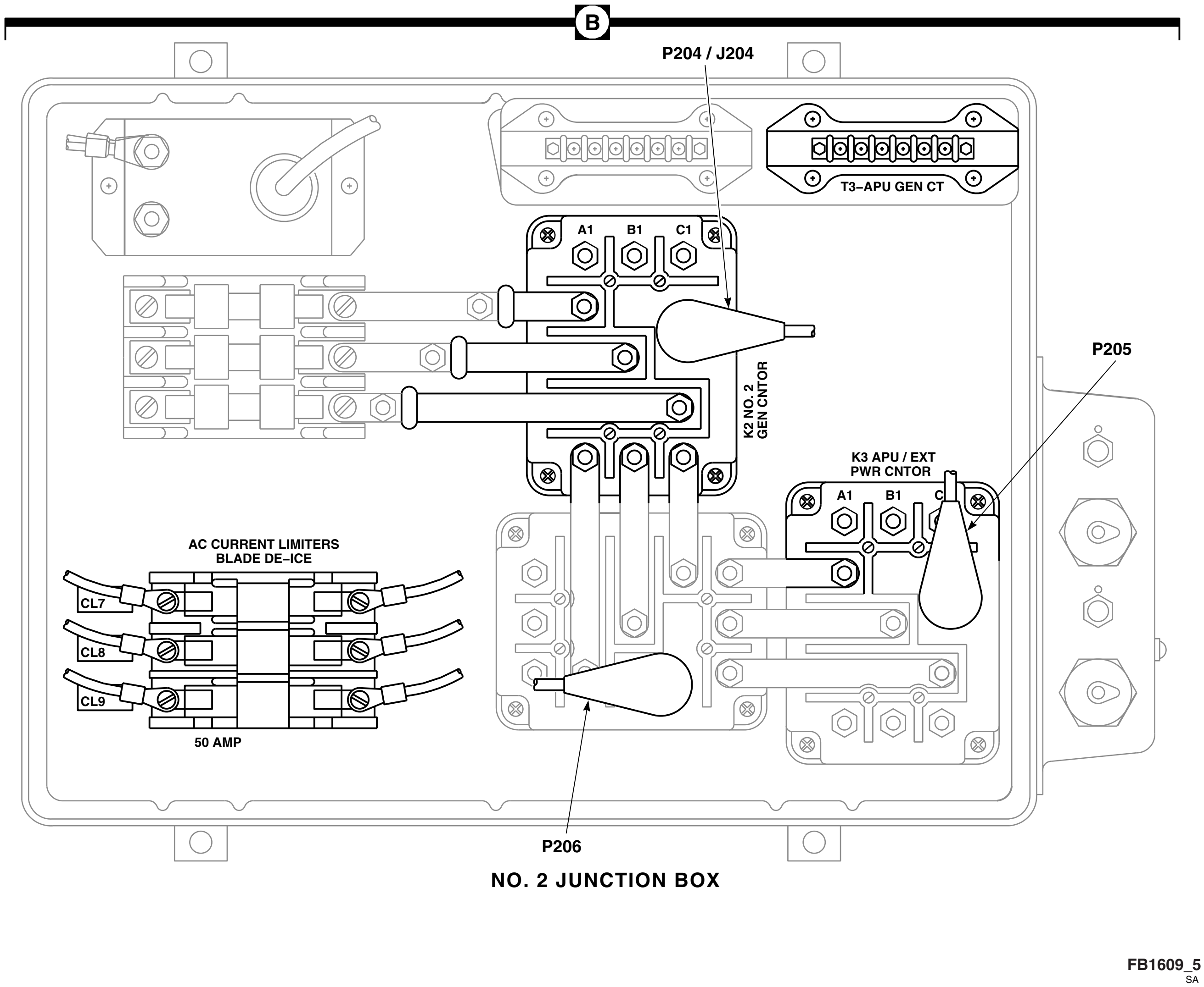
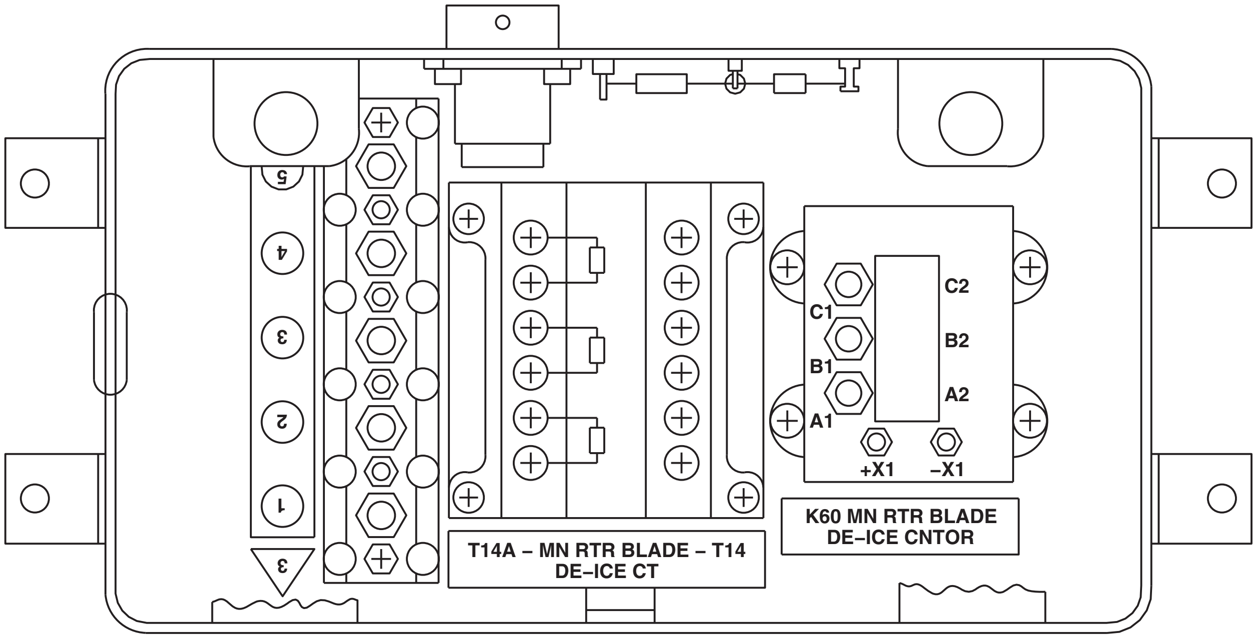


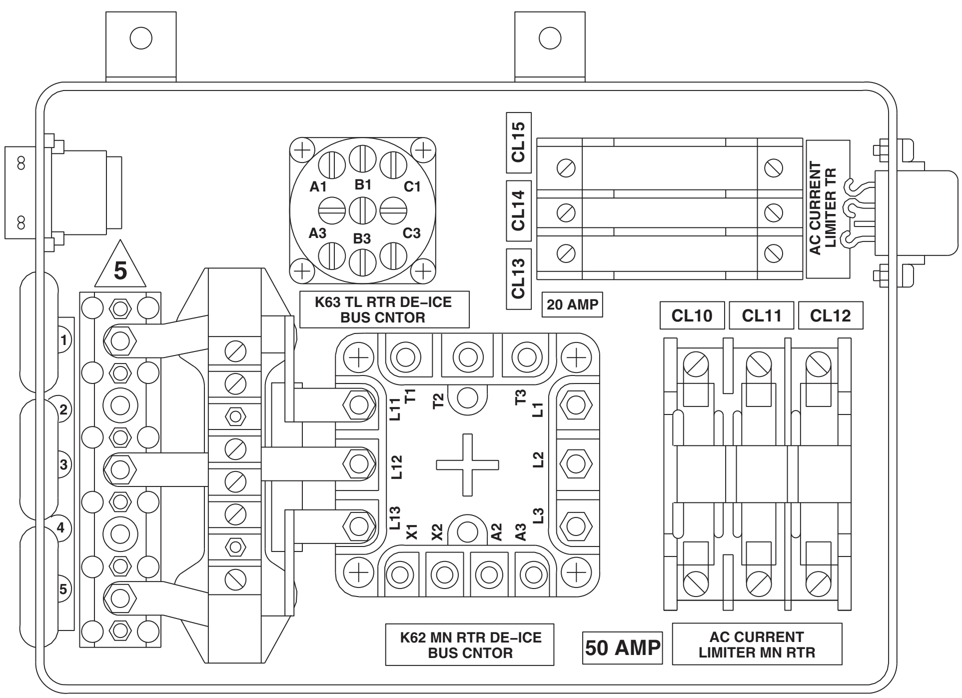
FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 5 OF 10)



MAIN ROTOR DEICE JUNCTION BOX

C

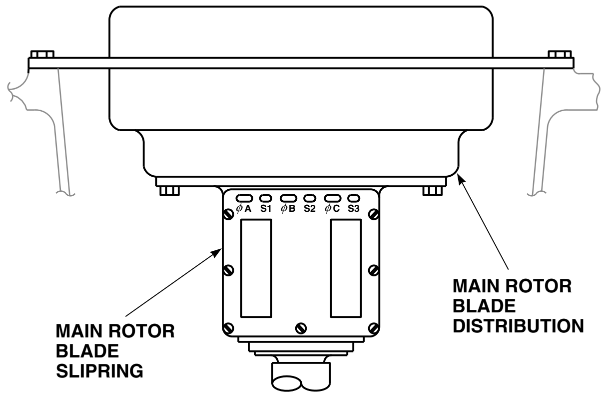
D



AUXILIARY AC JUNCTION BOX

FB1609_6
SA

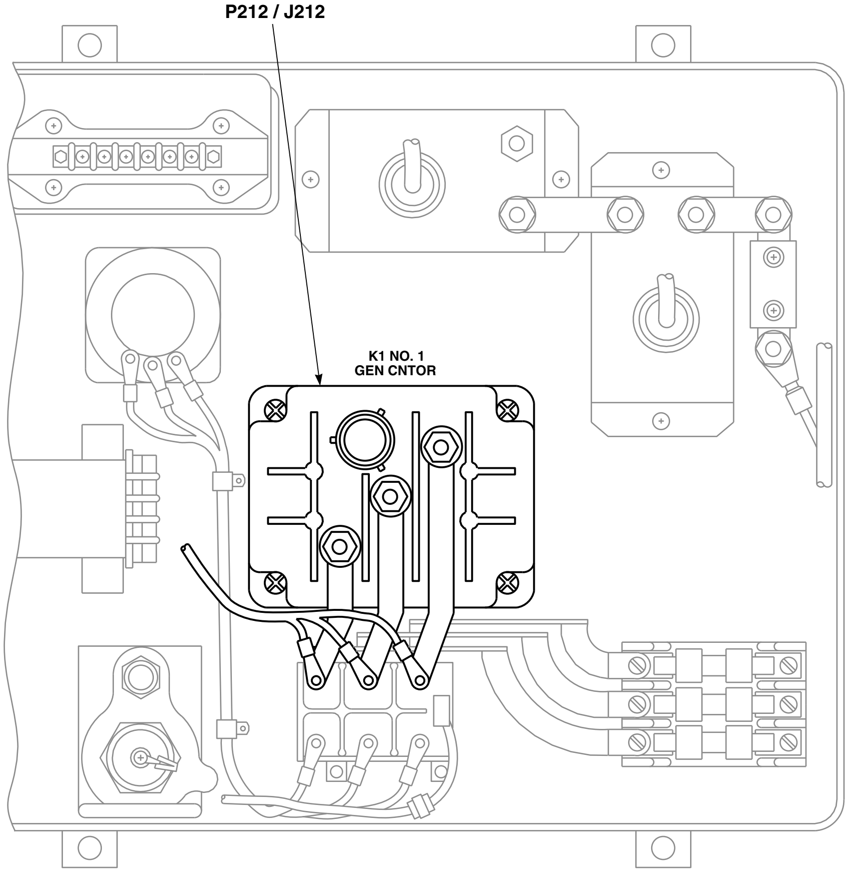
FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 6 OF 10)



MAIN ROTOR DISTRIBUTION
AND SLIPRINGS

E

F



NO. 1 JUNCTION BOX

FB1609_7
SA

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 7 OF 10)

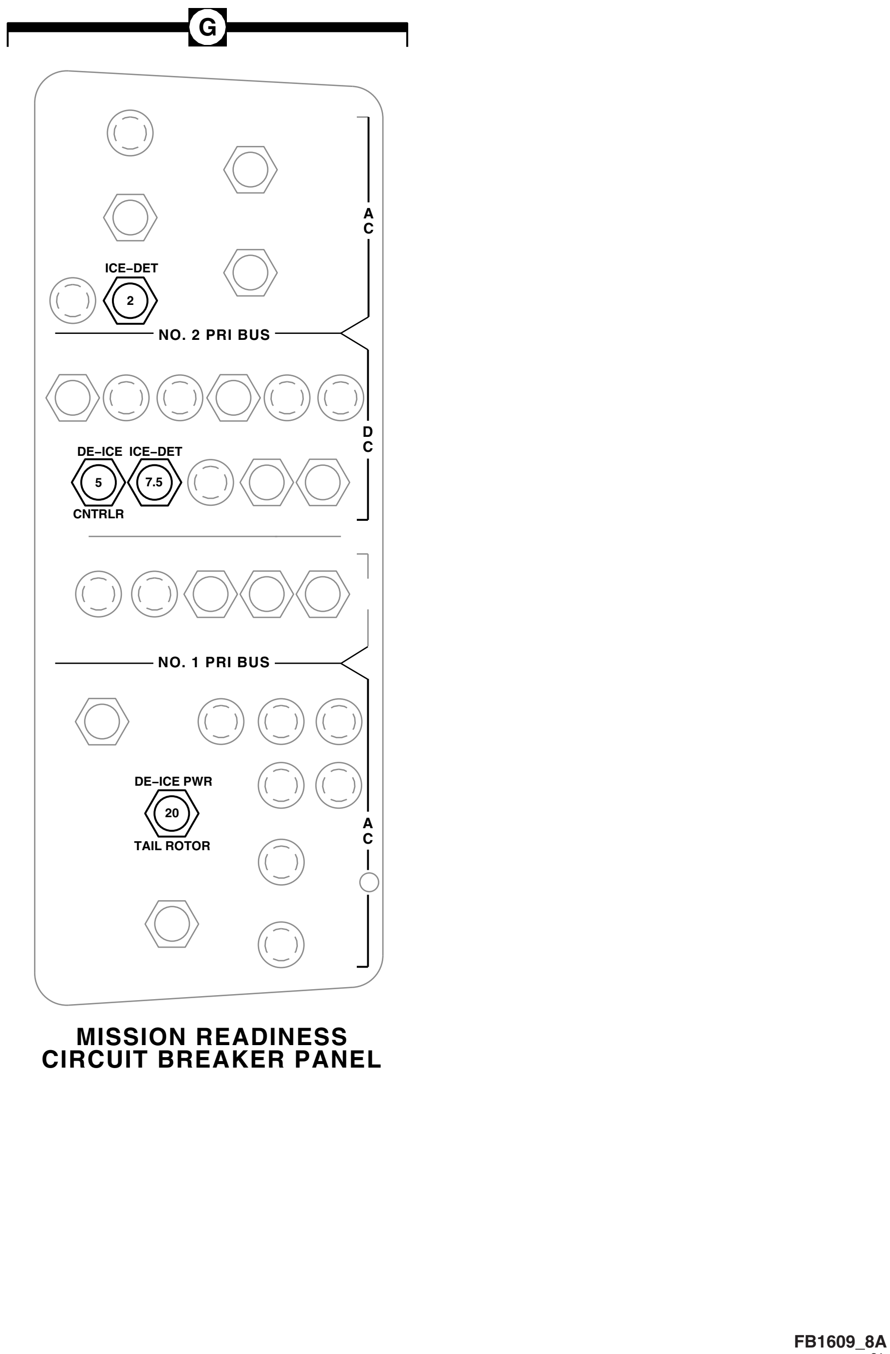
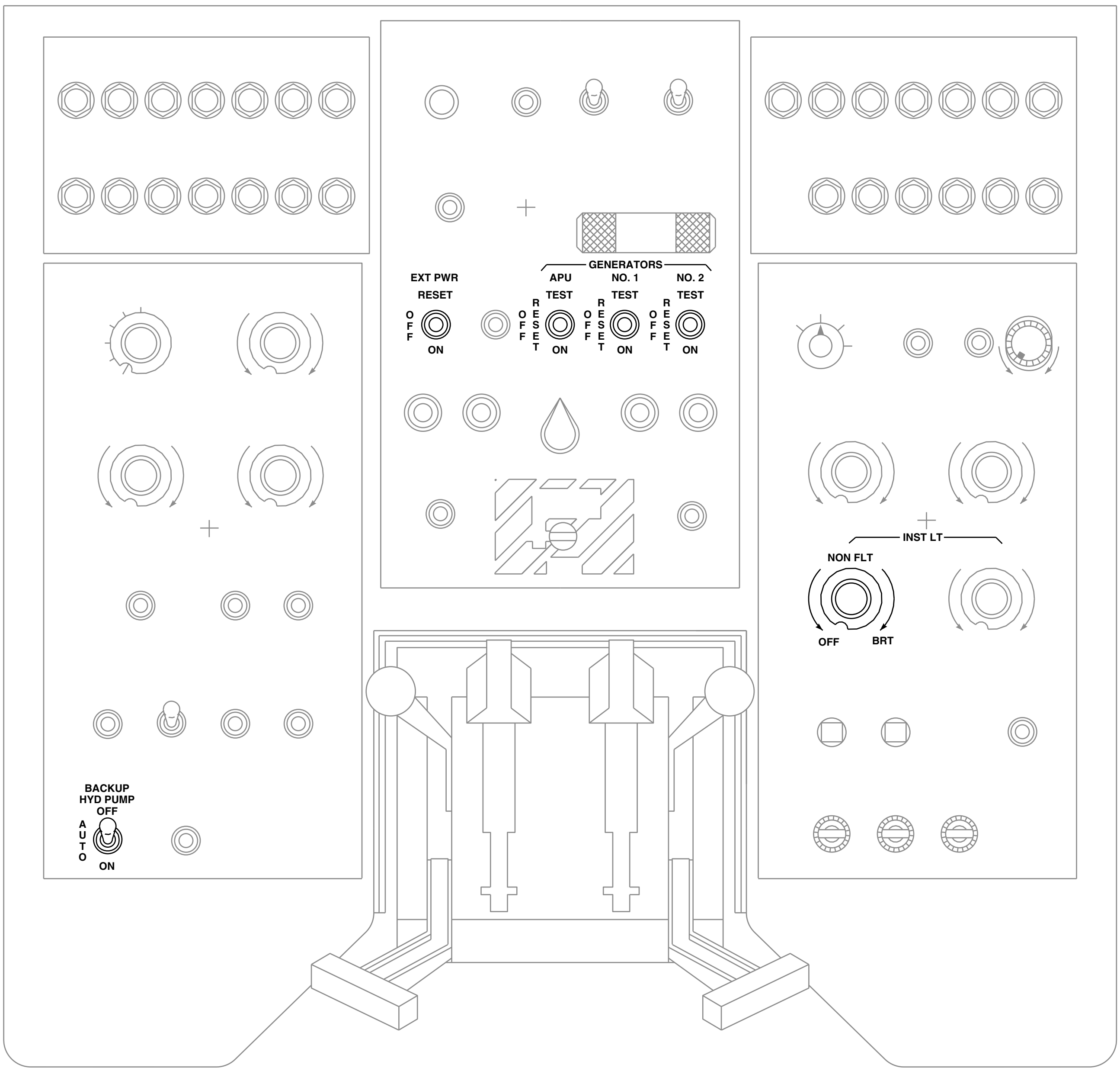


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 8 OF 10)

H



UPPER CONSOLE

FT1278_9A
SA

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 9 OF 10)

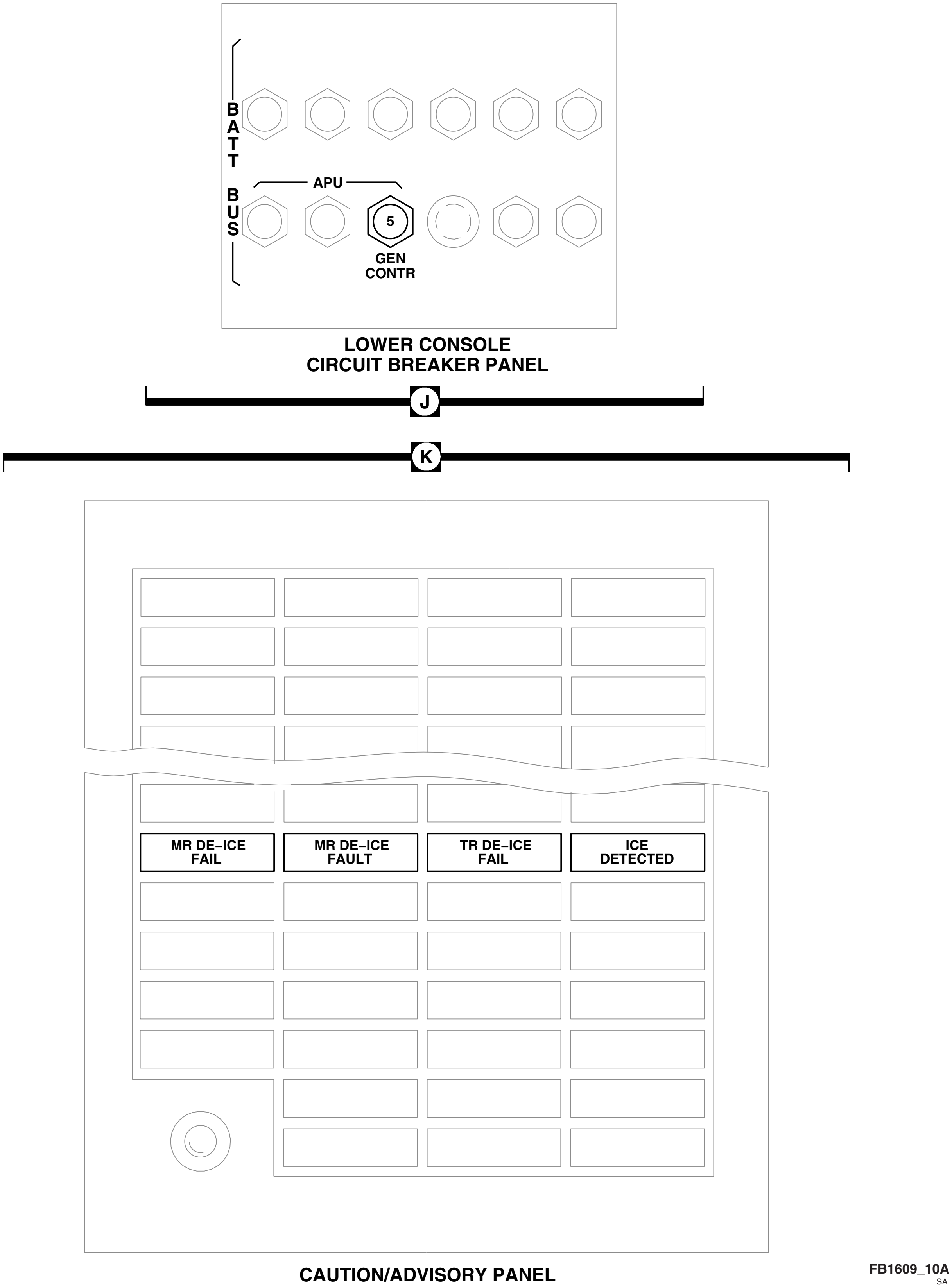
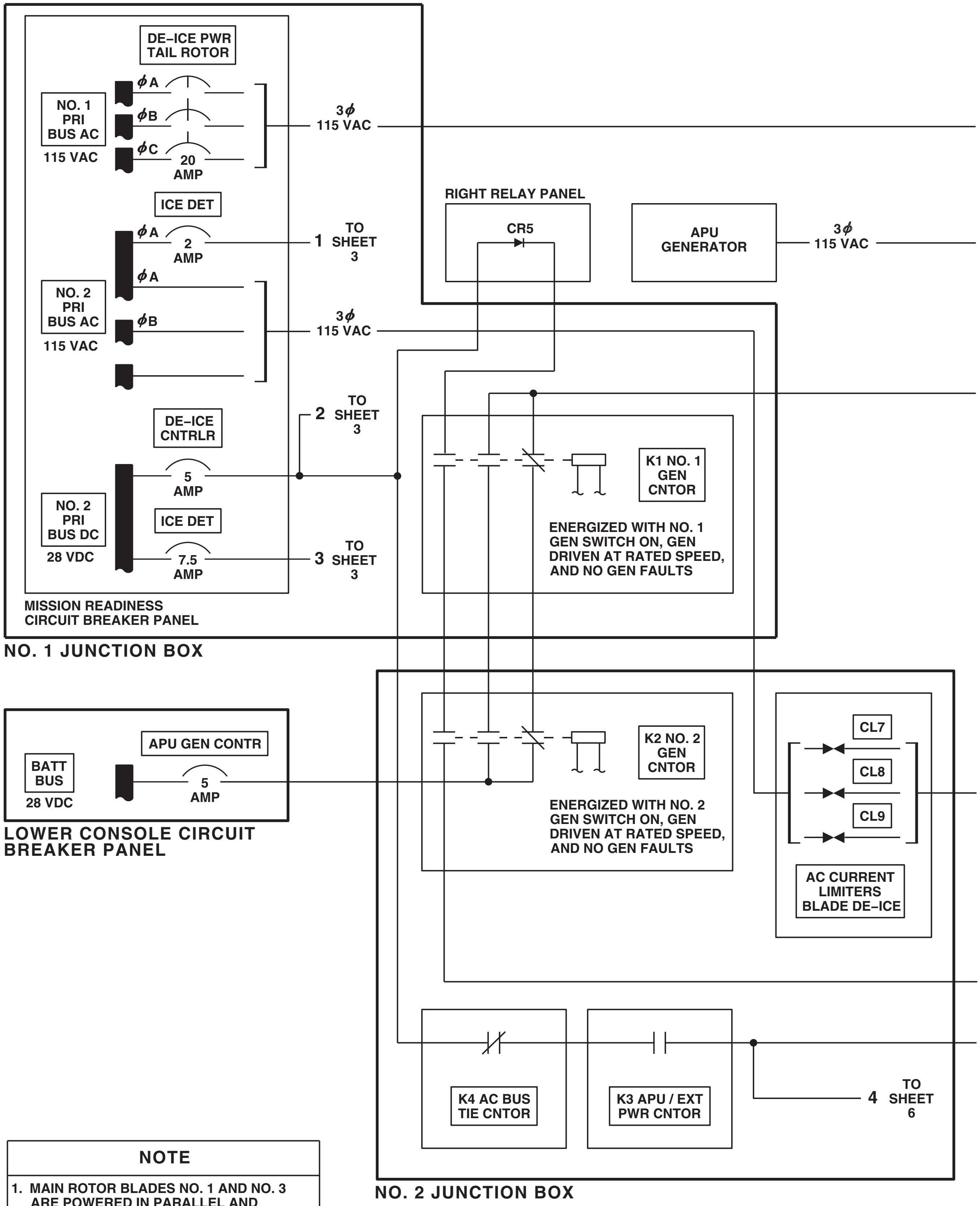
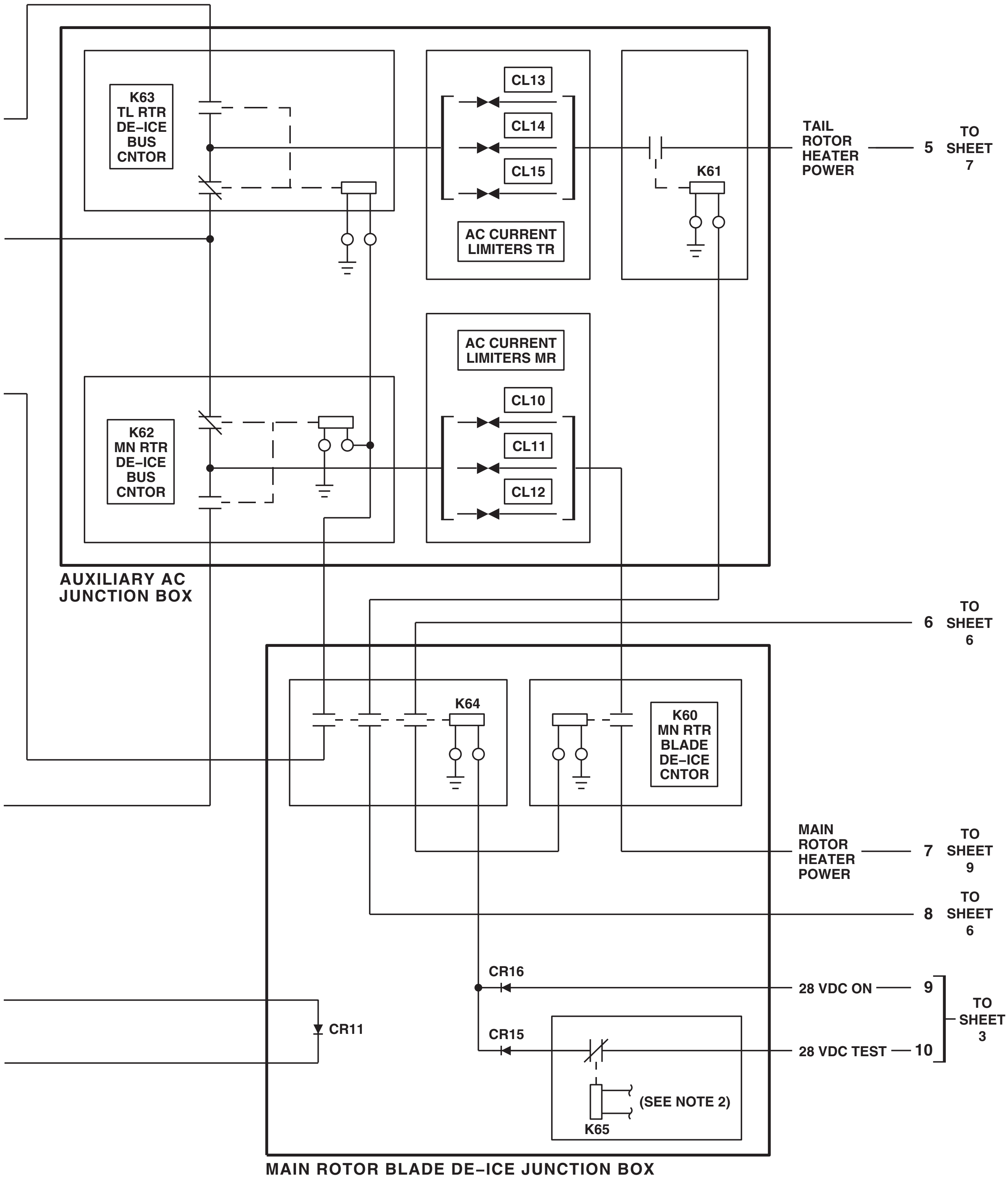


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM LOCATION DIAGRAM (SHEET 10 OF 10)



NOTE
1. MAIN ROTOR BLADES NO. 1 AND NO. 3 ARE POWERED IN PARALLEL AND BLADES NO. 2 AND NO. 4 ARE POWERED IN PARALLEL.

FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 1 OF 11)



NOTE

2. K65 CONTACTS OPEN WHEN HYDRAULIC BACKUP PUMP GOES ON.

FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 2 OF 11)

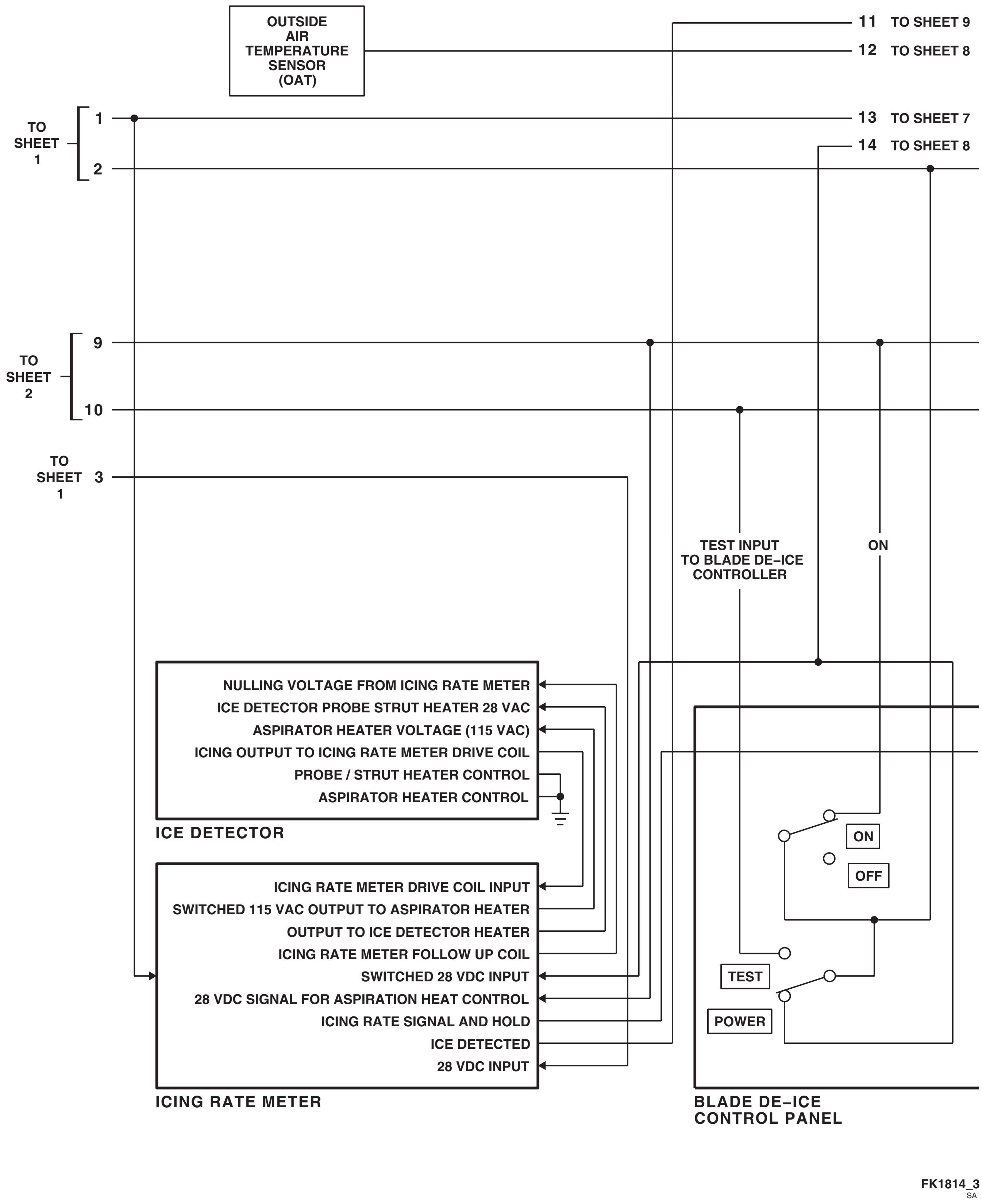


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 3 OF 11)

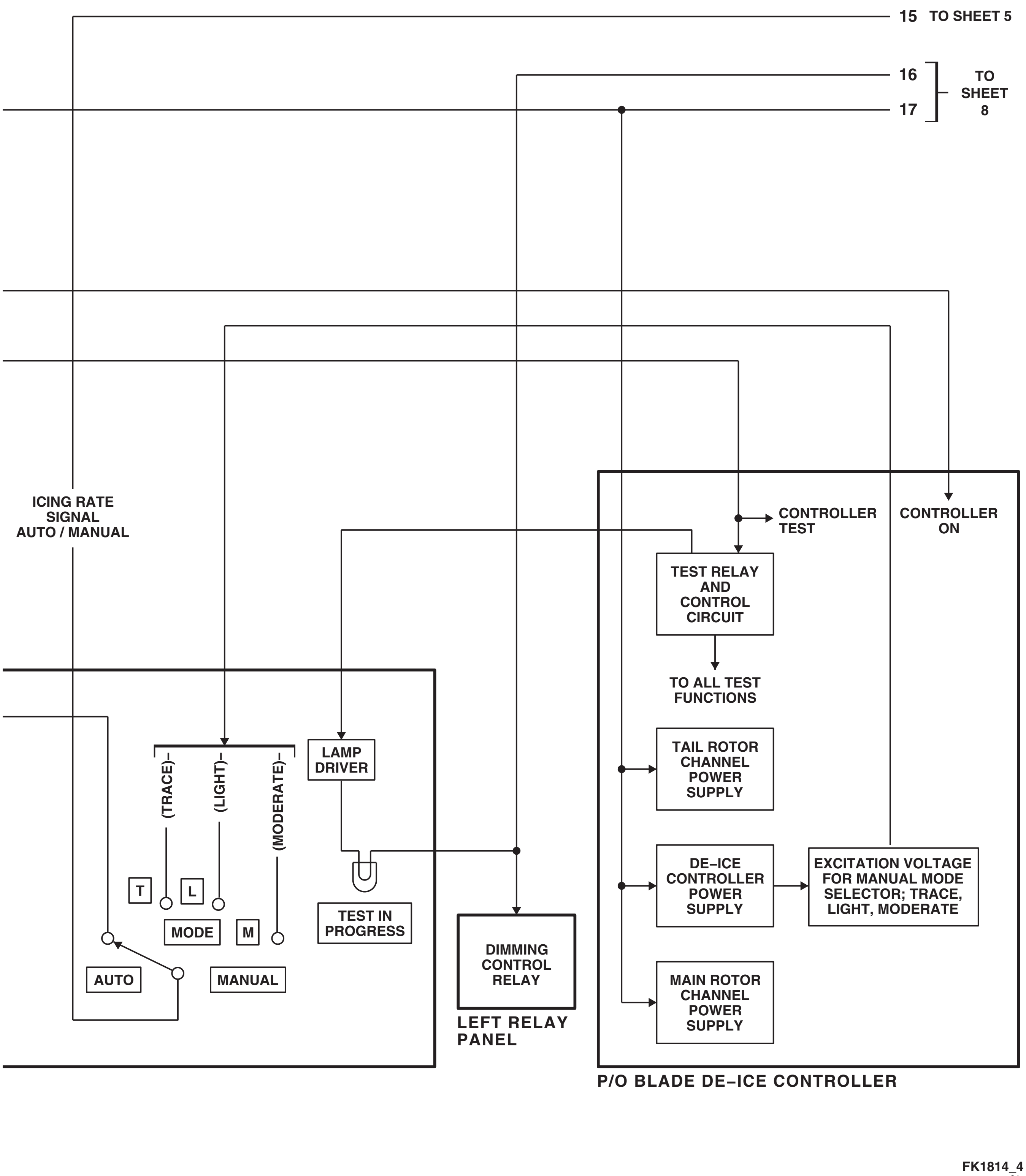


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 4 OF 11)

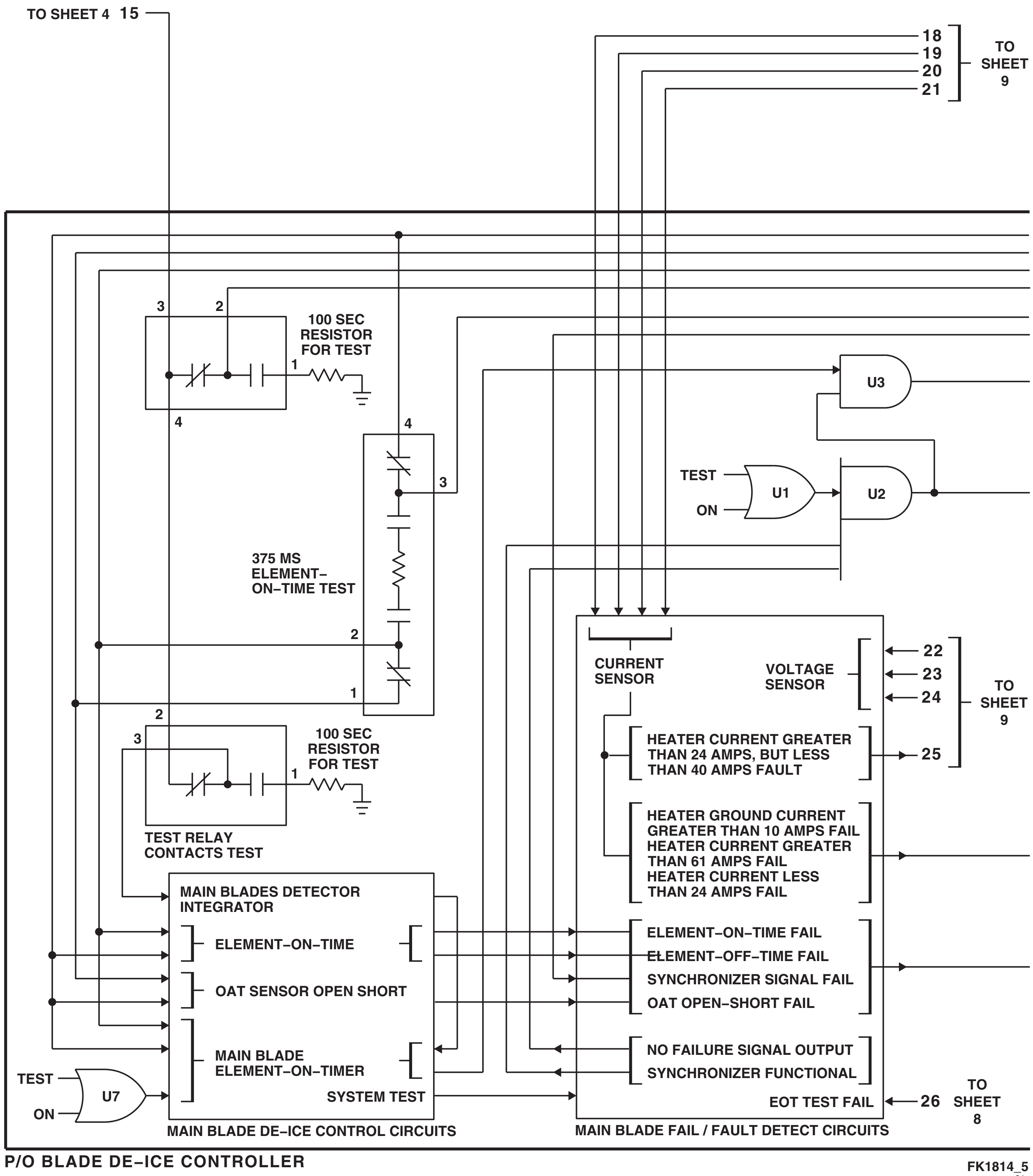


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 5 OF 11)

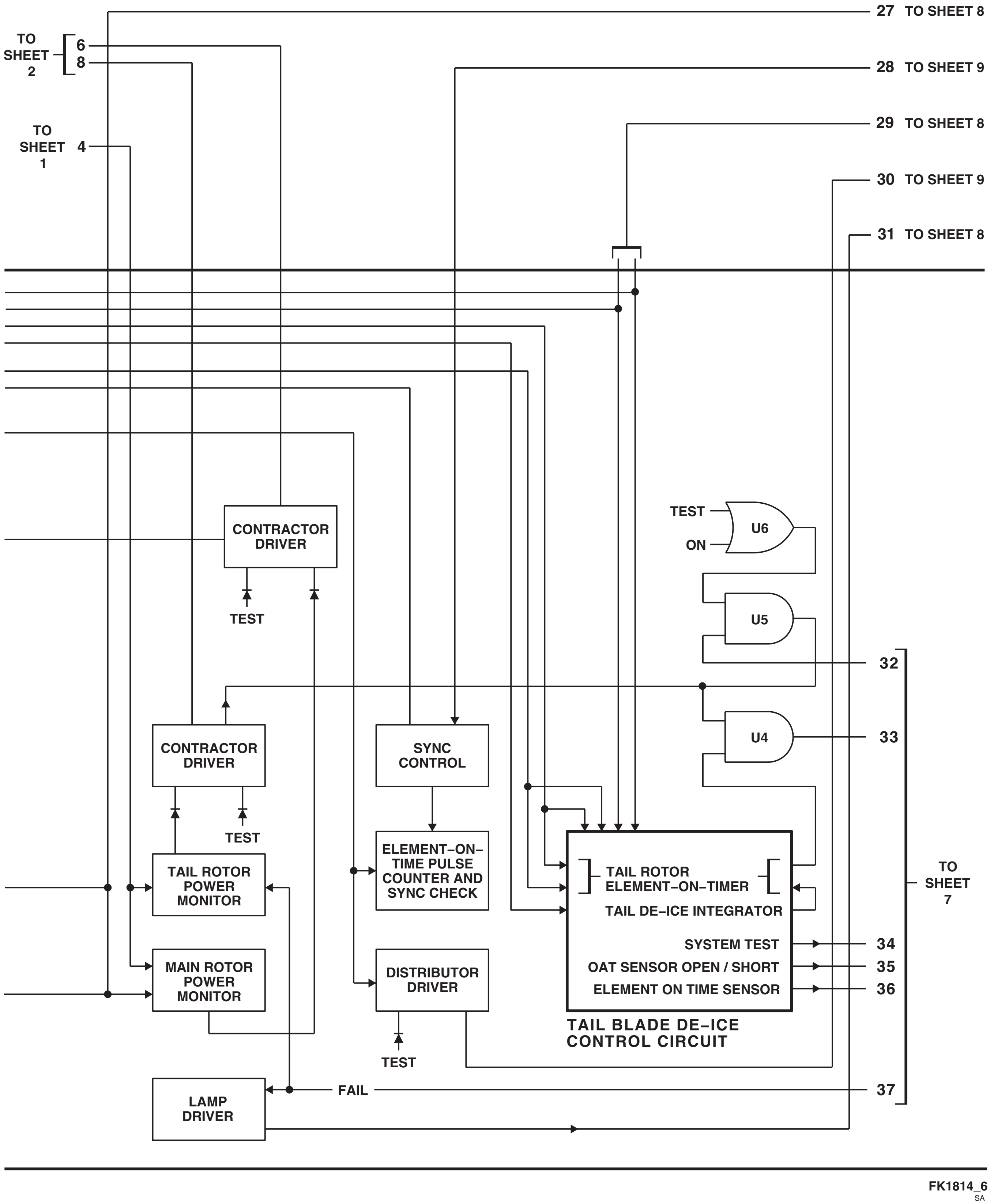


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 6 OF 11)

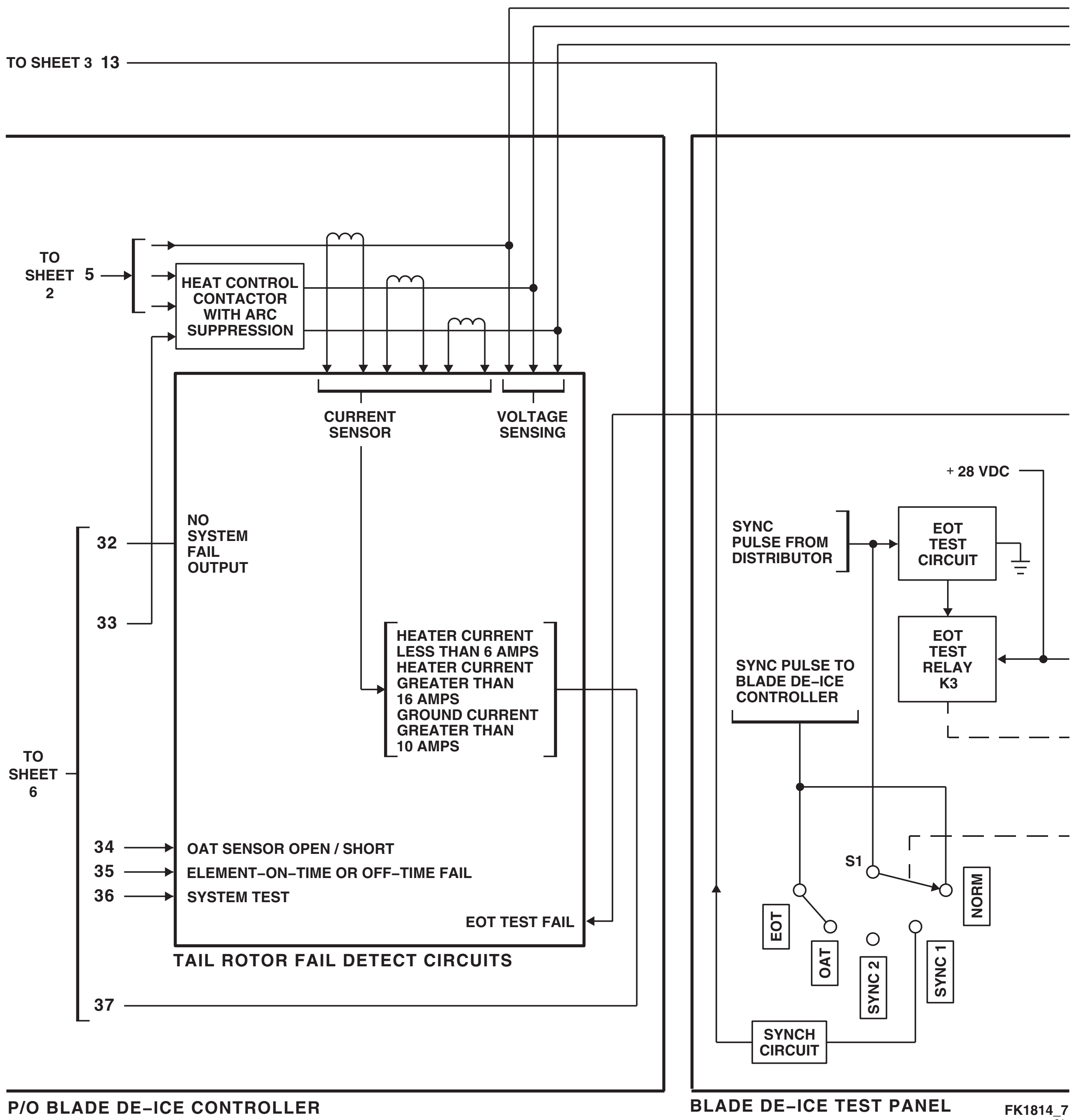


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 7 OF 11)

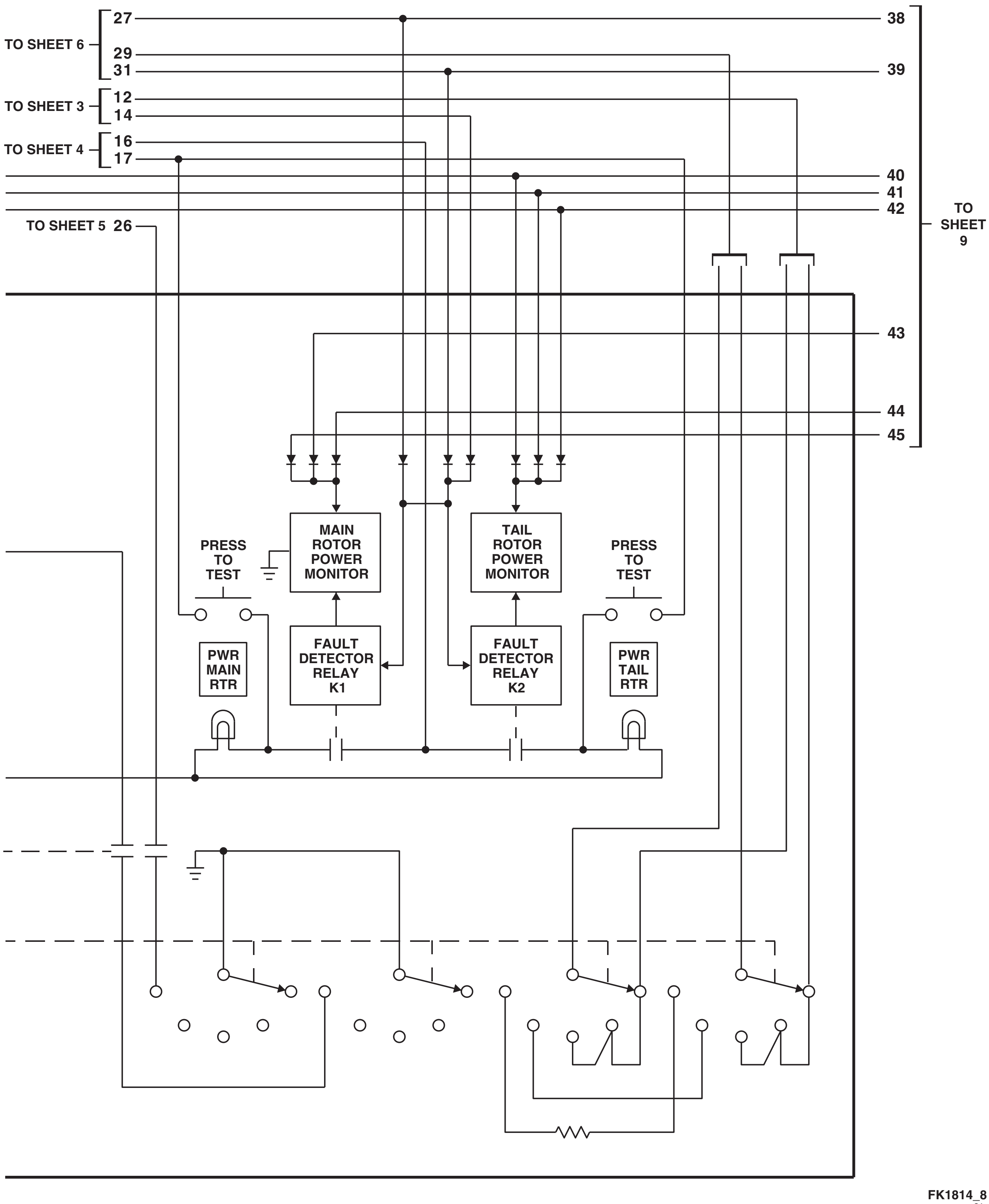


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 8 OF 11)

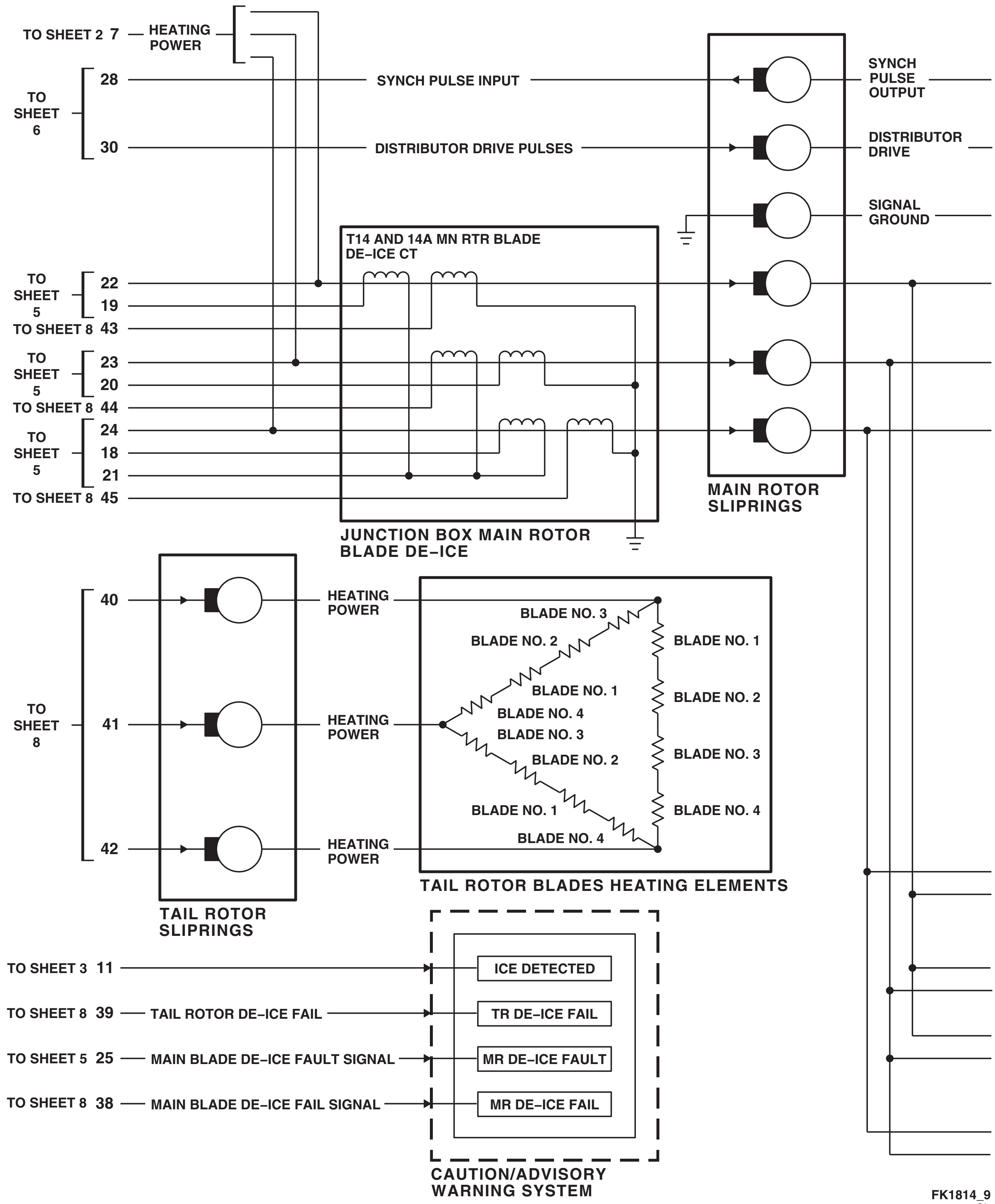


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 9 OF 11)

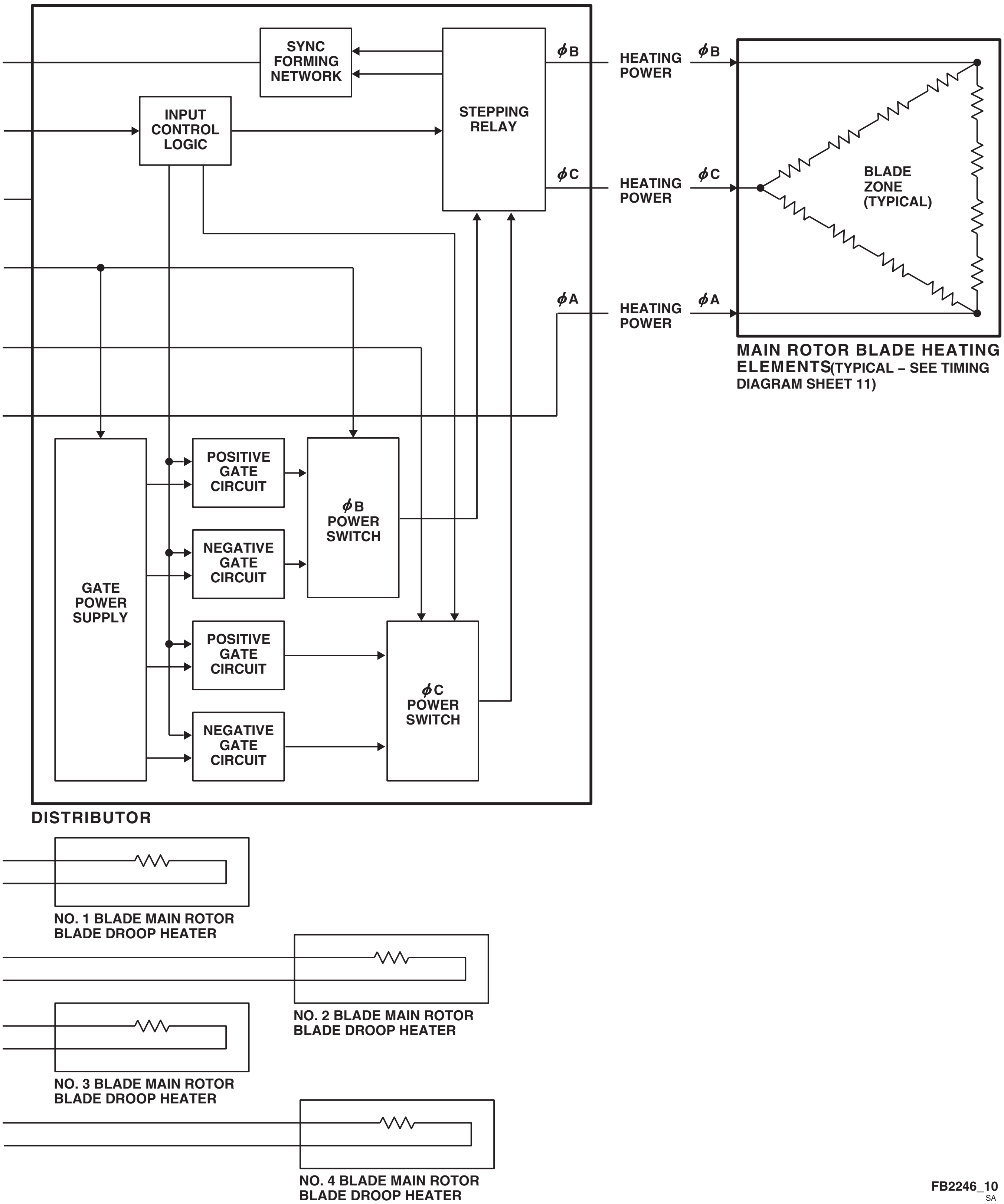


FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 10 OF 11)

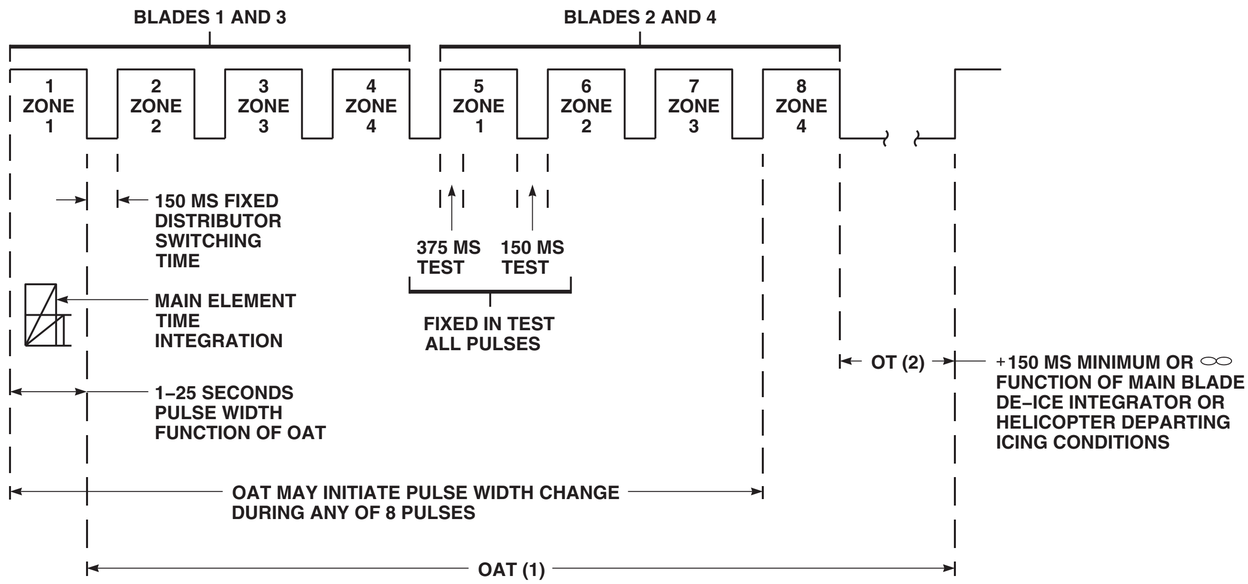
TABLE 1

TL RTR DE–ICE AND MAIN BLADE DE–ICE CONTACTOR CONTROL

	K62 MN RTR DE–ICE BUS CNTOR	K63 TL RTR DE–ICE BUS CNTOR	K64 BLADE DE–ICE PWR CNTOR (SEE NOTE 3)	BLADE DE–ICE SUPPLY
BOTH GENERATORS OPERATIVE	X	X	X	HELICOPTER MAIN GENERATORS
SINGLE GENERATOR OPERATIVE	O	O	X	NOT SUPPLIED
SINGLE GENERATOR AND APU GENERATOR OPERATIVE	O	O	X	APU GENERATOR
BOTH GENERATORS INOPERATIVE	X	X	X	EXTERNAL POWER OR APU TEST ONLY

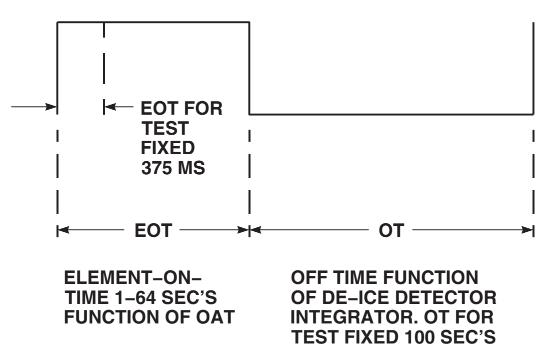
X ENERGIZED
O DE–ENERGIZED

MAIN BLADE DE–ICE TIMING AND DISTRIBUTION DIAGRAM



MAIN BLADE DE–ICE DETECTOR INTEGRATOR CAN GENERATE NEW OT (2) DURING TIME BETWEEN TRAILING EDGES OF PULSES 1 AND 8

TAIL BLADES EOT TEST TIMING DIAGRAM



NOTE
3. K64 IS ENERGIZED WITH BLADE DE–ICE SWITCH IN ON POSITION AND POWER APPLIED TO NO. 2 DC PRIMARY BUS. K64 IS DEENERGIZED BY K65 WHEN HYDRAULIC BACKUP PUMP GOES ON.

FK1814_11
SA

FIGURE 12-1-5-2. BLADE DE-ICING SYSTEM BLOCK DIAGRAM (SHEET 11 OF 11)

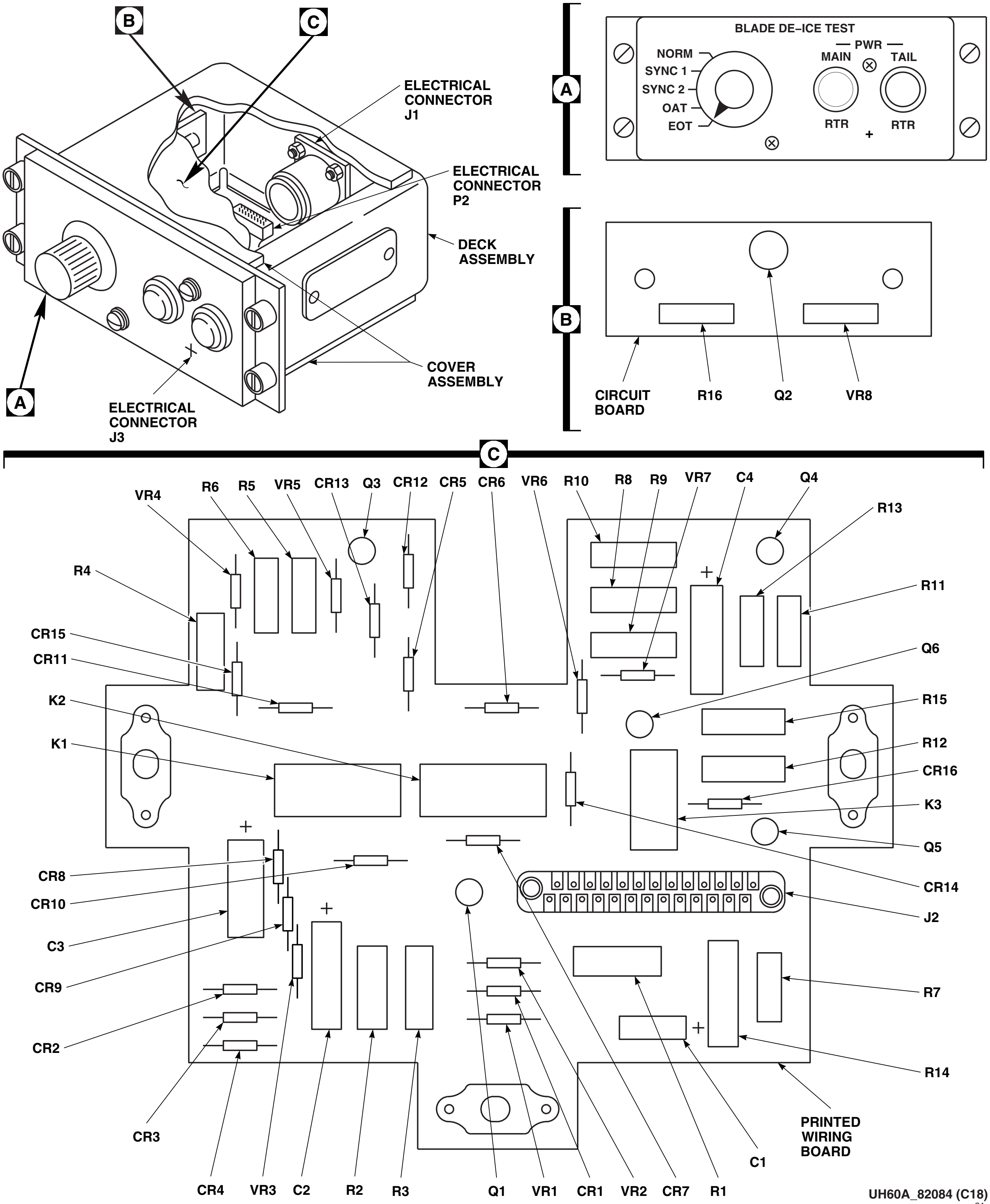


FIGURE 12-1-5-3. BLADE DE-ICE TEST PANEL LOCATION DIAGRAM

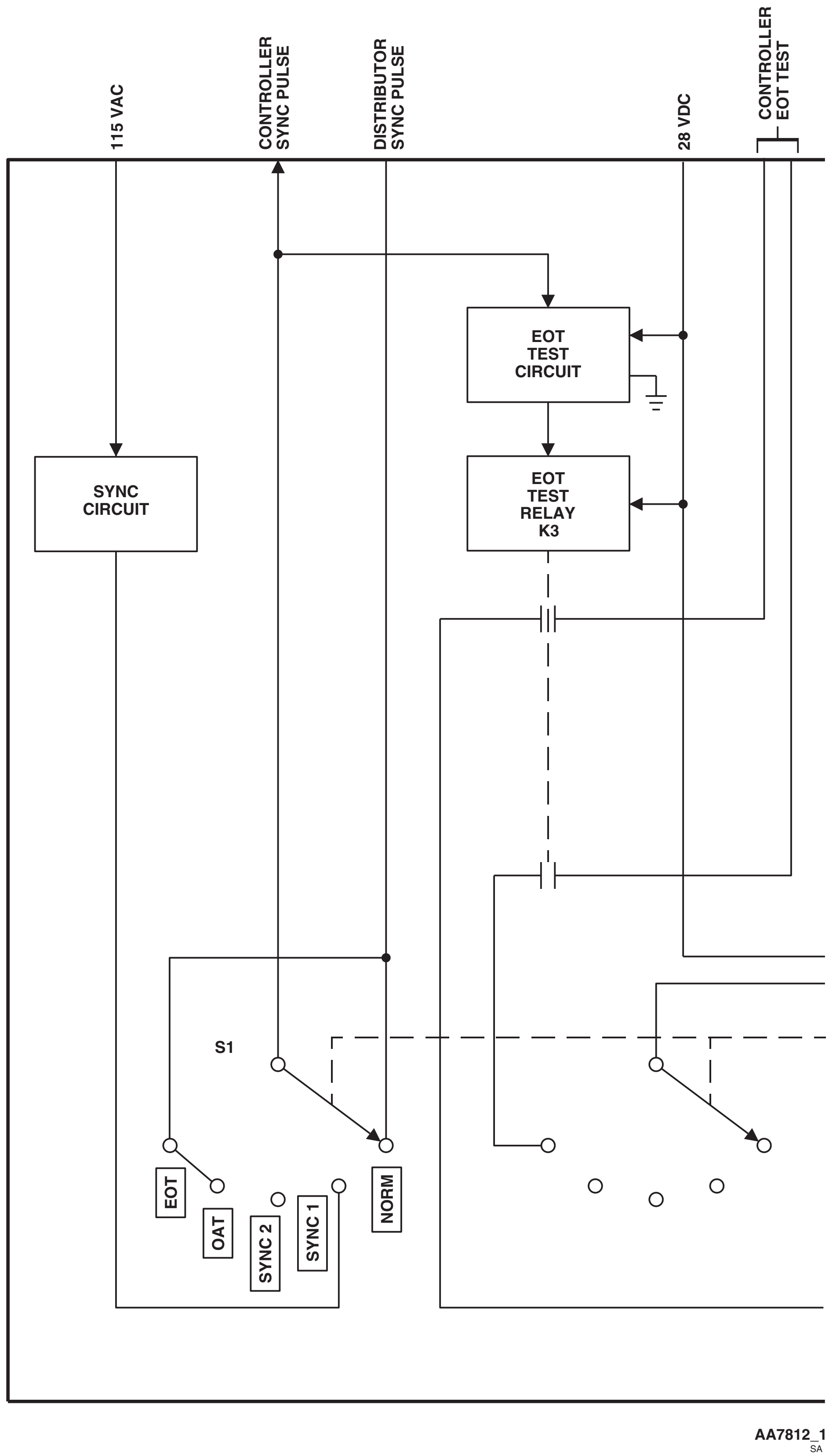


FIGURE 12-1-5-4. BLADE DE-ICE TEST PANEL BLOCK DIAGRAM (SHEET 1 OF 2)

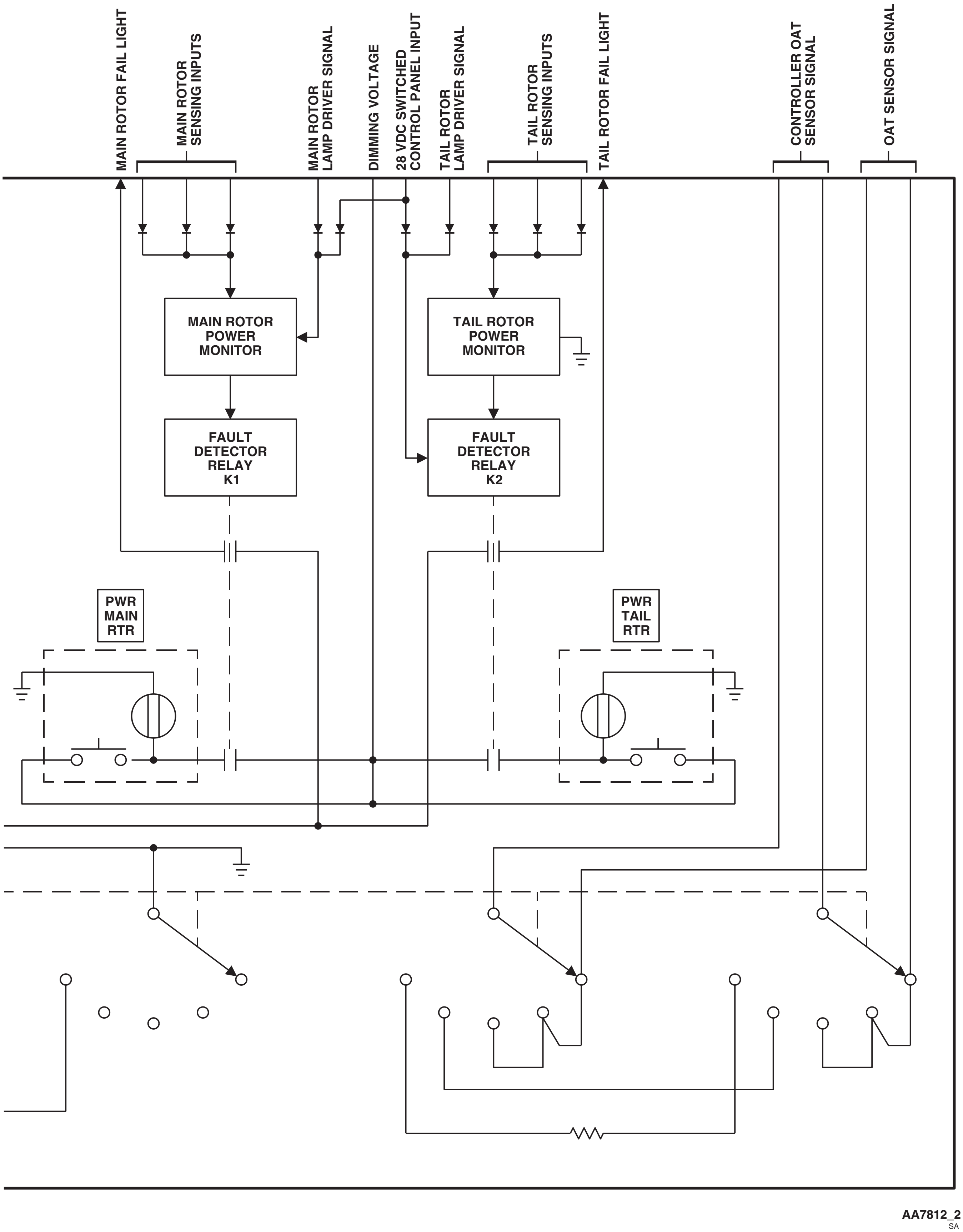


FIGURE 12-1-5-4. BLADE DE-ICE TEST PANEL BLOCK DIAGRAM (SHEET 2 OF 2)

12-1-6 CARGO HOOK SYSTEM DESCRIPTION AND DATA.

12-1-6.1 Cargo Hook System.

The cargo hook system consists of a 9000-pound-capacity hook and electrical circuits that control it (Figure 12-1-6-1). The cargo hook is located in the well area underneath the cabin floor. The electrical controls of the cargo hook system consist of the following: a CARGO HOOK ARMING switch labeled SAFE and ARMED; a CARGO HOOK CONTR switch labeled CKPT and ALL; CARGO HOOK EMERG REL switch labeled NORM, OPEN, and SHORT; a TEST light; CARGO REL switches on the pilot's and copilot's cyclic stick; a switch labeled CARGO HOOK NORMAL RLSE on the crewman's pendant; an emergency hook release button on the pilot's and copilot's collective sticks; and an EMER RLSE button on the crewman's cargo hook pendant.

The system incorporates three modes of load release:

(1) A normal release powered from the No. 2 dc primary bus through the CARGO HOOK CONTR and CARGO HOOK PWR circuit breakers.

(2) A manual release worked by the crewmember through a covered hatch in the cabin floor or by personnel on the ground.

(3) An emergency release system (cockpit or cabin controlled) using an electrically activated explosive charge. When 28 vdc is supplied to the cartridge it explodes, driving a piston inside the hook into the load arm lock. The load beam will not support a load, and the CARGO HOOK OPEN light will stay on until the old explosive charge has been replaced. Power to operate the emergency release system is by the dc essential bus through the CARGO HOOK EMER circuit breaker.

The cargo hook can be placed in a stowed position by opening the cargo hook access cover in the cabin floor and pulling the hook to the right and up. The cargo hook shall be maintained in the stow position while not in use. When the hook is in the stowed position, the load beam rests on a spring-loaded latch assembly and is prevented from vibrating by a Teflon bumper applying downward pressure on the load beam. To release the hook from its stowed position, downward pressure is placed on the latch assembly lever, retracting the latch from beneath the load beam, allowing the cargo hook to swing into the operating position.

12-1-6.2 Crewman's Pendant Control.

The crewman's cargo hook pendant consists of two normally-open pushbutton switches marked NORMAL RLSE and EMER RLSE. The cargo hook pendant (Figure 12-1-6-2) is for helicopters without the external hydraulic rescue hoist. The switches control the release of the cargo hook under normal and emergency conditions. Guards are mounted over each switch to prevent accidental cargo release. The pendant electrically interfaces with the helicopter system through a six-foot cable assembly. The pendant can be attached to the crewman by way of a strap assembly. Refer to PARA 14-1-1.3.3 for a description of the pendant control on helicopters equipped with the external hydraulic rescue hoist.

12-1-6.3 Power Distribution.

The cargo hook system gets dc electrical power from the pilot's circuit breaker panel. Electrical power of 28 vdc is supplied by the No. 2 dc primary bus through the CARGO HOOK PWR circuit breaker to the cargo hook relay and through the CARGO HOOK CONTR circuit breaker to the CARGO HOOK CONTR switch, CARGO HOOK ARMING switch, pilot's and copilot's CARGO REL. switches and to the load beam open switch.

12-1-6.4 Normal Release.

Placing the CARGO HOOK ARMING switch on the upper console to ARMED completes the circuit from the CARGO REL. switches on the pilot's and copilot's cyclic sticks to the hook. At the same time, the HOOK ARMED advisory light on the caution/advisory panel goes on. Pressing either CARGO REL. switch opens the cargo hook load arm and releases the load. At the same time the load arm starts to open, the CARGO HOOK OPEN advisory light goes on. When the load is released, the load arm will swing up and latch, and the CARGO HOOK OPEN advisory light will go off. If the CARGO HOOK CONTR switch is placed to ALL, the crewman's pendant can be used to release the load by pressing the CARGO HOOK NORMAL RLSE button (Figure 12-1-6-2).

12-1-6.5 Manual Release.

The cargo hook can be opened manually either through the cargo hook access panel in the cabin floor or from outside the helicopter. Pushing the manual release lever down, on the right side of the hook, opens the cargo hook and releases the load. If electrical power is on and the CARGO HOOK ARMING switch is at ARMED, the CARGO HOOK OPEN advisory light will go on and then off as the hook opens and closes.

12-1-6.6 Emergency Release.

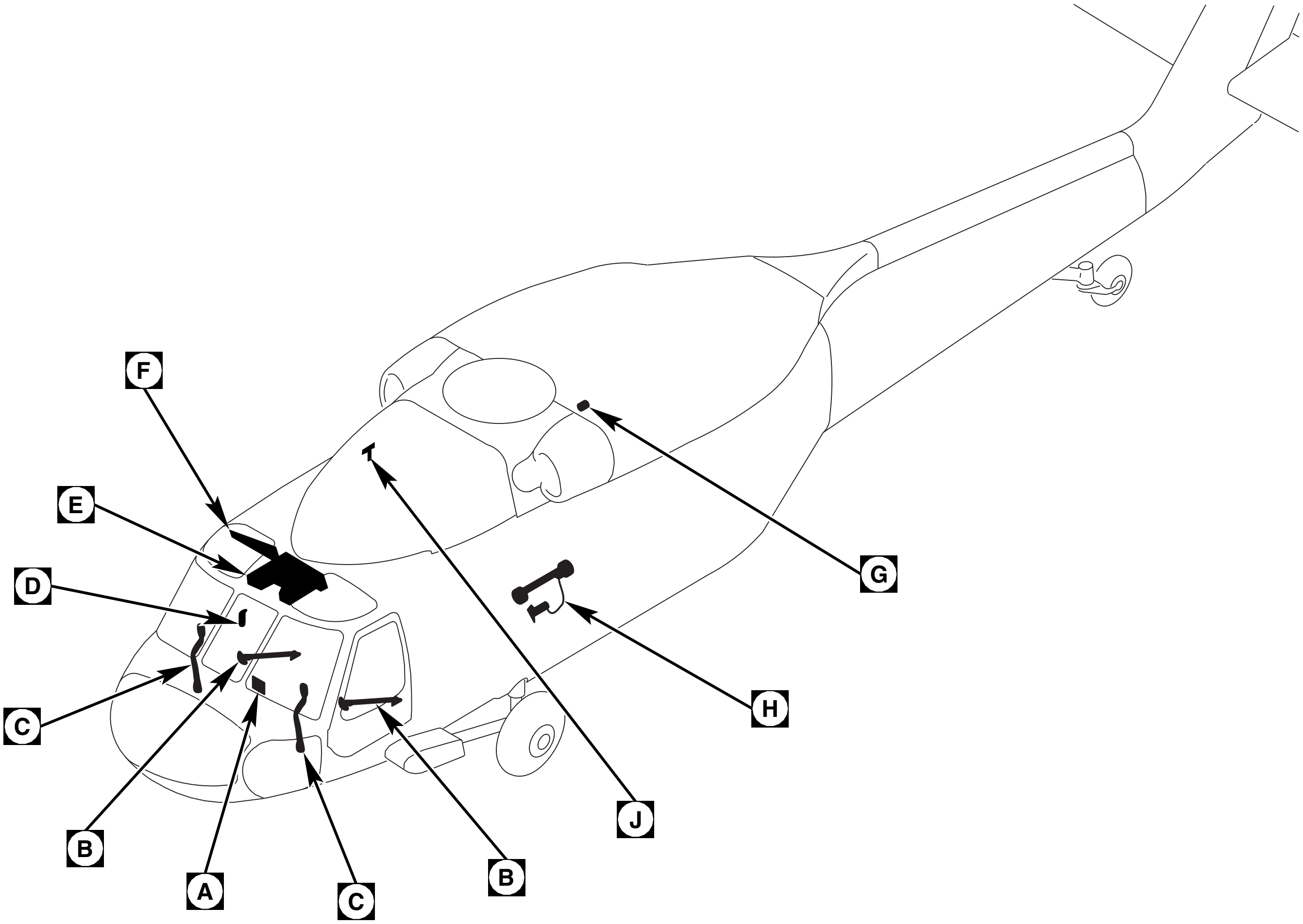
Cargo hook emergency release power is provided by the dc essential bus through the CARGO HOOK EMER circuit breaker. When the crewman's cargo hook pendant EMER RLSE button or the pilot's or copilot's collective stick emergency hook release switch is pressed, dc power is supplied through R4 to the EMERG REL switch on the upper console. With the EMERG REL switch placed to NORM, dc power is routed to the pressure cartridge (squib). The cartridge explodes and the pressure from the explosion drives a piston into the lock, releasing the load arm (Figure 12-1-6-3).

12-1-6.7 Test Function.

The test function of the cargo hook emergency release system checks the associated circuitry for open and short conditions. When the EMERG REL switch, on the upper console, is placed to OPEN or SHORT, the EMERG REL TEST light goes on if there are no open or short circuits in the emergency release circuitry. Power is routed from the left relay panel to one side of the EMERG REL TEST light. The press-to-test light circuit is completed to ground when the EMERG REL TEST light is pressed. Placing the EMERG REL switch to OPEN, and pressing the emergency hook release switch on the pilot's or copilot's collective stick, or the EMER RLSE button on the crewman's cargo hook pendant, causes the EMERG REL TEST light to go on if no open conditions exist in the emergency release circuitry. Placing the EMERG REL switch to SHORT and pressing the pilot's, copilot's, or crewman's emergency release button, causes the EMERG REL TEST light to go on if there are no short circuits in the emergency release test circuitry. The weight of the load causes the hook to open and the CARGO HOOK OPEN advisory light to go on. The CARGO HOOK OPEN light will remain on until the explosive charge has been replaced.

12-1-6.8 Cargo Hook Lights.

Lighting of the cargo hook well area is provided by three lights. For a further description of the cargo hook lighting, refer to PARA 9-1-9.



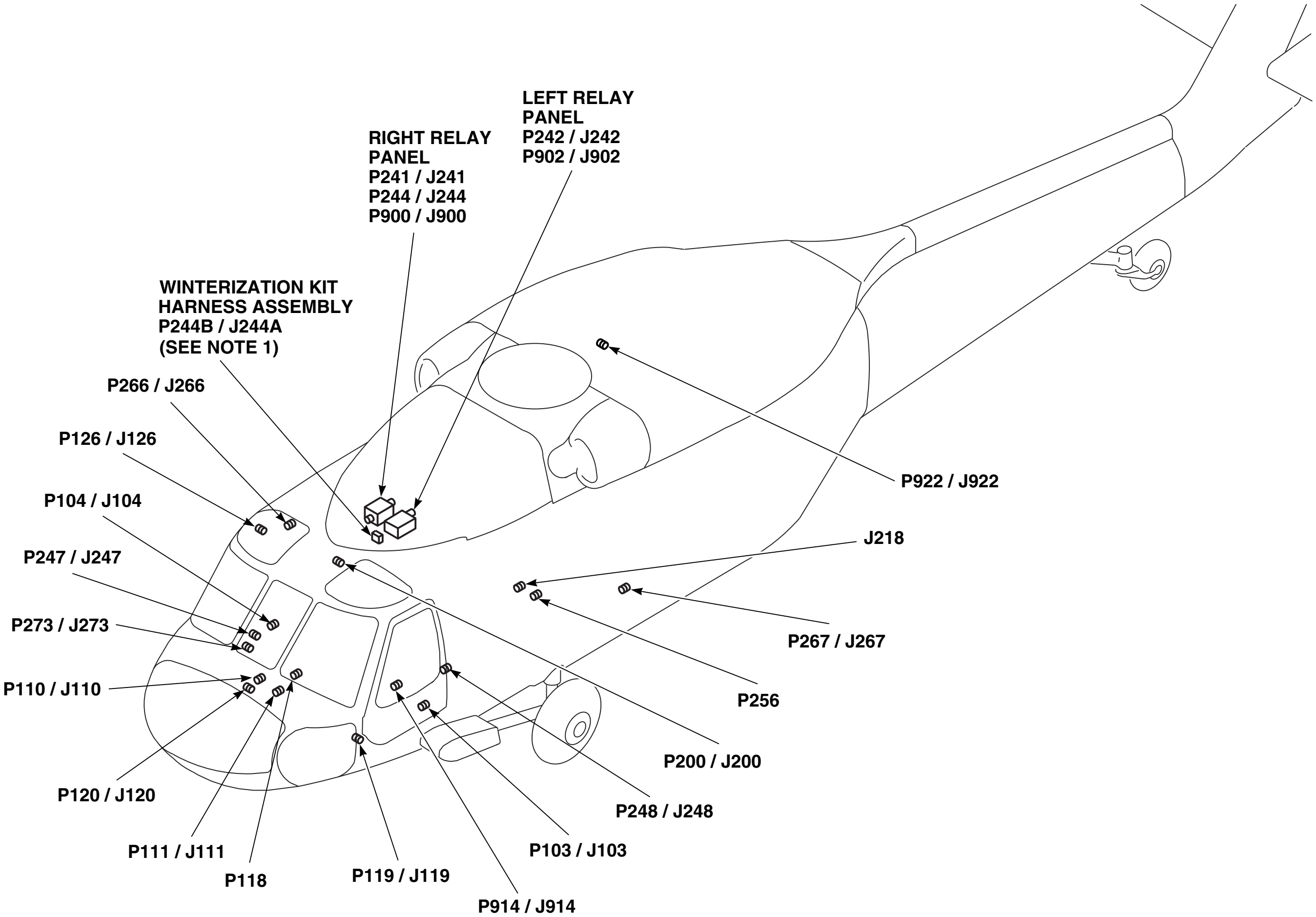
NOTES

1. WHEN WINTERIZATION KIT IS IN– STALLED, P244 IS CONNECTED TO J244A AND P244B IS CONNECTED TO RIGHT RELAY PANEL J244. WHEN WINTERIZATION KIT IS NOT IN– STALLED, P244 IS CONNECTED DI– RECTLY TO RIGHT RELAY PANEL J244.
2. RES HST
3. CYLIC GRIP ASSEMBLY PART NO. 70400–87261–102.
4. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–101.
5. ON GRIP ASSEMBLIES 70400–01405–103 AND –104, HOOK EMER REL IS EMERG HOOK REL .

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J218	CABIN TUB BL 9 RH, STA 343
P103 / J103	COCKPIT TUB BL 32 LH, STA 247
P104 / J104	COCKPIT TUB BL 15 RH, STA 247
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P118	CAUTION / ADVISORY PANEL
P119 / J119	COCKPIT TUB BL 30 LH, STA 220
P120 / J120	COCKPIT TUB BL 30 RH, STA 220

FB2005_1A
SA

FIGURE 12-1-6-1. CARGO HOOK SYSTEM LOCATION DIAGRAM (SHEET 1 OF 5)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P200 / J200	COCKPIT CEILING BL 13 RH, STA 246
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P244 / J244	RIGHT RELAY PANEL
P244B / J244A (SEE NOTE 1)	WINTERIZATION KIT HARNESS ASSEMBLY
P247 / J247	CABIN BL 40 RH, STA 248
P248 / J248	CABIN BL 40 LH, STA 248
P256	CABIN TUB BL 15 RH, STA 342

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P267 / J267	CABIN BL 14 LH, STA 381
P273 / J273	CABIN BL 40 RH, STA 248
P900 / J900	RIGHT RELAY PANEL
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT BL 24 LH, STA 247
P922 / J922 (SEE NOTE 2)	CABIN CEILING BL 10.7 LH, STA 380
P968 / J968 (SEE NOTE 2)	RESCUE HOIST / CARGO HOOK PENDANT

FB2005_2A
SA

FIGURE 12-1-6-1. CARGO HOOK SYSTEM LOCATION DIAGRAM (SHEET 2 OF 5)

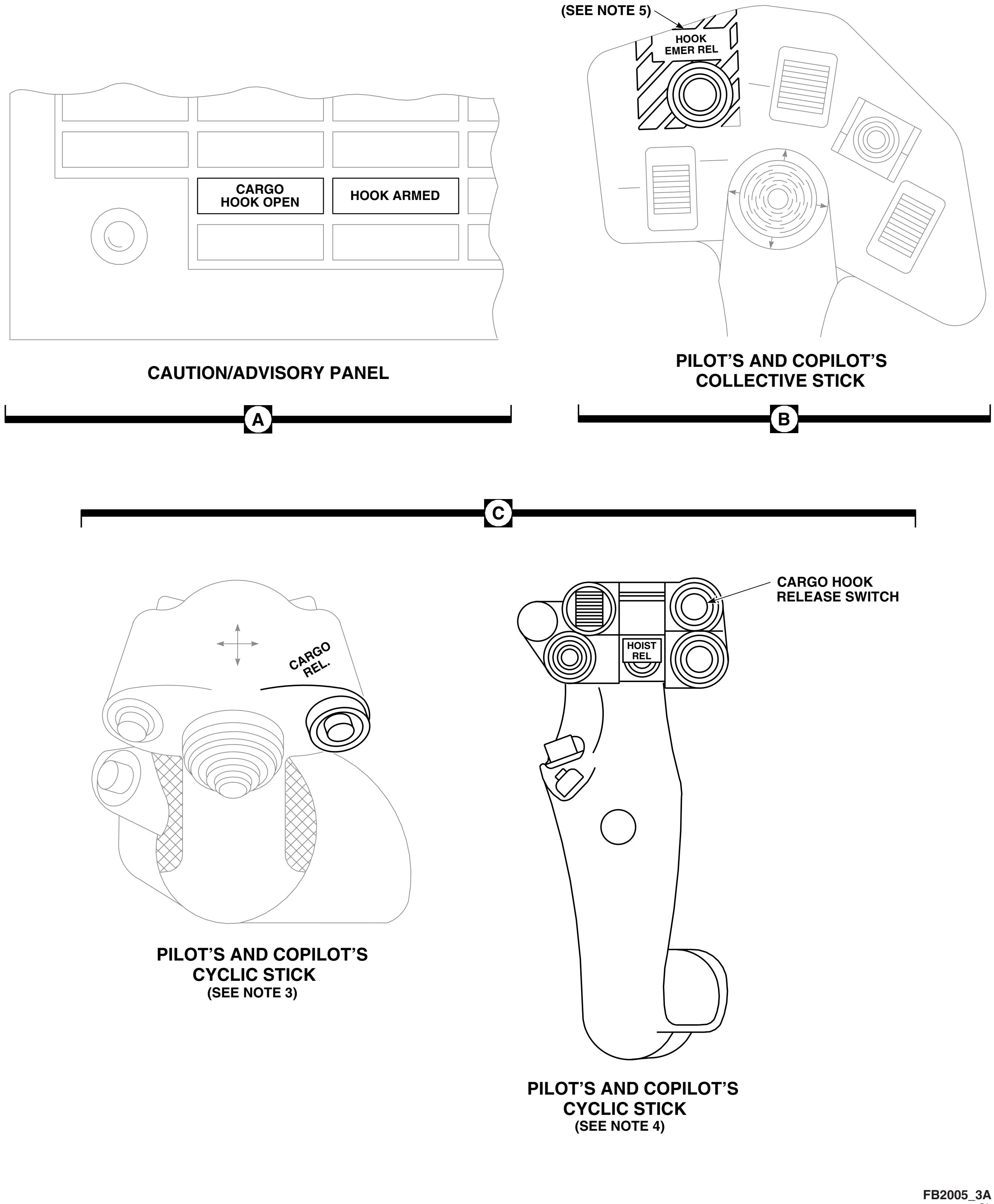
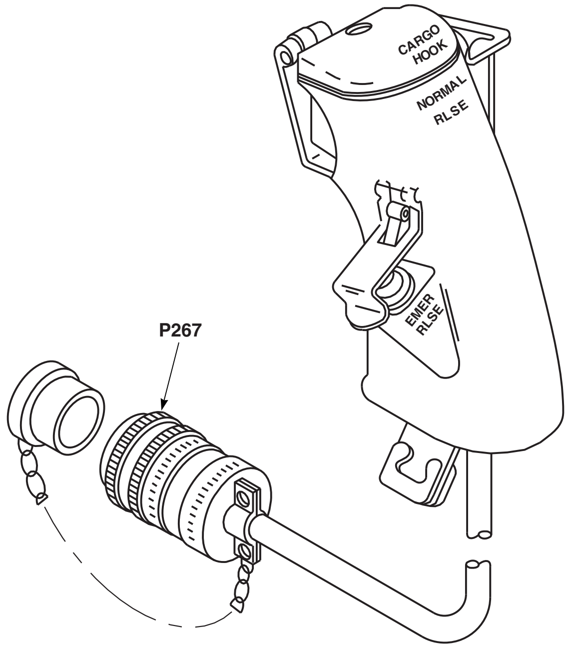


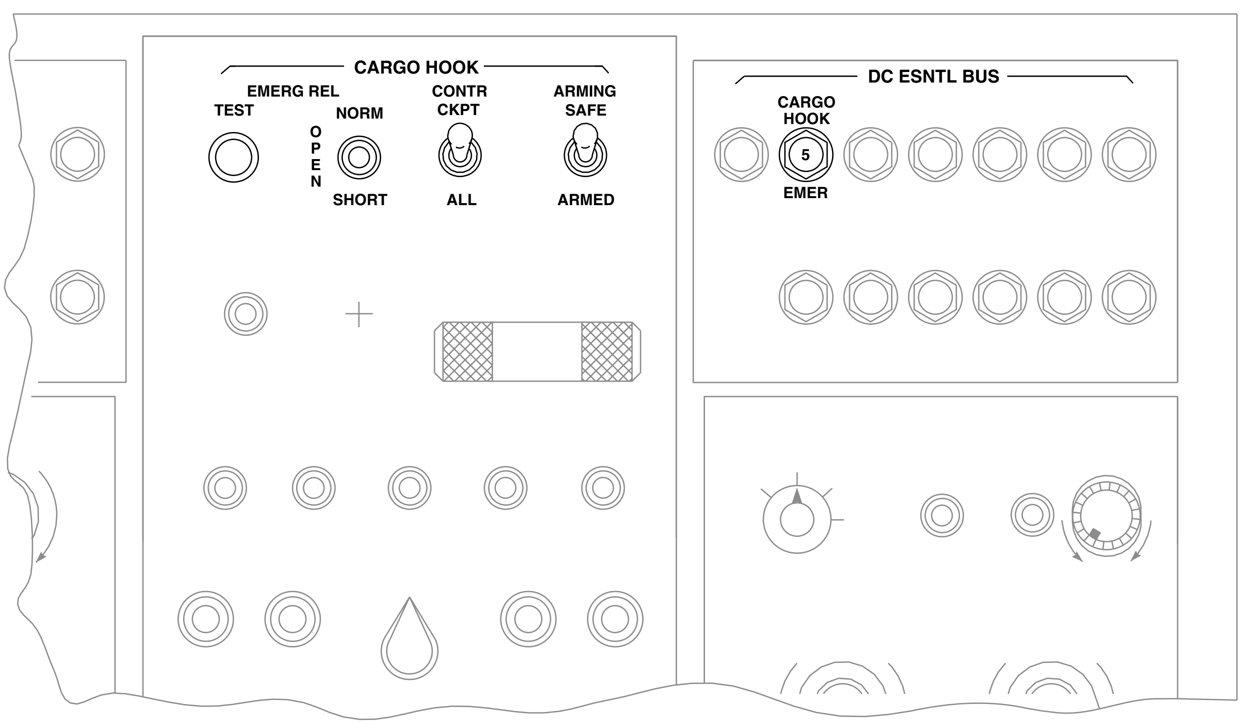
FIGURE 12-1-6-1. CARGO HOOK SYSTEM LOCATION DIAGRAM (SHEET 3 OF 5)



CREWMAN'S CARGO HOOK PENDANT

D

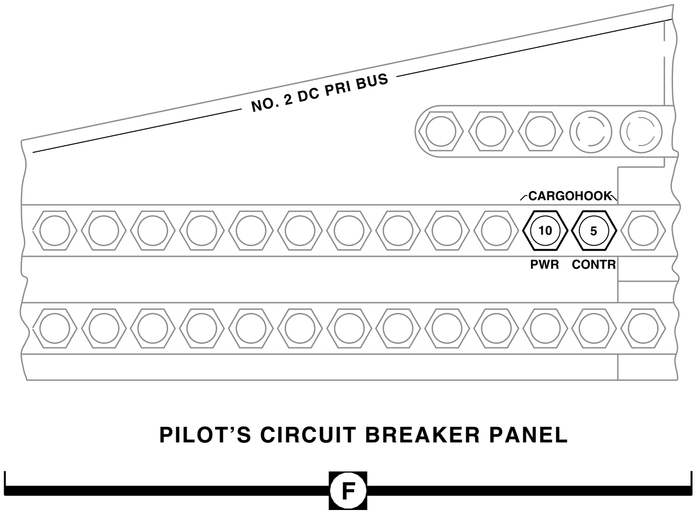
E

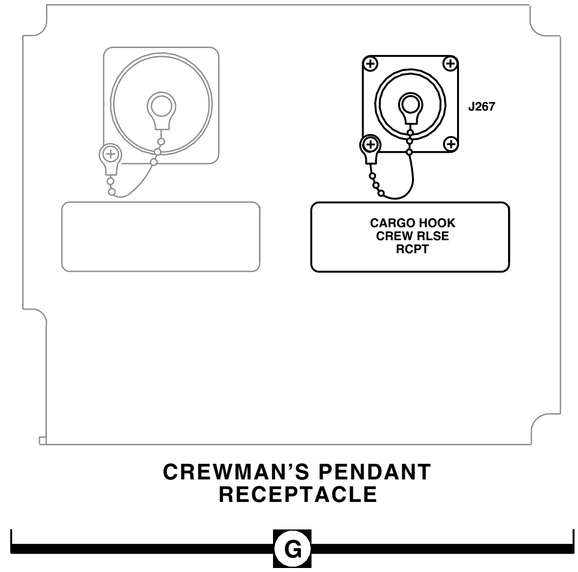


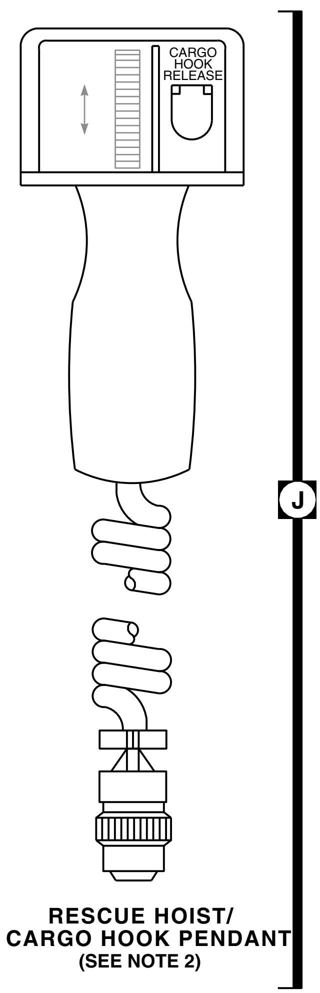
UPPER CONSOLE

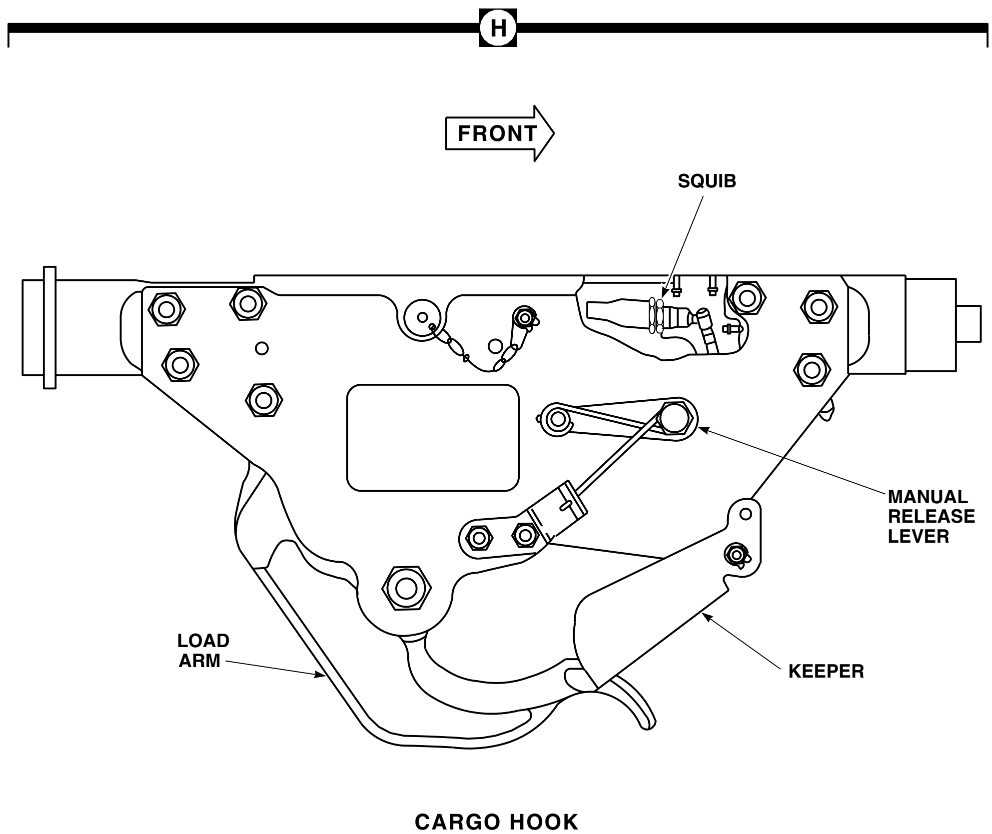
FB2005_4
SA

FIGURE 12-1-6-1. CARGO HOOK SYSTEM LOCATION DIAGRAM (SHEET 4 OF 5)









FB2005_5
SA

FIGURE 12-1-6-1. CARGO HOOK SYSTEM LOCATION DIAGRAM (SHEET 5 OF 5)

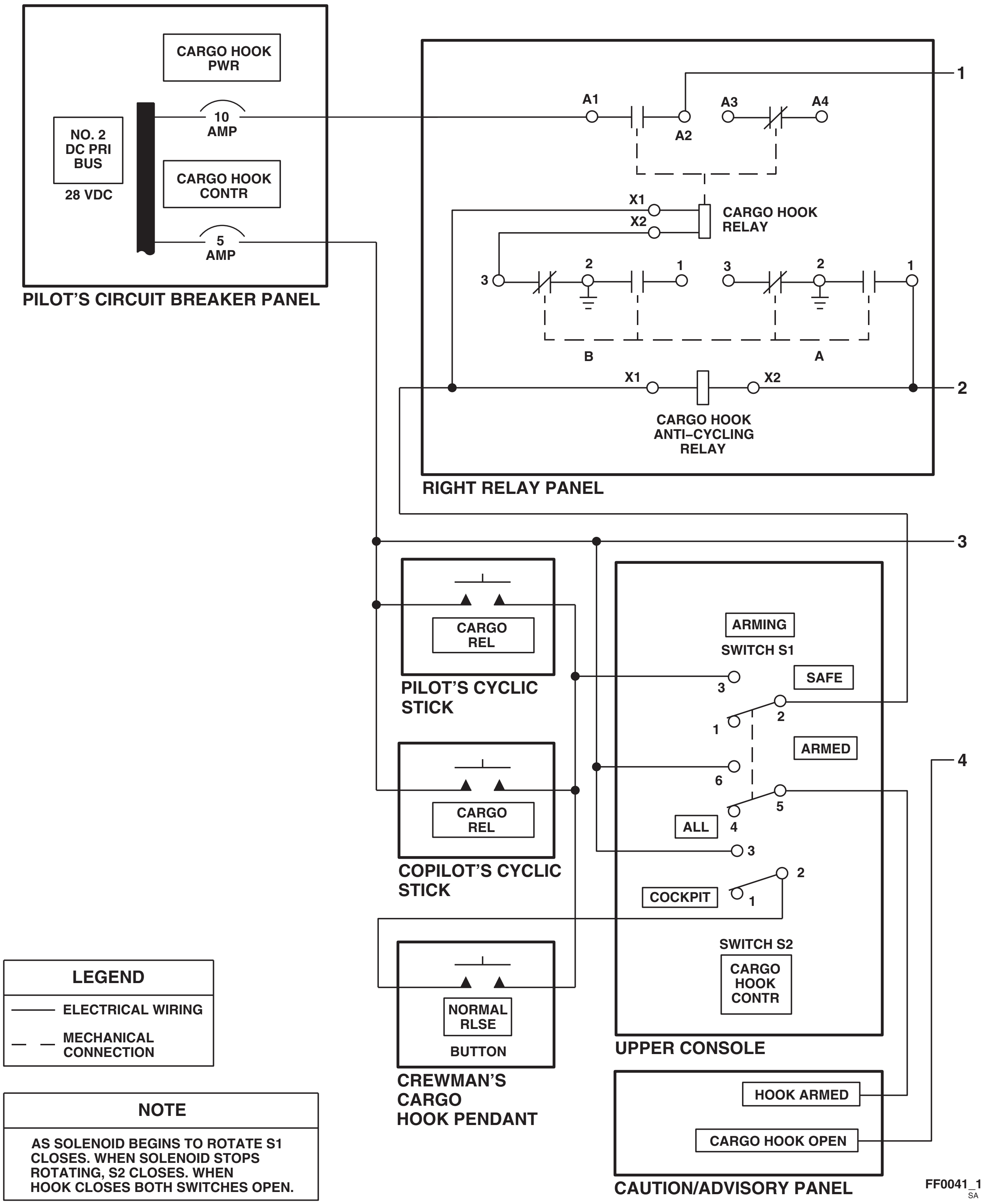


FIGURE 12-1-6-2. CARGO HOOK SYSTEM NORMAL RELEASE BLOCK DIAGRAM (SHEET 1 OF 2)

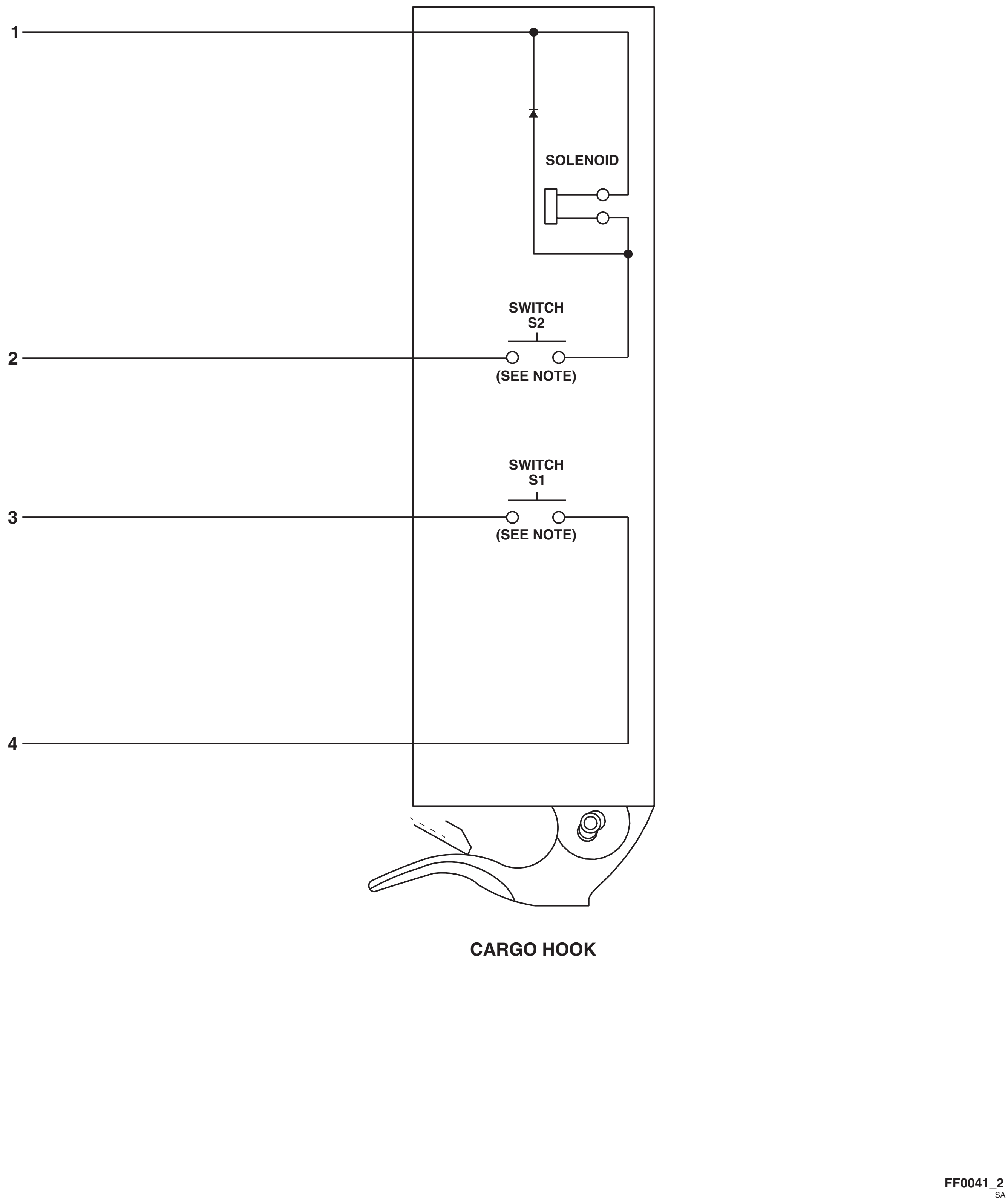


FIGURE 12-1-6-2. CARGO HOOK SYSTEM NORMAL RELEASE BLOCK DIAGRAM (SHEET 2 OF 2)

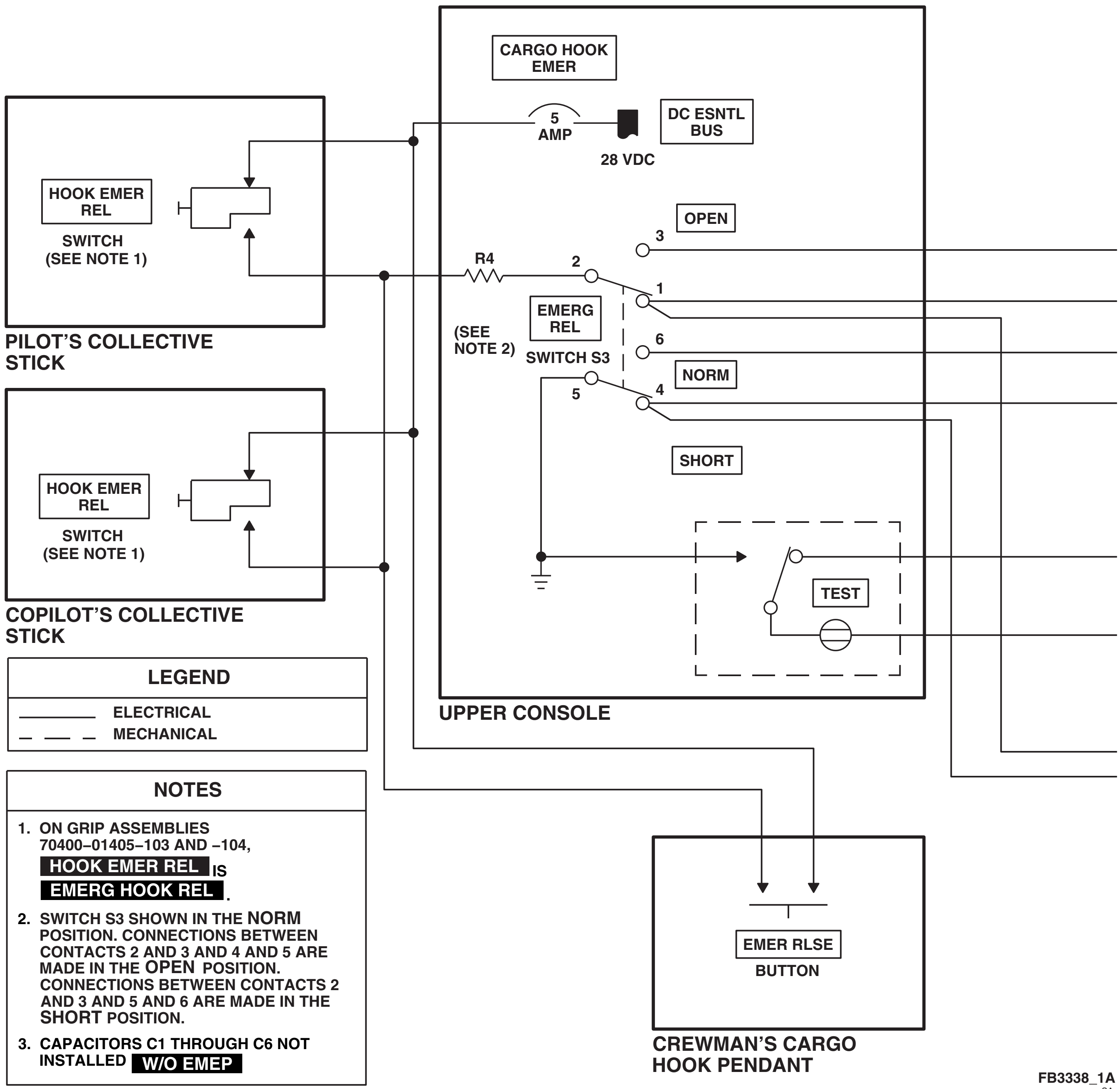


FIGURE 12-1-6-3. CARGO HOOK SYSTEM EMERGENCY RELEASE/TEST BLOCK DIAGRAM (SHEET 1 OF 2)

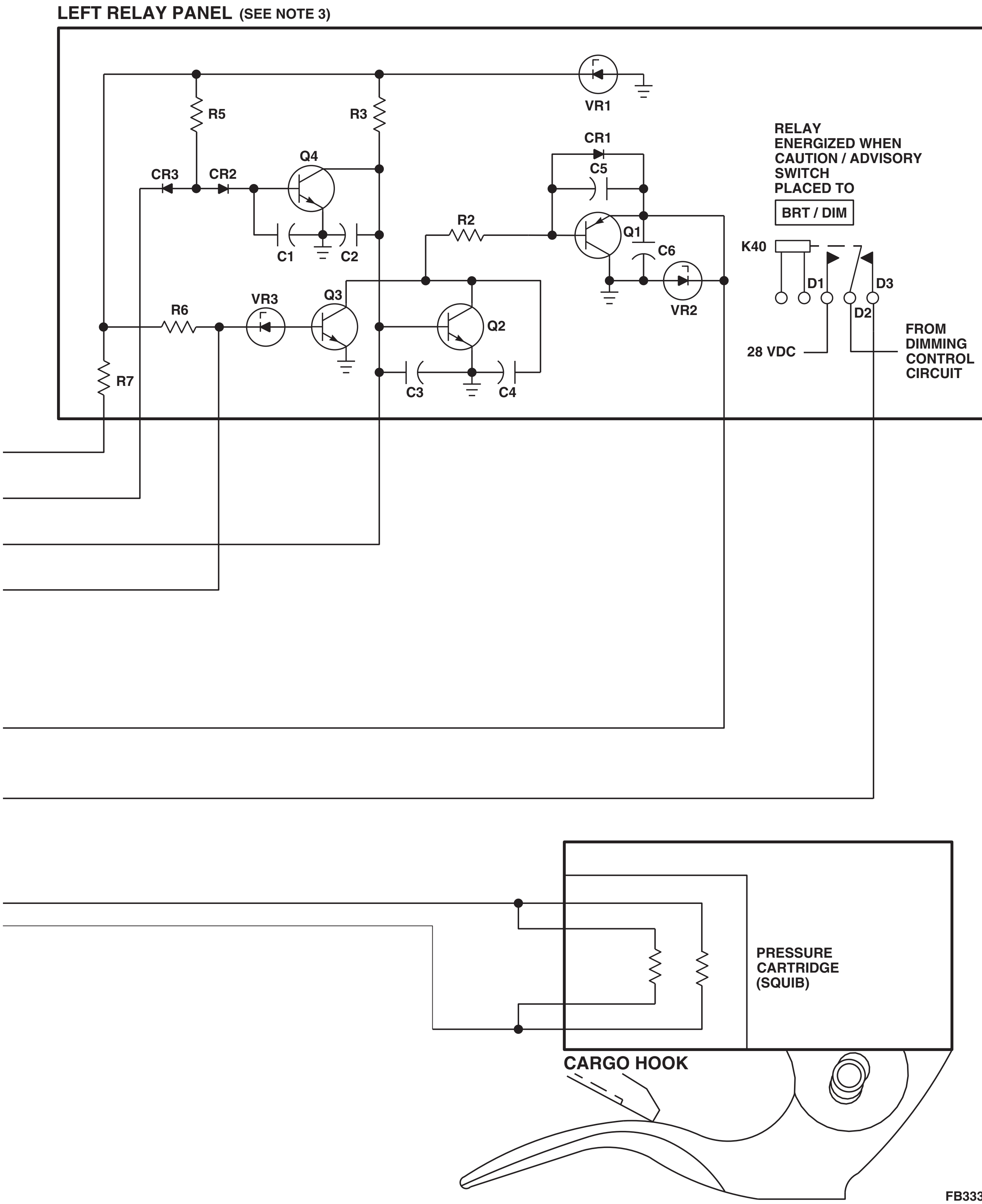


FIGURE 12-1-6-3. CARGO HOOK SYSTEM EMERGENCY RELEASE/TEST BLOCK DIAGRAM (SHEET 2 OF 2)

SECTION 12-2

TROUBLESHOOTING

SECTION OVERVIEW

12-2-1 FIRE DETECTION SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- Battery Power Available
- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

The photosensitive material in the fire detectors is sensitive to red light. Sunlight filtered through smoke or haze, or at sunrise or sunset, may contain greater amounts of red light. This could trigger the fire detectors and cause a false fire indication. For this reason, fire detectors must be shielded from sunlight.

Special Instructions

The fire detection system operational/troubleshooting procedure is subdivided as follows:

12-2-1.1 Quick Check

12-2-1.2 Detailed Check

12-2-1.1 Quick Check.

a. Inspect all external lines and fittings on the APU Combustion Section for signs of fluid leakage.

RESULT: No leakage shall be found.

- If result is not as specified, check coupling nuts and fittings for tightness.

  (a) If coupling nuts or fittings are loose, torque coupling nuts or fittings as required (PARAs 7-4-61, 15-4-9, and 15-4-11).

b. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

c. Place upper console BATT switch to ON.

d. Turn FIRE DET TEST switch to position 1.

RESULT:

(1) #1 ENG EMER OFF T-handle lamps shall be on.

(2) #2 ENG EMER OFF T-handle lamps shall be on.

(3) APU T-handle lamps shall be on.

- If results are not as specified, do Detailed Check

e. Turn FIRE DET TEST switch to position 2.

RESULT:

(1) #1 ENG EMER OFF T-handle lamps shall be on.

(2) #2 ENG EMER OFF T-handle lamps shall be on.

(3) APU T-handle lamps shall be off.

- If results are not as specified, do Detailed Check

f. Turn FIRE DET TEST switch to OPER.

12-2-1.2 Detailed Check.

a. If not already done, inspect all external lines and fittings on the APU Combustion Section for signs of fluid leakage.

RESULT: No leakage shall be found.

- If result is not as specified, check coupling nuts and fittings for tightness.

(a) If coupling nuts or fittings are loose, torque coupling nuts or fittings as required (PARAs 7-4-61, 15-4-9, and 15-4-11).

b. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

c. Turn on electrical power (PARA 1-6-16).

d. Make sure upper console FIRE DET TEST switch is at OPER.

RESULT:

(1) #1 ENG EMER OFF T-handle lamps shall be off.

(2) #2 ENG EMER OFF T-handle lamps shall be off.

(3) APU T-handle lamps shall be off.

- If result (1) is not as specified, go to TABLE NO. 12-2-1-1.

- If result (2) is not as specified, go to TABLE NO. 12-2-1-2.

- If result (3) is not as specified, check for less than 9.0 VDC between P285-1 and P285-10.

(a) If voltage is less than 9.0 VDC, replace APU fire detector control amplifier (PARA 12-4-11).

(b) If voltage is greater than 9.0 VDC, replace APU fire detector (PARA 12-4-12).

e. Pull out the following circuit breakers.

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
FIRE DET NO. 2 ENG	DC ESNTL BUS - Pilot's Side Upper Console
APU FIRE DET	BATT BUS - Lower console

f. Turn FIRE DET TEST switch to position 1.

RESULT:

(1) #1 ENG EMER OFF T-handle lamps and pilot's and copilot's master warning FIRE capsules shall go on.

(2) #2 ENG EMER OFF and APU T-handle lamps shall be off.

- If result (1) is not as specified, go to TABLE NO. 12-2-1-3.

- If both #1 ENG EMER OFF T-handle lamps or both master warning panel FIRE capsules are off, replace left relay panel (PARA 9-4-2).

- If pilot's master warning panel FIRE capsule is off, go to TABLE NO. 12-2-1-4.

- If copilot's master warning panel FIRE capsule is off, go to TABLE NO. 12-2-1-5.

- If result (2) is not as specified, replace left relay panel (PARA 9-4-2).

g. Turn FIRE DET TEST switch to position 2.

RESULT: #1 ENG EMER OFF T-handle lamps, and pilot's and copilot's master warning panel FIRE capsules shall be on.

- If result is not as specified, go to TABLE NO. 12-2-1-6.

h. Pull out FIRE DET NO. 1 ENG circuit breaker (DC ESNTL BUS - pilot's side upper console).

i. Push in FIRE DET NO. 2 ENG circuit breaker (DC ESNTL BUS - pilot's side upper console).

j. Turn FIRE DET TEST switch to position 1.

RESULT:

(1) #2 ENG EMER OFF T-handle lamps and pilot's and copilot's master warning panel FIRE capsules shall go on.

(2) #1 ENG EMER OFF T-handle lamps shall be off.

- If result (1) is not as specified, go to TABLE NO. 12-2-1-7.
- If both #2 ENG EMER OFF T-handle lamps or both master warning panel FIRE capsules are off, replace left relay panel (PARA 9-4-2).
- If result (2) is not as specified, replace left relay panel (PARA 9-4-2).

k. Turn FIRE DET TEST switch to position 2.

RESULT: #2 ENG EMER OFF T-handle lamps and pilot's and copilot's master warning panel FIRE capsules shall be on.

- If result is not as specified, go to TABLE NO. 12-2-1-8.

l. Pull out FIRE DET NO. 2 ENG circuit breaker (DC ESNTL BUS - pilot's side upper console).

m. Push in APU FIRE DET circuit breaker (BATT BUS - lower console).

n. Turn FIRE DET TEST switch to position 1.

RESULT: APU FIRE EXTGH T-handle lamps and pilot's and copilot's master warning panel FIRE capsules shall go on.

- If result is not as specified, go to TABLE NO. 12-2-1-9.
- If both master warning panel FIRE capsules are off, replace left relay panel (PARA 9-4-2).
- If both master warning panel FIRE capsules are on, but APU T-handle lamps are off, go to TABLE NO. 12-2-1-10.

o. Turn FIRE DET TEST switch to OPER.

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

p. Shine a flashlight with a red or blue filter on APU fire detector.

RESULT: APU T-handle lamps shall go on.

- If result is not as specified, check for 28 VDC between P414-A and P414-B.

  (a) If voltage is as specified, replace APU fire detector (PARA 12-4-12).

  (b) If voltage is not as specified, repair/replace wiring (Appendix F).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

q. Shine a flashlight with other than a red or blue filter on APU fire detector.

RESULT: APU T-handle lamps shall not go on.

- If result is not as specified, replace APU fire detector (PARA 12-4-12).

r. Push in FIRE DET NO. 1 ENG circuit breaker (DC ESNTL BUS - pilot's side upper console).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

s. Shine a flashlight with a red or blue filter on No. 1 engine firewall fire detector.

RESULT: #1 ENG EMER OFF T-handle lamps shall go on.

- If result is not as specified, check for 28 VDC between P413-A and P413-B.

  (a) If voltage is as specified, replace No. 1 engine firewall fire detector (PARA 12-4-12).

  (b) If voltage is not as specified, repair/replace wiring (Appendix F).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

t. Shine a flashlight with other than a red or blue filter on No. 1 engine firewall fire detector.

RESULT: #1 ENG EMER OFF T-handle lamps shall not go on.

- If result is not as specified, replace No. 1 engine fire detector (PARA 12-4-12).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

u. Shine a flashlight with a red or blue filter on No. 1 engine deck fire detector.

RESULT: #1 ENG EMER OFF T-handle lamps shall go on.

- If result is not as specified, check for 28 VDC between P286-A and P286-B.

  (a) If voltage is as specified, replace No. 1 engine deck fire detector (PARA 12-4-12).

  (b) If voltage is not as specified, repair/replace wiring (Appendix F).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

v. Shine a flashlight with other than a red or blue filter on No. 1 engine deck fire detector.

RESULT: #1 ENG EMER OFF T-handle lamps shall not go on.

- If result is not as specified, replace No. 1 engine deck fire detector (PARA 12-4-12).

w. Push in FIRE DET NO. 2 ENG circuit breaker (DC ESNTL BUS - pilot's side upper console).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

x. Shine a flashlight with a red or blue filter on No. 2 engine firewall fire detector.

RESULT: #2 ENG EMER OFF T-handle lamps shall go on.

- If result is not as specified, check for 28 VDC between P415-A and P415-B.

  (a) If voltage is as specified, replace No. 2 engine firewall fire detector (PARA 12-4-12).

  (b) If voltage is not as specified, repair/ replace wiring (Appendix F).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

y. Shine a flashlight with other than a red or blue filter on No. 2 engine firewall fire detector.

RESULT: #2 ENG EMER OFF T-handle lamps shall not go on.

- If result is not as specified, replace No. 2 engine firewall fire detector (PARA 12-4-12).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

z. Shine a flashlight with a red or blue filter on No. 2 engine deck fire detector.

RESULT: #2 ENG EMER OFF T-handle lamps shall go on.

- If result is not as specified, check for 28 VDC between P287-A and P287-B.

  (a) If voltage is as specified, replace No. 2 engine deck fire detector (PARA 12-4-12).

  (b) If voltage is not as specified, repair/ replace wiring (Appendix F).

NOTE

The flashlight must be positioned between 6 and 8 inches from the fire detector, with the flashlight beam aimed directly at the fire detector.

aa Shine a flashlight with other than a red or blue filter on No. 2 engine deck fire detector.

RESULT: #2 ENG EMER OFF T-handle lamps shall not go on.

- If result is not as specified, replace No. 2 engine deck fire detector (PARA 12-4-12).

ab. Turn FIRE DET TEST switch to position 1.

RESULT: #1 ENG EMER OFF, #2 ENG EMER OFF, and APU T-handles lamps shall go on.

- If result is not as specified, repeat operational/troubleshooting procedure.

ac. Rotate INST LT PILOT FLT fully counterclockwise, then rotate 1/4 turn clockwise.

ad. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM.

RESULT: #1 ENG EMER OFF, #2 ENG EMER OFF, and APU T-handles lamps shall go on dimly.

- If all T-handles remain bright, go to TABLE NO. 12-2-1-11.

- If any T-handle remains bright, replace left relay panel (PARA 9-4-2).

ae. Turn FIRE DET TEST switch to OPER.

af. Turn off electrical power (PARA 1-6-16).

TABLE NO. 12-2-1-1. #1 ENG EMER OFF T-handle Lamps Are On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for less than 9.0 VDC between P283-1 and P283-10.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, replace No. 1 engine firewall fire detector (PARA 12-4-12). Go to **3.**

2. Check for less than 9.0 VDC between P283-2 and P283-10.

Step 1. If voltage is as specified, replace No. 1 engine fire detector control amplifier (PARA 12-4-11). Go to **3.**
Step 2. If voltage is not as specified, replace No. 1 engine deck fire detector (PARA 12-4-12). Go to **3.**

3. Procedure completed.

TABLE NO. 12-2-1-2. #2 ENG EMER OFF T-handle Lamps Are On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for less than 9.0 VDC between P284-1 and P284-10.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, replace No. 2 engine firewall fire detector (PARA 12-4-12). Go to **3.**

2. Check for less than 9.0 VDC between P284-2 and P284-10.

Step 1. If voltage is as specified, replace No. 2 engine fire detector control amplifier (PARA 12-4-11). Go to **3.**
Step 2. If voltage is not as specified, replace No. 2 engine deck fire detector (PARA 12-4-12). Go to **3.**

TABLE NO. 12-2-1-2. #2 ENG EMER OFF T-handle Lamps Are On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

3. **Procedure completed.**

TABLE NO. 12-2-1-3. #1 ENG EMER OFF T-handle Lamps And Pilot's And Copilot's Master Warning FIRE Capsules Are Off In Position 1.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Set FIRE DET TEST switch to position 2 and observe that No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules go on.**

   Step 1. If No. 1 engine T-handle warning lamps and master warning panel FIRE capsules go on, go to **2.**
   Step 2. If No. 1 engine T-handle warning lamps and master warning panel FIRE capsules do not go on, go to **6.**

2. **Set FIRE DET TEST switch to position 1 and check continuity between FIRE DET TEST switch terminals BC1 and B2.**

   Step 1. If continuity is present, go to **3.**
   Step 2. If continuity is not present, replace FIRE DET TEST switch (PARA 12-4-13). Go to **15.**

3. **Check for 15 VDC between P283-1 and P283-10.**

   Step 1. If voltage is as specified, replace No. 1 engine fire detector control amplifier (PARA 12-4-11). Go to **15.**
   Step 2. If voltage is not specified, go to **4.**

4. **Check for 15 VDC between P413-E and ground.**

   Step 1. If voltage is as specified, go to **5.**
   Step 2. If voltage is not as specified, repair/replace wiring between FIRE DET TEST switch terminal BC1 and P413-E (Appendix F). Go to **15.**

5. **Check continuity between P413-C and P283-1.**

   Step 1. If continuity is present, replace No. 1 engine firewall fire detector (PARA 12-4-12). Go to **15.**

TABLE NO. 12-2-1-3. #1 ENG EMER OFF T-handle Lamps And Pilot's And Copilot's Master Warning FIRE Capsules Are Off In Position 1. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **15.**

6. Check for 28 VDC between P283-11 and P283-10.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **13.**

7. Set FIRE DET TEST switch to position 1 and check for 15 VDC between P283-1 and P283-10.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **10.**

8. Install an insulated jumper wire between P283-11 and P283-9 and observe that No. 1 engine T-handle warning lamps and master warning panel FIRE capsules go on.

Step 1. If No. 1 engine T-handle warning lamps and master warning panel FIRE capsules go on, remove jumper wire and replace No. 1 engine fire detector control amplifier (PARA 12-4-11). Go to **15.**
Step 2. If No. 1 engine T-handle warning lamps and master warning panel FIRE capsules do not go on, remove jumper wire and go to **9.**

9. Check continuity between P283-9 and P902-M.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **15.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **15.**

10. Check for 15 VDC between FIRE DET TEST switch terminal B2 and ground.

Step 1. If voltage is as specified, replace FIRE DET TEST switch (PARA 12-4-13). Go to **15.**
Step 2. If voltage is not as specified, go to **11.**

11. Check for 28 VDC between P902-P and ground.

Step 1. If voltage is as specified, go to **12.**
Step 2. If voltage is not as specified, go to **14.**

12. Check for 9,800 to 10,200 ohms between J902-F and J902-P.

TABLE NO. 12-2-1-3. #1 ENG EMER OFF T-handle Lamps And Pilot's And Copilot's Master Warning FIRE Capsules Are Off In Position 1. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If resistance is as specified, repair/replace wiring between P902-F and FIRE DET TEST switch terminal B2 (Appendix F). Go to **15.**
Step 2. If resistance is not as specified, replace left relay panel (PARA 9-4-2). Go to **15.**

13. Check continuity between P283-10 and ground.

Step 1. If continuity is present, go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **15.**

14. Check for 28 VDC between terminal 1 (FIRE DET NO. 1 ENG circuit breaker, DC ESNTL BUS - pilot's side upper console) and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P283-11 (Appendix F). Go to **15.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **15.**

15. Procedure completed.

TABLE NO. 12-2-1-4. Pilot's Master Warning Panel FIRE Capsule Does Not Go On During Fire Detector Test.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for burnt-out lamp.

Step 1. If lamp is good, go to **2.**
Step 2. If lamp is not good, replace lamp (PARA 8-4-25). Go to **7.**

2. Check continuity between J243-G and J902-H.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, replace left relay panel (PARA 9-4-2). Go to **7.**

3. Check helicopter configuration.

TABLE NO. 12-2-1-4. Pilot's Master Warning Panel FIRE Capsule Does Not Go On During Fire Detector Test. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. EMEP , go to **4.**
Step 2. W/O EMEP , go to **6.**

EMEP **4. Remove and check pin filter adapter (PFA) from P130 (PARA 9-4-98).**

Step 1. If PFA is good, go to **5.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **7.**

EMEP **5. Install PFA (PARA 9-4-98). Go to 6.**

6. Check continuity between P130-4 and P243-G.

Step 1. If continuity is present, replace master warning panel (PARA 8-4-24). Go to **7.**
Step 2. If trouble remains, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **7.**
Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-1-5. Copilot's Master Warning Panel FIRE Capsule Does Not Go On During Fire Detector Test.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for burnt-out lamp.

Step 1. If lamp is good, go to **2.**
Step 2. If lamp is not good, replace lamp (PARA 8-4-25). Go to **7.**

2. Check continuity between J243-G and J902-H.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, replace left relay panel (PARA 9-4-2). Go to **7.**

3. Check helicopter configuration.

TABLE NO. 12-2-1-5. Copilot's Master Warning Panel FIRE Capsule Does Not Go On During Fire Detector Test. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. EMEP , go to **4.**
Step 2. W/O EMEP , go to **6.**

EMEP **4. Remove and check PFA from P131 (PARA 9-4-98).**

Step 1. If PFA is good, go to **5.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **7.**

EMEP **5. Install PFA (PARA 9-4-98). Go to 6.**

6. Check continuity between P131-4 and P902-H.

Step 1. If continuity is present, replace master warning panel (PARA 8-4-24). Go to **7.**
Step 2. If trouble remains, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **7.**
Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-1-6. #1 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 2.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Set FIRE DET TEST switch to position 2 and check continuity between FIRE DET TEST switch terminals BC2 and B8.

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, replace FIRE DET TEST switch (PARA 12-4-13). Go to **7.**

2. Check for 15 VDC between P286-E and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **6.**

TABLE NO. 12-2-1-6. #1 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 2. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

3. Check continuity between:
P286-B and ground
P286-D and ground
P286-F and ground

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **7.**

4. Check continuity between P286-C and P283-2.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

5. Check for 15 VDC between P283-2 and ground.

Step 1. If voltage is as specified, replace No. 1 engine fire detector control amplifier (PARA 12-4-11). Go to **7.**
Step 2. If voltage is not as specified, replace No. 1 engine deck fire detector (PARA 12-4-12). Go to **7.**

6. Check continuity between FIRE DET TEST switch terminals B2 and B8.

Step 1. If continuity is present, repair/replace wiring between FIRE DET TEST switch terminal BC2 and P286-E (Appendix F). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-1-7. #2 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 1.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Set FIRE DET TEST switch to position 2 and observe that No. 2 engine T-handle warning lamps and master warning panel FIRE capsules go on.

TABLE NO. 12-2-1-7. #2 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 1. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If No. 2 engine T-handle warning lamps and master warning panel FIRE capsules go on, go to **2.**
Step 2. If No. 2 engine T-handle warning lamps and master warning panel FIRE capsules do not go on, go to **6.**

2. Set FIRE DET TEST switch to position 1 and check continuity between FIRE DET TEST switch terminals AC1 and A2.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, replace FIRE DET TEST switch (PARA 12-4-13). Go to **15.**

3. Check for 15 VDC between P284-1 and P284-10.

Step 1. If voltage is as specified, replace No. 2 engine fire detector control amplifier (PARA 12-4-11). Go to **15.**
Step 2. If voltage is not as specified, go to **4.**

4. Check for 15 VDC between P415-E and ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, repair/replace wiring between FIRE DET TEST switch terminal AC1 and P415-E (Appendix F). Go to **15.**

5. Check continuity between P415-C and P284-1.

Step 1. If continuity is present, replace No. 2 engine firewall fire detector (PARA 12-4-12). Go to **15.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **15.**

6. Check for 28 VDC between P284-11 and P284-10.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **13.**

7. Set FIRE DET TEST switch to position 1 and check for 15 VDC between P284-1 and P284-10.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **10.**

TABLE NO. 12-2-1-7. #2 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 1. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Install an insulated jumper wire between P284-9 and P284-11 and observe that No. 2 engine T-handle warning lamps and master warning panel FIRE capsules go on.

Step 1. If No. 2 engine T-handle warning lamps and master warning panel FIRE capsules go on, remove jumper wire and replace No. 2 engine fire detector control amplifier (PARA 12-4-11). Go to **15.**
Step 2. If No. 2 engine T-handle warning lamps and master warning panel FIRE capsules do not go on, remove jumper wire and go to **9.**

9. Check continuity between P284-9 and P243-E.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **15.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **15.**

10. Check for 15 VDC between FIRE DET TEST switch terminal A2 and ground.

Step 1. If voltage is as specified, replace FIRE DET TEST switch (PARA 12-4-13). Go to **15.**
Step 2. If voltage is not as specified, go to **11.**

11. Check for 28 VDC between P902-R and ground.

Step 1. If voltage is as specified, go to **12.**
Step 2. If voltage is not as specified, go to **14.**

12. Check for 9,800 to 10,200 ohms between J902-E and J902-R.

Step 1. If resistance is as specified, repair/replace wiring between P902-E and FIRE DET TEST switch terminal A2 (Appendix F). Go to **15.**
Step 2. If resistance is not as specified, replace left relay panel (PARA 9-4-2). Go to **15.**

13. Check continuity between P284-10 and ground.

Step 1. If continuity is present, go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **15.**

14. Check for 28 VDC between terminal 1 (FIRE DET NO. 2 ENG circuit breaker, DC ESNTL BUS - pilot's side upper console) and ground.

TABLE NO. 12-2-1-7. #2 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 1. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P284-11 (Appendix F). Go to **15.**

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **15.**

15. Procedure completed.

TABLE NO. 12-2-1-8. #2 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 2.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Set FIRE DET TEST switch to position 2 and check continuity between FIRE DET TEST switch terminals AC2 and A8.

Step 1. If continuity is present, go to **2.**

Step 2. If continuity is not present, replace FIRE DET TEST switch (PARA 12-4-13). Go to **7.**

2. Check for 15 VDC between P287-E and ground.

Step 1. If voltage is as specified, go to **3.**

Step 2. If voltage is not as specified, go to **6.**

3. Check continuity between:

P287-B and ground
P287-D and ground
P287-F and ground

Step 1. If continuity is present, go to **4.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **7.**

4. Check continuity between P287-C and P284-2.

Step 1. If continuity is present, go to **5.**

TABLE NO. 12-2-1-8. #2 ENG EMER OFF T-handle Lamps And Both Master Warning Panel FIRE Capsules Are Off In Position 2. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

5. Check for 15 VDC between P284-2 and ground.

Step 1. If voltage is as specified, replace No. 2 engine fire detector control amplifier (PARA 12-4-11). Go to **7.**
Step 2. If voltage is not as specified, replace No. 2 engine deck fire detector (PARA 12-4-12). Go to **7.**

6. Check continuity between FIRE DET TEST switch terminals A2 and A8.

Step 1. If continuity is present, repair/replace wiring between FIRE DET TEST switch terminal AC2 and P287-E (Appendix F). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-1-9. APU T-handle Lamps And Both Master Warning Panel FIRE Capsules Do Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Set FIRE DET TEST switch to position 1 and check for 15 VDC between FIRE DET TEST switch terminal CC1 and ground.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **7.**

2. Check for 28 VDC between P285-11 and P285-10.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **6.**

3. Check for 15 VDC between P285-1 and P285-10.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **10.**

TABLE NO. 12-2-1-9. APU T-handle Lamps And Both Master Warning Panel FIRE Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. **Install an insulated jumper wire between P285-11 and P285-9 and observe that APU T-handle warning lamps and master warning panel FIRE capsules go on.**

   Step 1. If APU T-handle warning lamps and master warning panel FIRE capsules go on, remove jumper wire and replace APU fire detector control amplifier (PARA 12-4-11). Go to **13.**
   Step 2. If APU T-handle warning lamps and master warning panel FIRE capsules do not go on, remove jumper wire and go to **5.**

5. **Check continuity between P285-9 and P902-L.**

   Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **13.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

6. **Check continuity between P285-10 and ground.**

   Step 1. If continuity is present, go to **12.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

7. **Check for 15 VDC between FIRE DET TEST switch terminal C2 and ground.**

   Step 1. If voltage is as specified, replace FIRE DET TEST switch (PARA 12-4-13). Go to **13.**
   Step 2. If voltage is not as specified, go to **8.**

8. **Check for 28 VDC between P902-N and ground.**

   Step 1. If voltage is as specified, go to **9.**
   Step 2. If voltage is not as specified, go to **12.**

9. **Check for 9,800 to 10,200 ohms between J902-G and J902-N.**

   Step 1. If resistance is as specified, repair/replace wiring between P902-G and FIRE DET TEST switch terminal C2 (Appendix F). Go to **13.**
   Step 2. If resistance is not as specified, replace left relay panel (PARA 9-4-2). Go to **13.**

10. **Check for 15 VDC between P414-E and ground.**

TABLE NO. 12-2-1-9. APU T-handle Lamps And Both Master Warning Panel FIRE Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **11.**
Step 2. If voltage is not as specified, repair/replace wiring between FIRE DET TEST switch terminal CC1 and P414-E (Appendix F). Go to **13.**

11. Check continuity between P414-C and P285-1.

Step 1. If continuity is present, replace APU fire detector (PARA 12-4-12). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring between P414-C and P285-1 (Appendix F). Go to **13.**

12. Check for 28 VDC between (APU FIRE DET circuit breaker, BATT BUS - lower console circuit breaker) and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and (Appendix F). Go to **13.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **13.**

13. Procedure completed.

TABLE NO. 12-2-1-10. Both Master Warning Panel FIRE Capsules Are On But APU T-handle Lamps Are Off.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between:
P902-K and TB8 terminal 1
TB8 terminal 2 and ground

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **2.**
Step 2. If trouble remains, replace APU T-handle (PARA 12-4-7). Go to **2.**
Step 3. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **2.**

2. Procedure completed.

TABLE NO. 12-2-1-11. All T-handles Remain Bright.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Do caution/advisory capsules go dim?

Step 1. If caution/advisory capsules are dim, go to **2.**
Step 2. If caution/advisory capsules are not dim, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **7.**

2. Do instrument panel and lower console lamps go dim?

Step 1. If lamps are dim, replace left relay panel (PARA 9-4-2). Go to **7.**
Step 2. If lamps are not dim, go to **3.**

3. Check helicopter configuration.

Step 1. **EMEP**, go to **4.**
Step 2. **W/O EMEP**, go to **6.**

EMEP **4. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **5.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **7.**

EMEP **5. Install PFA (PARA 9-4-98). Go to 6.**

6. Check continuity between P902-q and P118-T.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **7.**
Step 2. If trouble remains, replace caution/advisory panel (PARA 8-4-19). Go to **7.**
Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

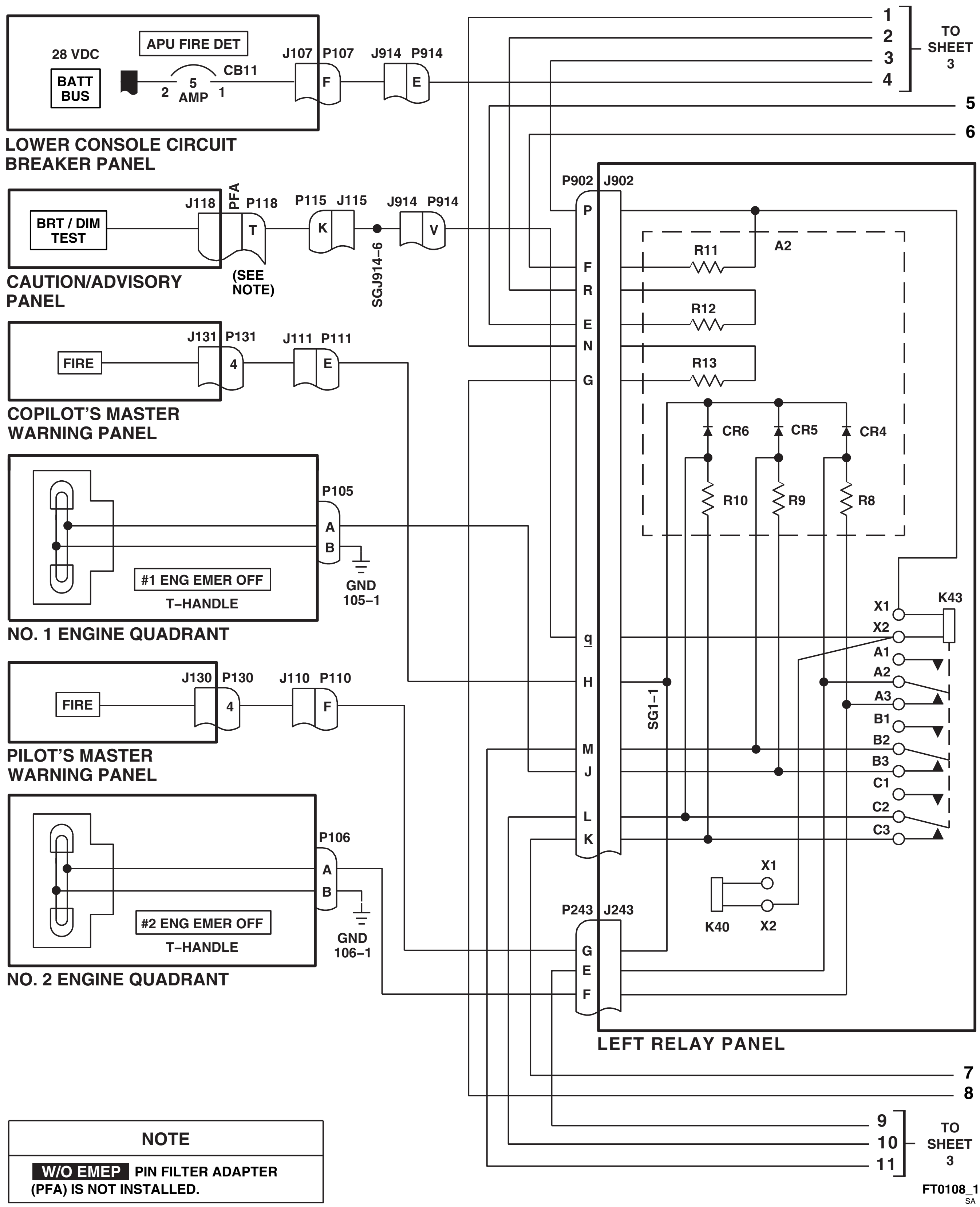


FIGURE 12-2-1-1. FIRE DETECTION SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

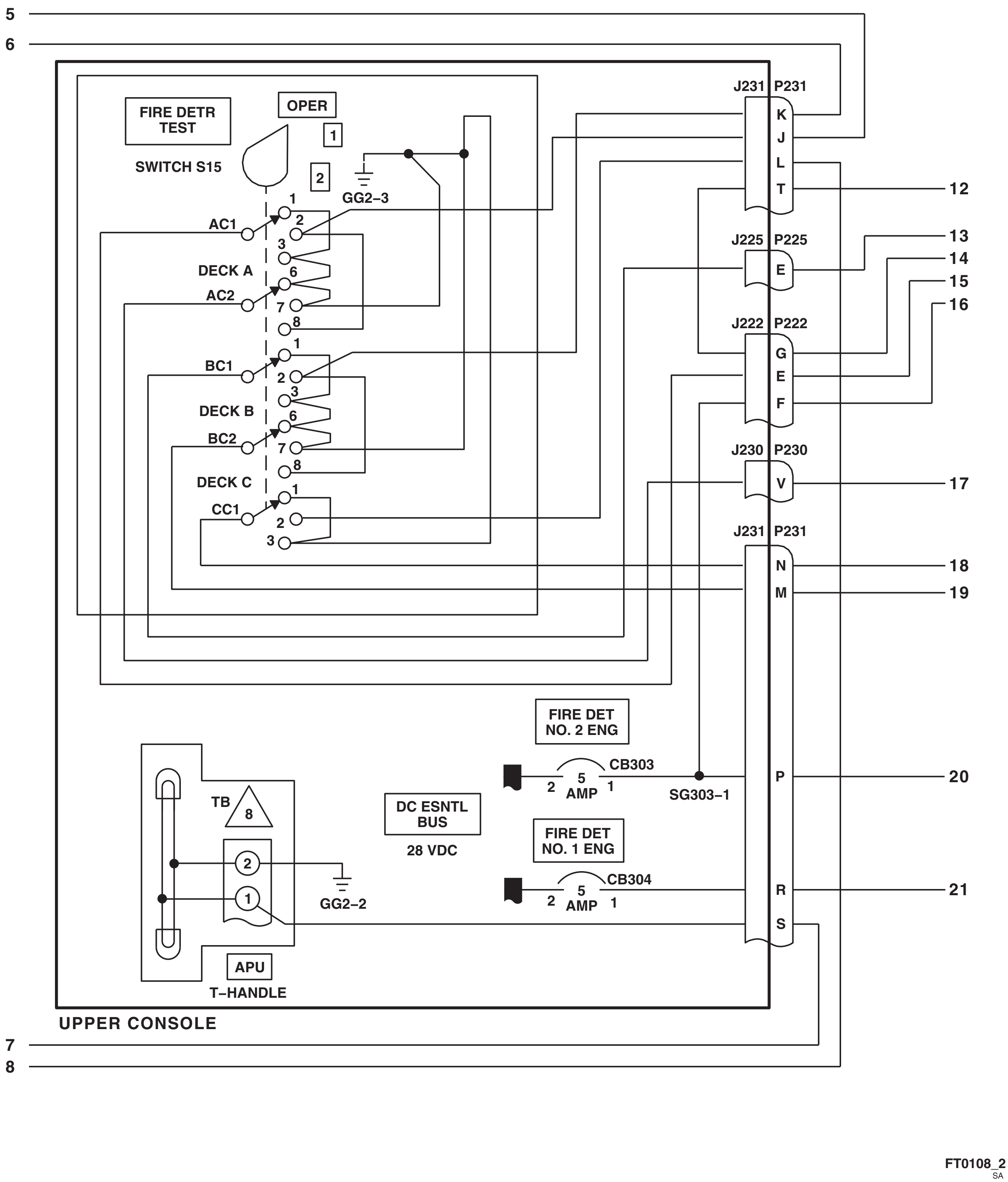


FIGURE 12-2-1-1. FIRE DETECTION SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

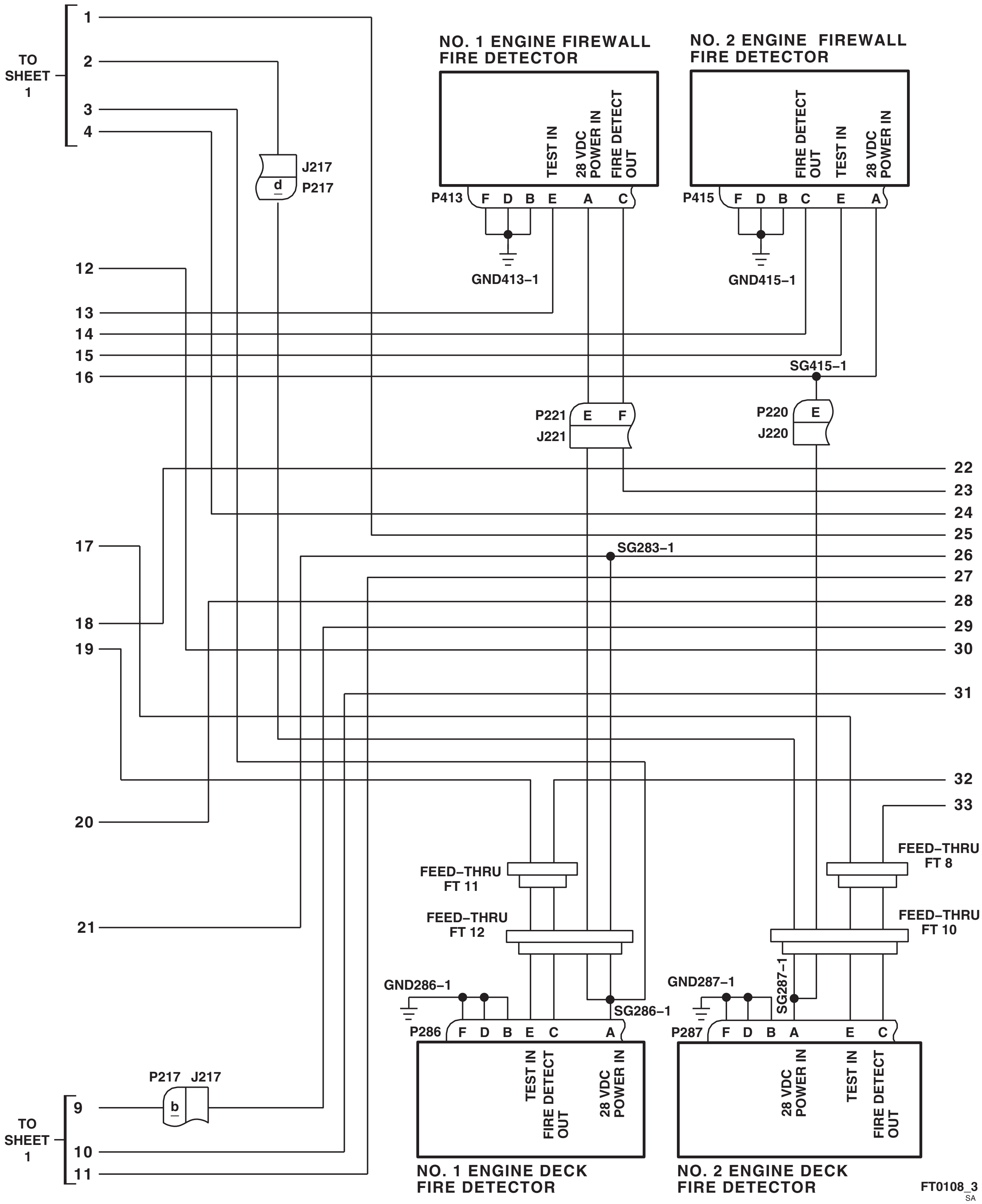


FIGURE 12-2-1-1. FIRE DETECTION SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)

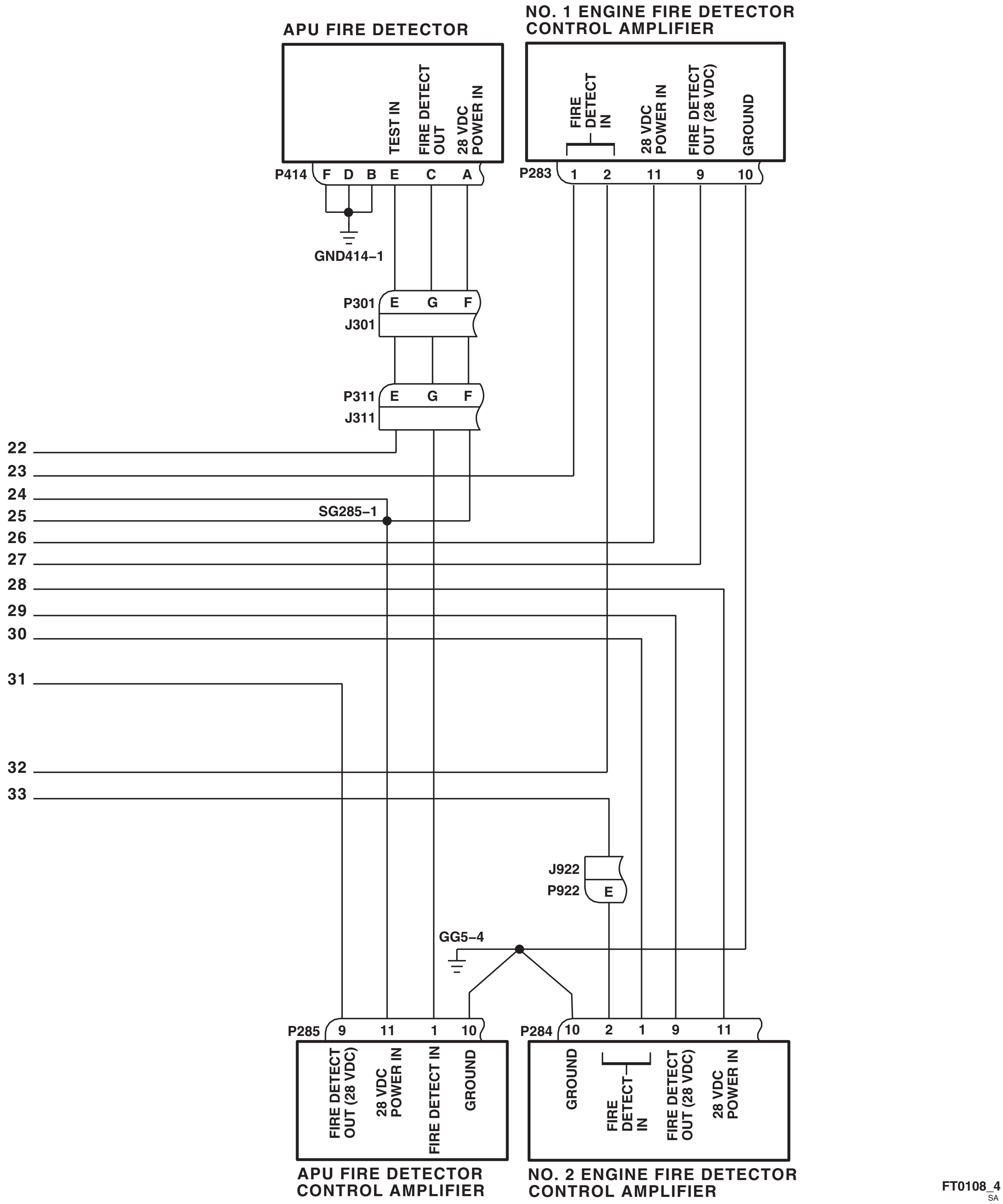


FIGURE 12-2-1-1. FIRE DETECTION SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

12-2-2 FIRE EXTINGUISHING SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16
- 70583–702374 PARA 9-4-34
- 702375 – SUBQ PARA 9-4-106
- PARA 12-4-1

Equipment Conditions

- External Electrical Power Available

WARNING

To prevent accidental firing of an explosive squib, observe all safety precautions regarding the handling and storage of explosives.

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

NOTE

When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always refer to Figure 12-2-2-1, Figure 12-2-2-2, and Figure 12-2-2-3 for proper connections.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The fire extinguishing system operational/troubleshooting procedure is subdivided as follows:

(1) Setup

(2) No. 1 Main Fire Extinguishing Bottle

(3) No. 2 Main Fire Extinguishing Bottle

(4) No. 1 Reserve Fire Extinguishing Bottle

(5) No. 2 Reserve Fire Extinguishing Bottle

(6) Directional Control Valve Check

(7) Shutdown

12-2-2.1 Setup.

a. Turn off external electrical power (PARA 1-6-16).

b. Disconnect battery by removing cover and disconnecting electrical connectors (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

NOTE

Make sure wiring terminals are inside rubber boots, and do not allow them to contact airframe.

c. Disconnect all wire terminals from main and reserve fire extinguishing bottles (PARA 12-4-1).

d. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

12-2-2.2 No. 1 Main Fire Extinguishing Bottle.

a. Do Setup, if not already done.

b. **FIRE** Connect multimeter leads to wires marked W3163A16 and W3162A16N.

c. **W/O FIRE** Connect multimeter leads to wires marked W3163A20 and W3162A20N.

d. Connect battery (**70583–702374** PARA 9-4-34 or **702375 – SUBQ** PARA 9-4-106).

e. Turn on external electrical power (PARA 1-6-16).

f. Pull engine controls quadrant #1 ENG EMER OFF T-handle back.

g. Place upper console FIRE EXTGH switch to MAIN, then release to OFF.

RESULT: Multimeter shall indicate 28 vdc when switch is placed to MAIN.

- If result is not as specified, go to TABLE NO. 12-2-2-5.

h. Push #1 ENG EMER OFF T-handle forward.

NOTE

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
- After completing each step, return all T-handles to the OFF position.

i. Perform a No. 1 Main Fire Extinguishing Bottle priority check by performing the operational checks in TABLE NO. 12-2-2-1.

TABLE NO. 12-2-2-1. No. 1 Main Fire Extinguishing Bottle.

FIRE BOTTLE TESTED	STEP NUMBER	CHECK FIRST	CHECK SECOND	CHECK THIRD	SWITCH POSITION
#1 Main	1	#2 ENG*	#1 ENG	-	MAIN
#1 Main	2	#1 ENG	#2 ENG*	-	MAIN
#1 Main	3	#1 ENG	APU	-	MAIN
#1 Main	4	#2 ENG	APU	-	MAIN
#1 Main	5	APU	#2 ENG*	#1 ENG	MAIN
#1 Main	6	#1 ENG	#2 ENG	APU	MAIN
#1 Main	7	APU	#1 ENG	-	MAIN
#1 Main	8	APU	#2 ENG *	-	MAIN

TABLE NO. 12-2-2-1. No. 1 Main Fire Extinguishing Bottle. (Cont)

FIRE BOTTLE TESTED	STEP NUMBER	CHECK FIRST	CHECK SECOND	CHECK THIRD	SWITCH POSITION

* Pull engine T-handle back, then push forward.

RESULT: Multimeter shall indicate 28 vdc.

- If results are not as specified, replace right relay panel (PARA 9-4-2).

j. Pull upper console APU T-handle down. Visually check directional control valve.

RESULT: Sound of directional control valve being energized shall be heard and position indicator shall pivot.

- If result is not as specified, go to TABLE NO. 12-2-2-6.

k. Place FIRE EXTGH switch to MAIN, then release to OFF.

RESULT: Multimeter shall indicate 28 vdc when switch is placed to MAIN.

- If result is not as specified, go to TABLE NO. 12-2-2-7.

l. Push APU T-handle up while assistant is observing directional control valve.

RESULT: Sound of directional control valve de-energizing shall be heard and position indicator shall turn.

- If result is not as specified, replace directional control valve (PARA 12-4-6).

m. Turn off external electrical power (PARA 1-6-16).

n. Disconnect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

o. Disconnect P243 and P902 from left relay panel.

p. Install an insulated jumper wire between P243-Z and P902-g (simulates actuation of impact switch).

q. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

RESULT: Meter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 12-2-2-8.

r. Disconnect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

s. Remove jumper wire from P243 and P902. Reconnect to left relay panel.

t. Disconnect multimeter leads from No. 1 main wire terminals.

u. If no other check is to be done, go to Shutdown.

12-2-2.3 No. 2 Main Fire Extinguishing Bottle.

a. Do Setup, if not already done.

b. Connect multimeter leads to wires labeled W3164A20 and W3165A20N.

c. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

d. Turn on external electrical power (PARA 1-6-16).

e. Pull engine controls quadrant #2 ENG EMER OFF T-handle back.

f. Place FIRE EXTGH switch to MAIN, then release to OFF.

RESULT: Multimeter shall indicate 28 vdc when switch is placed to MAIN.

- If result is not as specified, go to TABLE NO. 12-2-2-9.

g. Push #2 ENG EMER OFF T-handle forward.

NOTE

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
- After completing each step, return all T-handles to the OFF position.

h. Perform a No. 2 Main Fire Extinguishing Bottle priority check by performing the operational checks in TABLE NO. 12-2-2-2.

TABLE NO. 12-2-2-2. No. 2 Main Fire Extinguishing Bottle.

FIRE BOTTLE TESTED	STEP NUMBER	CHECK FIRST	CHECK SECOND	CHECK THIRD	SWITCH POSITION
#2 Main	1	#2 ENG	#1 ENG*	-	MAIN
#2 Main	2	#1 ENG	#2 ENG	-	MAIN
#2 Main	3	APU	#2 ENG	-	MAIN
#2 Main	4	APU	#2 ENG	#1 ENG*	MAIN
#2 Main	5	APU	#1 ENG	#2 ENG	MAIN
#2 Main	6	#1 ENG	#2 ENG	APU**	MAIN

* Pull engine T-handle back, then push forward

** Pull APU T-handle down, then push up

RESULT: Multimeter shall indicate 28 vdc.

- If results are not as specified, replace right relay panel (PARA 9-4-2).

i. Disconnect multimeter leads from No. 2 main wire terminals.

j. If no other check is to be done, go to Shutdown.

12-2-2.4 No. 1 Reserve Fire Extinguishing Bottle.

a. Do Setup, if not already done.

b. Connect multimeter leads to wires labeled W3167A20 and W3166A20N.

c. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

d. Turn on external electrical power (PARA 1-6-16).

e. Pull engine controls quadrant #1 ENG EMER OFF T-handle back.

f. Place FIRE EXTGH switch to RESERVE, then release to OFF.

RESULT: Multimeter shall indicate 28 vdc when switch is placed to RESERVE.

- If result is not as specified, go to TABLE NO. 12-2-2-10.

g. Push #1 ENG EMER OFF T-handle forward.

NOTE

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
- After completing each step, return all T-handles to the OFF position.

h. Perform a No. 1 Reserve Fire Extinguishing Bottle priority check by performing the operational checks in TABLE NO. 12-2-2-3.

TABLE NO. 12-2-2-3. No. 1 Reserve Fire Extinguishing Bottle.

FIRE BOTTLE TESTED	STEP NUMBER	CHECK FIRST	CHECK SECOND	CHECK THIRD	SWITCH POSITION
#1 Reserve	1	#2 ENG*	#1 ENG	-	RESERVE
#1 Reserve	2	APU	#1 ENG	-	RESERVE
#1 Reserve	3	APU	#1 ENG*	-	RESERVE
#1 Reserve	4	#1 ENG	APU	-	RESERVE
#1 Reserve	5	#2 ENG	APU	-	RESERVE
#1 Reserve	6	APU	#2 ENG*	#1 ENG	RESERVE
#1 Reserve	7	#1 ENG	#2 ENG	APU	RESERVE

* Pull engine T-handle back, then push forward.

RESULT: Multimeter shall indicate 28 vdc.

- If results are not as specified, replace right relay panel (PARA 9-4-2).

i. Pull APU T-handle down, and place FIRE EXTGH switch to RESERVE, then release to OFF.

RESULT: Multimeter shall indicate 28 vdc when switch is placed to RESERVE.

- If result is not as specified, go to TABLE NO. 12-2-2-11.

j. Push APU T-handle up. Visually check directional control valve.

RESULT: Sound of directional control valve de-energizing shall be heard and position indicator shall turn.

- If result is not as specified, replace directional control valve (PARA 12-4-6).

k. Disconnect multimeter leads from No. 1 reserve wire terminals.

l. If no other check is to be done, go to Shutdown.

12-2-2.5 No. 2 Reserve Fire Extinguishing Bottle.

a. Do Setup, if not already done.

b. FIRE Connect multimeter leads to wires marked W3168A16 and W3169A16N.

c. W/O FIRE Connect multimeter leads to wires marked W3168A20 and W3169A20N.

d. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

e. Turn on external electrical power (PARA 1-6-16).

f. Pull engine controls quadrant #2 ENG EMER OFF T-handle back.

g. Place FIRE EXTGH switch to RESERVE, then release to OFF.

RESULT: Multimeter shall indicate 28 vdc when switch is placed to RESERVE.

- If result is not as specified, go to TABLE NO. 12-2-2-12.

h. Push #2 ENG EMER OFF T-handle forward.

NOTE

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
- After completing each step, return all T-handles to the OFF position.

i. Perform a No. 2 Reserve Fire Extinguishing Bottle priority check by performing the operational checks in TABLE NO. 12-2-2-4.

TABLE NO. 12-2-2-4. No. 2 Reserve Fire Extinguishing Bottle.

FIRE BOTTLE TESTED	STEP NUMBER	CHECK FIRST	CHECK SECOND	CHECK THIRD	SWITCH POSITION
#2 Reserve	1	#1 ENG	#2 ENG	-	RESERVE
#2 Reserve	2	APU	#2 ENG	-	RESERVE
#2 Reserve	3	APU	#1 ENG	#2 ENG	RESERVE

RESULT: Multimeter shall indicate 28 vdc.

- If results are not as specified, replace right relay panel (PARA 9-4-2).

j. Turn off external power (PARA 1-6-16).

k. Disconnect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

l. Disconnect P243 and P902 from the left relay panel.

m. Install an insulated jumper wire across P243-Z and P902-g.

n. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 12-2-2-8.

o. Disconnect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

p. Remove jumper wires from P243 and P902.

q. Connect P243 and P902 to left relay panel.

r. Disconnect multimeter leads from No. 2 reserve wire terminals.

s. If no other check is to be done, go to Shutdown.

12-2-2.6 Directional Control Valve Check.

a. Do Setup, if not already done.

b. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

c. Turn on external electrical power (PARA 1-6-16).

d. On upper console, pull down APU T-handle while listening to directional control valve.

RESULT: Sound of directional control valve being energized shall be heard and position indicator shall pivot from neutral position to energized position.

- If result is not as specified, go to TABLE NO. 12-2-2-13.

e. On upper console, push in APU T-handle while listening to directional control valve.

RESULT: Sound of directional control valve de-energizing shall be heard and position indicator shall pivot back to neutral position.

- If result is not as specified, go to TABLE NO. 12-2-2-13.

f. If no other check is to be done, go to Shutdown.

12-2-2.7 Shutdown.

WARNING

To prevent accidental firing of an explosive squib, observe all safety precautions regarding the handling and storage of explosives.

a. Turn off and disconnect external electrical power (PARA 1-6-16).

b. Disconnect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

WARNING

To prevent injury to personnel, with APU T-handle pushed up and #1 and #2 ENG EMER OFF T-handles pushed forward, make sure there is no voltage on terminals before connecting wires to fire extinguisher bottles.

c. Connect helicopter wiring to fire extinguishing bottles (PARA 12-4-1).

d. Connect battery (70583–702374 PARA 9-4-34 or 702375 – SUBQ PARA 9-4-106).

TABLE NO. 12-2-2-5. Multimeter Connected To No. 1 Main Fire Bottle Wiring Has No Voltage Indication When #1 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure #1 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch is placed to MAIN. Go to 2.

2. Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **5.**

3. Check continuity between:
J313-B and J900-M
J313-B and J900-N

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, replace right relay panel (PARA 9-4-2). Go to **16.**

4. Check continuity between:
P900-M and ground
P900-N and ground

Step 1. If continuity is present, repair/replace wiring between P313-B and terminal of ground wire removed from squib's ground terminal (Appendix F). Go to **16.**
Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **16.**

5. Check continuity between No. 1 main squib wire terminal 1 and P313-C.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **16.**

6. Check continuity between #1 ENG EMER OFF T-handle switch terminals C and NC.

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, replace engine fire extinguisher arming switch (PARA 4-5-16). Go to **16.**

7. Check for 28 vdc between #1 ENG EMER OFF T-handle switch terminal C and ground.

TABLE NO. 12-2-2-5. Multimeter Connected To No. 1 Main Fire Bottle Wiring Has No Voltage Indication When #1 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **15.**

8. With FIRE EXTGH switch placed to MAIN, check for 28 vdc between FIRE EXTGH switch S20 terminal 4 and ground.

Step 1. If voltage is as specified, go to **9.**
Step 2. If voltage is not as specified, go to **13.**

9. With FIRE EXTGH switch placed to MAIN, check for 28 vdc between P900-U and ground.

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, repair/replace wiring between P900-U and FIRE EXTGH switch terminal 4 (Appendix F). Go to **16.**

10. Check helicopter configuration.

Step 1. WINTER , go to **11.**
Step 2. W/O WINTER , go to **12.**

WINTER **11. Check for 28 vdc between P244B-P and ground.**

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **16.**
Step 2. If voltage is not as specified, replace winterization kit (PARA 16-6-2). Go to **16.**
Step 3. If trouble remains, go to **12.**

12. Check for 28 vdc between P244-P and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **16.**
Step 2. If voltage is not as specified, repair/replace wiring between #1 ENG EMER OFF T-handle switch terminal NC and P244-P (Appendix F). Go to **16.**

13. Check for 28 vdc between of FIRE EXTGH circuit breaker terminal 1 (pilot's circuit breaker panel) and ground.

Step 1. If voltage is as specified, go to **14.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **16.**

TABLE NO. 12-2-2-5. Multimeter Connected To No. 1 Main Fire Bottle Wiring Has No Voltage Indication When #1 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

14. Check continuity between FIRE EXTGH circuit breaker terminal 1 (pilot's circuit breaker panel) and FIRE EXTGH switch terminal 5.

Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to **16.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **16.**

15. Check for 28 vdc between FIRE EXTGH circuit breaker terminal 1 (on lower console) and ground.

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and terminal C of #1 ENG EMER OFF T-handle switch (Appendix F). Go to **16.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **16.**

16. Procedure completed.

TABLE NO. 12-2-2-6. Directional Control Valve Does Not Operate When No. 1 Main Is Used For APU.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Pull APU T-handle down and place FIRE EXTGH switch to MAIN. Go to 2.

2. Check for 28 vdc between directional control valve terminals 1 and 2.

Step 1. If voltage is as specified, replace directional control valve (PARA 12-4-6). Go to **7.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between directional control valve terminal 2 and ground.

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

4. Check helicopter configuration.

Step 1. **WINTER** , go to **5.**

TABLE NO. 12-2-2-6. Directional Control Valve Does Not Operate When No. 1 Main Is Used For APU. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. W/O WINTER , go to **6.**

WINTER **5. Check for 28 vdc between J244A-N and ground.**

Step 1. If voltage is as specified, replace winterization kit (PARA 16-6-2). Go to **7.**
Step 2. If voltage is not as specified, go to **TABLE NO. 12-2-2-7.**
Step 3. If trouble remains, repair/replace wiring between P244-N and directional control valve terminal 1 (Appendix F). Go to **7.**

6. Check for 28 vdc between J244-N and ground.

Step 1. If voltage is as specified, repair/replace wiring between P244-N and directional control valve terminal 1 (Appendix F). Go to **7.**
Step 2. If voltage is not as specified, go to **TABLE NO. 12-2-2-7.**

7. Procedure completed.

TABLE NO. 12-2-2-7. Multimeter Connected To No. 1 Main Fire Bottle Wiring Has No Voltage Indication When APU T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure APU T-handle is pulled down and FIRE EXTGH switch is placed to MAIN. Go to 2.

2. Check for 28 vdc between P900-X and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **5.**

3. Check for 28 vdc between P900-T and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **7.**
Step 2. If voltage is not as specified, go to **4.**

4. With FIRE EXTGH switch placed to MAIN, check for 28 vdc between FIRE EXTGH switch S20 terminal 1 and ground.

TABLE NO. 12-2-2-7. Multimeter Connected To No. 1 Main Fire Bottle Wiring Has No Voltage Indication When APU T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 1 and P900-T (Appendix F). Go to **7.**
Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **7.**

5. Check for 28 vdc between APU T-handle terminal NO and ground.

Step 1. If voltage is as specified, repair/replace wiring between APU T-handle terminal NO and P900-X (Appendix F). Go to **7.**
Step 2. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between APU T-handle terminal C and ground.

Step 1. If voltage is as specified, replace T-handle (PARA 12-4-7). Go to **7.**
Step 2. If voltage is not as specified, repair/replace wiring between T-handle terminal C and FIRE EXTGH switch terminal 2 (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-2-8. Multimeter Has No Voltage Indication When Actuation Of Impact Switch Is Simulated.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Place an insulated jumper wire between P902-g and P243-Z. Go to 2.

2. Check for 28 vdc between P900-S and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **9.**
Step 2. If voltage is not as specified, go to **3.**

3. Remove jumper wire from P902 and P243. Go to 4.

4. Check continuity between P900-S and P243-Z.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

TABLE NO. 12-2-2-8. Multimeter Has No Voltage Indication When Actuation Of Impact Switch Is Simulated. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

5. Check helicopter configuration.

Step 1. WINTER , go to **6.**
Step 2. W/O WINTER , go to **7.**

WINTER **6. Check for 28 vdc between P244B-W and ground.**

Step 1. If voltage is as specified, replace winterization kit (PARA 16-6-2). Go to **9.**
Step 2. If trouble remains, repair/replace wiring between P244-W and FIRE EXTGH circuit breaker terminal 1 (on lower console) (Appendix F). Go to **9.**
Step 3. If voltage is not as specified, go to **8.**

W/O WINTER **7. Check for 28 vdc between P244-W and ground.**

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **9.**
Step 2. If trouble remains, repair/replace wiring between P244-W and FIRE EXTGH circuit breaker terminal 1 (on lower console) (Appendix F). Go to **9.**
Step 3. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between P902-g and ground.

Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to **9.**
Step 2. If voltage is not as specified, repair/replace wiring between P902-g and FIRE EXTGH circuit breaker terminal 1 (on lower console) (Appendix F). Go to **9.**

9. Procedure completed.

TABLE NO. 12-2-2-9. Multimeter Connected To No. 2 Main Fire Bottle Wiring Has No Voltage Indication When #2 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure #2 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch is placed to MAIN. Go to 2.

2. Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.

TABLE NO. 12-2-2-9. Multimeter Connected To No. 2 Main Fire Bottle Wiring Has No Voltage Indication When #2 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To MAIN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **4.**

3. Check continuity between P313-E and terminal of ground wire removed from squib's ground terminal.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

4. Check continuity between No. 2 main squib wire terminal 3 and P313-D.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

5. Check continuity between #2 ENG EMER OFF T-handle terminals C and NC.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, replace T-handle (PARA 4-5-11). Go to **8.**

6. Check for 28 vdc between #2 ENG EMER OFF T-handle terminal C and ground.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, repair/replace wiring between FIRE EXTGH switch terminal 5 and T-handle terminal C (Appendix F). Go to **8.**

7. Check for 28 vdc between P900-P and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **8.**
Step 2. If voltage is not as specified, repair/replace wiring between #2 ENG EMER OFF T-handle terminal NC and P900-P (Appendix F). Go to **8.**

8. Procedure completed.

TABLE NO. 12-2-2-10. Multimeter Connected To No. 1 Reserve Fire Bottle Wiring Has No Voltage Indication When #1 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To RESERVE.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Make sure #1 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch is placed to RESERVE. Go to 2.**

2. **Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, go to **4.**

3. **Check continuity between P313-F and terminal of ground wire removed from squib's ground terminal.**

   Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **8.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

4. **Check continuity between No. 1 reserve squib wire terminal 1 and P313-G.**

   Step 1. If continuity is present, go to **5.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

5. **With FIRE EXTGH switch placed to RESERVE, check for 28 vdc between FIRE EXTGH switch S20 terminal 3 and ground.**

   Step 1. If voltage is as specified, go to **6.**
   Step 2. If voltage is not as specified, go to **7.**

6. **Check for 28 vdc between P900-W and ground.**

   Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **8.**
   Step 2. If voltage is not as specified, repair/replace wiring between P900-W and switch S20 terminal 3 (Appendix F). Go to **8.**

7. **Check for 28 vdc between FIRE EXTGH switch terminal 2 and ground.**

   Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **8.**
   Step 2. If voltage is not as specified, repair/replace wiring between switch terminal 2 and FIRE EXTGH circuit breaker terminal 1 (on lower console) (Appendix F). Go to **8.**

TABLE NO. 12-2-2-10. Multimeter Connected To No. 1 Reserve Fire Bottle Wiring Has No Voltage Indication When #1 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To RESERVE. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Procedure completed.

TABLE NO. 12-2-2-11. Multimeter Connected To No. 1 Reserve Fire Bottle Wiring Has No Voltage Indication When APU T-handle Is Used And FIRE EXTGH Switch Is Placed To RESERVE.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure APU T-handle is pulled down and FIRE EXTGH switch is placed to RESERVE. Go to 2.

2. Check for 28 vdc between P900-V and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **4.**
Step 2. If voltage is not as specified, go to **3.**

3. With FIRE EXTGH switch placed to RESERVE, check for 28 vdc between FIRE EXTGH switch S20 terminal 6 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 6 and P900-V (Appendix F). Go to **4.**
Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-2-12. Multimeter Connected To No. 2 Reserve Fire Bottle Wiring Has No Voltage Indication When #2 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To RESERVE.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure #2 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch is placed to RESERVE. Go to 2.

2. Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.

TABLE NO. 12-2-2-12. Multimeter Connected To No. 2 Reserve Fire Bottle Wiring Has No Voltage Indication When #2 ENG EMER OFF T-handle Is Used And FIRE EXTGH Switch Is Placed To RESERVE. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **6.**

3. Check helicopter configuration.

Step 1. FIRE , go to **4.**
Step 2. W/O FIRE , go to **5.**

FIRE **4. Check continuity between P313-A and terminal of ground wire removed from squib's ground terminal.**

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

W/O FIRE **5. Check continuity between P313-J and terminal ground wire removed from squib's ground terminal.**

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

6. Check continuity between No. 2 reserve squib wire terminal 3 and P313-H.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-2-13. Directional Control Valve Does Not Operate When APU T-handle Is Pulled Down.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 vdc between directional control valve terminals 1 and 2.

Step 1. If voltage is as specified, replace directional control valve (PARA 12-4-6). Go to **11.**

TABLE NO. 12-2-2-13. Directional Control Valve Does Not Operate When APU T-handle Is Pulled Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, go to **2.**

2. Check continuity between directional control valve terminal 2 and ground.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **11.**

3. Check helicopter configuration.

Step 1. WINTER , go to **4.**
Step 2. W/O WINTER , go to **5.**

WINTER **4. Check for 28 vdc between J244A-N and ground.**

Step 1. If voltage is as specified, replace winterization kit (PARA 16-6-2).
Step 2. If trouble remains, repair/replace wiring between P244-N and directional control valve terminal 1 (Appendix F). Go to **11.**
Step 3. If voltage is not as specified, go to **6.**

W/O WINTER **5. Check for 28 vdc between J244-N and ground.**

Step 1. If voltage is as specified, repair/replace wiring between P244-N and directional control valve terminal 1 (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, go to **6.**

6. Check for 28 vdc between P900-X and ground.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **9.**

7. With FIRE EXTGH switch placed to MAIN, check for 28 vdc between P900-T and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **11.**
Step 2. If voltage is not as specified, go to **8.**

8. Check for 28 vdc between FIRE EXTGH switch terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 1 and P900-T (Appendix F). Go to **11.**

TABLE NO. 12-2-2-13. Directional Control Valve Does Not Operate When APU T-handle Is Pulled Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **11.**

9. Check for 28 vdc between APU T-handle terminal NO and ground.

Step 1. If voltage is as specified, repair/replace wiring between APU T-handle terminal NO and P900-X (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, go to **10.**

10. Check for 28 vdc between APU T-handle terminal C and ground.

Step 1. If voltage is as specified, replace T-handle (PARA 12-4-7). Go to **11.**
Step 2. If voltage is not as specified, repair/replace wiring between T-handle terminal C and FIRE EXTGH switch terminal 1 (on lower console) (Appendix F). Go to **11.**

11. Procedure completed.

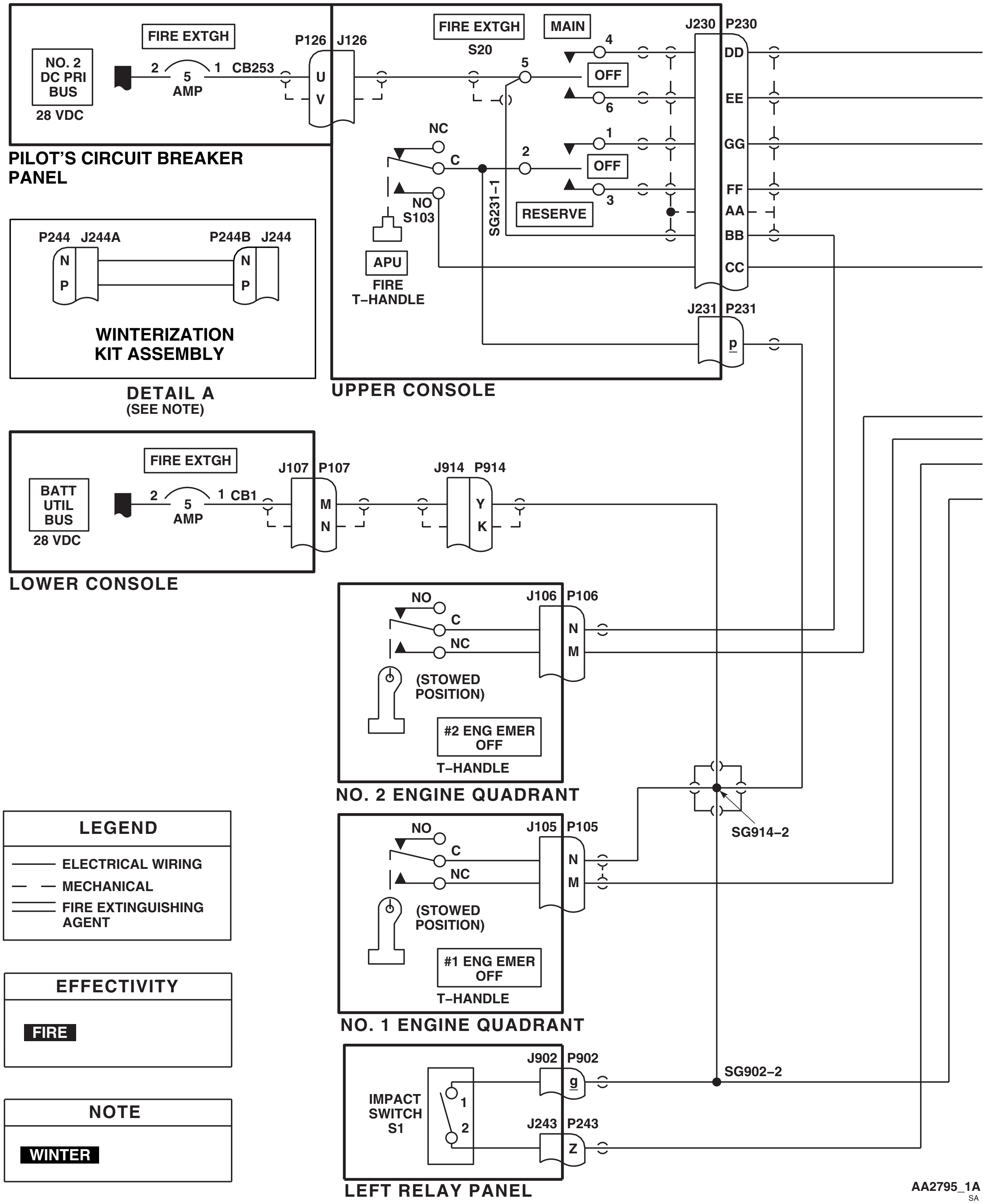


FIGURE 12-2-2-1. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

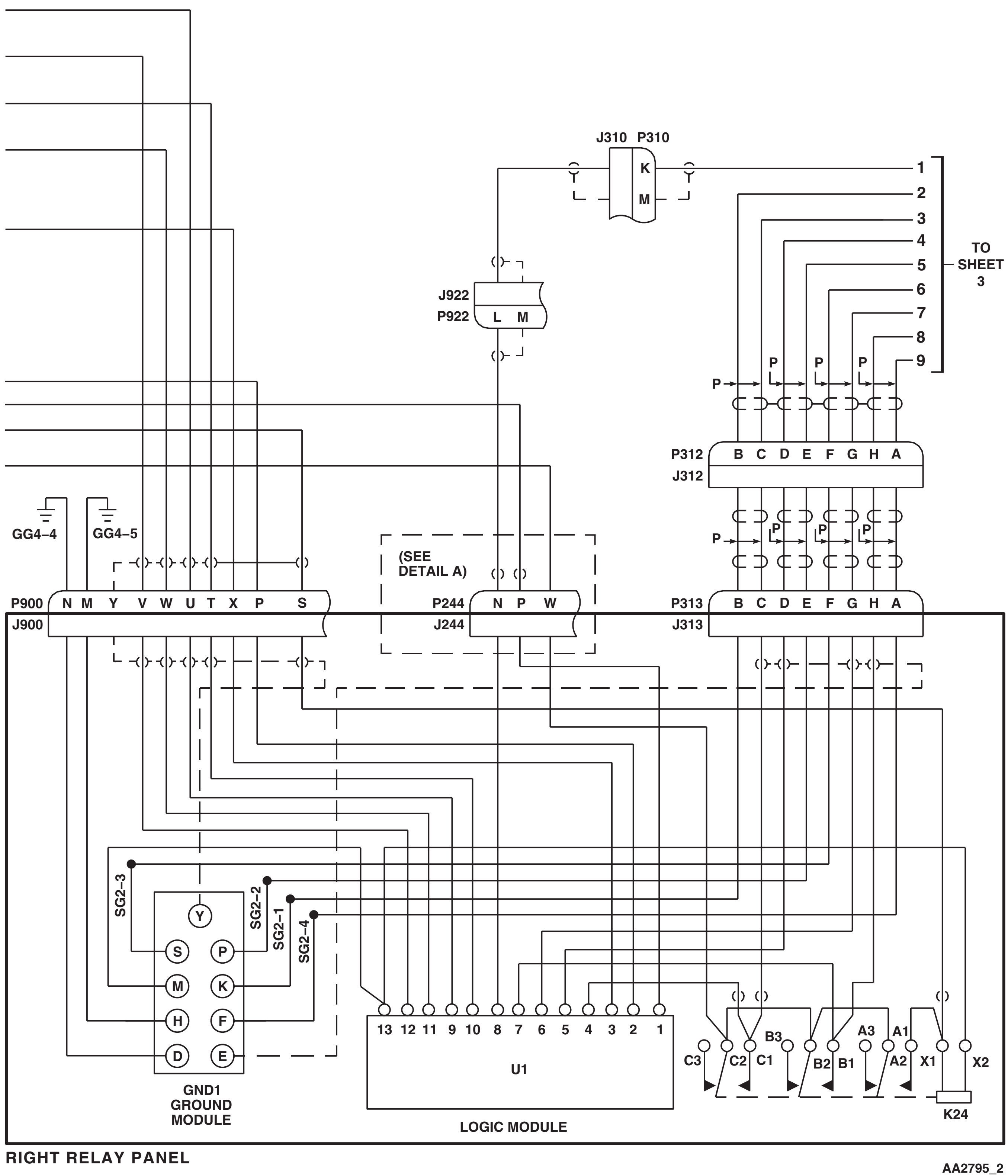


FIGURE 12-2-2-1. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

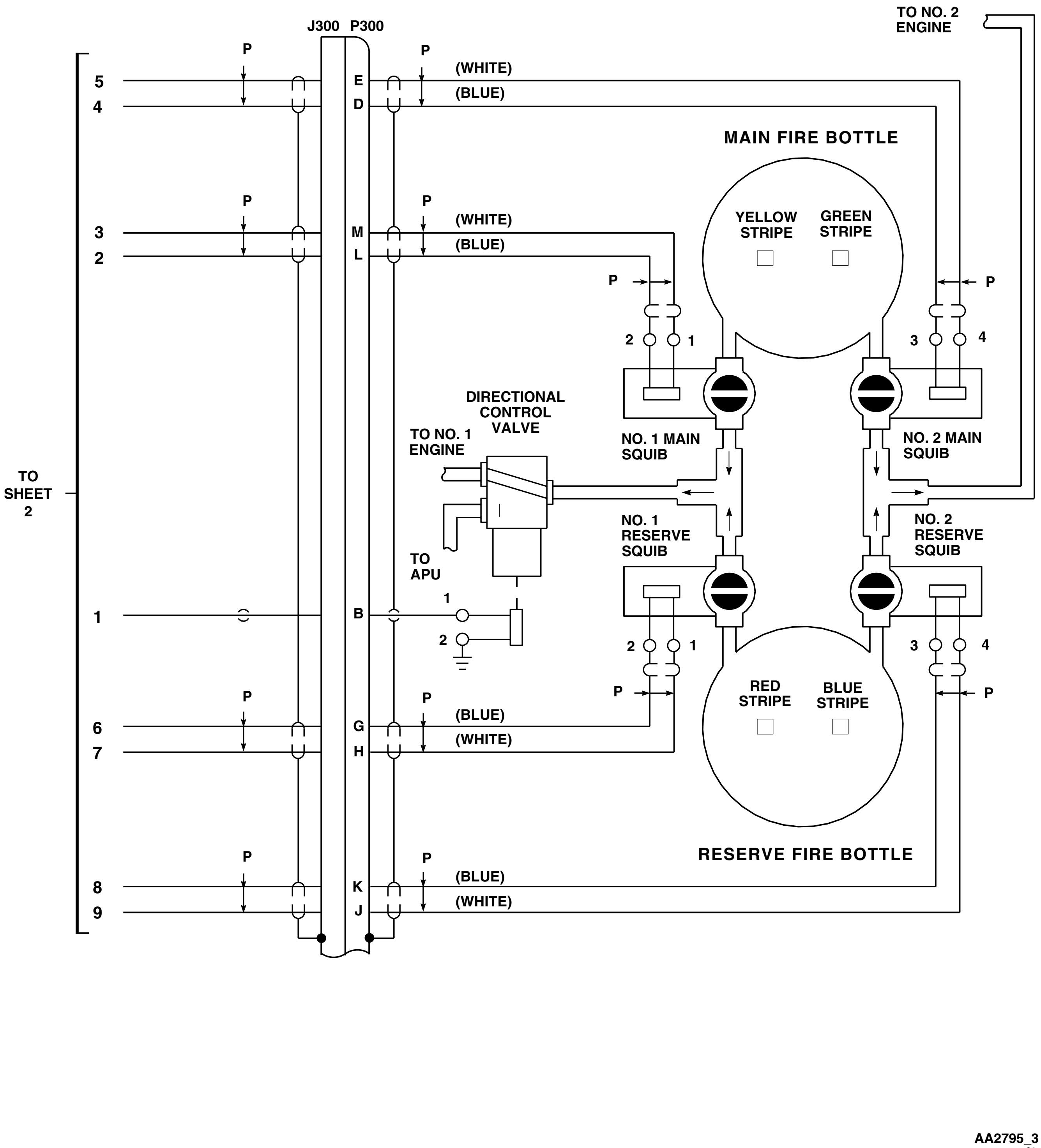


FIGURE 12-2-2-1. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)

FIRE BOTTLE SQUIB WIRING CONNECTION CHART				
FIRE BOTTLE SQUIB	WIRE CODE (PRINTED ON WIRE)	WIRE COLOR CODE	CONNECTION POINT TO SQUIB	FIRE BOTTLE COLOR CODE
MAIN (NO. 1)	W3163A16 W3162A16N	WHITE BLUE	#6 STUD #8 SCREW	YELLOW STRIPE
MAIN (NO. 2)	W3164A20 W3165A20N	BLUE WHITE	#6 STUD #8 SCREW	GREEN STRIPE
RESERVE (NO. 1)	W3167A20 W3166A20N	WHITE BLUE	#6 STUD #8 SCREW	RED STRIPE
RESERVE (NO. 2)	W3168A16 W3169A16N	BLUE WHITE	#6 STUD #8 SCREW	BLUE STRIPE

AA2795_4
SA

FIGURE 12-2-2-1. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

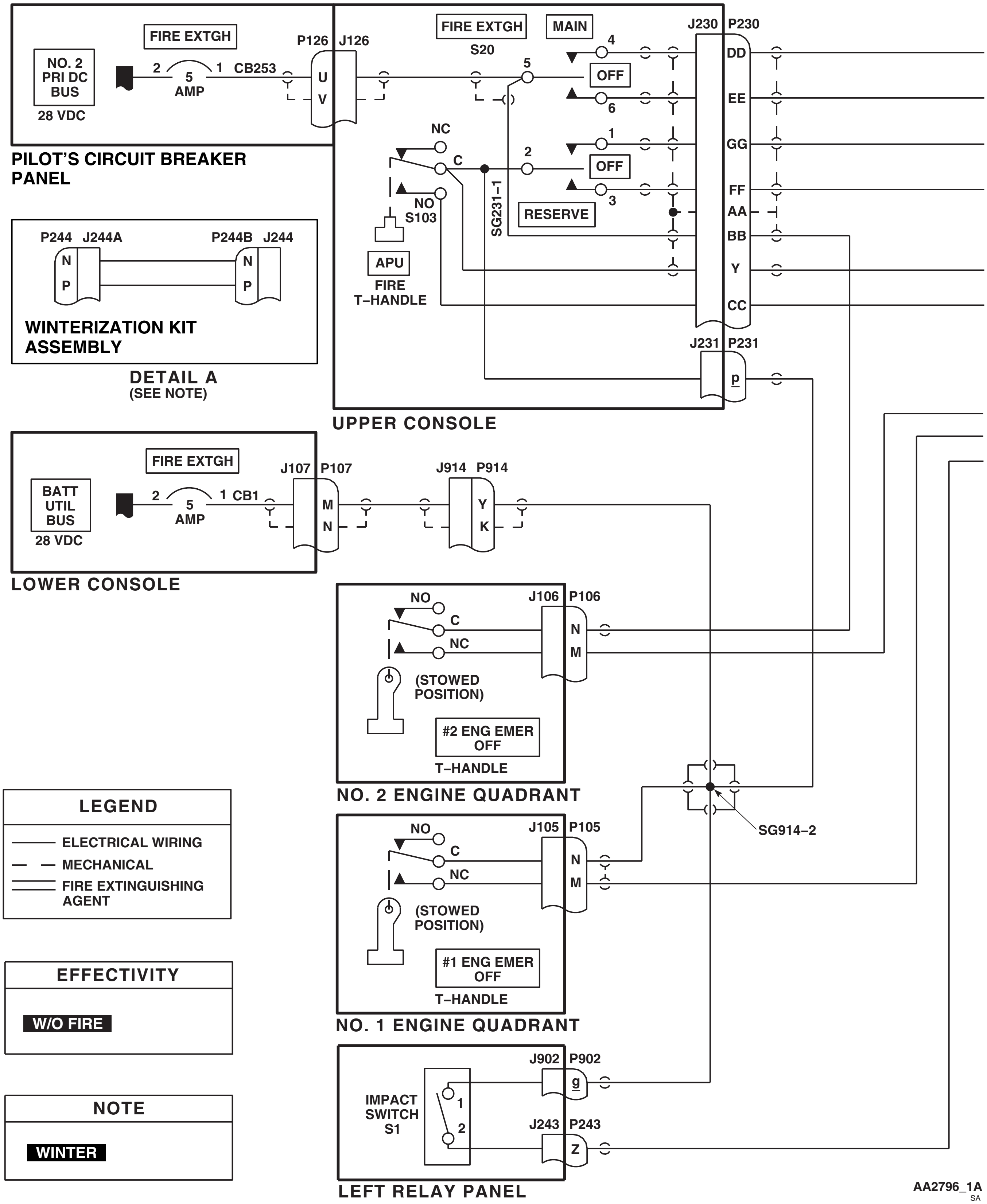


FIGURE 12-2-2-2. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

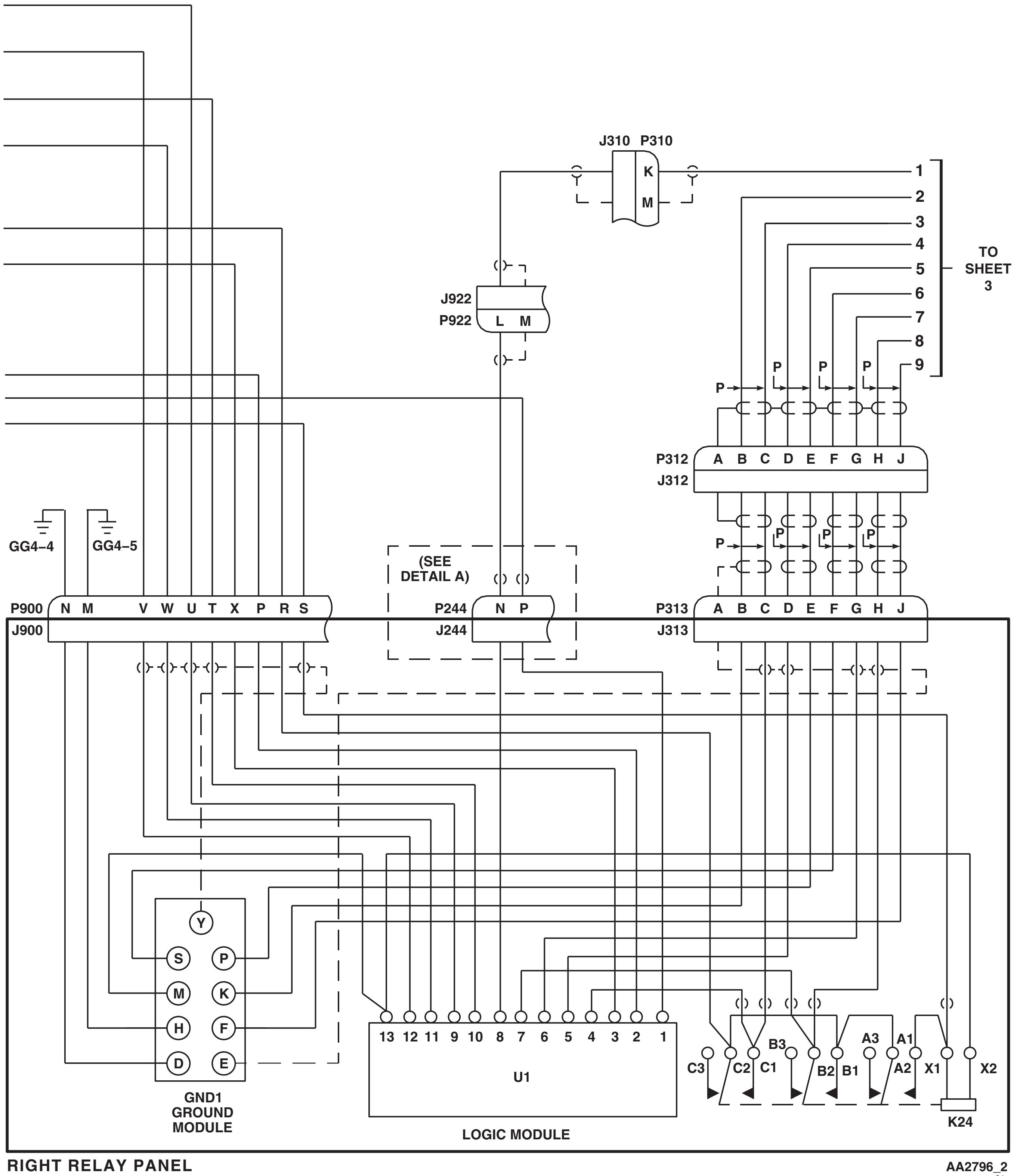


FIGURE 12-2-2-2. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

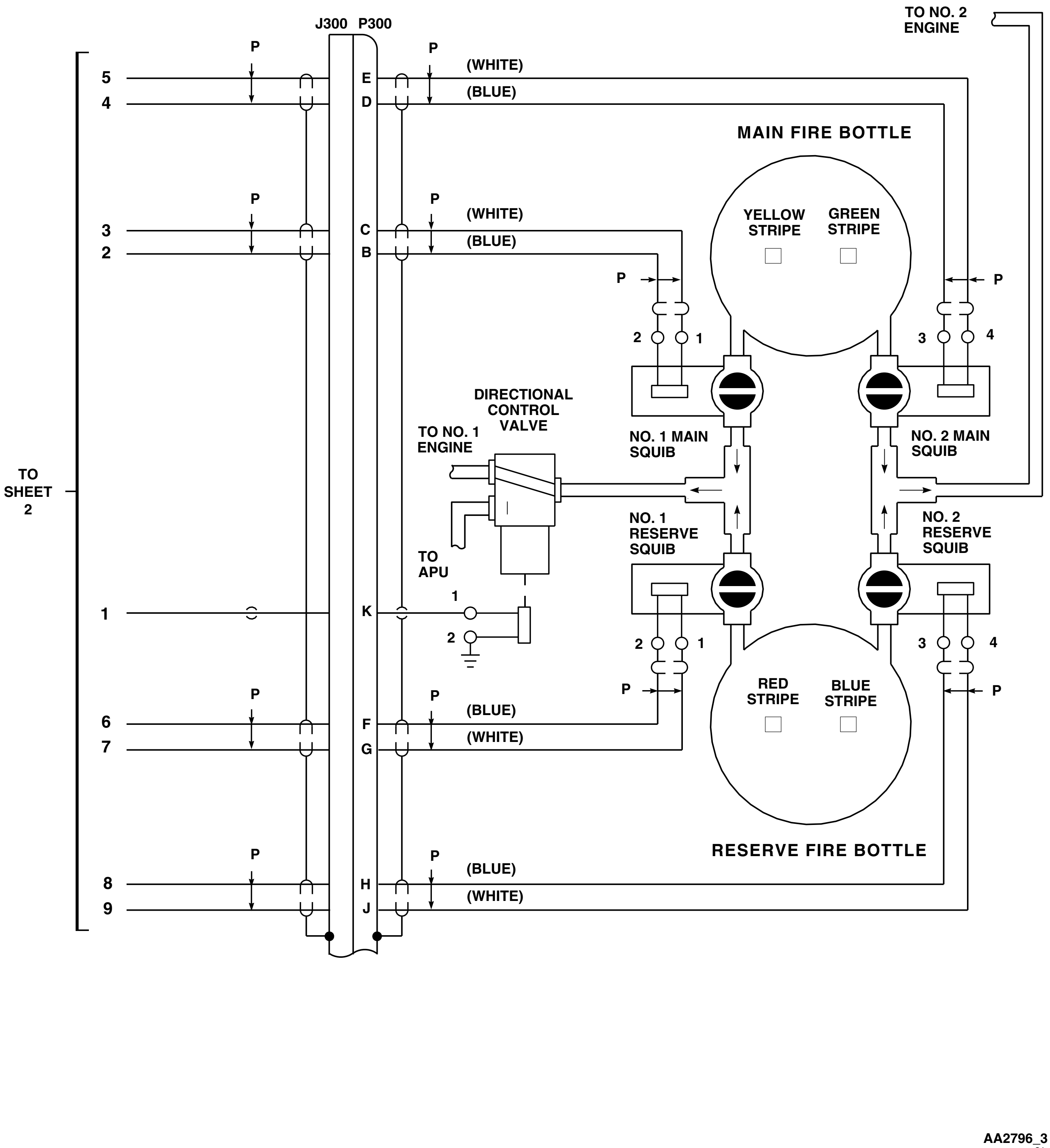


FIGURE 12-2-2-2. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)

FIRE BOTTLE SQUIB WIRING CONNECTION CHART				
FIRE BOTTLE SQUIB	WIRE CODE (PRINTED ON WIRE)	WIRE COLOR CODE	CONNECTION POINT TO SQUIB	FIRE BOTTLE COLOR CODE
MAIN (NO. 1)	W3163A20 W3162A20N	WHITE BLUE	#6 STUD #8 SCREW	YELLOW STRIPE
MAIN (NO. 2)	W3164A20 W3165A20N	BLUE WHITE	#6 STUD #8 SCREW	GREEN STRIPE
RESERVE (NO. 1)	W3167A20 W3166A20N	WHITE BLUE	#6 STUD #8 SCREW	RED STRIPE
RESERVE (NO. 2)	W3168A20 W3169A20N	BLUE WHITE	#6 STUD #8 SCREW	BLUE STRIPE

AA2796_4
SA

FIGURE 12-2-2-2. FIRE EXTINGUISHING SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

12-2-3 WINDSHIELD WIPER SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

CAUTION

To prevent scratching windshield, make sure windshield is kept wet during operational/troubleshooting procedure or install cotter pin in holdoff pin hole.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

12-2-3.1 Operational/Troubleshooting Procedure.

a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

b. Turn on electrical power (PARA 1-6-16).

NOTE

Windshield wiper controls consist of a rotary WINDSHIELD WIPER switch with PARK, OFF, LOW, and HI positions.

c. Turn WINDSHIELD WIPER switch to LOW and then to HI.

RESULT: Both pilot's and copilot's wipers shall operate in both LOW and HI.

- If result is not as specified, go to TABLE NO. 12-2-3-1.
- If only one wiper operates, go to TABLE NO. 12-2-3-2.
- If both wipers do not operate in LOW only, go to TABLE NO. 12-2-3-3.
- If both wipers do not operate in HI only, go to TABLE NO. 12-2-3-4.

d. Turn WINDSHIELD WIPER switch to LOW.

RESULT: Observe that both wipers shall produce the proper sweep and effectively wipe windshield.

- If wipers do not produce proper sweep, adjust converter (PARA 12-4-14).
- If wipers do not wipe effectively, check wiper arm for 4 to 5 pounds of tension.
    - (a) If tension is correct, replace wiper blades (PARA 12-4-17).
    - (b) If tension is not correct, adjust wiper arm tension (PARA 12-3-2).

e. Turn WINDSHIELD WIPER switch to OFF and then to PARK.

RESULT: Both wipers shall return to the normal park position (wiper blades parallel to inboard edge of windshield).

- If result is not as specified, go to TABLE NO. 12-2-3-5.

f. Turn off electrical power (PARA 1-6-16).

g. If necessary, remove cotter pins from holdoff pin holes.

TABLE NO. 12-2-3-1. Wipers Do Not Operate.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 115 vac between WINDSHIELD WIPER switch terminals:
11 and ground
21 and ground
31 and ground

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **5.**

2. Check flexible drive shaft for kinks.

Step 1. If kinks are present, replace flexible drive shaft (PARA 12-4-16). Go to **6.**
Step 2. If kinks are not present, go to **3.**

3. Check converter for binding (disconnect flexible drive shaft and turn converter drive shaft).

Step 1. If binding is present, replace converter (PARA 12-4-14). Go to **6.**
Step 2. If binding is not present, go to **4.**

4. Make sure shaft couplings are installed.

Step 1. If shaft couplings are installed, replace motor (PARA 12-4-15). Go to **6.**
Step 2. If shaft couplings are not installed, install couplings (PARA 12-4-16). Go to **6.**

5. Check for 115 vac between WSHLD WIPER circuit breaker:
Terminal A2 and ground
Terminal B2 and ground
Terminal C2 and ground

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and WINDSHIELD WIPER switch (Appendix F), as required. Go to **6.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **6.**

TABLE NO. 12-2-3-1. Wipers Do Not Operate. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

6. Procedure completed.

TABLE NO. 12-2-3-2. Only One Wiper Operates.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for disconnected or broken flexible drive shaft.

Step 1. If flexible drive shaft is disconnected or broken, connect or replace flexible drive shaft (PARA 12-4-16). Go to **3.**
Step 2. If flexible drive shaft is not disconnected or broken, go to **2.**

2. Check missing couplings.

Step 1. If couplings are missing, replace couplings (PARA 12-4-16). Go to **3.**
Step 2. If couplings are not missing, replace converter (PARA 12-4-14). Go to **3.**

3. Procedure completed.

TABLE NO. 12-2-3-3. Wipers Do Not Operate In LOW.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Turn WINDSHIELD WIPER switch to LOW. Go to 2.

2. Check for 115 vac between:
P108-C and ground
P108-E and ground
P108-G and ground

Step 1. If voltage is as specified, replace motor (PARA 12-4-15). Go to **4.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between WINDSHIELD WIPER switch terminals:

TABLE NO. 12-2-3-3. Wipers Do Not Operate In LOW. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

12 and P108-G
22 and P108-E
32 and P108-C

Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-3-4. Wipers Do Not Operate In HI.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Turn WINDSHIELD WIPER switch to HI. Go to 2.

2. Check for 115 vac between:
P108-B and ground
P108-D and ground
P108-F and ground

Step 1. If voltage is as specified, replace motor (PARA 12-4-15). Go to **4.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between WINDSHIELD WIPER switch terminals:
13 and P108-B
23 and P108-F
33 and P108-D

Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-3-5. Wipers Do Not Return To Park.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Turn WINDSHIELD WIPER switch to PARK. Go to 2.

2. Check for 115 vac between:
P108-A and ground
P108-H and ground

Step 1. If voltage is as specified, replace motor (PARA 12-4-15). Go to **4.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between WINDSHIELD WIPER switch terminals:
18 and P108-H
28 and P108-A

Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **4.**

4. Procedure completed.

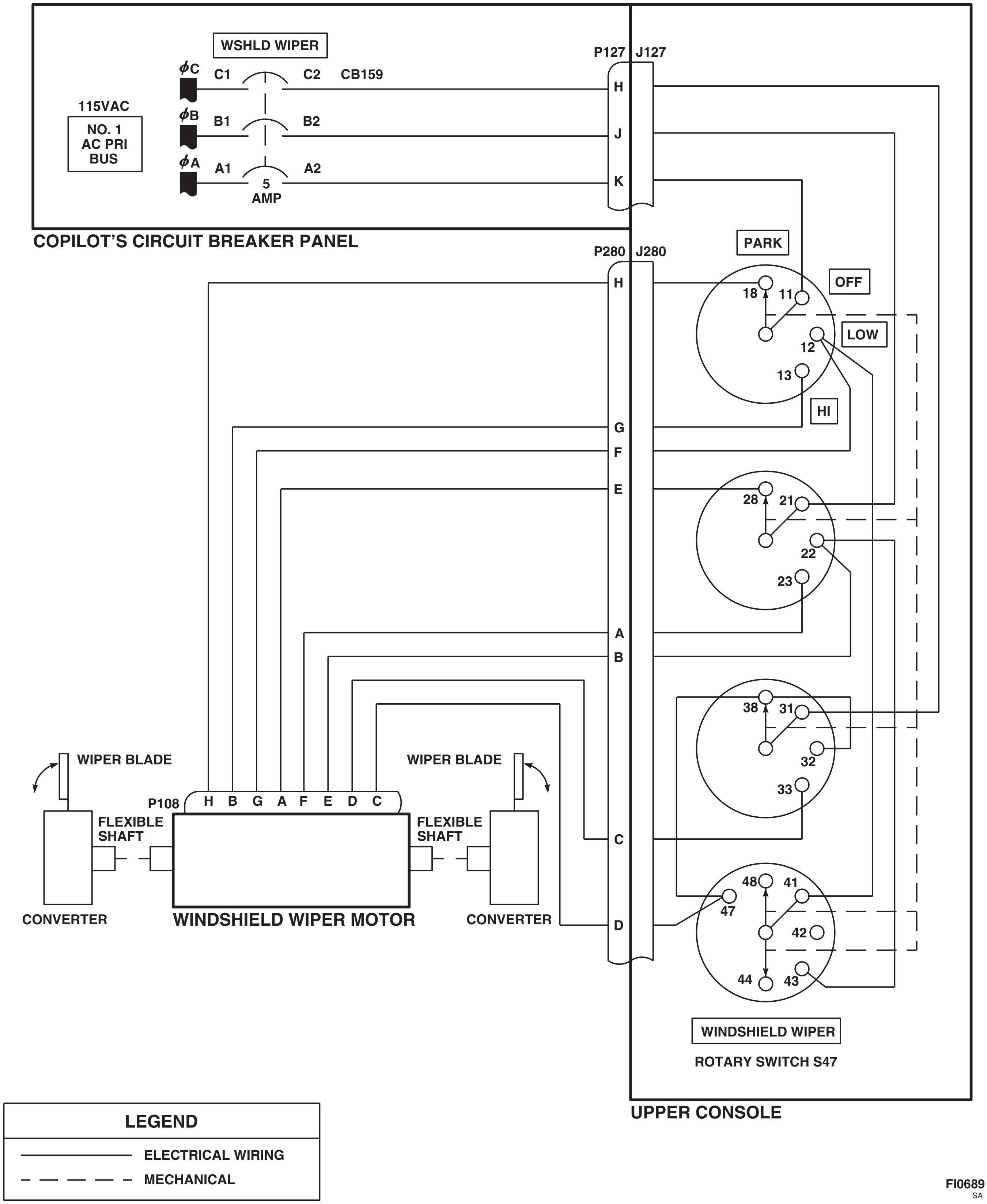


FIGURE 12-2-3-1. WINDSHIELD WIPER SYSTEM SCHEMATIC DIAGRAM

12-2-4 WINDSHIELD ANTI-ICE SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit
- Temperature Probe, 0790
- Thermometer, 389-3L

References

- PARA 1-6-16

Equipment Conditions

- Ambient Temperature Below 100°F (37.7°C)
- External Electrical Power Available or APU Operational

NOTE

Windshield Anti-Ice System cannot be checked out in ambient temperatures of 100°F (37.7°C) or higher. System also may not go on if windshield is in direct sunlight.

NOTE

The Windshield Anti-Ice System is inoperative and cannot be checked when all of these conditions exist: Both primary AC generators are off, the APU generator is on, and the backup hydraulic pump is running. These conditions are indicated by lighting of the APU GEN ON and BACK-UP PUMP ON caution/advisory panel capsules. When these capsules are on, do not perform the operational/troubleshooting procedure.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

For helicopters equipped with the environmental control system, when any WINDSHIELD ANTI-ICE switch is set to ON, the environmental control system is disabled.

12-2-4.1 Operational/Troubleshooting Procedure.

a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

b. Turn on electrical power (PARA 1-6-16).

c. Place upper console WINDSHIELD ANTI-ICE COPILOT switch to ON.

RESULT: Heat shall be felt on copilot's windshield in about five minutes.

- If result is not as specified, go to TABLE NO. 12-2-4-1.

d. Connect temperature probe to thermometer.

e. Place probe in contact with copilot's windshield in area of either sensor and check windshield temperature.

RESULT: Windshield temperature shall be between 23.9°C (75°F) and 46.1°C (115°F).

- If result is not as specified, go to TABLE NO. 12-2-4-2.

f. Place upper console WINDSHIELD ANTI-ICE COPILOT switch to OFF.

g. Place upper console WINDSHIELD ANTI-ICE PILOT switch to ON.

RESULT: Heat shall be felt on pilot's windshield in about five minutes.

- If result is not as specified, go to TABLE NO. 12-2-4-3.

h. Connect temperature probe to thermometer.

i. Place probe in contact with pilot's windshield in area of either sensor and check windshield temperature.

RESULT: Windshield temperature shall be between 23.9°C (75°F) and 46.1°C (115°F).

- If result is not as specified, go to TABLE NO. 12-2-4-4.

j. Place upper console WINDSHIELD ANTI-ICE PILOT switch to OFF.

k. Place upper console WINDSHIELD ANTI-ICE CTR switch to ON.

RESULT: Heat shall be felt on center windshield in about five minutes.

- If result is not as specified go to TABLE NO. 12-2-4-5.

l. Connect temperature probe to thermometer.

m. Place probe in contact with center windshield in area of either sensor and check windshield temperature.

RESULT: Windshield temperature shall be between 23.9°C (75°F) and 46.1°C (115°F).

- If result is not as specified, go to TABLE NO. 12-2-4-6.

n. Place upper console WINDSHIELD ANTI-ICE CTR switch to OFF.

o. Turn off electrical power (PARA 1-6-16).

TABLE NO. 12-2-4-1. Copilot's Windshield Does Not Heat.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that resistance between P275-A and P275-E is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, go to **2.**
Step 2. If resistance is not within limits, go to **20.**

2. Check for 115 VAC between:
Terminal 1 (copilot's windshield heater) and ground
Terminal 2 (copilot's windshield heater) and ground
Terminal 6 (copilot's windshield heater) and ground

Step 1. If voltage is as specified, replace copilot's windshield (PARA 2-4-39). Go to **22.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between:

TABLE NO. 12-2-4-1. Copilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P274-A and terminal 2 (copilot's windshield heater)
P274-G and terminal 1 (copilot's windshield heater)
P274-F and terminal 6 (copilot's windshield heater)

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **22.**

4. Check for 115 VAC between:
P274-B (phase A) and ground
P274-C (phase B) and ground
P274-E (phase C) and ground

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **19.**

5. Check continuity between P274-D and ground.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **22.**

6. Check for 28 VDC between P275-C and ground.

Step 1. If voltage is as specified, replace copilot's anti-ice controller (PARA 12-4-20). Go to **22.**
Step 2. If voltage is not as specified, go to **7.**

7. Check helicopter configuration.

Step 1. WINTER , go to **8.**
Step 2. W/O WINTER , go to **9.**

WINTER **8. Check for 28 VDC between P244B-G and ground.**

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, replace winterization kit (PARA 16-6-2). Go to **22.**
Step 3. If trouble remains, go to **16.**

9. Check for 28 VDC between P244-G and ground.

TABLE NO. 12-2-4-1. Copilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, go to **16.**

10. Check helicopter configuration.

Step 1. WINTER , go to **11.**
Step 2. W/O WINTER , go to **12.**

WINTER **11. Check for 28 VDC between P244B-E and ground.**

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **22.**
Step 2. If voltage is not as specified, go to **13.**

12. Check for 28 VDC between P244-E and ground.

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **22.**
Step 2. If voltage is not as specified, go to **13.**

13. Check helicopter configuration.

Step 1. WINTER , go to **14.**
Step 2. W/O WINTER , go to **15.**

WINTER **14. Check continuity between P244B-F and P275-C.**

Step 1. If continuity is present, troubleshoot right relay panel (PARA 9-2-19). Go to **22.**
Step 2. If continuity is not present, replace winterization kit (PARA 16-6-2). Go to **22.**

15. Check continuity between P244-F and P275-C.

Step 1. If continuity is present, troubleshoot right relay panel (PARA 9-2-19). Go to **22.**
Step 2. If continuity is not present, repair/replace wiring between P244-F and P275-C (Appendix F). Go to **22.**

TABLE NO. 12-2-4-1. Copilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

16. Check for 28 VDC between WINDSHIELD ANTI-ICE COPILOT switch S22 terminal 3 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch S22 terminal 3 and P244-G (Appendix F). Go to **22.**
Step 2. If voltage is not as specified, go to **17.**

17. Check for 28 VDC between WINDSHIELD ANTI-ICE COPILOT switch S22 terminal 2 and ground.

Step 1. If voltage is as specified, replace switch (PARA 12-4-21). Go to **22.**
Step 2. If voltage is not as specified, go to **18.**

18. Check continuity between terminal 1 (CPLT WSHLD ANTI-ICE circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and WINDSHIELD ANTI-ICE COPILOT switch S22 terminal 2.

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **22.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **22.**

19. Check continuity between:

P274-B and terminal A2 (CPLT WSHLD ANTI-ICE circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel)
P274-C and terminal B2 (CPLT WSHLD ANTI-ICE circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel)
P274-E and terminal C2 (CPLT WSHLD ANTI-ICE circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel)

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **22.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **22.**

20. Check that resistance between copilot's windshield terminal block terminals 7 and 8 is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, repair/replace wiring (Appendix F), as required between, then go to **22:**
P275-A and terminal 7
P275-E and terminal 8
Step 2. If resistance is not within limits, go to **21.**

TABLE NO. 12-2-4-1. Copilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

21. Check continuity between copilot's windshield terminal block:
3 and 8
4 and 7

Step 1. If continuity is present, replace copilot's windshield (PARA 2-4-39). Go to **22.**
Step 2. If continuity is not present, repair/replace terminal block wiring (Appendix F), as required. Go to **22.**

22. Procedure completed.

TABLE NO. 12-2-4-2. Copilot's Windshield Temperature Is Not Between 23.9°C (75°F) And 46.1°C (115°F).

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that resistance between P275-A and P275-E is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, replace copilot's anti-ice controller (PARA 12-4-20). Go to **4.**
Step 2. If resistance is not within limits, go to **2.**

2. Check that resistance between terminals 7 and 8 of copilot's windshield heater terminal block is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, repair/replace wiring (Appendix F), as required between, then go to **4:**
P275-A and terminal 8
P275-E and terminal 7
Step 2. If resistance is not within limits, go to **3.**

3. Check continuity between copilot's windshield heater terminal block:
3 and 8
4 and 7

TABLE NO. 12-2-4-2. Copilot's Windshield Temperature Is Not Between 23.9°C (75°F) And 46.1°C (115°F). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace copilot's windshield (PARA 2-4-39). Go to **4.**
Step 2. If continuity is not present, repair/replace terminal block wiring (Appendix F), as required. Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-4-3. Pilot's Windshield Does Not Heat.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that resistance between P271-A and P271-E is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, go to **2.**
Step 2. If resistance is not within limits, go to **16.**

2. Check for 115 VAC between:
Terminal 1 (pilot's windshield heater) and ground
Terminal 2 (pilot's windshield heater) and ground
Terminal 6 (pilot's windshield heater) and ground

Step 1. If voltage is as specified, replace pilot's windshield (PARA 2-4-39). Go to **18.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between pilot's windshield:
Terminal 1 (pilot's windshield heater) and P270-G
Terminal 2 (pilot's windshield heater) and P270-A
Terminal 6 (pilot's windshield heater) and P270-F

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **18.**

4. Check for 115 VAC between:

TABLE NO. 12-2-4-3. Pilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P270-B (phase A) and ground
P270-C (phase B) and ground
P270-E (phase C) and ground

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **15.**

5. Check continuity between P270-D and ground.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **18.**

6. Check for 28 VDC between P271-C and ground.

Step 1. If voltage is as specified, replace pilot's anti-ice controller (PARA 12-4-20). Go to **18.**
Step 2. If voltage is not as specified, go to **7.**

7. Check for 28 VDC between P900-F and ground.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **12.**

8. Check helicopter configuration.

Step 1. WINTER , go to **9.**
Step 2. W/O WINTER , go to **10.**

WINTER **9. Check for 28 VDC between P244B-E and ground.**

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **18.**
Step 2. If voltage is not as specified, go to **11.**

W/O WINTER **10. Check for 28 VDC between P244-E and ground.**

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **18.**
Step 2. If voltage is not as specified, go to **11.**

TABLE NO. 12-2-4-3. Pilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

11. Check continuity between P900-E and P271-C.

Step 1. If continuity is present, troubleshoot right relay panel (PARA 9-2-19). Go to **18.**
Step 2. If continuity is not present, repair/replace wiring between P900-E and P271-C (Appendix F). Go to **18.**

12. Check for 28 VDC between WINDSHIELD ANTI-ICE PILOT switch S21 terminal 3 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch S21 terminal 3 and P900-F (Appendix F). Go to **18.**
Step 2. If voltage is not as specified, go to **13.**

13. Check for 28 VDC between WINDSHIELD ANTI-ICE PILOT switch terminal 2 and ground.

Step 1. If voltage is as specified, replace switch (PARA 12-4-21). Go to **18.**
Step 2. If voltage is not as specified, go to **14.**

14. Check continuity between terminal 1 (PILOT WSHLD ANTI-ICE circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and WINDSHIELD ANTI-ICE PILOT switch terminal 2.

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **18.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **18.**

15. Check continuity between PILOT WSHLD ANTI-ICE circuit breaker (NO. 2 DC PRI BUS - pilot's circuit breaker panel):
Terminal A2 and P270-B
Terminal B2 and P270-C
Terminal C2 and P270-E

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **18.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **18.**

16. Check that resistance between pilot's windshield terminal block terminals 7 and 8 is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, repair/replace wiring (Appendix F), as required between, then go to **18.**

TABLE NO. 12-2-4-3. Pilot's Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P271-A and terminal 7
P271-E and terminal 8

Step 2. If resistance is not within limits, go to **17.**

17. Check continuity between pilot's windshield terminal block:
3 and 8
4 and 7

Step 1. If continuity is present, replace pilot's windshield (PARA 2-4-39). Go to **18.**

Step 2. If continuity is not present, repair/replace terminal block wiring (Appendix F), as required. Go to **18.**

18. Procedure completed.

TABLE NO. 12-2-4-4. Pilot's Windshield Temperature Is Not Between 23.9°C (75°F) And 46.1°C (115°F).

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that resistance between P271-A and P271-E is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, replace pilot's anti-ice controller (PARA 12-4-20). Go to **4.**

Step 2. If resistance is not within limits, go to **2.**

2. Check that resistance between terminals 7 and 8 of pilot's windshield terminal block is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, repair/replace wiring (Appendix F), as required between, then go to **4:**
P271-A and terminal 8
P271-E and terminal 7

Step 2. If resistance is not within limits, go to **3.**

3. Check continuity between pilot's windshield terminal block:
3 and 8
4 and 7

TABLE NO. 12-2-4-4. Pilot's Windshield Temperature Is Not Between 23.9°C (75°F) And 46.1°C (115°F). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace pilot's windshield (PARA 2-4-39). Go to **4.**
Step 2. If continuity is not present, repair/replace terminal block wiring (Appendix F), as required. Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-4-5. Center Windshield Does Not Heat.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that resistance between P385-A and P384-E is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, go to **2.**
Step 2. If resistance is not within limits, go to **18.**

2. Check for 115 VAC between:
TB7-1 (center windshield heater) and ground
TB7-2 (center windshield heater) and ground
TB7-6 (center windshield heater) and ground

Step 1. If voltage is as specified, replace center windshield (PARA 2-4-40 or PARA 2-4-41). Go to **20.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between:
P383-A and TB7-2 (center windshield heater)
P383-G and TB7-5 (center windshield heater)
P383-F and TB7-6 (center windshield heater)

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **20.**

4. Check for 115 VAC between:

TABLE NO. 12-2-4-5. Center Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P383-B (phase A) and ground
P383-C (phase B) and ground
P383-E (phase C) and ground

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **17.**

5. Check continuity between P383-D and ground.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). **20.**

6. Check for 28 VDC between P384-C and ground.

Step 1. If voltage is as specified, replace center anti-ice controller (PARA 12-4-20). Go to **20.**
Step 2. If voltage is not as specified, go to **7.**

7. Check helicopter configuration.

Step 1. WINTER , go to **8.**
Step 2. W/O WINTER , go to **9.**

WINTER **8. Check for 28 VDC between P244B-G and ground.**

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, replace winterization kit (PARA 16-6-2). Go to **20.**
Step 3. If trouble remains, go to **9.**

9. Check for 28 VDC between P244-G and ground.

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, go to **14.**

10. Check helicopter configuration.

Step 1. WINTER , go to **11.**
Step 2. W/O WINTER , go to **12.**

WINTER **11. Check for 28 VDC between P244B-E and ground.**

TABLE NO. 12-2-4-5. Center Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **20.**
Step 2. If voltage is not as specified, go to **12.**

12. Check for 28 VDC between P244-E and ground.

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **20.**
Step 2. If voltage is not as specified, go to **13.**

13. Check continuity between P900-C and P384-C.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **20.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **20.**

14. Check for 28 VDC between WINDSHIELD ANTI-ICE CTR switch S66 terminal 3 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch S66 terminal 3 and P900-D (Appendix F). Go to **20.**
Step 2. If voltage is not as specified, go to **15.**

15. Check for 28 VDC between WINDSHIELD ANTI-ICE CTR switch terminal 2 and ground.

Step 1. If voltage is as specified, replace switch (PARA 12-4-21). Go to **20.**
Step 2. If voltage is not as specified, go to **16.**

16. Check continuity between terminal 1 (WINDSHIELD ANTI-ICE CTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and WINDSHIELD ANTI-ICE CTR switch terminal 2.

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **20.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **20.**

17. Check continuity between CTR WSHLD ANTI-ICE circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel):
Terminal A2 and P383-B
Terminal B2 and P383-C
Terminal C2 and P383-E

TABLE NO. 12-2-4-5. Center Windshield Does Not Heat. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

d

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **20.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **20.**

18. Check that resistance between center windshield terminal block terminals 7 and 8 is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, repair/replace wiring (Appendix F), as required between, then go to **20:**
P384-E and terminal 7
P384-A and terminal 8
Step 2. If resistance is not within limits, go to **19.**

19. Check continuity between center windshield terminal block:
3 and 8
4 and 7

Step 1. If continuity is present, replace center windshield (PARA 2-4-40 or PARA 2-4-41). Go to **20.**
Step 2. If continuity is not present, repair/replace terminal block wiring (Appendix F), as required. Go to **20.**

20. Procedure completed.

TABLE NO. 12-2-4-6. Center Windshield Temperature Is Not Between 23.9°C (75°F) And 46.1°C (115°F).

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that resistance between P384-A and P384-E is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

Step 1. If resistance is within limits, replace center anti-ice controller (PARA 12-4-20). Go to **4.**
Step 2. If resistance is not within limits, go to **2.**

2. Check that resistance between terminals 7 and 8 of center windshield terminal block is within limits shown in Figure 12-2-4-4 for windshield sensor temperature.

TABLE NO. 12-2-4-6. Center Windshield Temperature Is Not Between 23.9°C (75°F) And 46.1°C (115°F). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If resistance is within limits, repair/replace wiring (Appendix F), as required between, then go to **4:**
P384-A and terminal 8
P384-E and terminal 7

Step 2. If resistance is not within limits, go to **3.**

3. Check continuity between center windshield terminal block:
3 and 8
4 and 7

Step 1. If continuity is present, replace center windshield (PARA 2-4-40 or PARA 2-4-41). Go to **4.**

Step 2. If continuity is not present, repair/replace terminal block wiring (Appendix F), as required. Go to **4.**

4. Procedure completed.

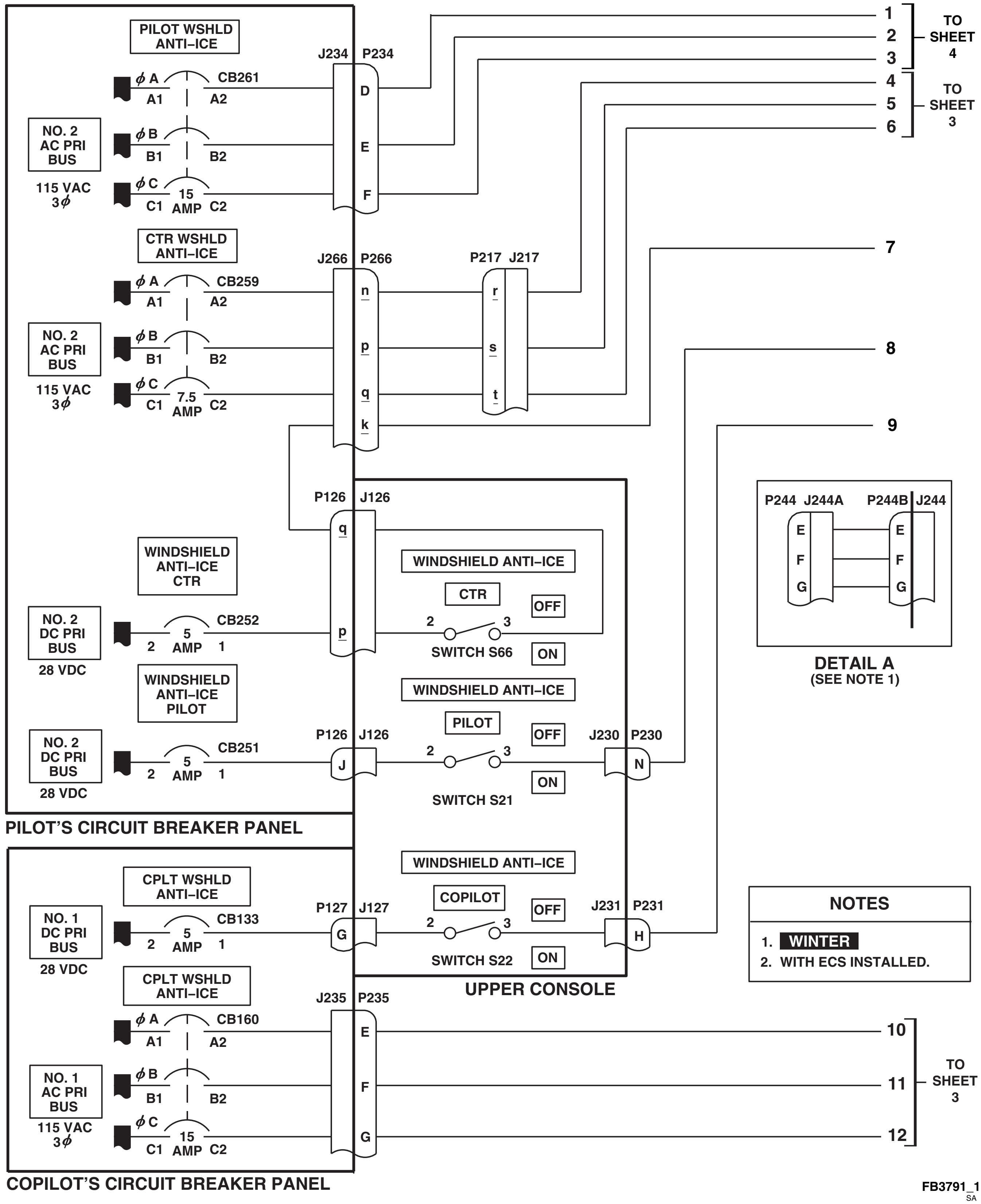


FIGURE 12-2-4-1. WINDSHIELD ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

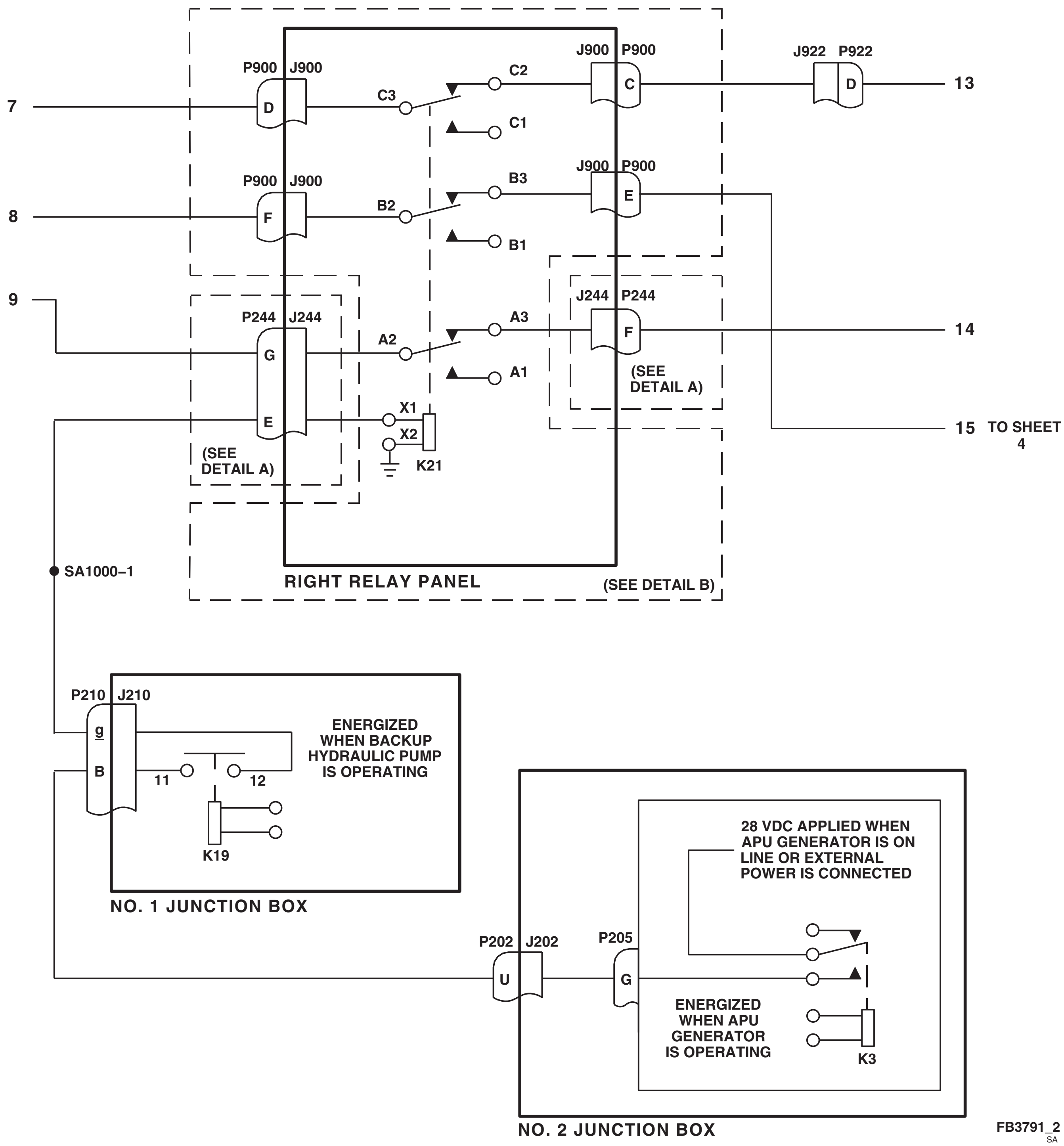


FIGURE 12-2-4-1. WINDSHIELD ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

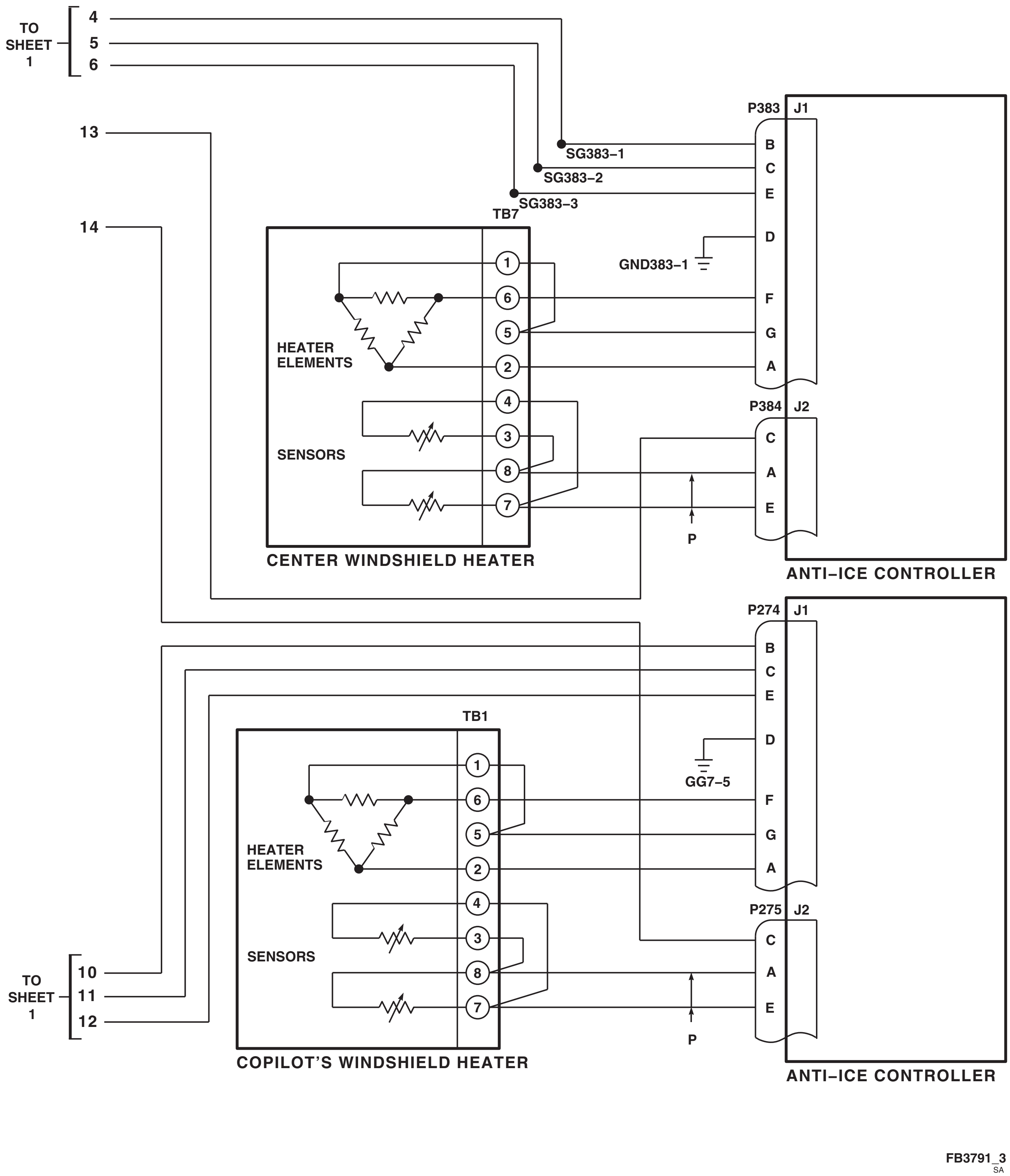
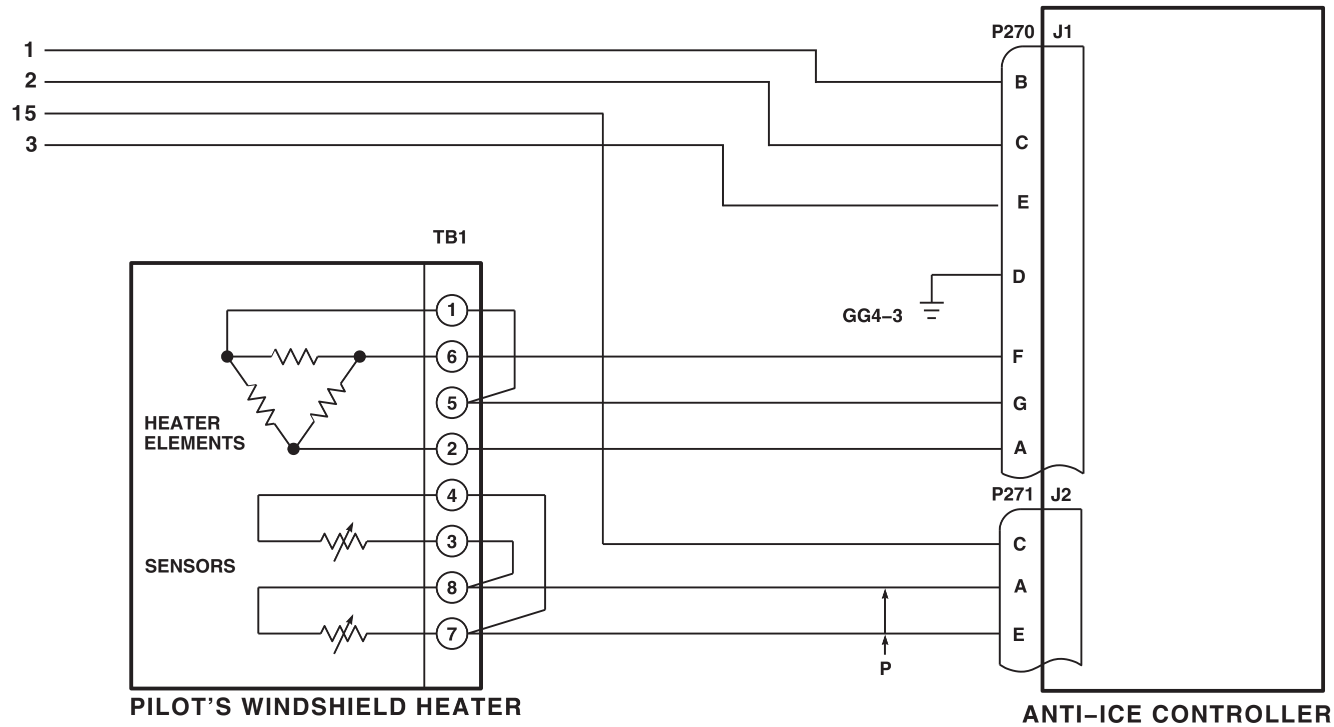
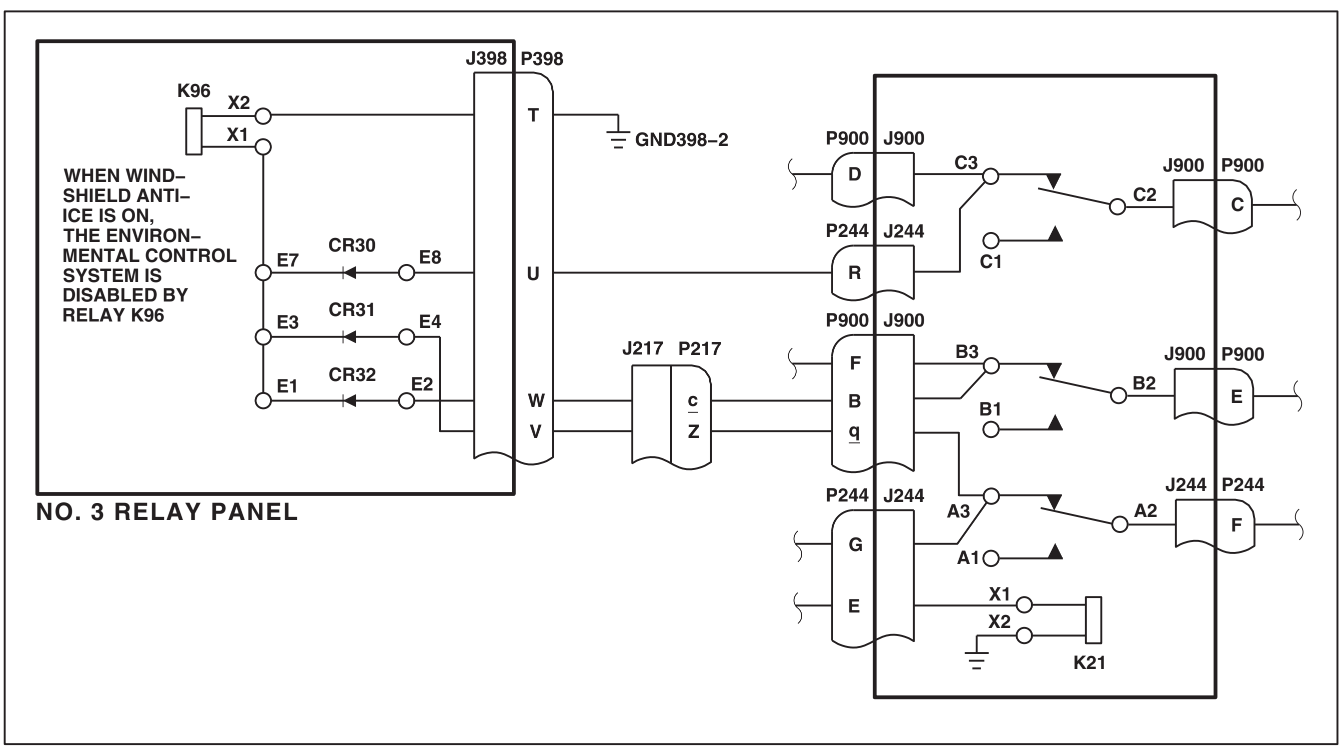


FIGURE 12-2-4-1. WINDSHIELD ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)





DETAIL B
(SEE NOTE 2)

FB3791_4
SA

FIGURE 12-2-4-1. WINDSHIELD ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

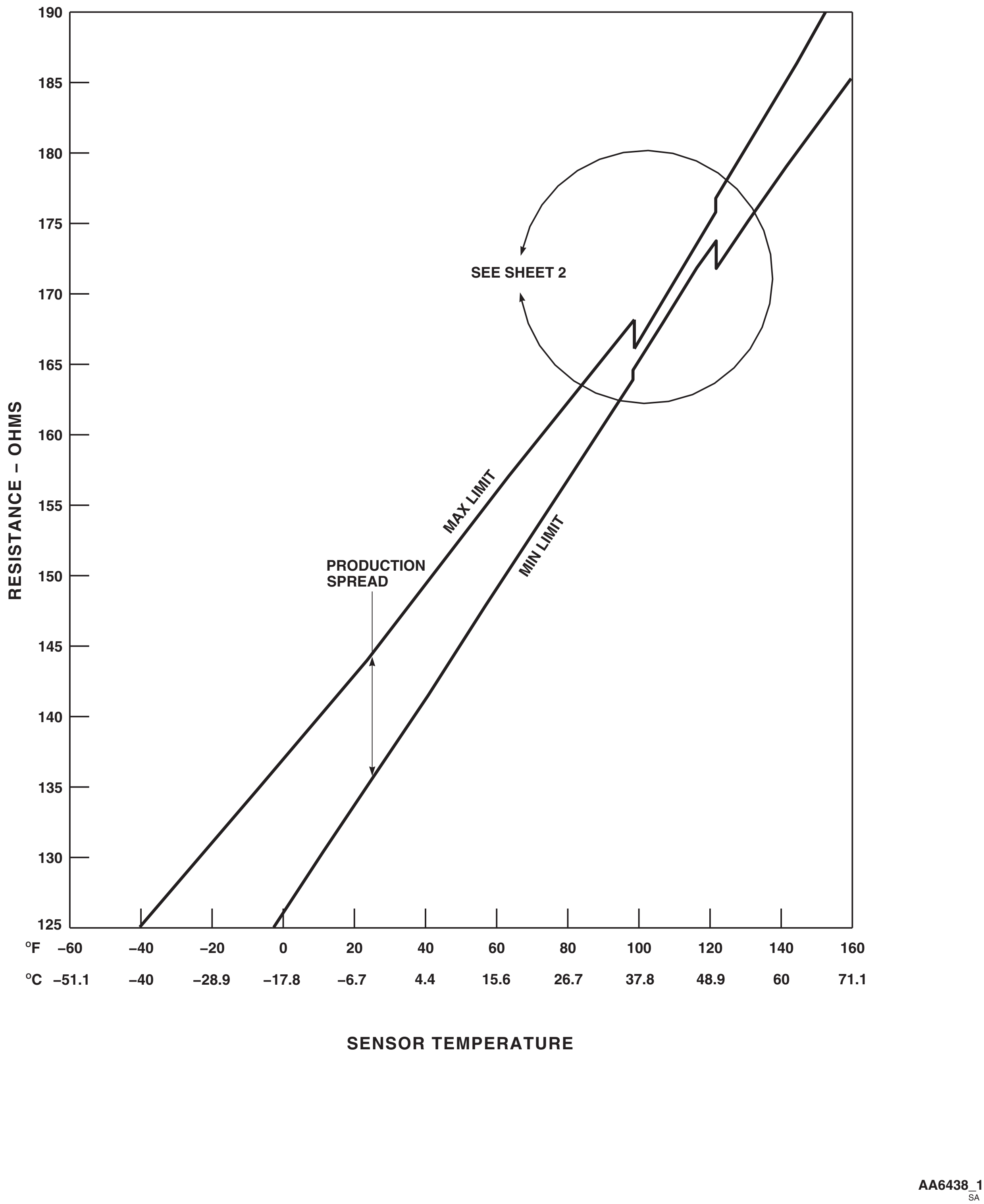


FIGURE 12-2-4-2. WINDSHIELD ANTI-ICE SYSTEM TEMPERATURE/RESISTANCE CHART (SHEET 1 OF 2)

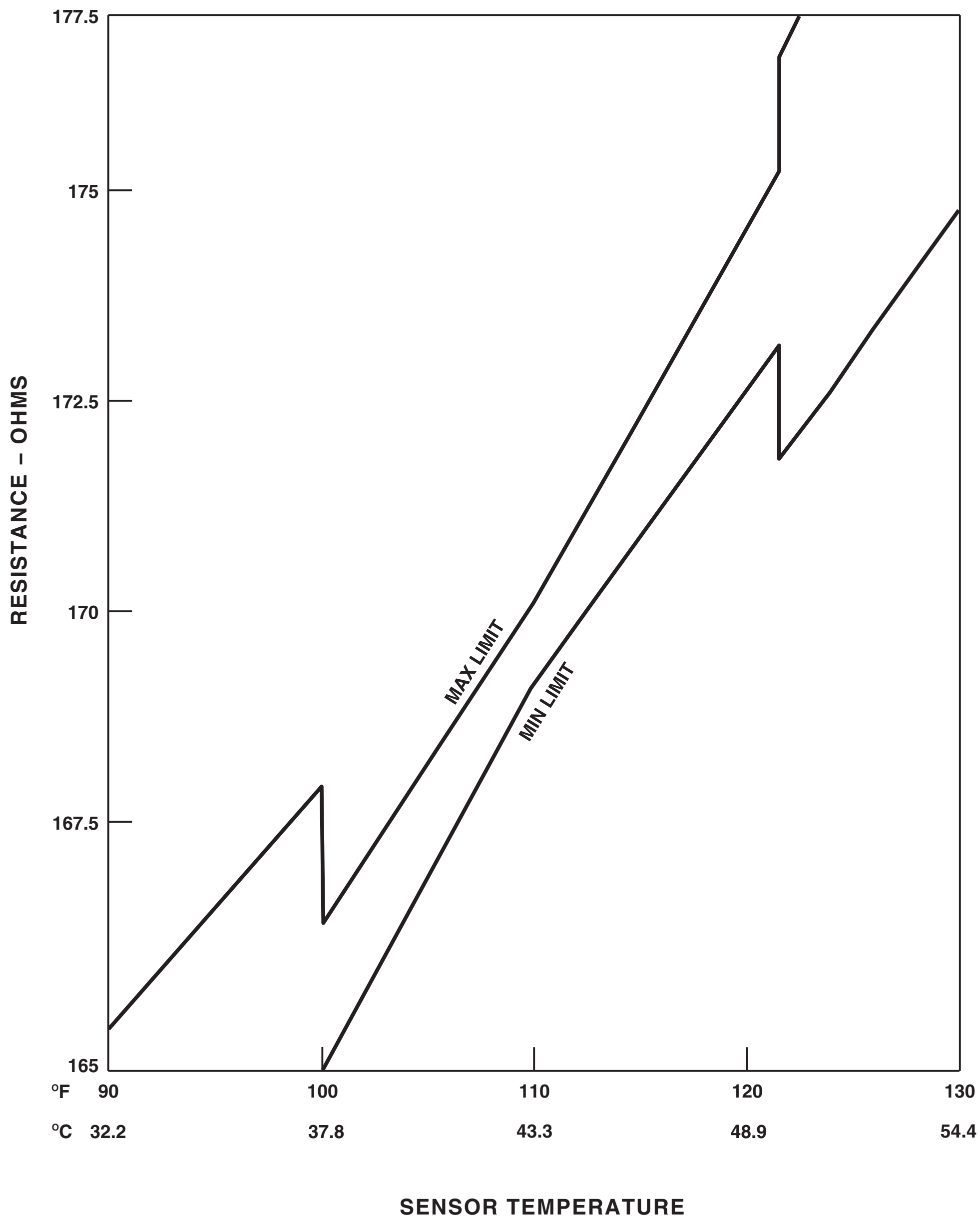


FIGURE 12-2-4-2. WINDSHIELD ANTI-ICE SYSTEM TEMPERATURE/RESISTANCE CHART (SHEET 2 OF 2)

12-2-5 BLADE DE-ICING SYSTEM.

INITIAL SETUP

Test Equipment

- Clamp-On Ammeter, 749
- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit
- Locally-Made Blade De-ice Test Harness (test harness), Figure H-147, Appendix H
- Locally-Made Blade De-ice Test Block (test block), Figure H-148, Appendix H

References

- Appropriate Flight Manual
- Appropriate MTF
- PARA 1-6-16
- PARA 9-4-98

Equipment Conditions

- External Electrical Power Available and No. 1 and No. 2 Generators Operational
- Rotor at 100%

CAUTION

To prevent damage to the blades, if PWR MAIN RTR or PWR TAIL RTR capsules go on for 15 seconds or more, immediately remove electrical power from helicopter.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

If main rotor blade erosion protection kit is installed, do not perform Blade De-icing System operational/ troubleshooting procedure.

The Blade De-icing System operational/ troubleshooting procedure shall be performed no more than one hour after passing a minimum of 1 ampere through all 12 legs of the blade heater. This can be done by performing the blade de-ice preflight test (Appropriate MTF).

If System was recently installed, do steps a. through bn. to check operation of Blade De-icing System. To check Blade De-icing System for unscheduled maintenance, do steps a. through bk.

12-2-5.1 Operational/Troubleshooting Procedure.

a. EMEP Remove and check pin filter adapters (PFAs) from P134, P324, and P325. Replace if necessary (PARA 9-4-98).

b. EMEP Install PFAs (PARA 9-4-98).

c. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

d. Place blade de-ice control panel POWER switch to OFF and MODE switch to AUTO.

e. Place blade de-ice test panel test function switch to NORM.

f. Make sure upper console BACKUP HYD PUMP switch is at OFF.

NOTE

Unless otherwise specified, ignore the PWR MAIN RTR and/or PWR TAIL RTR lamps if either goes on for less than 10 seconds.

g. Turn on electrical power (PARA 1-6-16).

h. Place caution/advisory panel BRT/DIM-TEST switch to TEST and hold.

RESULT: All caution/advisory capsules shall go on.

- If result is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

i. Release caution/advisory panel BRT/DIM-TEST switch.

RESULT: Caution/advisory panel #1 GEN, #2 GEN, or EXT PWR CONNECTED (if external power is connected) capsules shall remain on.

- If result is not as specified, troubleshoot AC Electrical System (PARA 9-2-1).

j. Pull out ICE-DET circuit breaker (NO. 2 PRI BUS DC - mission readiness circuit breaker panel).

RESULT: Icing rate meter shall indicate zero and FAIL flag shall appear.

- If result is not as specified, replace icing rate meter (PARA 12-4-27).

k. Push in ICE-DET circuit breaker (NO. 2 PRI BUS DC - mission readiness circuit breaker panel).

RESULT:

(1) Icing rate meter FAIL flag shall disappear from view.

(2) Icing rate meter needle shall deflect and return to zero.

- If results (1) and (2) are not as specified, go to TABLE NO. 12-2-5-1.

- If either result is not as specified, replace icing rate meter (PARA 12-4-27).

NOTE

- To avoid the possibility of obtaining a false indication, wait at least 20 seconds after power is turned on before pressing the icing rate meter PRESS TO START pushbutton.

- Allow at least 75 seconds between each PRESS TO START pushbutton actuation to allow test cycle to complete.

l. Press and release icing rate meter PRESS TO START pushbutton and simultaneously start timer (instrument panel clock). **RECORD RESULT**.

RESULT:

(1) Icing rate meter needle shall move to half scale (1.0 ± 1/8 inch).

(2) In 15 to 25 seconds after pressing PRESS TO TEST pushbutton, ICE DETECTED legends on pilot's and copilot's inboard MFDs shall go on.

(3) In 40 to 60 seconds after pressing PRESS TO TEST button, icing rate meter needle shall return to below zero, and ICE DETECTED legends shall go off.

(4) Within 75 seconds after pressing PRESS TO START button, icing rate meter needle shall move to zero to complete test.

- If result (1) is not as specified, replace icing rate meter (PARA 12-4-27).

- If result (2) is not as specified, go to TABLE NO. 12-2-5-2.

- If results (3) and (4) are not as specified, replace icing rate meter (PARA 12-4-27).

CAUTION

To prevent damage to main rotor blades, remove test block from ice detector when icing rate needle moves. When removing test block, be careful not to hit and damage sensing probe.

NOTE

- Icing rate meter needle may hold value for 50 to 70 seconds after test block is removed, and the ICE DETECTED capsule may go on.
- The greater the pressure on the test block, the greater the icing rate meter needle indication.

m. Carefully place test block against upper half of ice detector probe.

RESULT: Icing rate meter needle shall move to some value between zero and full scale.

- If result is not as specified, go to TABLE NO. 12-2-5-3.
- If the ICE DETECTED capsule came on in step m., wait up to 2 minutes after block is removed for ICE DETECTED capsule to go off before continuing. If capsule does not go off as specified, replace icing rate meter (PARA 12-4-27).

n. Reset instrument panel clock.

o. EMEP Remove and check PFA from P117. Replace if necessary (PARA 9-4-98).

p. EMEP Install PFA on P117 (PARA 9-4-98).

q. Make sure the following caution/advisory capsules are off:

#1 CONV
#2 CONV
MR DE-ICE FAIL
TR DE-ICE FAIL
MR DE-ICE FAULT
ICE DETECTED

RESULT: The specified capsules shall be off.

- If #1 CONV or #2 CONV capsule is on, troubleshoot DC Electrical System (PARA 9-2-2).
- If MR DE-ICE FAIL, TR DE-ICE FAIL or MR DE-ICE FAULT capsule is on, go to TABLE NO. 12-2-5-4.
- If ICE DETECTED capsule is on, check for 28 VDC between P117-63 and ground.

(a) If voltage is as specified, replace icing rate meter (PARA 12-4-27).

1 If trouble remains, replace ice detector (PARA 12-4-50).

(b) If voltage is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

r. Make sure blade de-ice test panel PWR MAIN RTR and PWR TAIL RTR lamps are off.

RESULT: The specified lamps shall be off.

- If both PWR MAIN RTR and PWR TAIL RTR lamps are on, check for 28 VDC between APU/generator interlock relay terminals K64-X1 and K64-X2.

(a) If voltage is as specified, replace blade de-ice control panel (PARA 12-4-29).

(b) If voltage is not as specified, replace K64 (PARA 12-4-37).

- If PWR MAIN RTR lamp is on, go to TABLE NO. 12-2-5-5.

- If PWR TAIL RTR lamp is on, go to TABLE NO. 12-2-5-6.

s. Momentarily press blade de-ice test panel PWR MAIN RTR lamp.

RESULT: PWR MAIN RTR lamp shall go on, then off.

- If result is not as specified, go to TABLE NO. 12-2-5-7.

t. Momentarily press blade de-ice test panel PWR TAIL RTR lamp.

RESULT: PWR TAIL RTR lamp shall go on, then off.

- If result is not as specified, unscrew lamp cap and check lamp.

  (a) If lamp is good, replace blade de-ice test panel (PARA 12-4-46).

  (b) If lamp is not good, replace lamp (PARA 12-4-47).

u. Momentarily press blade de-ice control panel TEST IN PROGRESS lamp.

RESULT: TEST IN PROGRESS lamp shall go on, then off.

- If result is not as specified, go to TABLE NO. 12-2-5-8.

CAUTION

To prevent damage to the blades, do not test Blade De-ice System more than once in 30 minutes when rotor head is not turning.

NOTE

Backup pump may run to recharge APU hydraulic start accumulator. Wait until BACKUP PUMP ON capsule goes off before starting step v. If backup pump goes on while doing steps v. through ah. and the APU generator is the only power source for the test, the test will automatically terminate. This will be indicated by the MR DE-ICE FAIL caution capsule and the BACKUP PUMP ON advisory capsule going on. If this occurs, place blade de-ice control panel POWER switch to OFF, wait until backup pump goes off, and then continue operational/troubleshooting procedure beginning with step v.

v. Place blade de-ice control panel POWER switch to TEST and simultaneously start instrument panel clock. **RECORD RESULT**.

RESULT:

(1) TEST IN PROGRESS lamp shall go on and stay on for 105 to 135 seconds.

(2) MR DE-ICE FAIL, TR DE-ICE FAIL, and MR DE-ICE FAULT capsules shall remain off for full 105 to 135 seconds of test.

(3) PWR MAIN RTR lamp shall go on for up to 11 seconds at end of test.

(4) PWR TAIL RTR lamp shall go on for up to 4 seconds at end of test.

- If result (1) is not as specified, go to TABLE NO. 12-2-5-9.

- If MR DE-ICE FAIL and MR DE-ICE FAULT capsules go on, go to TABLE NO. 12-2-5-10.

- If MR DE-ICE FAIL capsule goes on, go to TABLE NO. 12-2-5-11.

- If TR DE-ICE FAIL capsule goes on, go to TABLE NO. 12-2-5-12.

- If MR DE-ICE FAIL and TR DE-ICE FAIL capsules both go on, go to TABLE NO. 12-2-5-13.

- If MR DE-ICE FAULT capsule goes on, go to TABLE NO. 12-2-5-14.
- If PWR MAIN RTR lamp remains on, go to TABLE NO. 12-2-5-5.
- If PWR TAIL RTR lamp remains on, go to TABLE NO. 12-2-5-6.
- If result (3) or (4) is not as specified, go to TABLE NO. 12-2-5-15.

w. Reset instrument panel clock.

WARNING

A risk of low current shock exists when Blade De-ice System is turned ON for ground checks. Injury to personnel can result if contact is made with main rotor blade cuff, spindle, and droop stop cam while blade de-ice test is in progress. Clear aircraft upper deck of personnel during blade de-ice test until blade de-ice control panel TEST IN PROGRESS lamp is OFF and test switch is placed in OFF position.

x. Place blade de-ice control panel POWER switch to OFF.

WARNING

Droop stop hinge pins and cams may become very hot within 30 to 60 seconds after blade de-ice control panel POWER switch is placed to TEST. Use care when touching those components. A risk of low current shock exists when Blade De-icing System is turned ON for ground checks. Injury to personnel can result if contact is made with main rotor blade cuff, spindle, and droop stop cam while blade de-ice test is in progress. Do not touch droop stop cams until blade de-ice control panel TEST IN PROGRESS lamp is OFF and test switch is placed in the OFF position. Only momentarily touch droop stop cams.

y. Cautiously touch each droop stop cam.

RESULT: Droop stop cams shall be warm to touch.

- If result is not as specified, go to TABLE NO. 12-2-5-16.

z. Place blade de-ice test panel test function switch to SYNC 1.

aa. Place blade de-ice control panel POWER switch to TEST.

NOTE

MR DE-ICE FAULT capsule may go on.

RESULT: MR DE-ICE FAIL capsule shall go on.

- If result is not as specified, check for 115 VAC between P134-L and P134-T.

  (a) If voltage is as specified, replace blade de-ice test panel (PARA 12-4-46).

  (b) If voltage is not as specified, repair/replace wiring between P134-L and P906-W (Appendix F).

ab. Place blade de-ice control panel POWER switch to OFF.

RESULT: MR DE-ICE FAIL capsule shall go off.

- If result is not as specified, replace rotor blade de-ice controller (PARA 12-4-32).

ac. Place blade de-ice test panel test function switch to SYNC 2.

ad. Place blade de-ice control panel POWER switch to TEST.

NOTE

MR DE-ICE FAULT capsule may go on.

RESULT: MR DE-ICE FAIL capsule shall go on.

- If result is not as specified with test function switch in SYNC 2, check for open between P134-N and P134-P.
  - (a) If open is present, replace rotor blade de-ice controller (PARA 12-4-32).
  - (b) If open is not present, replace blade de-ice test panel (PARA 12-4-46).

ae. Place blade de-ice control panel POWER switch to OFF.

RESULT: MR DE-ICE FAIL capsule shall go off.

- If result is not as specified, replace rotor blade de-ice controller (PARA 12-4-32).

af. Place blade de-ice test panel test function switch to OAT.

ag. Place blade de-ice control panel POWER switch to TEST.

RESULT: MR DE-ICE FAIL and TR DE-ICE FAIL capsules shall go on.

- If MR DE-ICE FAIL capsule goes on but TR DE-ICE FAIL capsule does not, replace rotor blade de-ice controller (PARA 12-4-32).
- If TR DE-ICE FAIL capsule goes on but MR DE-ICE FAIL capsule does not, replace rotor blade de-ice controller (PARA 12-4-32).
- If neither capsule goes on with test function switch in OAT position, check continuity between J134-X and J134-W.
  - (a) If continuity is present, replace rotor blade de-ice controller (PARA 12-4-32).
  - (b) If continuity is not present, replace blade de-ice test panel (PARA 12-4-46).

ah. Place blade de-ice control panel POWER switch to OFF.

RESULT: MR DE-ICE FAIL and TR DE-ICE FAIL capsules shall go off.

- If result is not as specified, replace rotor blade de-ice controller (PARA 12-4-32).

ai. Place blade de-ice test panel test function switch to NORM.

aj. Start APU (Appropriate Flight Manual).

ak. With qualified personnel at controls, start engines and adjust rotor speed to 100% (Appropriate Flight Manual).

al. Place upper console GENERATORS - NO. 1 and NO. 2 switches to ON.

RESULT: #1 GEN and #2 GEN capsules shall be off.

- If result is not as specified, troubleshoot AC Electrical System (PARA 9-2-1).

CAUTION

To prevent rotor blades from overheating in ambient temperatures above 21°C (70°F), operate rotor at 100% for 5 minutes before doing de-ice EOT check. Do not do de-ice EOT check if FAT is above 38°C (100°F).

am. Place blade de-ice test panel test function switch to EOT.

an. Place blade de-ice control panel MODE switch to MANUAL-M.

NOTE

- PWR MAIN RTR lamp may flash at start of test.

- MR DE-ICE FAULT capsule may go on.

ao. Place blade de-ice control panel POWER switch to ON and simultaneously start instrument panel clock. **RECORD RESULT**.

RESULT:

(1) T/R DE-ICE FAIL legends shall go on 15 to 30 seconds after POWER switch is placed ON.

(2) MR DE-ICE FAIL capsule shall go on 50 to 70 seconds after POWER switch is placed ON.

- If TR DE-ICE FAIL capsule goes on, but MR DE-ICE FAIL capsule does not, check continuity between P325-W and P134-S.

  (a) If continuity is present, replace blade de-ice test panel (PARA 12-4-46).

  1 If trouble remains, replace rotor blade de-ice controller (PARA 12-4-32).

  (b) If continuity is not present, repair/replace wiring (Appendix F).

- If MR DE-ICE FAIL capsule goes on but TR DE-ICE FAIL capsule does not, check continuity between P325-V and P134-U.

- If neither capsule goes on, check continuity between P134-R and ground.

  (a) If continuity is present, replace blade de-ice test panel (PARA 12-4-46).

  1 If trouble remains, replace rotor blade de-ice controller (PARA 12-4-32).

  (b) If continuity is not present, repair/replace wiring (Appendix F).

ap. Place blade de-ice control panel POWER switch to OFF.

aq. Place blade de-ice test panel test function switch to NORM.

ar. Place blade de-ice control panel POWER switch to ON and wait 30 seconds.

RESULT: MR DE-ICE FAIL, TR DE-ICE FAIL, and MR DE-ICE FAULT capsules shall remain off.

- If MR DE-ICE FAIL capsule goes on, check for 115 VAC between MN RTR BUS CONT terminals L1, L2, L3, and ground.

  (a) If voltage is as specified, replace rotor blade de-ice controller (PARA 12-4-32).

  (b) If voltage is not as specified, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12).

- If TR DE-ICE FAIL capsule goes on, check for 115 VAC between TAIL ROTOR DE-ICE CONTACTOR terminals A1, B1, C1, and ground.

  (a) If voltage is as specified, replace rotor blade de-ice controller (PARA 12-4-32).

  (b) If voltage is not as specified, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11).

- If MR DE-ICE FAIL and TR DE-ICE FAIL capsules go on, go to TABLE NO. 12-2-5-17.

- If MR DE-ICE FAULT capsules go on, replace No. 2 generator contactor K2 (PARA 9-4-12).

as. Place blade de-ice control panel POWER switch to OFF.

at. Place upper console GENERATORS - NO. 1 switch to OFF and APU switch ON.

au. Place blade de-ice control panel POWER switch to ON and wait 30 seconds.

RESULT: MR DE-ICE FAIL, TR DE-ICE FAIL, and MR DE-ICE FAULT capsules shall remain off.

- If result is not as specified, go to TABLE NO. 12-2-5-18.

av. Place blade de-ice control panel POWER switch to OFF.

aw. Place upper console GENERATORS - NO. 1 switch to ON and NO. 2 switch to OFF.

ax. Place blade de-ice control panel POWER switch to ON, and wait 30 seconds.

RESULT: MR DE-ICE FAIL, TR DE-ICE FAIL, and MR DE-ICE FAULT capsules shall remain off.

- If result is not as specified, go to TABLE NO. 12-2-5-18.

ay. Place blade de-ice control panel POWER switch to OFF.

az. Place upper console GENERATORS - NO. 2 switch to ON and APU switch to OFF.

ba. Place blade de-ice control panel POWER switch to ON and wait 30 seconds.

RESULT: MR DE-ICE FAIL, TR DE-ICE FAIL, and MR DE-ICE FAULT capsules shall remain off.

- If result is not as specified, go to TABLE NO. 12-2-5-19.

bb. Place blade de-ice control panel POWER switch to OFF.

bc. Place upper console GENERATORS - NO. 2 switch to OFF.

CAUTION

To prevent damage to the blades, place blade de-ice control panel POWER switch to OFF if result of step bd. is not obtained within 30 seconds.

bd. Place blade de-ice control panel POWER switch to ON.

RESULT: MR DE-ICE FAIL and MR DE-ICE FAULT capsules shall go on immediately, and TR DE-ICE FAIL capsule shall go on after 30 seconds.

- If none of the capsules go on, go to TABLE NO. 12-2-5-20.

- If only MR capsules or TR capsule goes on, replace rotor blade de-ice controller (PARA 12-4-32).

be. Place blade de-ice control panel POWER switch to OFF.

bf. Place upper console GENERATORS - APU switch to ON and NO. 1 switch to OFF.

bg. Place blade de-ice control panel POWER switch to ON.

RESULT: MR DE-ICE FAIL and MR DE-ICE FAULT capsules shall go on immediately, and TR DE-ICE FAIL capsule shall go on after 20 seconds.

- If result is not as specified, go to TABLE NO. 12-2-5-20.

bh. Place blade de-ice control panel MODE switch to AUTO.

bi. Place blade de-ice control panel POWER switch to OFF.

bj. Shut down engines, rotor (Appropriate Flight Manual).

bk. Turn off electrical power (PARA 1-6-16).

NOTE

To obtain accurate resistance values of the rotor blade heater mat elements, a minimum of 1 ampere voltage current must be passed through blade heater mat elements for a minimum of 15 seconds or the blade de-ice operational/ troubleshooting procedure must be done.

bl. Select maximum resistance range on multimeter.

bm. Within one hour of operational/ troubleshooting procedure, check resistance between the following pins on J841, J842, J843, and J844:

1	and	2
2	and	3
1	and	3
4	and	5
5	and	6
4	and	6
7	and	8
8	and	9
7	and	9
10	and	11
11	and	12
10	and	12

RESULT: Resistance shall be between 7 and 10 ohms (Ω).

- If result is not as specified, check continuity between:
  P1001 and P841
  P1002 and P842
  P1003 and P843
  P1004 and P844

(a) If continuity is present, replace affected main rotor blade (PARA 5-4-31).

(b) If continuity is not present, replace affected harness (PARA 12-4-26).

bn. Within one hour of operational/ troubleshooting procedure, check resistance between the following pins on J611, J612, J613, and J614:

1	and	2
3	and	4
5	and	6

RESULT: Resistance shall be between 6 and 8 Ω.

- If result is not as specified, replace affected tail rotor blade (PARA 5-4-38).

TABLE NO. 12-2-5-1. Icing Rate Meter FAIL Flag In View And Meter Needle Did Not Deflect.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 VDC between P132-F and P132-G.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **6.**

2. Check for 115 VAC between P132-S and P132-G.

TABLE NO. 12-2-5-1. Icing Rate Meter FAIL Flag In View And Meter Needle Did Not Deflect. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **5.**

3. Check continuity between:
P132-A and P326-1
P132-B and P326-2
P132-C and P326-3
P132-D and P326-4
P132-E and P326-5
P132-T and P326-6

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **8.**

4. Check continuity between:
P326-8 and ground
P326-9 and ground

Step 1. If continuity is present, replace ice detector (PARA 12-4-50). Go to **8.**
Step 2. If trouble remains, replace icing rate meter (PARA 12-4-27). Go to **8.**
Step 3. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **8.**

5. Check for 115 VAC between terminal 1 (ICE-DET circuit breaker, NO. 2 PRI BUS AC - mission readiness circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P132-S (Appendix F). Go to **8.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

6. Check continuity between P132-G and ground.

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

7. Check for 28 VDC between terminal 1 (ICE-DET circuit breaker, NO. 2 PRI BUS DC - mission readiness circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between P132-F and terminal 1 (Appendix F). Go to **8.**

TABLE NO. 12-2-5-1. Icing Rate Meter FAIL Flag In View And Meter Needle Did Not Deflect. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. Procedure completed.

TABLE NO. 12-2-5-2. ICE DETECTED Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P117 (PARA 9-4-98). Go to 3.**

EMEP **3. Install good PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P117-63 and ground.

Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19). Go to **11.**
Step 2. If voltage is not as specified, go to **5.**

5. Check continuity between P117-63 and P132-J.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **11.**

6. Check for 28 VDC between P132-H and P132-G.

Step 1. If voltage is as specified, replace icing rate meter (PARA 12-4-27). Go to **11.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between P132-H and P133-T.

Step 1. If continuity is present, go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **11.**

TABLE NO. 12-2-5-2. ICE DETECTED Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Check for 28 VDC between P133-L and ground.

Step 1. If voltage is as specified, replace blade de-ice control panel (PARA 12-4-29). Go to **11.**
Step 2. If voltage is not as specified, go to **9.**

9. Check for 28 VDC between J906-N and ground.

Step 1. If voltage is as specified, repair/replace wiring between P906-N and P133-L (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, go to **10.**

10. Check for 28 VDC between terminal 1 (DE-ICE CNTRLR circuit breaker, NO. 2 PRI BUS DC - mission readiness circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and J906-N (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **11.**

11. Procedure completed.

TABLE NO. 12-2-5-3. Icing Rate Meter Needle Does Not Move.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **3.**

EMEP **2. Check for corresponding resistances between:**
J326-1 and J326-2 (Greater than 1000 Ω)
J326-3 and J326-4 (Greater than 1000 Ω)
J326-5 and J326-8 (4.9 to 5.9 Ω)
J326-6 and J326-9 (47 to 61 Ω)

TABLE NO. 12-2-5-3. Icing Rate Meter Needle Does Not Move. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If corresponding resistances are as specified, replace icing rate meter (PARA 12-4-27). Go to **4.**
Step 2. If corresponding resistances are not as specified, replace ice detector (PARA 12-4-50). Go to **4.**

W/O EMEP

3. Check for corresponding resistances between:
J326-1 and J326-2 (15 to 21 Ω)
J326-3 and J326-4 (15 to 21 Ω)
J326-5 and J326-8 (4.9 to 5.9 Ω)
J326-6 and J326-9 (47 to 61 Ω)

Step 1. If corresponding resistances are as specified, replace icing rate meter (PARA 12-4-27). Go to **4.**
Step 2. If corresponding resistances are not as specified, replace ice detector (PARA 12-4-50). Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-5-4. With Blade De-ice Control Panel Power Switch OFF, Caution/Advisory Panel MR DE-ICE FAIL, TR DE-ICE FAIL, Or MR DE-ICE FAULT Capsule Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. With power applied, pull out DE-ICE CNTRLR circuit breaker (NO. 2 PRI BUS DC - mission readiness circuit breaker panel). Go to 2.

2. Does caution/advisory capsule go off?

Step 1. If capsule is off, replace rotor blade de-ice controller (PARA 12-4-32). Go to **7.**
Step 2. If capsule is not off, go to **3.**

3. Check helicopter configuration.

Step 1. EMEP , go to **4.**
Step 2. W/O EMEP , go to **6.**

EMEP

4. Remove and check PFA P117 (PARA 9-4-98). Go to 5.

TABLE NO. 12-2-5-4. With Blade De-ice Control Panel Power Switch OFF, Caution/Advisory Panel MR DE-ICE FAIL, TR DE-ICE FAIL, Or MR DE-ICE FAULT Capsule Is On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

EMEP 5. **Install good PFA (PARA 9-4-98). Go to 6.**

6. **Check for 28 VDC between:**
   P117-15 and ground
   P117-31 and ground
   P117-47 and ground

   Step 1. If voltage is as specified, troubleshoot wiring for short circuit to 28 VDC (Appendix F). Go to **7.**
   Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to **7.**

7. **Procedure completed.**

TABLE NO. 12-2-5-5. PWR MAIN RTR Lamp Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check helicopter configuration.**

   Step 1. EMEP , go to **2.**
   Step 2. W/O EMEP , go to **4.**

EMEP 2. **Remove and check PFA P134 (PARA 9-4-98). Go to 3.**

EMEP 3. **Install good PFA (PARA 9-4-98). Go to 4.**

4. **Check for 13 VAC between:**
   P134-D and ground
   P134-E and ground
   P134-F and ground

   Step 1. If voltage is as specified, go to **5.**
   Step 2. If voltage is not as specified, replace blade de-ice test panel (PARA 12-4-46). Go to **6.**

TABLE NO. 12-2-5-5. PWR MAIN RTR Lamp Is On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

5. Check for 28 VDC between K60 main rotor contactor terminals X1 and X2.

Step 1. If voltage is as specified, replace K64 APU/generator interlock relay (PARA 12-4-37). Go to **6.**
Step 2. If voltage is not as specified, replace contactor (PARA 12-4-39). Go to **6.**

6. Procedure completed.

TABLE NO. 12-2-5-6. PWR TAIL RTR Lamp Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P134 (PARA 9-4-98). Go to 3.**

EMEP **3. Install good PFA (PARA 9-4-98). Go to 4.**

4. Check for 115 VAC between:
P134-A and ground
P134-B and ground
P134-C and ground

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace blade de-ice test panel (PARA 12-4-46). Go to **6.**

5. Check for 28 VDC between K61 tail rotor contactor terminals X1 and X2.

Step 1. If voltage is as specified, replace K64 APU/generator interlock relay (PARA 12-4-37). Go to **6.**
Step 2. If voltage is not as specified, replace contactor (PARA 12-4-36). Go to **6.**

6. Procedure completed.

TABLE NO. 12-2-5-7. PWR MAIN RTR Lamp Does Not Cycle On And Off And/Or PWR MAIN RTR Lamp Does Not Go On When Press-To-Test.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Momentarily press PWR MAIN RTR lamp. Go to 2.

2. Does PWR MAIN RTR lamp go on then off?

Step 1. If lamp did go on, then off, go to **3.**
Step 2. If lamp did not go on, then off, go to **9.**

3. Check helicopter configuration.

Step 1. EMEP , go to **4.**
Step 2. W/O EMEP , go to **6.**

EMEP **4. Remove and check PFA P134 (PARA 9-4-98). Go to 5.**

EMEP **5. Install good PFA (PARA 9-4-98). Go to 6.**

6. Check for 28 VDC between P134-H and P134-T on blade de-ice test panel.

Step 1. If voltage is as specified, replace blade de-ice test panel (PARA 12-4-46). Go to **10.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between P134-T and ground.

Step 1. If continuity is present, go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **10.**

8. Check continuity between P134-H and P902-h.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **10.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **10.**

9. Unscrew PWR MAIN RTR lamp cap and check for burnt-out lamp.

Step 1. If lamp is good, replace blade de-ice test panel (PARA 12-4-46). Go to **10.**
Step 2. If lamp is not good, replace lamp (PARA 12-4-47). Go to **10.**

TABLE NO. 12-2-5-7. PWR MAIN RTR Lamp Does Not Cycle On And Off And/Or PWR MAIN RTR Lamp Does Not Go On When Press-To-Test. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

10. Procedure completed.

TABLE NO. 12-2-5-8. TEST IN PROGRESS Lamp Does Not Cycle On And Off.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Unscrew lamp cap and check for burnt-out lamp.

Step 1. If lamp is good, go to **2.**
Step 2. If lamp is not good, replace lamp (PARA 12-4-30). Go to **5.**

2. Check for 28 VDC between P133-K and P133-D.

Step 1. If voltage is as specified, replace blade de-ice control panel (PARA 12-4-29). Go to **5.**
Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between P133-D and ground.

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **5.**

4. Check continuity between P133-K and P902-h.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **5.**

5. Procedure completed.

TABLE NO. 12-2-5-9. TEST IN PROGRESS LAMP Does Not Function As Specified.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check helicopter configuration.**

   Step 1. EMEP , go to **2.**
   Step 2. W/O EMEP , go to **4.**

EMEP 2. **Remove and check PFA P325 (PARA 9-4-98). Go to 3.**

EMEP 3. **Install good PFA (PARA 9-4-98). Go to 4.**

4. **Place POWER switch on blade de-ice control panel to TEST. Go to 5.**

5. **Check for 28 VDC between P325-Y and P325-H.**

   Step 1. If voltage is as specified, go to **6.**
   Step 2. If voltage is not as specified, go to **10.**

6. **Check continuity between P133-N and P325-T.**

   Step 1. If continuity is present, go to **7.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

7. **Check for 15 VDC between P133-N and ground.**

   Step 1. If voltage is as specified, go to **8.**
   Step 2. If voltage is not as specified, replace rotor blade de-ice controller (PARA 12-4-32). Go to **12.**

8. **Check for 28 VDC between P133-K and ground.**

   Step 1. If voltage is as specified, replace rotor blade de-ice controller (PARA 12-4-32). Go to **12.**
   Step 2. If voltage is not as specified, go to **9.**

9. **Check continuity between P133-K and P902-h.**

   Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **12.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

10. **Check continuity between P325-H and P133-U.**

    Step 1. If continuity is present, go to **11.**

TABLE NO. 12-2-5-9. TEST IN PROGRESS LAMP Does Not Function As Specified. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

11. Check for 28 VDC between P133-R and P133-D.

Step 1. If voltage is as specified, replace blade de-ice control panel (PARA 12-4-29). Go to **12.**
Step 2. If voltage is not as specified, repair/replace wiring between P133-R and J121-A (Appendix F). Go to **12.**

12. Procedure completed.

TABLE NO. 12-2-5-10. MR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect P841 and P1001. Go to 2.

2. Check continuity between:
P841-1 and connector shell
P841-2 and connector shell
P841-3 and connector shell
P841-4 and connector shell
P841-5 and connector shell
P841-6 and connector shell
P841-7 and connector shell
P841-8 and connector shell
P841-9 and connector shell
P841-10 and connector shell
P841-11 and connector shell
P841-12 and connector shell

Step 1. If continuity is present, replace harness (PARA 12-4-26). Go to **25.**
Step 2. If continuity is not present, go to **3.**

3. Check for less than 1 Ω resistance between:

TABLE NO. 12-2-5-10. MR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P841-1 and P1001-1
P841-2 and P1001-2
P841-3 and P1001-3
P841-5 and P1001-4
P841-6 and P1001-5
P841-8 and P1001-6
P841-9 and P1001-7
P841-11 and P1001-8
P841-12 and P1001-9

Step 1. If resistance is as specified, go to **4.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**

4. Check continuity between:
P841-1 and P841-4
P841-4 and P841-7
P841-7 and P841-10

Step 1. If continuity is present, reconnect P841. Go to **5.**
Step 2. If continuity is not present, replace harness (PARA 12-4-26). Go to **25.**

5. Disconnect P845. Go to 6.

6. Check for less than 1 Ω resistance between:
P845-1 and P1001-10
P845-2 and P1001-11

Step 1. If resistance is as specified, reconnect P845 and P1001. Go to **7.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**
Step 3. If trouble remains, replace droop stop heater pin (PARA 5-4-10). Go to **25.**

7. Disconnect P842 and P1002. Go to 8.

8. Check continuity between:

TABLE NO. 12-2-5-10. MR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P842-1 and connector shell
P842-2 and connector shell
P842-3 and connector shell
P842-4 and connector shell
P842-5 and connector shell
P842-6 and connector shell
P842-7 and connector shell
P842-8 and connector shell
P842-9 and connector shell
P842-10 and connector shell
P842-11 and connector shell
P842-12 and connector shell

Step 1. If continuity is present, replace harness (PARA 12-4-26). Go to **25.**
Step 2. If continuity is not present, go to **9.**

9. Check for less than 1 Ω resistance between:

P842-1 and P1002-1
P842-2 and P1002-2
P842-3 and P1002-3
P842-5 and P1002-4
P842-6 and P1002-5
P842-8 and P1002-6
P842-9 and P1002-7
P842-11 and P1002-8
P842-12 and P1002-9

Step 1. If resistance is as specified, go to **10.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**

10. Check continuity between:

P842-1 and P842-4
P842-4 and P842-7
P842-7 and P842-10

Step 1. If continuity is present, reconnect P842. Go to **11.**
Step 2. If continuity is not present, replace harness (PARA 12-4-26). Go to **25.**

11. Disconnect P846. Go to 12.

12. Check for less than 1 Ω resistance between:

TABLE NO. 12-2-5-10. MR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P846-1 and P1002-10
P846-2 and P1002-11

Step 1. If resistance is as specified, reconnect P846 and P1002. Go to **13.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**
Step 3. If trouble remains, replace droop stop heater pin (PARA 5-4-10). Go to **25.**

13. Disconnect P843 and P1003. Go to 14.

14. Check continuity between:

P843-1 and connector shell
P843-2 and connector shell
P843-3 and connector shell
P843-4 and connector shell
P843-5 and connector shell
P843-6 and connector shell
P843-7 and connector shell
P843-8 and connector shell
P843-9 and connector shell
P843-10 and connector shell
P843-11 and connector shell
P843-12 and connector shell

Step 1. If continuity is present, replace harness (PARA 12-4-26). Go to **25.**
Step 2. If continuity is not present, go to **15.**

15. Check for less than 1 Ω resistance between:

P843-1 and P1003-1
P843-2 and P1003-2
P843-3 and P1003-3
P843-5 and P1003-4
P843-6 and P1003-5
P843-8 and P1003-6
P843-9 and P1003-7
P843-11 and P1003-8
P843-12 and P1003-9

Step 1. If resistance is as specified, go to **16.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**

16. Check continuity between:

TABLE NO. 12-2-5-10. MR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P843-1 and P843-4
P843-4 and P843-7
P843-7 and P843-10

Step 1. If continuity is present, reconnect P843. Go to **17.**
Step 2. If continuity is not present, replace harness (PARA 12-4-26). Go to **25.**

17. Disconnect P847. Go to 18.

18. Check for less than 1 Ω resistance between:
P847-1 and P1003-10
P847-2 and P1003-11

Step 1. If resistance is as specified, reconnect P847 and P1003. Go to **19.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**
Step 3. If trouble remains, replace droop stop heater pin (PARA 5-4-10). Go to **25.**

19. Disconnect P844 and P1004. Go to 20.

20. Check continuity between:
P844-1 and connector shell
P844-2 and connector shell
P844-3 and connector shell
P844-4 and connector shell
P844-5 and connector shell
P844-6 and connector shell
P844-7 and connector shell
P844-8 and connector shell
P844-9 and connector shell
P844-10 and connector shell
P844-11 and connector shell
P844-12 and connector shell

Step 1. If continuity is present, replace harness (PARA 12-4-26). Go to **25.**
Step 2. If continuity is not present, go to **21.**

21. Check for less than 1 Ω resistance between:

TABLE NO. 12-2-5-10. MR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P844-1 and P1004-1
P844-2 and P1004-2
P844-3 and P1004-3
P844-5 and P1004-4
P844-6 and P1004-5
P844-8 and P1004-6
P844-9 and P1004-7
P844-11 and P1004-8
P844-12 and P1004-9

Step 1. If resistance is as specified, go to **22.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**

22. Check continuity between:
P844-1 and P844-4
P844-4 and P844-7
P844-7 and P844-10

Step 1. If continuity is present, reconnect P844. Go to **23.**
Step 2. If continuity is not present, replace harness (PARA 12-4-26). Go to **25.**

23. Disconnect P848. Go to 24.

24. Check for less than 1 Ω resistance between:
P848-1 and P1004-10
P848-2 and P1004-11

Step 1. If resistance is as specified, replace droop stop heater pin (PARA 5-4-10). Go to **25.**
Step 2. If resistance is not as specified, replace harness (PARA 12-4-26). Go to **25.**

25. Procedure completed.

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Does capsule go on immediately after placing blade de-ice control panel POWER switch to TEST?**

   Step 1. If capsule is on immediately, go to **2.**
   Step 2. If capsule is not on immediately, go to **9.**

2. **Check helicopter configuration.**

   Step 1. EMEP , go to **3.**
   Step 2. W/O EMEP , go to **5.**

EMEP 3. **Remove and check PFA P134 (PARA 9-4-98). Go to 4.**

EMEP 4. **Install good PFA (PARA 9-4-98). Go to 5.**

5. **Check for resistance proportional to outside air temperature between P134-Y and P134-Z.**

   Step 1. If resistance is as specified, go to **6.**
   Step 2. If resistance is not as specified, go to **8.**

6. **Check for resistance between 7 and 10 Ω between the following pins on J841, J842, J843, and J844:**

   1 and 2
   2 and 3
   1 and 3
   4 and 5
   5 and 6
   4 and 6
   7 and 8
   8 and 9
   7 and 9
   10 and 11
   11 and 12
   10 and 12

   Step 1. If resistance is as specified, go to **7.**
   Step 2. If resistance is not as specified, replace malfunctioning main rotor blade (PARA 5-4-31). Go to **87.**

7. **Disconnect P325 from rotor blade de-ice controller. Does capsule go off?**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If capsule is off, replace rotor blade de-ice controller (PARA 12-4-32). Go to **87.**

Step 2. If capsule is not off, replace blade de-ice test panel (PARA 12-4-46). Go to **87.**

8. Check continuity between:

P122-A and P134-Y
P122-B and P134-Z
P122-C and P134-Z

Step 1. If continuity is present, replace OAT sensor (PARA 12-4-49). Go to **87.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **87.**

9. Does PWR MAIN RTR lamp flash and MR DE-ICE FAIL capsule go on in 6 to 80 seconds or greater than 120 seconds after blade de-ice control panel POWER switch is placed to TEST?

Step 1. If PWR MAIN RTR lamp flashes and MR DE-ICE FAIL capsule is on, replace rotor blade de-ice controller (PARA 12-4-32). Go to **87.**

Step 2. If PWR MAIN RTR lamp does not flash and MR DE-ICE FAIL capsule is not on, go to **10.**

10. Place blade de-ice control panel POWER switch to OFF. Go to 11.

11. Turn off electrical power (PARA 1-6-16). Go to 12.

12. Disconnect wire H6373A20 from TB3-3. Go to 13.

13. Connect test harness P1 to DC utility receptacle and P2 to AC utility receptacle. Go to 14.

14. Connect DC alligator clip to TB3-3 and AC alligator clip to TB3-5. Go to 15.

15. Connect the negative lead of a multimeter, set on the 50 VDC scale, to TB3-1 and the multimeter positive lead to ground. Go to 16.

16. Place test harness switch to OFF. Go to 17.

17. Disconnect P841, P842, P843, and P844 from rotor blades. Go to 18.

18. Turn on electrical power (PARA 1-6-16). Go to 19.

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION

19. Place test harness switch alternately to ON and OFF. Go to 20.

20. Multimeter shall indicate alternately between 0 and -30 VDC. Go to 21.

21. After a maximum of 8 on-off pulses of test harness switch, does multimeter indicate between -7 and -3 volts?

Step 1. If multimeter is as specified, go to **22.**
Step 2. If multimeter is not as specified, go to **42.**

22. Place test harness switch alternately to ON and OFF 7 times. Go to 23.

23. Voltage at TB3-1 shall be 0 VDC. Go to 24.

24. Turn off electrical power (PARA 1-6-16). Go to 25.

25. Disconnect test harness. Go to 26.

26. Connect wire H6373A20 to TB3-3. Go to 27.

27. Connect P841, P842, P843, and P844 to blades. Go to 28.

28. Turn on electrical power (PARA 1-6-16). Go to 29.

29. Place blade de-ice control panel POWER switch to TEST. Go to 30.

30. Check that multimeter at TB3-1 indicates alternately 0 and -30 VDC.

Step 1. If multimeter indicates alternately 0 VDC and -30 VDC, go to **31.**
Step 2. If multimeter does not indicate alternately 0 VDC and -30 VDC, go to **52.**

31. Place blade de-ice control panel POWER switch to OFF. Go to 32.

32. Turn off electrical power (PARA 1-6-16). Go to 33.

33. Place clamp-on ammeter around øA wire at TB3-2. Go to 34.

34. Turn on electrical power (PARA 1-6-16). Go to 35.

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

35. Monitor ammeter as blade de-ice control panel POWER switch is placed to TEST. Go to 36.

36. Check that current pulses are indicated on ammeter.

Step 1. If pulses are indicated, replace rotor blade de-ice controller (PARA 12-4-32). Go to **87.**

Step 2. If pulses are not indicated, go to **37.**

37. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 38.

38. Check for 115 VAC between:

TB3-2 and ground
TB3-4 and ground

Step 1. If voltage is as specified, go to **39.**

Step 2. If voltage is not as specified, go to **44.**

39. Place blade de-ice control panel POWER switch to OFF. Go to 40.

40. Turn off electrical power (PARA 1-6-16). Go to 41.

41. Check for resistance of 7 to 10 Ω between the following pins of J841, J842, J843, and J844:

1 and 2
2 and 3
1 and 3
4 and 5
5 and 6
4 and 6
7 and 8
8 and 9
7 and 9
10 and 11
11 and 12
10 and 12

Step 1. If resistance is as specified, go to **42.**

Step 2. If resistance is not as specified, replace malfunctioning main rotor blade (PARA 5-4-31). Go to **87.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION

42. Remove distributor and slipring from helicopter (PARA 12-4-26). Go to 43.

43. Check continuity between the following for slipring input wire and slipring output terminal strip respectively:

S1 and S1
S2 and S2
S3 and S3
øA and øA
øB and øB
øC and øC

Step 1. If continuity is present, replace main rotor blade de-ice distributor (PARA 12-4-26). Go to **87.**
Step 2. If continuity is not present, replace slipring brush blocks (PARA 12-4-26). Go to **87.**
Step 3. If trouble remains, replace harness (PARA 12-4-26), as required. Go to **87.**
Step 4. If trouble still remains, replace slipring assembly (PARA 12-4-26). Go to **87.**

44. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 45.

45. Check for 115 VAC between K60 main rotor contactor:

Terminal A1 and ground
Terminal B1 and ground

Step 1. If voltage is as specified, replace current transformer (PARA 12-4-43). Go to **87.**
Step 2. If voltage is not as specified, go to **46.**

46. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 47.

47. Check for 115 VAC between K60 main rotor contactor:

Terminal A2 and ground
Terminal B2 and ground

Step 1. If voltage is as specified, replace K60 main rotor contactor (PARA 12-4-39). Go to **87.**
Step 2. If voltage is not as specified, go to **48.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

48. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 49.

49. Check for 115 VAC between K62 main rotor de-ice bus contactor:
Terminal T1 and ground
Terminal T2 and ground

Step 1. If voltage is as specified, replace current limiter either CL10 or CL11, or both (PARA 12-4-34). Go to **87.**
Step 2. If voltage is not as specified, go to **50.**

50. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 51.

51. Check for 115 VAC between K62 main rotor de-ice bus contactor:
Terminal L1 and ground
Terminal L2 and ground

Step 1. If voltage is as specified, replace contactor (PARA 12-4-40). Go to **87.**
Step 2. If voltage is not as specified, replace current limiters CL7 and CL8 (PARA 12-4-34). Go to **87.**

52. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 53.

53. Check for 115 VAC between:
TB3-2 and ground
TB3-4 and ground
TB3-5 and ground

Step 1. If voltage is as specified, go to **54.**
Step 2. If voltage is not as specified, go to **60.**

54. Check helicopter configuration.

Step 1. **EMEP** , go to **55.**
Step 2. **W/O EMEP** , go to **57.**

EMEP **55. Remove and check PFA P325 (PARA 9-4-98). Go to 56.**

EMEP **56. Install good PFA (PARA 9-4-98). Go to 57.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

57. Check continuity between P325-a and TB3-3.

Step 1. If continuity is present, replace rotor blade de-ice controller (PARA 12-4-32). Go to **87.**
Step 2. If continuity is not present, go to **58.**

58. Check continuity between P325-J and P134-P.

Step 1. If continuity is present, go to **59.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

59. Check continuity between P134-N and TB3-1.

Step 1. If continuity is present, replace blade de-ice test panel (PARA 12-4-46). Go to **87.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

60. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 61.

61. Check for 115 VAC between K60 main rotor contactor:
Terminal A1 and ground
Terminal B1 and ground
Terminal C1 and ground

Step 1. If voltage is as specified, replace current transformer (PARA 12-4-43). Go to **87.**
Step 2. If voltage is not as specified, go to **62.**

62. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 63.

63. Check for 115 VAC between K60 main rotor contactor:
Terminal A2 and ground
Terminal B2 and ground
Terminal C2 and ground

Step 1. If voltage is as specified, go to **64.**
Step 2. If voltage is not as specified, go to **82.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

64. Check for 28 VDC between K60 main rotor contactor terminals X1 and X2.

Step 1. If voltage is as specified, replace contactor (PARA 12-4-39). Go to **87.**
Step 2. If voltage is not as specified, go to **65.**

65. Check continuity between K60 main rotor contactor terminal X2 and ground.

Step 1. If continuity is present, go to **66.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

66. Check continuity between K64 APU/generator interlock relay terminal A1 and K60 main rotor contactor terminal X1.

Step 1. If continuity is present, go to **67.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

67. Check for 28 VDC between K64 APU/generator interlock relay terminal A2 and ground.

Step 1. If voltage is as specified, go to **68.**
Step 2. If voltage is not as specified, go to **77.**

68. Check for 28 VDC between K64 APU/generator interlock relay terminals X1 and X2.

Step 1. If voltage is as specified, replace relay (PARA 12-4-37). Go to **87.**
Step 2. If voltage is not as specified, go to **69.**

69. Check continuity between K64 APU/generator interlock relay terminal X2 and ground.

Step 1. If continuity is present, go to **70.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

70. Check for 28 VDC between K65 backup pump interlock relay terminal A3 and ground.

Step 1. If voltage is as specified, repair/replace wiring (Appendix F). Go to **87.**
Step 2. If voltage is not as specified, go to **71.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

71. Check for 28 VDC between K65 backup pump interlock relay terminal A2 and ground.

Step 1. If voltage is as specified, go to **72.**
Step 2. If voltage is not as specified, go to **75.**

72. Check for 28 VDC between K65 backup pump interlock relay terminals X1 and X2.

Step 1. If voltage is as specified, go to **73.**
Step 2. If voltage is not as specified, replace relay (PARA 12-4-38). Go to **87.**

73. Check continuity between K65 backup pump interlock relay terminal X1 and J329-a.

Step 1. If continuity is present, go to **74.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

74. Check for 28 VDC between P329-a and ground.

Step 1. If voltage is as specified, troubleshoot AC Electrical System (PARA 9-2-1). Go to **87.**
Step 2. If voltage is not as specified, replace main rotor de-ice junction box (PARA 12-4-44). Go to **87.**

75. Check continuity between K65 backup pump interlock relay terminal A2 and J329-Z.

Step 1. If continuity is present, go to **76.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

76. Check continuity between P133-U and P329-Z.

Step 1. If continuity is present, replace blade de-ice control panel (PARA 12-4-29). Go to **87.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

77. Check continuity between K64 APU/generator interlock relay terminal A2 and J329-M.

Step 1. If continuity is present, repair/replace wiring (Appendix F). Go to **87.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, go to **78.**

78. Check helicopter configuration.

Step 1. EMEP , go to **79.**
Step 2. W/O EMEP , go to **81.**

EMEP **79. Remove and check PFA from P325 (PARA 9-4-98). Go to 80.**

EMEP **80. Install good PFA (PARA 9-4-98). Go to 81.**

81. Check continuity between P329-M and P325-h.

Step 1. If continuity is present, replace rotor blade de-ice controller (PARA 12-4-32). Go to **87.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

82. Place blade de-ice control panel POWER switch to OFF and then ON. Go to 83.

83. Check for 115 VAC between K62 main rotor de-ice bus contactor:
Terminal T1 and ground
Terminal T2 and ground
Terminal T3 and ground

Step 1. If voltage is as specified, replace current limiters CL10, CL11, and CL12 (PARA 12-4-34). Go to **87.**
Step 2. If trouble remains, repair/replace wiring (Appendix F), as required between the following, then go to **87:**
K62-T1 and K60-A2
K62-T2 and K60-B2
K62-T3 and K60-C2
Step 3. If voltage is not as specified, go to **84.**

84. Check for 115 VAC between K62 main rotor de-ice bus contactor:
Terminal L1 and ground
Terminal L2 and ground
Terminal L3 and ground

Step 1. If voltage is as specified, go to **85.**
Step 2. If voltage is not as specified, replace current limiters CL7, CL8, and CL9 (PARA 12-4-34). Go to **87.**

TABLE NO. 12-2-5-11. MR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

85. Check for 28 VDC between K62 main rotor de-ice bus contactor terminals X1 and X2.

Step 1. If voltage is as specified, replace contactor (PARA 12-4-40). Go to **87.**
Step 2. If voltage is not as specified, go to **86.**

86. Check continuity between K62 main rotor de-ice bus contactor terminal X2 and ground.

Step 1. If continuity is present, repair/replace wiring between J333-A and contactor terminal X1 (Appendix F). Go to **87.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **87.**

87. Procedure completed.

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does TR DE-ICE FAIL capsule go on immediately after placing blade de-ice control panel POWER switch to TEST?

Step 1. If capsule goes on, go to **2.**
Step 2. If capsule does not go on, go to **6.**

2. Check helicopter configuration.

Step 1. EMEP , go to **3.**
Step 2. W/O EMEP , go to **5.**

EMEP **3. Remove and check PFA P325 (PARA 9-4-98). Go to 4.**

EMEP **4. Install good PFA (PARA 9-4-98). Go to 5.**

5. Disconnect P325 from rotor de-ice controller. Does TR DE-ICE FAIL capsule go off?

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If capsule is off, replace rotor blade de-ice controller (PARA 12-4-32). Go to **53.**
Step 2. If capsule is not off, replace blade de-ice test panel (PARA 12-4-46). Go to **53.**

6. Connect P325 to rotor de-ice controller. Go to 7.

7. Does PWR-TAIL RTR lamp flash and TR DE-ICE FAIL capsule go on in less than 75 seconds or more than 120 seconds?

Step 1. If lamp flashes and capsule is on in less than 75 seconds or more than 120 seconds, replace blade de-ice test panel (PARA 12-4-46). Go to **53.**
Step 2. If lamp does not flash and capsule is not on in less than 75 seconds or more than 120 seconds, go to **8.**

8. Turn off electrical power (PARA 1-6-16). Go to 9.

9. Place 20 amp clamp-on ammeter around wire H6389A14 connected to K61 tail rotor contactor terminal A2. Go to 10.

10. Turn on electrical power (PARA 1-6-16). Go to 11.

11. Place blade de-ice control panel POWER switch to TEST. Go to 12.

12. Check that between 75 to 105 seconds after blade de-ice control panel POWER switch is placed to TEST, a current pulse occurs.

Step 1. If pulse is present, replace rotor blade de-ice controller (PARA 12-4-32). Go to **53.**
Step 2. If pulse is not present, go to **13.**

13. Place blade de-ice control panel POWER switch to OFF. Go to 14.

14. Turn off electrical power (PARA 1-6-16). Go to 15.

15. Disconnect P611, P612, P613, and P614. Go to 16.

16. Turn on electrical power (PARA 1-6-16). Go to 17.

17. Place blade de-ice control panel POWER switch to OFF and then TEST. Go to 18.

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

18. Check for 115 VAC between K61 tail rotor contactor:
Terminal A2 and ground
Terminal B2 and ground
Terminal C2 and ground

Step 1. If voltage is as specified, go to **19.**
Step 2. If voltage is not as specified, go to **37.**

19. Check helicopter configuration.

Step 1. EMEP , go to **20.**
Step 2. W/O EMEP , go to **22.**

EMEP **20. Remove and check PFA P324 (PARA 9-4-98). Go to 21.**

EMEP **21. Install good PFA (PARA 9-4-98). Go to 22.**

22. Check continuity between K61 tail rotor contactor:
Terminal A2 and P324-A
Terminal B2 and P324-B
Terminal C2 and P324-C

Step 1. If continuity is present, go to **23.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **53.**

23. Disconnect P134. Go to 24.

24. Check for resistance of 16 to 22 Ω between:
P324-D and P324-E
P324-E and P324-H
P324-D and P324-H

Step 1. If resistance is as specified, replace rotor blade de-ice controller (PARA 12-4-32). Go to **53.**
Step 2. If resistance is not as specified, go to **25.**

25. Check continuity between:
P324-D and P327-1
P324-E and P327-2
P324-H and P327-3

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **26.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **53.**

26. Connect P134. Go to 27.

27. Check for resistance of 6 to 9 Ω between:
J611-1 and J611-2
J611-3 and J611-4
J611-5 and J611-6
J612-1 and J612-2
J612-3 and J612-4
J612-5 and J612-6
J613-1 and J613-2
J613-3 and J613-4
J613-5 and J613-6
J614-1 and J614-2
J614-3 and J614-4
J614-5 and J614-6

Step 1. If resistance is as specified, go to **28.**
Step 2. If resistance is not as specified, replace malfunctioning tail rotor blade (PARA 5-4-38). Go to **53.**

28. Check continuity between tail rotor brush assembly connector pin 1 and P614-2.

Step 1. If continuity is present, go to **29.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

29. Check continuity between P611-3 and P614-2.

Step 1. If continuity is present, go to **30.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**
Step 3. If trouble remains, replace tail rotor slipring rotor (PARA 12-4-52). Go to **53.**

30. Check continuity between tail rotor brush assembly connector pin 2 and P611-1.

Step 1. If continuity is present, go to **31.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

31. Check continuity between P611-1 and P613-6.

Step 1. If continuity is present, go to **32.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**
Step 3. If trouble remains, replace tail rotor slipring rotor (PARA 12-4-52). Go to **53.**

32. Check continuity between tail rotor brush assembly connector pin 3 and P614-4.

Step 1. If continuity is present, go to **33.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**
Step 3. If trouble remains, replace tail rotor slipring rotor (PARA 12-4-52). Go to **53.**

33. Check continuity between P614-4 and P614-5.

Step 1. If continuity is present, go to **34**.
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

34. Check continuity between:
P611-2 and P612-1
P611-4 and P612-3
P611-5 and P614-6
P611-6 and P612-5
P612-2 and P613-1
P612-4 and P613-3
P612-6 and P613-5
P613-2 and P614-1
P613-4 and P614-3

Step 1. If continuity is present, go to **35.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **53.**
Step 3. If trouble remains, replace tail rotor slipring brush assembly (PARA 12-4-54). Go to **53.**

35. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 36.

36. Check for 115 VAC between K61 tail rotor contactor:

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Terminal A1 and ground
Terminal B1 and ground
Terminal C1 and ground

Step 1. If voltage is as specified, go to **37.**
Step 2. If voltage is not as specified, go to **48.**

37. Check for 28 VDC between K61 tail rotor contactor terminals X1 and X2.

Step 1. If voltage is as specified, replace contactor (PARA 12-4-36). Go to **53.**
Step 2. If voltage is not as specified, go to **38.**

38. Check continuity between K61-X2 and ground.

Step 1. If continuity is present, go to **39.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

39. Check continuity between P329-V and K61-X1.

Step 1. If continuity is present, go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

40. Check continuity between K64 APU/generator interlock relay terminal B1 and J329-V.

Step 1. If continuity is present, go to **41.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

41. Place blade de-ice control panel POWER switch to OFF then to TEST. Go to 42.

42. Check for 28 VDC between K64 APU/generator interlock relay terminal B2 and ground.

Step 1. If voltage is as specified, replace relay (PARA 12-4-37). Go to **53.**
Step 2. If voltage is not as specified, go to **43.**

43. Check continuity between K64 APU/generator interlock relay terminal B2 and J329-U.

Step 1. If continuity is present, go to **44.**

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

44. Check helicopter configuration.

Step 1. EMEP , go to **45.**
Step 2. W/O EMEP , go to **47.**

EMEP **45. Remove and check PFA P325 (PARA 9-4-98). Go to 46.**

EMEP **46. Install good PFA (PARA 9-4-98). Go to 47.**

47. Check continuity between P329-U and P325-j.

Step 1. If continuity is present, replace rotor blade de-ice controller (PARA 12-4-32). Go to **53.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

48. Check for 115 VAC between K63 tail rotor de-ice contactor:
Terminal A2 and ground
Terminal B2 and ground
Terminal C2 and ground

Step 1. If voltage is as specified, replace current limiters CL13, CL14, and CL15 (PARA 12-4-34). Go to **53.**
Step 2. If voltage is not as specified, go to **49.**

49. Check for 115 VAC between K63 tail rotor de-ice contactor:
Terminal A1 and ground
Terminal B1 and ground
Terminal C1 and ground

Step 1. If voltage is as specified, go to **50.**
Step 2. If voltage is not as specified, go to **52.**

50. Check for 28 VDC between K63 tail rotor de-ice contactor terminals X1 and X2.

Step 1. If voltage is as specified, replace contactor (PARA 12-4-41). Go to **53.**
Step 2. If voltage is not as specified, go to **51.**

51. Check continuity between K63 tail rotor de-ice contactor terminal X2 and ground.

TABLE NO. 12-2-5-12. TR DE-ICE FAIL Capsule Goes On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION

Step 1. If continuity is present, repair/replace wiring between J333-A and contactor terminal X1 (Appendix F). Go to **53.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **53.**

52. Check for 115 VAC between (DE-ICE PWR TAIL ROTOR circuit breaker, NO. 1 PRI BUS AC - mission readiness circuit breaker panel):

Terminal A1 and ground
Terminal B1 and ground
Terminal C1 and ground

Step 1. If voltage is as specified, repair/replace wiring (Appendix F), as required between the following, then go to **53:**

Terminal A1 and K61-C1
Terminal B1 and K61-B1
Terminal C1 and K61-A1

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **53.**

53. Procedure completed.

TABLE NO. 12-2-5-13. MR DE-ICE FAIL And TR DE-ICE FAIL Capsules Are Both On.

TEST OR INSPECTION CORRECTIVE ACTION

1. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 2.

2. Check for 28 VDC between P333-A and ground.

Step 1. If voltage is as specified, repair/replace wiring (Appendix F), as required between the following, then go to **19:**

J333-A and K62-X1
J333-A and K63-X1

Step 2. If voltage is not as specified, go to **3.**

3. Check continuity between P333-A and P329-X.

TABLE NO. 12-2-5-13. MR DE-ICE FAIL And TR DE-ICE FAIL Capsules Are Both On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

4. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 5.

5. Check for 28 VDC between K64 APU/generator interlock relay terminal C1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between relay terminal C1 and J329-X (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, go to **6.**

6. Check for 28 VDC between K64 APU/generator interlock relay terminal C2 and ground.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **12.**

7. Check for 28 VDC between K64 APU/generator interlock relay terminals X1 and X2.

Step 1. If voltage is as specified, replace relay (PARA 12-4-37). Go to **19.**
Step 2. If voltage is not as specified, go to **8.**

8. Check continuity between K64 APU/generator interlock relay terminal X2 and ground.

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

9. Check for 28 VDC between main rotor blade de-ice junction box terminal E4 and ground.

Step 1. If voltage is as specified, replace diode CR15 (PARA 12-4-45). Go to **19.**
Step 2. If voltage is not as specified, go to **10.**

10. Check continuity between main rotor blade de-ice junction box terminal E4 and J329-Z.

TABLE NO. 12-2-5-13. MR DE-ICE FAIL And TR DE-ICE FAIL Capsules Are Both On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, repair/replace wiring between P329-Z and P133-U (Appendix F). Go to **19.**
Step 2. If continuity is not present, go to **11.**

11. Check continuity between main rotor blade de-ice junction box terminal E4 and backup pump interlock relay terminal A3.

Step 1. If continuity is present, replace relay (PARA 12-4-38). Go to **19.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

12. Check for 28 VDC between P329-W and ground.

Step 1. If voltage is as specified, repair/replace wiring between K64 APU/generator interlock relay terminal C2 and J329-W (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, go to **13.**

13. Check continuity between P329-W and P212-H.

Step 1. If continuity is present, go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

14. Check for 28 VDC between P212-J and ground.

Step 1. If voltage is as specified, replace K1 No. 1 generator line contactor (PARA 9-4-11). Go to **19.**
Step 2. If voltage is not as specified, go to **15.**

15. Check continuity between P212-J and P204-J.

Step 1. If continuity is present, go to **16.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

16. Check for 28 VDC between P204-H and ground.

Step 1. If voltage is as specified, replace K2 No. 2 generator line contactor (PARA 9-4-12). Go to **19.**
Step 2. If voltage is not as specified, go to **17.**

17. Check for 28 VDC between J107-K and ground.

TABLE NO. 12-2-5-13. MR DE-ICE FAIL And TR DE-ICE FAIL Capsules Are Both On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between P107-K and P204-H (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, go to **18.**

18. Check continuity between J107-K and terminal 1 (APU GEN CONTR circuit breaker, BATT BUS - lower console circuit breaker panel).

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **19.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

19. Procedure completed.

TABLE NO. 12-2-5-14. MR DE-ICE FAULT Capsule Goes On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does MR DE-ICE FAULT capsule go on immediately after blade de-ice control panel POWER switch placed to TEST?

Step 1. If capsule goes on, replace rotor blade de-ice controller (PARA 12-4-32). Go to **18.**
Step 2. If capsule does not go on, go to **2.**

2. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 3.

3. Check for 115 VAC between:
TB3-2 and ground
TB3-4 and ground

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **11.**

4. Check continuity between:
T14-T1 and P325-K
T14-T2 and P325-L
T14-T3 and P325-N

TABLE NO. 12-2-5-14. MR DE-ICE FAULT Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **18.**

5. Check for resistance of 85 to 95 Ω between:
T14-T1 and T14-T4
T14-T2 and T14-T5
T14-T3 and T14-T6

Step 1. If resistance is as specified, go to **6.**
Step 2. If resistance is not as specified, replace transformer T14 (PARA 12-4-43). Go to **18.**

6. Place blade de-ice control panel POWER switch to OFF. Go to 7.

7. Turn off electrical power (PARA 1-6-16). Go to 8.

8. Check for resistance of 7 to 10 Ω between the following pins of J841, J842, J843, and J844:
1 and 2
2 and 3
1 and 3
4 and 5
5 and 6
4 and 6
7 and 8
8 and 9
7 and 9
10 and 11
11 and 12
10 and 12

Step 1. If resistance is as specified, go to **9.**
Step 2. If resistance is not as specified, replace malfunctioning main rotor blade (PARA 5-4-31). Go to **18.**

9. Remove distributor and slipring from helicopter (PARA 12-4-26). Go to 10.

10. Check continuity between the following for slipring input wire and slipring output terminal strip respectively:

TABLE NO. 12-2-5-14. MR DE-ICE FAULT Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

S1 and S1
S2 and S2
S3 and S3
øA and øA
øB and øB
øC and øC

Step 1. If continuity is present, replace main rotor blade de-ice distributor (PARA 12-4-26). Go to **18.**
Step 2. If continuity is not present, replace slipring brush blocks (PARA 12-4-26). Go to **18.**
Step 3. If trouble remains, replace harness (PARA 12-4-26), as required. Go to **18.**
Step 4. If trouble still remains, replace slipring (PARA 12-4-26). Go to **18.**

11. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 12.

12. Check for 115 VAC between K60 main rotor contactor:
Terminal A1 and ground
Terminal B1 and ground

Step 1. If voltage is as specified, replace transformer T14 (PARA 12-4-43). Go to **18.**
Step 2. If voltage is not as specified, go to **13.**

13. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 14.

14. Check for 115 VAC between K60 main rotor contactor:
Terminal A2 and ground
Terminal B2 and ground

Step 1. If voltage is as specified, replace contactor (PARA 12-4-39). Go to **18.**
Step 2. If voltage is not as specified, go to **15.**

15. Place blade de-ice control panel POWER switch to OFF and then to TEST. Go to 16.

16. Check for 115 VAC between K62 main rotor de-ice bus contactor:
Terminal T1 and ground
Terminal T2 and ground

TABLE NO. 12-2-5-14. MR DE-ICE FAULT Capsule Goes On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring (Appendix F), as required between the following, then go to **18:**
K62-T1 and K62-A2
K62-T2 and K62-B2

Step 2. If voltage is not as specified, go to **17.**

17. Check for 115 VAC between K62 main rotor de-ice bus contactor:
Terminal L1 and ground
Terminal L2 and ground

Step 1. If voltage is as specified, replace contactor (PARA 12-4-40). Go to **18.**

Step 2. If voltage is not as specified, replace current limiter CL7 or CL8 (PARA 12-4-34), as required. Go to **18.**

18. Procedure completed.

TABLE NO. 12-2-5-15. PWR MAIN RTR And PWR TAIL RTR Lamps Do Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect P324 and P325. Go to 2.

2. Check helicopter configuration.

Step 1. EMEP , go to **3.**

Step 2. W/O EMEP , go to **5.**

EMEP **3. Remove and check PFA P134 (PARA 9-4-98). Go to 4.**

EMEP **4. Install good PFA (PARA 9-4-98). Go to 5.**

5. Check for 28 VDC between P134-M and ground.

Step 1. If voltage is as specified, replace blade de-ice test panel (PARA 12-4-46). Go to **9.**

Step 2. If voltage is not as specified, go to **6.**

6. Check continuity between P133-T and P134-M.

TABLE NO. 12-2-5-15. PWR MAIN RTR And PWR TAIL RTR Lamps Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

7. Check continuity between J133-L and J133-T.

Step 1. If continuity is present, go to **8.**
Step 2. If continuity is not present, replace blade de-ice control panel (PARA 12-4-29). Go to **9.**

8. Check for 28 VDC between terminal 1 (DE-ICE CNTRLR circuit breaker, NO. 2 PRI BUS DC - mission readiness circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P133-L (Appendix F). Go to **9.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **9.**

9. Procedure completed.

TABLE NO. 12-2-5-16. Droop Stop Cams Are Not Warm When Blade De-ice Control Panel POWER Switch Is In TEST.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **5.**

EMEP **2. Is more than one droop stop heater cool?**

Step 1. If more than one droop stop heater is cool, go to **4.**
Step 2. If no more than one droop stop heater is cool, go to **3.**

EMEP **3. Check for 115 VAC between pins 1 and 2 of affected droop stop heater.**

Step 1. If voltage is as specified, replace affected droop stop heater pin assembly (PARA 5-4-10). Go to **7.**

TABLE NO. 12-2-5-16. Droop Stop Cams Are Not Warm When Blade De-ice Control Panel POWER Switch Is In TEST. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace harness (PARA 12-4-26). Go to **4.**

EMEP

4. Check continuity between:

P845-1 and P1001-10
P845-2 and P1001-11
P846-1 and P1002-10
P846-2 and P1002-11
P847-1 and P1003-10
P847-2 and P1003-11
P848-1 and P1004-10
P848-2 and P1004-11

Step 1. If continuity is present, replace main rotor blade de-ice distributor (PARA 12-4-26). Go to **7.**
Step 2. If continuity is not present, replace harness (PARA 12-4-26), as required. Go to **7.**

5. Is more than one droop stop heater cool?

Step 1. If more than one droop stop heater is cool, replace main rotor blade de-ice distributor (PARA 12-4-26). Go to **7.**
Step 2. If not more than one droop stop heater is cool, go to **6.**

6. Check for 115 VAC between pins 1 and 2 of affected droop stop heater.

Step 1. If voltage is as specified, replace affected droop stop heater pin assembly (PARA 5-4-10). Go to **7.**
Step 2. If voltage is not as specified, replace main rotor blade de-ice distributor (PARA 12-4-26). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-5-17. MR DE-ICE FAIL And TR DE-ICE FAIL Capsules Are On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. With blade de-ice control panel POWER switch at ON, check for 28 VDC between P329-Y and ground.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **11.**

2. Check for 28 VDC between main rotor blade de-ice junction box terminal E6 and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, repair/replace wiring between terminal E6 and J329-Y (Appendix F). Go to **12.**

3. Check helicopter configuration.

Step 1. **EMEP** , go to **4.**
Step 2. **W/O EMEP** , go to **6.**

EMEP **4. Remove and check PFA P325 (PARA 9-4-98). Go to 5.**

EMEP **5. Install good PFA (PARA 9-4-98). Go to 6.**

6. With GENERATORS NO. 1 and NO. 2 switches ON and rotor at 100%, check for 28 VDC between P325-P and ground.

Step 1. If voltage is as specified, replace rotor blade de-ice controller (PARA 12-4-32). Go to **12.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between P205-F and P325-P.

Step 1. If continuity is present, go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

8. Check for 28 VDC between P205-E and ground.

Step 1. If voltage is as specified, replace K3 APU/EXT PWR contactor (PARA 9-4-17). Go to **12.**
Step 2. If voltage is not as specified, go to **9.**

9. Check continuity between P205-E and P206-L.

TABLE NO. 12-2-5-17. MR DE-ICE FAIL And TR DE-ICE FAIL Capsules Are On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **10.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

10. Check for 28 VDC between P206-M and ground.

Step 1. If voltage is as specified, replace K4 ac bus tie contactor (PARA 9-4-21). Go to **12.**
Step 2. If voltage is not as specified, repair/replace wiring between P206-M and terminal 1 (DE-ICE CNTRLR circuit breaker, NO. 2 PRI BUS DC - mission readiness circuit breaker panel) (Appendix F). Go to **12.**

11. Check continuity between P329-Y and P133-J.

Step 1. If continuity is present, replace blade de-ice control panel (PARA 12-4-29). Go to **12.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

12. Procedure completed.

TABLE NO. 12-2-5-18. MR DE-ICE FAIL, TR DE-ICE FAIL And/Or MR DE-ICE FAULT Capsules Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Did MR DE-ICE FAIL, TR DE-ICE FAIL and MR DE-ICE FAULT capsules go on?

Step 1. If capsules are on, go to **2.**
Step 2. If capsules are not on, go to **13.**

2. Check wiring to TB5-1, TB5-3, and TB5-5.

Step 1. If wiring is good, go to **3.**
Step 2. If wiring is not good, repair/replace wiring (Appendix F), as required. Go to **16.**

3. Check helicopter configuration.

TABLE NO. 12-2-5-18. MR DE-ICE FAIL, TR DE-ICE FAIL And/Or MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. EMEP , go to **4.**
Step 2. W/O EMEP , go to **6.**

EMEP **4. Remove and check PFA P325 (PARA 9-4-98). Go to 5.**

EMEP **5. Install good PFA (PARA 9-4-98). Go to 6.**

6. With GENERATORS NO. 1 switch at OFF and GENERATORS NO. 2 and APU switches at ON, check for 28 VDC between P325-P and ground.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **8.**

7. Check wiring between T10 and T3.

Step 1. If wiring is good, replace current transformer T10 (PARA 12-4-42). Go to **16.**
Step 2. If trouble remains, replace transformer T3 (PARA 9-4-14). Go to **16.**
Step 3. If wiring is not good, repair/replace wiring (Appendix F), as required. Go to **16.**

8. Check continuity between P325-P and P205-F.

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **16.**

9. Check for 28 VDC between P205-E and ground.

Step 1. If voltage is as specified, replace K3 APU/EXT PWR contactor (PARA 9-4-17). Go to **16.**
Step 2. If voltage is not as specified, go to **10.**

10. Check continuity between P205-E and P206-L.

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **16.**

11. Check for 28 VDC between P206-M and ground.

TABLE NO. 12-2-5-18. MR DE-ICE FAIL, TR DE-ICE FAIL And/Or MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, replace K4 ac bus tie contactor (PARA 9-4-21). Go to **16.**

Step 2. If voltage is not as specified, go to **12.**

12. Check for 28 VDC between terminal 1 (DE-ICE CNTRLR circuit breaker, NO. 2 PRI BUS DC - mission readiness circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between P206-M and terminal 1 (Appendix F). Go to **16.**

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **16.**

13. Did TR DE-ICE FAIL capsule go on?

Step 1. If capsule is on, go to **14.**

Step 2. If capsule is not on, go to **15.**

14. Check wiring to K63 tail rotor de-ice contactor.

Step 1. If wiring is good, replace contactor (PARA 12-4-41). Go to **16.**

Step 2. If wiring is not good, repair/replace wiring (Appendix F). Go to **16.**

15. Check wiring to K62 main rotor de-ice bus contactor.

Step 1. If wiring is good, replace contactor (PARA 12-4-40). Go to **16.**

Step 2. If wiring is not good, repair/replace wiring (Appendix F). Go to **16.**

16. Procedure completed.

TABLE NO. 12-2-5-19. MR DE-ICE FAIL, TR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between P325-P and P329-L.

Step 1. If continuity is present, go to **2.**

TABLE NO. 12-2-5-19. MR DE-ICE FAIL, TR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

2. **With rotor at 100% and GENERATORS NO. 1 and NO. 2 switches at ON, check for 28 VDC between main rotor blade de-ice junction box terminal E1 and ground.**

Step 1. If voltage is as specified, replace diode CR11 (PARA 12-4-45). Go to **13.**
Step 2. If voltage is not as specified, go to **3.**

3. **Check continuity between P204-K and E1 in main rotor blade de-ice junction box.**

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

4. **With rotor at 100% and GENERATORS NO. 1 and NO. 2 switches at ON, check for 28 VDC between P204-L and ground.**

Step 1. If voltage is as specified, replace No. 2 generator line contactor (PARA 9-4-12). Go to **13.**
Step 2. If voltage is not as specified, go to **5.**

5. **Check continuity between P204-L and P212-L.**

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

6. **Check for 28 VDC between P212-K and ground.**

Step 1. If voltage is as specified, replace No. 1 generator line contactor (PARA 9-4-11). Go to **13.**
Step 2. If voltage is not as specified, go to **7.**

7. **Check helicopter configuration.**

Step 1. WINTER , go to **8.**
Step 2. W/O WINTER , go to **9.**

WINTER 8. **Check continuity between P244B-L and P212-K.**

TABLE NO. 12-2-5-19. MR DE-ICE FAIL, TR DE-ICE FAIL And MR DE-ICE FAULT Capsules Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, replace winterization kit (PARA 16-6-2). Go to **13.**
Step 3. If trouble remains, go to **9.**

9. Check continuity between P244-L and P212-K.

Step 1. If continuity is present, go to **10.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

10. Check helicopter configuration.

Step 1. WINTER , go to **11.**
Step 2. W/O WINTER , go to **12.**

WINTER **11. Check for 28 VDC between P244B-V and ground.**

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **13.**
Step 2. If voltage is not as specified, replace winterization kit (PARA 16-6-2). Go to **13.**
Step 3. If trouble remains, go to **12.**

12. Check for 28 VDC between P244-V and ground.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **13.**
Step 2. If voltage is not as specified, repair/replace wiring between P244-V and terminal 1 (DE-ICE CNTRLR circuit breaker, NO. 2 PRI BUS DC - mission readiness circuit breaker panel) (Appendix F). Go to **13.**

13. Procedure completed.

TABLE NO. 12-2-5-20. MR DE-ICE FAIL, TR DE-ICE FAIL And MR DE-ICE FAULT Capsules Do Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

TABLE NO. 12-2-5-20. MR DE-ICE FAIL, TR DE-ICE FAIL And MR DE-ICE FAULT Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P325 (PARA 9-4-98). Go to 3.**

EMEP **3. Install good PFA (PARA 9-4-98). Go to 4.**

4. With GENERATORS NO. 1, GENERATORS NO. 2, and APU switches at OFF, check for 28 VDC between P325-P and ground.

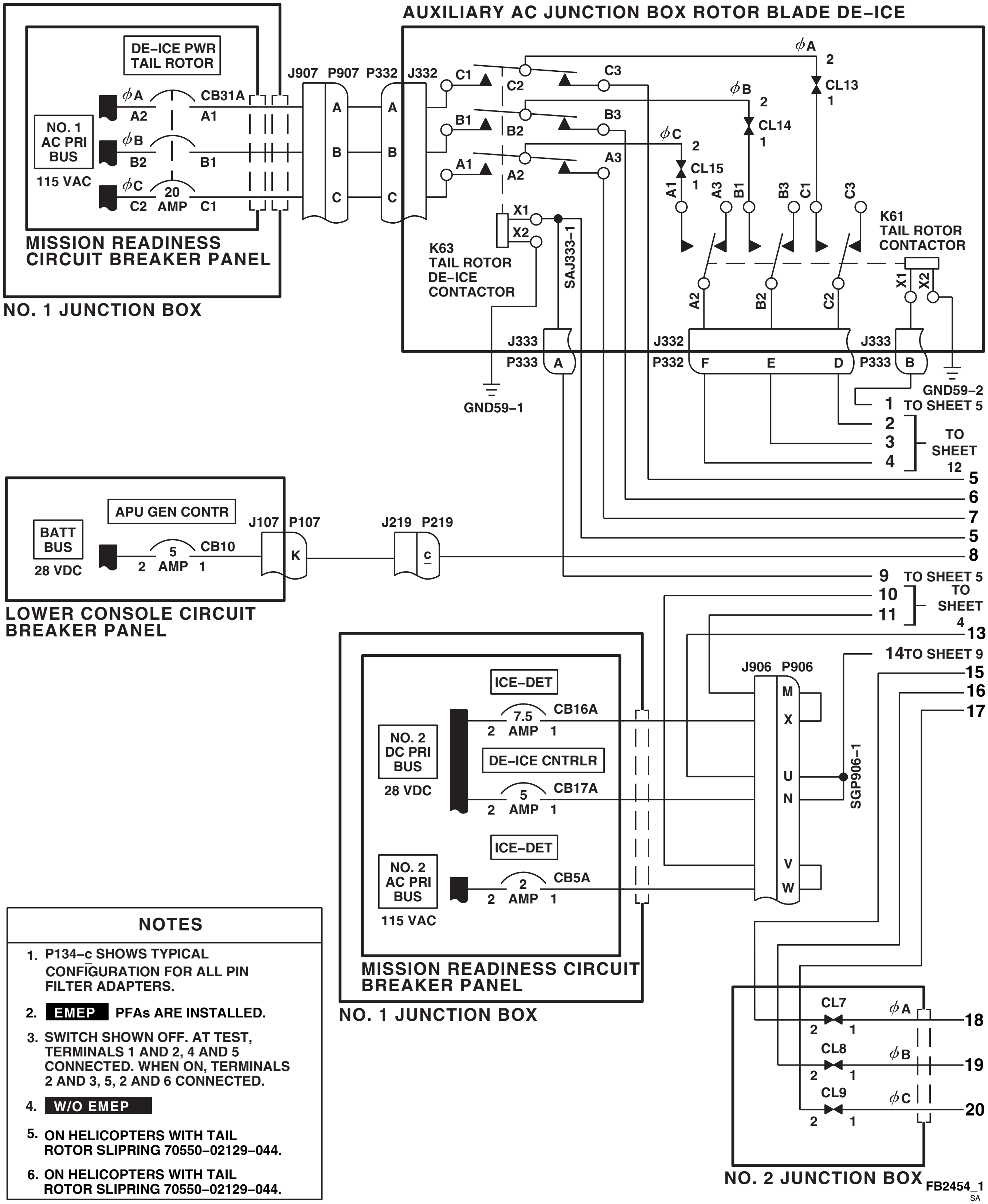
Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace rotor blade de-ice controller (PARA 12-4-32). Go to **7.**

5. Disconnect P205 from K3 APU/EXT PWR contactor. Go to 6.

6. Check for 28 VDC between P325-P and ground.

Step 1. If voltage is as specified, replace K2 No. 2 generator line contactor (PARA 9-4-12). Go to **7.**
Step 2. If voltage is not as specified, replace K3 APU/EXT PWR contactor (PARA 9-4-17). Go to **7.**

7. Procedure completed.



NOTES

1. P134–c SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTER ADAPTERS.
2. EMEP PFAs ARE INSTALLED.
3. SWITCH SHOWN OFF. AT TEST, TERMINALS 1 AND 2, 4 AND 5 CONNECTED. WHEN ON, TERMINALS 2 AND 3, 5, 2 AND 6 CONNECTED.
4. W/O EMEP
5. ON HELICOPTERS WITH TAIL ROTOR SLIPRING 70550–02129–044.
6. ON HELICOPTERS WITH TAIL ROTOR SLIPRING 70550–02129–044.

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 18)

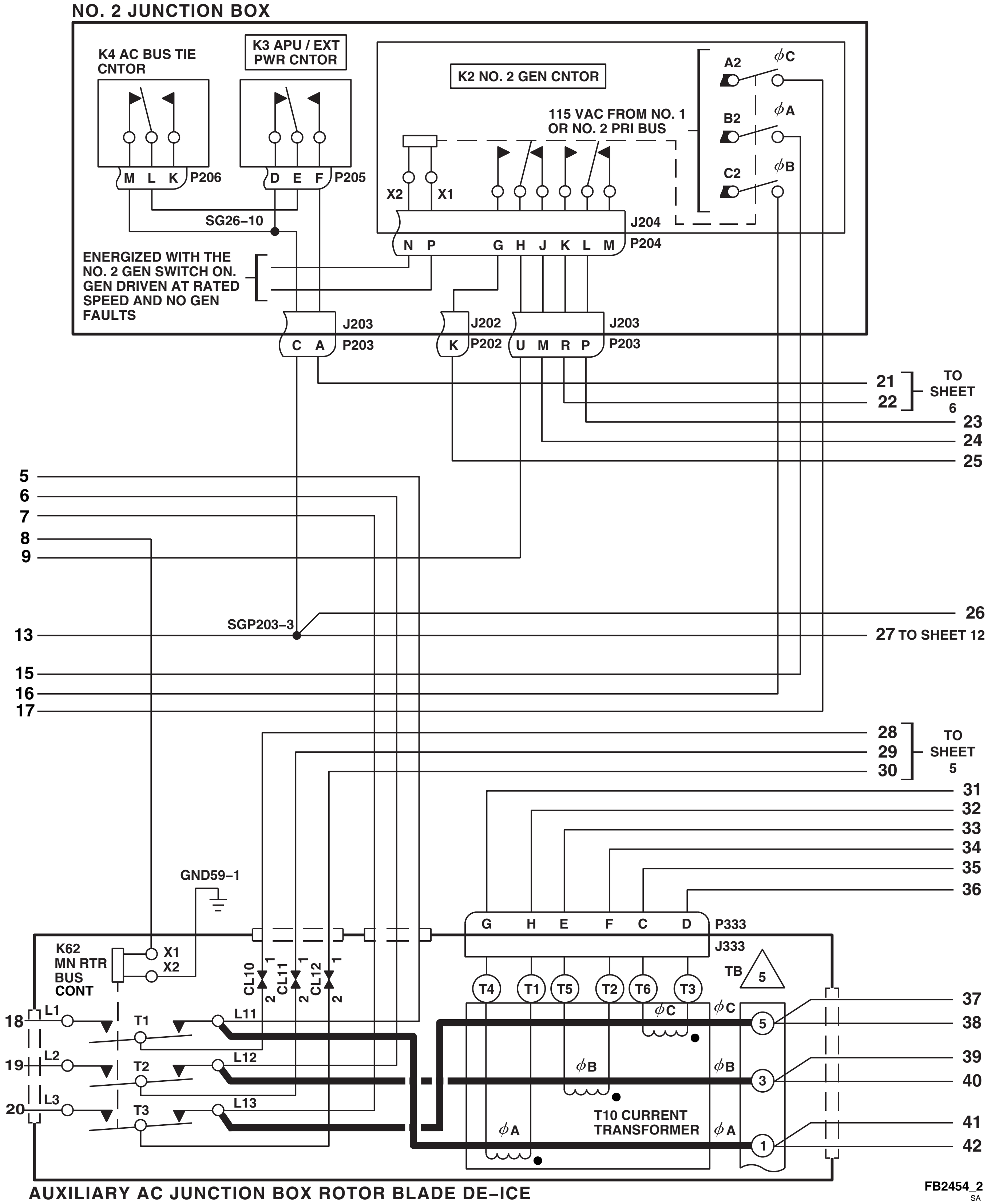


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 18)

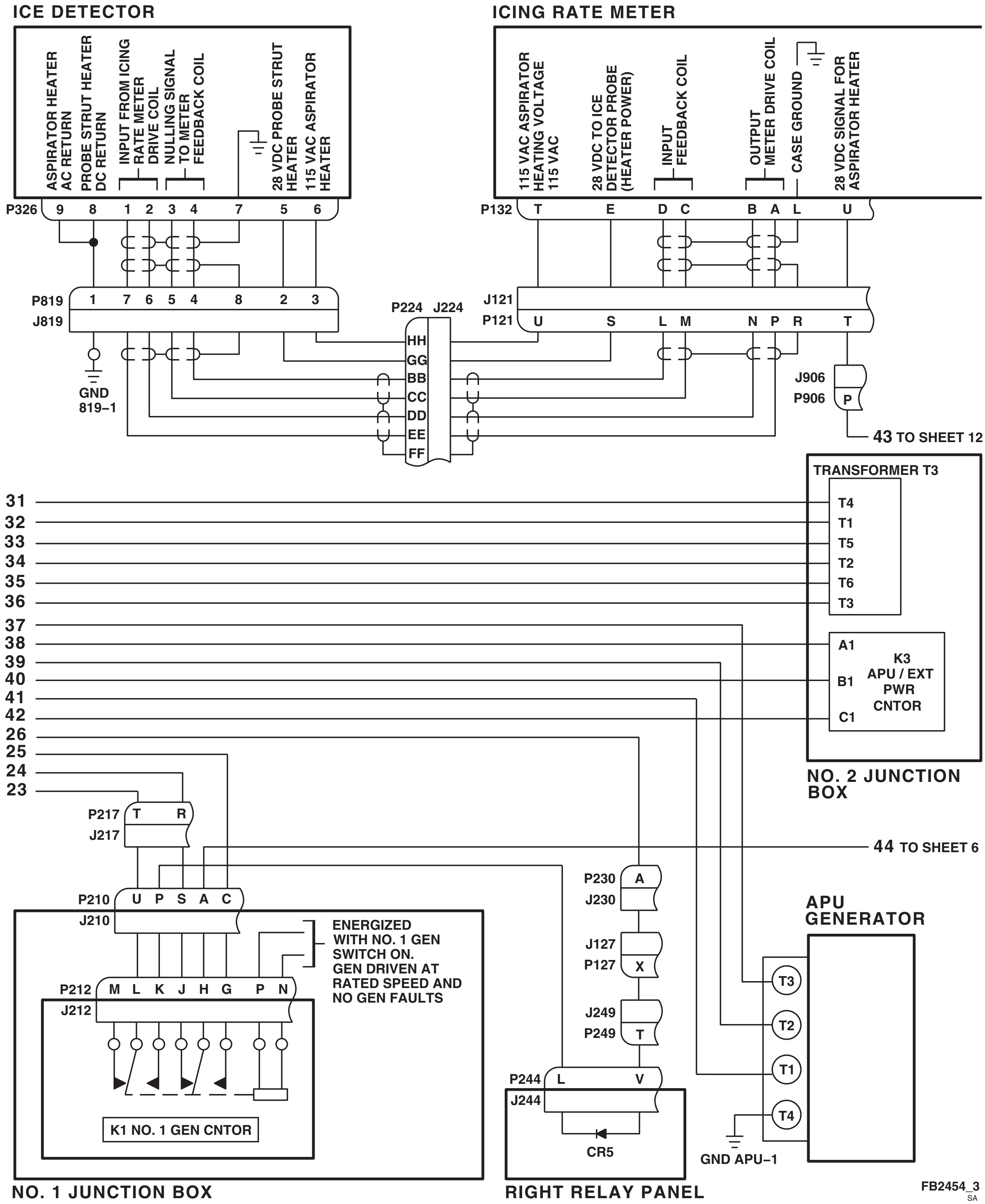


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 18)

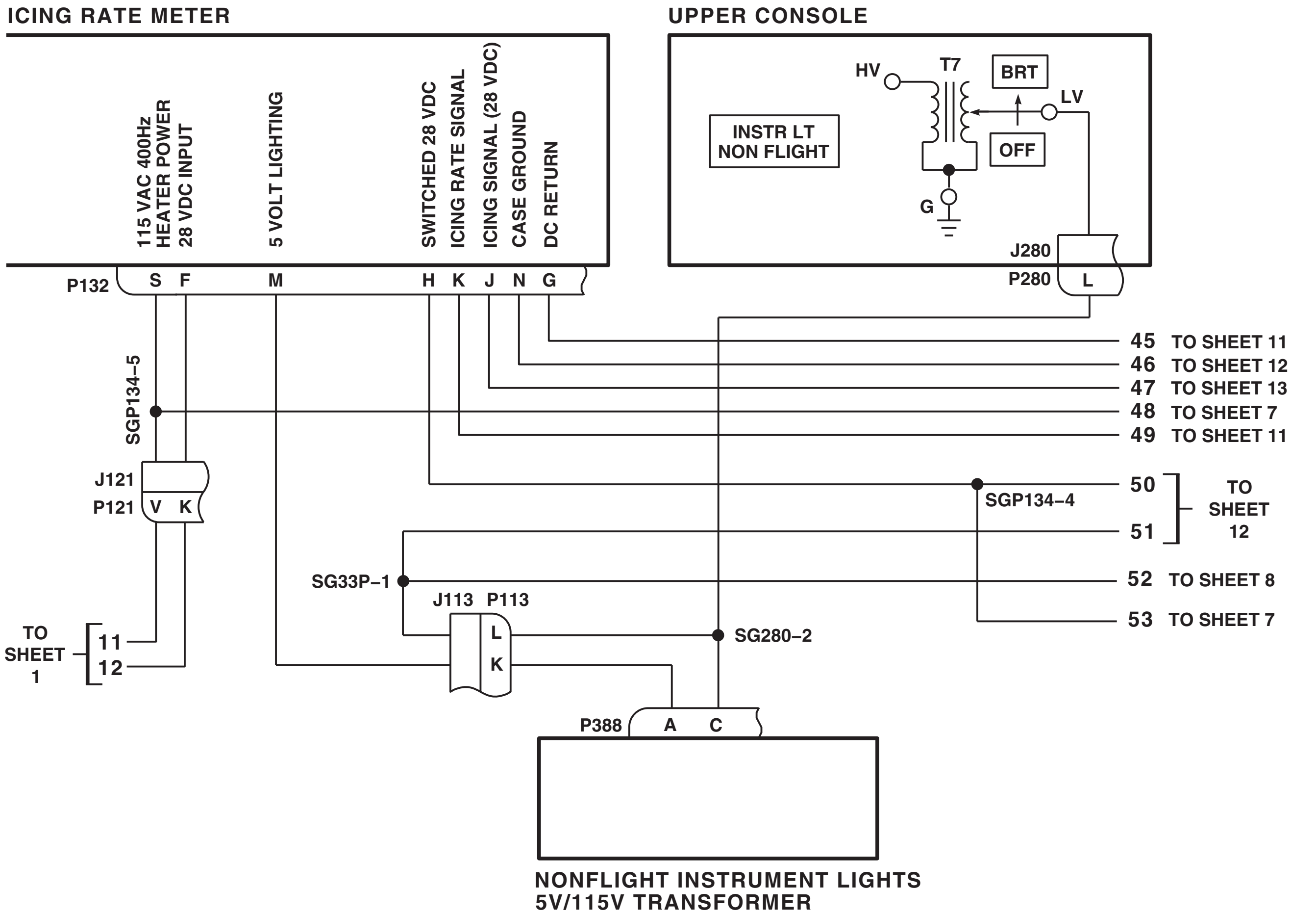


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 18)

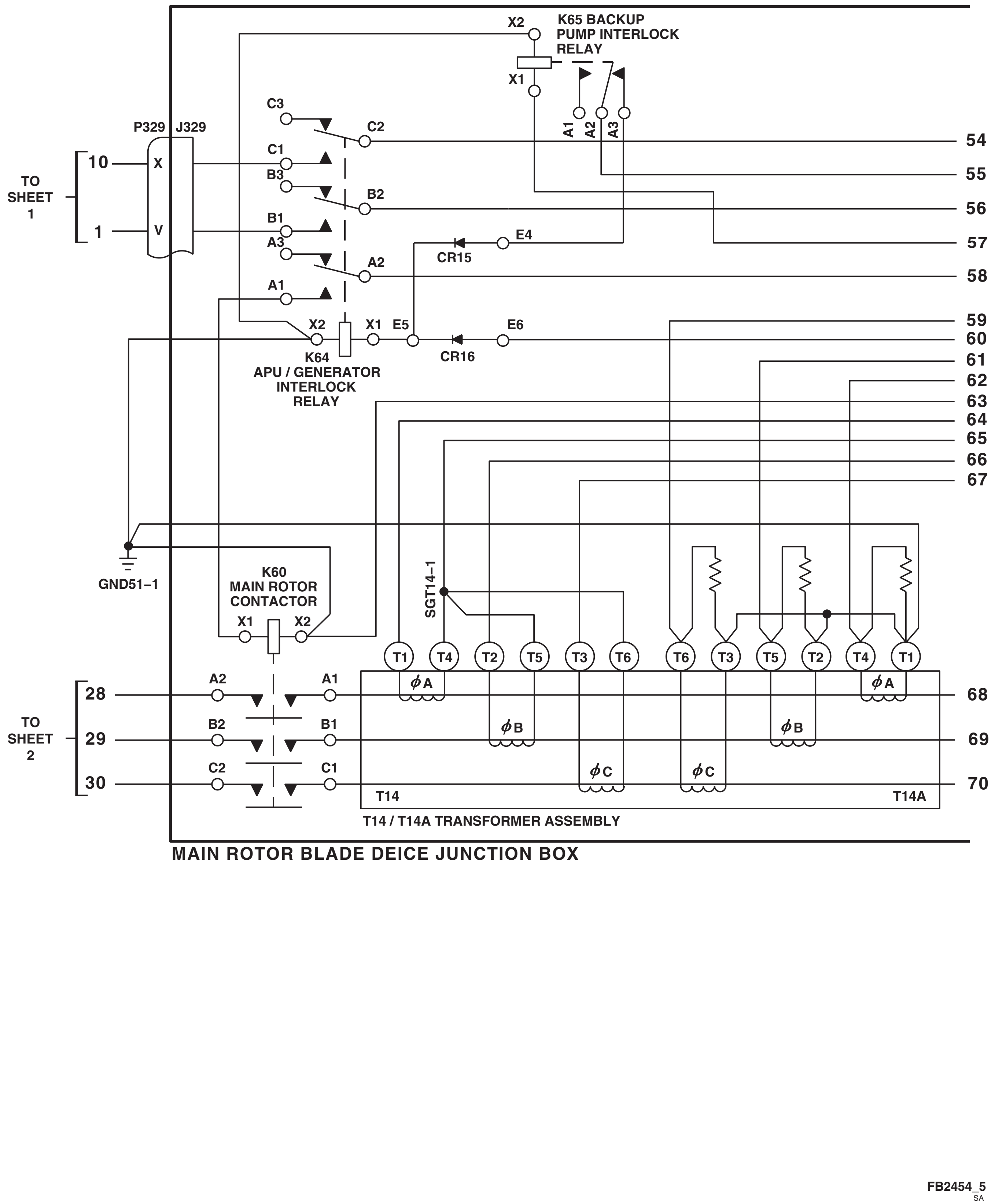


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 18)

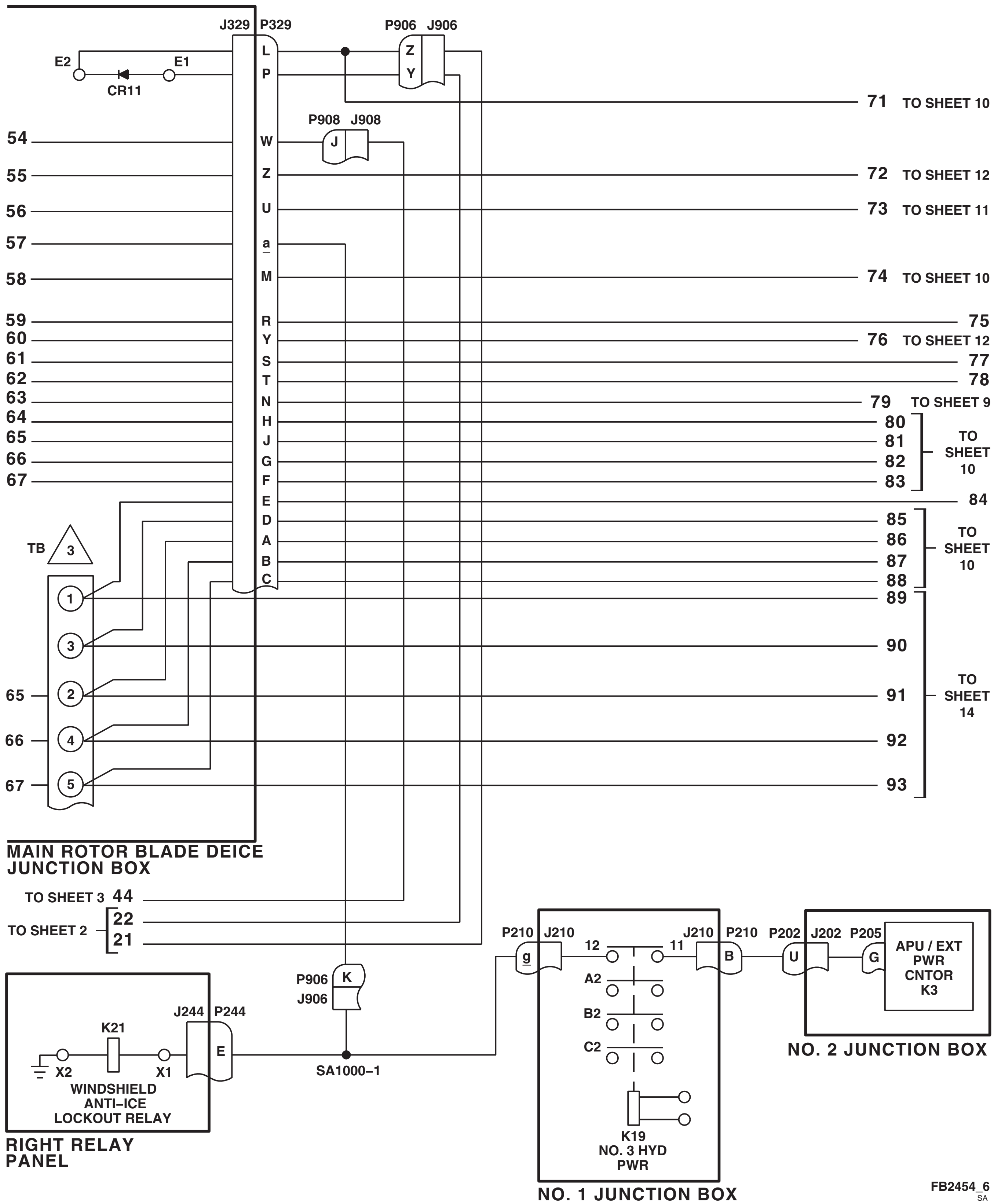


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 18)

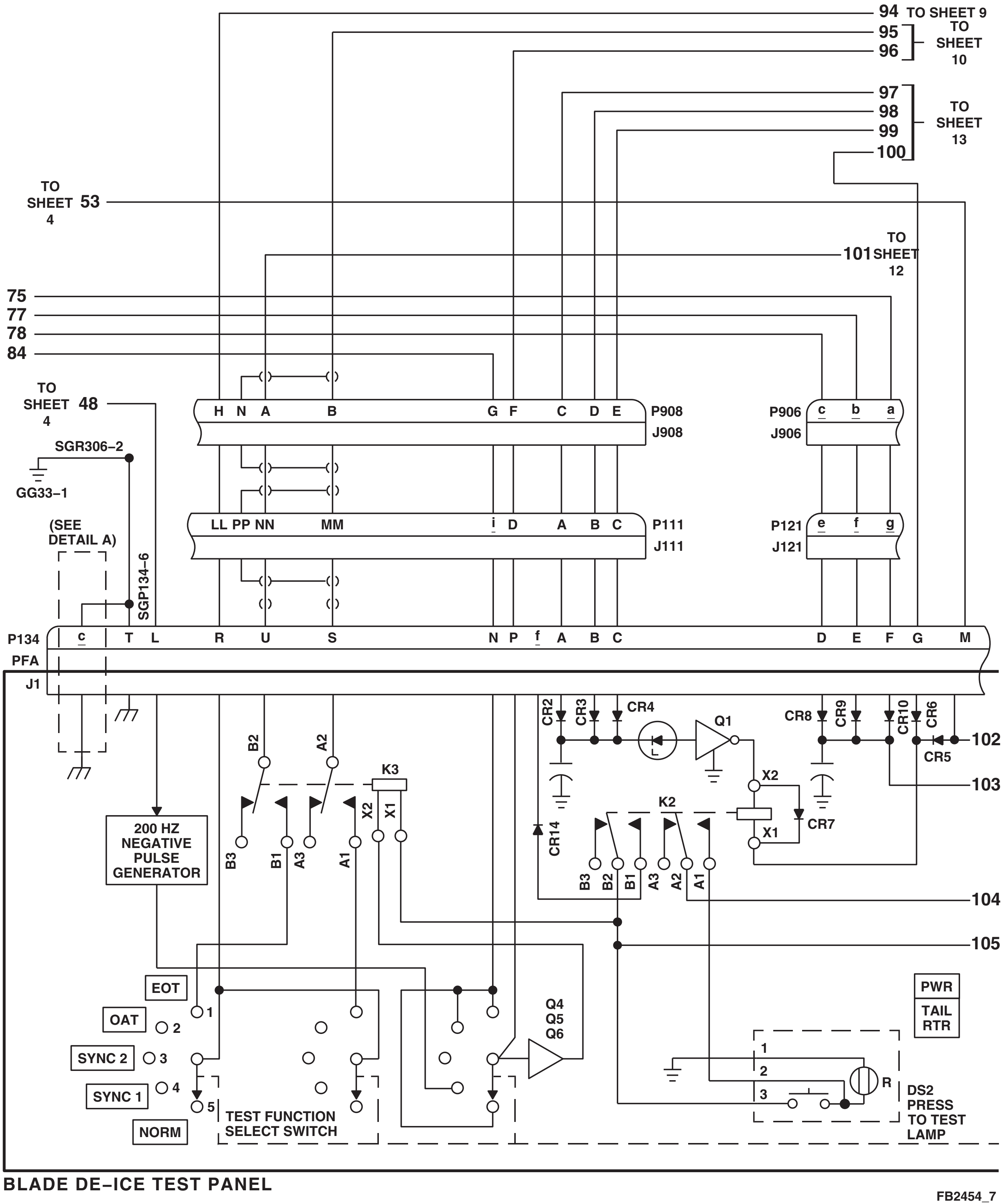


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 7 OF 18)

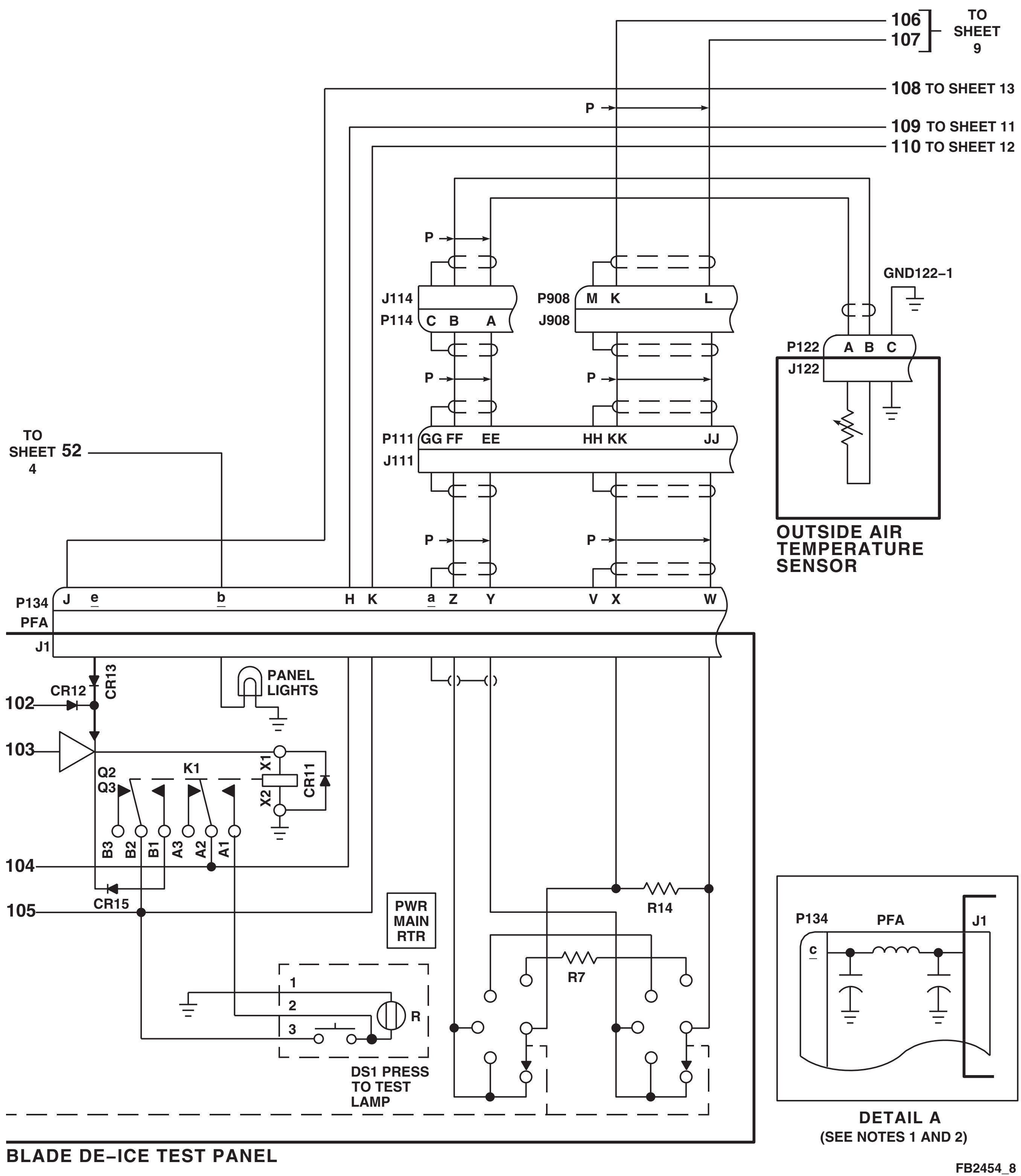


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 8 OF 18)

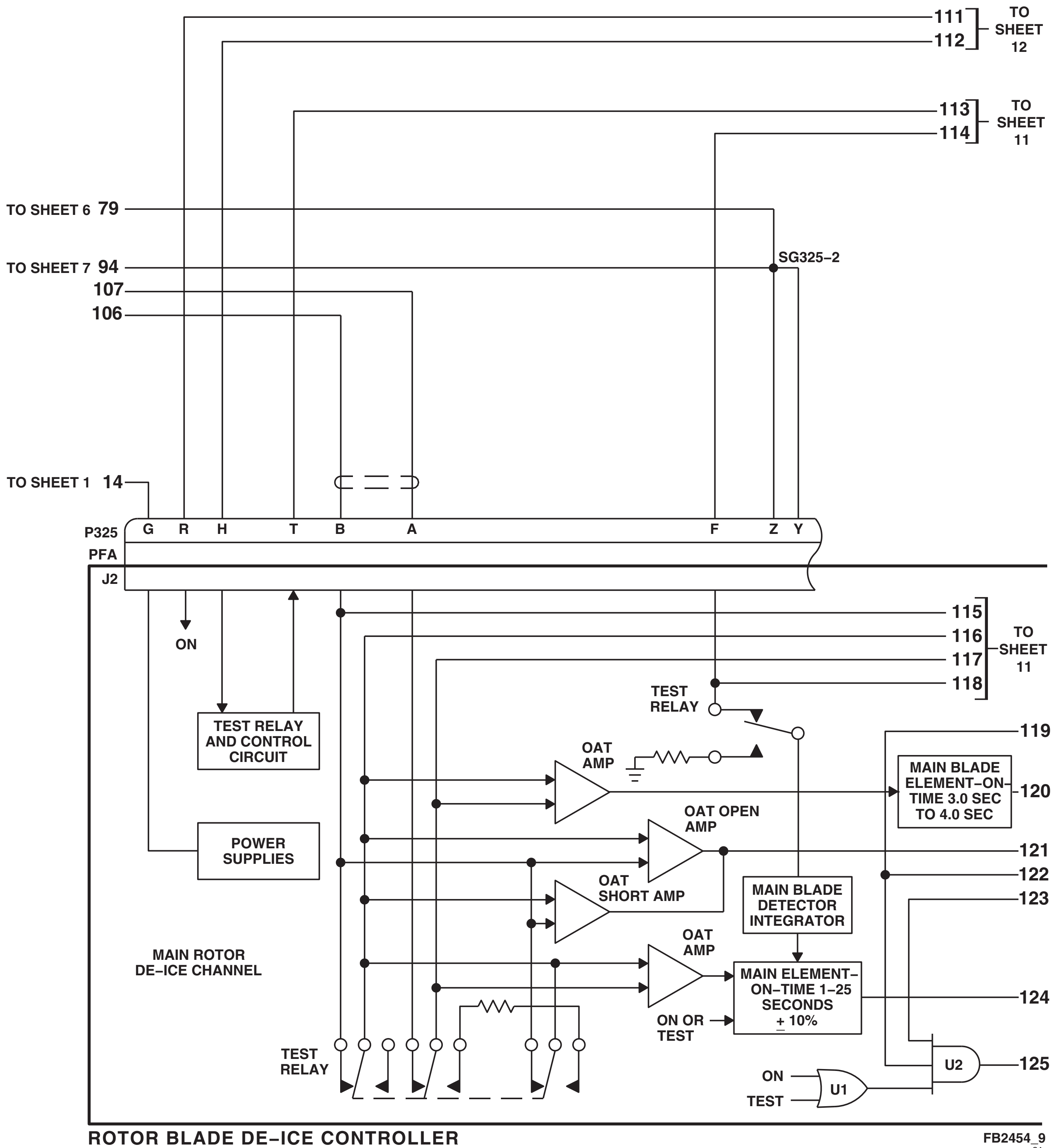


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 9 OF 18)

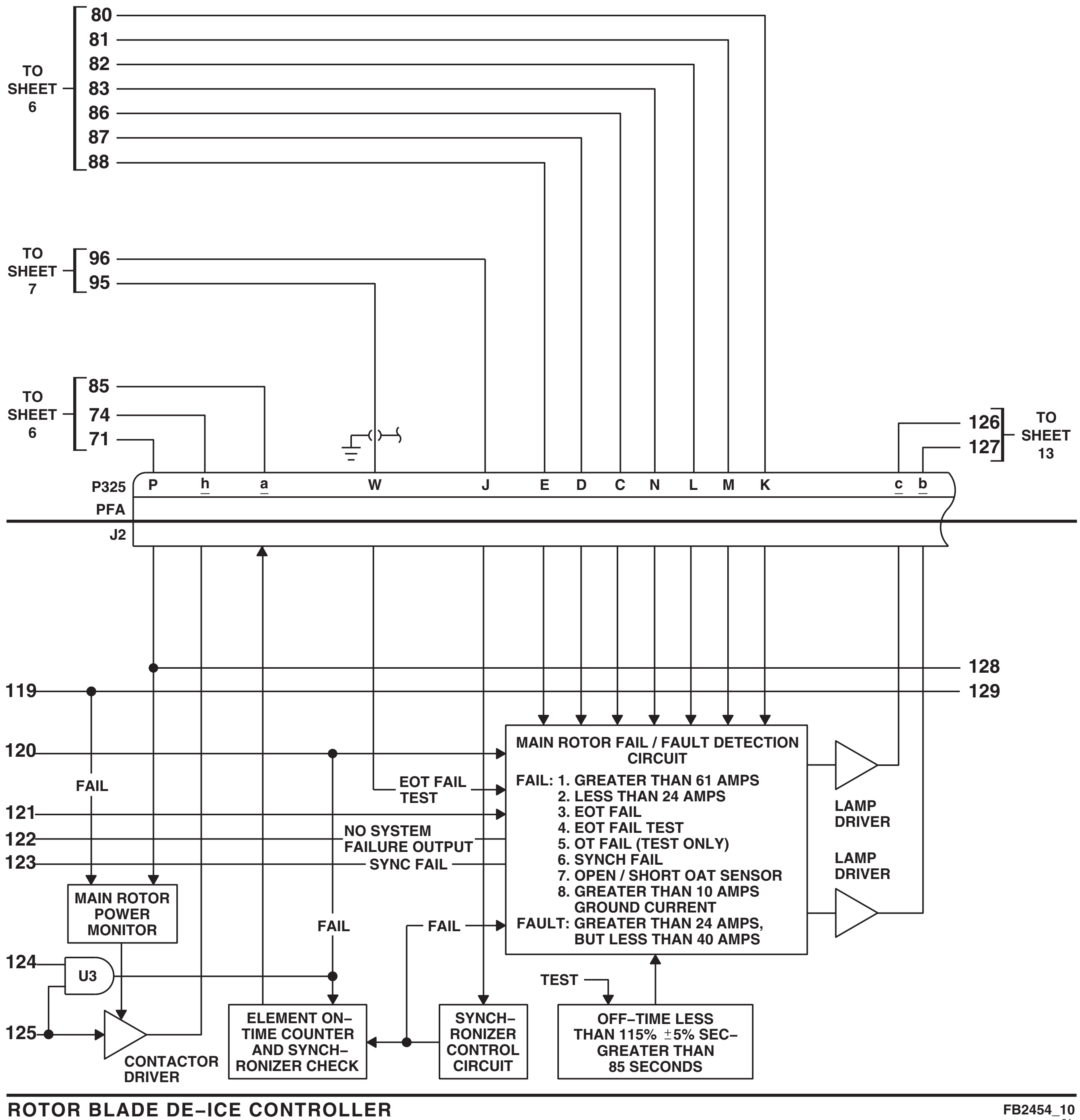


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 10 OF 18)

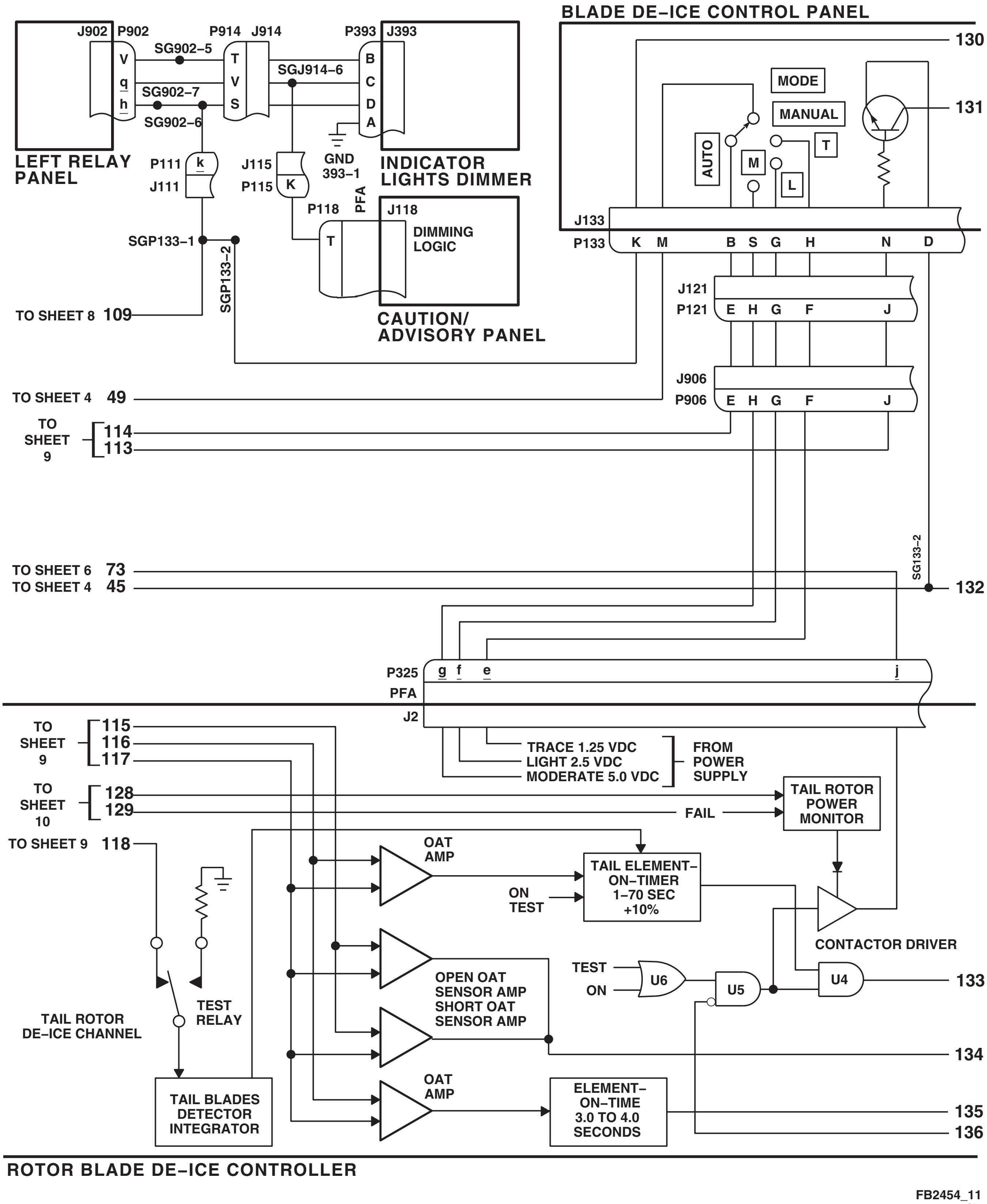


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 11 OF 18)

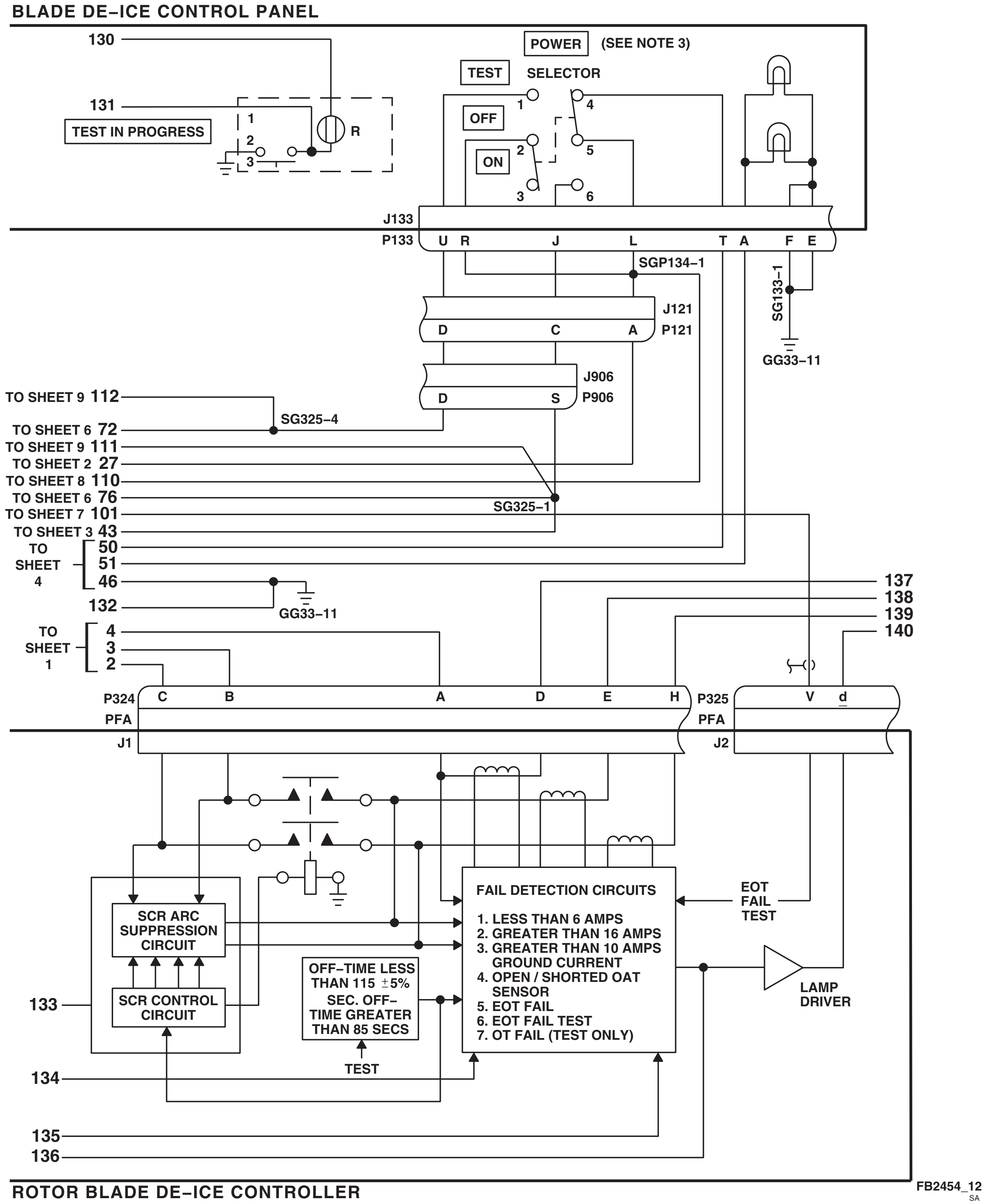


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 12 OF 18)

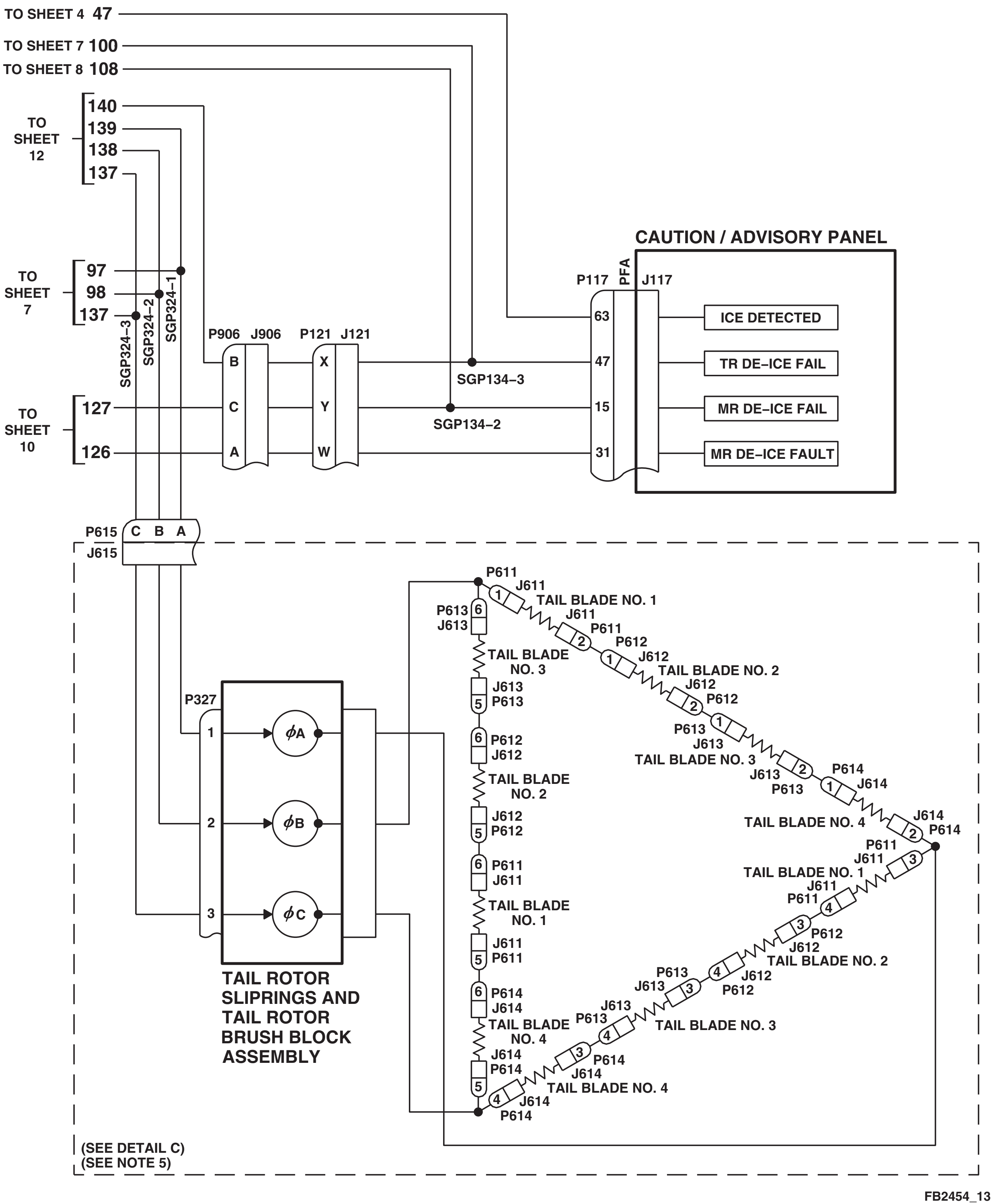


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 13 OF 18)

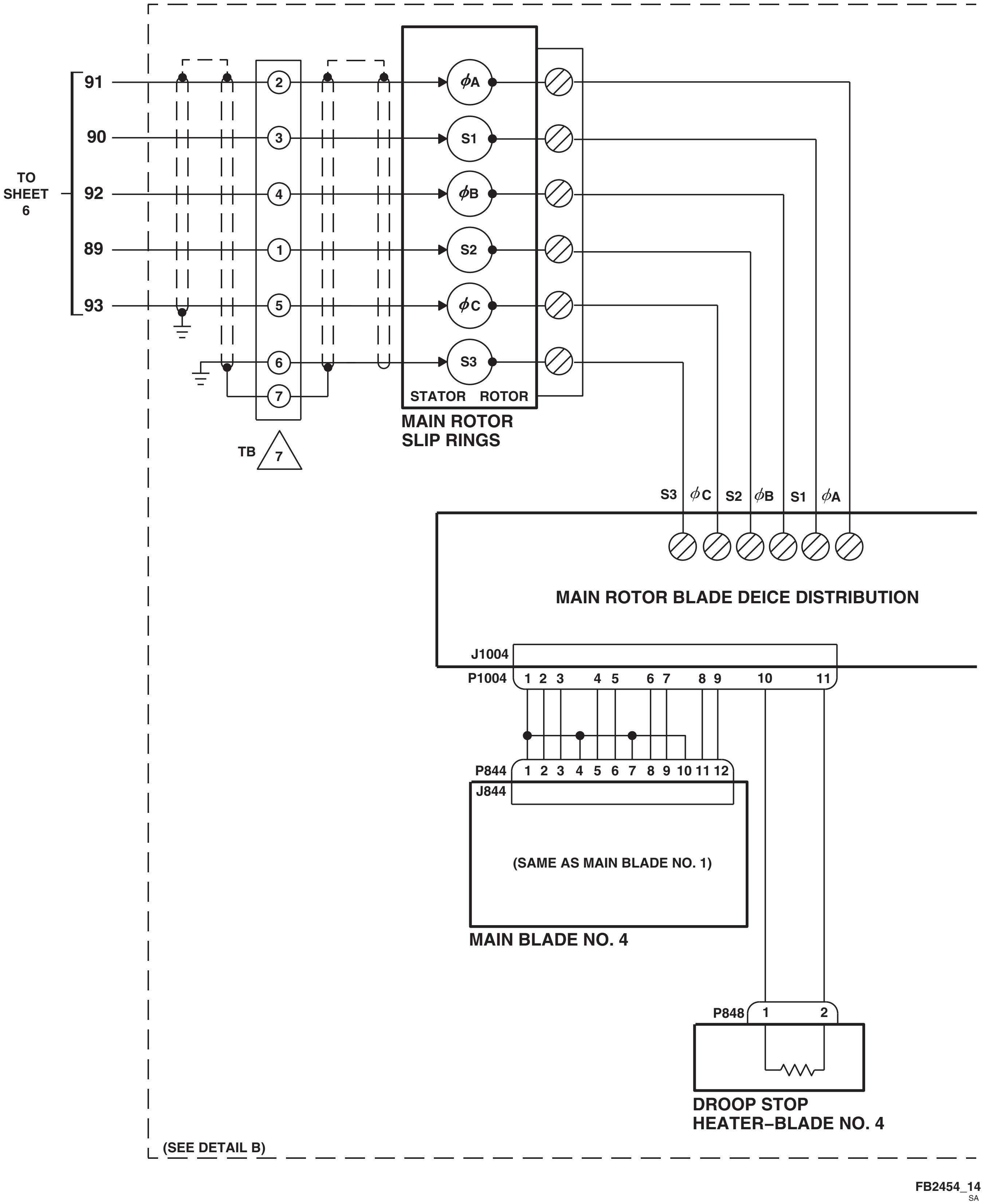


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 14 OF 18)

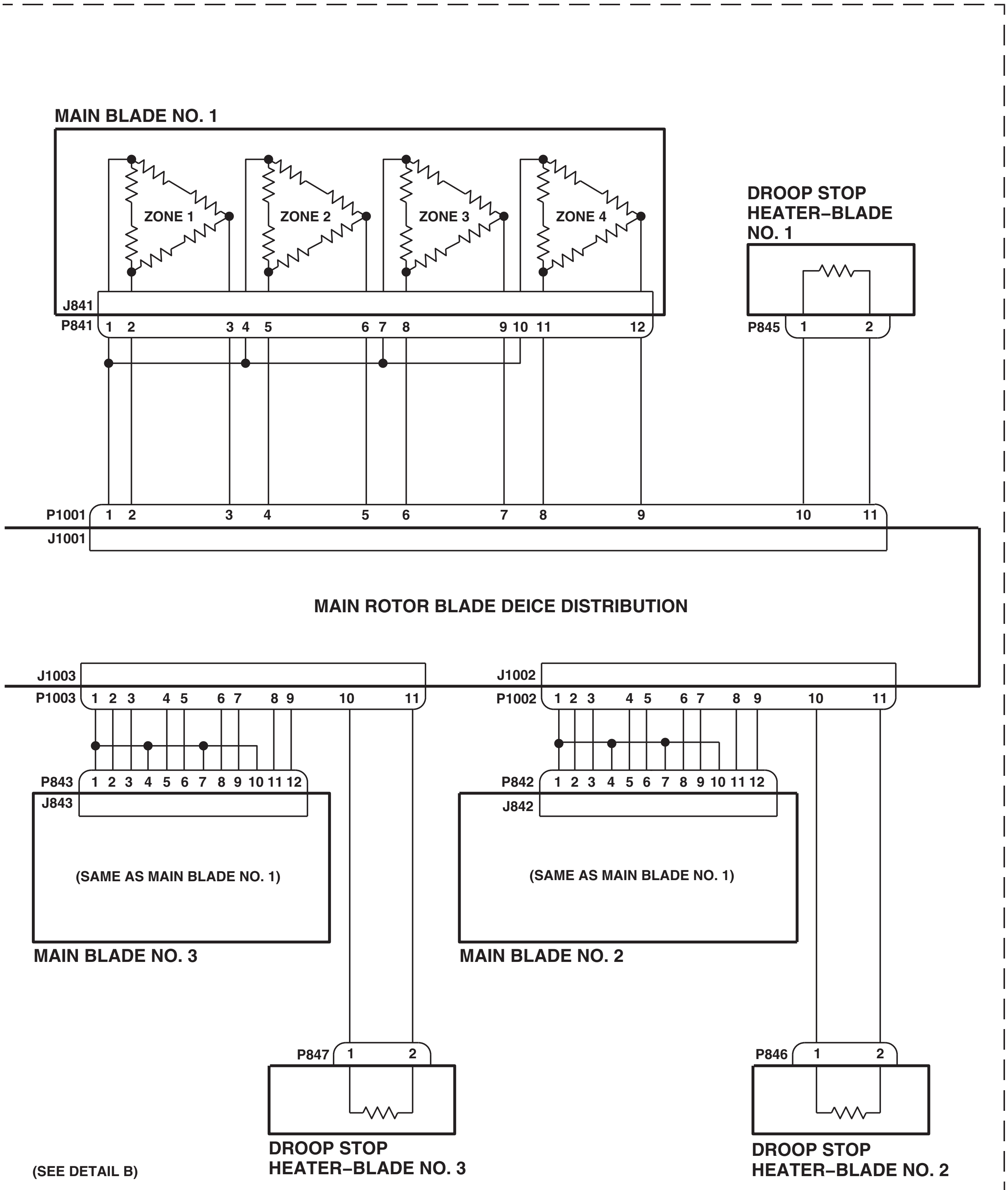
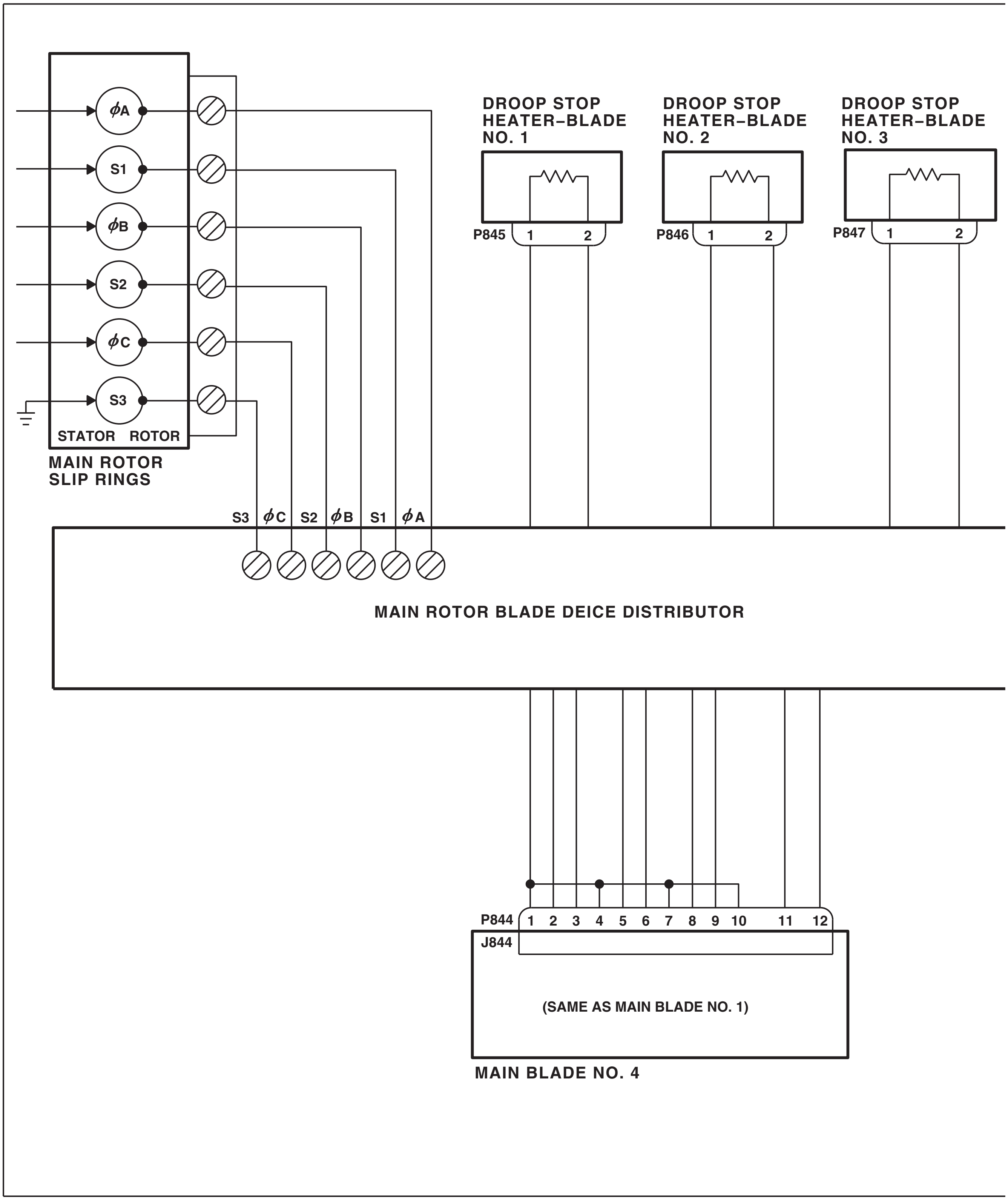


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 15 OF 18)



DETAIL B
(SEE NOTE 4)

FB2454_16
SA

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 16 OF 18)

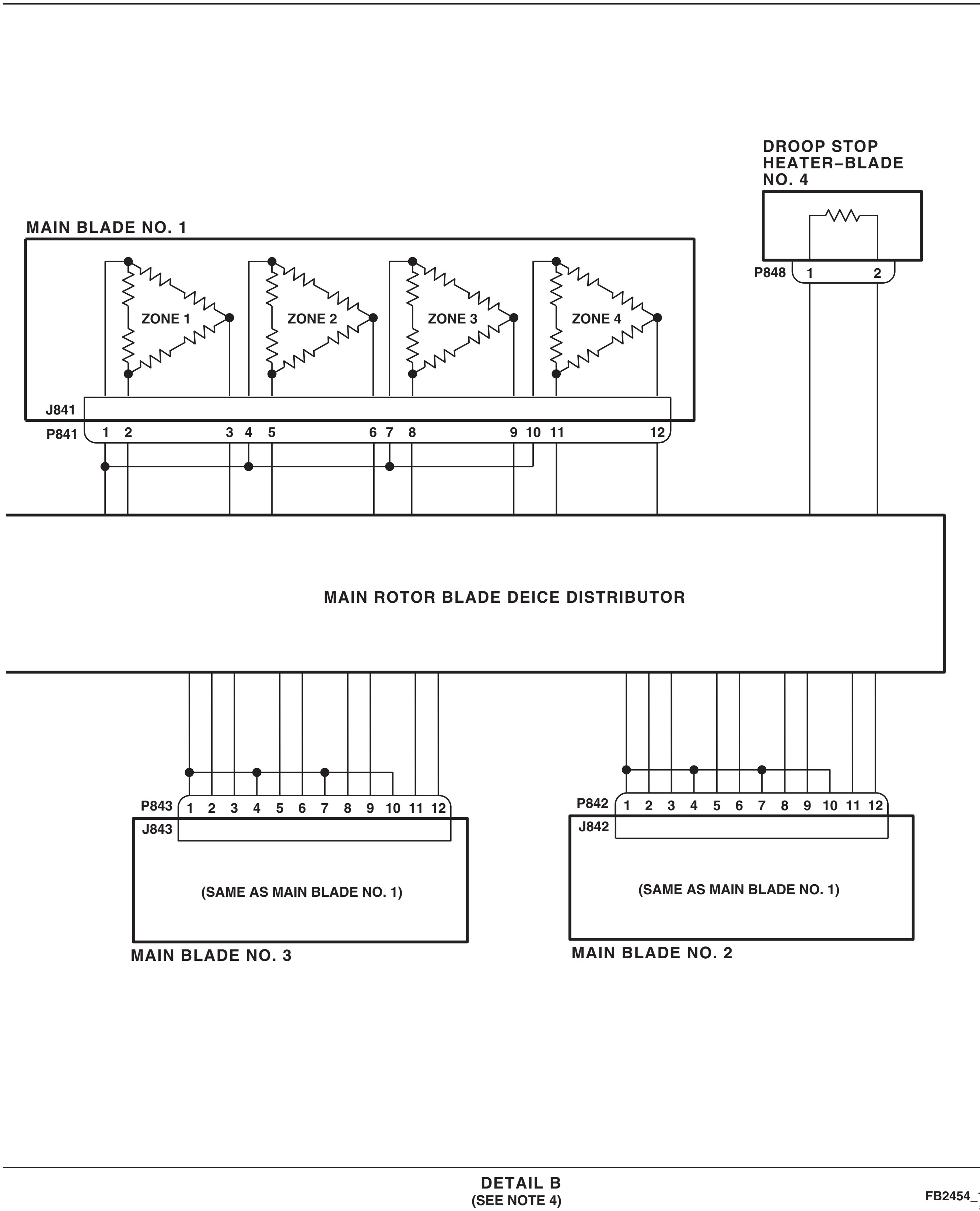
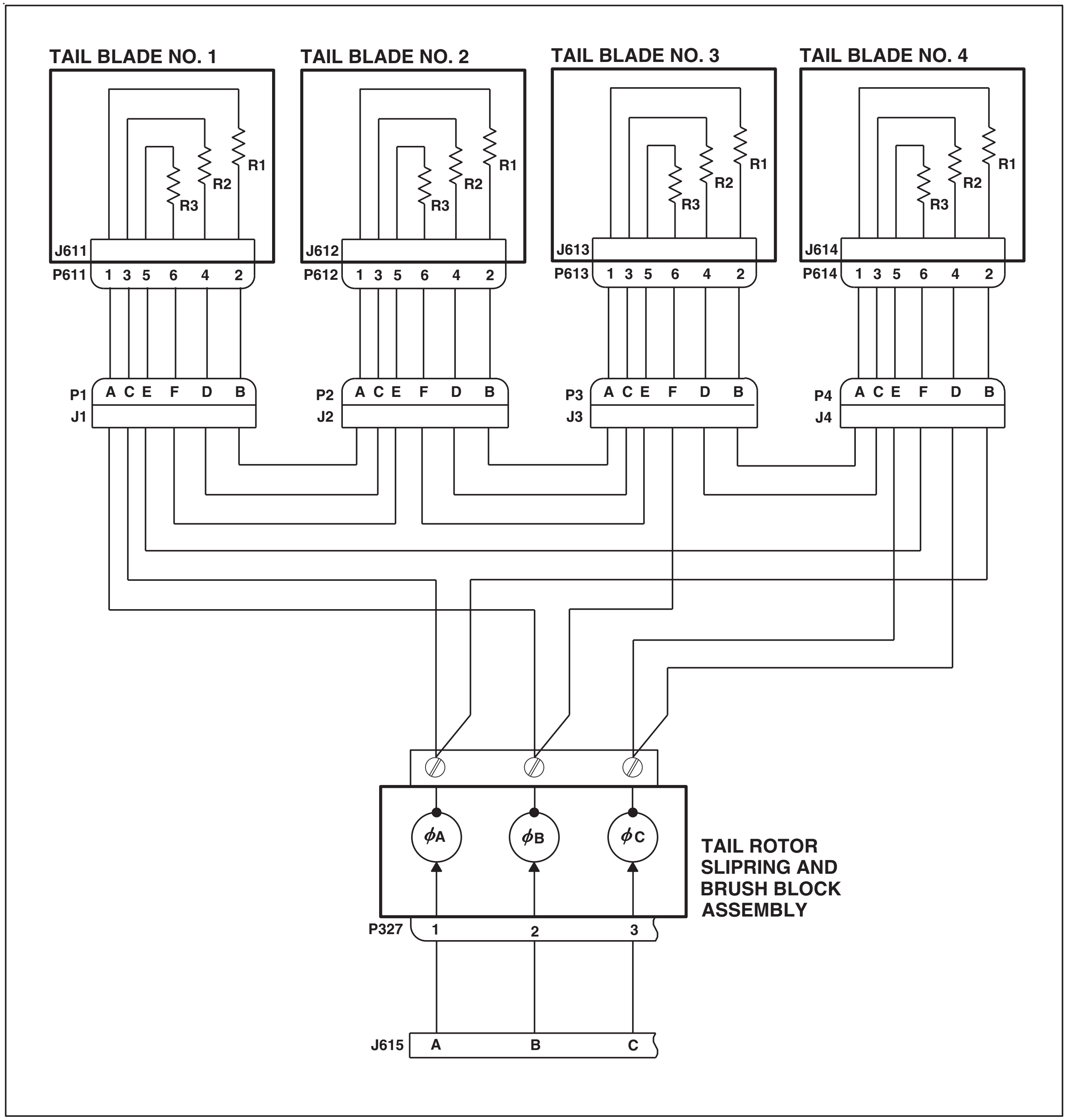


FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 17 OF 18)



DETAIL C
(SEE NOTE 6)

FB2454_18
SA

FIGURE 12-2-5-1. BLADE DE-ICING SYSTEM SCHEMATIC DIAGRAM (SHEET 18 OF 18)

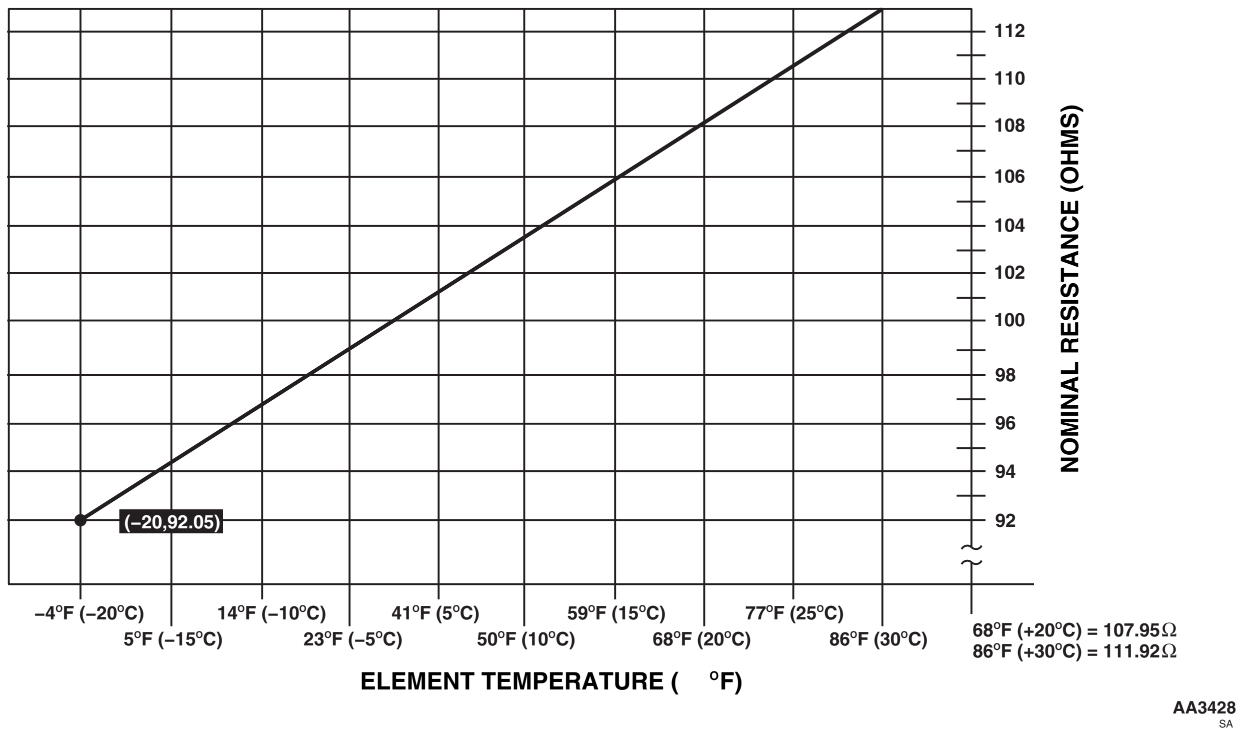


FIGURE 12-2-5-2. OUTSIDE AIR TEMPERATURE/RESISTANCE CHART

12-2-6 CARGO HOOK SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

Materials/Parts

- 20 Pound (or more) Weight

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

CAUTION

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

Steps a. through r. check out the normal release system. Steps t. through ae. check out the emergency release test circuit.

12-2-6.1 Operational/Troubleshooting Procedure.

a. Disconnect the CARGO HOOK CAD (Cartridge Actuated Device).

b. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

c. Make sure upper console panel switches are placed as follows:

SWITCH	POSITION
CARGO HOOK ARMING	SAFE
CARGO HOOK CONTR	CKPT
CARGO HOOK EMERG REL	NORM

d. On cabin floor, open access panel 4B-24. Push lever to allow cargo hook to swing down.

e. Turn on electrical power (PARA 1-6-16).

f. Install weight on cargo hook.

g. Place upper console CARGO HOOK ARMING switch to ARMED.

RESULT: HOOK ARMED capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 12-2-6-1.

CAUTION

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

h. Press pilot's cyclic stick CARGO REL button until weight has dropped, then release.

RESULT:

(1) Cargo hook shall open and release weight.

(2) CARGO HOOK OPEN capsule on caution/advisory panel shall go on.

(3) CARGO HOOK OPEN capsule on caution/advisory panel shall go off (momentary light).

- If result (1) is not as specified, go to TABLE NO. 12-2-6-2.
- If result (2) is not as specified, go to TABLE NO. 12-2-6-3.
- If result (3) is not as specified, go to TABLE NO. 12-2-6-4.
- If cargo hook 70800-02503-107 is used and CARGO HOOK OPEN capsule goes on, but hook does not open, replace cargo hook (PARA 12-4-57).

i. Install weight on cargo hook.

CAUTION

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

j. Press copilot's cyclic stick CARGO REL button until weight has dropped, then release.

RESULT:

(1) Cargo hook shall open and release weight.

(2) CARGO HOOK OPEN capsule on caution/advisory panel shall go on.

(3) CARGO HOOK OPEN capsule on caution/advisory panel shall go off (momentary light).

- If result (1) is not as specified, go to TABLE NO. 12-2-6-2.
- If result (2) is not as specified, go to TABLE NO. 12-2-6-3.
- If result (3) is not as specified, go to TABLE NO. 12-2-6-4.
- If cargo hook 70800-02503-107 is used and CARGO HOOK OPEN capsule goes on, but hook does not open, replace cargo hook (PARA 12-4-57).

k. Install weight on cargo hook.

l. Place upper console CARGO HOOK CONTR switch to ALL.

m. **RES HST** Press CARGO HOOK RELEASE button on rescue hoist/cargo hook pendant.

RESULT: Cargo hook shall open and release weight. CARGO HOOK OPEN capsule on caution/advisory panel shall go on and then off (momentary light).

- If result is not as specified, go to TABLE NO. 12-2-6-5.

n. Install weight on cargo hook.

o. Remove crewman's cargo hook pendant from behind pilot's seat. Connect pendant cord to J267.

CAUTION

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

p. Press crewman's pendant NORMAL RLSE button until weight has dropped, then release.

RESULT:

(1) Cargo hook shall open and release weight.

(2) CARGO HOOK OPEN capsule on caution/advisory panel shall go on.

(3) CARGO HOOK OPEN capsule on caution/advisory panel shall go off (momentary light).

- If result (1) is not as specified, go to TABLE NO. 12-2-6-2.
- If result (2) is not as specified, go to TABLE NO. 12-2-6-3.
- If result (3) is not as specified, go to TABLE NO. 12-2-6-4.
- If cargo hook 70800-02503-107 is used and CARGO HOOK OPEN capsule goes on, but hook does not open, replace cargo hook (PARA 12-4-57).

q. Place upper console CARGO HOOK ARMING switch to SAFE.

RESULT: HOOK ARMED capsule on caution/advisory panel shall go off.

- If result is not as specified, replace cargo hook arming switch S1 (PARA 9-4-5).

r. Press pilot's cyclic stick CARGO REL button.

RESULT: Cargo hook shall not open.

- If result is not as specified, check continuity between J120-E and P200-B (for cyclic stick grip serial no. 70400-87261-102) or J120-F and P200-B (for cyclic stick grip serial no. 70400-87261-101), and between J120-F and terminal 1 of CARGO HOOK CONTR circuit breaker.

  (a) If continuity is present, troubleshoot pilot's cyclic stick (PARA 11-2-1).

  (b) If continuity is not present, repair/replace wiring (Appendix F), as required.

s. Remove weight from under helicopter.

t. Install insulated jumper wire between pins 2 and 3 of connector P256.

u. Press face of upper console CARGO HOOK EMERG REL TEST light.

RESULT: TEST light shall go on.

- If result is not as specified, go to TABLE NO. 12-2-6-6.

v. Place upper console CARGO HOOK EMERG REL switch to OPEN.

w. Press crewman's pendant EMER RLSE button until TEST light goes on, then release button. Push pilot's and copilot's collective sticks emergency hook release button until TEST light goes on, then release button.

RESULT: Whenever any emergency release button is pressed, TEST light shall go on.

- If result is not as specified, go to TABLE NO. 12-2-6-7.

x. Place upper console CARGO HOOK EMERG REL switch to SHORT.

y. Press crewman's pendant EMER RLSE button until TEST light goes on, then release button.

On pilot's and copilot's collective sticks, push emergency hook release button until TEST light goes on, then release button.

RESULT: Whenever any emergency release button is pressed, TEST light shall go on.

- If result is not as specified, go to TABLE NO. 12-2-6-7.

z. Remove jumper wire from pins 2 and 3 of connector P256.

aa. Install the CARGO HOOK CAD (Cartridge Actuated Device).

ab. Place upper console CARGO HOOK EMERG REL switch to OPEN.

ac. Install crewman's cargo hook pendant behind pilot's seat.

ad. Turn off electrical power (PARA 1-6-16).

ae. If necessary, swing cargo hook up and latch in place. Close and latch access panel.

TABLE NO. 12-2-6-1. HOOK ARMED Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Place caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.

2. Check that HOOK ARMED capsule goes on.

Step 1. If HOOK ARMED capsule is on, go to **3.**
Step 2. If HOOK ARMED capsule is not on, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to **11.**

3. With CARGO HOOK ARMING switch placed at ARMED, check continuity between J126-W and J200-A.

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, go to **10.**

4. Check for 28 vdc between P126-W and ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **9.**

5. Check helicopter configuration.

Step 1. EMEP , go to **6.**
Step 2. W/O EMEP , go to **8.**

EMEP **6. Remove and check pin filtered adapter from P118 (PARA 9-4-98).**

Step 1. If pin filtered adapter is good, go to **7.**

TABLE NO. 12-2-6-1. HOOK ARMED Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to **11.**

EMEP

7. Install pin filtered adapter (PARA 9-4-98). Go to 8.

8. Check for 28 vdc between P118-M and ground.

Step 1. If voltage is as specified, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to **11.**
Step 2. If voltage is not as specified, repair/replace wiring between P200-A and P118-M (Appendix F). Go to **11.**

9. Check for 28 vdc between CARGO HOOK CONTR circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between P126-W and circuit breaker terminal 1 (Appendix F). Go to **11.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **11.**

10. Check continuity between:
J126-W and S1-6
J200-A and S1-5

Step 1. If continuity is present, replace switch (PARA 12-4-63). Go to **11.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **11.**

11. Procedure completed.

TABLE NO. 12-2-6-2. Cargo Hook Does Not Open.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does cargo hook open with pilot's cyclic, copilot's cyclic, or crewman's pendant?

Step 1. If cargo hook does open, go to **2.**
Step 2. If cargo hook does not open, go to **5.**

TABLE NO. 12-2-6-2. Cargo Hook Does Not Open. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Does cargo hook open using crewman's pendant?

Step 1. If cargo hook does open, go to **3.**
Step 2. If cargo hook does not open, go to **19.**

3. Does cargo hook open using copilot's cyclic CARGO REL button?

Step 1. If cargo hook does open, go to **4.**
Step 2. W/O RES HST If cargo hook does not open, go to **24.**
Step 3. RES HST If cargo hook does not open, go to **25.**

4. Does cargo hook open using pilot's cyclic CARGO REL button?

Step 1. If cargo hook does open, system is operational. Go to **30.**
Step 2. W/O RES HST If cargo hook does not open, go to **27.**
Step 3. RES HST If cargo hook does not open, go to **28.**

5. With any CARGO REL button pressed, check for 28 vdc between J218-A and J218-B.

Step 1. If voltage is as specified, replace cargo hook (PARA 12-4-57). Go to **30.**
Step 2. If voltage is not as specified, go to **6.**

6. With any CARGO REL button pressed, check for 28 vdc between J218-A and ground.

Step 1. If voltage is as specified, repair/replace wiring between J218-B and ground (Appendix F). Go to **30.**
Step 2. If voltage is not as specified, go to **7.**

7. With any CARGO REL button pressed, check for 28 vdc between P244-R and ground.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **17.**

8. Check for 28 vdc between P900-r and ground.

Step 1. If voltage is as specified, go to **9.**
Step 2. If voltage is not as specified, go to **16.**

TABLE NO. 12-2-6-2. Cargo Hook Does Not Open. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Check continuity between P241-q and P900-j.

Step 1. If continuity is present, go to **10.**
Step 2. If continuity is not present, repair/replace ground wire (Appendix F). Go to **30.**

10. Install an insulated jumper wire between P900-m and P900-r. Go to 11.

11. Check for 28 vdc between J218-A and ground.

Step 1. If voltage is as specified, go to **12.**
Step 2. If voltage is not as specified, go to **14.**

12. Remove jumper wire from P900. Go to 13.

13. Replace right relay panel (PARA 9-4-2). Go to 30.

14. Remove jumper wire from P900. Go to 15.

15. Repair/replace wiring between P900-m and J218-A (Appendix F). Go to 30.

16. Check for 28 vdc between CARGO HOOK PWR circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between P266-m and P900-r (Appendix F). Go to **30.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **30.**

17. With CARGO HOOK ARMING switch placed to ARMED, check continuity between J200-B and J200-C.

Step 1. If continuity is present, go to **18.**
Step 2. If continuity is not present, replace switch (PARA 12-4-63). Go to **30.**

18. With CARGO REL button pressed, check for 28 vdc between P200-B and ground.

Step 1. If voltage is as specified, repair/replace wiring between P200-C and P244-R (Appendix F). Go to **30.**

TABLE NO. 12-2-6-2. Cargo Hook Does Not Open. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, repair/replace wiring between P200-B and J248-R (Appendix F). Go to **30.**

19. With crewman's pendant NORMAL RLSE button pressed, check continuity between P267-B and P267-F.

Step 1. If continuity is present, go to **20.**
Step 2. If continuity is not present, replace crewman's cargo hook pendant (PARA 12-4-62). Go to **30.**

20. With CARGO HOOK CONTR switch placed to ALL, check for 28 vdc between J267-F and ground.

Step 1. If voltage is as specified, repair/replace wiring between J267-B and J248-R (Appendix F). Go to **30.**
Step 2. If voltage is not as specified, go to **21.**

21. With CARGO HOOK CONTR switch placed to ALL, check for 28 vdc between J200-G and ground.

Step 1. If voltage is as specified, repair/replace wiring between P200-G and J267-F (Appendix F). Go to **30.**
Step 2. If voltage is not as specified, go to **22.**

22. Check continuity between J200-G and CARGO HOOK CONTR switch S2-2.

Step 1. If continuity is present, go to **23.**
Step 2. If continuity is not present, repair/replace wiring between J200-G and S2-2 (Appendix F). Go to **30.**

23. With CARGO HOOK CONTR switch placed to ALL, check continuity between S2-2 and S2-3.

Step 1. If continuity is present, repair/replace wiring between S2-3 and ARMING switch S1-6 (Appendix F). Go to **30.**
Step 2. If continuity is not present, replace switch (PARA 12-4-63). Go to **30.**

W/O RES HST **24. For copilot's cyclic grip serial No. 70400-87261-102, with CARGO REL button pressed, check continuity between P119-E and P119-F.**

Step 1. If continuity is present, go to **26.**

TABLE NO. 12-2-6-2. Cargo Hook Does Not Open. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, troubleshoot copilot's cyclic stick (PARA 11-2-1). Go to **30.**

RES HST **25. For copilot's cyclic grip serial No. 70400-87261-101, with CARGO REL button pressed, check continuity between P119-H and P119-F.**

Step 1. If continuity is present, go to **26.**
Step 2. If continuity is not present, troubleshoot copilot's cyclic stick (PARA 11-2-1). Go to **30.**

26. Check for 28 vdc between J119-F and ground.

Step 1. If voltage is as specified, repair/replace wiring between W/O RES HST J119-E and P248-R RES HST J119-H and P248-R (Appendix F). Go to **30.**
Step 2. If voltage is not as specified, repair/replace wiring between J119-F and CARGO HOOK CONTR circuit breaker terminal 1 (Appendix F). Go to **30.**

W/O RES HST **27. For pilot's cyclic grip serial No. 70400-87261-102, with CARGO REL button pressed, check continuity between P120-E and P120-F.**

Step 1. If continuity is present, go to **29.**
Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to **30.**

RES HST **28. For pilot's cyclic grip serial No. 70400-87261-101, with CARGO REL button pressed, check continuity between P120-H and P120-F.**

Step 1. If continuity is present, go to **29.**
Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to **30.**

29. Check for 28 vdc between J120-F and ground.

Step 1. If voltage is as specified, repair/replace wiring between W/O RES HST J120-E and P248-R RES HST J120-H and P248-R (Appendix F). Go to **30.**
Step 2. If voltage is not as specified, repair/replace wiring between J120-F and CARGO HOOK CONTR circuit breaker terminal 1 (Appendix F). Go to **30.**

30. Procedure completed.

TABLE NO. 12-2-6-3. Cargo Hook Opens But CARGO HOOK OPEN Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Place caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.**

2. **Check that CARGO HOOK OPEN capsule goes on.**

   Step 1. If CARGO HOOK OPEN capsule goes on, go to **3.**
   Step 2. If CARGO HOOK OPEN capsule does not go on, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to **9.**

3. **With load beam on cargo hook pulled down, check continuity between P218-C and P218-D.**

   Step 1. If continuity is present, go to **4.**
   Step 2. If continuity is not present, replace cargo hook (PARA 12-4-57). Go to **9.**

4. **Check for 28 vdc between J218-D and ground.**

   Step 1. If voltage is as specified, go to **5.**
   Step 2. If voltage is not as specified, repair/replace wiring between J218-D and J119-F (Appendix F). Go to **9.**

5. **Check helicopter configuration.**

   Step 1. EMEP , go to **6.**
   Step 2. W/O EMEP , go to **8.**

EMEP 6. **Remove and check pin filtered adapter from P118 (PARA 9-4-98).**

   Step 1. If pin filtered adapter is good, go to **7.**
   Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to **9.**

EMEP 7. **Install pin filtered adapter (PARA 9-4-98). Go to 8.**

8. **With load beam pulled down, check for 28 vdc between P118-G and ground.**

   Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19). Go to **9.**
   Step 2. If voltage is not as specified, repair/replace wiring between J218-C and P118-G (Appendix F). Go to **9.**

TABLE NO. 12-2-6-3. Cargo Hook Opens But CARGO HOOK OPEN Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. **Procedure completed.**

TABLE NO. 12-2-6-4. CARGO HOOK OPEN Capsule Stays On After Load Is Released.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Make sure no CARGO REL buttons are pressed. Go to 2.**

2. **Place a 20-pound (or more) weight on cargo hook. Go to 3.**

3. **Check that hook stays closed and holds weight.**

   Step 1. If hook stays closed and holds weight, go to **4.**
   Step 2. If hook does not stay closed and hold weight, replace cargo hook (PARA 12-4-57). Go to **5.**

4. **Check for 28 vdc between J218-A and ground.**

   Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **5.**
   Step 2. If voltage is not as specified, replace cargo hook (PARA 12-4-57). Go to **5.**

5. **Procedure completed.**

TABLE NO. 12-2-6-5. Cargo Hook Does Not Release When Rescue Hoist/Cargo Hook Pendant CARGO HOOK RELEASE Button Is Pressed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **With CARGO HOOK RELEASE button pressed, check continuity between P968-G and P968-H.**

   Step 1. If continuity is present, go to **2.**
   Step 2. If continuity is not present, replace rescue hoist/cargo hook pendant (PARA 12-4-62). Go to **3.**

TABLE NO. 12-2-6-5. Cargo Hook Does Not Release When Rescue Hoist/Cargo Hook Pendant CARGO HOOK RELEASE Button Is Pressed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Check continuity between J968-G and P200-B.

Step 1. If continuity is present, repair/replace wiring between J968-H and P200-G (Appendix F). Go to **3.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **3.**

3. Procedure completed.

TABLE NO. 12-2-6-6. CARGO HOOK EMERG REL TEST Light Does Not Go On When Pressed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for burnt-out lamp.

Step 1. If lamp is good, go to **2.**
Step 2. If lamp is not good, replace lamp in test light (PARA 9-4-6). Go to **5.**

2. Check for voltage between P200-S and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **4.**

3. Check continuity between TEST lamp pin 3 and ground.

Step 1. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming system (PARA 9-2-10). Go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **5.**

4. Check continuity between P902-h and TEST lamp pin 1.

Step 1. If continuity is present, replace test lamp (PARA 9-4-6). Go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **5.**

5. Procedure completed.

TABLE NO. 12-2-6-7. CARGO HOOK EMERG REL TEST Light Does Not Go On During OPEN Or SHORT Test.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does light go on using pilot's or copilot's collective stick grip, or crewman's cargo hook pendant?

Step 1. If light goes on, go to **2.**
Step 2. If light does not go on, go to **5.**

2. Does light go on using crewman's cargo hook pendant?

Step 1. If light goes on, go to **3.**
Step 2. If light does not go on, go to **17.**

3. Does light go on using copilot's collective stick grip?

Step 1. If light goes on, go to **4.**
Step 2. S70–I If light does not go on, go to **19.**
Step 3. S70–II If light does not go on, go to **21.**

4. Does light go on using pilot's collective stick grip?

Step 1. If light goes on, system is operational. Go to **27.**
Step 2. S70–I If light does not go on, go to **23.**
Step 3. S70–II If light does not go on, go to **25.**

5. Check for 28 vdc between upper console CARGO HOOK EMER circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **27.**

6. Check for 28 vdc between J267-E and ground.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, repair/replace wiring between J267-E and P200-J (Appendix F). Go to **27.**

7. With crewman's cargo hook pendant EMER RLSE button pressed, check for 28 vdc between TB1-5 and ground.

Step 1. If voltage is as specified, go to **8.**

TABLE NO. 12-2-6-7. CARGO HOOK EMERG REL TEST Light Does Not Go On During OPEN Or SHORT Test. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, repair/replace wiring between TB1-5 and J267-C (Appendix F). Go to **27.**

8. Check for 1.5 to 2.5 ohms between TB1-4 and TB1-5.

Step 1. If resistance is as specified, go to **9.**
Step 2. If resistance is not as specified, replace resistor R4 (PARA 9-4-54). Go to **27.**

9. Check continuity between TB1-4 and upper console EMERG REL switch S3-2.

Step 1. If continuity is present, go to **10.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **27.**

10. Check continuity between EMERG REL switch S3-2 and S3-3.

Step 1. If continuity is present, go to **11.**
Step 2. If continuity is not present, replace switch (PARA 12-4-63). Go to **27.**

11. Check continuity between EMERG REL switch S3-3 and P242-g.

Step 1. If continuity is present, go to **12.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **27.**

12. Check continuity between EMERG REL TEST switch terminal 2 and P242-d.

Step 1. If continuity is present, go to **13.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **27.**

13. Disconnect P256 on cargo hook and place an insulated jumper wire between P256-2 and P256-3. Go to 14.

14. Check continuity between P242-c and S3-1.

Step 1. If continuity is present, remove jumper wire from P256, then replace cargo hook squib (PARA 12-4-58). Go to **27.**
Step 2. If trouble remains, remove jumper wire from P256, then replace left relay panel (PARA 9-4-2). Go to **27.**
Step 3. If continuity is not present, go to **15.**

TABLE NO. 12-2-6-7. CARGO HOOK EMERG REL TEST Light Does Not Go On During OPEN Or SHORT Test. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

15. Remove jumper wire from P256. Go to 16.

16. Repair/replace wiring (Appendix F), as required between, then go to 27:
P242-c and S3-1
P256-3 and S3-1
P256-2 and S3-4

17. With crewman's cargo hook pendant EMER RLSE button pressed, check continuity between P267-C and P267-E.

Step 1. If continuity is present, go to **18.**
Step 2. If continuity is not present, replace crewman's cargo hook pendant (PARA 12-4-62). Go to **27.**

18. Check for 28 vdc between J267-E and ground.

Step 1. If voltage is as specified, repair/replace wiring between J267-C and J248-T (Appendix F). Go to **27.**
Step 2. If voltage is not as specified, repair/replace wiring between J267-E and J248-S (Appendix F). Go to **27.**

S70–I **19. With copilot's HOOK EMER REL button pressed, check continuity between P103-F and P103-G.**

Step 1. If continuity is present, go to **20.**
Step 2. If continuity is not present, troubleshoot copilot's collective stick grip (PARA 11-2-2). Go to **27.**

S70–I **20. Check for 28 vdc between J103-G and ground.**

Step 1. If voltage is as specified, repair/replace wiring between J103-F and P248-T (Appendix F). Go to **27.**
Step 2. If voltage is not as specified, repair/replace wiring between J103-G and P248-S (Appendix F). Go to **27.**

S70–II **21. With copilot's HOOK EMER REL switch pressed, check continuity between P103-F and P103-H.**

Step 1. If continuity is present, go to **22.**
Step 2. If continuity is not present, troubleshoot copilot's collective stick grip (PARA 11-2-2). Go to **27.**

TABLE NO. 12-2-6-7. CARGO HOOK EMERG REL TEST Light Does Not Go On During OPEN Or SHORT Test. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

S70–II **22. Check for 28 vdc between J103-H and ground.**

Step 1. If voltage is as specified, repair/replace wiring between J103-F and P248-T (Appendix F). Go to **27.**
Step 2. If voltage is not as specified, repair/replace wiring between J103-H and P248-S (Appendix F). Go to **27.**

S70–I **23. With pilot's HOOK EMER REL button pressed, check continuity between P104-F and P104-G.**

Step 1. If continuity is present, go to **24.**
Step 2. If continuity is not present, troubleshoot pilot's collective stick grip (PARA 11-2-2). Go to **27.**

S70–I **24. Check for 28 vdc between J104-G and ground.**

Step 1. If voltage is as specified, repair/replace wiring between J104-F and P248-T (Appendix F). Go to **27.**
Step 2. If voltage is not as specified, repair/replace wiring between J104-G and P248-S (Appendix F). Go to **27.**

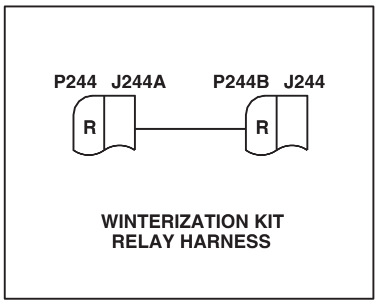
S70–II **25. With pilot's HOOK EMERG RELHOOK EMER REL switch pressed, check continuity between P104-F and P104-H.**

Step 1. If continuity is present, go to **26.**
Step 2. If continuity is not present, troubleshoot pilot's collective stick grip (PARA 11-2-2). Go to **27.**

S70–II **26. Check for 28 vdc between J104-H and ground.**

Step 1. If voltage is as specified, repair/replace wiring between J104-F and P248-T (Appendix F). Go to **27.**
Step 2. If voltage is not as specified, repair/replace wiring between J104-H and P248-S (Appendix F). Go to **27.**

27. Procedure completed.



DETAIL A
(SEE NOTE 4)

LEGEND

———	ELECTRICAL
— —	MECHANICAL

NOTES

1. SWITCH S3 IS SHOWN IN THE NORM POSITION. CONNECTIONS BETWEEN CONTACTS 2 AND 3, AND 4 AND 5 ARE MADE IN THE OPEN POSITION. CONNECTIONS BETWEEN CONTACTS 2 AND 3, AND 5 AND 6 ARE MADE IN THE SHORT POSITION.
2. RESISTORS R1 AND R4 PROVISIONAL.
3. W/O EMEP CAPACITORS C1 THROUGH C6 NOT INSTALLED.
4. WINTER
5. IF HOOK DOES NOT LATCH (LOAD BEAM SWINGS UP BUT CANNOT SUPPORT WEIGHT), LINKAGE SAFE SWITCH WILL STAY CLOSED CAUSING CARGO HOOK OPEN LIGHT TO STAY ON.
6. WHEN HOOK STARTS TO OPEN (LOAD BEAM SWINGS DOWN) LOAD BEAM OPEN SWITCH CLOSES.
7. AS SOLENOID BEGINS TO ROTATE, S1 CLOSES. WHEN SOLENOID STOPS ROTATING, S2 CLOSES. WHEN HOOK CLOSES, BOTH SWITCHES OPEN.
8. RES HST
9. S70–I
10. S70–II
11. W/O EMEP PIN FILTERED ADAPTERS (PFA) ARE NOT INSTALLED.
12. CYCLIC STICK GRIP SERIAL NO. 70400–87261–101.
13. CYCLIC STICK GRIP SERIAL NO. 70400–87261–102.

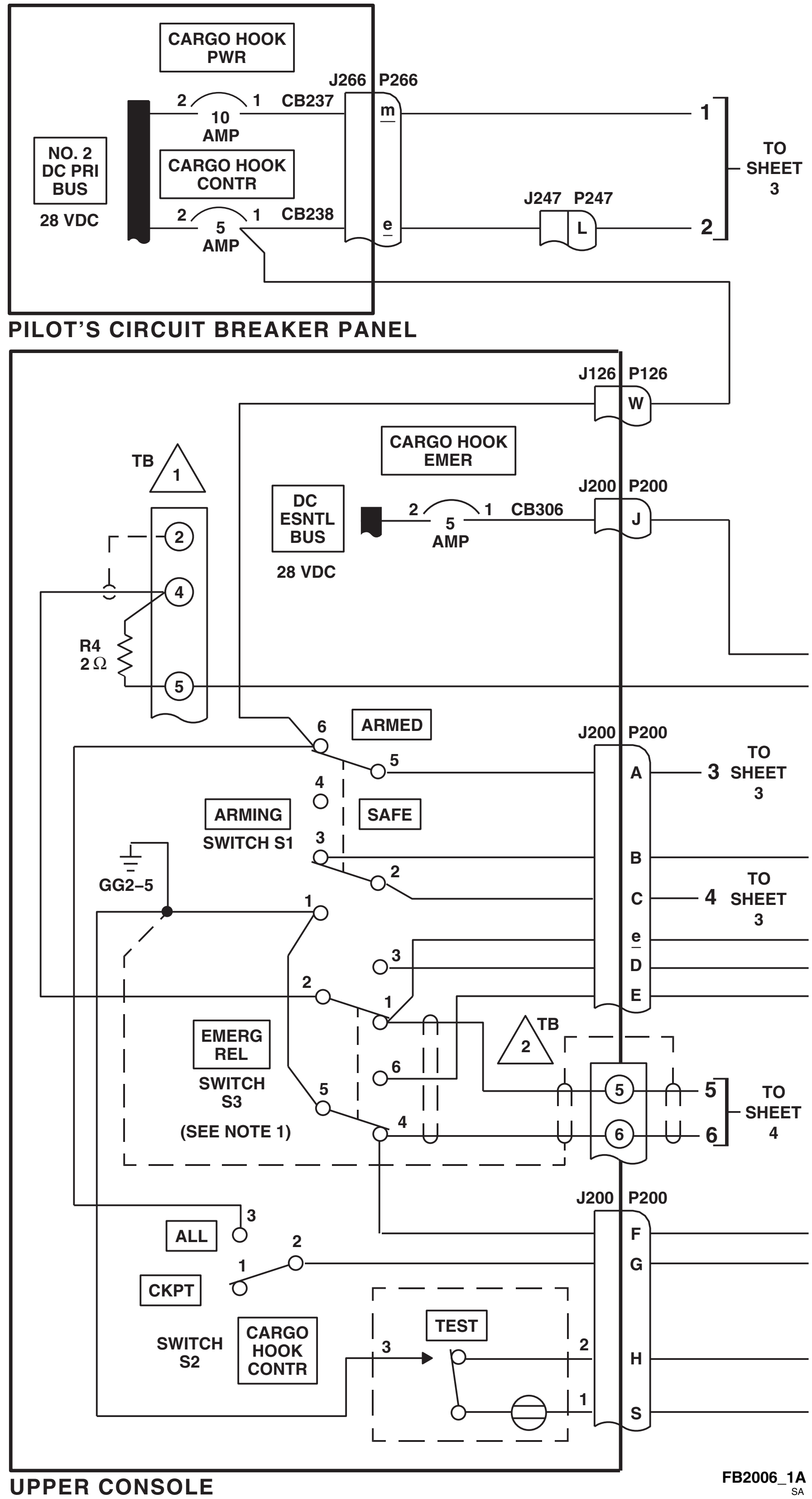


FIGURE 12-2-6-1. CARGO HOOK SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 5)

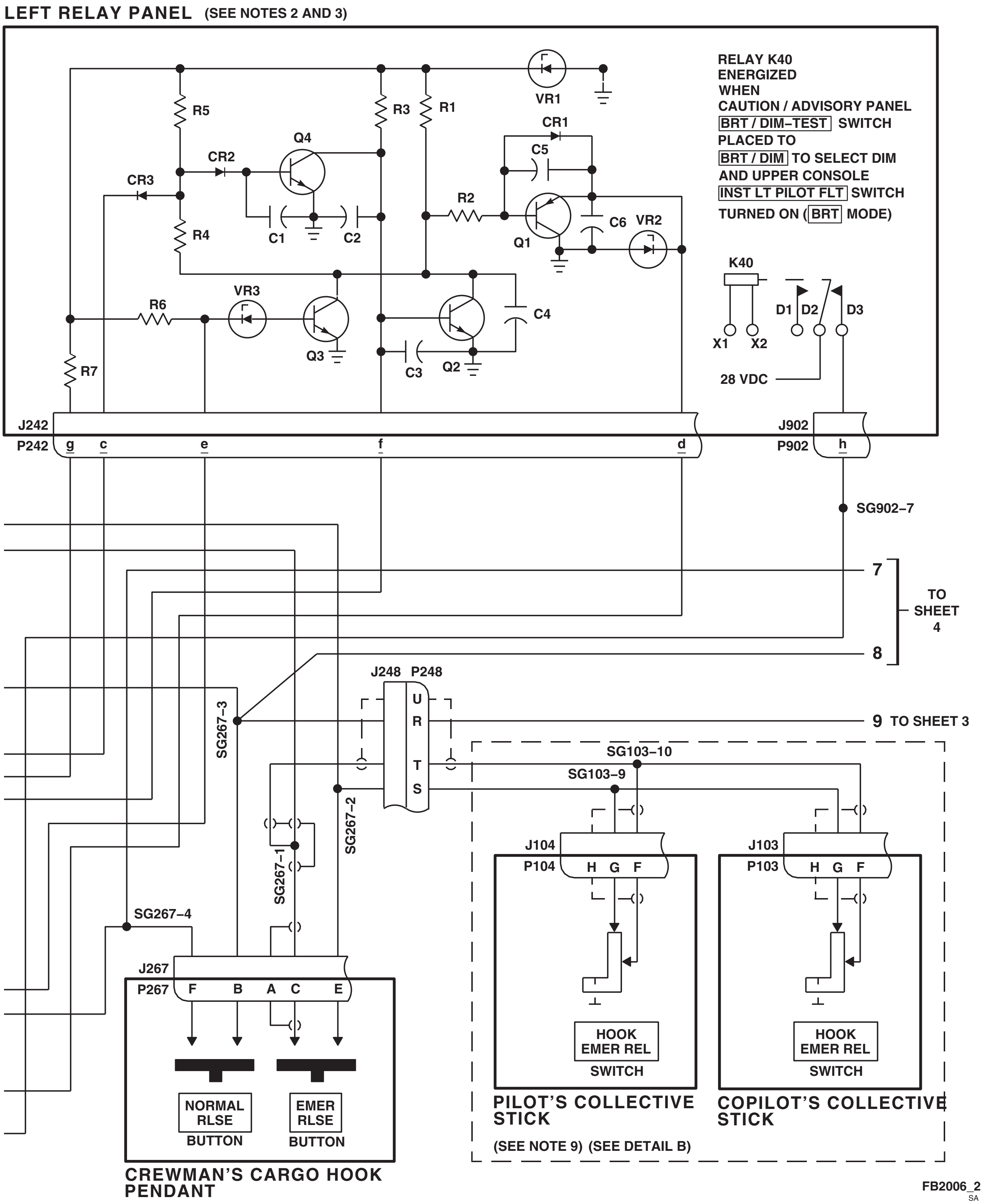


FIGURE 12-2-6-1. CARGO HOOK SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 5)

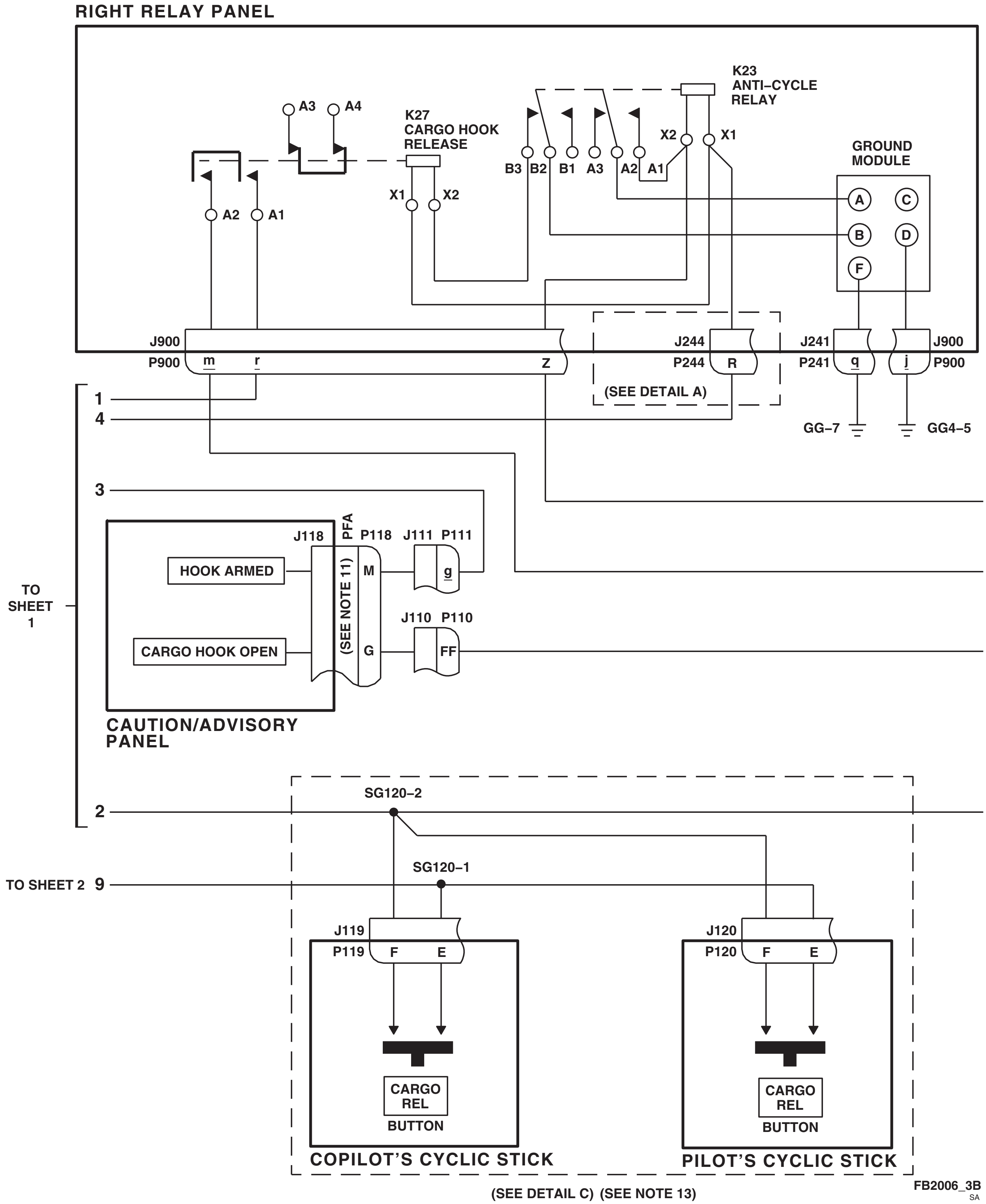


FIGURE 12-2-6-1. CARGO HOOK SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 5)

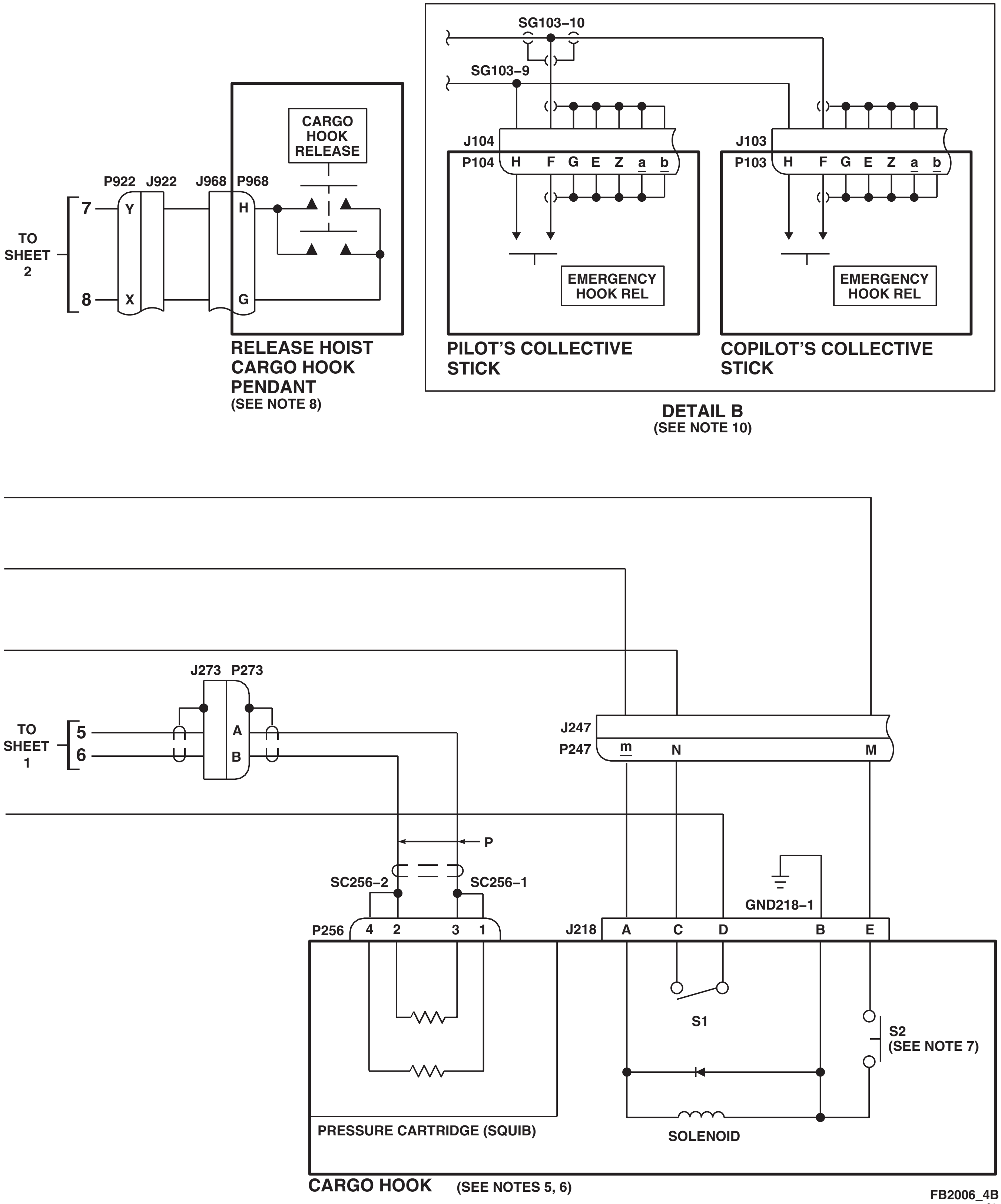
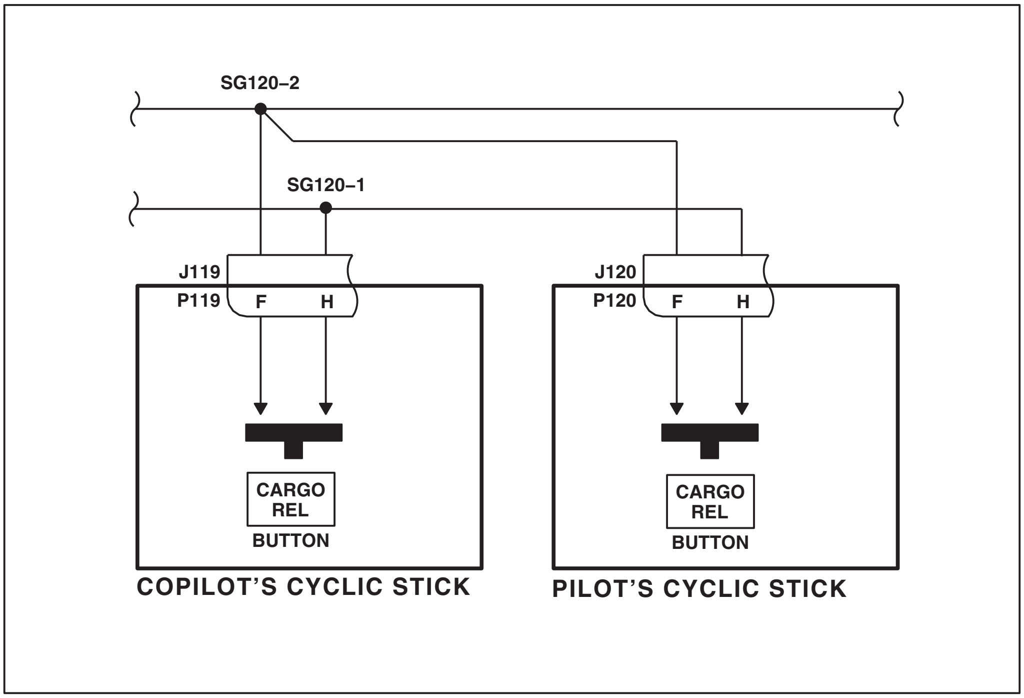


FIGURE 12-2-6-1. CARGO HOOK SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 5)



DETAIL C
(SEE NOTE 12)

FB2006_5
SA

FIGURE 12-2-6-1. CARGO HOOK SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 5)

12-2-7 CREWMAN'S CARGO HOOK PENDANT (BENCH).

INITIAL SETUP

Test Equipment

- Insulation Test Set, 412
- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

12-2-7.1 Operational/Troubleshooting Procedure.

a. Visually check grip for cracks.

RESULT: Grip shall not be cracked.

- If result is not as specified, replace pendant assembly (PARA 12-4-62).

b. Inspect switch guards for spring tension.

RESULT: Guard tension shall be enough to keep guard in place over switch.

- If result is not as specified, replace guard spring (PARA 12-4-63).

c. Remove cover from pendant assembly and inspect for broken or burnt wires, security of components, and general condition.

RESULT: Wires shall not be broken or burnt, and components shall be secure and in good condition.

- If result is not as specified, repair/replace, as required.

d. Measure resistance between P267-B and P267-F.

RESULT: Meter shall indicate infinite resistance.

- If result is not as specified, replace NORMAL RLSE switch (PARA 12-4-63).

e. With NORMAL RLSE switch pressed, measure resistance between P267-B and P267-F.

RESULT: Meter shall indicate continuity.

- If result is not as specified, check resistance between P267-B and NORMAL RLSE switch terminal 1, and between P267-F and NORMAL RLSE switch terminal 2.

  (a) If continuity is present, replace NORMAL RLSE switch (PARA 12-4-63).

  (b) If continuity is not present, repair/replace wiring, as required.

f. Check resistance between P267-C and P267-E.

RESULT: Meter shall indicate infinite resistance.

- If result is not as specified, replace EMER RLSE switch (PARA 12-4-63).

g. With EMER RLSE switch pressed, measure resistance between P267-C and P267-E.

RESULT: Meter shall indicate continuity.

- If result is not as specified, check resistance between P267-C and EMER RLSE switch terminal 1, and between P267-E and EMER RLSE switch terminal 2.

  (a) If continuity is present, replace EMER RLSE switch (PARA 12-4-63).

  (b) If continuity is not present, repair/replace wiring, as required.

h. Connect insulation test set to P267-B and P267-F and apply 1000 Vrms, 60 Hz.

RESULT: The insulation resistance meter shall indicate greater than 1000 megohms.

- If result is not as specified, replace NORMAL RLSE switch (PARA 12-4-63).
- If meter indicates greater than 100 megohms, but less than 1000 megohms, repair/replace wiring, as required.

i. Connect insulation test set to P267-C and P267-E and apply 1000 Vrms, 60 Hz.

RESULT: The insulation resistance meter shall indicate greater than 1000 megohms.

- If result is not as specified, replace EMER RLSE switch (PARA 12-4-63).
- If meter indicates greater than 100 megohms, but less than 1000 megohms, repair/replace wiring, as required.

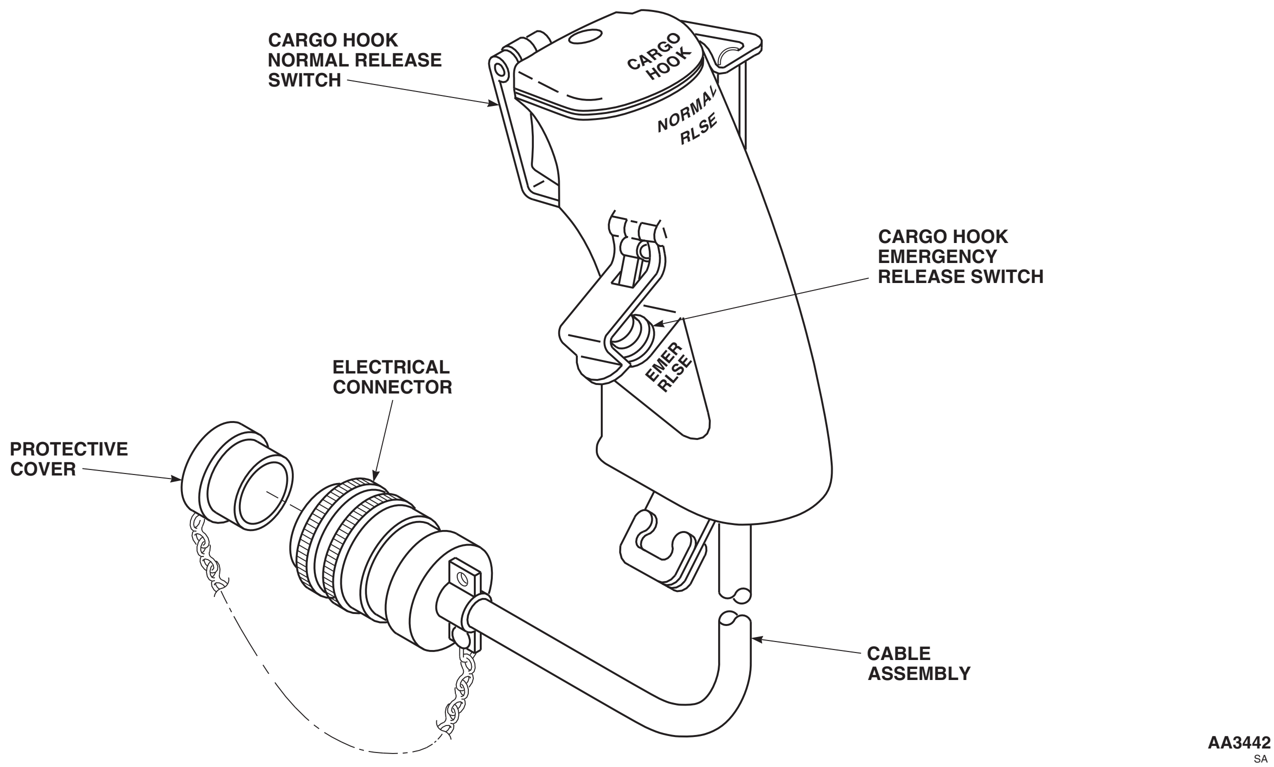
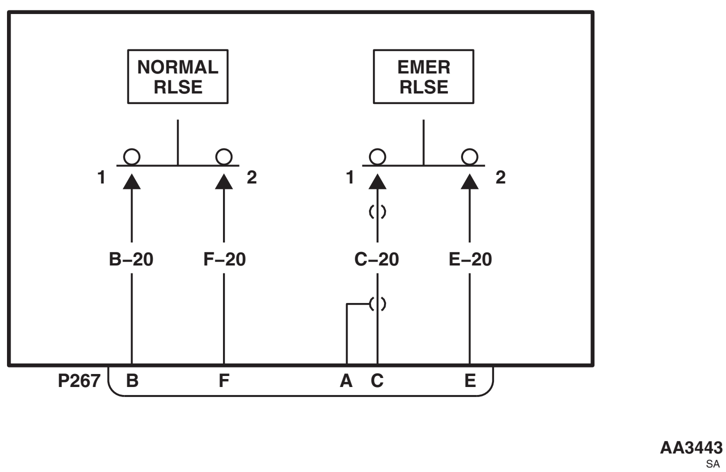


FIGURE 12-2-7-1. CREWMAN'S CARGO HOOK PENDANT



NOTE
SWITCHES ARE PUSHBUTTON, NORMALLY OPEN.

FIGURE 12-2-7-2. CREWMAN'S CARGO HOOK PENDANT SCHEMATIC DIAGRAM

12-2-8 BLADE DE-ICE TEST PANEL (BENCH).

INITIAL SETUP

Test Equipment

- Locally-Made Blade De-ice Test Set and Cables, Figure H-155, Appendix H
- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- 0 to 40 vdc Bench Power Available
- 115 vac, 400 Hz, 3ø Bench Power Available

Special Instructions

The blade de-ice test panel operational/troubleshooting procedure is subdivided as follows:

(1) Setup

(2) OAT Resistance

(3) SYNC/EOT

(4) Tail Rotor Fail

(5) Main Rotor Fail

(6) Shutdown

12-2-8.1 Setup.

a. Connect test cable to J3 of blade de-ice test set (test set) and J1 of blade de-ice test panel (test panel).

b. Place all test set switches to OFF or NORM, SYNC switch to 0V and test panel switch to NORM.

c. Connect locally-made 28 vdc cable to J1 of test set and 28 vdc bench power.

d. Connect locally-made 115 vac cable to J2 of test set and 115V, 3ø 400 Hz bench power.

12-2-8.2 OAT Resistance.

a. Do Setup, if not already done.

b. Measure resistance or check continuity per TABLE NO. 12-2-8-1.

TABLE NO. 12-2-8-1. OAT Resistance.

TEST PANEL SWITCH POSITION	TEST SET, TEST JACKS	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
NORM	W - X	13.167 - 13.433	Replace R14 (PARA 12-5-7)
SYNC 1	W - X	13.167 - 13.433	Replace S1 (PARA 12-5-2)
SYNC 2	W - X	13.167 - 13.433	Replace S1 (PARA 12-5-2)

TABLE NO. 12-2-8-1. OAT Resistance. (Cont)

TEST PANEL SWITCH POSITION	TEST SET, TEST JACKS	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
OAT	W - X	0	J1-W - S1D-C S1D-2 - S1E-2 S1E-C - J1-X
EOT	W - X	97.6 - 98.6	Replace R7 (PARA 12-5-7)
NORM	W - Z	0	J1-W - S1D-C S1D-5 - J1-Z
SYNC 1	W - Z	0	J1-W - S1D-C S1D-4 - S1D-5 S1D-5 - J1-Z
SYNC 2	W - Z	0	J1-W - S1D-C S1D-3 - S1D-4 S1D-4 - S1D-5 S1D-5 - J1-Z
OAT	W - Z	Infinity	Replace S1 (PARA 12-5-2)
EOT	W - Z	Infinity	Replace S1 (PARA 12-5-2)
NORM	X - Y	0	J1-X - S1E-C S1E-5 - J1-Y
SYNC 1	X - Y	0	J1-X - S1E-C S1E-4 - S1E-5 S1E-5 - J1-Y
SYNC 2	X - Y	0	J1-X - S1E-C S1E-3 - S1E-4 S1E-4 - S1E-5 S1E-5 - J1-Y
OAT	X - Y	Infinity	Replace S1 (PARA 12-5-2)
EOT	W - Z	Infinity	Replace S1 (PARA 12-5-2)

RESULT: Meter shall indicate corresponding resistance or continuity.

- If result is not as specified, do corresponding CORRECTIVE ACTION replacement or continuity check.

(a) If continuity is present, replace switch S1 (PARA 12-5-2).

(b) If continuity is not present, repair/replace wiring, as required.

c. If no other check is to be done, go to Shutdown.

12-2-8.3 SYNC/EOT.

a. Do Setup, if not already done.

b. Check continuity per TABLE NO. 12-2-8-2.

TABLE NO. 12-2-8-2. SYNC/EOT Continuity.

TEST PANEL SWITCH POSITION	TEST SET, TEST JACKS	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
NORM	N - P	0	J1-N - S1A-5 S1A-C - J1-P
OAT	N - P	0	J1-N - S1A-5 S1A-5 - S1A-1 S1A-1 - S1A-2 S1A-C - J1-P
EOT	N - P	0	J1-N - S1A-5 S1A-5 - S1A-1 S1A-1 - S1A-C S1A-C - J1-P
EOT	R - S	Infinity	Replace K3 (PARA 12-5-7)
EOT	R - U	Infinity	Replace K3 (PARA 12-5-7)

RESULT: Meter shall indicate corresponding resistance.

- If result is not as specified, do corresponding CORRECTIVE ACTION replacement or continuity check.

  (a) If continuity is present, replace switch S1 (PARA 12-5-2).

  (b) If continuity is not present, repair/ replace wiring, as required.

c. Place test panel switch to NORM.

d. Place test set POWER 28 VDC and 115 VAC switches to ON.

RESULT:

(1) Test set 28 VDC and 115 VAC indicator lamps shall go on.

(2) Test panel information plate lamps shall light evenly.

- If result (1) is not as specified, replace test set.

- If result (2) is not as specified, replace information plate lamp (PARA 12-4-48).

  (a) If trouble remains, replace information plate (PARA 12-4-48).

- If none of the information plate lamps light, check continuity between J1-b and P3-1, and between J1-c and P3-2.

(a) If continuity is present, replace information plate (PARA 12-4-48).

(b) If continuity is not present, repair/ replace wiring, as required.

e. Place test set FAULT SELECT ICE DET switch to FAULT.

f. Place test panel switch to SYNC 1.

g. Check for -33.0 to -27.0 vdc between test set TEST POINTS P and T.

RESULT: Meter shall indicate -33.0 to -27.0 vdc.

- If result is not as specified, go to TABLE NO. 12-2-8-3.

h. Place test set FAULT SELECT ICE DET switch to NORM.

i. Place test panel switch to EOT.

j. Check continuity between test set TEST POINTS R and S, and between R and U.

RESULT: Meter shall not indicate continuity.

- If result is not as specified, go to TABLE NO. 12-2-8-4.

k. Place test set SYNC switch to -5V.

l. Check resistance between test set TEST POINTS R and S, and between R and U.

RESULT: Meter shall indicate less than 0.5 ohm resistance.

- If result is not as specified, go to TABLE NO. 12-2-8-5.

m. Place test set SYNC switch to 0V.

n. Check continuity between test set TEST POINTS R and S, and between R and U.

RESULT: Meter shall not indicate continuity.

- If result is not as specified, replace relay K3 (PARA 12-5-7).

o. Place test set SYNC switch to -30V.

p. Check continuity between test set TEST POINTS R and S, and between R and U.

RESULT: Meter shall not indicate continuity.

- If result is not as specified, go to TABLE NO. 12-2-8-6.

q. Place test set SYNC switch to OV.

r. Place test panel switch to the following positions and check continuity between the following points:

SWITCH POSITION	CHECK CONTINUITY BETWEEN
NORM	TEST POINT R and S1B-5
	TEST POINT R and S1C-5
SYNC 1	TEST POINT R and S1B-4
	TEST POINT R and S1C-4
SYNC 2	TEST POINT R and S1B-3
	TEST POINT R and S1C-3
OAT	TEST POINT R and S1B-2
	TEST POINT R and S1C-2

RESULT: Meter shall indicate continuity.

- If result is not as indicated, replace switch S1 (PARA 12-5-2).

s. If no other check is to be done, go to Shutdown.

12-2-8.4 Tail Rotor Fail.

a. Do Setup, if not already done.

b. Momentarily press test panel PWR TAIL RTR lamp.

RESULT: Lamp shall light when pressed and shall go off when released.

- If result is not as specified, replace lamp DS2 (PARA 12-4-47).

(a) If trouble remains, check continuity between the following:

P2-23	and	DS2-2
P2-22	and	DS2-3
P2-14	and	DS2-1

1 If continuity is present, replace lamp switch assembly (PARA 12-5-3).

2 If continuity is not present, repair/replace wiring, as required.

c. Check voltage between test set TEST POINTS f and T.

RESULT: Meter shall indicate 0 vdc.

- If result is not as specified, replace relay K2 (PARA 12-5-7).

d. Place test set TAIL ROTOR Aø and Bø switches to ON.

e. Place test set FAULT SELECT CONT PNL switch to FAULT.

RESULT: Test panel PWR TAIL RTR lamp shall light.

- If result is not as specified, go to TABLE NO. 12-2-8-7.

f. Check voltage between test set TEST POINTS G and T, and between J and T.

RESULT:

(1) Meter shall indicate 0 vdc between G and T.

(2) Meter shall indicate 0 vdc between J and T.

- If result (1) is not as specified, replace diode CR6 (PARA 12-5-7).

- If result (2) is not as specified, replace diode CR13 (PARA 12-5-7).

g. Check voltage between test set TEST POINTS K and f.

RESULT: Meter shall indicate 0.6 to 0.8 vdc.

- If result is not as specified, check continuity between J1-K and E51, and between J1-f and E47.

(a) If continuity is present, replace diode CR14 (PARA 12-5-7).

(b) If continuity is not present, repair/replace wiring, as required.

h. Place test set FAULT SELECT CONT PNL switch to NORM.

RESULT: Test panel PWR TAIL RTR lamp shall go off.

- If result is not as specified, replace relay K2 (PARA 12-5-7).

i. Check voltage between test set TEST POINTS f and T.

RESULT: Meter shall indicate 0 vdc.

- If result is not as specified, replace relay K2 (PARA 12-5-7).

j. Place test set FAULT SELECT TR FAIL switch to FAULT.

RESULT: Test set PWR TAIL RTR lamp shall light.

- If result is not as specified, check continuity between J1-G and E45.

  (a) If continuity is present, replace diode CR6 (PARA 12-5-7).

  (b) If continuity is not present, repair/ replace wiring.

k. Place test set CURRENT TRANSFORMER Aø switch to ON.

RESULT:

(1) Test panel PWR TAIL RTR lamp shall remain on.

(2) Test panel PWR MAIN RTR lamp shall be off.

- If results are not as specified, replace diode CR5 (PARA 12-5-7).

l. Place test set CURRENT TRANSFORMER Aø switch to OFF.

m. Place test set TAIL ROTOR Aø and Bø switches to OFF.

RESULT: Test panel PWR TAIL lamp shall go off in less than 2 seconds.

- If result is not as specified, replace capacitor C2 (PARA 12-5-7).

n. Place test set FAULT SELECT TR FAIL switch to NORM.

o. Place test set TAIL ROTOR Bø and Cø switches to ON.

p. Place test set FAULT SELECT TR FAIL switch to FAULT.

RESULT: Test panel PWR TAIL RTR lamp shall light.

- If result is not as specified, go to TABLE NO. 12-2-8-8.

q. Place test set TAIL ROTOR Bø and Cø switches to OFF.

RESULT: Test panel PWR TAIL RTR lamp shall go off in less than 2 seconds.

- If result is not as specified, replace capacitor C2 (PARA 12-5-7).

r. Place test set FAULT SELECT TR FAIL switch to NORM.

s. Place test set TAIL ROTOR Aø and Cø switches to ON.

t. Place test set FAULT SELECT TR FAIL switch to FAULT.

RESULT: Test panel PWR TAIL RTR lamp shall light.

- If result is not as specified, go to TABLE NO. 12-2-8-9.

u. Place test set TAIL ROTOR Aø and Cø switches to OFF.

RESULT: Test panel PWR TAIL RTR lamp shall go off in less than 2 seconds.

- If result is not as specified, replace capacitor C2 (PARA 12-5-7).

v. Place test set FAULT SELECT TR FAIL switch to NORM.

w. If no other check is to be done, go to Shutdown.

12-2-8.5 Main Rotor Fail.

a. Do Setup, if not already done.

b. Momentarily press test panel PWR MAIN RTR lamp.

RESULT: Test panel PWR MAIN RTR lamp shall go on when pressed and go off when released.

- If result is not as specified, replace lamp DS1 (PARA 12-4-47).

(a) If trouble remains, check continuity between the following:

P2-21	and	DS1-3
P2-24	and	DS1-2
P2-15	and	DS1-1

1 If continuity is present, replace lamp switch assembly (PARA 12-5-3).

2 If continuity is not present, repair/replace wiring, as required.

c. Check voltage between test set TEST POINTS e and T.

RESULT: Meter shall indicate 0 vdc.

- If result is not as specified, replace relay K1 (PARA 12-5-7).

d. Place test set CURRENT TRANSFORMER Aø switch to ON.

e. Place test set FAULT SELECT CONT PNL switch to FAULT.

RESULT: Test panel PWR MAIN RTR lamp shall light.

- If result is not as specified, go to TABLE NO. 12-2-8-10.

f. Check for voltage between test set TEST POINTS K and e.

RESULT: Meter shall indicate between 0.6 and 0.8 vdc.

- If result is not as specified, replace diode CR15 (PARA 12-5-7).

g. Place test set FAULT SELECT CONT PNL switch to NORM.

RESULT: Test panel PWR MAIN RTR lamp shall go off.

- If result is not as specified, replace relay K1 (PARA 12-5-7).

h. Check for voltage between test set TEST POINTS e and T.

RESULT: Meter shall indicate 0 vdc.

- If result is not as specified, replace relay K1 (PARA 12-5-7).

i. Place test set FAULT SELECT MR FAIL switch to FAULT.

RESULT: Test panel PWR MAIN RTR lamp shall light.

- If result is not as specified, check continuity between J1-J and E54.

(a) If continuity is present, replace diode CR13 (PARA 12-5-7).

(b) If continuity is not present, repair/replace wiring.

j. Place test set TAIL ROTOR Aø and Bø switches to ON.

RESULT: Test panel PWR MAIN RTR lamp shall remain on and test panel PWR TAIL RTR lamp shall be off.

- If result is not as specified, replace diode CR12 (PARA 12-5-7).

k. Place test set TAIL ROTOR Aø and Bø switches to OFF.

l. Place test set CURRENT TRANSFORMER Aø switch to OFF.

RESULT: Test panel PWR MAIN RTR lamp shall go off in less than 5 seconds.

- If result is not as specified, replace capacitor C3 (PARA 12-5-7).

m. Place test set FAULT SELECT MR FAIL switch to NORM.

n. Place test set CURRENT TRANSFORMER Bø switch to ON.

o. Place test set FAULT SELECT MR FAIL switch to FAULT.

RESULT: Test panel PWR MAIN RTR lamp shall light.

- If result is not as specified, replace diode CR9 (PARA 12-5-7).

p. Place test set CURRENT TRANSFORMER Bø switch to OFF.

RESULT: Test panel PWR MAIN RTR lamp shall go off in less than 5 seconds.

- If result is not as specified, replace capacitor C3 (PARA 12-5-7).

q. Place test set FAULT SELECT MR FAIL switch to NORM.

r. Place test set CURRENT TRANSFORMER Cø switch to ON.

s. Place test set FAULT SELECT MR FAIL switch to FAULT.

RESULT: Test panel PWR MAIN RTR lamp shall light.

- If result is not as specified, replace diode CR10 (PARA 12-5-7).

t. Place test set CURRENT TRANSFORMER Cø switch to OFF.

RESULT: Test panel PWR MAIN RTR lamp shall go off in less than 5 seconds.

- If result is not as specified, replace capacitor C3 (PARA 12-5-7).

u. Place test set DETECT switch to ON.

v. Place test set FAULT SELECT MR FAIL switch to FAULT.

w. Place test set CURRENT TRANSFORMER Aø, Bø, and Cø switches to ON.

RESULT: Test panel PWR MAIN RTR lamp shall remain off.

- If result is not as specified, check for 95,000 to 105,000 ohms resistance across resistor R4.

  (a) If meter reading is not as specified, replace resistor R4 (PARA 12-5-7).

  (b) If meter reading is 100,000 ohms, replace zener diode VR8 (PARA 12-5-7).

x. If no other check is to be done, go to Shutdown.

12-2-8.6 Shutdown.

a. Place all test set switches to OFF or NORM.

b. Disconnect test set from test panel.

TABLE NO. 12-2-8-3. Voltage Output Measured Between Test Set TEST POINTS P And T Is Not Between -33.0 And -27.0 VDC When Test Set FAULT SELECT ICE DET Switch Is Placed To FAULT And Test Panel Switch Is Placed To SYNC 1.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for -33.0 to -27.0 vdc at zener diode VR1 anode.

Step 1. If voltage is as specified, repair/replace wiring, as required. Go to **7.**
Step 2. If voltage is not as specified, go to **2.**

TABLE NO. 12-2-8-3. Voltage Output Measured Between Test Set TEST POINTS P And T Is Not Between -33.0 And -27.0 VDC When Test Set FAULT SELECT ICE DET Switch Is Placed To FAULT And Test Panel Switch Is Placed To SYNC 1. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Is voltage output zero?

Step 1. If voltage output is zero, go to **3.**
Step 2. If voltage output is not zero, go to **6.**

3. Check diode CR1 for open circuit.

Step 1. If open circuit is present, replace diode CR1 (PARA 12-5-7). Go to **7.**
Step 2. If open circuit is not present, go to **4.**

4. Check resistor R1 for 41,000 to 45,000 ohms.

Step 1. If resistance is as specified, go to **5.**
Step 2. If resistance is not as specified, replace resistor R1 (PARA 12-5-7). Go to **7.**

5. Check capacitor C1 for continuity.

Step 1. If continuity is present, replace capacitor C1 (PARA 12-5-7). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring. Go to **7.**

6. Check resistor R1 for 41,000 to 45,000 ohms.

Step 1. If resistance is as specified, replace zener diode VR1 (PARA 12-5-7). Go to **7.**
Step 2. If resistance is not as specified, replace resistor R1 (PARA 12-5-7). Go to **7.**

7. Procedure completed.

TABLE NO. 12-2-8-4. Open Circuit Does Not Exist Between Test Set TEST POINTS R And S, And R And U When FAULT SELECT ICE DET Switch Is Placed To NORM And Test Panel Switch Is Placed To EOT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between TEST POINT R and switch S1B-C.

Step 1. If continuity is present, go to **2.**

TABLE NO. 12-2-8-4. Open Circuit Does Not Exist Between Test Set TEST POINTS R And S, And R And U When FAULT SELECT ICE DET Switch Is Placed To NORM And Test Panel Switch Is Placed To EOT. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, repair/replace wiring. Go to **15.**

2. Check continuity between S1B-C and S1B-1.

Step 1. If continuity is present, go to **3.**
Step 2. If continuity is not present, replace test panel switch (PARA 12-5-2). Go to **15.**

3. Check continuity between S1B-1 and P2-20.

Step 1. If continuity is present, go to **4.**
Step 2. If continuity is not present, repair/replace wiring. Go to **15.**

4. Check continuity between S1B-C and S1C-C.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring. Go to **15.**

5. Check continuity between S1C-C and S1C-1.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, replace test panel switch (PARA 12-5-2). Go to **15.**

6. Check continuity between S1C-1 and P2-19.

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, repair/replace wiring. Go to **15.**

7. Check for 26 to 28 vdc between S1A-1 and ground.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, repair/replace wiring. Go to **15.**

8. Check for 14.5 to 15.5 vdc between transistor Q6 emitter and ground.

Step 1. If voltage is as specified, go to **9.**
Step 2. If voltage is not as specified, replace zener diode VR7 (PARA 12-5-7). Go to **15.**

TABLE NO. 12-2-8-4. Open Circuit Does Not Exist Between Test Set TEST POINTS R And S, And R And U When FAULT SELECT ICE DET Switch Is Placed To NORM And Test Panel Switch Is Placed To EOT. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Check for 0.6 to 1.0 vdc between transistor Q4 base and ground.

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, go to **14.**

10. Check for 0.5 to 1.5 vdc between transistor Q4 collector and ground.

Step 1. If voltage is as specified, go to **11.**
Step 2. If voltage is not as specified, replace transistor Q4 (PARA 12-5-7). Go to **15.**

11. Check for 26 to 28 vdc between transistor Q5 collector and ground.

Step 1. If voltage is as specified, go to **12.**
Step 2. If voltage is not as specified, replace transistor Q5 (PARA 12-5-7). Go to **15.**

12. Check diode CR16 for short circuit.

Step 1. If short circuit is present, replace diode CR16 (PARA 12-5-7). Go to **15.**
Step 2. If short circuit is not present, go to **13.**

13. Check continuity between:
Relay K3-A2 and test point S
Relay K3-B2 and test point U

Step 1. If continuity is present, replace relay K3 (PARA 12-5-7). Go to **15.**
Step 2. If continuity is not present, repair/replace wiring, as required. Go to **15.**

14. Check for 13 to 14 vdc between transistor Q6 collector and ground.

Step 1. If voltage is as specified, replace transistor Q4 and/or associated circuitry (PARA 12-5-7). Go to **15.**
Step 2. If voltage is not as specified, replace transistor Q5 and/or associated circuitry (PARA 12-5-7). Go to **15.**

15. Procedure completed.

TABLE NO. 12-2-8-5. Resistance Check For Less Than 0.5 Ohms Does Not Exist Between Test Points R And S, And R And U When Test Set SYNC Switch Is Placed To -5V, FAULT SELECT ICE DET Switch Is Placed To NORM, And Test Panel Switch Is Placed To EOT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 2.0 to 4.0 vdc between transistor Q4 collector and ground.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **4.**

2. Check for 2 to 3 vdc between transistor Q5 collector and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, replace transistor Q5 (PARA 12-5-7). Go to **6.**

3. Check diode CR16 for short or open circuit.

Step 1. If short or open circuit is present, replace diode CR16 (PARA 12-5-7). Go to **6.**
Step 2. If short or open circuit is not present, replace relay K3 (PARA 12-5-7). Go to **6.**

4. Check for -3 to -1 vdc between transistor Q4 base and ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace transistor Q4 and/or associated components (PARA 12-5-7). Go to **6.**

5. Check for 15 vdc between zener diode VR7 cathode and ground.

Step 1. If voltage is as specified, replace transistor Q5 and/or associated components (PARA 12-5-7). Go to **6.**
Step 2. If voltage is not as specified, replace zener diode VR7 (PARA 12-5-7). Go to **6.**

6. Procedure completed.

TABLE NO. 12-2-8-6. Open Circuit Does Not Exist Between Test Set TEST POINTS R And S, And R And U When Test Set SYNC Switch Is Placed To -30V, FAULT SELECT ICE DET Switch Is Placed To NORM, And Test Panel Switch Is Placed To EOT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for -50 to -45 vdc across zener diode VR1.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, replace capacitor C1 and/or associated circuitry (PARA 12-5-7). Go to **8.**

2. Check for 14.5 to 15.5 vdc between transistor Q6 emitter and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, replace zener diode VR7 (PARA 12-5-7). Go to **8.**

3. Check for 0.6 to 1.0 vdc between transistor Q4 base and ground.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **7.**

4. Check for 0.5 to 1.5 vdc between transistor Q4 collector and ground.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, replace transistor Q4 (PARA 12-5-7). Go to **8.**

5. Check for 26 to 28 vdc between transistor Q5 collector and ground.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, replace transistor Q5 (PARA 12-5-7). Go to **8.**

6. Check diode CR16 for short or open circuit.

Step 1. If short or open circuit is present, replace diode CR16 (PARA 12-5-7). Go to **8.**
Step 2. If short or open circuit is not present, replace relay K3 (PARA 12-5-7). Go to **8.**

7. Check for 13 to 14 vdc between transistor Q6 collector and ground.

Step 1. If voltage is as specified, replace transistor Q4 and/or associated circuitry (PARA 12-5-7). Go to **8.**

TABLE NO. 12-2-8-6. Open Circuit Does Not Exist Between Test Set TEST POINTS R And S, And R And U When Test Set SYNC Switch Is Placed To -30V, FAULT SELECT ICE DET Switch Is Placed To NORM, And Test Panel Switch Is Placed To EOT. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace transistor Q6 and/or associated circuitry (PARA 12-5-7). Go to **8.**

8. Procedure completed.

TABLE NO. 12-2-8-7. Test Panel PWR TAIL RTR Lamp Does Not Light When Test Set TAIL ROTOR Aø And Bø Switches Are Placed To ON And FAULT SELECT CONT PNL Switch Is Placed To FAULT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 1 to 3 vdc between transistor Q1 collector and ground.

Step 1. If voltage is as specified, replace transistor Q1 and/or associated circuitry (PARA 12-5-7). Go to **4.**
Step 2. If voltage is not as specified, go to **2.**

2. Check capacitor C2 for short circuit.

Step 1. If short circuit is present, replace capacitor C2 (PARA 12-5-7). Go to **4.**
Step 2. If short circuit is not present, go to **3.**

3. Check diode CR7 for short or open circuit.

Step 1. If short or open circuit is present, replace diode CR7 (PARA 12-5-7). Go to **4.**
Step 2. If short or open circuit is not present, replace relay K2 (PARA 12-5-7). Go to **4.**

4. Procedure completed.

TABLE NO. 12-2-8-8. Test Panel PWR TAIL RTR Lamp Does Not Light When Test Set TAIL ROTOR Bø And Cø Switches Are Placed To ON And FAULT SELECT TR FAIL Switch Is Placed To FAULT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between:
J1-B and terminal E42
J1-C and terminal E43

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

2. Check diode CR3 for open circuit.

Step 1. If open circuit is present, replace diode CR3 (PARA 12-5-7). Go to **3.**
Step 2. If open circuit is not present, replace diode CR4 (PARA 12-5-7). Go to **3.**

3. Procedure completed.

TABLE NO. 12-2-8-9. Test Panel PWR TAIL RTR Lamp Does Not Light When Test Set TAIL ROTOR Aø And Cø Switches Are Placed To ON And FAULT SELECT TR FAIL Switch Is Placed To FAULT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between:
J1-A and terminal E41
J1-C and terminal E43

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, repair/replace wiring. Go to **3.**

2. Check diode CR2 for open circuit.

Step 1. If open circuit is present, replace diode CR2 (PARA 12-5-7). Go to **3.**
Step 2. If open circuit is not present, replace diode CR4 (PARA 12-5-7). Go to **3.**

3. Procedure completed.

TABLE NO. 12-2-8-10. Test Panel PWR MAIN RTR Lamp Does Not Light When Test Set CURRENT TRANSFORMER Aø Switch Is Placed To ON And FAULT SELECT CONT PNL Switch Is Placed To FAULT.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 2.5 to 4.5 vdc between transistor Q2 base and ground.**

   Step 1. If voltage is as specified, go to **2.**
   Step 2. If voltage is not as specified, replace capacitor C3 (PARA 12-5-7). Go to **5.**

2. **Check for 3.7 to 5.7 vdc between transistor Q2 collector and ground.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, replace transistor Q2 and/or associated circuitry (PARA 12-5-7). Go to **5.**

3. **Check for 26 to 27 vdc between transistor Q3 collector and ground.**

   Step 1. If voltage is as specified, go to **4.**
   Step 2. If voltage is not as specified, replace transistor Q3 and/or associated circuitry (PARA 12-5-7). Go to **5.**

4. **Check diode CR11 for short circuit.**

   Step 1. If short circuit is present, replace diode CR11 (PARA 12-5-7). Go to **5.**
   Step 2. If short circuit is not present, replace relay K1 (PARA 12-5-7). Go to **5.**

5. **Procedure completed.**

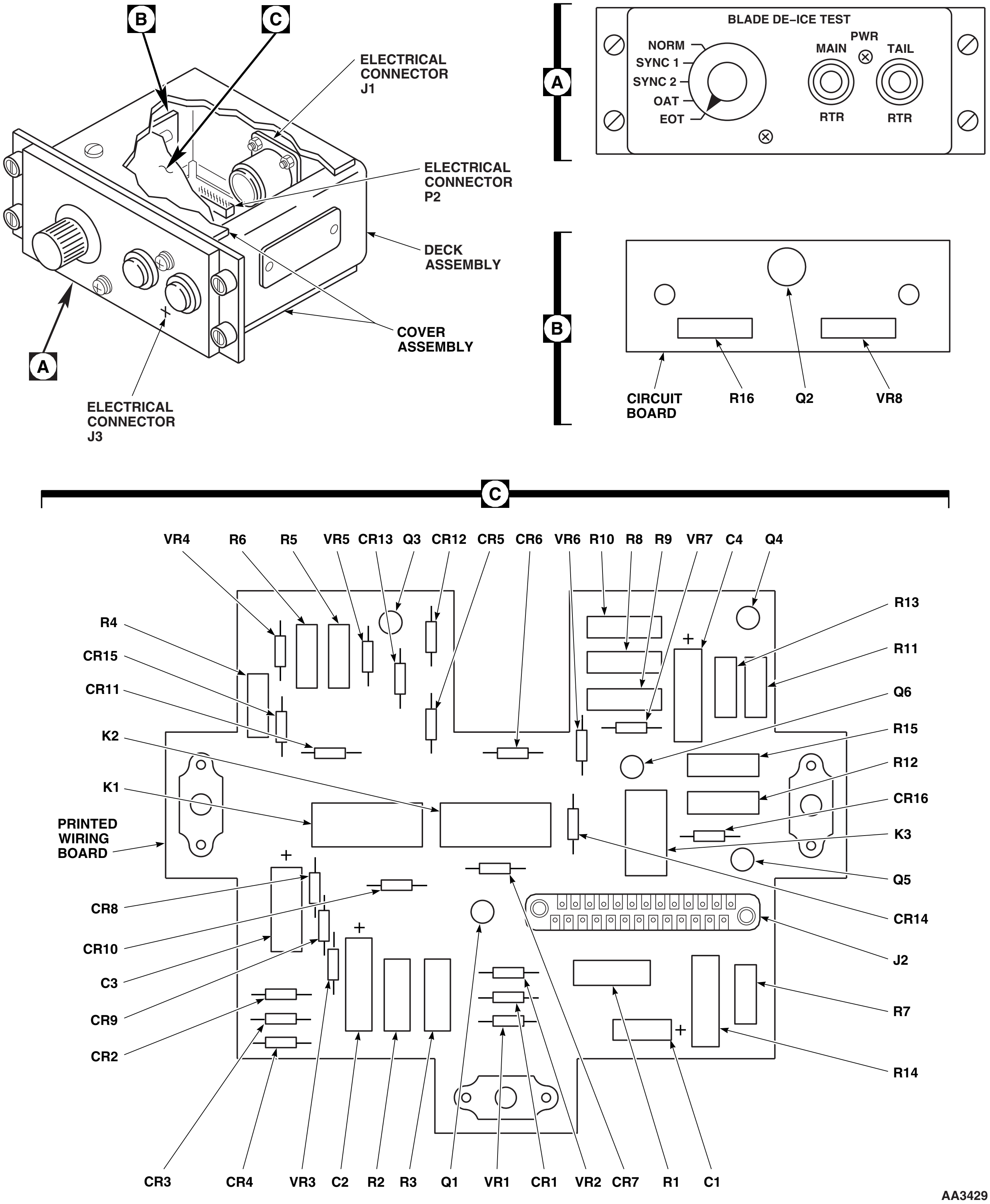
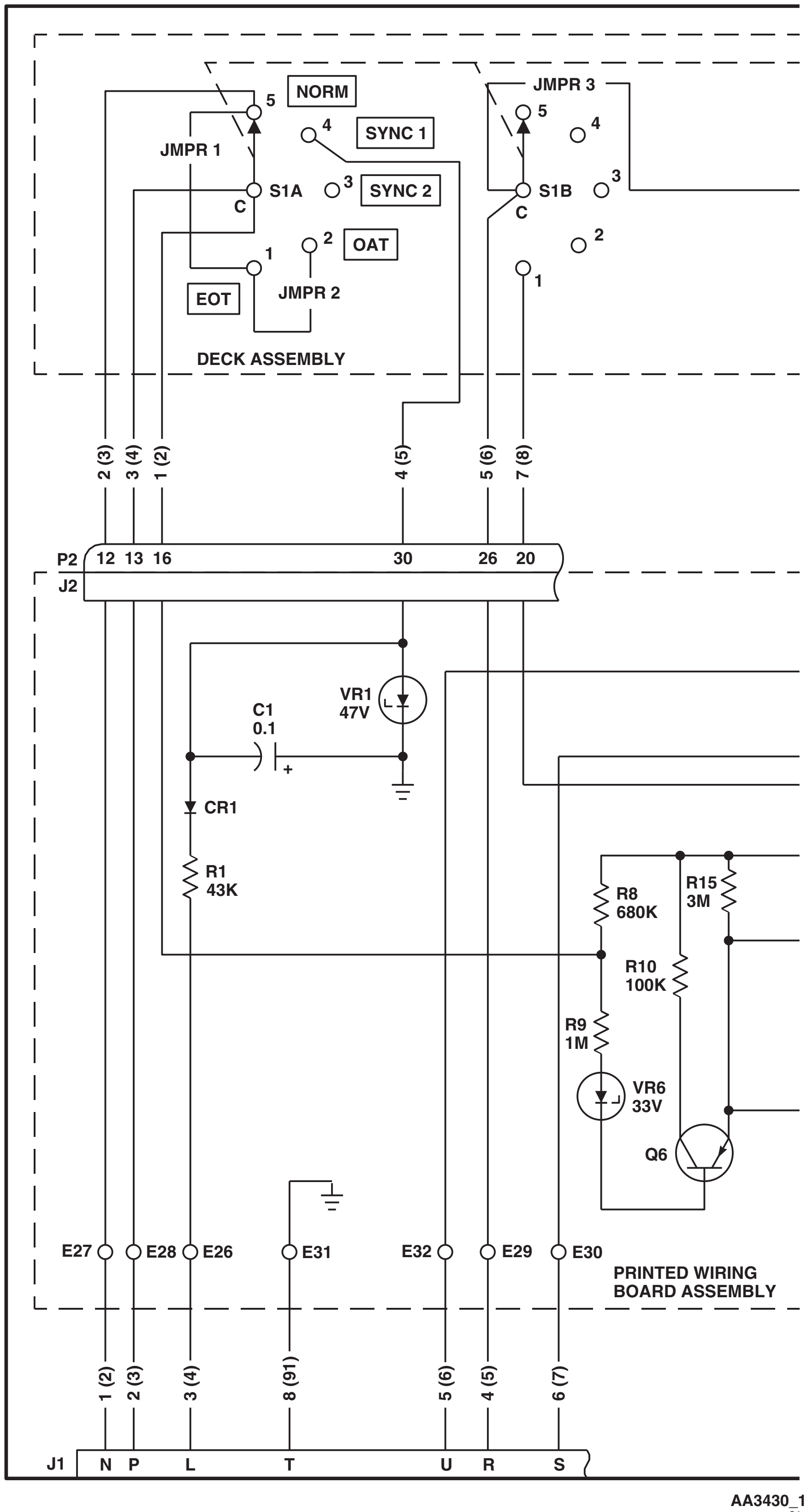


FIGURE 12-2-8-1. BLADE DE-ICE TEST PANEL



LEGEND
CASE GROUND

NOTES

1. ALL WIRES ARE 22 GAUGE.
2. WIRE CODES ARE DESIGNATED AS FOLLOWS:
   COLOR CODE
   12 (9 0 2)
   WIRE NUMBER
   BASE COLOR
   FIRST TRACER
   SECOND TRACER
3. ALL RESISTOR VALUES ARE IN OHMS.
4. ALL CAPACITORS ARE IN MICRO–FARADS UNLESS OTHERWISE SPECI–FIED. POLARITY AS INDICATED (+).
5. ALL RELAYS ARE SHOWN DE–ENERGIZED.

FIGURE 12-2-8-2. BLADE DE-ICE TEST PANEL SCHEMATIC DIAGRAM (SHEET 1 OF 4)

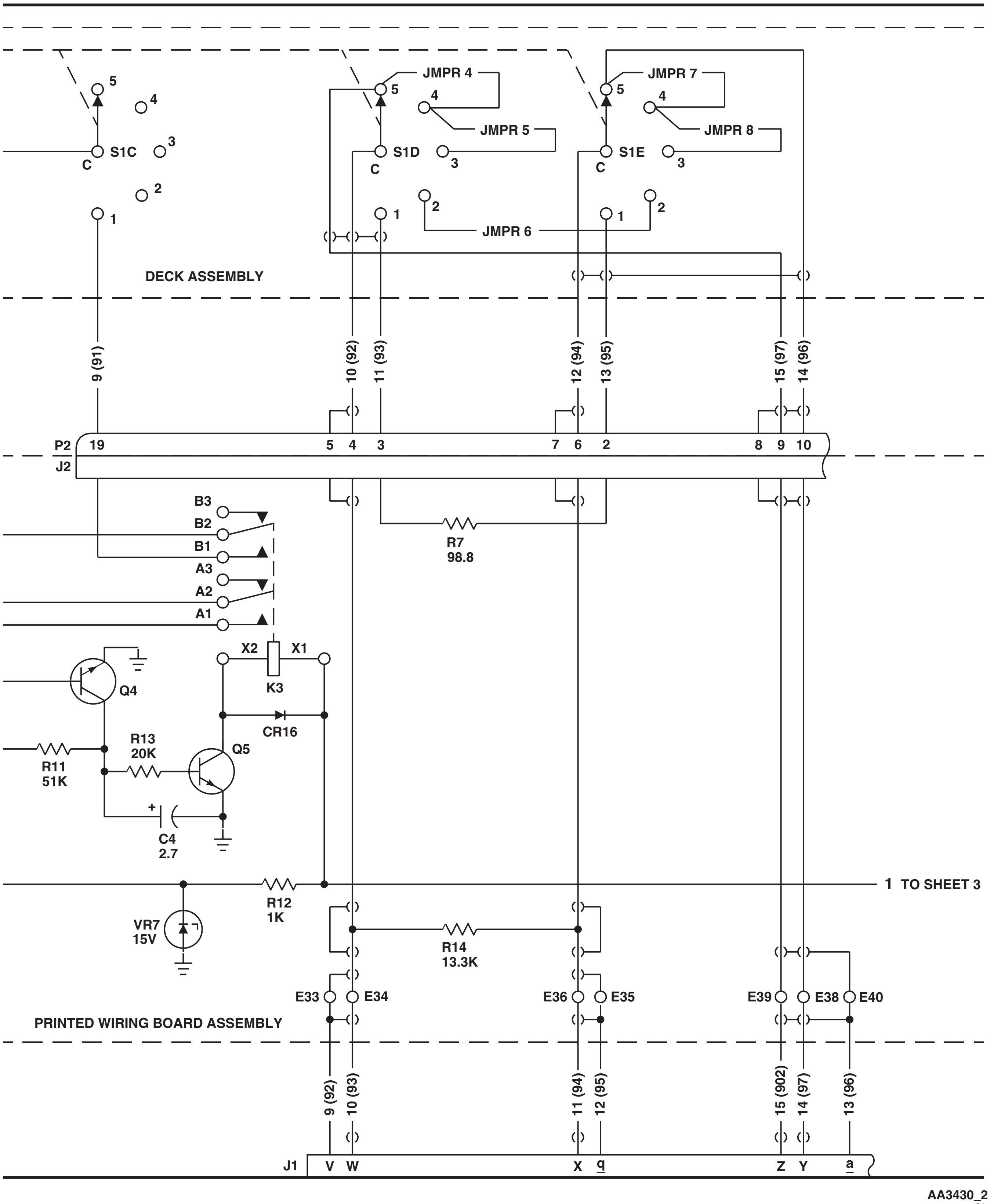


FIGURE 12-2-8-2. BLADE DE-ICE TEST PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 4)

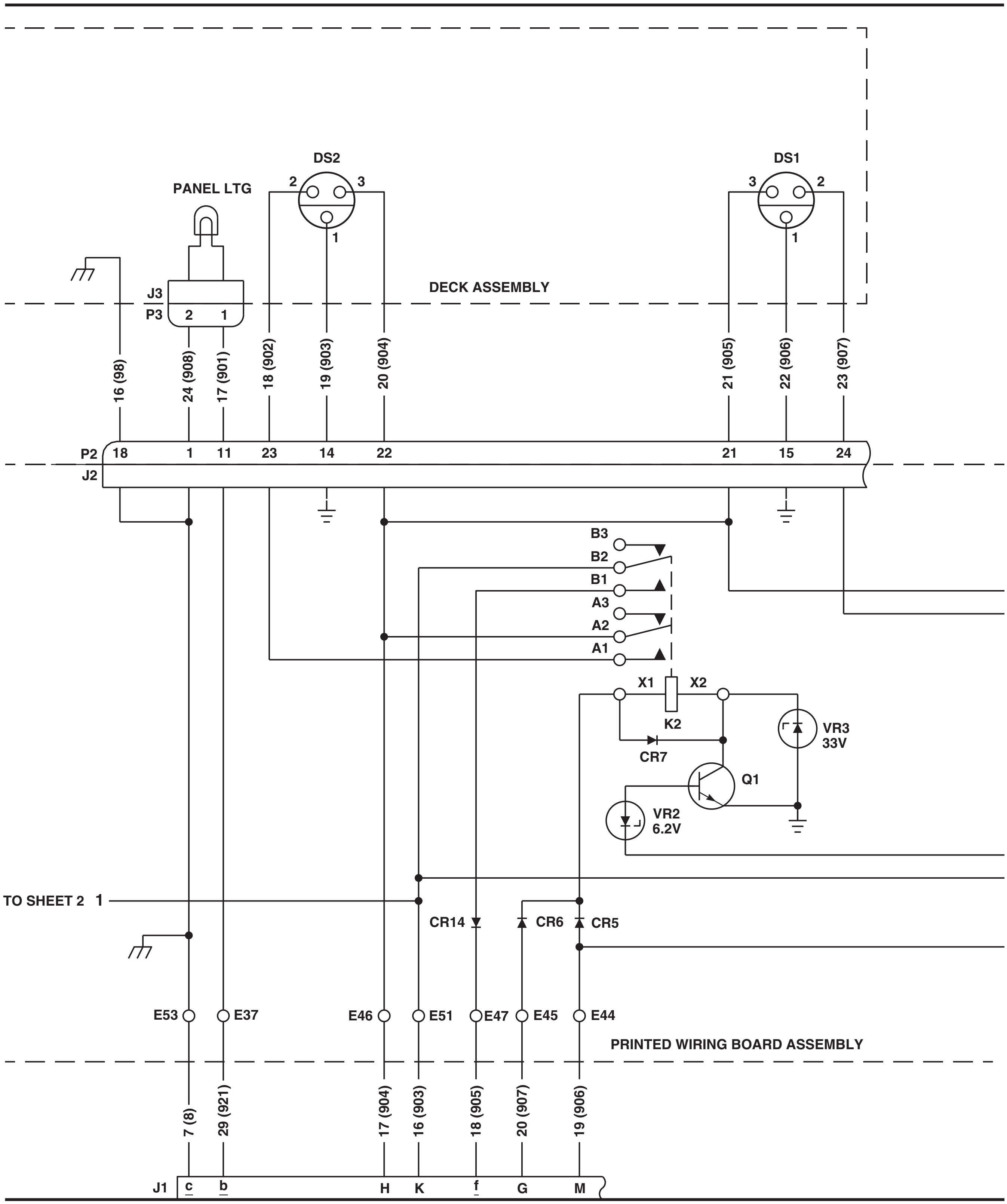


FIGURE 12-2-8-2. BLADE DE-ICE TEST PANEL SCHEMATIC DIAGRAM (SHEET 3 OF 4)

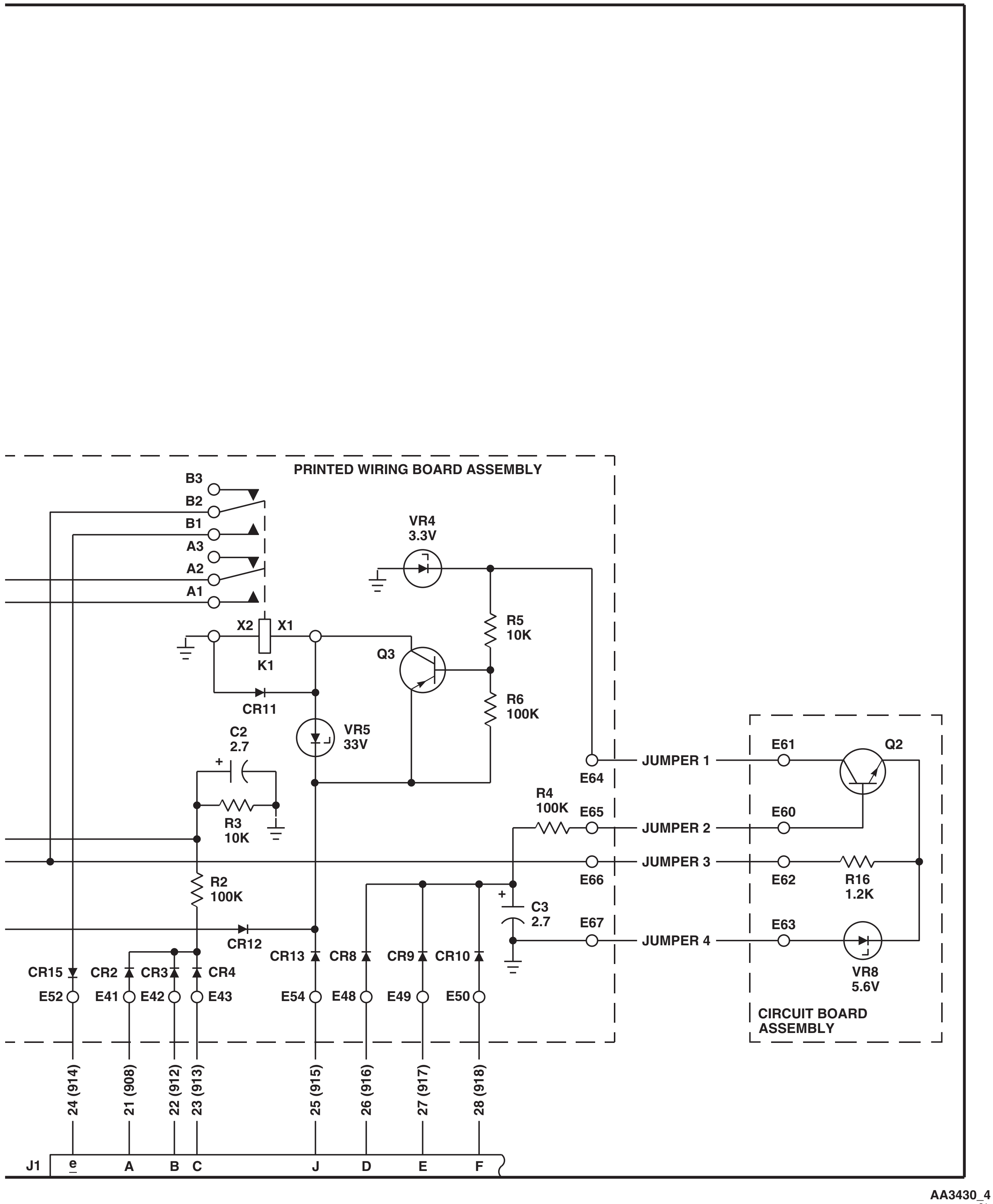


FIGURE 12-2-8-2. BLADE DE-ICE TEST PANEL SCHEMATIC DIAGRAM (SHEET 4 OF 4)

SECTION 12-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

12-3-1 FIRE EXTINGUISHING SYSTEM INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 12-4-5

12-3-1.1 Discharge Tubes Inspection.

Check for dents or distortion in tubes. **DENTS SHALL NOT EXCEED 0.080-INCH DEEP OR BE WITHIN 4 INCHES OF EACH OTHER.** If limits are exceeded, replace tube(s) (PARA 12-4-5).

12-3-2 WINDSHIELD WIPER INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Spanner Wrench
- Spring Scale

Materials/Parts

- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 12-4-19

12-3-2.1 Windshield Wiper Arm and Link Inspection.

a. Check wiper arm and link for cracks and corrosion. **NONE ALLOWED.** If cracked, replace wiper arm and/or link (PARA 12-4-19).

b. Using scouring pad, Item 270, Appendix D, remove corrosion.

c. Check wiper link rod end bearing for binding and play. **NO BINDING ALLOWED. PLAY NOT TO EXCEED 0.015-INCH.** If binding is evident or play exceeds limit, replace rod end bearing.

12-3-2.2 Check/Adjust Windshield Wiper Arm Pressure.

a. Hook spring scale under wiper arm at wiper blade attachment point (Figure 12-3-2-1).

b. Check wiper arm pressure. **SCALE READING SHALL BE BETWEEN 4 AND 5 POUNDS AT POINT WHERE WIPER BLADE LEAVES WINDSHIELD.**

c. If adjustment is required, use spanner wrench to turn pressure adjustment screw counterclockwise to increase pressure, or clockwise to decrease pressure (Figure 12-3-2-1, Detail A).

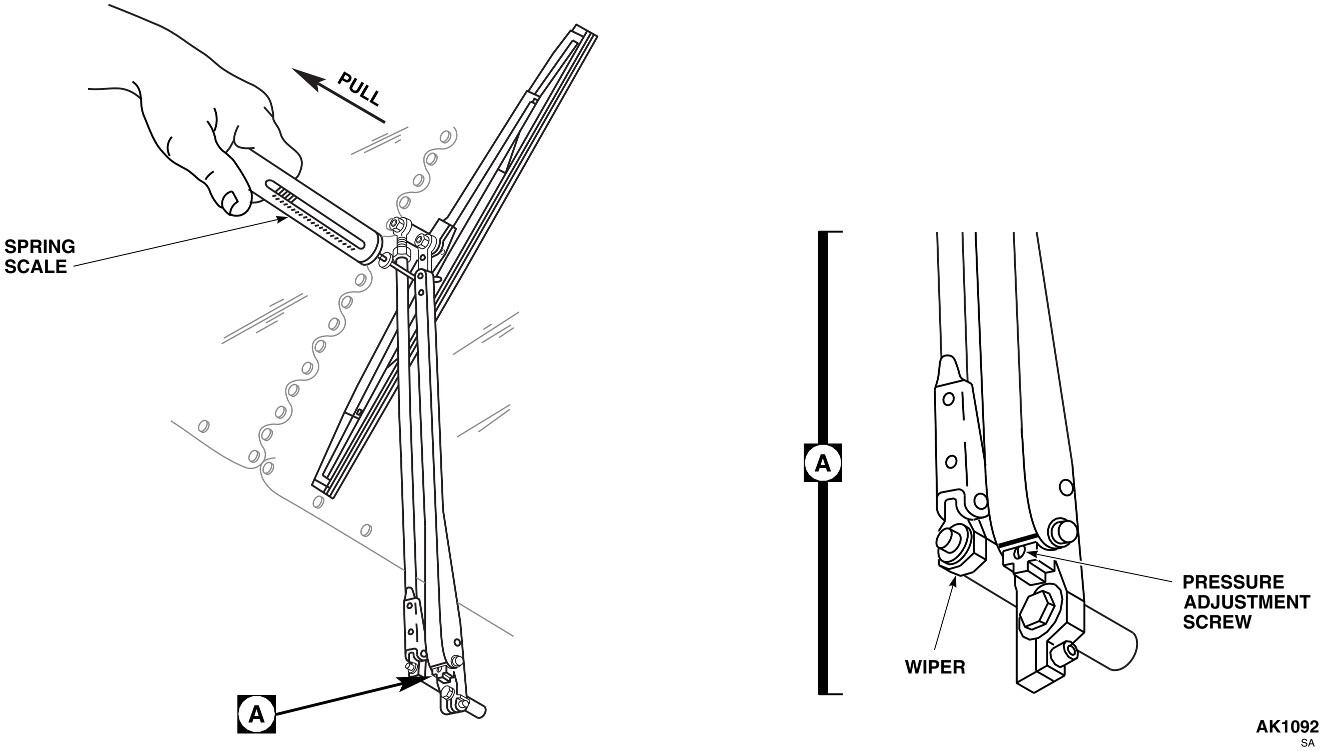


FIGURE 12-3-2-1. CHECK/ADJUST WINDSHIELD WIPER ARM PRESSURE

12-3-3 TAIL ROTOR BLADE DEICE INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-4-15
- PARA 1-6-16
- PARA 12-4-52
- PARA 12-4-54
- PARA 12-4-55

12-3-3.1 Tail Rotor Stator Brush Inspection.

a. Turn off all electrical power (PARA 1-6-16).

b. Visually inspect brushes for wear.

c. If top of one or more brushes is even with brush holder, replace brushes (PARA 12-4-55).

d. If brush spring tension seems weak or brush springs are broken, replace brush retainer board (PARA 12-4-55).

e. If brush assembly cover is dirty and visual inspection cannot be done, perform the following:

(1) Remove tail rotor blade deice slipring stator brush assembly (PARA 12-4-54).

(2) Lift cover from brush box and clean cover using dishwashing compound, Item 122, Appendix D, and cold water.

(3) Inspect brushes for wear. If top of one or more brushes is even with brush holder, replace brushes (PARA 12-4-55).

(4) Install tail rotor blade deice slipring stator brush assembly (PARA 12-4-54).

f. Remove screws securing slipring stator access cover and remove access cover.

g. Clean slipring rotor (PARA 1-4-15).

h. Slowly turn tail rotor blades and inspect slipring rotor contact rings for wear. If white surface has worn through to a yellow substrate, replace slipring rotor (PARA 12-4-52).

i. Remove all access cover gasket material from access cover and slipring rotor.

j. Position access cover and gasket on slipring rotor and secure with screws.

k. Lockwire, Item 196, Appendix D, screws.

12-3-3.2 Tail Rotor Blade Deice Slipring Cables Inspections.

NOTE

- Currently there is **NO ACCEPTABLE REPAIR CRITERIA. IF ANY REPAIRS TO THE SLIPRING CABLE ARE FOUND, THE REPAIR MUST BE REMOVED AND SLIPRING CABLE INSPECTED.**
- If aircraft is equipped with slipring without removable slipring cables, **THE ENTIRE SLIPRING ASSEMBLY MUST BE REPLACED.** If aircraft is equipped with removable slipring cables, individual slipring cables may be replaced (PARA 12-4-52).

a. Inspect all areas of the slipring cable. If any sections of the jacket show evidence of wear, cracking, or splitting, use the following criteria to determine the necessary action. **IF ANY OF THE FOLLOWING CONDITIONS EXIST, SLIPRING CABLE MUST BE REPLACED** (PARA 12-4-52):

(1) Helical wire core of the cable assembly sleeve is visible, broken, bent, nicked, or exhibits any sharp edges.

(2) Continuity/electrical check shows any signs of leakage or breakdown.

(3) Any convoluted (twisted) tube is cracked radially.

(4) Any convoluted (twisted) tube is cracked longitudinally.

(5) The jacket is separated from the mold or fittings at either end of the cable.

(6) Any damage which could allow moisture or any other form of contamination to enter into the tube.

(7) Any cracking of the metal backshell or flanges.

(8) If the metal backshell does not freely swivel through 270°.

12-3-3.3 Polyurethane Mold Inspection.

a. **IF ANY OF THE MOLDED SECTIONS SHOWS SIGNS OF CRACKING OR EXCESSIVE SEPARATION, AS DESCRIBED BELOW, SLIPRING CABLE MUST BE REPLACED** (PARA 12-4-52):

(1) Cracking: Any crack in the mold that is visible to the naked eye.

(2) Excessive separation: Greater than ⅛-inch between the cable segment and the molded section when the adjacent slipring cable is bent to a 6-inch radius.

SECTION 12-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

12-4-1 FIRE EXTINGUISHING SYSTEM AGENT CONTAINER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Shorting Jumper
- Spring Scale
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 0 - 300 ft lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Carton
- Lubricant, Water Displacing, Item 201B, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 12-2-2
- PARA 12-4-2

12-4-1.1 Replace Fire Extinguishing System Agent Container.

NOTE

Replacement of both main (right) and reserve (left) fire extinguisher system agent containers is the same.

12-4-1.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open APU compartment access door.

WARNING

Injury to personnel and damage to equipment will result if cartridge does not have shorting jumper installed before being removed. Cartridge could detonate. Make sure shorting jumper is installed before removing cartridge.

c. Slide boot down to expose power terminal on main cartridge. Loosen nut on power terminal and connect shorting jumper between power terminal and ground terminal or grounding lug assembly (Figure 12-4-1-1, Sheet 2, Detail B).

WARNING

Injury to personnel and damage to equipment will result if electrical wires are crossed. Make sure electrical wires are tagged with correct identification.

d. Disconnect wiring from main cartridge as follows:

(1) On HTL fire extinguishing container, perform the following:

(a) Tag wires.

(b) Remove screw, nut, and washers, and braided and ground wires from power and ground terminals.

(2) On Walter Kidde fire extinguishing container, perform the following:

(a) Tag wires.

(b) Remove nuts and washers, and braided and ground wires from power terminal and grounding lug assembly.

(c) Tag manifold tube to discharge outlet with same identification as removed wiring (Figure 12-4-1-1, Sheet 1, Detail A and Sheet 2, Detail B).

WARNING

Injury to personnel and damage to equipment will result if cartridge does not have shorting jumper installed before being removed. Cartridge could detonate. Make sure shorting jumper is installed before removing cartridge.

e. Slide boot down to expose power terminal on reserve cartridge. Loosen nut on power terminal and connect shorting jumper between power terminal and ground terminal or grounding lug assembly (Figure 12-4-1-1, Sheet 2, Detail B).

f. Disconnect wiring from reserve cartridge as follows:

WARNING

Injury to personnel and damage to equipment will result if electrical wires are crossed. Make sure electrical wires are tagged with correct identification.

(1) On HTL fire extinguishing container, perform the following:

(a) Tag wires.

(b) Remove screw, nut, and washers, and braided and ground wires from power and ground terminals.

(2) On Walter Kidde fire extinguishing container, perform the following:

(a) Tag wires.

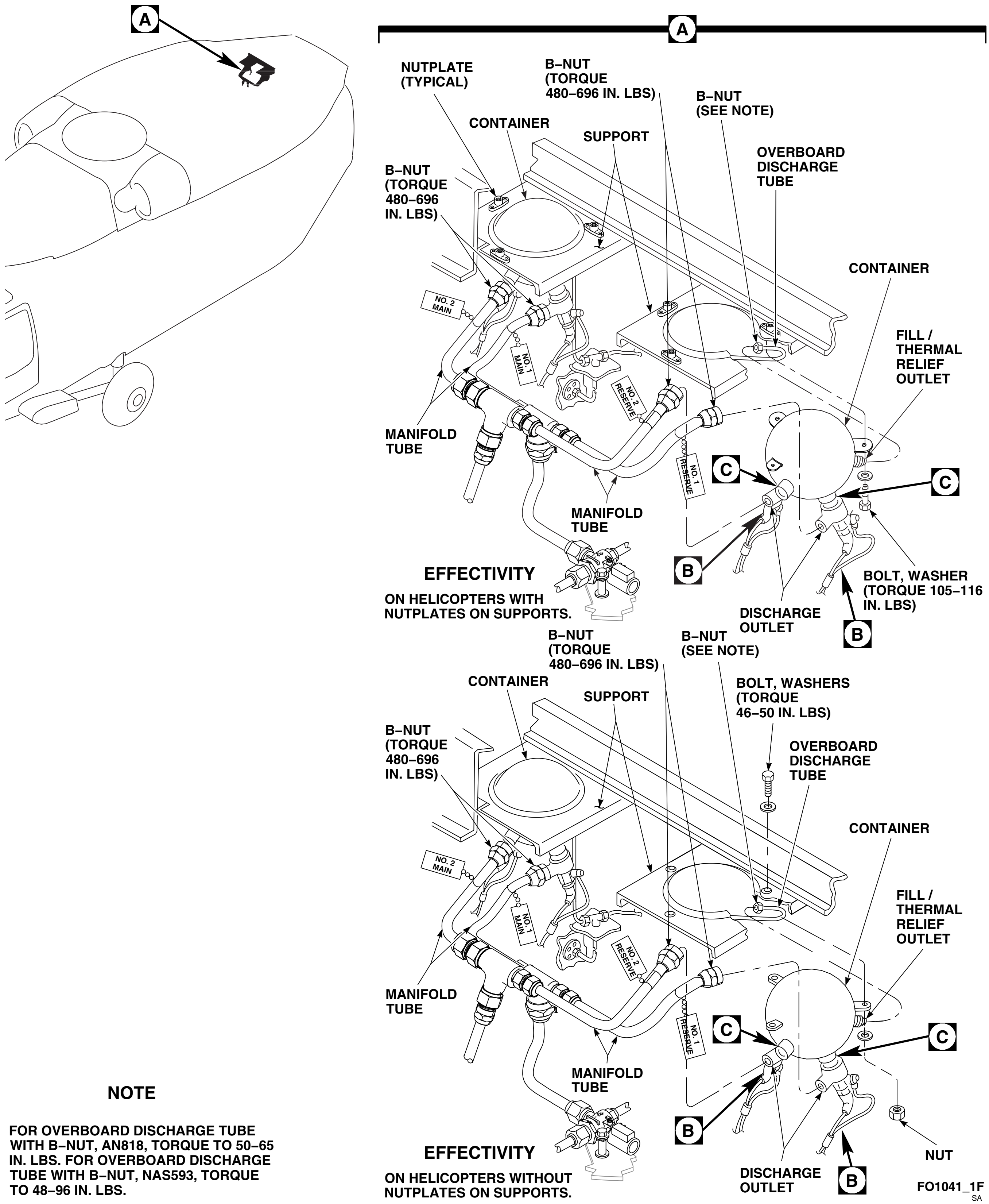
(b) Remove nuts and washers, and braided and ground wires from power terminal and grounding lug assembly.

(c) Tag manifold tube to discharge outlet with same identification as removed wiring (Figure 12-4-1-1, Sheet 1, Detail A and Sheet 2, Detail B).

WARNING

Injury to personnel and damage to equipment will result if fill/thermal relief valve is not secured. To prevent injury to personnel from gas discharge, secure fill/thermal relief valve when disconnecting overboard discharge tube.

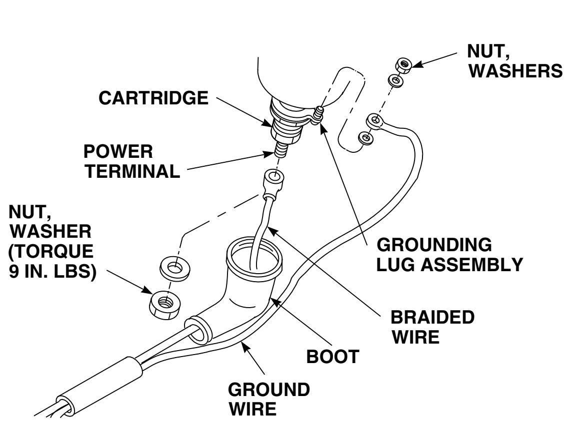
g. Disconnect overboard discharge tube from fill/thermal relief outlet (Figure 12-4-1-1, Sheet 1, Detail A). Plug tube.

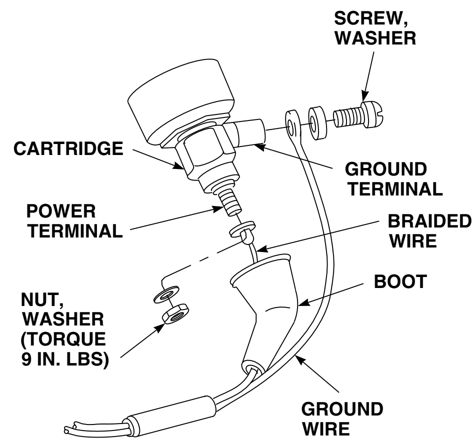


NOTE

FOR OVERBOARD DISCHARGE TUBE WITH B–NUT, AN818, TORQUE TO 50–65 IN. LBS. FOR OVERBOARD DISCHARGE TUBE WITH B–NUT, NAS593, TORQUE TO 48–96 IN. LBS.

FIGURE 12-4-1-1. REPLACE FIRE EXTINGUISHING SYSTEM AGENT CONTAINER (SHEET 1 OF 2)





EFFECTIVITY

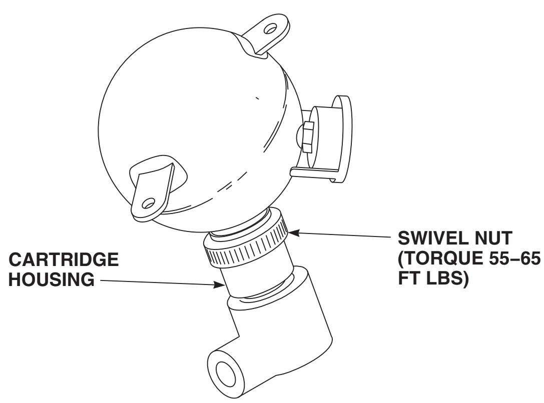
WALTER KIDDE FIRE EXTINGUISHING CONTAINER, 899153.

EFFECTIVITY

HTL FIRE EXTINGUISHING CONTAINER, 30400004.

B

C



FO1041_2C
SA

FIGURE 12-4-1-1. REPLACE FIRE EXTINGUISHING SYSTEM AGENT CONTAINER (SHEET 2 OF 2)

h. Disconnect manifold tubes from discharge outlets. Plug tubes.

i. Remove bolts, washers, and nuts (if installed) securing container to support.

j. Remove container from support. Cap discharge outlets and fill/thermal relief outlet.

k. Preserve container as follows:

(1) Remove container from helicopter.

WARNING

Injury to personnel or damage to equipment will result if cartridge is not removed from container prior to packing.

(2) Remove fire extinguishing agent container cartridges (PARA 12-4-2).

(3) Cap cartridge housing assemblies and fill/ thermal relief outlet on container.

(4) Plug firing cartridge hole.

(5) Pack in carton large enough to allow for 1 inch of cushioning around container.

12-4-1.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install fire extinguishing agent container cartridges (PARA 12-4-2).

c. Using spring scale, weigh container with cartridges installed. **WEIGHT WITH EXTINGUISHING AGENT AND CARTRIDGES MUST BE A MINIMUM OF 5.5 POUNDS TO A MAXIMUM OF 5.9 POUNDS.** If weight is not within limits, replace container.

d. Install container as follows (Figure 12-4-1-1, Sheet 1, Detail A):

(1) On helicopters with nutplates on support, install container to support and secure with bolts and washers. **TORQUE BOLTS TO 105 - 116 INCH-POUNDS.**

(2) On helicopters without nutplates on support, install container to support and secure with bolts, washers, and nuts. **TORQUE BOLTS TO 46 - 50 INCH-POUNDS.**

e. Remove caps from container.

f. Loosen swivel nuts and align cartridge housing assemblies with manifold tubes (Figure 12-4-1-1, Sheet 2, Detail C).

g. Hold cartridge housing assemblies in place and secure with swivel nuts. **TORQUE SWIVEL NUTS TO 55 - 65 FOOT-POUNDS.**

h. Coat threads on discharge outlets and fill/ thermal relief outlet with antiseize compound, Item 74, Appendix D.

i. Unplug and connect manifold tubes to discharge outlets (Figure 12-4-1-1, Sheet 1, Detail A). **TORQUE B-NUTS TO 480 - 696 INCH-POUNDS.**

j. Unplug and connect overboard discharge tube to fill/thermal relief outlet. Torque B-nut as follows:

(1) For overboard discharge tube with B-nut, AN818, **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**

(2) For overboard discharge tube with B-nut, NAS593, **TORQUE B-NUT TO 48 - 96 INCH-POUNDS.**

k. Perform operational check of fire extinguishing system (PARA 12-2-2).

l. Apply water displacing lubricant, Item 201B, Appendix D, to power terminal and ground terminal/grounding lug assembly (Figure 12-4-1-1, Sheet 2, Detail B).

WARNING

Injury to personnel and damage to equipment will result if electrical wires are crossed. Pay particular attention to information noted on tags prior to connecting wiring.

m. Connect wiring as follows:

(1) Match NO. 1 MAIN or NO. 1 RESERVE electrical wiring to manifold tube tagged with same identification.

(2) On HTL fire extinguishing container, install braided and ground wires on power and ground terminals and secure with screw, nut, and washers.

(3) On Walter Kidde fire extinguishing container, install braided and ground wires on power terminal and grounding lug assembly and secure with nuts and washers.

(4) Remove tags.

n. Remove shorting jumper between power terminal and ground terminal.

o. Tighten nut securing braided wire to power terminal. **TORQUE NUT ON POWER TERMINAL TO 9 INCH-POUNDS.** Install boot.

p. Repeat step k. through step p. to connect NO. 2 MAIN or NO. 2 RESERVE electrical wiring to cartridge.

q. Remove tags from manifold tubes.

r. Make sure area is clean and free of foreign material.

s. Close APU compartment access door.

12-4-2 FIRE EXTINGUISHING AGENT CONTAINER CARTRIDGE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Open End Wrench, 1 1⁄16 x 1 1⁄4-in.
- Protective Cover
- Shorting Jumper
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricant, Water Displacing, Item 201B, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-8-1
- PARA 9-4-34
- PARA 9-4-106
- PARA 12-2-2

12-4-2.1 Replace Fire Extinguishing Agent Container Cartridge.

NOTE

Replacement of all four cartridges is the same.

12-4-2.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

c. Open APU compartment access door.

WARNING

Injury to personnel and damage to equipment will result if cartridge does not have shorting jumper installed before being removed. Cartridge could detonate. Make sure shorting jumper is installed before removing cartridge.

d. Slide boot down to expose power terminal on cartridge. Loosen nut on power terminal and connect shorting jumper between power terminal and ground terminal or grounding lug assembly (Figure 12-4-2-1, Detail A).

e. Disconnect wiring from cartridge as follows:

WARNING

Injury to personnel and damage to equipment will result if electrical wires are crossed. Make sure electrical wires are tagged with correct identification.

(1) On HTL fire extinguishing container, perform the following:

(a) Tag wires.

(b) Remove screw, nut, and washers, and braided and ground wires from power and ground terminals.

(2) On Walter Kidde fire extinguishing container, perform the following:

(a) Tag wires.

(b) Remove nuts and washers, and braided and ground wires from power terminal and grounding lug assembly.

WARNING

Injury to personnel and/or damage to equipment will result if cartridge housing assembly and/or swivel discharge outlet are allowed to turn and loosen on container while container is pressurized. Do not remove lockwire from the cartridge housing assembly to the container. Cartridge may be removed with pressure on the container as long as cartridge housing assembly and swivel discharge outlet are held in place by a wrench.

f. Hold cartridge housing assembly with 1 1⁄16 x 1 1⁄4-inch open end wrench and remove cartridge without loosening cartridge housing assembly. On HTL fire extinguishing container, remove and discard packing.

g. Install protective cover on cartridge.

12-4-2.1.2 Inspect.

Inspect braided and ground wires for fraying, security, and general condition. **NO DAMAGE ALLOWED.** If damaged, replace wires.

12-4-2.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

- Injury to personnel or damage to equipment will result if protective cover is removed from cartridge before installation in cartridge housing assembly. Make sure protective cover is not removed before cartridge is installed.

- Injury to personnel and damage to equipment may result if wrong cartridge is used with fire extinguisher bottle. Use cartridge, 70310-03101-105, with fire extinguisher bottle, 70310-03101-106. Use cartridge, 70310-03101-107, with fire extinguisher bottle, 70310-03101-108.

b. Check cartridge for installation, manufacturing date, and serviceability (PARA 1-8-1).

c. Hold cartridge housing assembly with 1 1⁄16 x 1 1⁄4-inch open end wrench.

d. On HTL fire extinguishing container, perform the following (Figure 12-4-2-1, Detail A):

(1) Install packing on cartridge.

(2) Install cartridge in cartridge housing assembly on swivel discharge outlet. **TORQUE CARTRIDGE TO 95 - 105 INCH-POUNDS.**

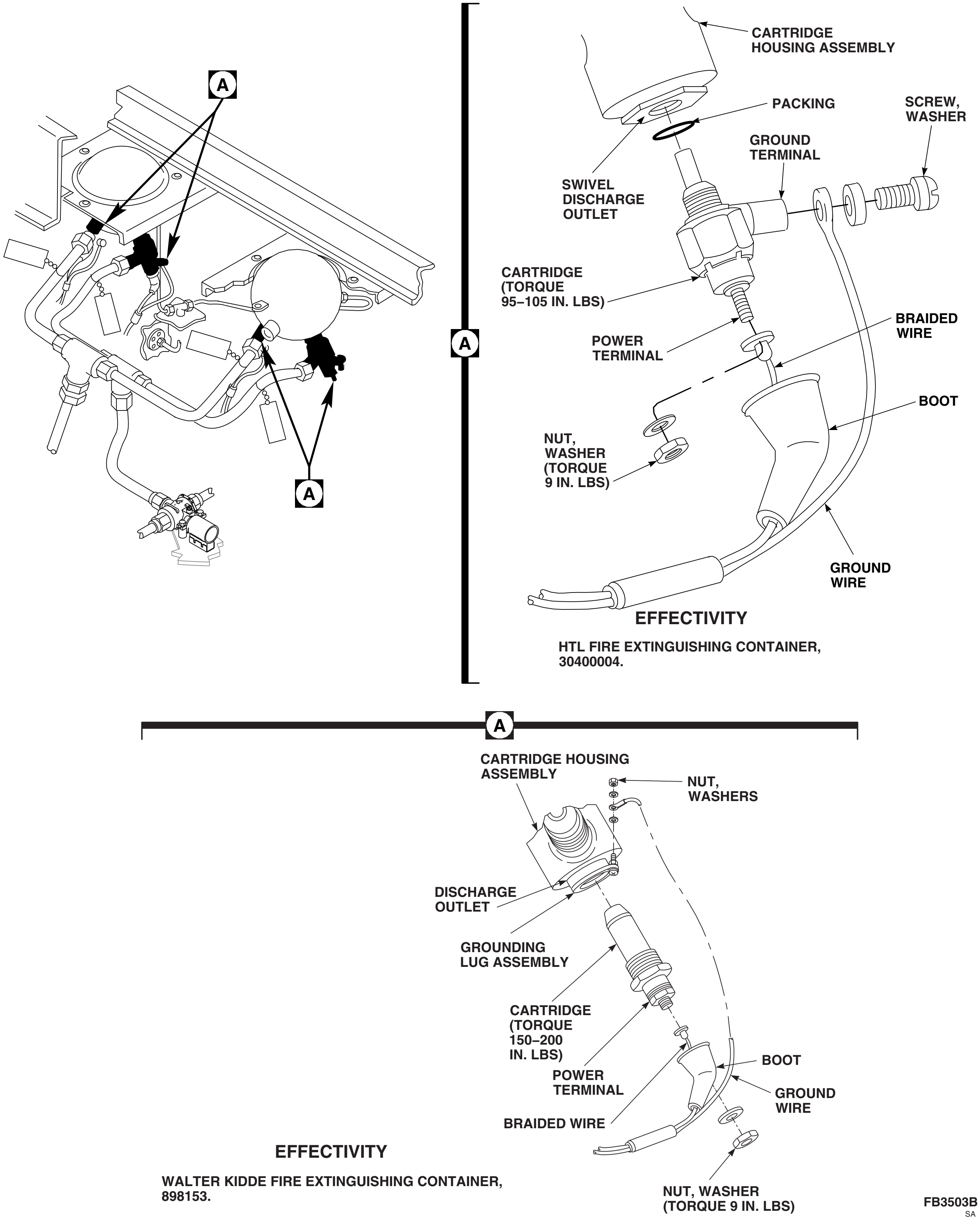


FIGURE 12-4-2-1. REPLACE FIRE EXTINGUISHING AGENT CONTAINER CARTRIDGE

e. On Walter Kidde fire extinguishing container, install cartridge in cartridge housing assembly on discharge outlet. **TORQUE CARTRIDGE TO 150 - 200 INCH-POUNDS.**

f. Lockwire, Item 197, Appendix D, cartridge to cartridge housing assembly. Remove protective cover.

g. Perform operational check of fire extinguishing system (PARA 12-2-2).

h. Apply water displacing lubricant, Item 201B, Appendix D, to power terminal and ground terminal/grounding lug assembly.

i. Connect wiring as follows:

WARNING

Injury to personnel and damage to equipment will result if electrical wires are crossed. Pay particular attention to information noted on tags prior to connecting wiring.

(1) On HTL fire extinguishing container, install braided and ground wires on power and ground terminals and secure with screw, nut, and washers.

(2) On Walter Kidde fire extinguishing container, install braided and ground wires on power terminal and ground lug assembly and secure with nuts and washers.

(3) Remove tags.

j. Remove shorting jumper between power terminal and ground terminal.

k. Tighten nut securing braided wire to power terminal. **TORQUE NUT ON POWER TERMINAL TO 9 INCH-POUNDS.** Install boot.

l. Make sure area is clean and free of foreign material.

m. Close and latch APU compartment access door.

12-4-3 FIRE EXTINGUISHING AGENT CONTAINER CHECK VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16

12-4-3.1 Replace Fire Extinguishing Agent Container Check Valve.

NOTE

Replacement of both check valves is the same.

12-4-3.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open right APU access door.

c. Disconnect manifold tubes from check valve and reducer. Plug tubes (Figure 12-4-3-1, Detail A).

d. Remove nut and reducer from check valve.

12-4-3.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install reducer and nut on check valve (Figure 12-4-3-1, Detail A). **TORQUE NUT TO 500 - 700 INCH-POUNDS.**

WARNING

If both check valves are being replaced, make sure fire extinguishing lines are properly routed and connected. Failure to do so may result in injury or death to personnel as a result of fire extinguishing agent being sent to incorrect compartment.

c. Unplug and connect manifold tubes to check valve. **TORQUE B-NUTS TO 480 - 696 INCH-POUNDS.**

d. Unplug and connect manifold tube to reducer. **TORQUE B-NUT TO 216 - 540 INCH-POUNDS.**

e. Make sure area is clean and free of foreign material.

f. Close right APU access door.

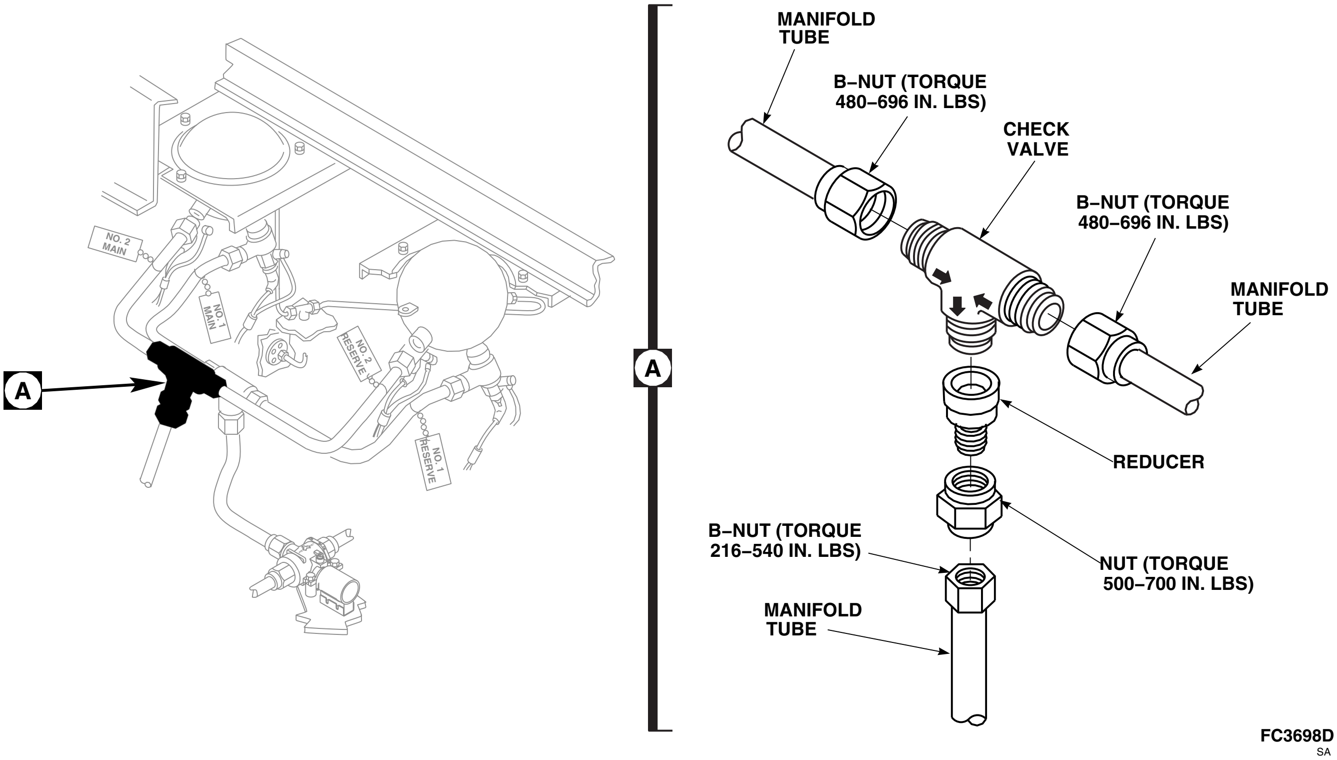


FIGURE 12-4-3-1. REPLACE FIRE EXTINGUISHING AGENT CONTAINER CHECK VALVE

12-4-4 FIRE EXTINGUISHING AGENT CONTAINER THERMAL DISC.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

12-4-4.1 Replace Fire Extinguishing Agent Container Thermal Disc.

12-4-4.1.1 Remove.

Remove retaining ring and thermal disc from discharge port of indicator body on right side of helicopter (Figure 12-4-4-1, Detail A).

12-4-4.1.2 Install.

Install new thermal disc and retaining ring in discharge port of indicator body (Figure 12-4-4-1, Detail A).

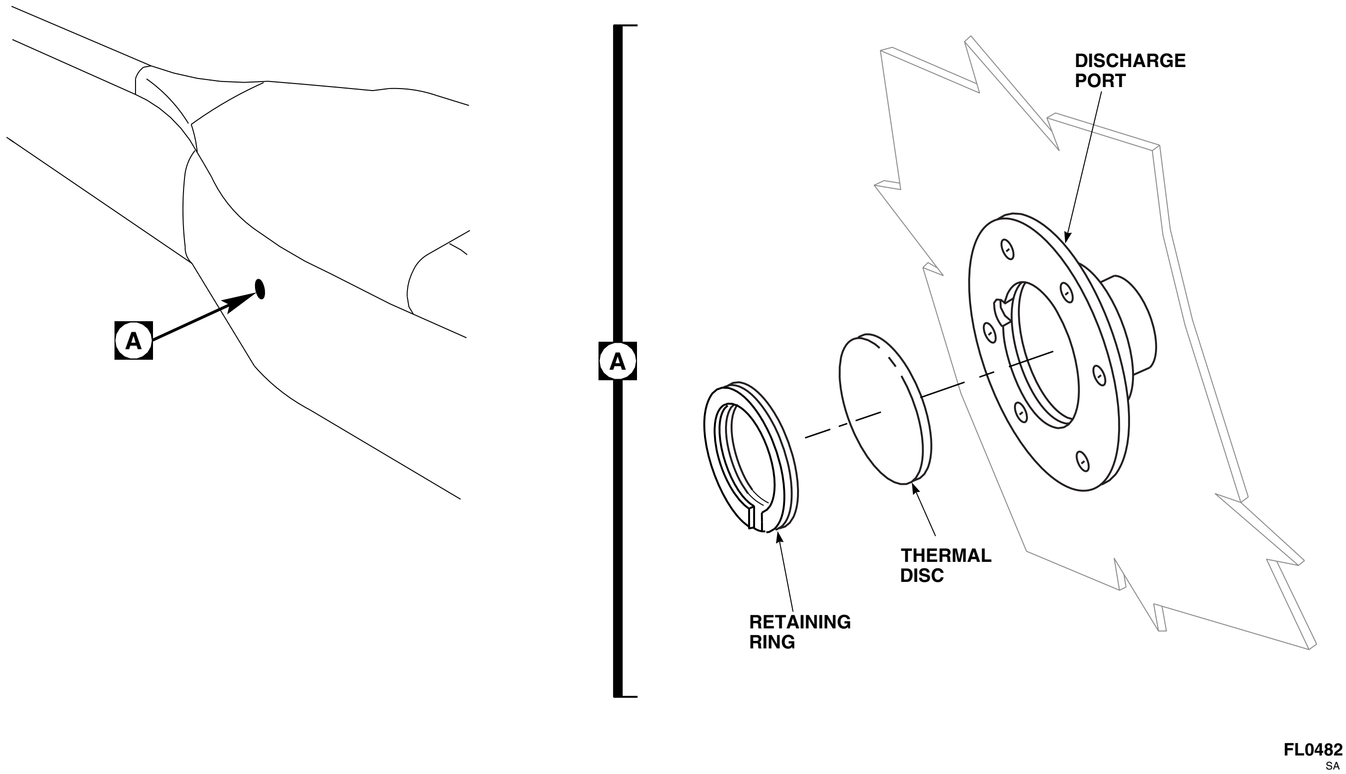


FIGURE 12-4-4-1. REPLACE FIRE EXTINGUISHING AGENT CONTAINER THERMAL DISC

12-4-5 FIRE EXTINGUISHING SYSTEM DISCHARGE TUBES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 9-4-34
- PARA 9-4-106

12-4-5.1 Replace Fire Extinguishing System Discharge Tubes.

12-4-5.1.1 Remove.

a. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

b. Open engine cowlings, oil cooler, and APU compartment access doors as required.

NOTE

Cap or plug all tubes, ports, and fittings as parts are removed and disconnected.

c. Disconnect tube. If installed, remove ring.

d. If required, remove screws, spacers (if installed), washers, and nuts (if installed) securing clamps to brackets or airframe. Remove clamps (Figure 12-4-5-1, Sheet 2, Detail B, and Sheet 3, Detail C and Detail D).

e. If required, remove jamnuts, washers, and spacers (if installed) securing elbows, unions, or reducers to airframe.

12-4-5.1.2 Inspect.

a. Inspect flared ends of tubes for damage. **NONE ALLOWED.** If damaged, replace tube.

b. If removed, inspect ring for damage. **NONE ALLOWED.** If damaged, replace ring.

12-4-5.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

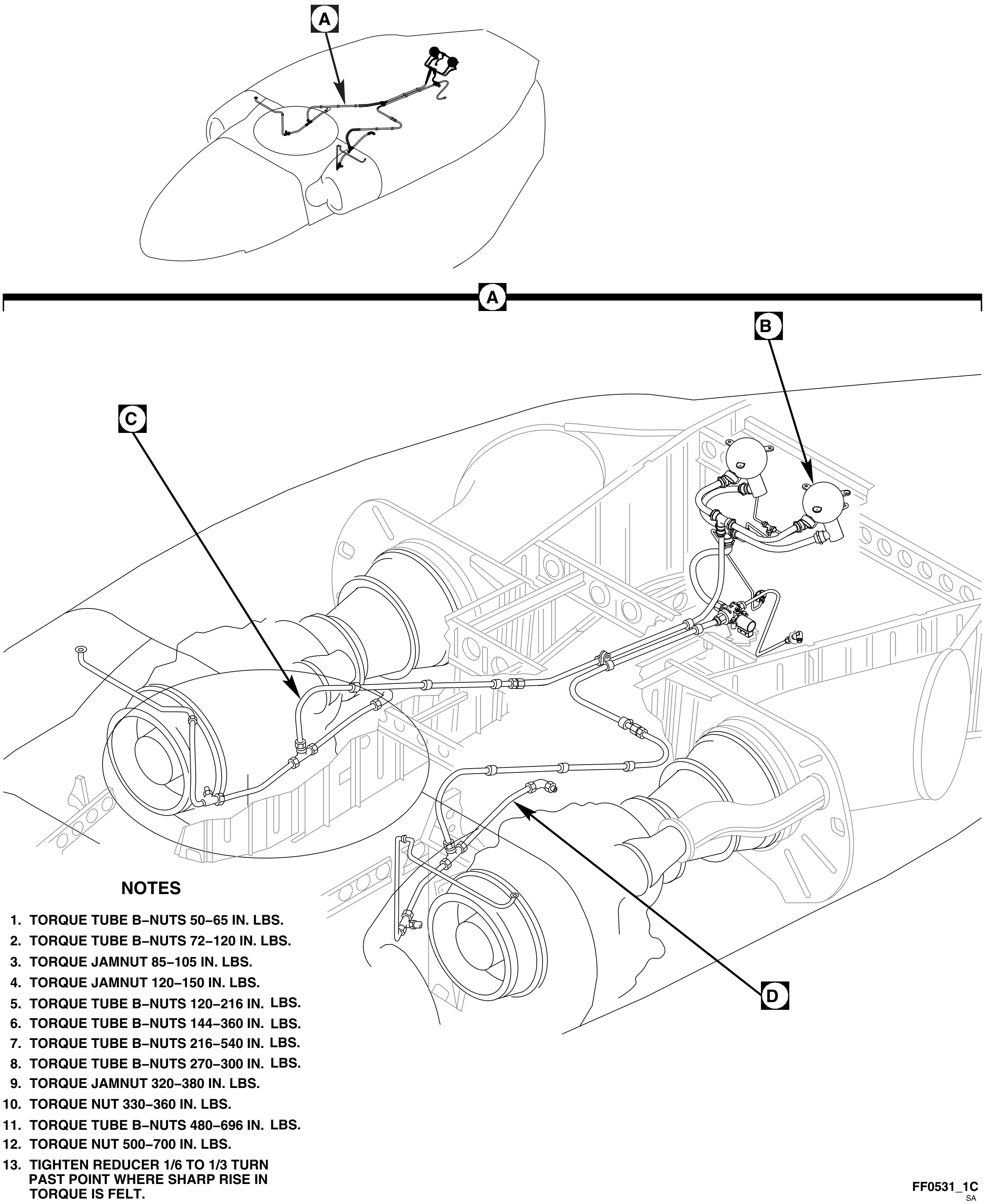


FIGURE 12-4-5-1. REPLACE FIRE EXTINGUISHING SYSTEM DISCHARGE TUBES (SHEET 1 OF 3)

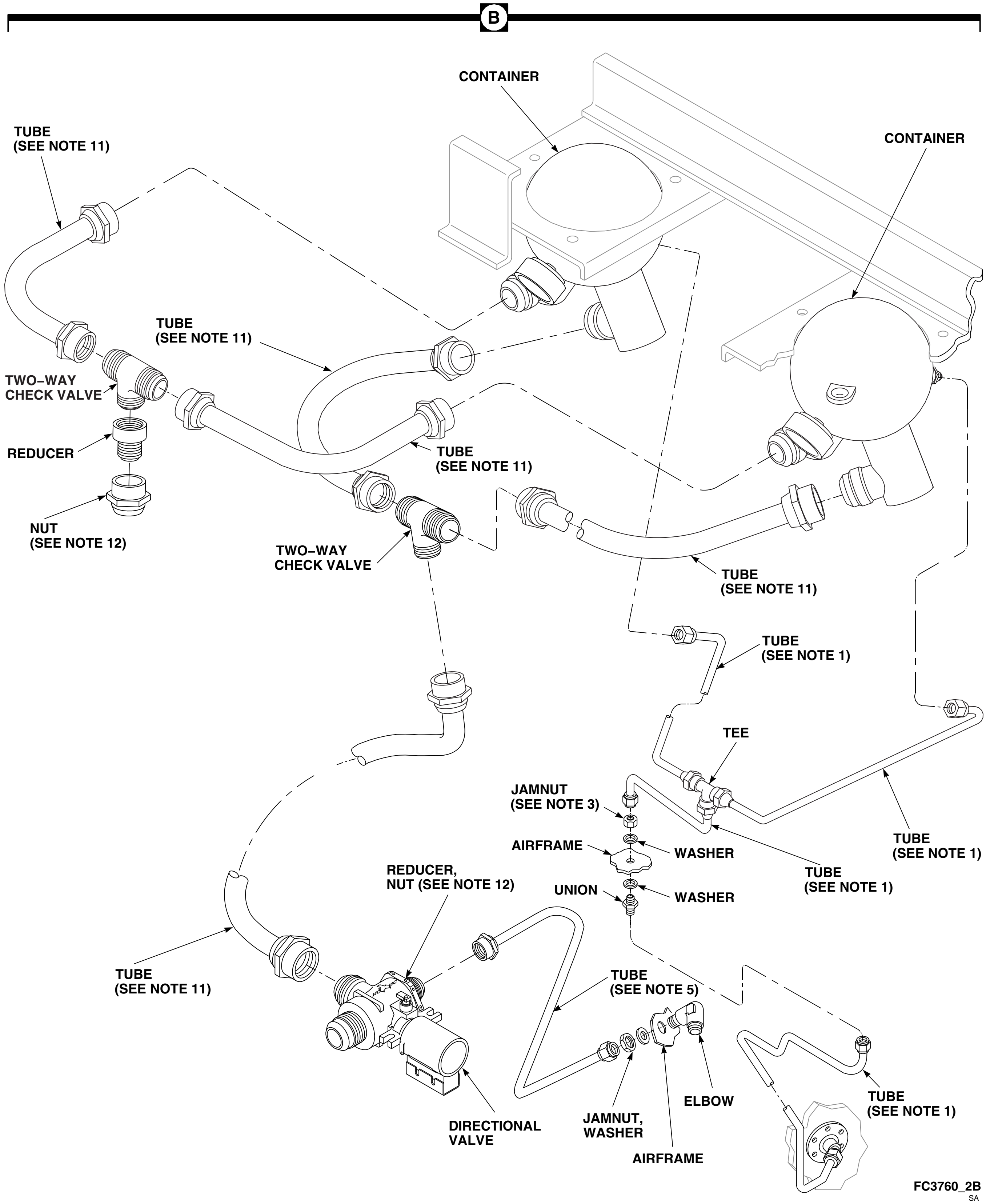
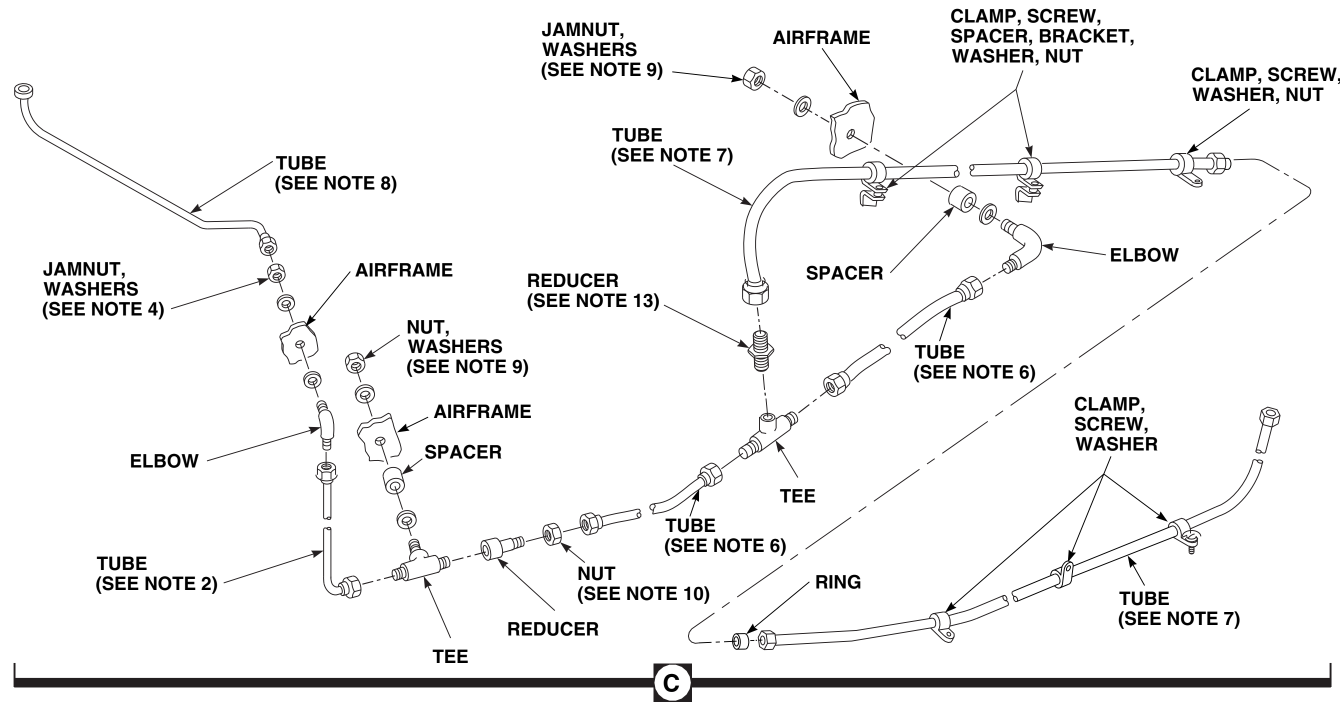


FIGURE 12-4-5-1. REPLACE FIRE EXTINGUISHING SYSTEM DISCHARGE TUBES
(SHEET 2 OF 3)



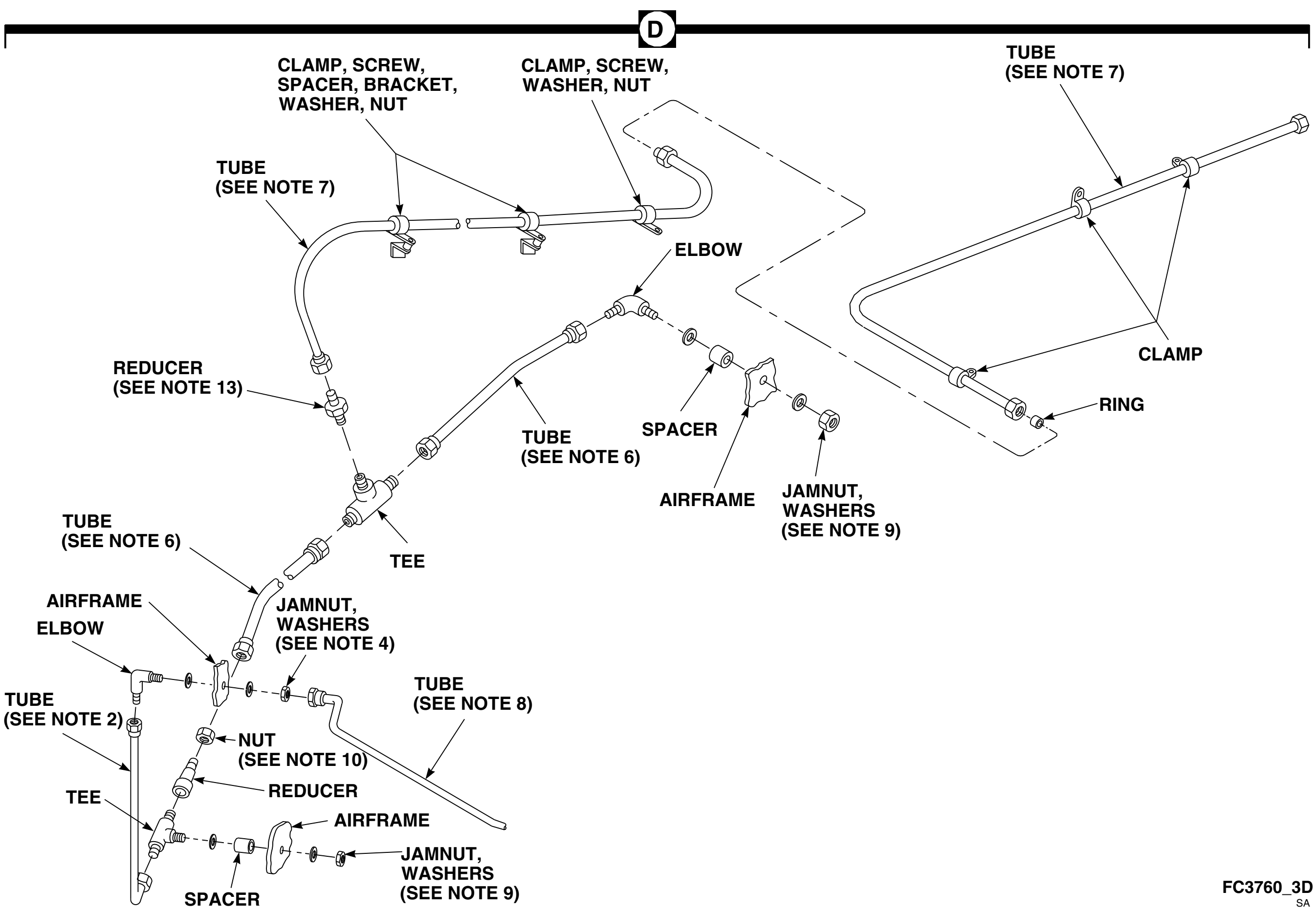


FIGURE 12-4-5-1. REPLACE FIRE EXTINGUISHING SYSTEM DISCHARGE TUBES
(SHEET 3 OF 3)

WARNING

Make sure fire extinguishing lines are properly routed and connected. Failure to do so may result in injury or death to personnel as a result of fire extinguishing agent being sent to incorrect compartment.

NOTE

Remove caps and plugs from tubes, ports, and fittings as parts are installed and connected.

b. If removed, install elbows, unions, or reducers to airframe using jamnuts, washers, and spacers (if required) (Figure 12-4-5-1, Sheet 2, Detail B, and Sheet 3, Detail C and Detail D). **TORQUE JAMNUTS TO APPLICABLE VALUE SPECIFIED IN FIGURE 12-4-5-1, SHEET 1, DETAIL A.**

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Using clean, dry compressed air, blow all tubes clear at installation.

c. If tube is equipped with ring, install ring as follows:

(1) Position ring against flared end of tube so that inside diameters of ring and tubing are coaxial.

(2) Make sure that ring does not obstruct fire extinguishing agent flow through discharge system.

(3) Place ring and first tube against flared end of second tube.

NOTE

When connecting fittings, continue to hold tubes together so ring does not reposition itself and obstruct agent flow through discharge system.

(4) Connect fittings. **TIGHTEN HANDTIGHT.**

(5) **TORQUE B-NUTS TO APPLICABLE VALUE SPECIFIED IN FIGURE 12-4-5-1, SHEET 1, DETAIL A.**

(6) Inspect ring for proper installation as follows:

(a) Installation with only one thread showing indicates a properly installed ring. No further action required.

(b) Installation with more than one visible thread indicates a ring that is not properly seated. If more than one thread is visible, remove ring from lines and inspect ring and flared ends of each line for damage. If damage is found on ring and/or fire extinguishing lines, replace as necessary.

(c) Installation with fittings bottomed out indicates a missing ring.

d. Install clamps to bracket or airframe using screws, spacers (if required), washers, and nuts (if required).

e. Make sure discharge tubes are positioned as follows:

(1) The target point at which left discharge tube should aim is 1 ½-inches up from engine-mounted fuel filter base and ⅜-

inch inboard of fuel filter centerline. The target zone is a 2-inch diameter circle around target point.

(2) The target point at which the right discharge tube should aim is 5 ½-inches to rear of air inlet firewall, measured along engine inlet duct bleed air line, and ⅜-inch inboard of bleed air line centerline. The target zone is a 2-inch diameter circle around target point.

f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

g. Make sure area is clean and free of foreign material.

h. Close engine cowlings, oil cooler, and APU compartment access doors, as required.

12-4-6 FIRE EXTINGUISHING SYSTEM DIRECTIONAL VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 12-2-2

12-4-6.1 Replace Fire Extinguishing System Directional Valve.

12-4-6.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unlatch and open APU compartment access door.

c. Remove screw, lockwasher, and washer securing terminal cover to terminal block (Figure 12-4-6-1, Detail B). Remove terminal cover.

d. Disconnect wires from terminal block as follows:

CAUTION

Damage to directional valve may result if lower nuts are rotated. Make sure lower nuts do not rotate when removing upper nuts or upper self-locking nuts.

(1) For terminal blocks with upper nuts, MS35649-262 and MS35650-302, perform the following:

NOTE

Upper nuts, MS35649-262 and MS35650-302, will not be reused.

(a) Tag wires and remove upper nuts, lockwashers, and washers securing wires to terminal block.

(b) Disconnect wires.

(2) For terminal blocks with upper self-locking nuts, MS21083N3 and MS21083N06, perform the following:

(a) Tag wires and remove upper self-locking nuts, lockwashers, and washers securing wires to terminal block.

(b) Disconnect wires.

e. Disconnect tubes from directional valve and reducer (Figure 12-4-6-1, Detail A). Plug tubes.

f. Remove bolts, washers, directional valve, and spacers from APU compartment deck.

g. Remove nut and reducer from directional valve.

12-4-6.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install reducer and nut on normally closed port (large diameter) of directional valve (Figure 12-4-6-1, Detail A). **TORQUE NUT TO 500 - 700 INCH-POUNDS.**

NOTE

Make sure that normally open port (small diameter) faces No. 1 engine tube, and normally closed port faces APU tube (reducer faces rear).

c. Install directional valve on APU compartment deck with spacers, washers, and bolts. **TORQUE BOLTS TO 46 - 50 INCH-POUNDS.**

d. Unplug and connect No. 1 engine tube to normally open port on directional valve. **TORQUE B-NUT TO 216 - 540 INCH-POUNDS.**

e. Unplug and connect inlet tube to inlet port on directional valve. **TORQUE B-NUT TO 480 - 696 INCH-POUNDS.**

f. Unplug and connect APU tube to reducer. **TORQUE B-NUT TO 120 - 216 INCH-POUNDS.**

g. Connect wires to terminal block as follows (Figure 12-4-6-1, Detail B):

CAUTION

Damage to directional valve may result if lower nuts are rotated. Make sure lower nuts do not rotate when torquing upper self-locking nuts.

(1) Connect wire on terminal 1 and secure with upper self-locking nut, MS21083N06, lockwasher, and washer. Remove tag. **TORQUE UPPER SELF-LOCKING NUT TO 5 - 7 INCH-POUNDS.**

(2) Connect wire on terminal 2 and secure with upper self-locking nut, MS21083N3, lockwasher, and washer. Remove tag. **TORQUE UPPER SELF-LOCKING NUT TO 13 - 15 INCH-POUNDS.**

h. Install terminal cover on terminal block and secure with screw, lockwasher, and washer.

i. Perform operational check of directional valve (PARA 12-2-2).

j. Make sure area is clean and free of foreign material.

k. Close and latch APU compartment access door.

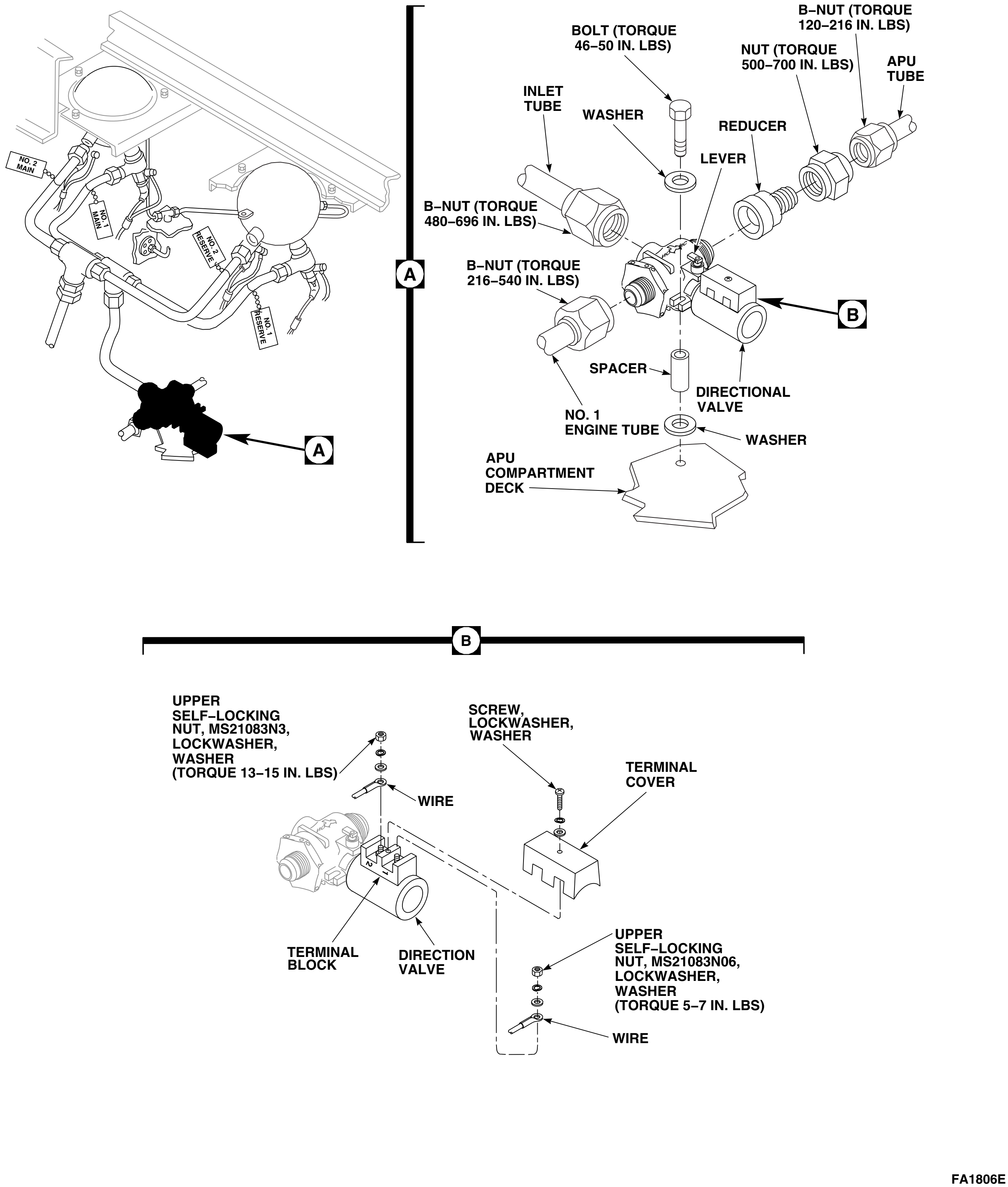


FIGURE 12-4-6-1. REPLACE FIRE EXTINGUISHING SYSTEM DIRECTIONAL VALVE

12-4-7 APU T-HANDLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 12-2-1

12-4-7.1 Replace APU T-Handle.

12-4-7.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screw and lockwasher from APU T-handle (Figure 12-4-7-1, Detail A).

c. Pull APU T-handle outward off shaft.

12-4-7.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Place APU T-handle on shaft and push it on (Figure 12-4-7-1, Detail A).

c. Line up holes and install screw and lock-washer.

d. Perform operational check of fire detection system (PARA 12-2-1).

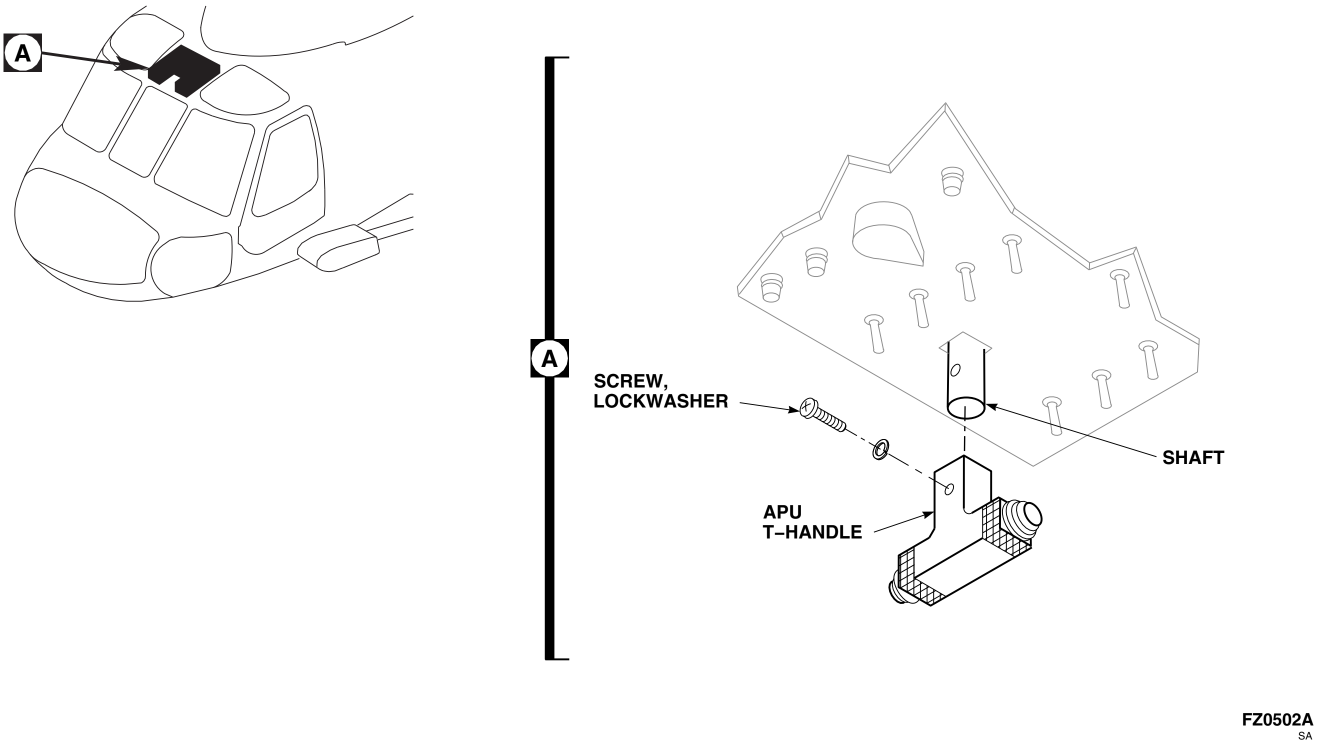


FIGURE 12-4-7-1. REPLACE APU T-HANDLE

12-4-8 APU T-HANDLE FIRE DETECTION LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 12-2-1

12-4-8.1 Replace APU T-Handle Fire Detection Lamps.

NOTE

Replacement of both lamps is the same.

12-4-8.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unscrew and remove red filter and preformed packing from T-handle (Figure 12-4-8-1, Detail B).

c. Remove lamp from filter.

12-4-8.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Insert lamp into filter (Figure 12-4-8-1, Detail B).

c. Install preformed packing and screw filter into T-handle.

d. Perform operational check of fire detection system (PARA 12-2-1).

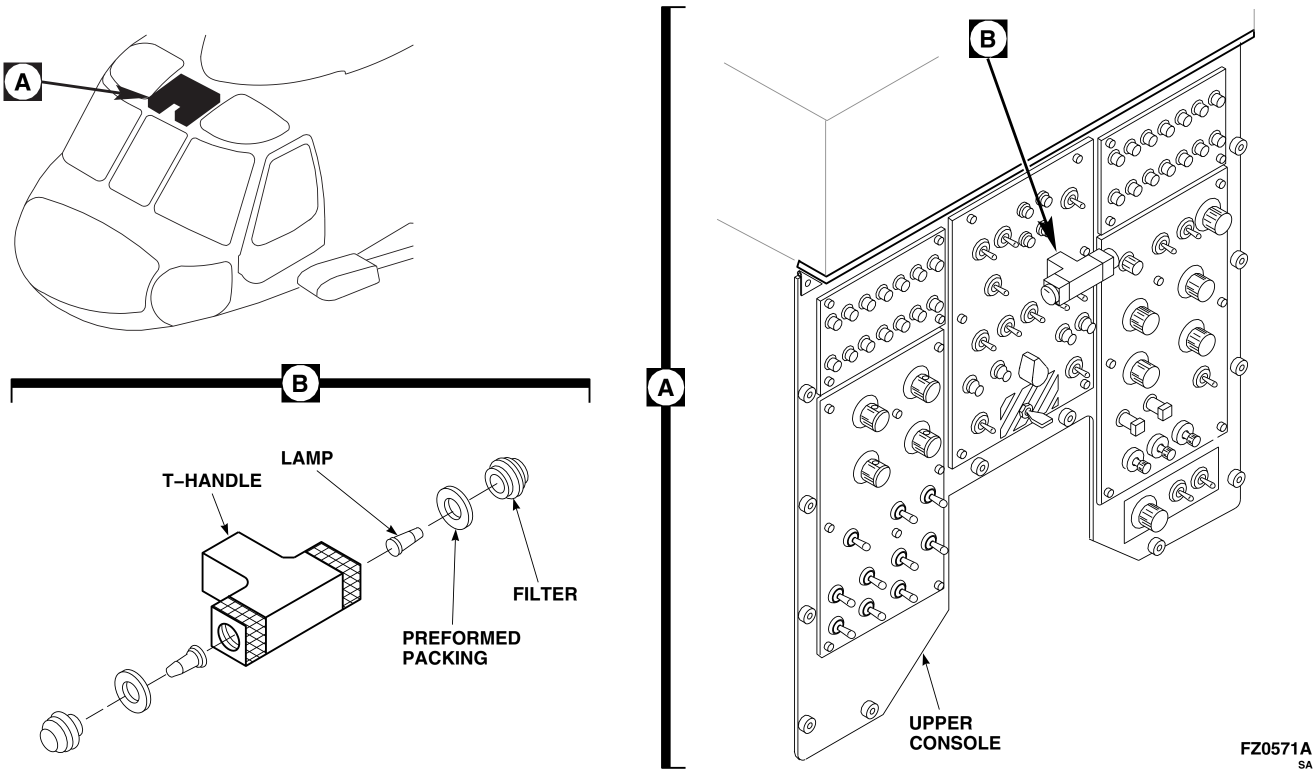


FIGURE 12-4-8-1. REPLACE APU T-HANDLE FIRE DETECTION LAMPS

12-4-9 APU T-HANDLE FIRE EXTINGUISHER SWITCH ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-4-6
- PARA 9-4-34
- PARA 9-4-106
- PARA 12-2-2

12-4-9.1 Replace APU T-Handle Fire Extinguisher Switch Assembly.

12-4-9.1.1 Remove.

a. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

b. Remove center information plate from upper console (PARA 9-4-6).

c. Loosen fasteners securing upper console and carefully lower (Figure 12-4-9-1, Detail A).

d. Remove screws and washers securing wires to APU T-handle fire extinguisher switch assembly (Figure 12-4-9-1, Detail B). Tag wires.

e. Remove screws securing APU fire extinguisher switch assembly to upper console. Remove APU fire extinguisher switch assembly.

12-4-9.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Insert APU fire extinguisher switch assembly control shaft through back of upper console and install screws (Figure 12-4-9-1, Detail B).

c. Connect wires to switch assembly with screws and washers. Remove tags.

d. Make sure area is clean and free of foreign material.

e. Carefully raise upper console and secure with fasteners (Figure 12-4-9-1, Detail A).

f. Install center information plate on upper console (PARA 9-4-6).

g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

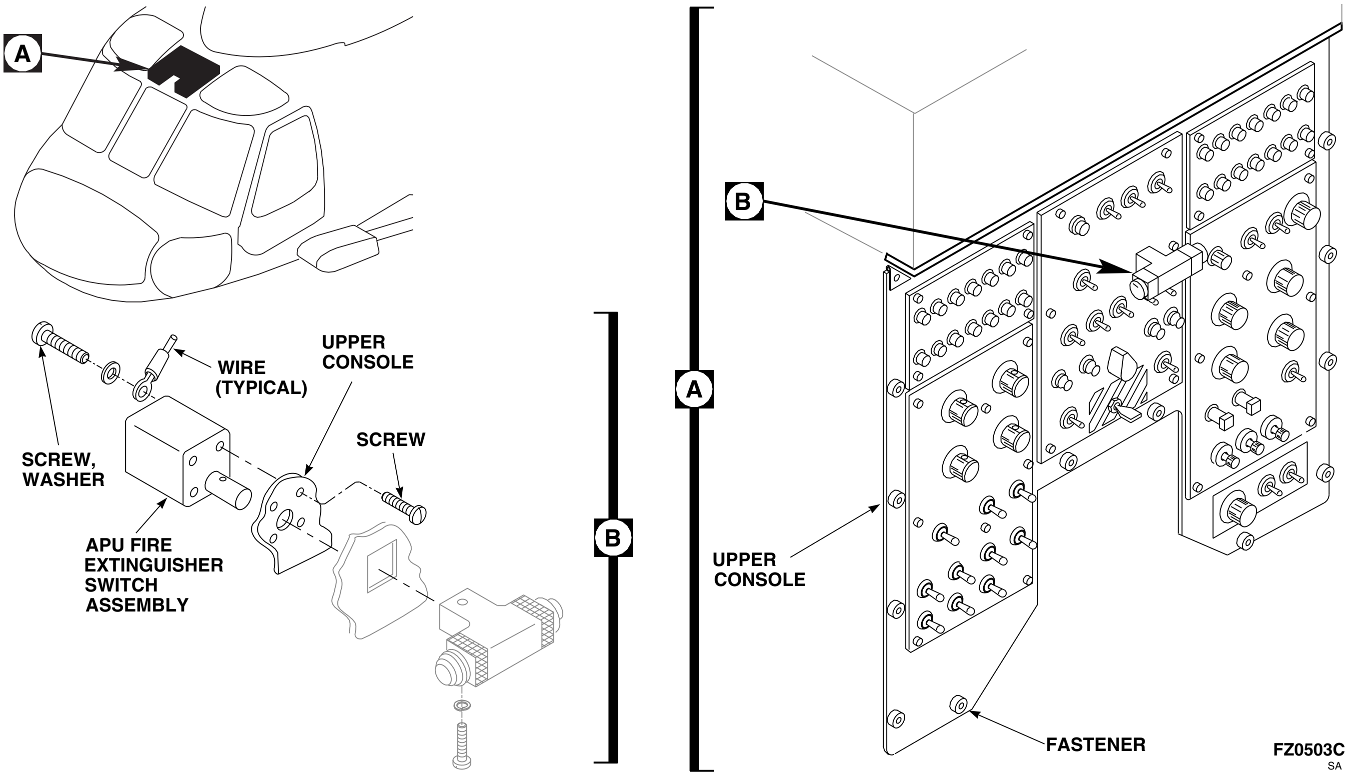


FIGURE 12-4-9-1. REPLACE APU T-HANDLE FIRE EXTINGUISHER SWITCH ASSEMBLY

h. Perform operational check of fire extinguishing system (PARA 12-2-2).

12-4-10 FIRE EXTINGUISHER T-HANDLE LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 12-2-2

12-4-10.1 Replace Fire Extinguisher T-Handle Lamps.

NOTE

- Replacement for No. 1 and No. 2 fire extinguisher T-handle lamps is the same.
- Each T-handle contains two lamps.

12-4-10.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove #1 ENG EMER OFF or #2 ENG EMER OFF lens from T-handle (Figure 12-4-10-1, Detail B).

c. Remove lamp from T-handle.

12-4-10.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install lamp in T-handle (Figure 12-4-10-1, Detail B).

c. Install #1 ENG EMER OFF or #2 ENG EMER OFF lens on T-handle.

d. Perform operational check of engine fire extinguisher system (PARA 12-2-2).

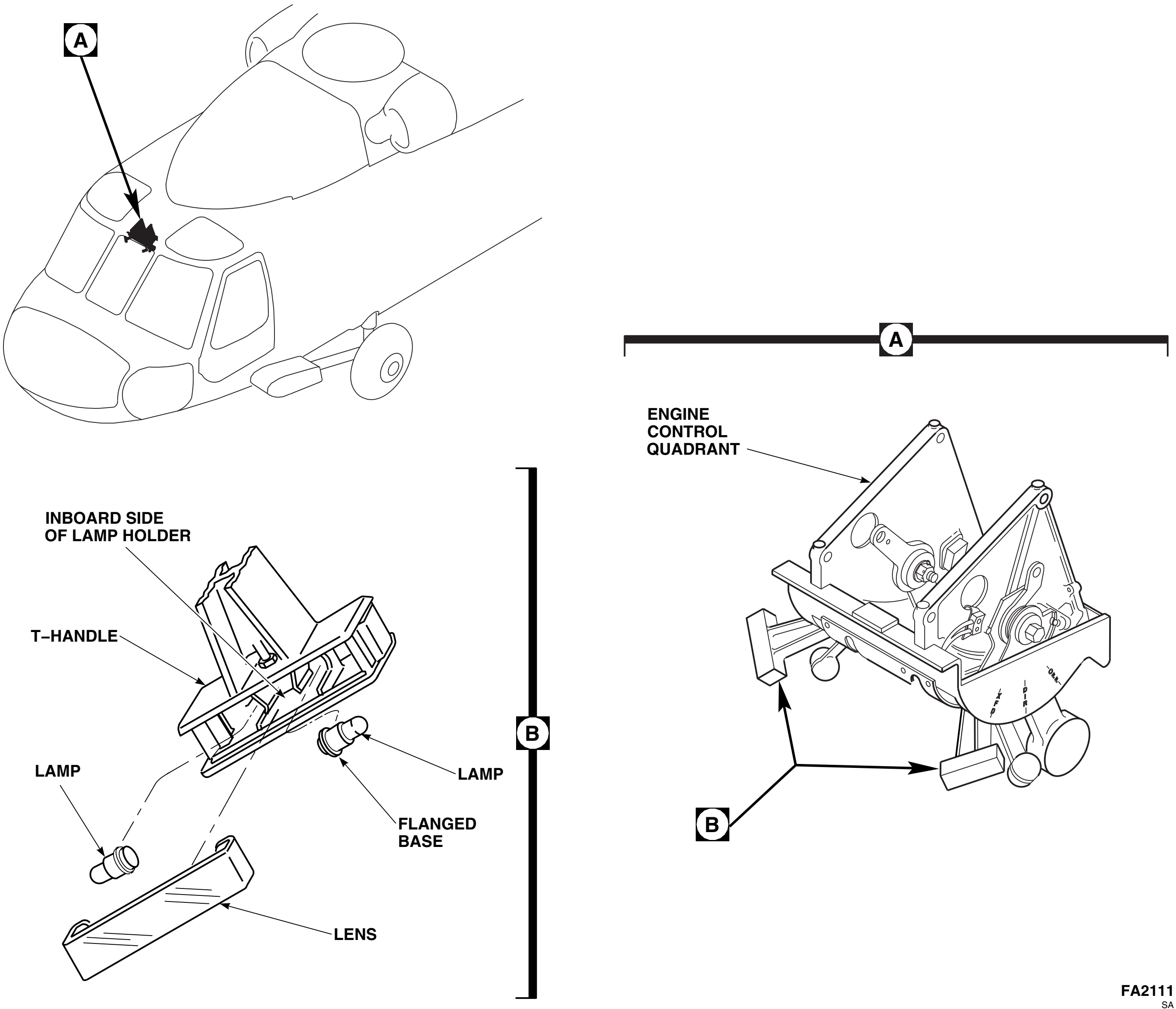


FIGURE 12-4-10-1. REPLACE FIRE EXTINGUISHER T-HANDLE LAMPS

12-4-11 FIRE DETECTION SYSTEM CONTROL AMPLIFIER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 12-2-1

12-4-11.1 Replace Fire Detection System Control Amplifier.

NOTE

Replacement of all three control amplifiers is the same, except as noted.

12-4-11.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Tag and disconnect electrical connectors, P283, P284, or P285, from control amplifier (Figure 12-4-11-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

CAUTION

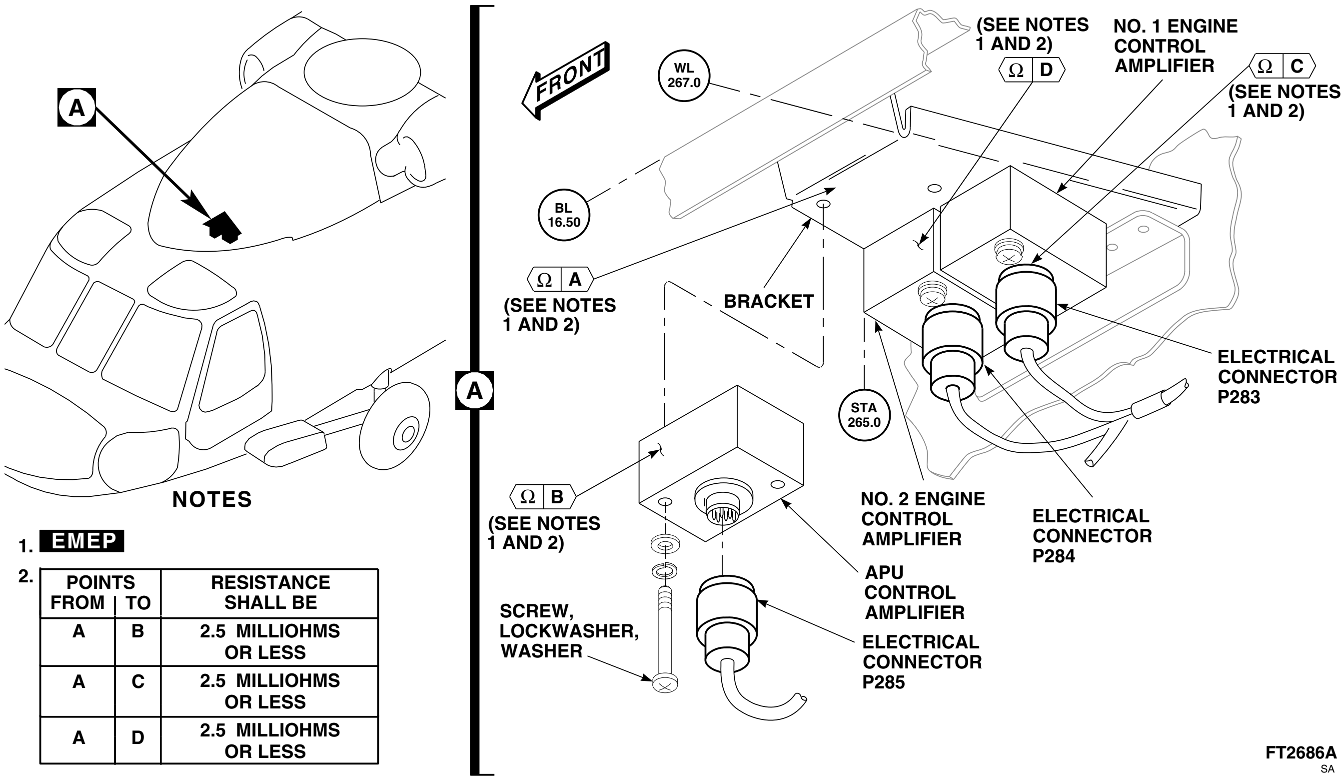
EMEP Control amplifier has chemical conversion coating on mounting surface for EMEP. Coating will be damaged if control amplifier is not handled carefully. Do not place component on its coated surface after removal.

e. Remove screws, lockwashers, washers, and control amplifier from bracket.

12-4-11.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. EMEP Clean surface around all mounting holes using cheesecloth, Item 90, Appendix D,



NOTES

1. EMEP

2.

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS
A	C	2.5 MILLIOHMS OR LESS
A	D	2.5 MILLIOHMS OR LESS

FIGURE 12-4-11-1. REPLACE FIRE DETECTION SYSTEM CONTROL AMPLIFIER

dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.

c. EMEP Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

d. Install control amplifier to bracket with screws, lockwashers, and washers (Figure 12-4-11-1, Detail A).

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connectors, P283, P284, or P285, to control amplifier. Remove tags.

h. EMEP Perform EME resistance check of control amplifier as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, measure resistance between control amplifier and bracket. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, remove amplifiers and apply chemical conversion coating (PARA 2-6-5).

i. Make sure area is clean and free of foreign material.

j. Install soundproofing panels (PARA 2-4-69).

k. Perform operational check of fire detection system (PARA 12-2-1).

12-4-12 FLAME DETECTORS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Protection, Eye, Skin, and Breathing

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 12-2-1

12-4-12.1 Replace Flame Detectors.

12-4-12.1.1 Remove.

WARNING

Cadmium and cadmium compounds are considered carcinogenic to humans. Cadmium sulphide from detector cell may escape if integrity of flame detector is breached (damaged housing or cracked lens). Wear suitable protective equipment when handling damaged detectors. Consult local procedures for hazardous substances management and disposal procedures.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Disconnect the following electrical connectors as necessary (Figure 12-4-12-1):

(1) P286 from J286 (deck detector, left side)

(2) P287 from J287 (deck detector, right side)

(3) P413 from J413 (engine firewall detector, left side)

(4) P414 from J414 (APU firewall detector)

(5) P415 from J415 (engine firewall detector, right side)

d. Install protective caps and plugs on electrical connectors.

e. If removing APU or engine firewall detector(s), remove screws, washers, and flame detector from firewall (Figure 12-4-12-1, Detail A).

f. If removing deck detector(s), remove screws, washers, nuts, shield, and flame detector from engine compartment deck (Figure 12-4-12-1, Detail B).

g. If removing deck detector(s), perform the following:

(1) Remove sealing compound from airframe structure using nonmetallic scraper.

(2) Clean sealing compound with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D.

12-4-12.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

For good electrical ground contact, before mounting flame detector, make certain that mounting holes are clean and free of paint.

b. If installing APU or engine firewall detector(s), install flame detector on firewall with screws and washers (Figure 12-4-12-1, Detail A).

c. If installing deck detector(s), install flame detector and shield to engine compartment deck with screws, washers, and nuts (Figure 12-4-12-1, Detail B).

d. For deck detectors, perform the following:

(1) Apply sealing compound, Item 292, Appendix D, around junction of airframe and flame detector shield.

(2) If necessary, clean off extra sealing compound with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, while compound is still soft.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect the following electrical connectors as necessary (Figure 12-4-12-1):

(1) P286 to J286 (deck detector, left side)

(2) P287 to J287 (deck detector, right side)

(3) P413 to J413 (engine firewall detector, left side)

(4) P414 to J414 (APU firewall detector)

(5) P415 to J415 (engine firewall detector, right side)

h. Make sure area is clean and free of foreign material.

i. Install soundproofing panels (PARA 2-4-69).

j. Perform operational check of fire detection system (PARA 12-2-1).

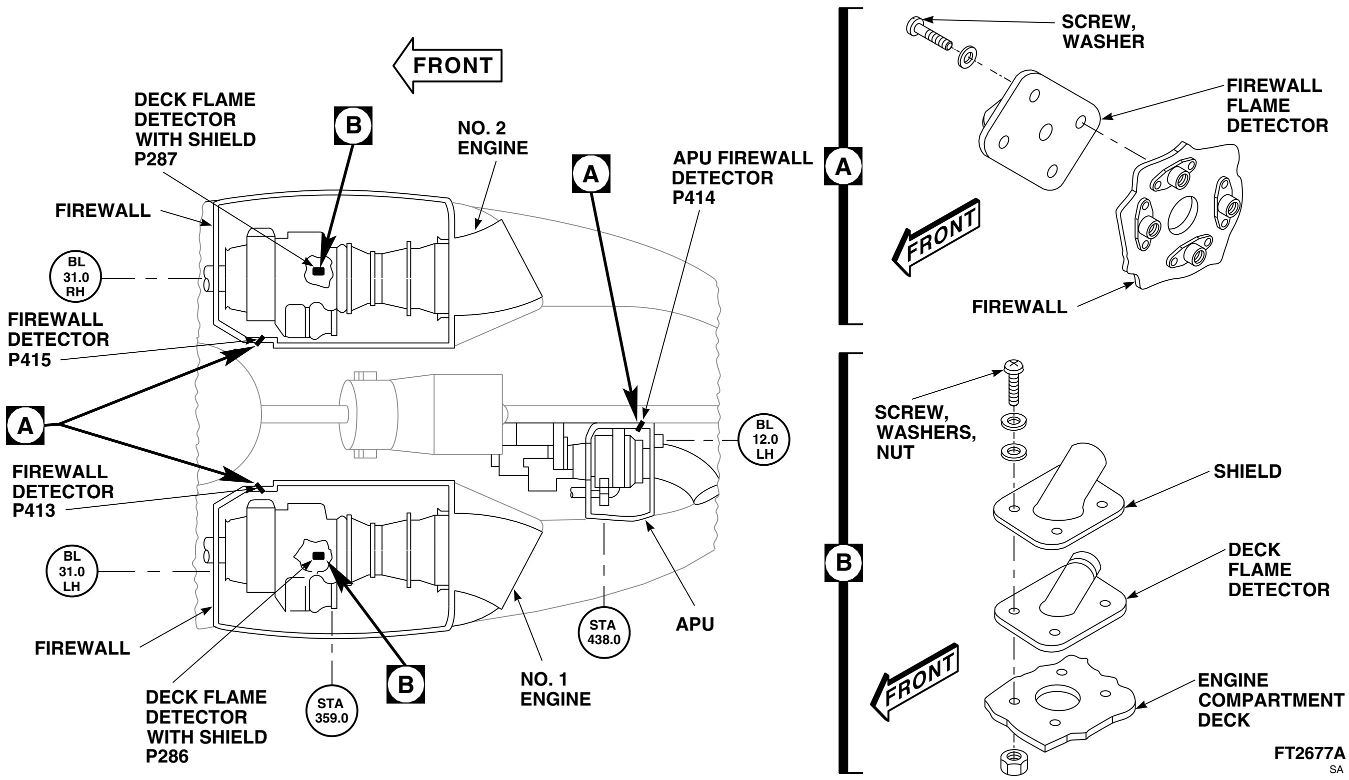


FIGURE 12-4-12-1. REPLACE FLAME DETECTORS

12-4-13 FIRE DETECTOR TEST SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Insulating Compound, Item 159, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-4-6
- PARA 12-2-1

12-4-13.1 Replace Fire Detector Test Switch.

12-4-13.1.1 Remove.

a. Remove center information plate (PARA 9-4-6).

b. Loosen fasteners that secure upper console and carefully lower (Figure 12-4-13-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Tag wires. Using soldering set, unsolder wires from fire detector test switch (Figure 12-4-13-1, Detail B).

d. Remove nut, lockwasher, and switch from upper console.

12-4-13.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Turn fire detector test switch shaft fully counterclockwise.

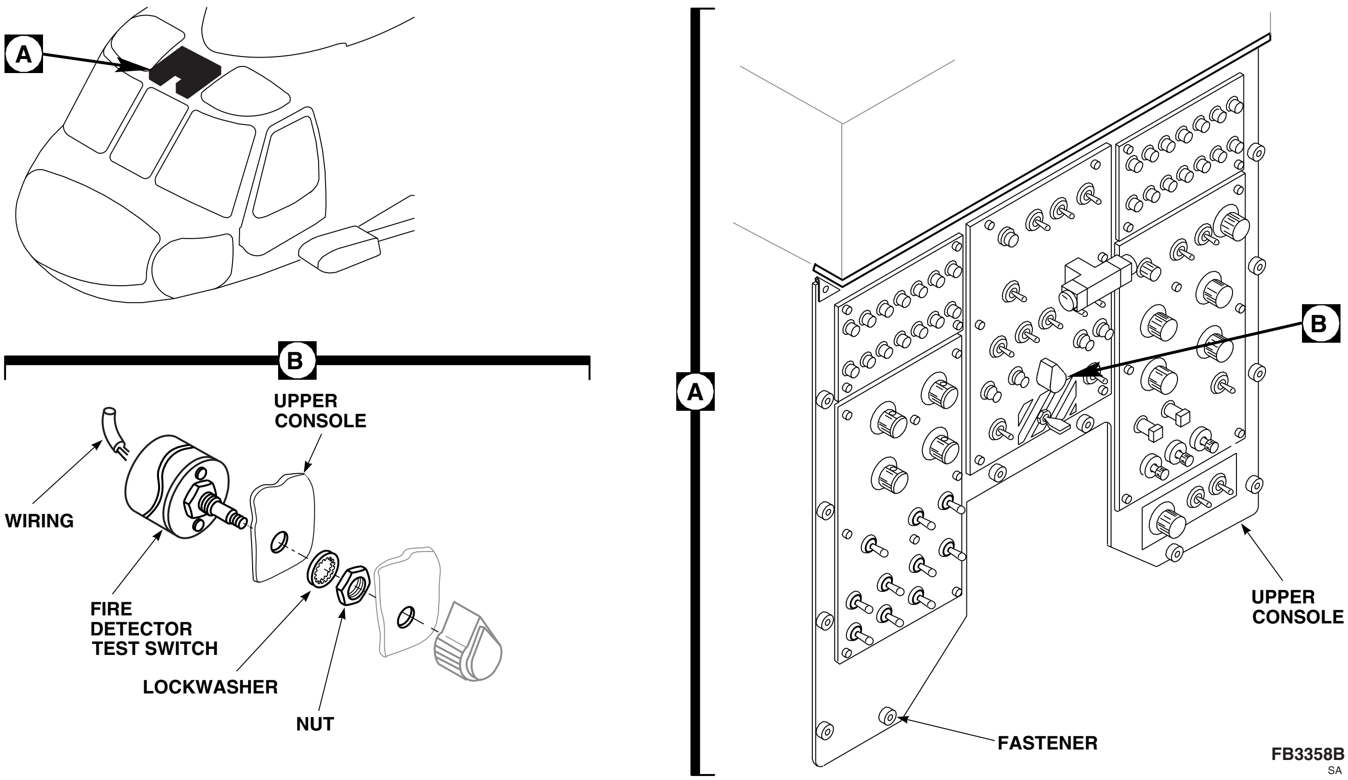


FIGURE 12-4-13-1. REPLACE FIRE DETECTOR TEST SWITCH

c. Insert switch shaft through back of upper console, and secure with nut and lockwasher (Figure 12-4-13-1, Detail B).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering set and solder, Item 313, Appendix D, solder wires to switch (Figure 12-4-13-1, Detail B). Untag wires.

e. Cover soldered connections with insulating compound, Item 159, Appendix D.

f. Make sure area is clean and free of foreign material.

g. Carefully raise upper console and secure with fasteners (Figure 12-4-13-1, Detail A).

h. Install center information plate (PARA 9-4-6).

i. Perform operational check of fire detection system (PARA 12-2-1).

12-4-14 WINDSHIELD WIPER CONVERTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Compound, Item 108, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-5-9
- PARA 1-6-16
- PARA 12-2-3
- PARA 12-3-2
- PARA 12-4-19

12-4-14.1 Replace Windshield Wiper Converter.

NOTE

Replacement of left and right windshield wiper converter is the same.

12-4-14.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove windshield wiper arm and link (PARA 12-4-19).

CAUTION

Damage may occur when disconnecting flexible drive shaft from converter. Make sure coupling does not fall from either drive shaft or converter into electronic equipment.

c. From behind instrument panel, disconnect flexible drive shaft from converter. Keep coupling with flexible drive shaft (Figure 12-4-14-1, Detail A).

d. From inside of cockpit, remove cotter pin from upper inboard pivot stud, and remove nut and washer. With aid of assistant outside of helicopter, remove pivot stud and washer from converter.

e. Remove screws, washers, and nuts, securing converter to skin. Remove bonding jumper from converter.

f. Remove converter from skin.

12-4-14.1.2 Disassemble.

a. Remove bolt, retaining ring, and nylon sleeve from wiper shaft (Figure 12-4-14-1, Detail B).

NOTE

Sealing compound is used to seal front cover to body. A nonmetallic scraper may be needed to loosen cover.

b. Remove screws, machine screw, washers, nut, and front cover from housing.

c. Remove end caps from housing.

d. Remove sleeve and packing from wiper shaft. Discard packing.

e. Remove converter linkage, as an assembly, from housing.

f. Remove screws and bearing end cover.

g. Remove screws and idler plate cover.

12-4-14.1.3 Inspect/Repair.

NOTE

Corrosion is indicated by red film or pitting.

a. Visually inspect pivot stud for corrosion. **PITS ON WEARING SURFACE SHALL NOT EXCEED 0.050-INCH IN DIAMETER.**

b. Clean grease from converter linkage, housing, front cover, bearing end cover, and idler plate cover using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, and wipe dry. Do not allow solvent to get into bearings.

c. Remove bearings and washers from converter linkage. Inspect washers for wear, and bearings for rough or binding rotation. **NONE ALLOWED.** If necessary, replace worn washers and rough or binding bearings.

d. Using nonmetallic scraper, clean remaining sealing compound from front cover and housing mating surfaces. Wipe area clean using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

e. Remove corrosion on pivot stud using scouring pad, Item 270, Appendix D.

f. Mix solution of one part corrosion compound, Item 108, Appendix D, to one part water.

g. Apply corrosion compound solution to corroded areas of pivot stud.

h. Rinse thoroughly with clean water.

i. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint with epoxy primer and polyurethane paint, Item 223, Appendix D.

j. Lubricate converter housing and gears (PARA 1-5-9).

12-4-14.1.4 Assemble.

a. Install converter linkage, as an assembly, in housing (Figure 12-4-14-1, Detail B).

b. Install sleeve and packing, lubricated with hydraulic fluid, Item 155, Appendix D, on wiper shaft.

c. Apply a thin even coat of sealing compound, Item 292, Appendix D, to mating surfaces of front cover and housing. Allow sealing compound to become tacky.

d. Install end caps on housing.

e. Install front cover on housing using screws, machine screw, washers, and nut.

f. Using lockwire, Item 197, Appendix D, lockwire screws to each other, and then the front screws to end cap.

g. Install bearing end cover using screws.

h. Install idler plate cover using screws.

i. Install nylon sleeve, retaining ring, and bolt on wiper shaft. **TORQUE BOLT TO 45 - 55 INCH-POUNDS.**

12-4-14.1.5 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Clean mating surface of converter where bonding jumper is installed as follows:

(1) Wipe area clean using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

(2) Remove paint from mounting area (just slightly larger than jumper wire lug).

(3) Clean area of abrasives or filings.

(4) Treat cleaned area using corrosion resistant coating, Item 118, Appendix D.

c. Install converter and bonding jumper on skin and secure with screws, washers, and nuts (Figure 12-4-14-1, Detail A).

d. Seal all gaps between wiper shaft, bonding jumper, attaching screws, and mounting points in skin with sealing compound, Item 292, Appendix D.

e. With aid of assistant outside of helicopter, install pivot stud and washer in converter. From inside cockpit, secure pivot stud to converter with washer, nut, and cotter pin.

f. Seal opening where pivot stud passes through skin with sealing compound, Item 292, Appendix D.

NOTE

Make sure coupling is installed in flexible drive shaft before connecting drive shaft to converter.

g. From behind instrument panel, install coupling in flexible drive shaft. Connect flexible drive shaft to converter.

h. Using lockwire, Item 197, Appendix D, lockwire coupling nut.

i. Make sure area is clean and free of foreign material.

j. Install windshield wiper arm and link (PARA 12-4-19).

k. If additional adjustment is required to bring wiper into exact PARK alignment, perform the following:

(1) Turn off all electrical power (PARA 1-6-16).

(2) Insert cotter pin in windshield wiper arm pin hole.

(3) Disconnect flexible drive shaft at motor.

(4) Rotate flexible drive shaft coupling by hand until wiper blade moves into precise PARK position.

(5) Connect flexible drive shaft to motor.

(6) Using lockwire, Item 197, Appendix D, lockwire coupling nut.

(7) Remove cotter pin from hole in windshield wiper arm and ease wiper blade back to windshield.

l. Check windshield wiper arm pressure (PARA 12-3-2).

m. Perform operational check of windshield wiper system (PARA 12-2-3).

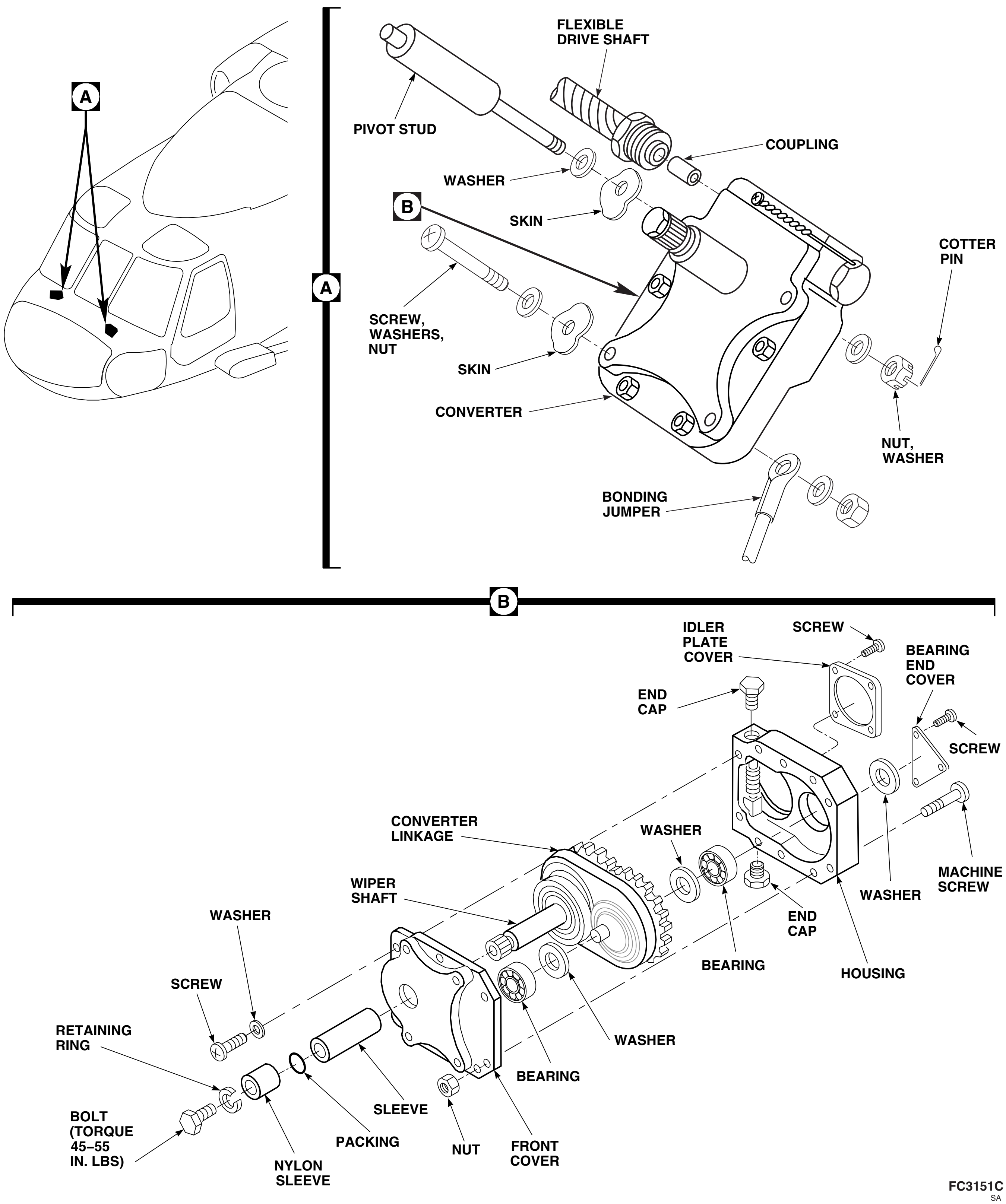


FIGURE 12-4-14-1. REPLACE WINDSHIELD WIPER CONVERTER

12-4-15 WINDSHIELD WIPER MOTOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-3
- TM 11-70-23

12-4-15.1 Replace Windshield Wiper Motor.

12-4-15.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open nose door.

c. Remove avionics components as required to access wiper motor (TM 11-70-23).

d. Disconnect electrical connector, P108, from wiper motor (Figure 12-4-15-1, Detail A).

e. Install protective caps and plugs on electrical connectors.

CAUTION

Damage may occur when disconnecting flexible drive shafts from motor. Make sure couplings do not fall from drive shaft casings or motor into electronic compartment.

f. Disconnect flexible drive shafts from wiper motor. Keep couplings with flexible drive shafts.

g. Remove screw, lockwasher, washers, and bonding jumper from wiper motor.

h. With aid of assistant holding wiper motor, remove screws and washers securing wiper motor to canopy.

i. Using nonmetallic scraper and dry-cleaning solvent, Item 124, Appendix D, remove sealing compound from mounting area.

12-4-15.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

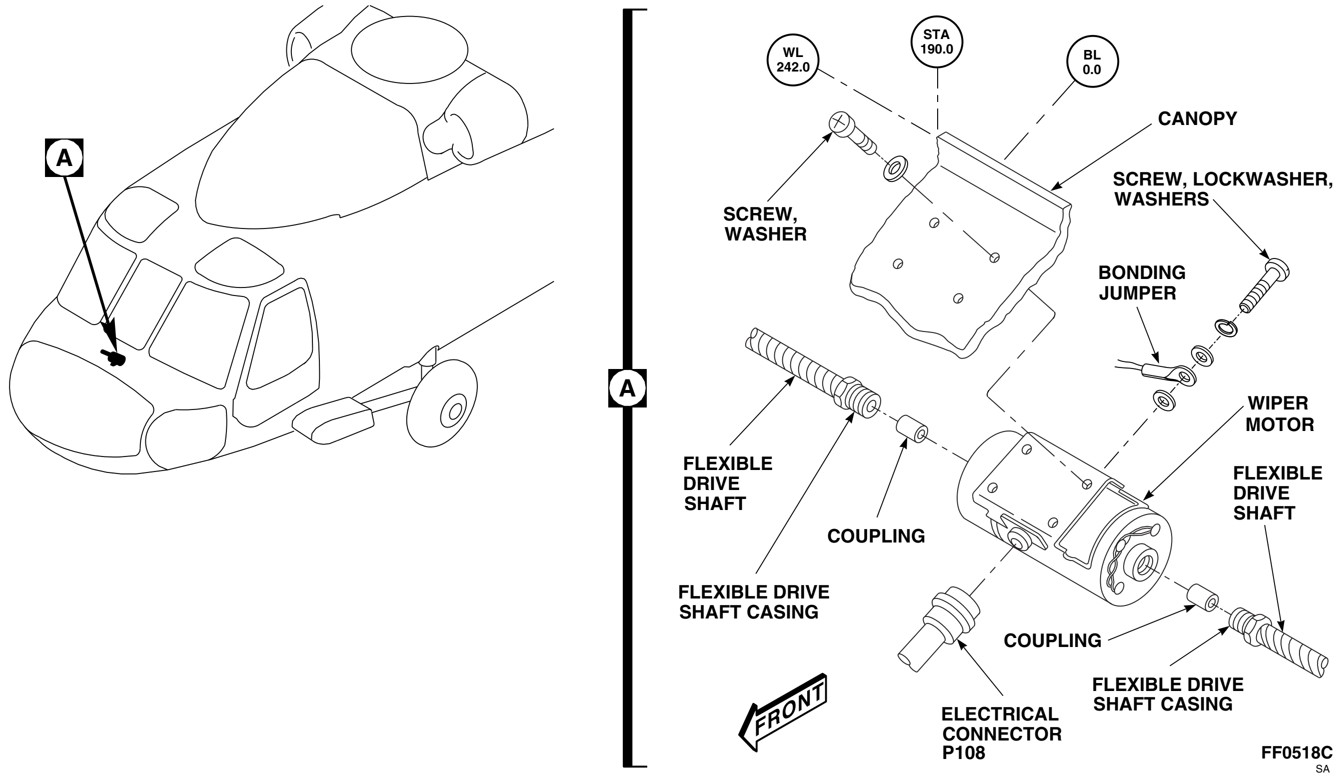


FIGURE 12-4-15-1. REPLACE WINDSHIELD WIPER MOTOR

b. Seal area around attaching screwholes on wiper motor with sealing compound, Item 292, Appendix D.

c. With aid of assistant positioning wiper motor, secure wiper motor to canopy with screws and washers (Figure 12-4-15-1, Detail A).

d. Seal area around screws with sealing compound, Item 292, Appendix D.

e. Install bonding jumper to wiper motor with screw, lockwasher, and washers.

NOTE

Make sure couplings are installed in flexible drive shaft casings before connecting drive shafts to wiper motor.

f. Install couplings in flexible drive shaft casings.

g. Connect flexible drive shafts to wiper motor.

h. Using lockwire, Item 197, Appendix D, lockwire coupling.

i. Remove protective caps and plugs from electrical connectors.

j. Inspect and treat electrical connectors (PARA 1-7-20).

k. Connect electrical connector, P108, to wiper motor.

l. Using lockwire, Item 197, Appendix D, lockwire electrical connector to wiper motor.

m. Perform operational check of windshield wiper system (PARA 12-2-3).

n. Install any previously removed avionics components (TM 11-70-23).

o. Make sure area is clean and free of foreign material.

p. Close nose door.

12-4-16 WINDSHIELD WIPER FLEXIBLE DRIVE SHAFT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 12-2-3
- TM 11-70-23

12-4-16.1 Replace Windshield Wiper Flexible Drive Shaft.

NOTE

Replacement of both flexible drive shafts is the same.

12-4-16.1.1 Remove.

a. Place wiper blades in PARK position.

b. Turn off all electrical power (PARA 1-6-16).

c. Open nose door.

d. Remove avionics components as required to access wiper motor (TM 11-70-23).

CAUTION

Damage may occur when disconnecting flexible drive shaft from converter and wiper motor. Make sure couplings do not fall from flexible drive shaft casings, converter, or wiper motor into electronic compartment.

e. Disconnect flexible drive shaft from wiper motor. Keep coupling with drive shaft (Figure 12-4-16-1, Detail A).

f. Disconnect flexible drive shaft from converter behind cockpit instrument panel. Keep coupling with drive shaft.

g. Remove screw, lockwasher, washers, and nut securing clamp and bonding jumper to beam.

h. Remove flexible drive shaft from helicopter.

i. Remove clamp from flexible drive shaft.

12-4-16.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

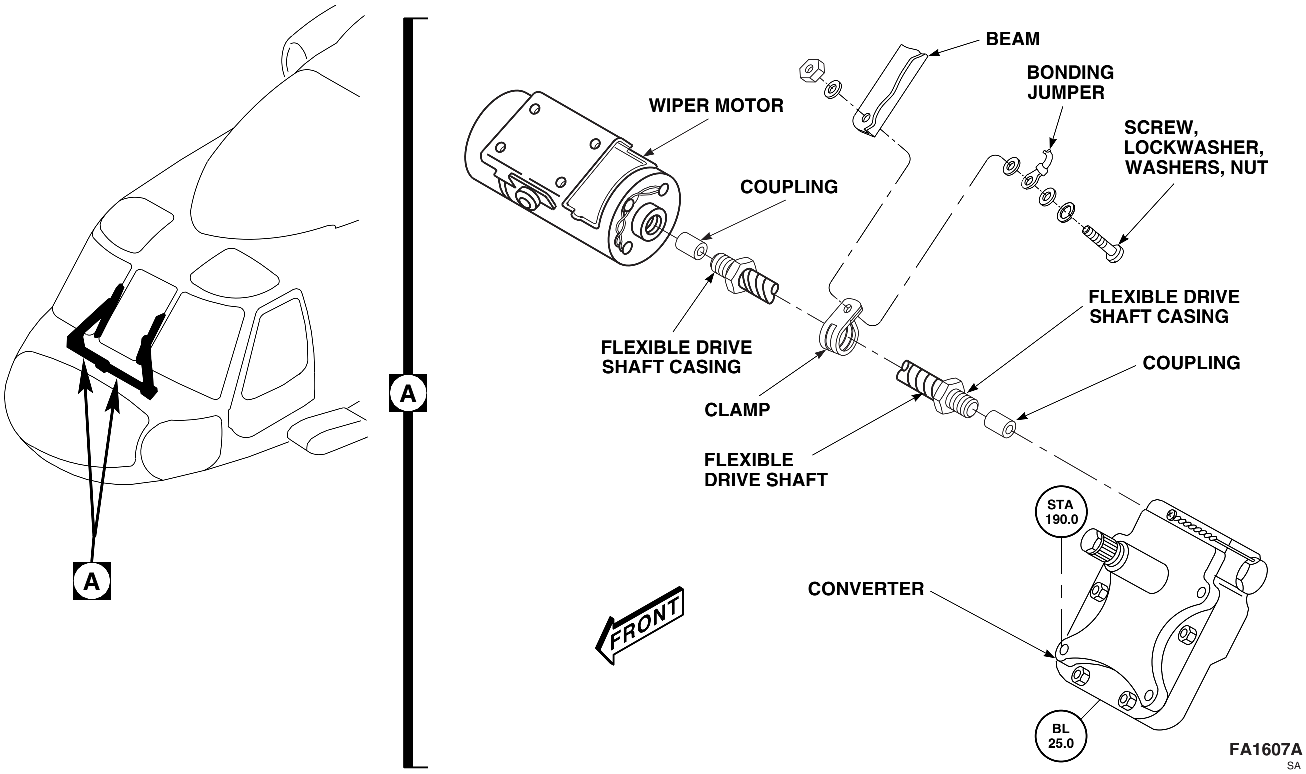


FIGURE 12-4-16-1. REPLACE WINDSHIELD WIPER FLEXIBLE DRIVE SHAFT

WARNING

Make sure spiral wrap is installed on flexible drive shaft. Make sure adequate clearance exists between flexible drive shaft and pitot static lines.

NOTE

Make sure couplings are installed in flexible drive shaft casings before connecting drive shaft to wiper motor and converter.

b. Install couplings in flexible drive shaft casings. Connect flexible drive shaft to wiper motor and converter (Figure 12-4-16-1, Detail A).

c. Using lockwire, Item 197, Appendix D, lockwire coupling nut.

d. Install clamp on flexible drive shaft.

e. Secure clamp and bonding jumper to beam with screw, lockwasher, washers, and nut.

f. Perform operational check of windshield wiper system (PARA 12-2-3).

g. Make sure area is clean and free of foreign material.

h. Install any previously removed avionics components (TM 11-70-23).

i. Close nose door.

12-4-17 WINDSHIELD WIPER BLADE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs

References

- PARA 1-6-16
- PARA 12-2-3

12-4-17.1 Replace Windshield Wiper Blade.

NOTE

Replacement of both windshield wiper blades is the same.

12-4-17.1.1 Remove.

a. Place wiper blades in PARK position.

b. Turn off all electrical power (PARA 1-6-16).

c. Hold wiper arm away from windshield. Insert cotter pin in holdoff pin hole in wiper arm to lock wiper blade away from windshield surface (Figure 12-4-17-1, Detail A).

d. Remove nuts and washers securing wiper arm and link to wiper blade. Remove wiper blade.

12-4-17.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install wiper arm and link on wiper blade and fasten with washers and nuts (Figure 12-4-17-1, Detail A). **TORQUE NUTS TO 19 - 20 INCH-POUNDS.**

c. Remove cotter pin from holdoff pin hole in wiper arm and ease wiper blade back to windshield.

d. Perform operational check of windshield wiper system (PARA 12-2-3).

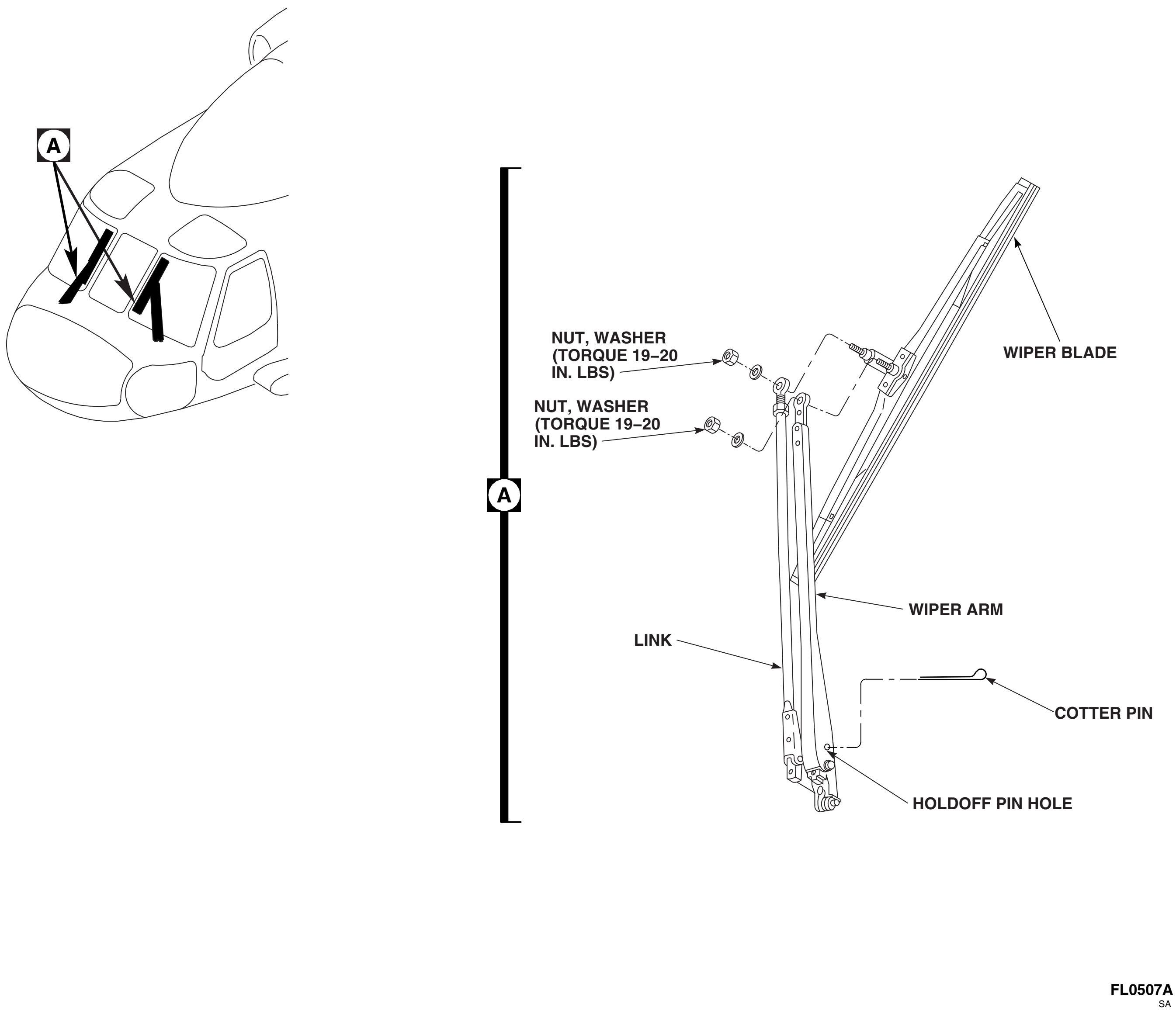


FIGURE 12-4-17-1. REPLACE WINDSHIELD WIPER BLADE

12-4-18 WINDSHIELD WIPER BLADE INSERT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Screwdriver, Thin-Bladed

References

- PARA 1-6-16
- PARA 12-4-17

12-4-18.1 Replace Windshield Wiper Blade Insert.

NOTE

Replacement of both windshield wiper blade inserts is the same.

12-4-18.1.1 Remove.

a. Remove windshield wiper blade (PARA 12-4-17).

b. Using thin-bladed screwdriver, pry between end clip and metal reinforcement strip at each end of wiper blade (Figure 12-4-18-1, Detail A) to unseat detents in end clip from holes in metal reinforcement strip (Figure 12-4-18-1, Detail B).

NOTE

Insert will have to be bowed slightly to remove it from frame.

c. With end clip unseated, pull insert and metal reinforcement strips out of frame (Figure 12-4-18-1, Detail A).

d. Remove metal reinforcement strips from insert.

12-4-18.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install metal reinforcement strips in replacement insert (Figure 12-4-18-1, Detail B).

NOTE

Insert will have to be bowed slightly to install it in frame.

c. Install insert and metal reinforcement strips into frame (Figure 12-4-18-1, Detail A).

d. Slide ends of end clip under lip of insert at ends of wiper blade.

e. Press end clips at ends of wiper blade towards each other at the same time to seat detents in end clips in holes in metal reinforcement strips

f. Install windshield wiper blade (PARA 12-4-17).

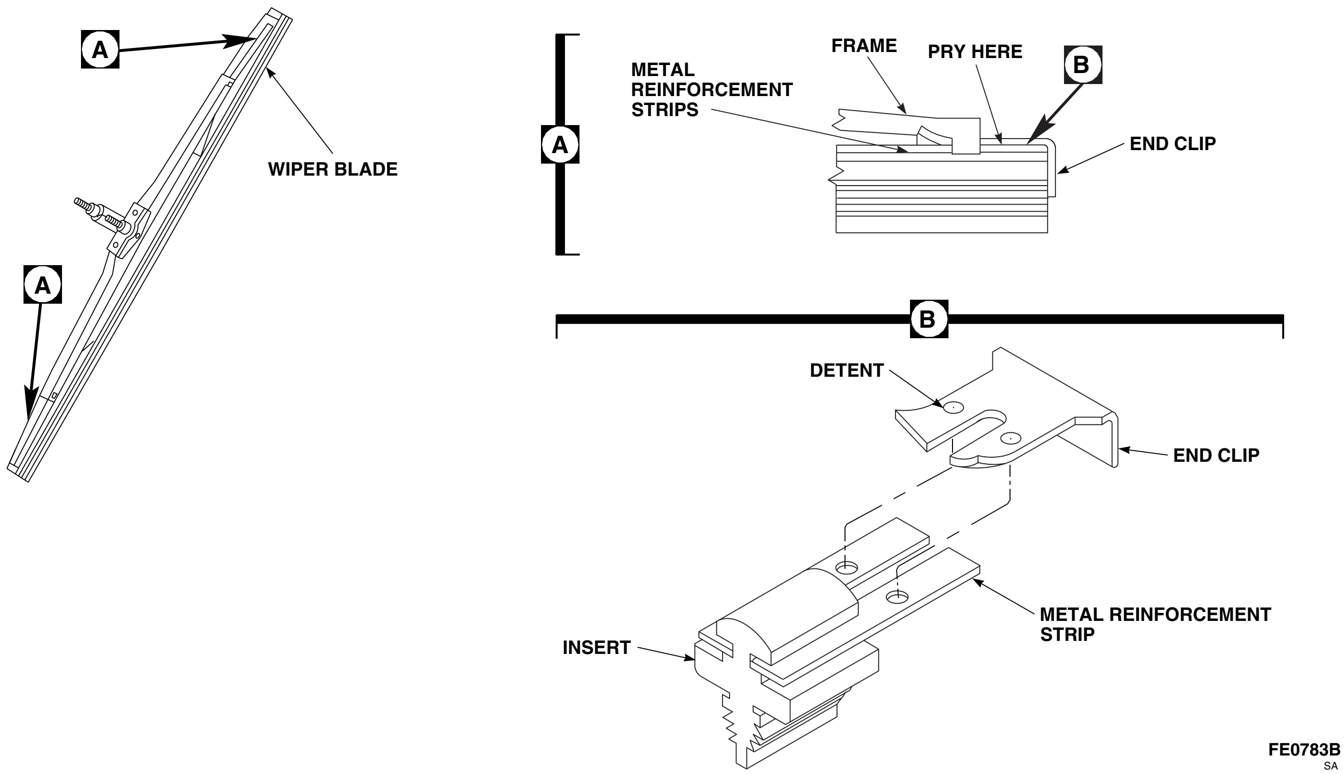


FIGURE 12-4-18-1. REPLACE WINDSHIELD WIPER BLADE INSERT

12-4-19 WINDSHIELD WIPER ARM AND LINK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Screwdriver
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 12-2-3
- PARA 12-3-2
- PARA 12-4-17

12-4-19.1 Replace Windshield Wiper Arm and Link.

NOTE

Replacement of both windshield wiper arms and links is the same.

12-4-19.1.1 Remove.

a. Remove windshield wiper blade (PARA 12-4-17).

b. Remove bolt and washer from converter shaft (Figure 12-4-19-1, Detail A).

c. Remove cotter pin and washer from pivot stud. Pull link from pivot stud (Figure 12-4-19-1, Detail A).

NOTE

If wiper arm is frozen on converter shaft, slightly spread bottom of arm using a common screwdriver.

d. Loosen wiper arm Allen-head screw. Pull wiper arm from converter shaft.

12-4-19.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove cotter pin from holdoff pin hole in wiper arm, and install in replacement wiper arm (Figure 12-4-19-1, Detail A).

c. If installed, remove bolt and washer from converter shaft prior to installing wiper arm. Place wiper arm on converter shaft with wiper arm in parked position, inboard about 40° from centerline of converter. W/O HCW (Alternate method of lining up wiper arm in proper position is a direct line-of-sight from converter shaft to 1 inch above FAT probe on center windshield frame) (Figure 12-4-19-1, Detail B).

d. Install bolt and washer on converter shaft. **TORQUE BOLT 45 - 55 INCH-POUNDS.**

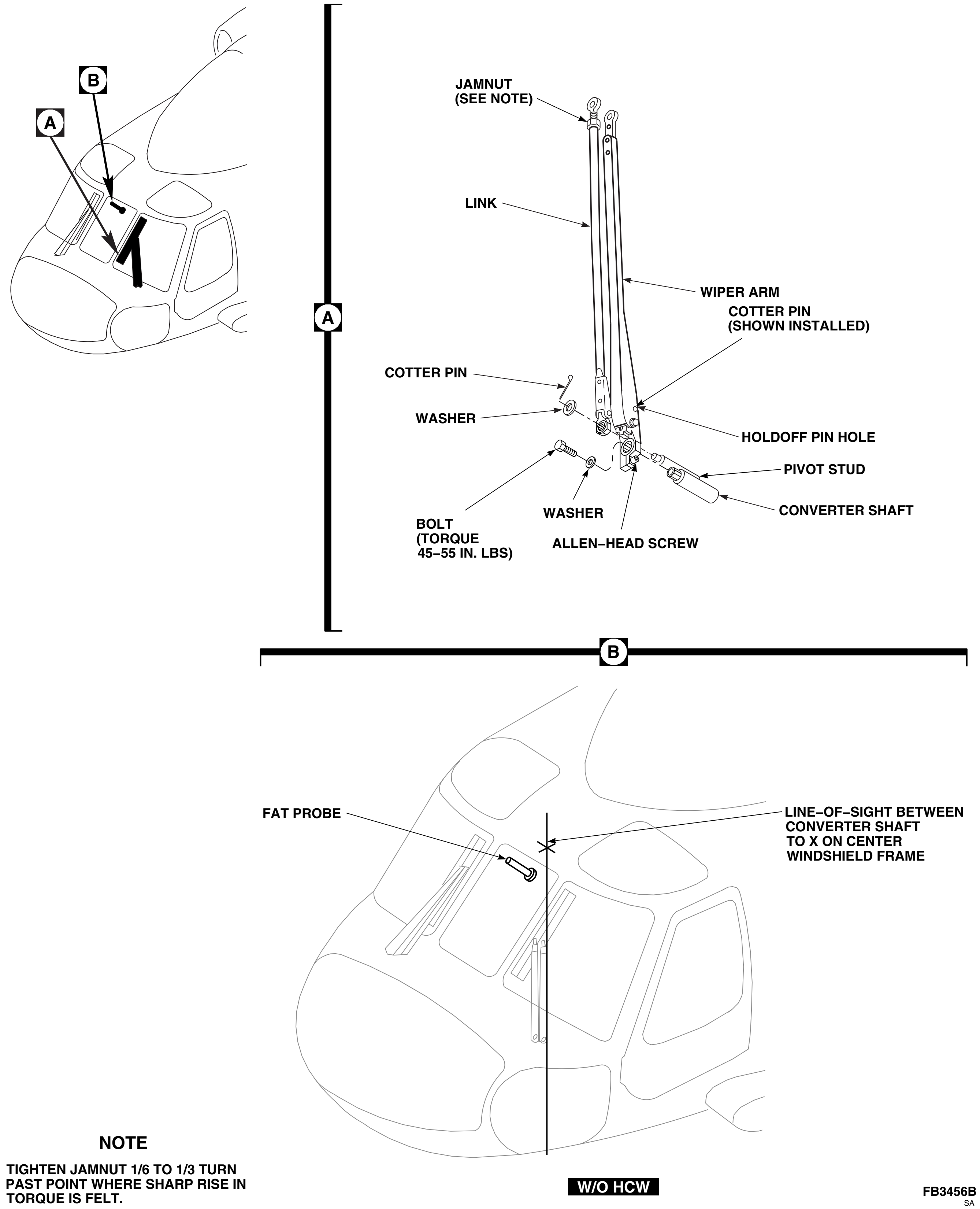


FIGURE 12-4-19-1. REPLACE WINDSHIELD WIPER ARM AND LINK

NOTE

Before tightening screw, make sure wiper blade arm and insert are against retaining bolt and washer.

e. **TIGHTEN WIPER ARM ALLEN-HEAD SCREW.**

f. Using lockwire, Item 197, Appendix D, lockwire bolt and Allen-head screw to wiper arm.

g. Secure link to pivot stud with washer and cotter pin.

h. Install windshield wiper blade (PARA 12-4-17).

i. Check alignment of wiper blade to edge of center windshield frame. **WIPER BLADE SHALL ALIGN PARALLEL TO CENTER WINDSHIELD FRAME.**

j. If adjustment is required, disconnect link from wiper blade, loosen jamnut and turn rod end ball bearing in or out to align wiper blade. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE A SHARP RISE IN TORQUE IS FELT.**

k. Check wiper arm pressure (PARA 12-3-2).

l. Perform operational check of windshield wiper system (PARA 12-2-3).

12-4-20 WINDSHIELD ANTI-ICE CONTROL UNIT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 12-2-4

12-4-20.1 Replace Windshield Anti-Ice Control Unit.

NOTE

- These steps apply to both pilot's and copilot's windshield anti-ice control units.
- HCW These steps also apply to center control unit.

12-4-20.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Disconnect electrical connectors from control unit as follows (Figure 12-4-20-1, Detail A and Detail B):

(1) P270 from J1, P271 from J2 (pilot's)

(2) P274 from J1, P275 from J2 (copilot's)

(3) P383 from J1, P384 from J2 (center)

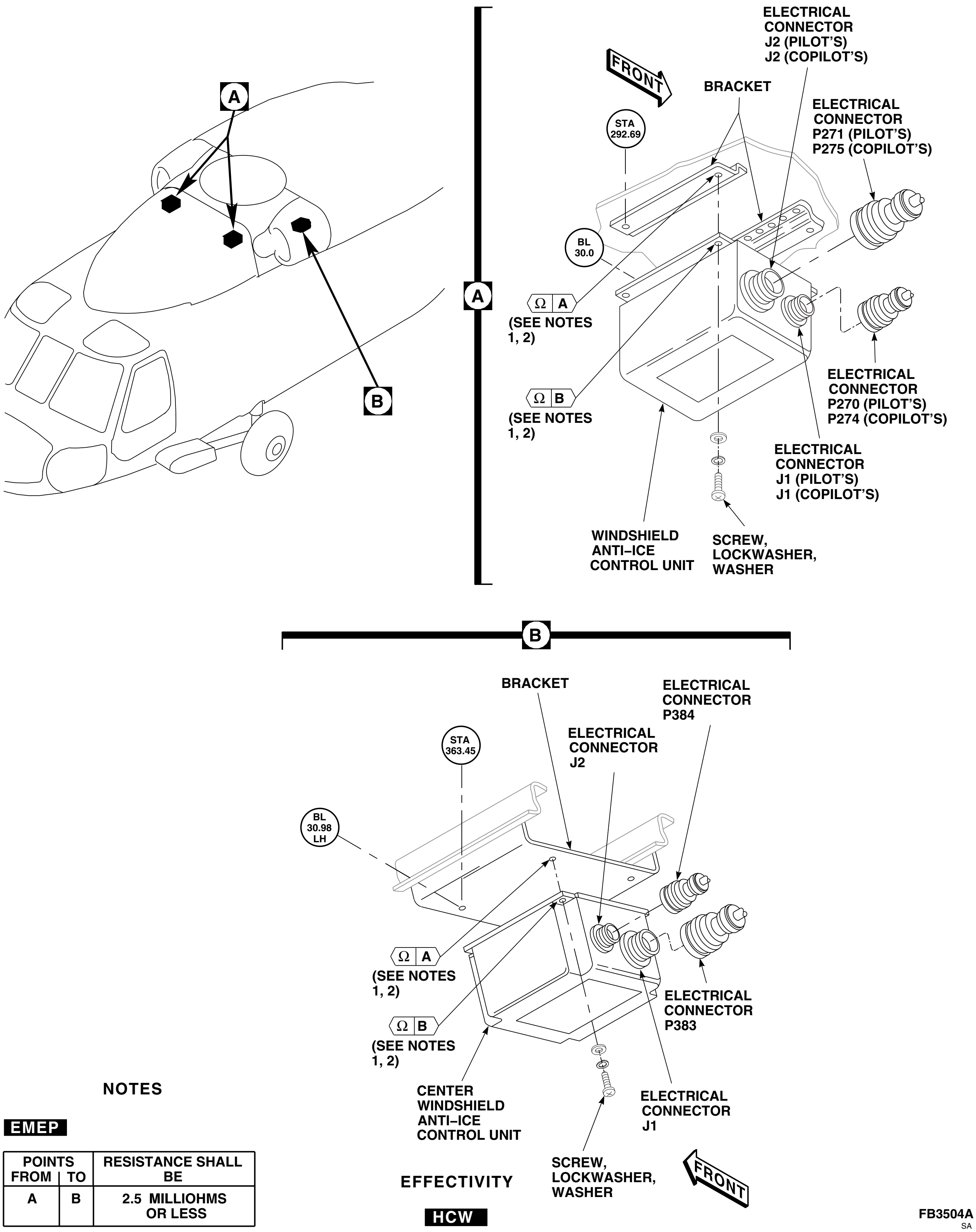
d. Install protective caps and plugs on electrical connectors.

e. Remove screws, lockwashers, and washers securing control unit to bracket. Remove control unit.

12-4-20.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Clean surface around all mounting holes using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.



NOTES

1. EMEP

2.

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS

FIGURE 12-4-20-1. REPLACE WINDSHIELD ANTI-ICE CONTROL UNIT

c. **EMEP** Inspect chemical conversion coated surfaces on center control unit for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

d. Fasten control unit to bracket with screws, lockwashers, and washers (Figure 12-4-20-1, Detail A and Detail B).

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connectors to control unit as follows:

(1) P270 to J1, P271 to J2 (pilot's)

(2) P274 to J1, P275 to J2 (copilot's)

(3) P383 to J1, P384 to J2 (center)

h. **EMEP** Perform EME resistance check of control unit as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, measure resistance between control unit and bracket surface. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, remove control unit and apply chemical conversion coating (PARA 2-6-5).

i. Perform operational check of windshield anti-ice system (PARA 12-2-4).

j. Make sure area is clean and free of foreign material.

k. Install soundproofing panels (PARA 2-4-69).

12-4-21 WINDSHIELD ANTI-ICE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

References

- PARA 9-4-5

12-4-21.1 Replace Windshield Anti-Ice Switch.

Replace pilot's, copilot's, and center windshield anti-ice switch (PARA 9-4-5).

12-4-22 ENGINE AIR INLET ANTI-ICING VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Soft Stick, Wooden or Plastic
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 4-2-7

12-4-22.1 Replace Engine Air Inlet Anti-Icing Valve.

NOTE

Replacement of No. 1 and No. 2 engine anti-icing valve is the same, except as noted.

12-4-22.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open engine cowling.

c. Remove screws, washers, and cover from engine air inlet (Figure 12-4-22-1, Detail A). Be careful not to damage temperature sensor tube when removing cover.

d. Disconnect temperature sensor tube from cover as follows:

(1) If temperature sensor hose, AE7012G0114-080, is installed, disconnect hose from fitting on cover.

(2) If temperature sensor tube, 70306-10019-041 or 70306-10019-042, is installed, slip tube out of grommet on cover. Inspect grommet for damage. **NONE ALLOWED.** If damaged, replace grommet.

e. Disconnect temperature sensor tube from anti-icing valve.

f. Keep temperature sensor tube with cover for installation.

g. Disconnect anti-icing valve electrical connector, P820 (No. 1 engine), or P822 (No. 2 engine).

h. Install protective caps and plugs on electrical connectors.

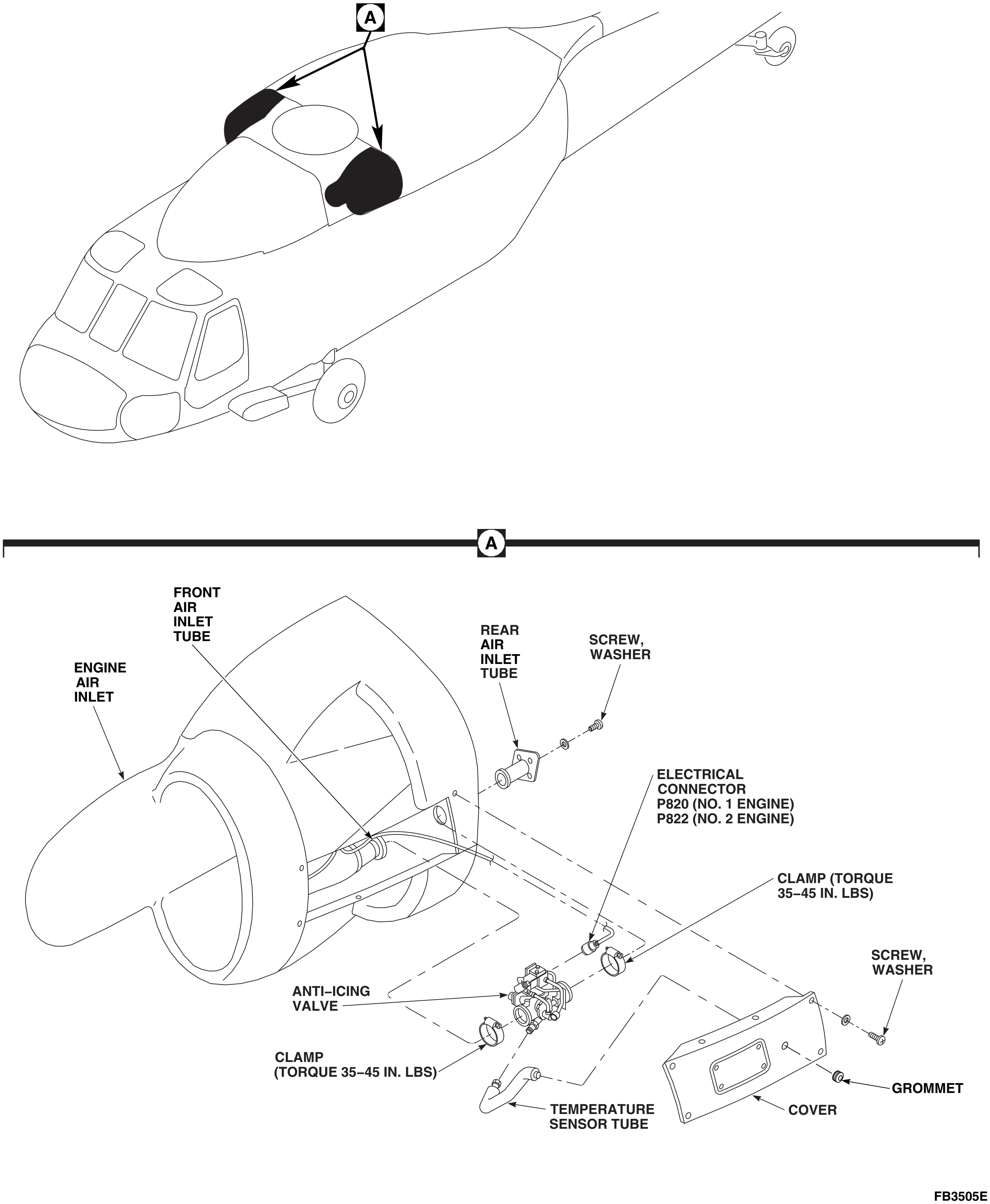


FIGURE 12-4-22-1. REPLACE ENGINE AIR INLET ANTI-ICING VALVE

i. Unclamp anti-icing valve from air inlet tubes.

j. Remove screws, washers, and rear air inlet tube from engine air inlet. Remove anti-icing valve from engine air inlet.

12-4-22.1.2 Inspect.

Inspect grommet for damage (Figure 12-4-22-1, Detail A). **NONE ALLOWED.** If damaged, replace grommet.

12-4-22.1.3 Clean.

a. Clean solenoid valve as follows:

(1) Remove screws and rim clenching clamps securing solenoid valve to valve housing assembly (Figure 12-4-22-2, Detail A). Remove solenoid valve. Remove and discard packings.

(2) Remove screws, tube retainers, and tube from housings (Figure 12-4-22-2, Detail E). Remove and discard packings.

(3) Clean valve housing, solenoid valve, and tube using dry-cleaning solvent, Item 124, Appendix D.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(4) Dry valve housing, solenoid valve, and tube with clean, filtered, low-pressure compressed air (15 psig maximum).

(5) Install packings onto tube and install tube with screws and tube retainers.

(6) Install packings onto solenoid valve and install solenoid valve into valve housing assembly. Secure with rim clenching clamps and screws (Figure 12-4-22-2, Detail A).

b. Clean nozzle, tube nipple, sleeve spacer, and nozzle housing assembly as follows:

(1) Remove screws and washers securing nozzle housing assembly to valve housing assembly (Figure 12-4-22-2, Detail B). Remove nozzle housing assembly and sleeve spacer.

(2) Remove tube nipple from valve housing assembly.

(3) Clean nozzle, tube nipple, sleeve spacer, and nozzle housing assembly using dry-cleaning solvent, Item 124, Appendix D.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(4) Dry nozzle, tube nipple, sleeve spacer, and nozzle housing assembly with clean, dry, filtered, low-pressure compressed air (15 psig maximum).

(5) Install tube nipple into valve housing assembly.

(6) Install sleeve spacer and nozzle housing assembly into valve housing assembly with washers and screws.

c. Clean valve body as follows:

(1) Remove screws, washers, and nuts securing valve cover assembly to housing as-

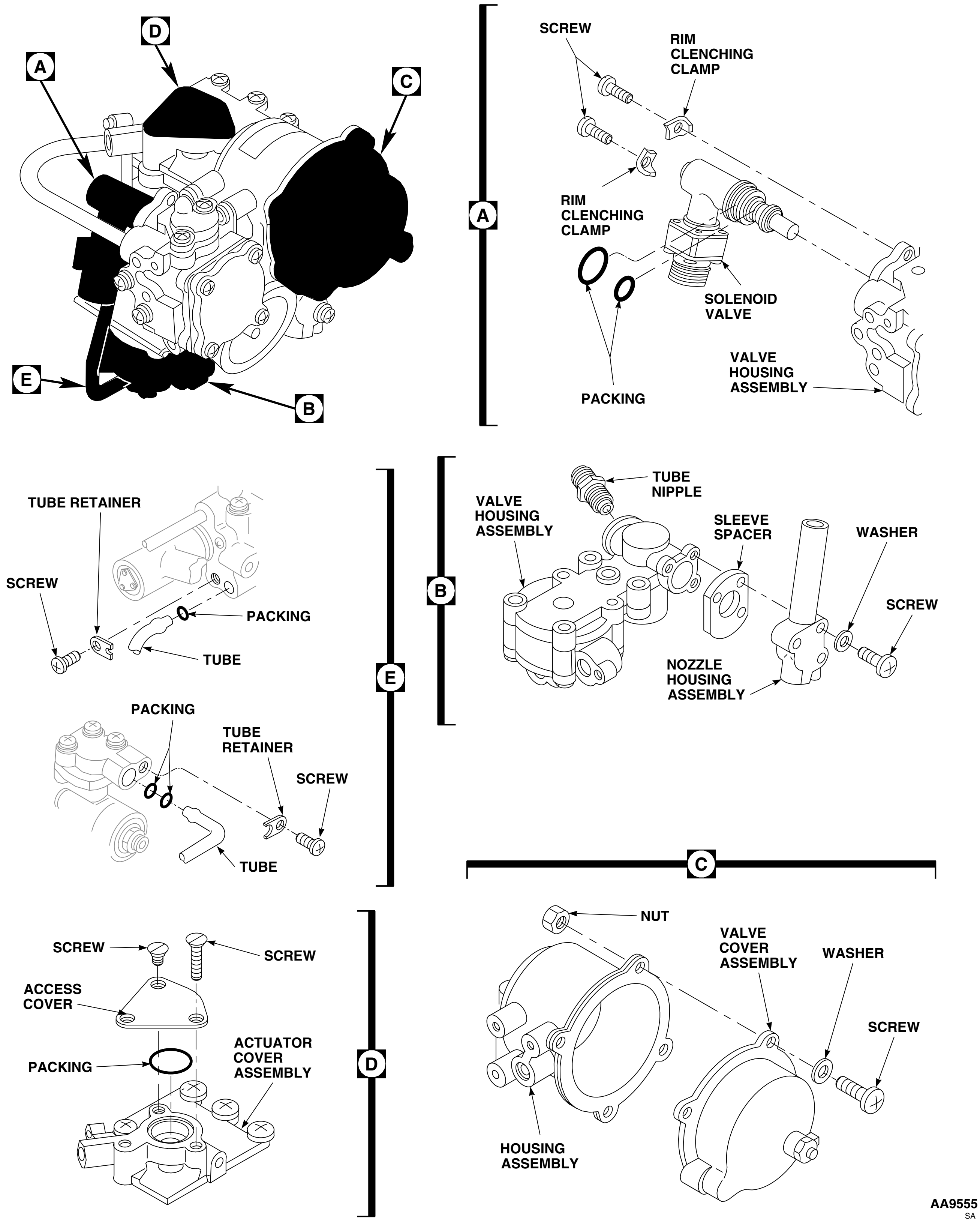


FIGURE 12-4-22-2. CLEAN ENGINE AIR INLET ANTI-ICING VALVE sembly (Figure 12-4-22-2, Detail C). Remove valve cover assembly.

(2) Hold housing assembly upside down and clean inside of housing assembly and air diaphragm using dry-cleaning solvent, Item 124, Appendix D. Make sure air diaphragm is free of dirt and sand.

(3) Spray dry-cleaning solvent, Item 124, Appendix D, inside valve body while gently pushing spring in housing assembly in and out.

(4) Air-dry housing assembly at room temperature.

CAUTION

Damage will occur to anodized coating of valve body if anything other than a soft wooden or plastic stick is used to hold open valve.

(5) Push actuator spring down to open valve inside of valve body and insert a soft wooden or plastic stick inside valve to keep it open.

(6) Install valve cover assembly. Make sure diaphragm seats correctly into housing assembly diaphragm slot. Install screws, washers, and nuts.

CAUTION

Damage will occur to anodized coating of valve body if care is not used in removing wooden or plastic stick.

(7) Slowly remove wooden or plastic stick from inside valve.

d. Clean actuator cover assembly as follows:

(1) Remove screws securing access cover to actuator cover assembly (Figure 12-4-22-2, Detail D). Remove access cover and discard packing.

(2) Clean actuator cover assembly using dry-cleaning solvent, Item 124, Appendix D.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(3) Dry actuator cover assembly with clean, dry, filtered, low-pressure compressed air.

(4) Install packing and access cover with screws.

e. Clean temperature sensor tube using dry-cleaning solvent, Item 124, Appendix D.

12-4-22.1.4 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Place anti-icing valve in engine inlet and install rear air inlet tube with screws and washers (Figure 12-4-22-1, Detail A).

NOTE

Make sure flow arrow on valve is pointing in direction of airflow.

c. Connect anti-icing valve to front and rear air inlet tubes with clamps. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

d. Connect temperature sensor tube to cover as follows:

(1) If temperature sensor hose, AE7012G0114-080, is installed, connect hose to anti-icing valve and fitting on cover.

(2) If temperature sensor tube, 70306-10019-041 or 70306-10019-042, is installed, connect tube to anti-icing valve. When cover is installed, make sure that end of tube is in position to slip into grommet on cover.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connector, P820 (No. 1 engine), or P822 (No. 2 engine), to anti-icing valve.

h. Make sure area is clean and free of foreign material.

NOTE

Make sure that end of temperature sensor tube is in position to slip into grommet on cover.

i. Install cover on engine air inlet with washers and screws.

j. Close engine cowling.

k. Perform operational check of engine anti-ice system (PARA 4-2-7).

12-4-23 ENGINE AIR INLET ANTI-ICE THERMAL SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 4-2-7

12-4-23.1 Replace Engine Air Inlet Anti-Ice Thermal Switch.

NOTE

Replacement of No. 1 and No. 2 engine thermal switch is the same, except as noted.

12-4-23.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and washers from cover on engine air inlet (Figure 12-4-23-1, Detail A).

c. Disconnect temperature sensor tube from cover as follows:

(1) If temperature sensor hose, AE7012G0114-080, is installed, disconnect hose from fitting on cover.

(2) If temperature sensor tube, 70306-10019-041 or 70306-10019-042, is installed, slip tube out of grommet on cover.

d. Disconnect electrical connector, P821 (No. 1 engine) or P828 (No. 2 engine), from thermal switch.

e. Install protective caps and plugs on electrical connectors.

f. Remove thermal switch from engine air inlet.

12-4-23.1.2 Inspect.

Inspect grommet for damage (Figure 12-4-23-1, Detail A). **NONE ALLOWED.** If damaged, replace grommet.

12-4-23.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install thermal switch in engine air inlet (Figure 12-4-23-1, Detail A).

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P821 (No. 1 engine) or P828 (No. 2 engine), to thermal switch.

f. Make sure area is clean and free of foreign material.

g. Connect temperature sensor tube from cover as follows:

(1) If temperature sensor hose, AE7012G0114-080, is installed, connect hose to fitting on cover.

(2) If temperature sensor tube, 70306-10019-041 or 70306-10019-042, is installed, slip tube into grommet on cover.

h. Install cover on engine air inlet with temperature sensor tube positioned in grommet in cover. Secure cover with screws and washers.

i. Perform operational check of engine anti-ice system (PARA 4-2-7).

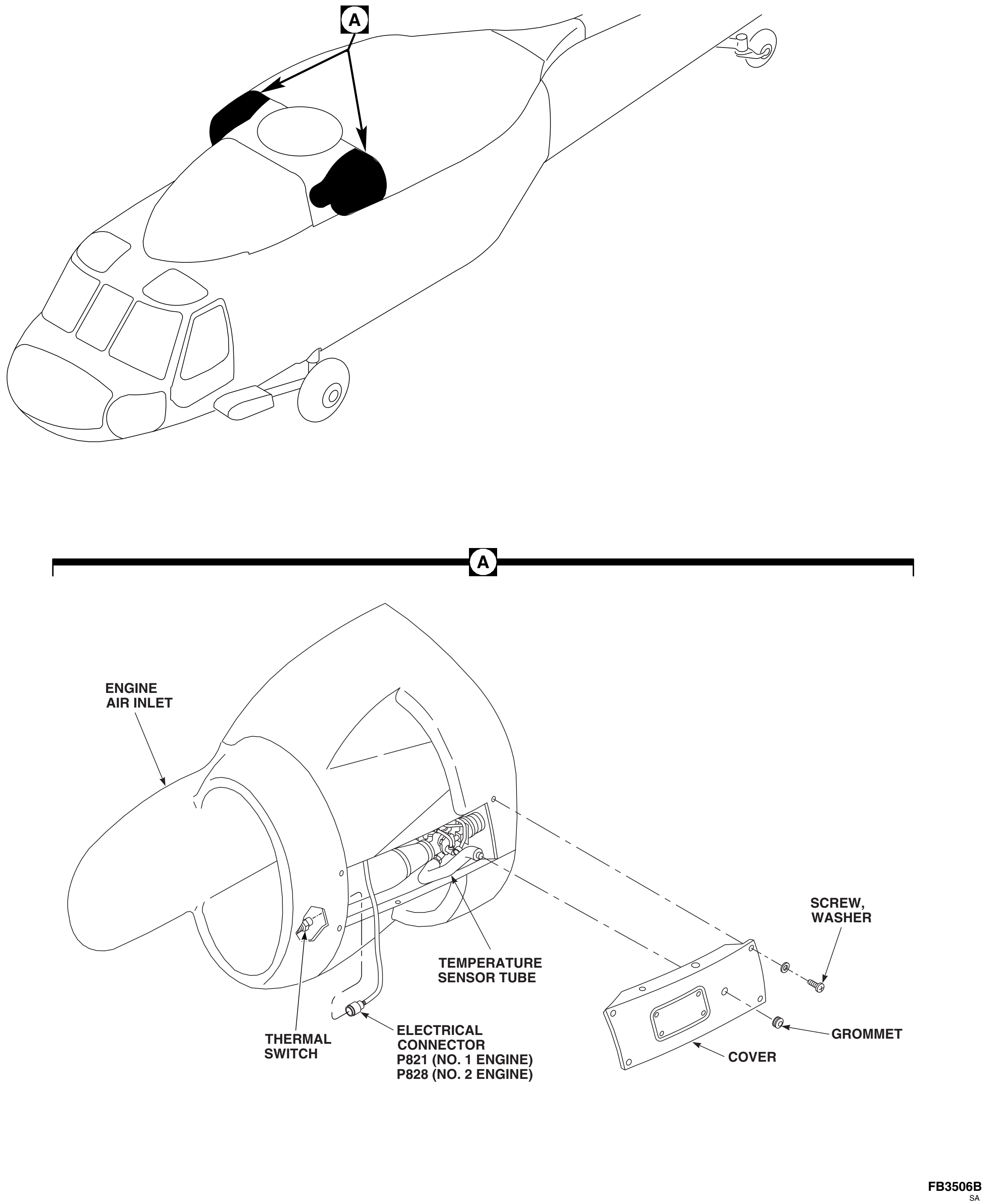


FIGURE 12-4-23-1. REPLACE ENGINE AIR INLET ANTI-ICE THERMAL SWITCH

12-4-24 BLADE DEICE FLANGE SEAL AND PACKING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 2-4-70
- PARA 12-2-5
- PARA 12-4-26

12-4-24.1 Replace Blade Deice Flange Seal and Packing.

12-4-24.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove main transmission drip pan (PARA 2-4-70).

c. Drain main transmission (PARA 1-3-12).

d. Remove distributor and slipring/brush assembly (PARA 12-4-26).

e. Remove nuts, washers, and sump bracket from flange seal (Figure 12-4-24-1, Detail A).

f. Remove flange seal from main transmission.

g. Remove and discard packing.

12-4-24.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install packing, lubricated with lubricating oil, Item 205 or Item 206, Appendix D (Figure 12-4-24-1, Detail A).

c. Coat end of flange seal, opposite flange, with lubricating oil, Item 205 or Item 206, Appendix D.

CAUTION

A spring-loaded oil seal in the rotor shaft can be damaged if flange seal is installed crooked. Make sure to install flange seal straight into sump.

d. Carefully install flange seal on transmission. If binding occurs, flange seal is not lined up with

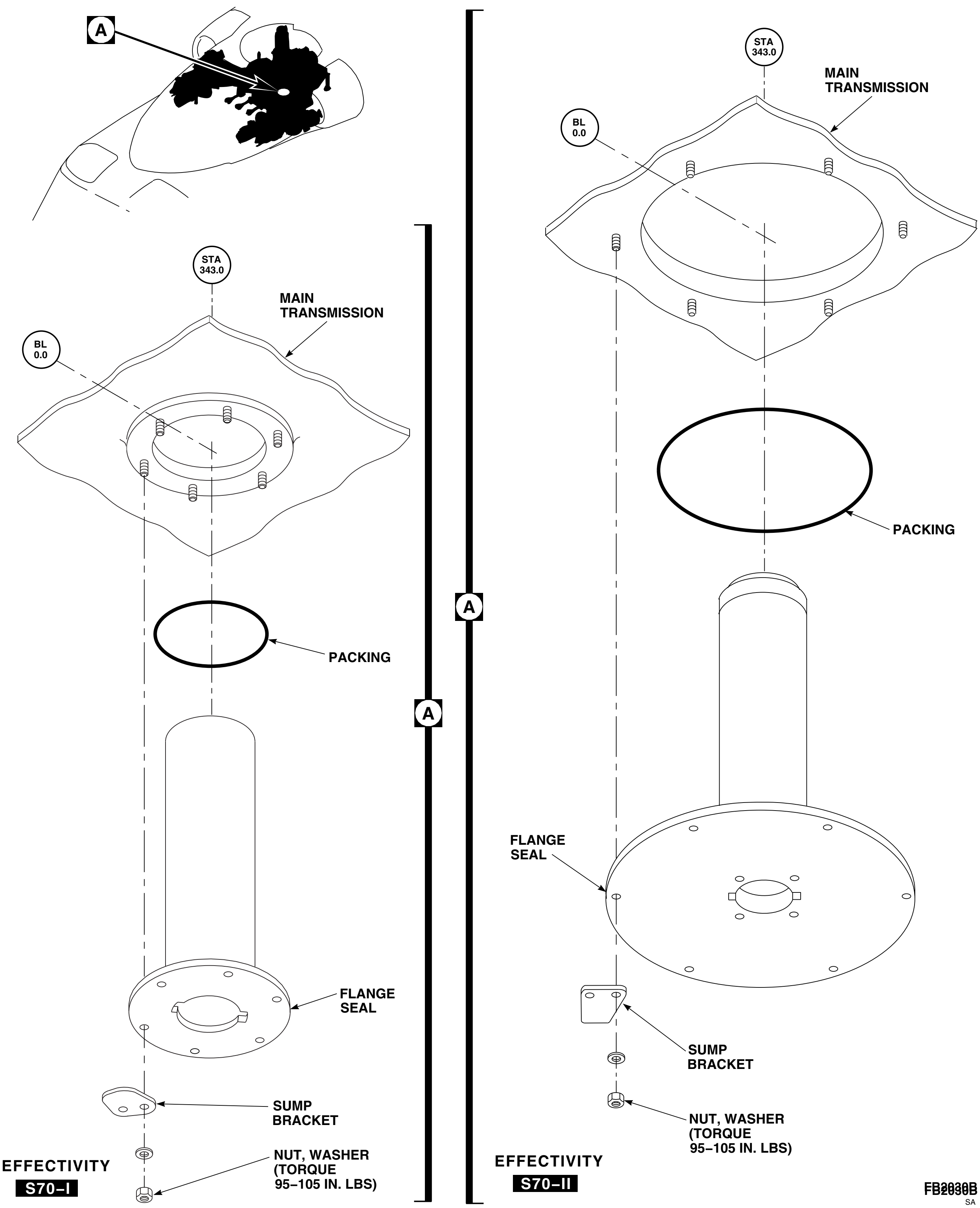


FIGURE 12-4-24-1. REPLACE BLADE DEICE FLANGE SEAL AND PACKING rotor shaft oil seal. Straighten flange seal and slowly turn flange while pressing it into rotor shaft.

e. Secure flange seal using nuts, washers, and sump bracket.

f. **TORQUE FLANGE SEAL NUTS TO 95 - 105 INCH-POUNDS.**

g. Apply sealing compound, Item 292, Appendix D, around flange seal edges and mounting hardware.

h. Install distributor and slipring/brush assembly (PARA 12-4-26).

i. Service main transmission (PARA 1-3-12).

j. Perform operational check of blade deicing system (PARA 12-2-5).

k. Install main transmission drip pan (PARA 2-4-70).

12-4-25 BLADE DEICE TUBE ASSEMBLY AND PACKING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-70
- PARA 12-2-5
- PARA 12-4-26

12-4-25.1 Replace Blade Deice Tube Assembly and Packing.

12-4-25.1.1 Remove.

a. Remove main rotor blade deice distributor and slipring/brush assembly (PARA 12-4-26).

b. Reach inside main rotor shaft and remove tube assembly (Figure 12-4-25-1, Detail A).

c. Remove and discard packing at bottom outer edge of tube assembly.

12-4-25.1.2 Inspect/Repair.

a. Clean tube assembly using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

b. Inspect antitorque collar for loose bolts and bent or cracked disc pack (Figure 12-4-25-1, Detail B).

c. Inspect tube assembly as follows (Figure 12-4-25-2):

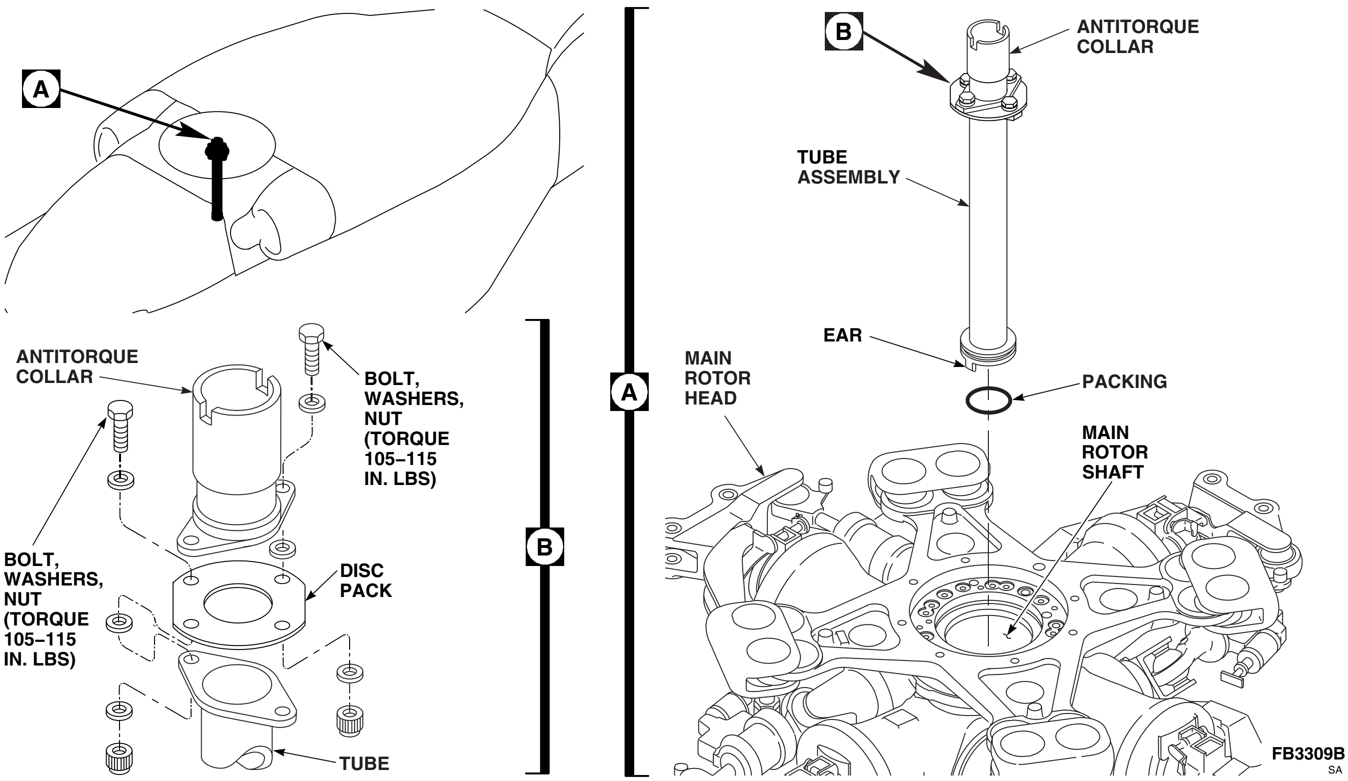


FIGURE 12-4-25-1. REPLACE BLADE DEICE TUBE ASSEMBLY AND PACKING

(1) Repair damage to tube, antitorque collar, and fitting that is within limits by blending and polishing as follows:

(a) Polish scratches, nicks, or scoring using abrasive cloth, Item 6, Appendix D.

(b) Blend radius on tube outside diameter shall be ½-inch or greater.

(c) Fine scratches not penetrating through paint do not need to be polished.

(d) Touch up aluminum surfaces with corrosion resistant coating, Item 118, Appendix D.

(2) **AREA A.** Inspect tube for scratches, nicks, or scoring. **DAMAGE SHALL NOT EXCEED 0.020-INCH IN DEPTH OR EXCEED 20% OF TOTAL AREA.** If damage exceeds limits, replace tube (PARA 12-4-25.1.3).

(3) **AREA B.**

(a) Inspect antitorque collar for scratches, nicks, or scoring. **DAMAGE SHALL NOT EXCEED 0.010-INCH IN DEPTH OR EXCEED 10% OF TOTAL AREA.** If damage exceeds limits, replace antitorque collar (PARA 12-4-25.1.3).

(b) Inspect antitorque collar bolthole for elongation and wear. **DIAMETER SHALL NOT EXCEED 0.275-INCH.** If damage exceeds limits, replace antitorque collar (PARA 12-4-25.1.3).

(4) **AREA C.**

(a) Inspect fitting for scratches, nicks, or scoring. **DAMAGE SHALL NOT EXCEED 0.010-INCH IN DEPTH OR EXCEED 10% OF TOTAL AREA.** If damage exceeds limits, replace tube (PARA 12-4-25.1.3).

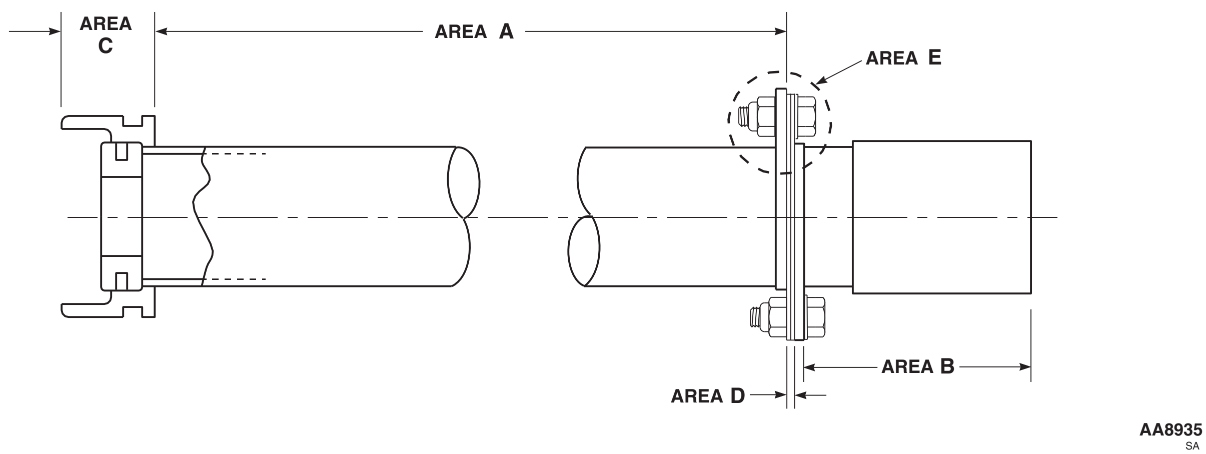


FIGURE 12-4-25-2. INSPECT TUBE ASSEMBLY

(b) Inspect packing groove in fitting for scratches, nicks, or scoring. **NONE ALLOWED.** If damage exceeds limits, replace tube (PARA 12-4-25.1.3).

(5) **AREA D.**

(a) Inspect tube flange for scratches, nicks, or scoring. **DAMAGE SHALL NOT EXCEED 0.010-INCH IN DEPTH OR EXCEED 10% OF TOTAL AREA.** If damage exceeds limits, replace tube (PARA 12-4-25.1.3).

(b) Inspect tube flange bolthole for elongation and wear. **DIAMETER SHALL NOT EXCEED 0.275-INCH.** If damage exceeds limits, replace tube flange collar (PARA 12-4-25.1.3).

(6) **AREA E.**

(a) Inspect disc pack for corrosion. **PITTING SHALL NOT EXCEED 0.0005-INCH IN DEPTH.** Remove corrosion using abrasive cloth, Item 10, Appendix D. If damage exceeds limit, replace disc pack (PARA 12-4-25.1.3).

(b) Inspect disc pack for dents and nicks on edges of individual disks. **DAMAGE SHALL NOT EXCEED 0.020-INCH IN DEPTH.** Remove dents and nicks using abrasive cloth, Item 6, Appendix D. Polish repaired area using abrasive cloth, Item 10, Appendix D. If damage exceeds limit, replace disc pack (PARA 12-4-25.1.3).

d. Inspect bottom interior of tube for damaged or missing grommet. If grommet is damaged or missing, replace tube.

12-4-25.1.3 Replace Tube Components.

a. Remove bolts, washers, nuts, disc pack, and antitorque collar from tube (Figure 12-4-25-1, Detail B).

b. Install disc pack and antitorque collar on tube with bolts, washers, and nuts. **TORQUE NUTS 105 - 115 INCH-POUNDS.**

12-4-25.1.4 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install packing and lubricate with lubricating oil, Item 205 or Item 206, Appendix D, on tube assembly (Figure 12-4-25-1, Detail A).

c. Place tube assembly in main rotor shaft with antitorque collar up.

d. Turn tube assembly until ears on bottom of tube assembly line up with slots on flange seal. Press tube down until seated.

e. Install main rotor blade deice distributor and slipring/brush assembly (PARA 12-4-26).

12-4-26 MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit
- Heat Gun
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 56, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 163, Appendix D
- Lockwire, Item 194, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Spiral Wrap, Item 314, Appendix D
- Tags
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-4-70
- PARA 2-6-5
- PARA 12-2-5

12-4-26.1 Replace Main Rotor Blade Deice Distributor and Slipring/Brush Assembly.

12-4-26.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel and damage to equipment will occur if bifilar turns while maintenance is being performed in this area. Personnel or tools can come in contact with moving parts. Make sure gust lock is engaged before beginning maintenance on this area.

b. If not already done, engage gust lock (PARA 1-6-18).

c. Remove main transmission drip pan (PARA 2-4-70).

d. W/O EMEP Disconnect slipring harness as follows:

(1) Remove soundproofing panels as required (PARA 2-4-69).

(2) Tag wire and remove screw, lockwasher, washers, and nut, and disconnect ground wire, 1-20-S3, from airframe (Figure 12-4-26-1, Sheet 1, Detail A).

(3) Loosen fasteners securing cover on main rotor blade deice junction box and let cover hang by chain (Figure 12-4-26-1, Sheet 1, Detail B).

(4) Tag wires and remove nuts, lockwashers, and washers and disconnect slipring harness wires from terminal board 3.

(5) Pull wires out through grommet.

e. EMEP Disconnect slipring and deice harnesses as follows:

(1) Remove soundproofing panels as required (PARA 2-4-69).

(2) At STA 345.0, RBL 16.5, loosen thumb screws and remove cover from terminal board 7 (Figure 12-4-26-1, Sheet 3, Detail F).

(3) Tag wires and remove nuts, lockwashers, and washers and disconnect slipring and deice harnesses from terminal board 7.

(4) Tag wires and remove screw, lockwasher, washers, and nut, and disconnect ground wire, GNDS3-1, and alpha braid, GNDS3-1, from airframe (Figure 12-4-26-1, Sheet 1, Detail A).

(5) Loosen fasteners that secure cover on main rotor blade deice junction box and let cover hang by chain (Figure 12-4-26-1, Sheet 1, Detail B).

(6) Tag wires and remove nuts, lockwashers, and washers and disconnect slipring harness wires from terminal board 3.

(7) Pull wires out through grommet.

f. Remove screws, washers, nuts, and clamps securing slipring harness, as required, to free harness (Figure 12-4-26-1, Sheet 2, Detail C and Sheet 3, Detail E).

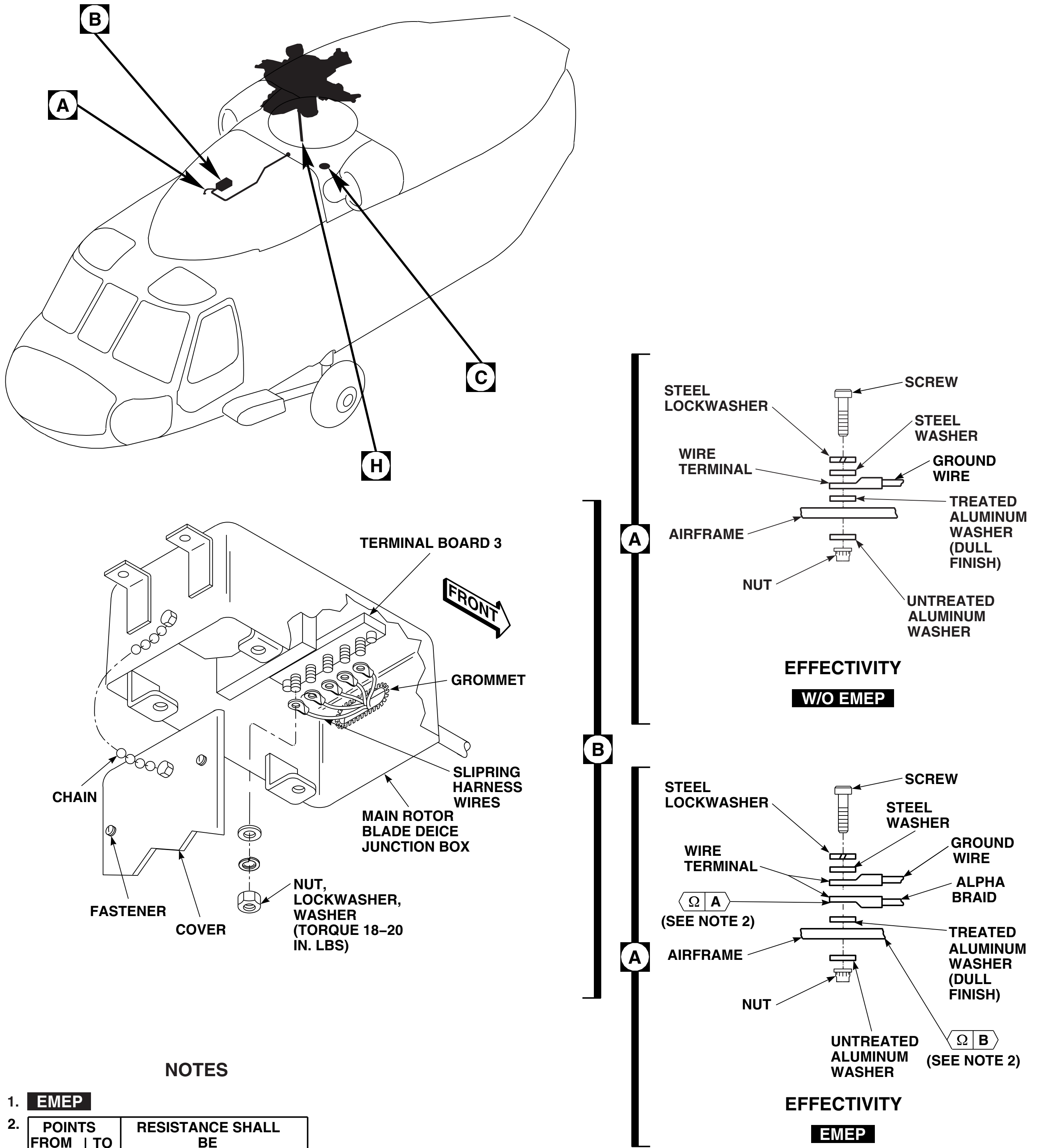
g. Remove anti-chafe strip from slipring harness.

h. Pull slipring harness through clamps or grommet until it hangs down at flange seal. Remove slipring harness tiedown straps as required to free harness.

i. W/O EMEP Clean sealing compound from slipring harness and grommet with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

j. W/O EMEP S70–I Remove nuts and washers securing conduit bracket to flange seal and remove from studs (Figure 12-4-26-1, Sheet 4, Detail G).

k. W/O EMEP S70–II Remove bolts and washers securing conduit bracket to flange seal (Figure 12-4-26-1, Sheet 4, Detail G).



NOTES

1. EMEP
2.

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS

3. REFER TO INSTALLATION PROCEDURE FOR CORRECT PLACEMENT OF WIRE ENDS TO TERMINALS.
4. NUT, LOCKWASHER, WASHER, AND WIRE SHOWN REMOVED FOR CLARITY.

FC3699_1F
SA

FIGURE 12-4-26-1. REPLACE MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY (SHEET 1 OF 6)

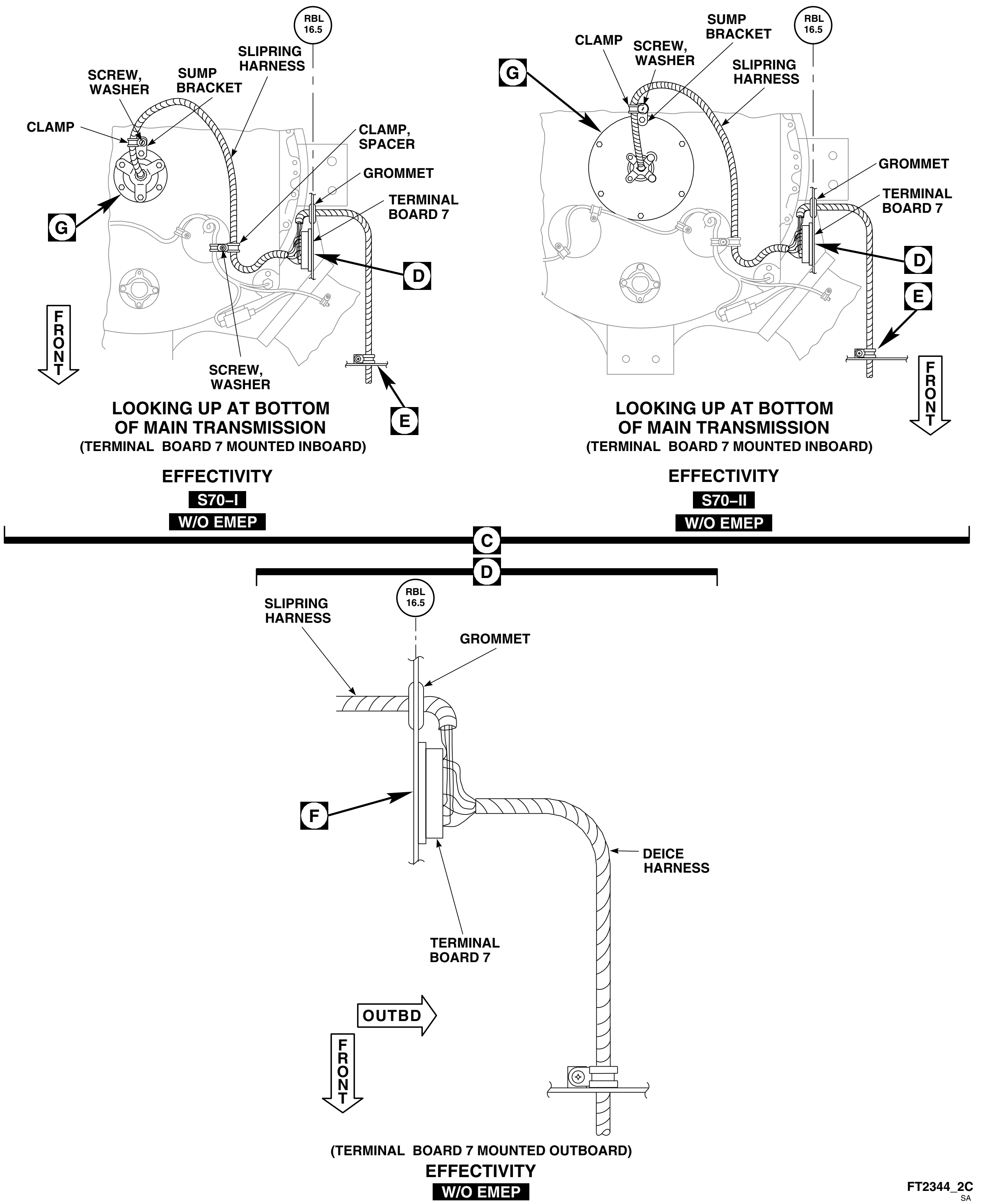


FIGURE 12-4-26-1. REPLACE MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY (SHEET 2 OF 6)

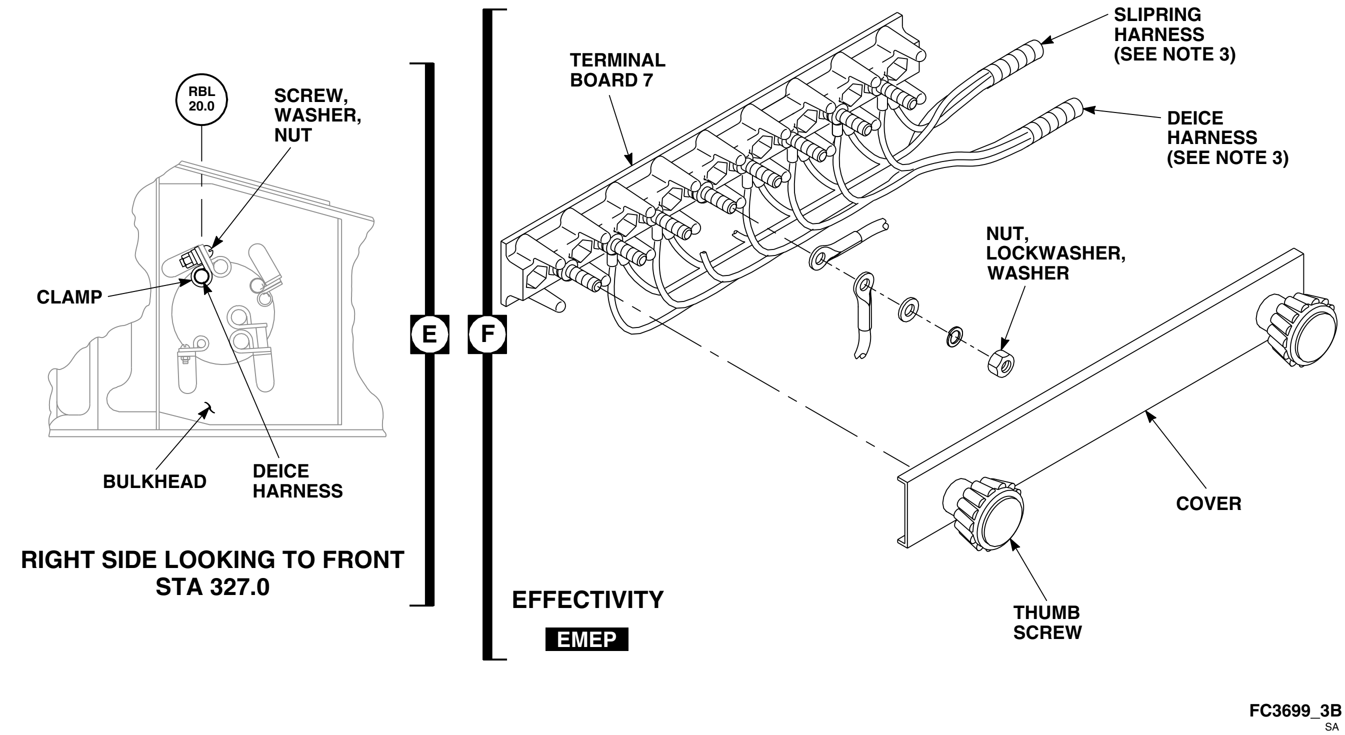


FIGURE 12-4-26-1. REPLACE MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY (SHEET 3 OF 6)

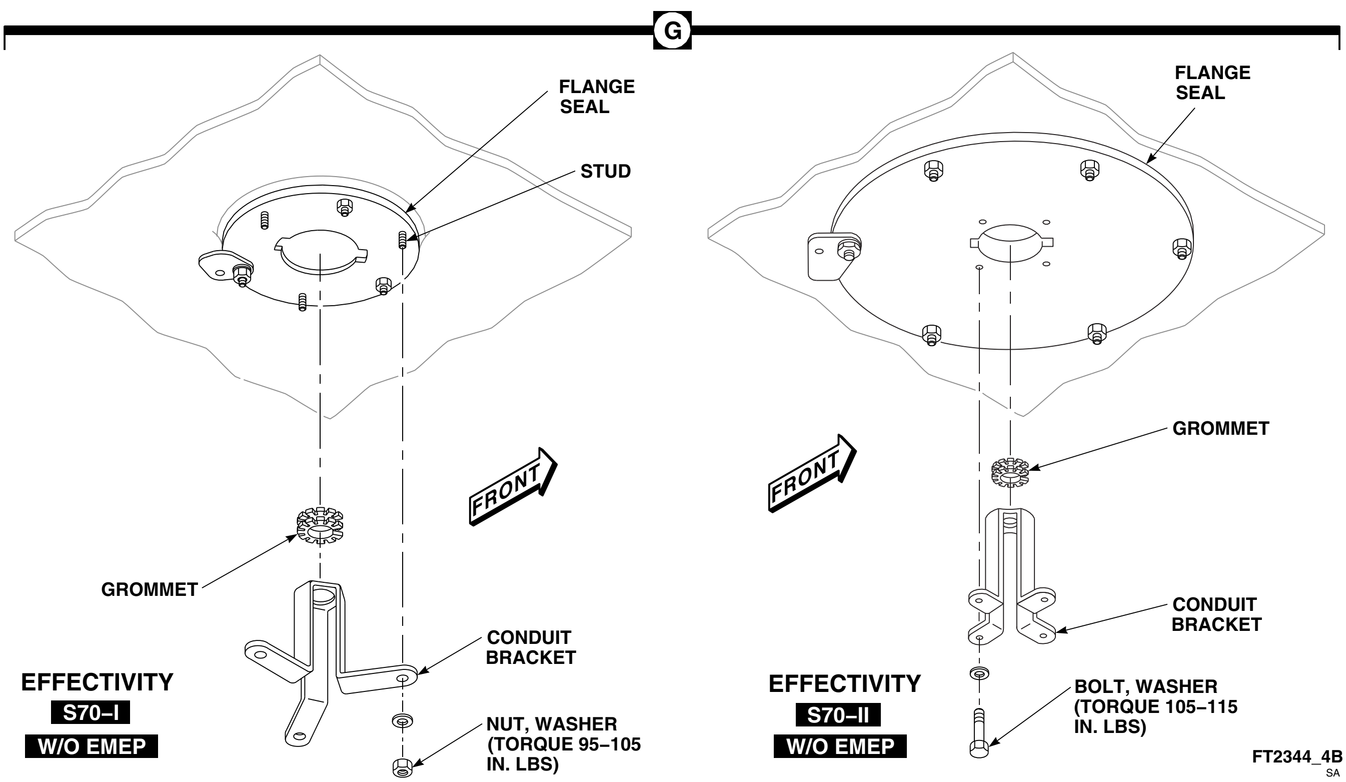
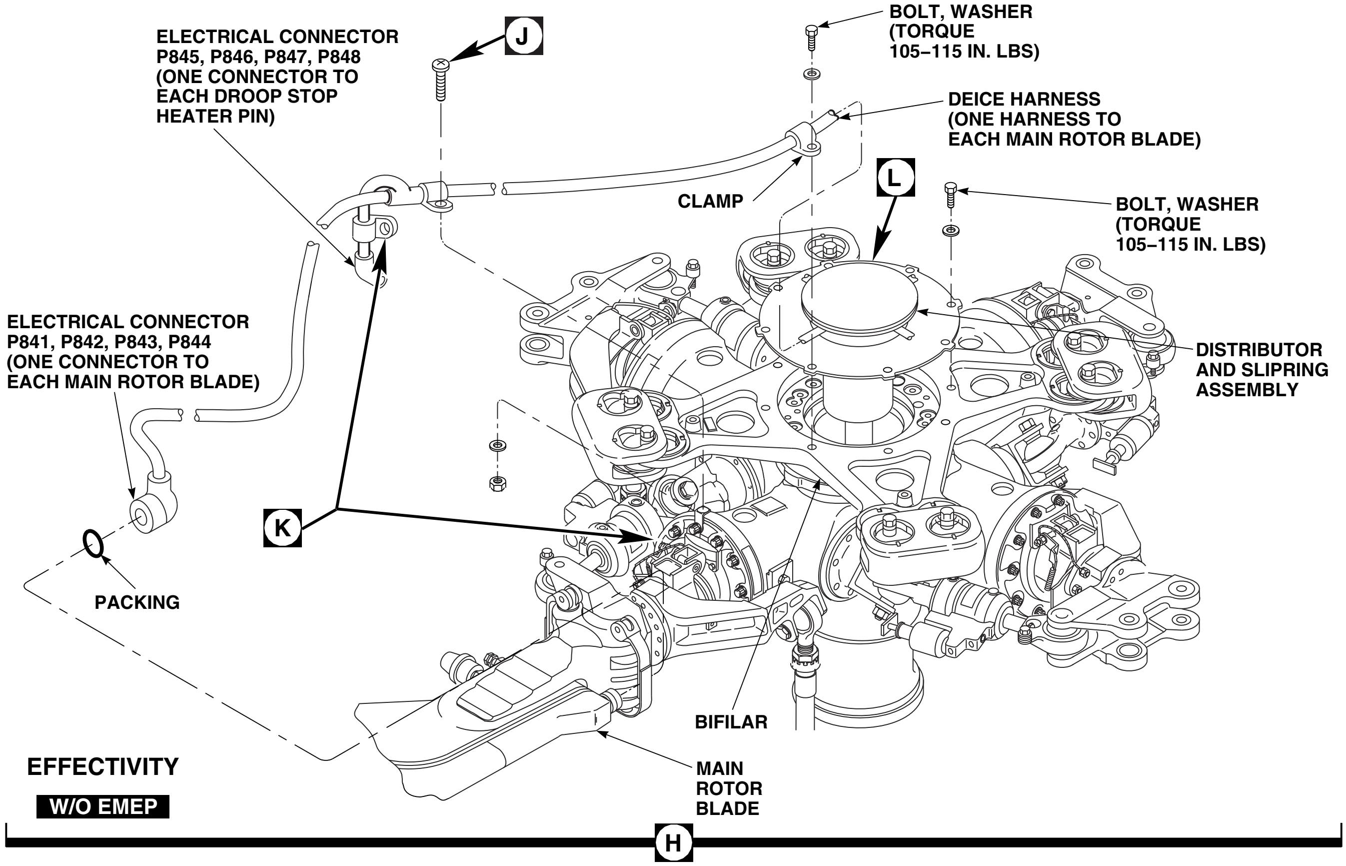


FIGURE 12-4-26-1. REPLACE MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY (SHEET 4 OF 6)



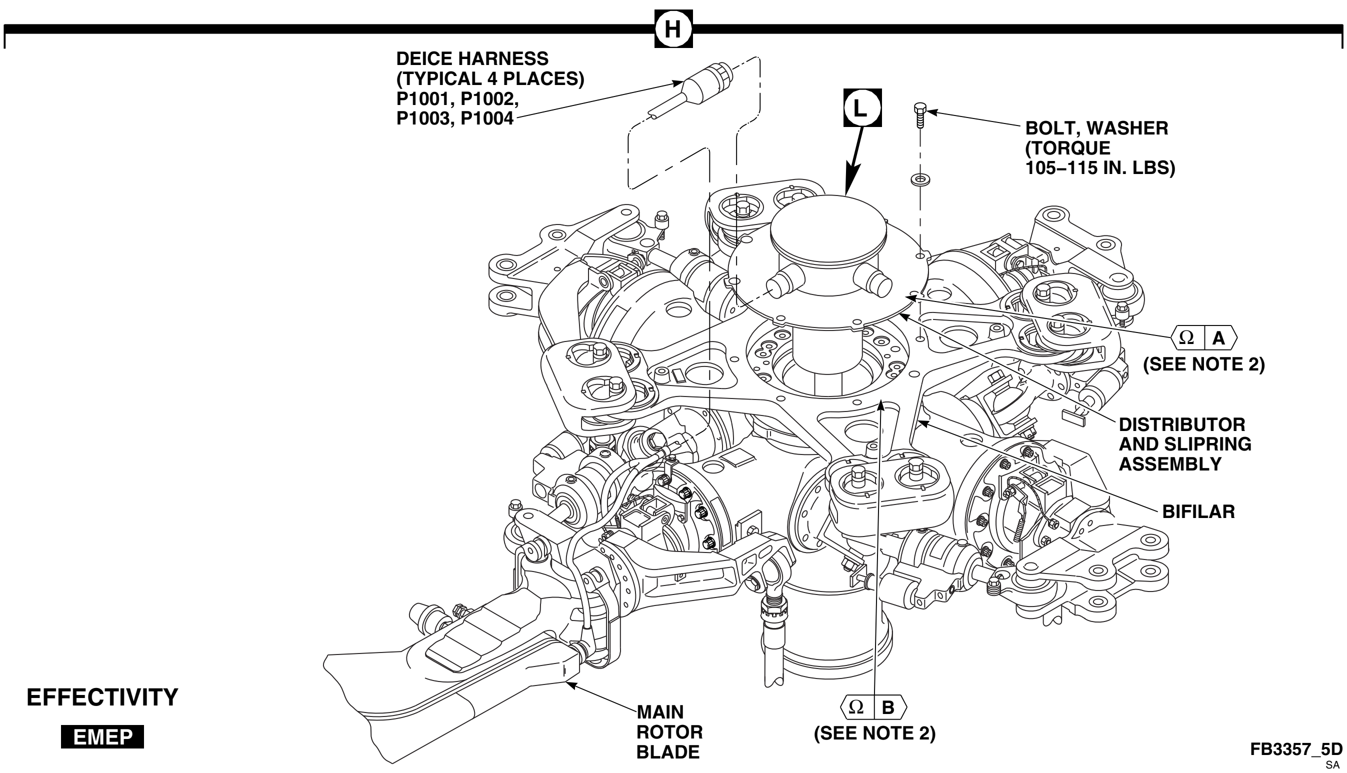


FIGURE 12-4-26-1. REPLACE MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY (SHEET 5 OF 6)

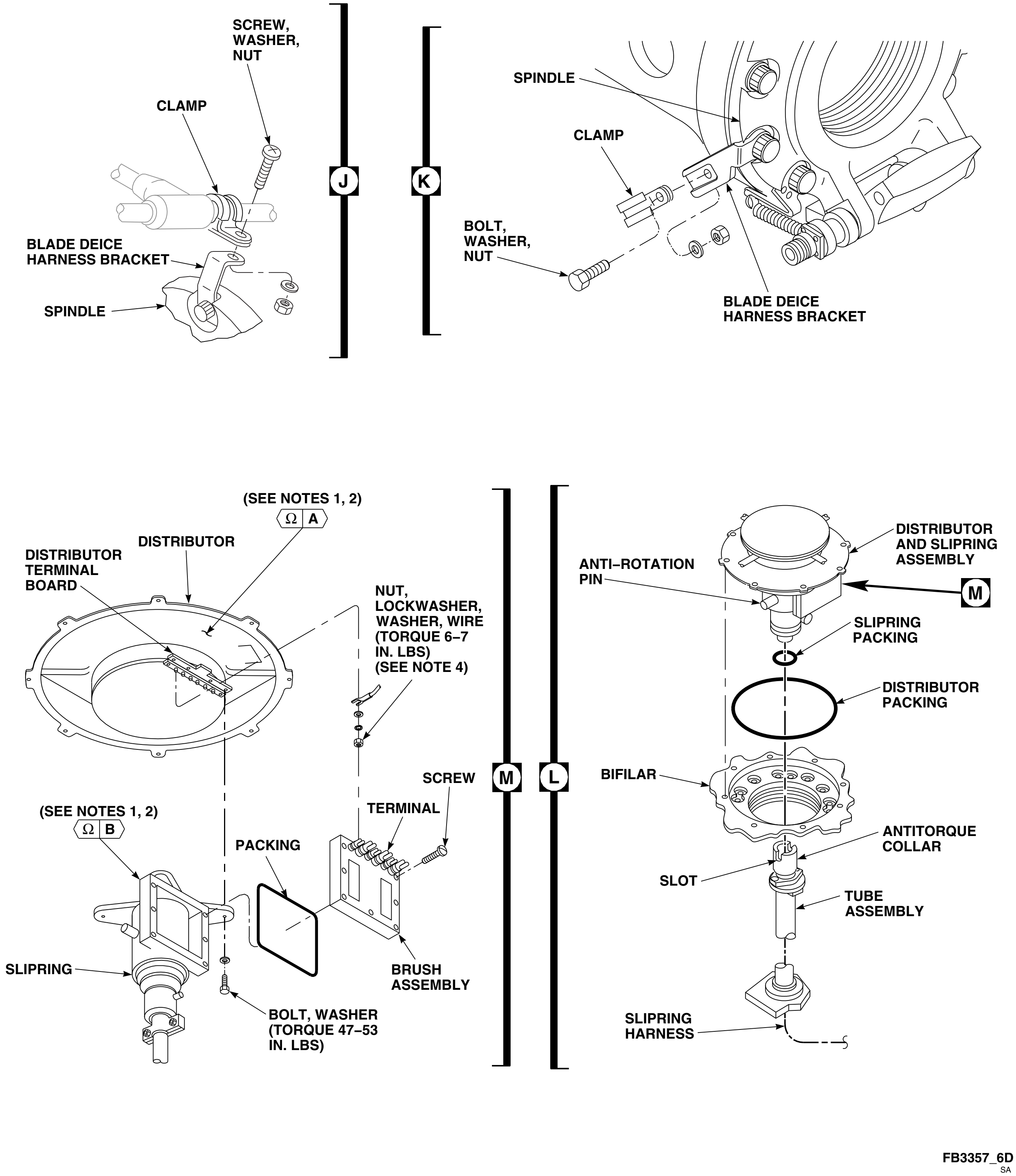


FIGURE 12-4-26-1. REPLACE MAIN ROTOR BLADE DEICE DISTRIBUTOR AND SLIPRING/ BRUSH ASSEMBLY (SHEET 6 OF 6)

l. W/O EMEP Slide conduit bracket off slipring harness.

m. W/O EMEP Disconnect deice harnesses as follows:

(1) Tag and disconnect electrical connectors, P841, P842, P843, and P844, from main rotor blades. Remove and discard packings (Figure 12-4-26-1, Sheet 5, Detail H).

(2) Disconnect electrical connectors, P845, P846, P847, and P848, from droop stop heater pins.

(3) Install protective caps and plugs on electrical connectors.

NOTE

Surface discoloration of screws securing clamps to spindle bracket is not cause for replacement.

(4) Remove bolts, screws, washers, nuts, and clamps securing deice harness to blade deice harness brackets on spindles (Figure 12-4-26-1, Sheet 5, Detail H, and Sheet 6, Detail J and Detail K).

n. EMEP Disconnect deice harness as follows:

(1) Tag and disconnect electrical connectors as follows (Figure 12-4-26-1, Sheet 5, Detail H):

(a) P1001 from J1001

(b) P1002 from J1002

(c) P1003 from J1003

(d) P1004 from J1004

(2) Install protective caps and plugs on electrical connectors.

o. Remove bolts and washers securing distributor and slipring assembly to bifilar (Figure 12-4-26-1, Sheet 5, Detail H).

p. EMEP Remove grommet from flange seal and slipring harness (Figure 12-4-26-1, Sheet 4, Detail G).

q. Carefully lift distributor and slipring assembly from bifilar. Have assistant in cabin feed slipring harness through tube assembly (Figure 12-4-26-1, Sheet 6, Detail L). Remove distributor and slipring packings. Discard packings.

12-4-26.1.2 Disassemble.

a. Remove bolts and washers securing slipring to distributor (Figure 12-4-26-1, Sheet 6, Detail M).

b. Loosen nuts on distributor terminal board and remove slipring from distributor.

c. Remove screws securing brush assembly to slipring. Remove brush assembly and packing. Discard packing.

12-4-26.1.3 Clean/Inspect.

a. W/O EMEP Inspect distributor as follows:

(1) Inspect distributor packing for cracks and decay. **NONE ALLOWED.** If cracked or decayed, replace packing (Figure 12-4-26-1, Sheet 6, Detail L).

(2) Check bifilar and distributor mating surfaces for cleanness.

(3) Clean slipring and distributor mounting surfaces using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

(4) Inspect each deice harness for cracks and damage. **NONE ALLOWED.** If cracked or damaged, replace distributor.

(5) Inspect conduit bracket for loose or missing grommet. If grommet is loose or missing, replace or install grommet as follows:

(a) If necessary, cut grommet to length for conduit bracket.

(b) Apply coat of adhesive, Item 56, Appendix D, to grommet and conduit bracket. Allow adhesive to air-dry for 10 minutes, until it becomes tacky.

(c) Firmly press grommet in place (Figure 12-4-26-1, Sheet 4, Detail G). Allow to cure for 24 hours.

b. EMEP Inspect distributor as follows:

(1) Inspect distributor packing for cracks and decay. **NONE ALLOWED.** If cracked or decayed, replace packing (Figure 12-4-26-1, Sheet 6, Detail L).

(2) Inspect chemical conversion coating on mating surfaces of bifilar and distributor for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

(3) Inspect chemical conversion coating on mating surfaces of slipring and distributor for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

(4) Inspect each deice harness for cracks and damage. **NONE ALLOWED.** If cracked or damaged, replace deice harness.

c. Inspect deice harnesses for cracked or missing heat-shrinkable insulation sleeving. **NONE ALLOWED.** If insulation sleeving is cracked or missing, replace insulation sleeving as follows:

(1) Remove damage insulation sleeving as required.

(2) Clean hard epoxy back of electrical connector shell and about 3 to 3 ½-inches of convoluted harness with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(3) Install 2-inch long piece of heat-shrinkable insulation sleeving, Item 163, Appendix D, on electrical connector and deice harness, ⅛-inch from band. Heat-shrink insulation sleeving using heat gun.

(4) Install 2 ¾-inch long piece of heat-shrinkable insulation sleeving, Item 163, Appendix D, over 2-inch piece. Heat-shrink 2 ¾-inch sleeving insulation sleeving using heat gun.

d. Inspect slipring housing for evidence of metal shavings. **NONE ALLOWED.** If necessary, clean metal shavings from slipring housing using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

12-4-26.1.4 Assemble.

a. Carefully install brush assembly and packing on slipring and secure with screws (Figure 12-4-26-1, Sheet 6, Detail M).

b. Position brush assembly terminals on distributor terminal board between washer and wire. Secure slipring to distributor with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

c. Tighten nuts securing brush assembly terminals and wire terminals to distributor terminal board. **TORQUE NUTS TO 6 - 7 INCH-POUNDS.**

d. EMEP Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, check resistance from slipring to distributor. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, perform the following:

(a) Remove slipring from distributor (PARA 12-4-26.1.2).

(b) Apply chemical conversion coating (PARA 2-6-5).

(c) Install slipring to distributor (PARA 12-4-26.1.4).

12-4-26.1.5 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

Do not apply adhesive around entire perimeter of distributor housing. Area must be clean for bonding.

b. Install distributor packing on distributor using four small equally spaced drops of adhesive, Item 56, Appendix D, on distributor (Figure 12-4-26-1, Sheet 6, Detail L).

c. Install slipring packing lubricated with lubricating oil, Item 205 or Item 206, Appendix D, on distributor and slipring assembly.

d. Feed slipring harness through tube assembly. Have assistant in cabin pull wire harness through tube assembly while distributor and slipring assembly are set in place on bifilar.

CAUTION

Damage to tube assembly and main rotor shaft will occur if distributor and slipring assembly is improperly installed. Anti-rotation pin on slipring and distributor assembly must engage with slot on anti-rotation collar of tube assembly. Make sure the anti-rotation pin will engage with slot on antitorque collar of tube assembly before lowering distributor on bifilar.

e. Line up anti-rotation pin on distributor and slipring assembly with slot on antitorque collar of tube assembly.

f. Install distributor and slipring assembly on bifilar with blade deice harnesses lined up with rotor blades (Figure 12-4-26-1, Sheet 5, Detail H).

g. Secure distributor and slipring assembly to bifilar with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

h. Remove protective caps and plugs from electrical connectors.

i. Inspect and treat electrical connectors (PARA 1-7-20).

j. W/O EMEP Connect deice harnesses as follows:

(1) Install clamps securing deice harness to blade deice harness brackets on spindles and secure with screws, bolts, washers, and nuts (Figure 12-4-26-1, Sheet 5, Detail H and Sheet 6, Detail J and Detail K).

(2) Install packings and connect electrical connectors, P841, P842, P843, and P844, to main rotor blades (Figure 12-4-26-1, Sheet 5, Detail H). Remove tags.

(3) Connect electrical connectors, P845, P846, P847, and P848, to droop stop heater pins.

k. EMEP Connect electrical connectors as follows (Figure 12-4-26-1, Sheet 5, Detail H):

(1) P1001 to J1001

(2) P1002 to J1002

(3) P1003 to J1003

(4) P1004 to J1004

(5) Remove tags

l. EMEP Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, check resistance from distributor to bifi-

lar. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, perform the following:

(a) Remove distributor from slipring (PARA 12-4-26.1.2).

(b) Apply chemical conversion coating (PARA 2-6-5).

(c) Install distributor to slipring (PARA 12-4-26.1.4).

m. Apply bead of sealing compound, Item 292, Appendix D, around distributor.

n. **W/O EMEP** Install conduit bracket as follows (Figure 12-4-26-1, Sheet 4, Detail G):

(1) Feed slipring harness through grommet in conduit bracket.

(2) **S70–I** Position conduit bracket on studs of flange seal and secure with nuts and washers. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

(3) **S70–II** Position conduit bracket on flange seal and secure with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Using lockwire, Item 194, Appendix D, lockwire bolts.

o. Install grommet onto slipring harness and into flange seal.

p. Apply sealing compound, Item 292, Appendix D, around flange seal hardware.

q. Install spiral wrap, Item 314, Appendix D, on slipring harness. Begin installation at conduit bracket grommet.

r. Install slipring harness to brackets with screws, washers, and clamps. Make sure clamps secure anti-chafe strip and that slipring harness is centered within flange seal (Figure 12-4-26-1, Sheet 2, Detail C).

s. **W/O EMEP** Connect and install slipring harness as follows:

(1) Clean sealing compound from slipring harness and grommet, located at STA 340.0, RBL 16.5, using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D (Figure 12-4-26-1, Sheet 2, Detail C).

(2) Feed slipring harness through grommet at STA 340.0, RBL 16.5.

(3) Feed slipring harness through bulkhead at STA 327.0, RBL 20.0. Clamp harness to side of bulkhead with screw, washer, clamp, and nut (Figure 12-4-26-1, Sheet 3, Detail E).

(4) Loosen fasteners that secure cover on main rotor blade deice junction box and let cover hang by chain (Figure 12-4-26-1, Sheet 1, Detail B).

(5) Carefully feed slipring harness wires through grommet in main rotor blade deice junction box.

(6) Connect slipring harness wires to terminal board 3 with nuts, lockwashers, and washers as follows, then remove tags:

WIRE CODE	TO	TERMINAL NO.
1-20-S2		1
1-10-A		2
1-20-S1		3
1-10-B		4
1-10-C		5

(7) **TORQUE NUTS TO 18 - 20 INCH-POUNDS.**

(8) Connect ground wire, 1-20-S3, to airframe with screw, lockwasher, washers, and nut (Figure 12-4-26-1, Sheet 1, Detail A). Remove tag.

(9) Install cover on main rotor blade deice junction box and secure with fasteners (Figure 12-4-26-1, Sheet 1, Detail B).

(10) Install tiedown straps, Item 329, Appendix D, as required to secure slipring harness.

(11) Clamp slipring harness to stringer with screw, washer, clamp, and nut(Figure 12-4-26-1, Sheet 2, Detail C).

(12) Apply a bead of sealing compound, Item 292, Appendix D, around slipring harness, where it passes through grommet.

t. EMEP Connect and install slipring and deice harnesses as follows:

(1) Clean sealing compound from slipring and deice harnesses, and grommet, located at STA 340.0, RBL 16.5, using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D (Figure 12-4-26-1, Sheet 2, Detail C and Detail D).

NOTE

When feeding slipring and deice harnesses, make sure that wire ends marked TB7-# are routed to terminal board 7 and wire ends marked TB3-# and GNDS3-1 are routed to terminal board 3, located in the main rotor blade deice junction box.

(2) Route slipring harness as follows:

(a) Feed slipring harness through grommet at STA 340.0, RBL 16.5 (Figure 12-4-26-1, Sheet 2, Detail C and Detail D).

(b) Feed slipring harness through bulkhead at STA 343.0, RBL 20.0 to terminal board 7. Loosely clamp slipring harness to bulkhead with screw, washer, clamp, and nut (Figure 12-4-26-1, Sheet 3, Detail E). **DO NOT TIGHTEN CLAMP AT THIS TIME.**

(c) Feed deice harness through bulkhead at STA 327.0, RBL 20.0 and STA 343.0, RBL 20.0. Loosely clamp deice harness to bulkhead with screw, washer, clamp, and nut. **DO NOT TIGHTEN CLAMPS AT THIS TIME.**

(3) Connect deice and slipring harness wires to terminal board 7 with nuts, lockwashers, and washers as follows (Figure 12-4-26-1, Sheet 3, Detail F), then remove tags:

WIRE CODE	TO	TERMINAL NO.
TB7-1		S2
TB7-2		PH-A
TB7-3		S1
TB7-4		PH-B
TB7-5		PH-C
TB7-6		S3
TB7-7 (ALPHA BRAID)		S4

(4) Install cover on terminal board 7 and tighten thumb screws.

(5) Carefully feed deice harness wires through grommet in main rotor deice junction box (Figure 12-4-26-1, Sheet 1, Detail B).

(6) Connect slipring harness wires to terminal board 3 with nuts, lockwashers, and washers as follows, then remove tags:

WIRE CODE	TO	TERMINAL NO.
TB3-1		1
TB3-2		2
TB3-3		3
TB3-4		4
TB3-5		5

(7) **TORQUE NUTS TO 18 - 20 INCH-POUNDS.**

(8) Connect ground wire, GNDS3-1, and alpha braid, GNDS3-1, to airframe with screw, lockwasher, washers, and nut (Figure 12-4-26-1, Sheet 1, Detail A). Remove tags.

(9) Install cover on main rotor blade deice junction box and secure with fasteners (Figure 12-4-26-1, Sheet 1, Detail B).

(10) **TIGHTEN ALL CLAMPS SECURING SLIPRING AND DEICE HARNESSES.**

(11) Install tiedown straps, Item 329, Appendix D, as required to secure slipring and deice harnesses.

(12) Apply a bead of sealing compound, Item 292, Appendix D, around slipring and deice harnesses where they pass through grommet, STA 340.0, RBL 16.5 (Figure 12-4-26-1, Sheet 2, Detail D).

u. **EMEP** Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Check resistance of deice harness using digital low-resistance ohmmeter. Place lead A on alpha braid at ground point and lead B on airframe a minimum of 12 inches away from lead A (Figure 12-4-26-1, Sheet 1, Detail A). **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, perform the following:

(a) Tag wires and remove screw, lockwasher, washers, and nut, and disconnect ground wire, GNDS3-1, and alpha braid, GNDS3-1, from airframe.

(b) Apply chemical conversion coating to ground point (PARA 2-6-5).

(c) Connect ground wire, GNDS3-1, and alpha braid, GNDS3-1, to airframe with screw, lockwasher, washers, and nut. Remove tags.

(d) Repeat step u. (1) and step u. (2) until resistance is 2.5 milliohms or less.

v. Perform operational check of blade deicing system (PARA 12-2-5).

w. Install soundproofing panels (PARA 2-4-69).

x. Install main transmission drip pan (PARA 2-4-70).

y. If required, disengage gust lock (PARA 1-6-18).

12-4-27 ICING RATE METER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 12-2-5

12-4-27.1 Replace Icing Rate Meter.

12-4-27.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen fasteners and slowly lift icing rate meter from instrument panel until electrical connector, P132, is exposed (Figure 12-4-27-1, Detail A).

c. Disconnect electrical connector, P132, from electrical connector, J132, and remove icing rate meter.

d. Install protective caps and plugs on electrical connectors.

12-4-27.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. EMEP Perform the following:

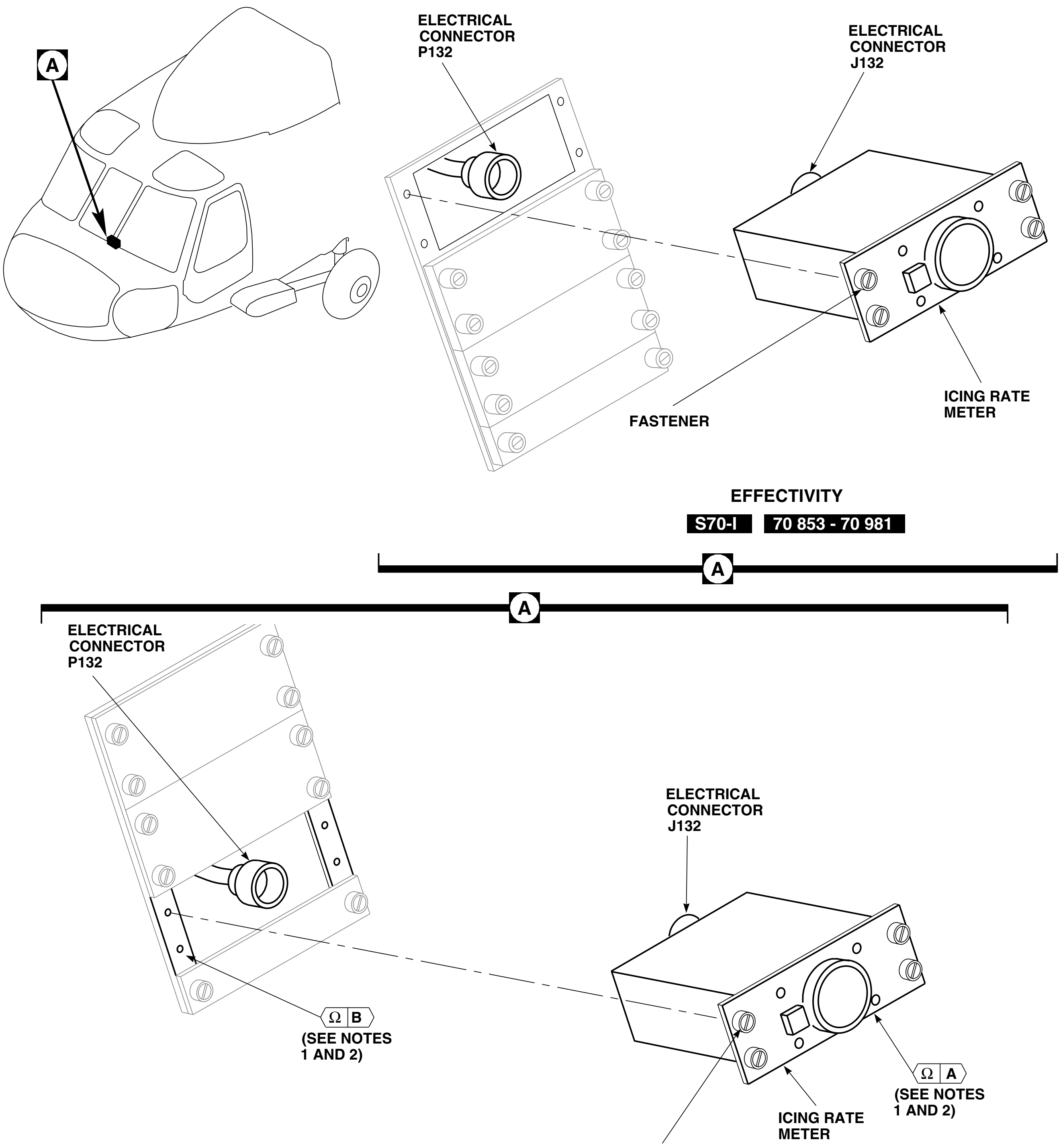
(1) Clean surface around all mounting holes with cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.

(2) Inspect chemical conversion coated mating surfaces of icing rate meter and instrument panel for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches apply chemical conversion coating (PARA 2-6-5).

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P132, to electrical connector, J132, on icing rate meter (Figure 12-4-27-1, Detail A).



NOTES

1. EMEP

2.

POINTS FROM	TO	READING MUST NOT BE GREATER THAN
A	B	2.5 MILLIOHMS

FB2032C
SA

FIGURE 12-4-27-1. REPLACE ICING RATE METER

f. Make sure area is clean and free of foreign material.

g. Carefully place meter into instrument panel and secure with fasteners.

h. EMEP Perform EME resistance check of icing rate meter as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Check resistance measured from face of meter and instrument panel using digital low-resistance ohmmeter. **RESISTANCE MUST NOT BE GREATER THAN 2.5 MILLIOHMS.**

(2) If resistance is greater than 2.5 milliohms, remove meter and apply chemical conversion coating (PARA 2-6-5).

i. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-28 ICING RATE METER LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 12-2-5

12-4-28.1 Replace Icing Rate Meter Lamps.

12-4-28.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Grip edges of PRESS TO TEST switch button assembly and pull assembly off switch to expose lamp (Figure 12-4-28-1, Detail A).

c. Pull lamp from switch.

d. If post light is installed, remove lens hood and lamp.

12-4-28.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Push lamp into lamp socket on switch (Figure 12-4-28-1, Detail A).

c. Press switch button assembly onto PRESS TO TEST switch.

d. If post light is installed, push lamp into lens hood and push into post light.

e. Perform operational check of instrument panel lights (PARA 12-2-5).

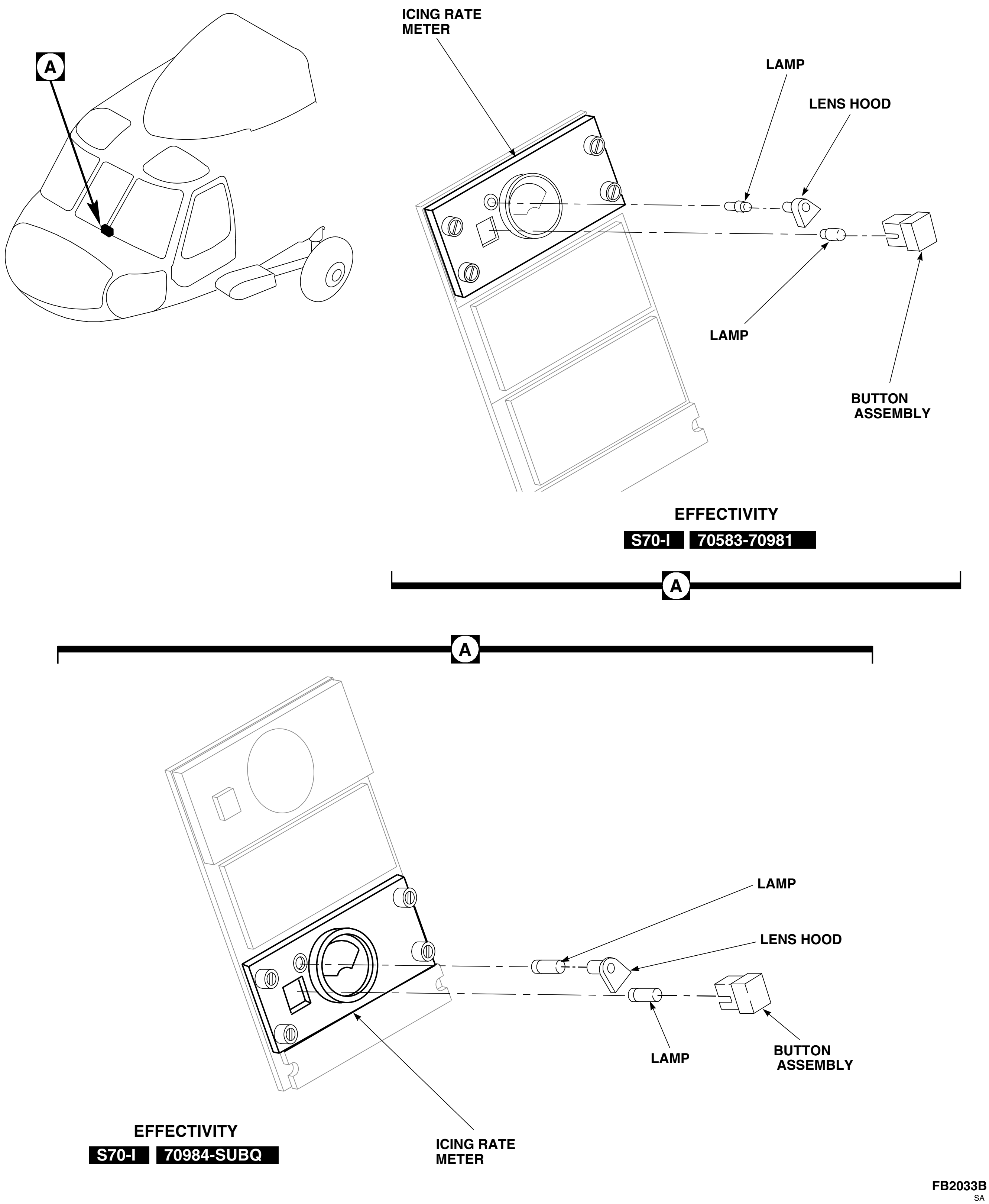


FIGURE 12-4-28-1. REPLACE ICING RATE METER LAMPS

12-4-29 BLADE DEICE CONTROL PANEL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 12-2-5

12-4-29.1 Replace Blade Deice Control Panel.

12-4-29.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen fasteners and slowly lift blade deice control panel from instrument panel until electrical connector, P133, is exposed (Figure 12-4-29-1, Detail A).

c. Disconnect electrical connector, P133 from J133, and remove control panel.

d. Install protective caps and plugs on electrical connectors.

12-4-29.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. EMEP Perform the following:

(1) Clean surface around all mounting holes with cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.

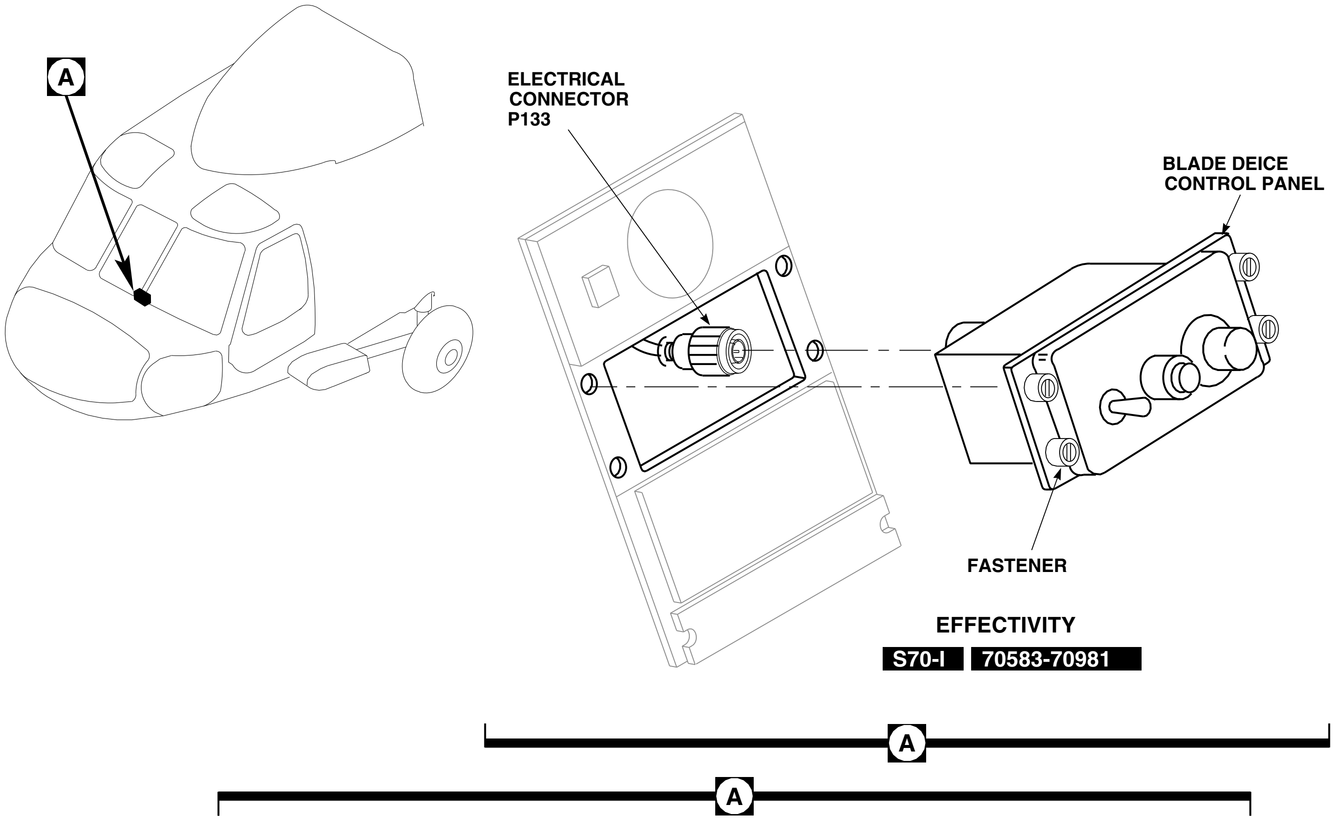
(2) Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches apply chemical conversion coating (PARA 2-6-5).

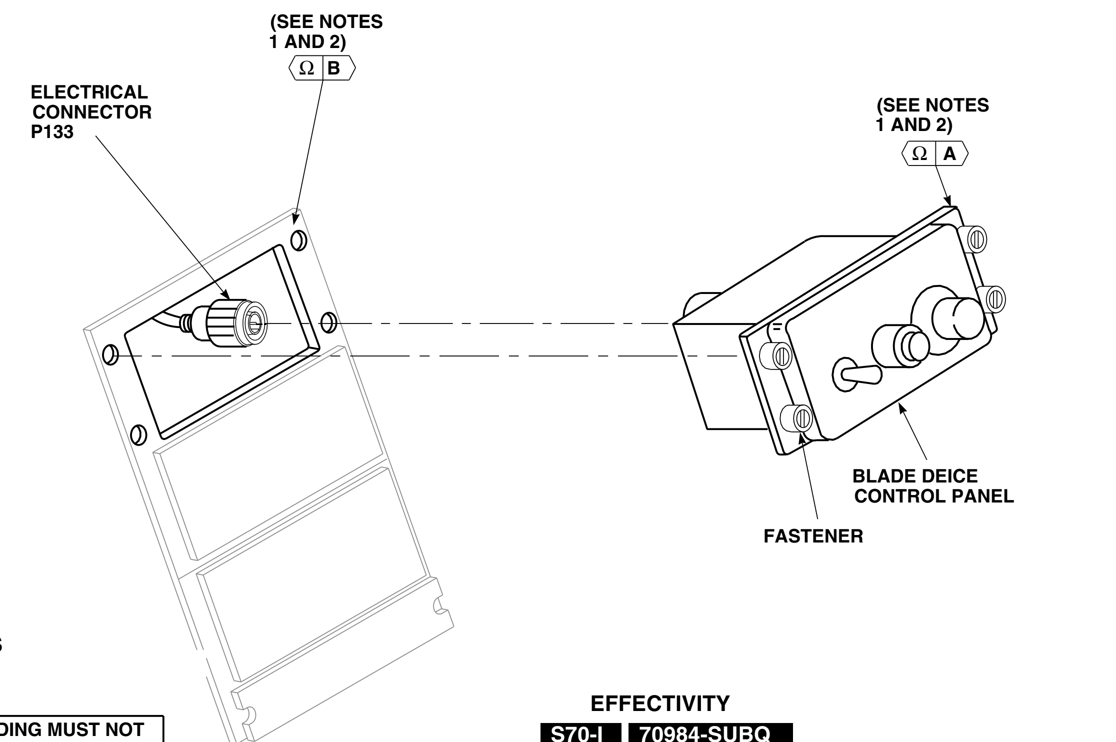
c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P133 to J133, on the rear of the control panel (Figure 12-4-29-1, Detail A).

f. Make sure area is clean and free of foreign material.





NOTES

1. EMEP

2.

POINTS FROM	TO	READING MUST NOT BE GREATER THAN
A	B	2.5 MILLIOHMS

FB2034C
SA

FIGURE 12-4-29-1. REPLACE BLADE DEICE CONTROL PANEL

g. Carefully place control panel into instrument panel and secure with fasteners.

h. EMEP Perform EME resistance check of deice control panel as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Check resistance measured from face of control panel and instrument panel using digital low-resistance ohmmeter. **RESISTANCE MUST NOT BE GREATER THAN 2.5 MILLIOHMS.**

(2) If resistance is greater than 2.5 milliohms, remove control panel and apply chemical conversion coating (PARA 2-6-5).

i. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-30 BLADE DEICE CONTROL PANEL TEST-IN-PROGRESS LAMP.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 12-2-5

12-4-30.1 Replace Blade Deice Control Panel TEST-IN-PROGRESS Lamp.

12-4-30.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unscrew lens assembly from blade deice control panel (Figure 12-4-30-1, Detail A).

c. Pull lamp from lens assembly.

12-4-30.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Push lamp into lens assembly (Figure 12-4-30-1, Detail A).

c. Screw lens assembly into blade deice control panel.

d. Turn on electrical power (PARA 1-6-16).

e. Make sure LIGHTS ADVSY circuit breaker is pushed in.

f. Press TEST-IN-PROGRESS indicator and check that indicator lamp lights go on. If result is not as specified, perform operational check (PARA 12-2-5).

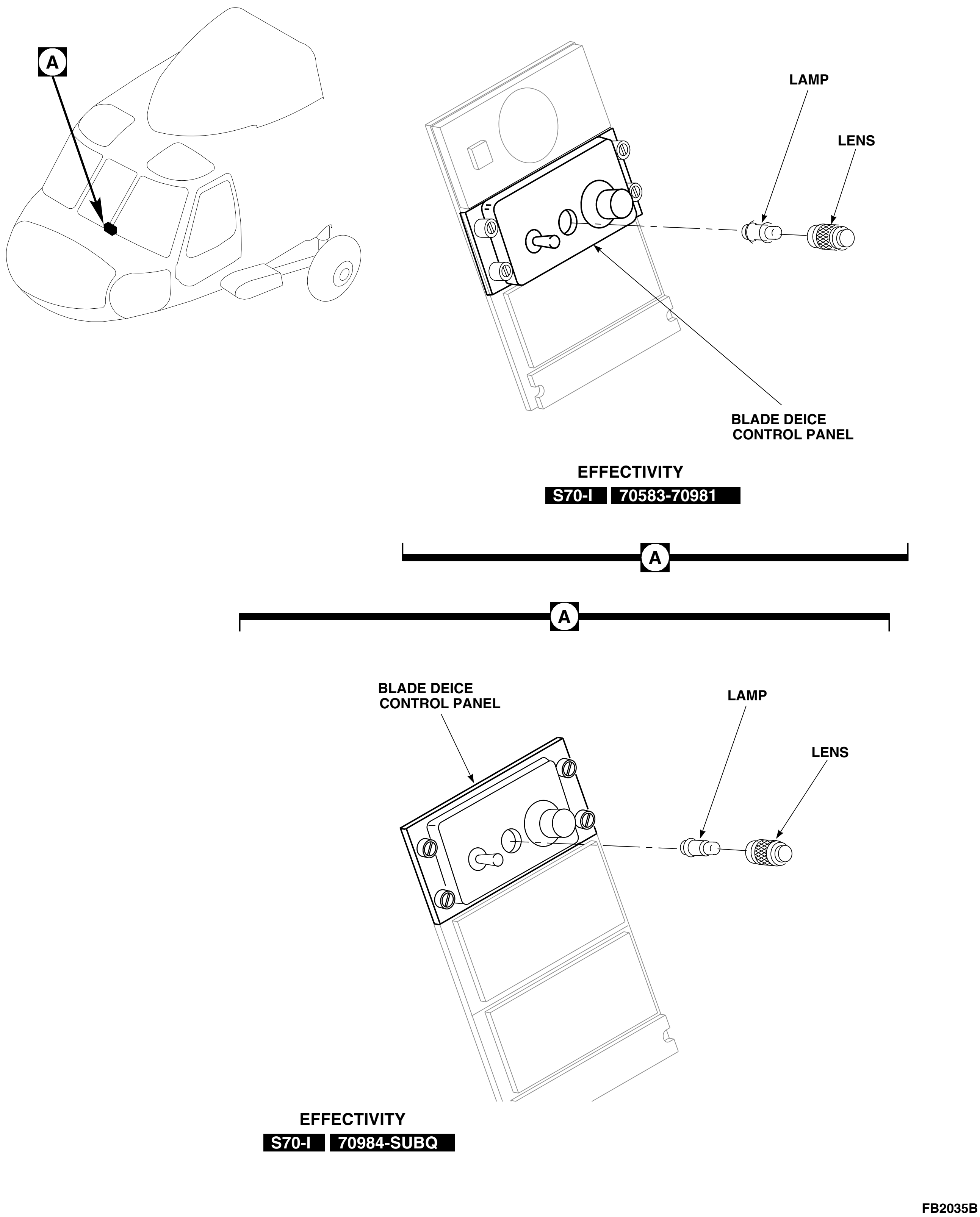


FIGURE 12-4-30-1. REPLACE BLADE DEICE CONTROL PANEL TEST-IN-PROGRESS LAMP

12-4-31 BLADE DEICE CONTROL PANEL INFORMATION PLATE AND LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Cap and Plug

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-5

12-4-31.1 Replace Blade Deice Control Panel Information Plate and Lamps.

12-4-31.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen two setscrews and remove knob from MODE rotary switch (Figure 12-4-31-1, Detail A).

c. Unscrew (counterclockwise) TEST-IN-PROGRESS indicator lens cap, and remove cap and lamp assembly.

d. Remove three mounting screws from information plate.

e. Disconnect information plate electrical connector, P1, from control panel assembly connector, J1.

f. Install protective cap and plug on electrical connectors.

g. Remove moisture seal from lamp to be replaced (Figure 12-4-31-2).

h. Turn slotted lamp until slot lines up with guides in information plate.

i. Remove lamp from information plate.

12-4-31.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Place lamp in information plate receptacle (Figure 12-4-31-2).

c. Line up slot in lamp with information plate guides.

d. Turn lamp clockwise to lock in place.

e. Seal lamp receptacle by applying moisture seal.

f. Remove protective cap and plug from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Carefully connect information plate electrical connector, P1, to control panel assembly

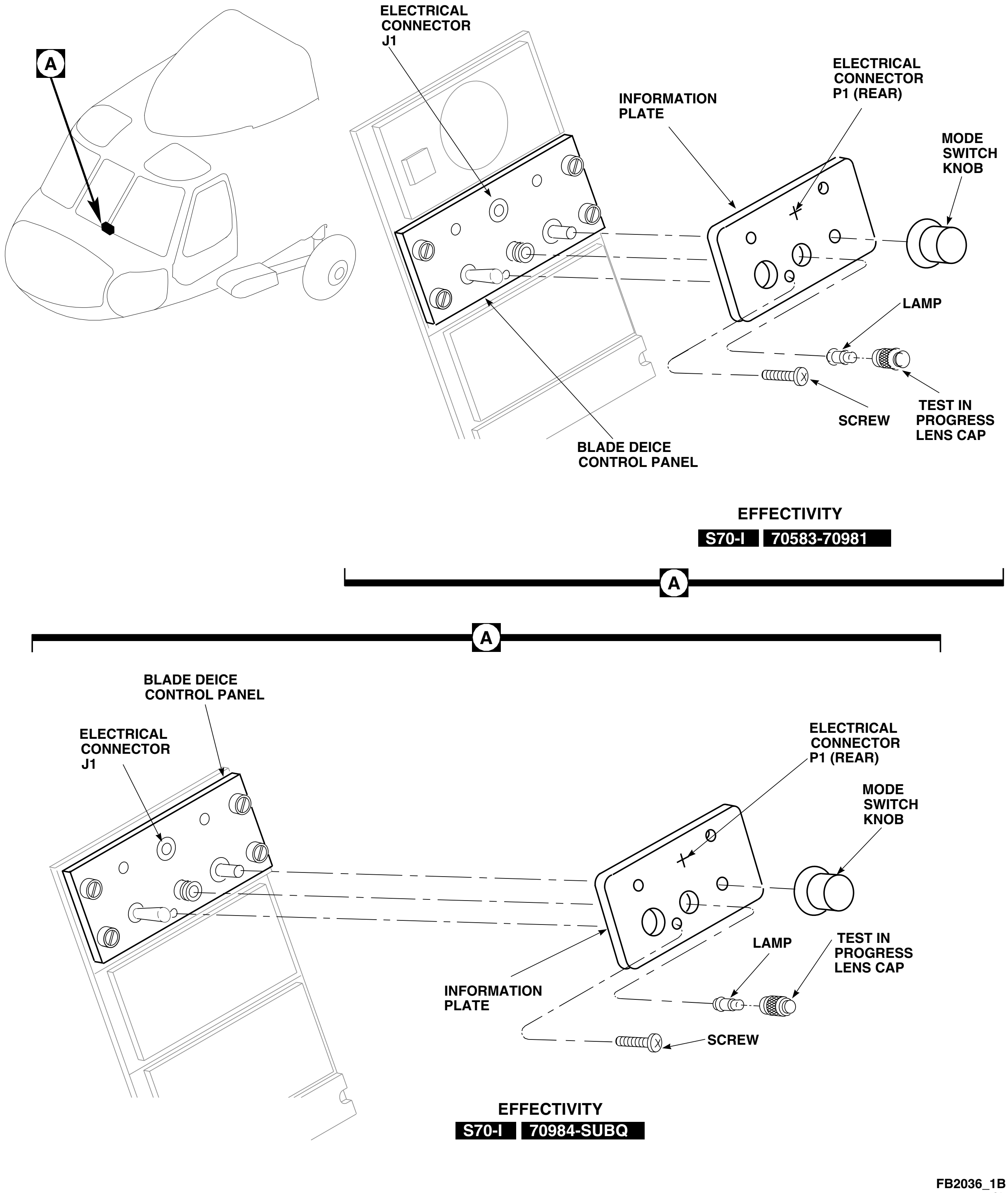


FIGURE 12-4-31-1. REPLACE BLADE DEICE CONTROL PANEL INFORMATION PLATE AND LAMPS

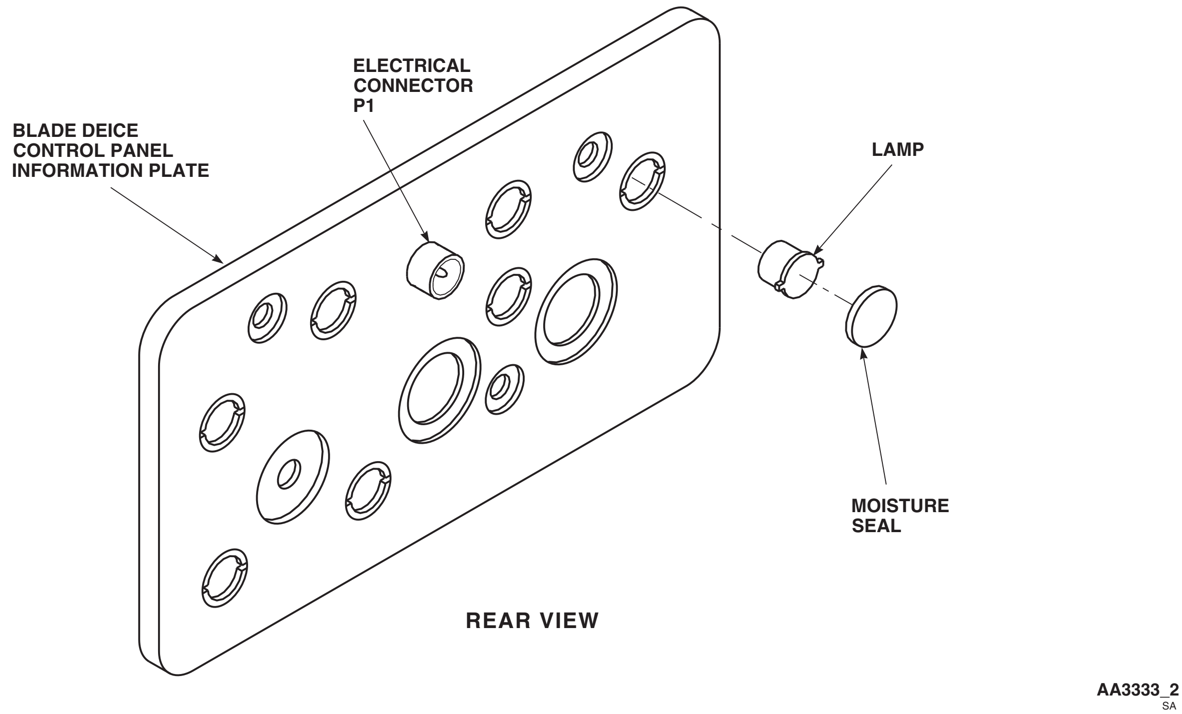


FIGURE 12-4-31-2. REPLACE BLADE DEICE CONTROL PANEL INFORMATION PLATE AND LAMPS

electrical connector, J1 (Figure 12-4-31-1, Detail A).

i. Secure information plate to control panel with screws.

j. Screw (clockwise) lens cap and lamp assembly onto TEST-IN-PROGRESS indicator.

k. Line up MODE rotary switch knob setscrews with flat surfaces on shaft of MODE rotary switch and push knob onto shaft. Make sure knob pointer is lined up with information plate stenciling. **TIGHTEN SETSCREWS.**

l. Make sure area is clean and free of foreign material.

m. Turn on electrical power (PARA 1-6-16).

n. Make sure LOWER CONSOLE LT circuit breaker is in.

o. Turn CONSOLE LT LOWER control to BRT. All nonflight instrument panel lights should go on. If result is not as specified, perform operational check of blade deicing system (PARA 12-2-5).

12-4-32 BLADE DEICE CONTROLLER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 12-2-5

12-4-32.1 Replace Blade Deice Controller.

12-4-32.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

NOTE

EMEP Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapter from electrical connectors, P324 and P325 (Figure 12-4-32-1, Detail B).

c. Tag and disconnect electrical connectors, P324 from J2, and P325 from J1, on blade deice controller (Figure 12-4-32-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

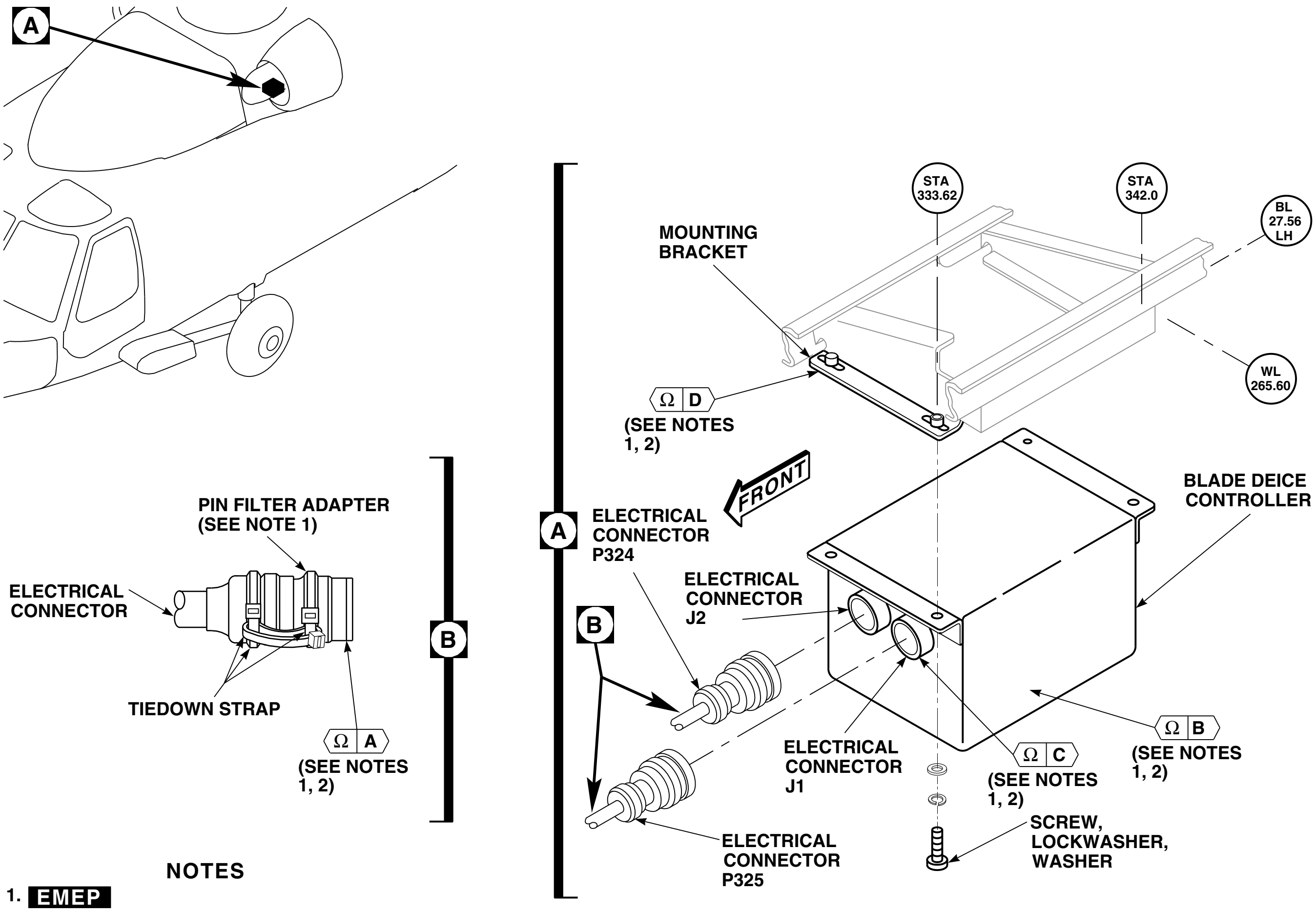
e. Remove screws, lockwashers, and washers securing blade deice controller to mounting bracket, and remove controller.

12-4-32.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. EMEP Clean surface around all mounting holes with cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.

c. EMEP Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or



NOTES

1. EMEP

2.

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	5.0 MILLIOHMS OR LESS
A	C	2.5 MILLIOHMS OR LESS
B	D	2.5 MILLIOHMS OR LESS

FB3317B
SA

FIGURE 12-4-32-1. REPLACE BLADE DEICE CONTROLLER

scratches, apply chemical conversion coating (PARA 2-6-5).

d. Install blade deice controller to mounting bracket with screws, lockwashers, and washers (Figure 12-4-32-1, Detail A).

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connector, P324 to J2, and P325 to J1, on blade deice controller. Remove tags.

h. EMEP Perform EME resistance check of pin filter adapter as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, measure resistance between shell of each pin filter adapter and blade deice controller. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**

(2) If resistance is greater than 5.0 milliohms, measure resistance between shell of pin

filter adapter and electrical connector on blade deice controller. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(3) If resistance is still beyond limits, clean surface between pin filter adapter and blade deice controller.

i. EMEP Perform EME resistance check of deice controller as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, measure resistance between mounting bracket and blade deice controller. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, remove blade deice controller and apply chemical conversion coating (PARA 2-6-5).

j. Make sure area is clean and free of foreign material.

k. Install soundproofing panels (PARA 2-4-69).

l. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-33 BLADE DEICE AUXILIARY JUNCTION BOX.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tiedown Straps, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 12-2-5

12-4-33.1 Replace Blade Deice Auxiliary Junction Box.

12-4-33.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Disconnect electrical connectors, P332 from J332, and P333 from J333, on blade deice auxiliary junction box (Figure 12-4-33-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Loosen fasteners that secure cover on junction box and let cover hang by chain.

f. Remove screws securing terminal shield to contactor, K62, and remove shield.

g. Remove nuts, lockwashers, and washers securing wires to terminals, L1, L2, and L3, of contactor, K62. Remove wires (Figure 12-4-33-1, Detail B).

h. Remove screws and washers securing wires to terminal 1, of current limiters, CL10, CL11, and CL12. Remove wires.

i. Remove nuts, lockwashers, and washers securing wires to terminals 1, 3, and 5, of terminal board 5. Remove wires.

j. Carefully pull all disconnected wires out through junction box grommets. Cut tiedown straps as required to free wires.

k. Install cover on junction box and secure with fasteners.

l. Remove bolt, lockwashers, washers, and nut, and disconnect ground wires, H6434A20N, H6434B20N, and H6435A20N, from airframe at STA 352.0 (Figure 12-4-33-1, Detail C and Detail D).

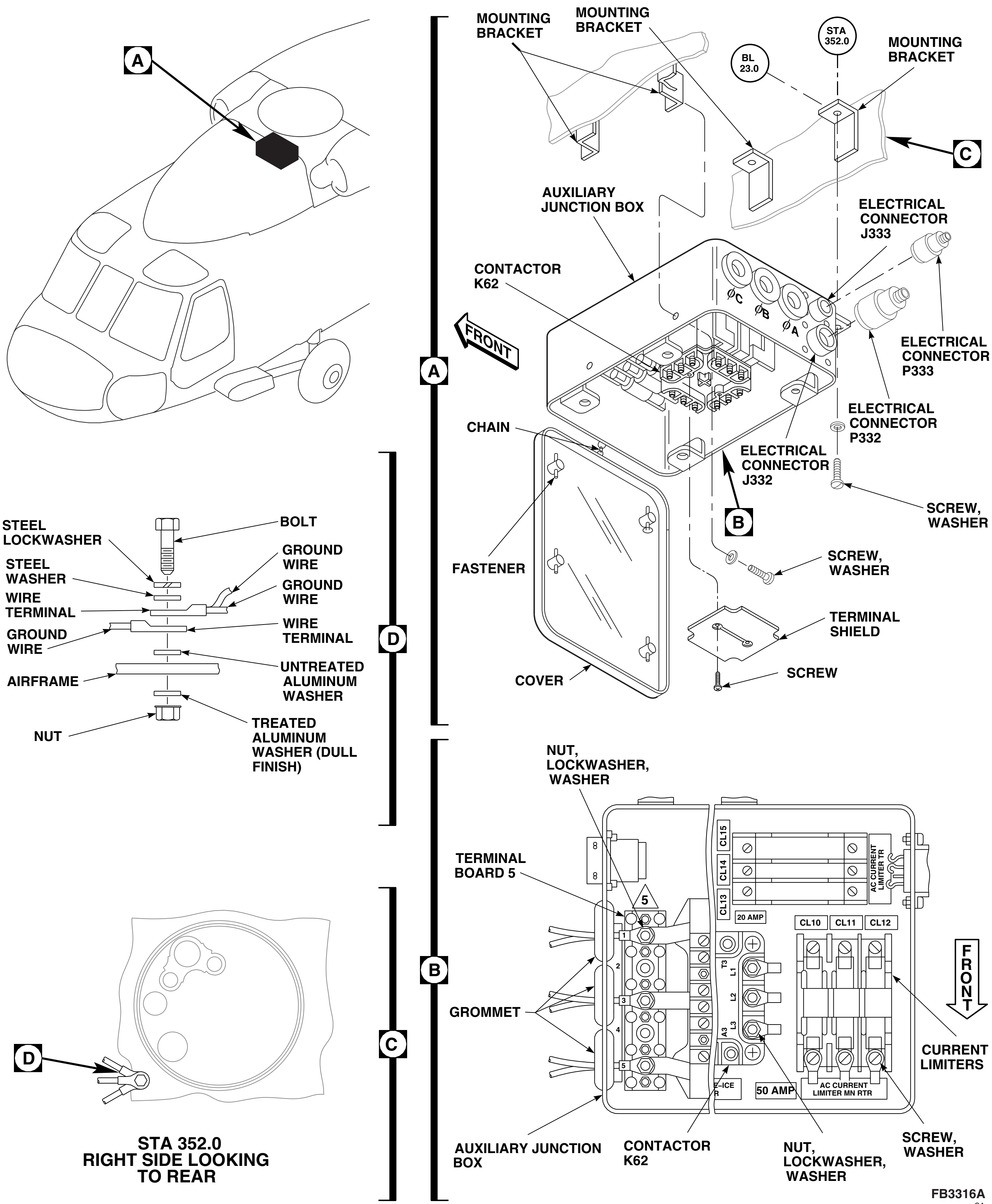


FIGURE 12-4-33-1. REPLACE BLADE DEICE AUXILIARY JUNCTION BOX

m. Remove screws and washers securing junction box to mounting brackets and remove junction box (Figure 12-4-33-1, Detail A).

12-4-33.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install blade deice auxiliary junction box on mounting brackets in cabin ceiling and secure with screws and washers (Figure 12-4-33-1, Detail A).

c. Loosen fasteners that secure cover on junction box and let cover hang by chain.

d. Carefully feed wires listed below through junction box grommets marked A, B, C and connect to terminal board 5, using washers, lockwashers, and nuts (Figure 12-4-33-1, Detail B):

GROMMET	WIRE NO.	TERMINAL BOARD 5
A	X8D06A	1
A	X8E06A	1
B	X9D06B	3
B	X9E06B	3
C	X10D06C	5
C	X10E06C	5

e. Carefully feed remaining wires through unmarked junction box grommet and connect to contactor, K62, and current limiters, CL10, CL11, and CL12, as follows:

WIRE NO.	CONNECT TO	ATTACHING HARDWARE
H6431A10A	CL10-1	Screw, Washer
H6432A10B	CL11-1	Screw, Washer
H6433A10C	CL12-1	Screw, Washer
H6379A10A	K62-L1	Nut, Lockwasher, Washer
H6380A10B	K62-L2	Nut, Lockwasher, Washer
H6381A10C	K62-L3	Nut, Lockwasher, Washer

f. Route ground wires, H6434A20N, H6434B20N, and H6435A20N, out of unmarked junction box grommet.

g. Connect ground wires, H6434A20N, H6434B20N, and H6435A20N, to airframe at STA 352.0 with bolt, lockwasher, washers, and nut (Figure 12-4-33-1, Detail C and Detail D).

h. Form wires into bundles using tiedown straps, Item 329, Appendix D.

i. Remove protective caps and plugs from electrical connectors.

j. Inspect and treat electrical connectors (PARA 1-7-20).

k. Connect electrical connectors, P332 to J332, and P333 to J333, on junction box (Figure 12-4-33-1, Detail A).

l. Install terminal shield on contactor, K62, and secure with screws.

m. Install cover on junction box and secure with fasteners.

n. Perform operational check of blade deicing system (PARA 12-2-5).

o. Make sure area is clean and free of foreign material.

p. Install soundproofing panels (PARA 2-4-69).

12-4-34 BLADE DEICE CURRENT LIMITERS, CL10 THROUGH CL15.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 9-4-34
- PARA 9-4-106
- PARA 12-2-5

12-4-34.1 Replace Blade Deice Current Limiters, CL10 through CL15.

NOTE

Replacement of all current limiters is the same.

12-4-34.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

c. Remove soundproofing panels (PARA 2-4-69).

d. Unfasten and hang auxiliary junction box cover by chain.

e. Pull current limiter from current limiter holder (Figure 12-4-34-1, Detail A).

12-4-34.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Press current limiter into current limiter holder (Figure 12-4-34-1, Detail A).

c. Place auxiliary junction box cover on auxiliary junction box and fasten.

d. Make sure area is clean and free of foreign material.

e. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

f. Perform operational check of blade deicing system (PARA 12-2-5).

g. Install soundproofing panels (PARA 2-4-69).

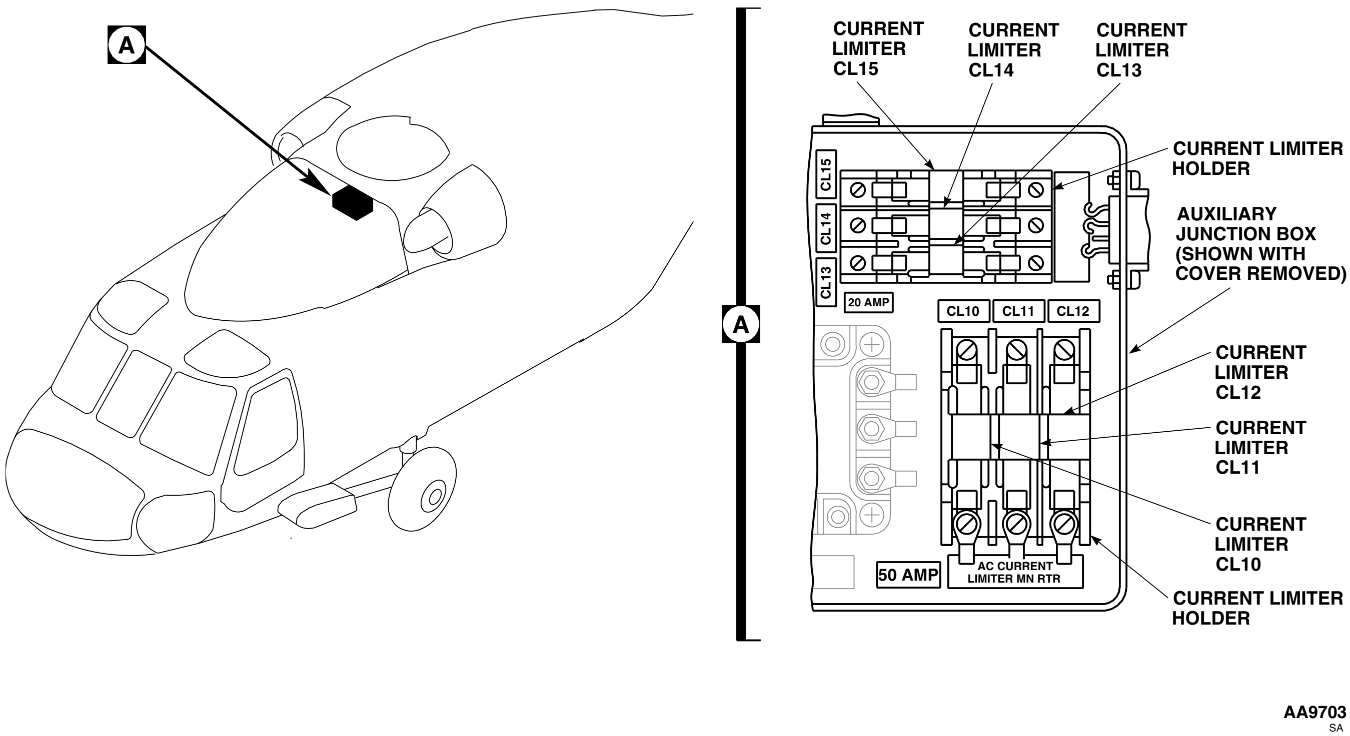


FIGURE 12-4-34-1. REPLACE BLADE DEICE CURRENT LIMITERS, CL10 THROUGH CL15

12-4-35 BLADE DEICE CURRENT LIMITER HOLDER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 12-4-34

12-4-35.1 Replace Blade Deice Current Limiter Holder.

NOTE

Replacement of all blade deice current limiter holders is the same.

12-4-35.1.1 Remove.

a. Remove current limiters (PARA 12-4-34).

b. Tag and remove screws, washers, and wires from holder (Figure 12-4-35-1, Detail A).

c. Remove screws and lift holder from auxiliary junction box.

12-4-35.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install holder in auxiliary junction box with screws (Figure 12-4-35-1, Detail A).

c. Connect wires to holder with screws and washers. Remove tags.

d. Install current limiters (PARA 12-4-34).

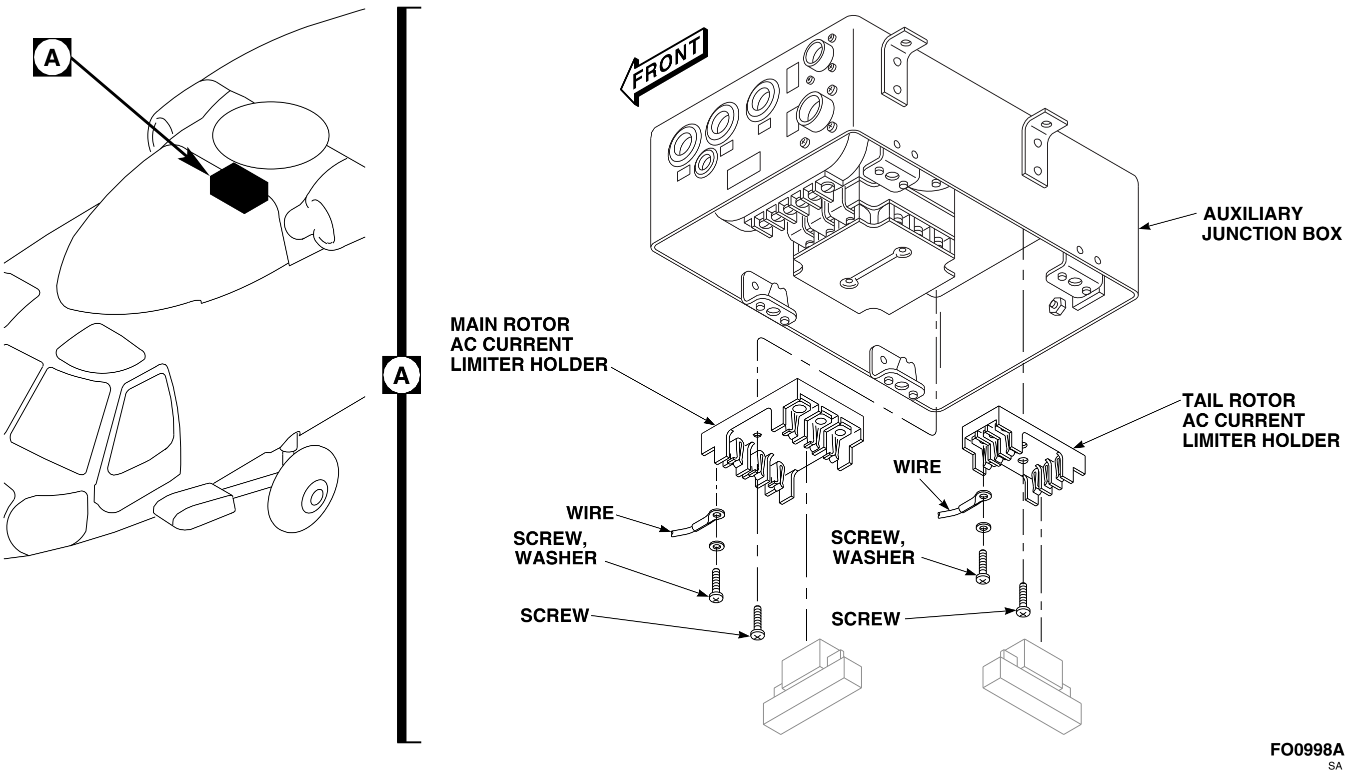


FIGURE 12-4-35-1. REPLACE BLADE DEICE CURRENT LIMITER HOLDER

12-4-36 BLADE DEICE RELAY, K61.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 163, Appendix D
- Protective Caps and Plugs
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 12-2-5

12-4-36.1 Replace Blade Deice Relay, K61.

12-4-36.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Disconnect electrical connectors, P332 from J332 and P333 from J333, from auxiliary junction box (Figure 12-4-36-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Unfasten and hang auxiliary junction box cover by chain.

f. Remove auxiliary junction box screws and washers, and hang auxiliary junction box by external wiring.

g. Slide insulation sleeving off terminals of relay.

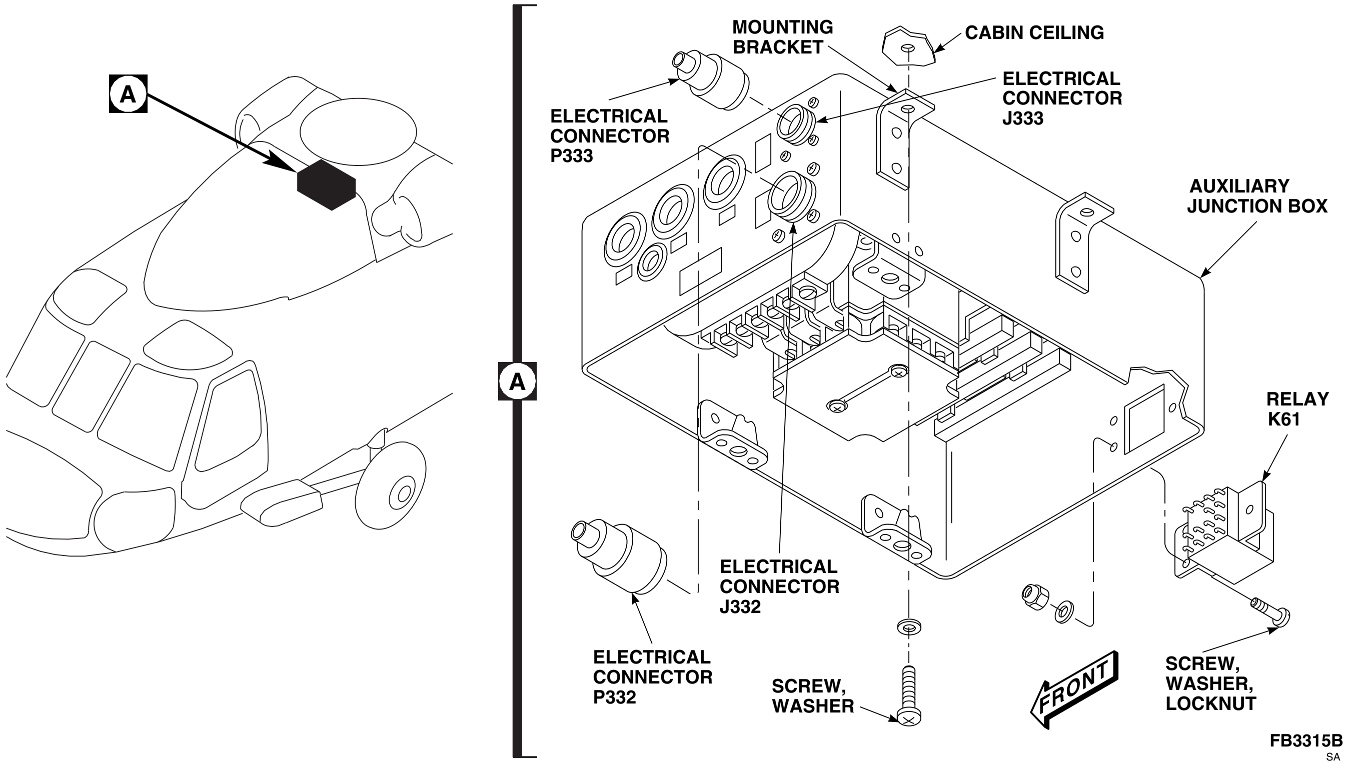


FIGURE 12-4-36-1. REPLACE BLADE DEICE RELAY, K61

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

h. Tag wires. Using soldering set, unsolder wires at terminals of relay.

i. Remove screws, washers, and locknuts from relay and remove relay from auxiliary junction box.

12-4-36.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install relay in auxiliary junction box using screws, washers, and locknuts (Figure 12-4-36-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Using soldering set, solder, Item 313, Appendix D, wires to terminals of relay and remove tags.

d. Slide insulation sleeving, Item 163, Appendix D, over soldered terminals of relay.

e. Position auxiliary junction box on mounting brackets in cabin ceiling and secure with screws and washers.

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connectors, P332 to J332 and P333 to J333, to auxiliary junction box.

i. Install auxiliary junction box cover on auxiliary junction box using cover fasteners.

j. Perform operational check of blade deicing system (PARA 12-2-5).

k. Make sure area is clean and free of foreign material.

l. Install soundproofing panels (PARA 2-4-69).

12-4-37 BLADE DEICE RELAY, K64.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 163, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-37.1 Replace Blade Deice Relay, K64.

12-4-37.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Unfasten and hang main rotor blade deice junction box cover by chain.

d. Slide insulating sleeving off terminals of relay (Figure 12-4-37-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Tag wires. Using soldering set, unsolder wires at terminals of relay.

f. Remove screws and washers and let main rotor blade deice junction box hang by external wiring.

g. Remove screws, washers, and locknuts from relay and remove from main rotor blade deice junction box.

12-4-37.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install relay in main rotor blade deice junction box using screws, washers, and locknuts (Figure 12-4-37-1, Detail A).

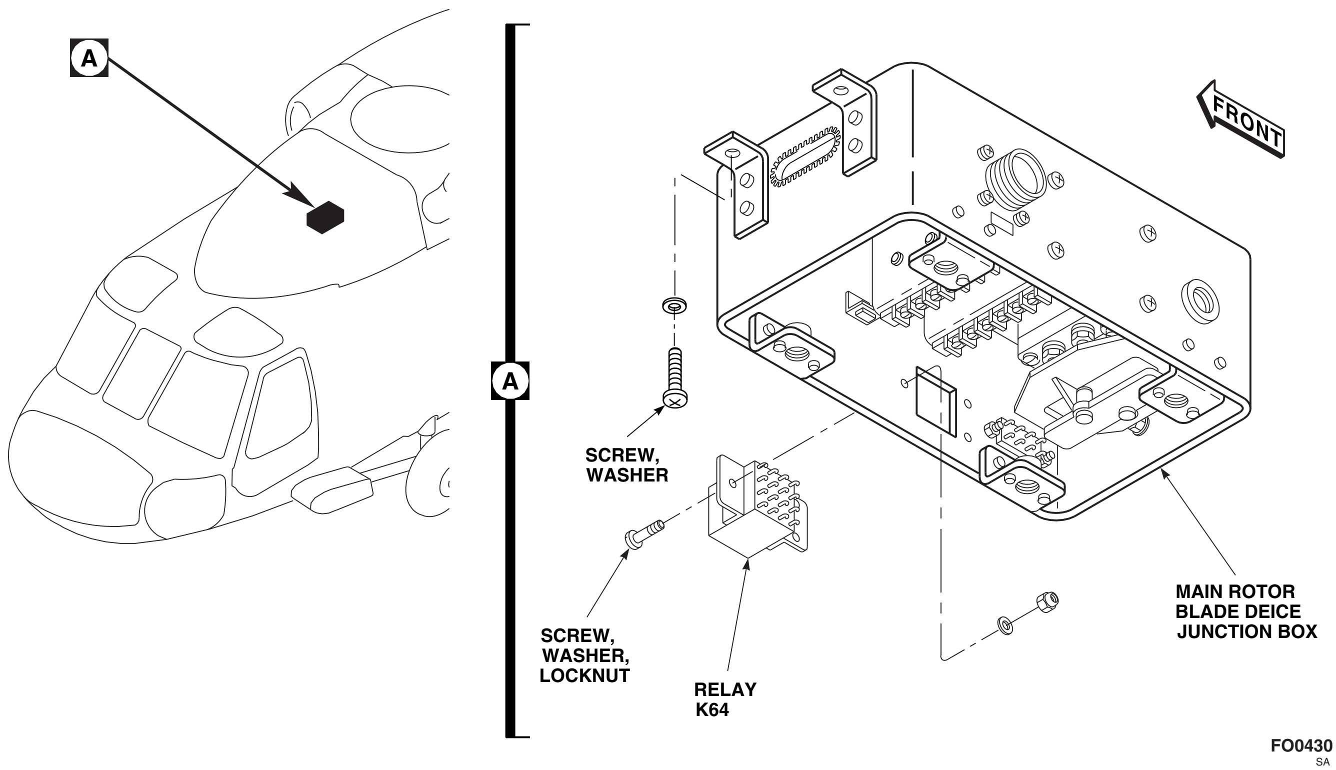


FIGURE 12-4-37-1. REPLACE BLADE DEICE RELAY, K64

c. Position main rotor blade deice junction box on mounting brackets and secure with screws and washers.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering set, solder, Item 313, Appendix D, wires to terminals of relay.

e. Remove tags and slide insulation sleeving, Item 163, Appendix D, over terminals of relay.

f. Install main rotor blade deice junction box cover on main rotor blade deice junction box using cover fasteners.

g. Perform operational check of blade deicing system (PARA 12-2-5).

h. Make sure area is clean and free of foreign material.

i. Install soundproofing panels (PARA 2-4-69).

12-4-38 BLADE DEICE RELAY, K65.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 163, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-38.1 Replace Blade Deice Relay, K65.

12-4-38.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Unfasten and hang main rotor blade deice junction box cover by chain.

d. Slide heat-shrinkable insulating sleeving off terminals of relay (Figure 12-4-38-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Tag wires. Using soldering set, unsolder wires at terminals of relay.

f. Remove screws and washers and let main rotor blade deice junction box hang by external wiring.

g. Remove screws, washers, and locknuts from relay and remove relay from main rotor blade deice junction box.

12-4-38.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install relay in main rotor blade deice junction box using screws, washers, and locknuts (Figure 12-4-38-1, Detail A).

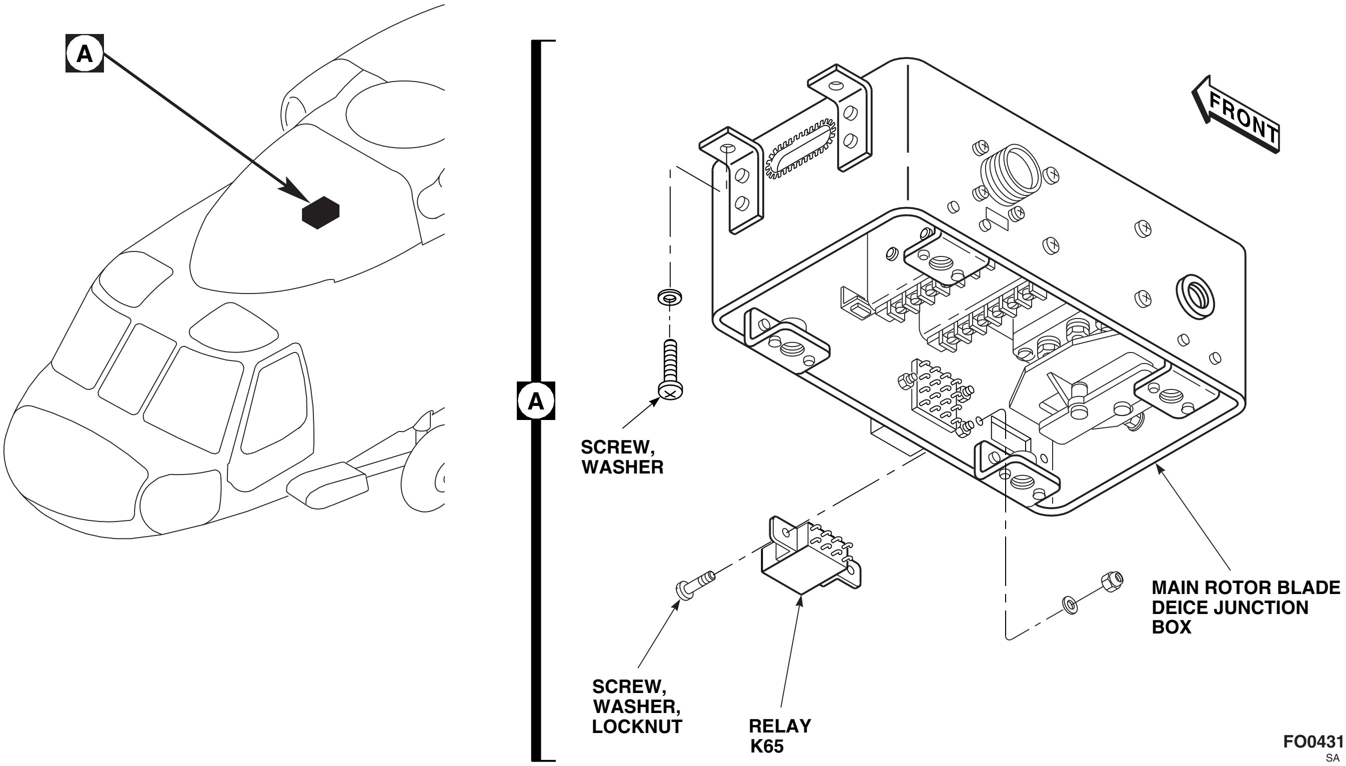


FIGURE 12-4-38-1. REPLACE BLADE DEICE RELAY, K65

c. Position main rotor blade deice junction box on mounting brackets and secure with screws and washers.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering set, solder, Item 313, Appendix D, wires to terminals of relay.

e. Remove tags and slide heat-shrinkable insulation sleeving, Item 163, Appendix D, over terminals of relay.

f. Install main rotor blade deice junction box cover on main rotor blade deice junction box using cover fasteners.

g. Perform operational check of blade deicing system (PARA 12-2-5).

h. Make sure area is clean and free of foreign material.

i. Install soundproofing panels (PARA 2-4-69).

12-4-39 BLADE DEICE CONTACTOR, K60.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-39.1 Replace Blade Deice Contactor, K60.

12-4-39.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Unfasten and hang main rotor blade deice junction box cover by chain.

d. Remove screws securing terminal shield No. 1 to contactor. Remove terminal shield (Figure 12-4-39-1, Detail A).

e. Tag and remove nuts, lockwasher, washers, and wires from terminals, A2, B2, and C2, of contactor.

f. Remove terminal shield No. 2 from contactor.

g. Tag and remove nuts, lockwashers, washers, and wires from terminals, A1, B1, C1, X1, and X2, of contactor.

h. Remove screws and washers from contactor and remove contactor from main rotor blade deice junction box.

12-4-39.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position contactor over mounting holes in main rotor blade deice junction box and secure with screws and washers (Figure 12-4-39-1, Detail A).

c. Connect wires to terminals, A1, B1, and C1 of contactor using nuts, lockwashers, and washers. **TORQUE NUTS TO 18 - 20 INCH-POUNDS.** Remove tags.

d. Connect wires to terminals, X1 and X2, of contactor using nuts, lockwashers, and washers. Remove tags.

e. Place terminal shield No. 2 over terminals, A1, B1, C1, X1, and X2 of contactor.

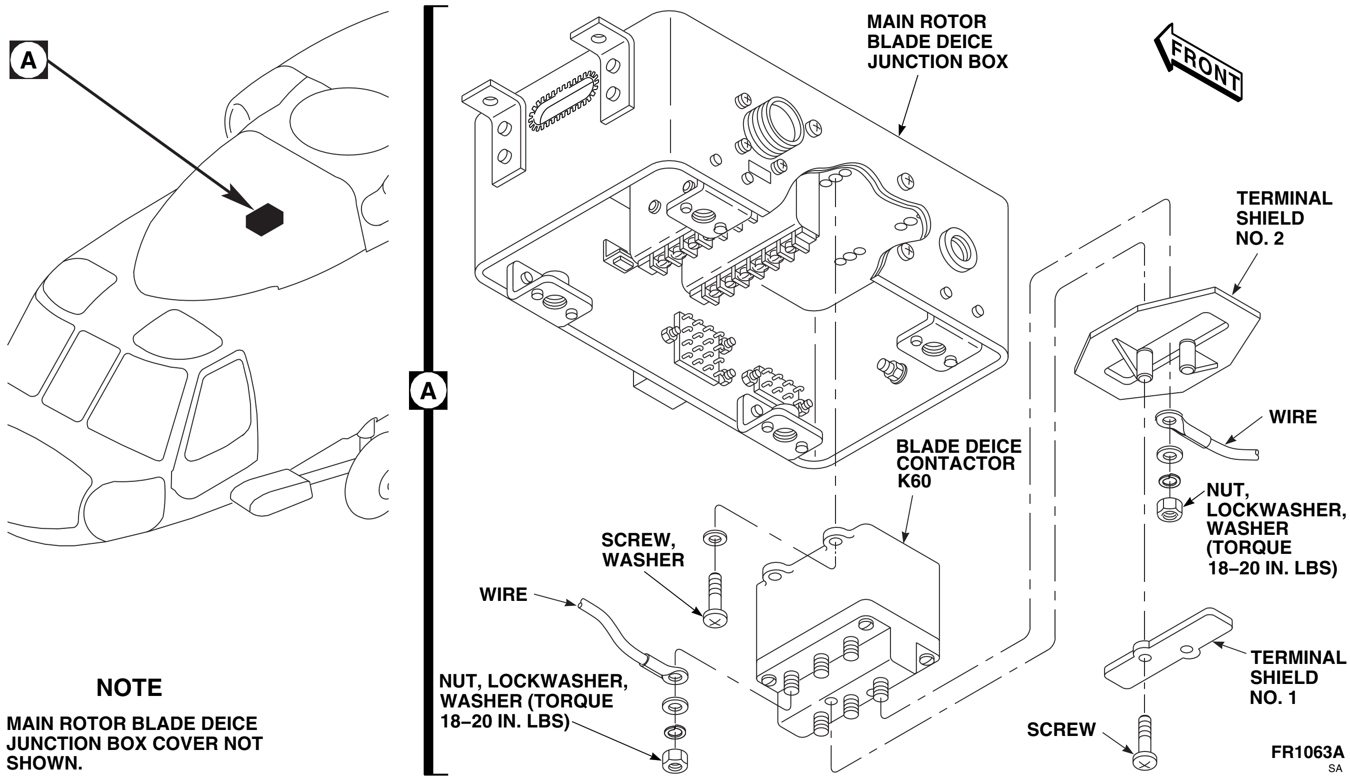


FIGURE 12-4-39-1. REPLACE BLADE DEICE CONTACTOR, K60

f. Connect wires to terminals, A2, B2, and C2, of contactor, using nuts, lockwashers, and washers. **TORQUE NUTS TO 18 - 20 INCH-POUNDS.** Remove tags.

g. Install terminal shield No. 1, on contactor, using screws.

h. Install main rotor blade deice junction box cover on main rotor blade deice junction box using cover fasteners.

i. Perform operational check of blade deicing system (PARA 12-2-5).

j. Make sure area is clean and free of foreign material.

k. Install soundproofing panels (PARA 2-4-69).

12-4-40 BLADE DEICE CONTACTOR, K62.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-40.1 Replace Blade Deice Contactor, K62.

12-4-40.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Unfasten and hang auxiliary junction box cover by chain.

d. Remove screws and terminal shield from contactor (Figure 12-4-40-1, Detail A).

e. Tag wires and remove nuts, lockwashers, washers, and wires from terminals of contactor.

f. Remove screws and washers from contactor and slide toward current limiter block until free of bus bars.

g. Remove contactor from auxiliary junction box.

12-4-40.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position contactor in auxiliary junction box with terminals, L11, L12, and L13, facing bus bars (Figure 12-4-40-1, Detail A).

c. Slide contactor toward bus bars until slots on bus bars engage terminals, L11, L12, and L13.

d. Secure contactor using screws and washers.

e. Connect wires to terminals of contactor, using nuts, lockwashers, and washers. Remove tags.

f. Install terminal shield on contactor using screws.

g. Install auxiliary junction box cover on auxiliary junction box using cover fasteners.

h. Perform operational check of blade deicing system (PARA 12-2-5).

i. Make sure area is clean and free of foreign material.

j. Install soundproofing panels (PARA 2-4-69).

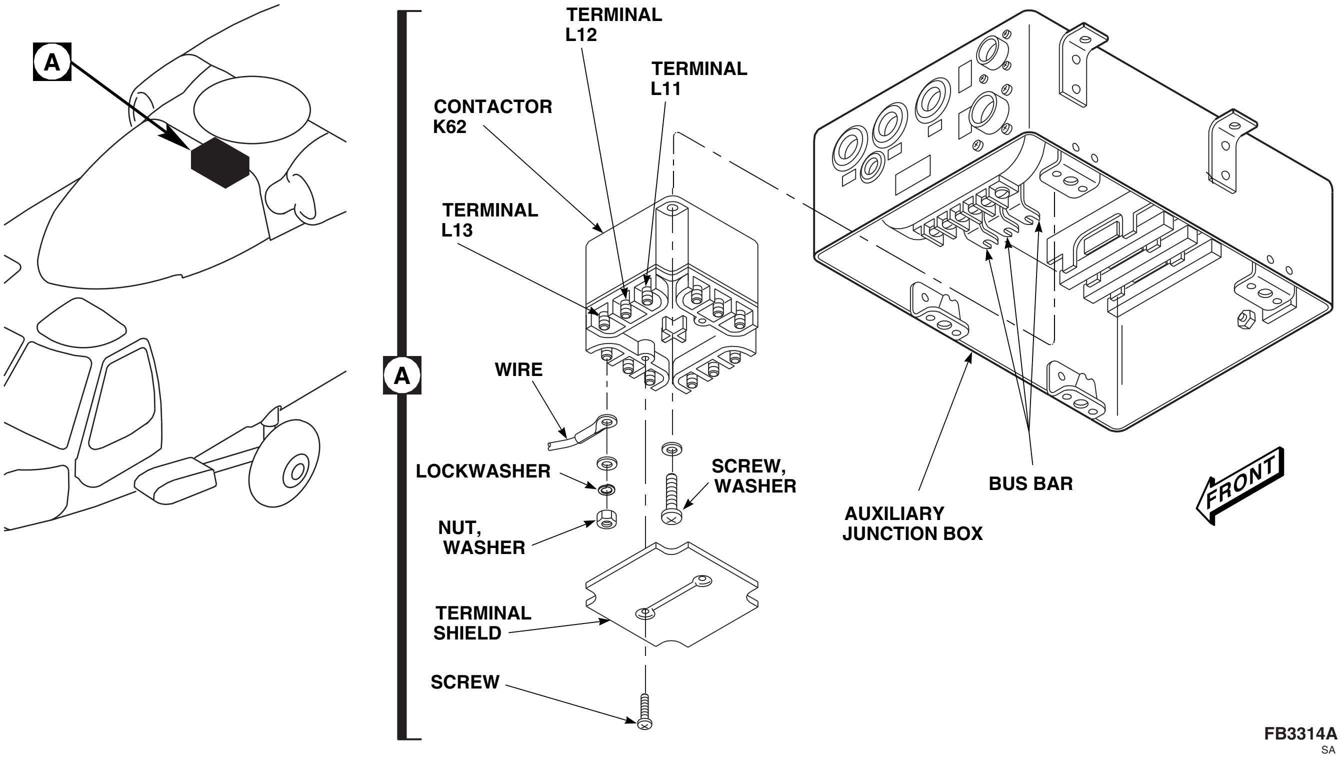


FIGURE 12-4-40-1. REPLACE BLADE DEICE CONTACTOR, K62

12-4-41 BLADE DEICE CONTACTOR, K63.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 163, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-41.1 Replace Blade Deice Contactor, K63.

12-4-41.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Unfasten and hang auxiliary junction box cover by chain.

d. Slide insulating sleeving off terminals of contactor (Figure 12-4-41-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Tag wires. Using soldering set, unsolder wires at terminals of contactor.

f. Remove screws and washers from contactor and remove from auxiliary junction box.

12-4-41.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position contactor, K63, in auxiliary junction box with terminals, A3, B3, and C3, turned toward contactor, K62 (Figure 12-4-41-1, Detail A).

c. Secure contactor with screws and washers.

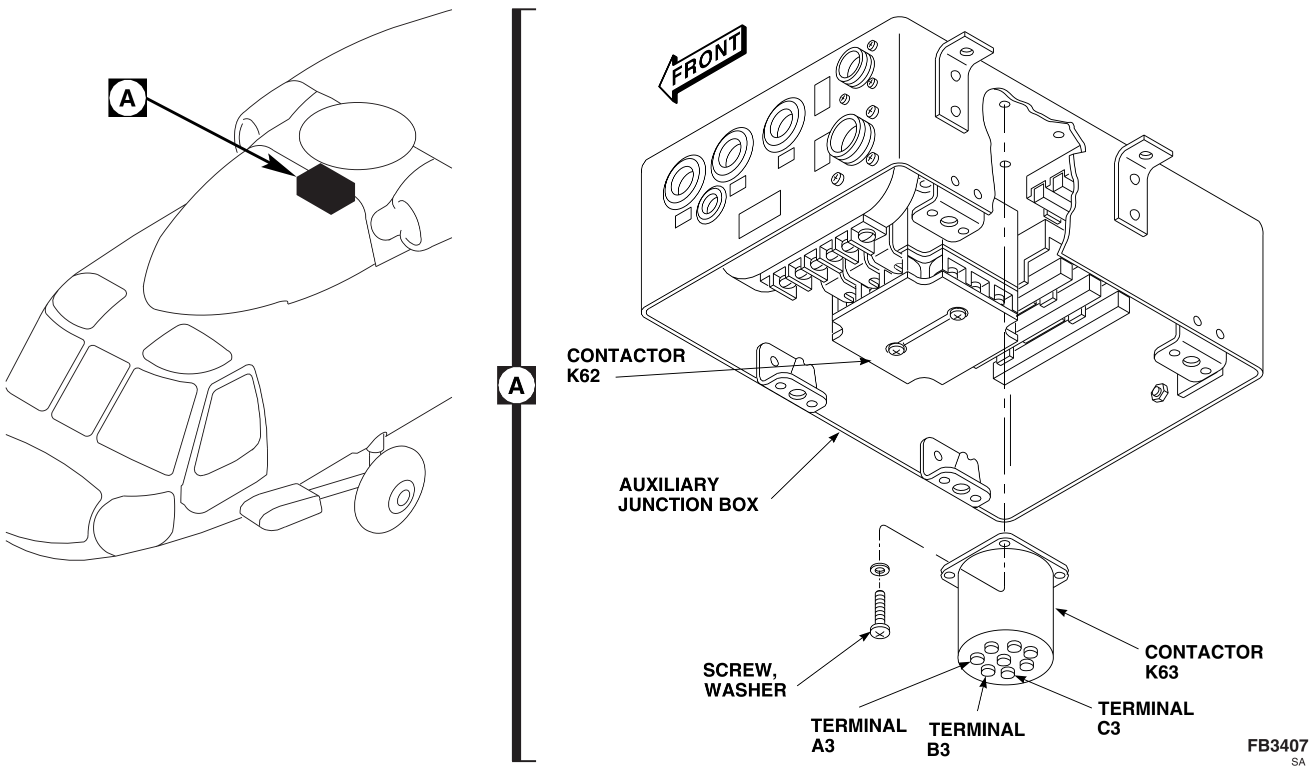


FIGURE 12-4-41-1. REPLACE BLADE DEICE CONTACTOR, K63

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering set, solder, Item 313, Appendix D, wires to terminals of contactor.

e. Remove tags and slide insulation sleeving, Item 163, Appendix D, over soldered terminals of contactor.

f. Install auxiliary junction box cover on auxiliary junction box using cover fasteners.

g. Perform operational check of blade deicing system (PARA 12-2-5).

h. Make sure area is clean and free of foreign material.

i. Install soundproofing panels (PARA 2-4-69).

12-4-42 BLADE DEICE CURRENT TRANSFORMER, T10.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 12-4-40

12-4-42.1 Replace Blade Deice Current Transformer, T10.

12-4-42.1.1. Remove.

a. Remove blade deice contactor, K62 (PARA 12-4-40).

b. Tag wires and remove screws, washers, and wires from transformer, T10, terminal strip (Figure 12-4-42-1, Detail A).

c. Tag wires and remove nuts, lockwashers, washers, and wires from terminal board 5.

d. Tag and pull bus bars through transformer, T10.

e. Remove screws and washers securing transformer, T10, to auxiliary junction box. Remove transformer.

12-4-42.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install current transformer, T10, using screws and washers (Figure 12-4-42-1, Detail A).

c. Slide bus bars through transformer, T10, and align slotted end of bus bars with terminals 1, 3, and 5, of terminal board 5. Remove tags.

d. Connect wires and bus bars to terminal board 5 as follows: washer, bus bar, wire terminal lug, washer, lockwasher, and nut. Remove tags.

e. Connect wires to transformer, T10, terminal strip with screws and washers. Remove tags.

f. Install blade deice contactor, K62 (PARA 12-4-40).

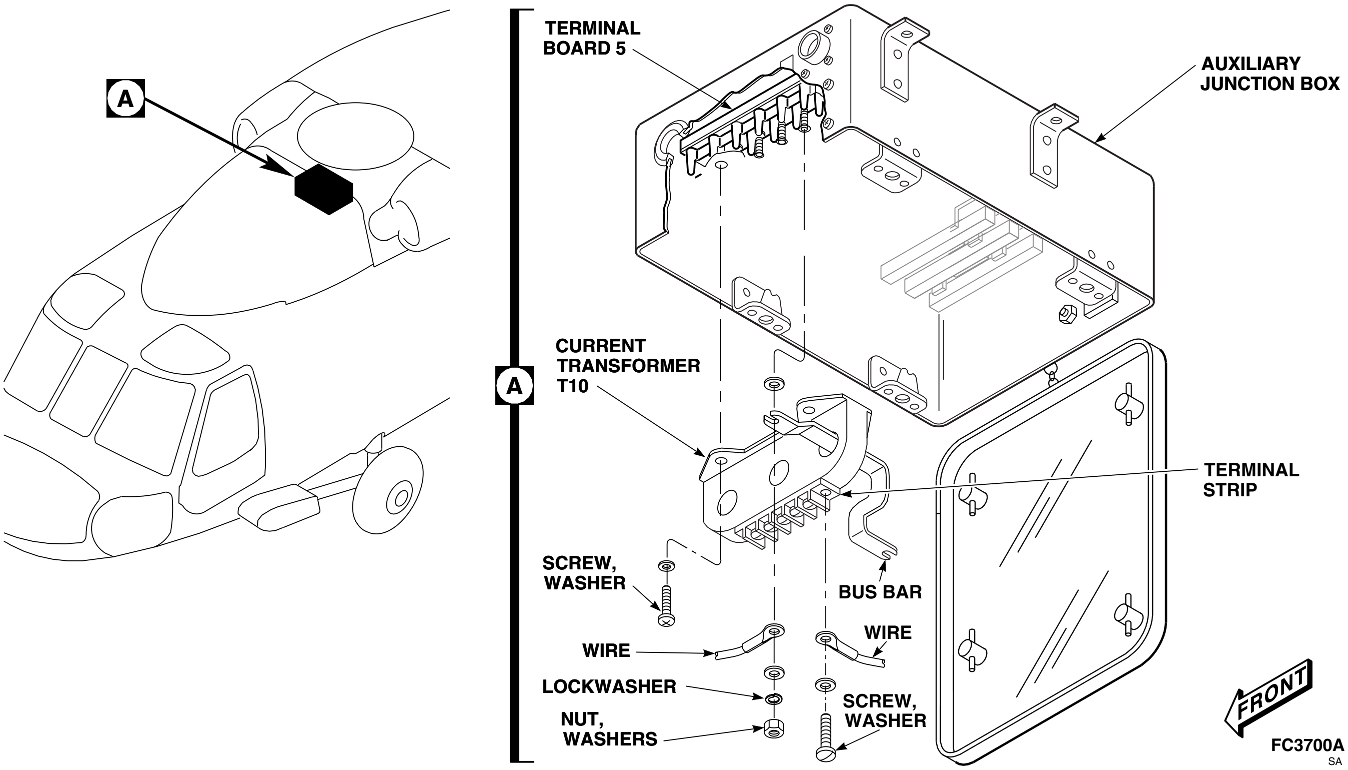


FIGURE 12-4-42-1. REPLACE BLADE DEICE CURRENT TRANSFORMER, T10

12-4-43 BLADE DEICE CURRENT TRANSFORMER, T14.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-43.1 Replace Blade Deice Current Transformer, T14.

12-4-43.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Unfasten and hang main rotor blade deice junction box cover by chain.

d. Remove nuts, lockwashers, washers, and wires from terminals, 2, 4, and 5 of terminal board 3 (Figure 12-4-43-1, Detail A and Detail B).

NOTE

The existing red, white, and black wires on the input side of terminal board 3 will be reconnected when the current transformer wires are installed.

e. Remove screws and terminal shields No. 1 and No. 2 from contactor, K60 (Figure 12-4-43-1, Detail A).

f. Remove nuts, lockwashers, washers, and wires from terminals, A1, B1, and C1, of contactor, K60 (Figure 12-4-43-1, Detail A and Detail B).

g. Tag wires and remove screws, washers, and wires from terminals of current transformer, T14 and T14A. Remove and save three resistors from terminals of current transformer, T14A (Figure 12-4-43-1, Detail A).

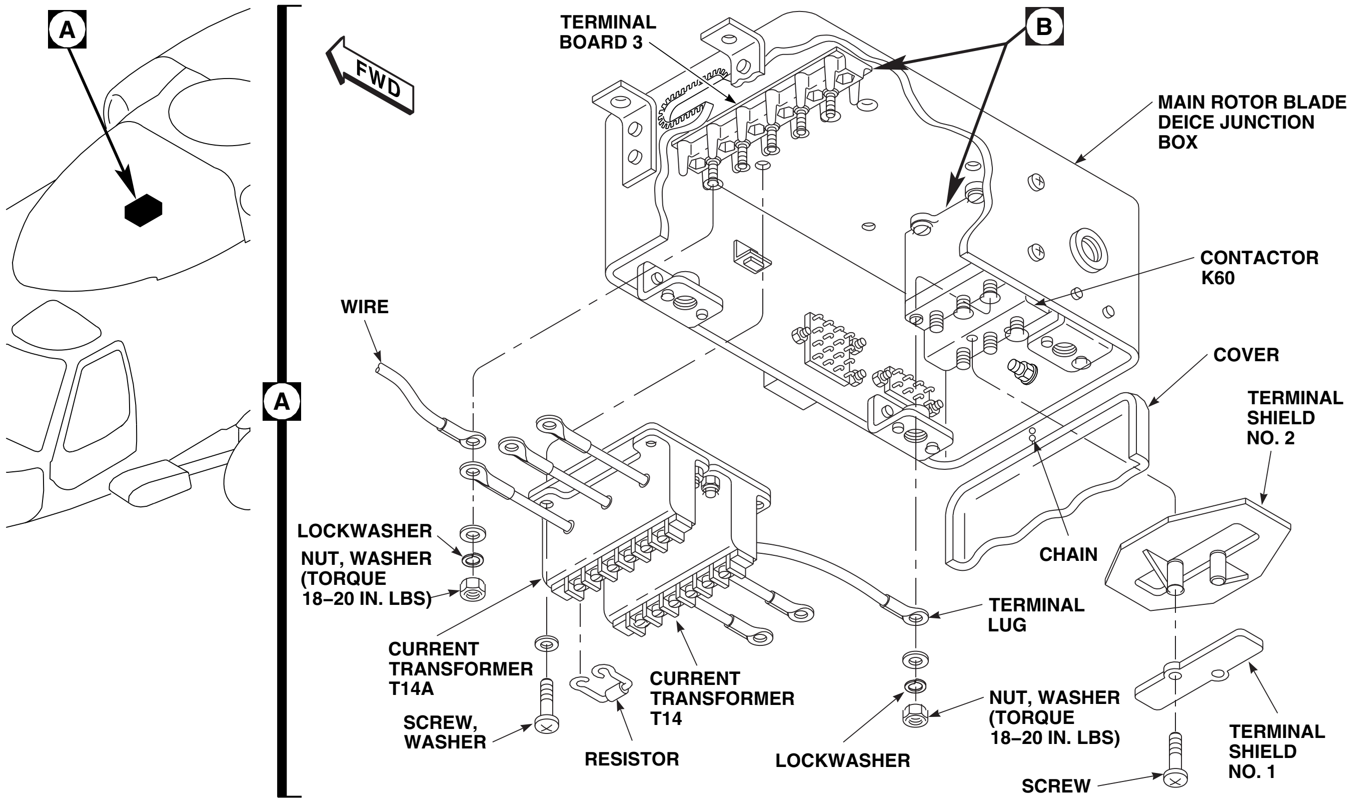
h. Remove screws and washers from current transformer, and carefully remove transformer and disconnected wires from main rotor blade deice junction box.

12-4-43.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Cut red, white, and black wires on current transformer being installed to same lengths as wires being replaced.

c. Install new terminal lugs on ends of current transformer wires.



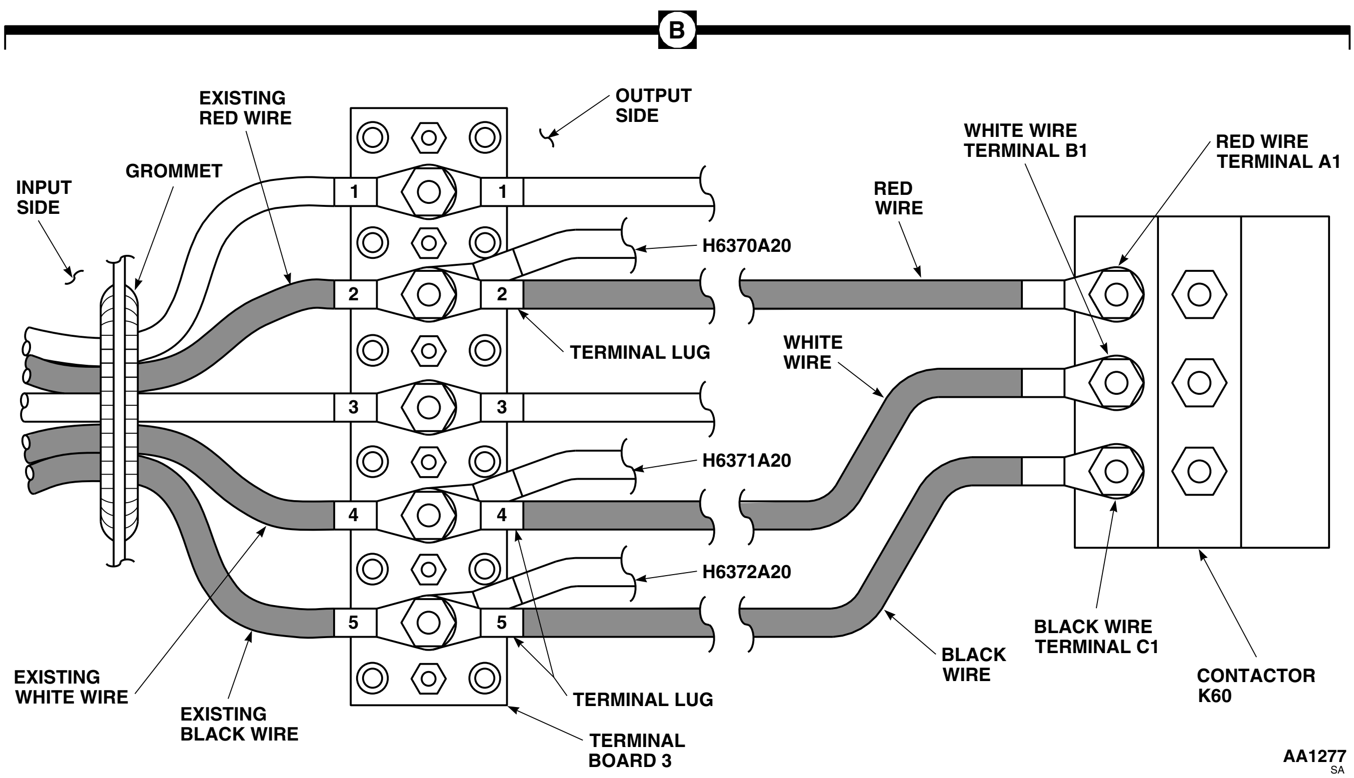


FIGURE 12-4-43-1. REPLACE BLADE DEICE CURRENT TRANSFORMER, T14

d. Position transformer over mounting holes and secure transformer with screws and washers (Figure 12-4-43-1, Detail A).

e. Using nuts, lockwashers, and washers, connect wires to terminal board 3 of junction box as follows (Figure 12-4-43-1, Detail A and Detail B):

WIRE CODE	TO	TERMINAL NO.
(Input Side) Red		2
(Output Side) Red		2
(Input Side) White		4
(Output Side) White		4
(Input Side) Black		5
(Output Side) Black		5

f. Using nuts, lockwashers, and washers, connect wires to contactor, K60, as follows:

WIRE CODE	TO	TERMINAL NO.
Red		A1
White		B1
Black		C1

g. **TORQUE NUTS TO 18 - 20 INCH-POUNDS** (Figure 12-4-43-1, Detail A).

h. Using screws, install terminal shields, No. 1 and No. 2, on contactor, K60.

i. Install resistors between terminals, T1-T4, T2-T5, and T3-T6, on current transformer, T14A.

j. Connect wires to terminals of current transformer, T14 and T14A. Secure with screws and washers. Remove tags.

k. Install main rotor blade deice junction box cover on main rotor blade deice junction box using cover fasteners.

l. Perform operational check of blade deicing system (PARA 12-2-5).

m. Make sure area is clean and free of foreign material.

n. Install soundproofing panels (PARA 2-4-69).

12-4-44 MAIN ROTOR BLADE DEICE JUNCTION BOX.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 12-2-5

12-4-44.1 Replace Main Rotor Blade Deice Junction Box.

12-4-44.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Disconnect electrical connector, P329 from J329, on main rotor blade deice junction box (Figure 12-4-44-1, Sheet 1, Detail A).

d. Install protective cap and plug on electrical connectors.

e. Loosen fasteners that secure cover on main rotor blade deice junction box and let cover hang by chain.

f. Tag wires and remove nuts, lockwashers, and washers and disconnect slipring harness wires from terminal board 3, and pull wires out through grommet (Figure 12-4-44-1, Sheet 1, Detail B).

g. Remove screws and terminal shield, No. 1, from contactor, K60.

NOTE

Contactor terminal shield, No. 2, may fall off contactor, K60, when wires are removed from terminals, A2, B2, and B3.

h. Tag wires and disconnect nuts, lockwashers, washers, and wires from terminals, A2, B2, and C2 of contactor, K60, and pull wires out through grommet.

i. W/O EMEP Tag wire and remove screw, lockwasher, washers, and nut, and disconnect ground wire from airframe (Figure 12-4-44-1, Sheet 2, Detail C and Detail D).

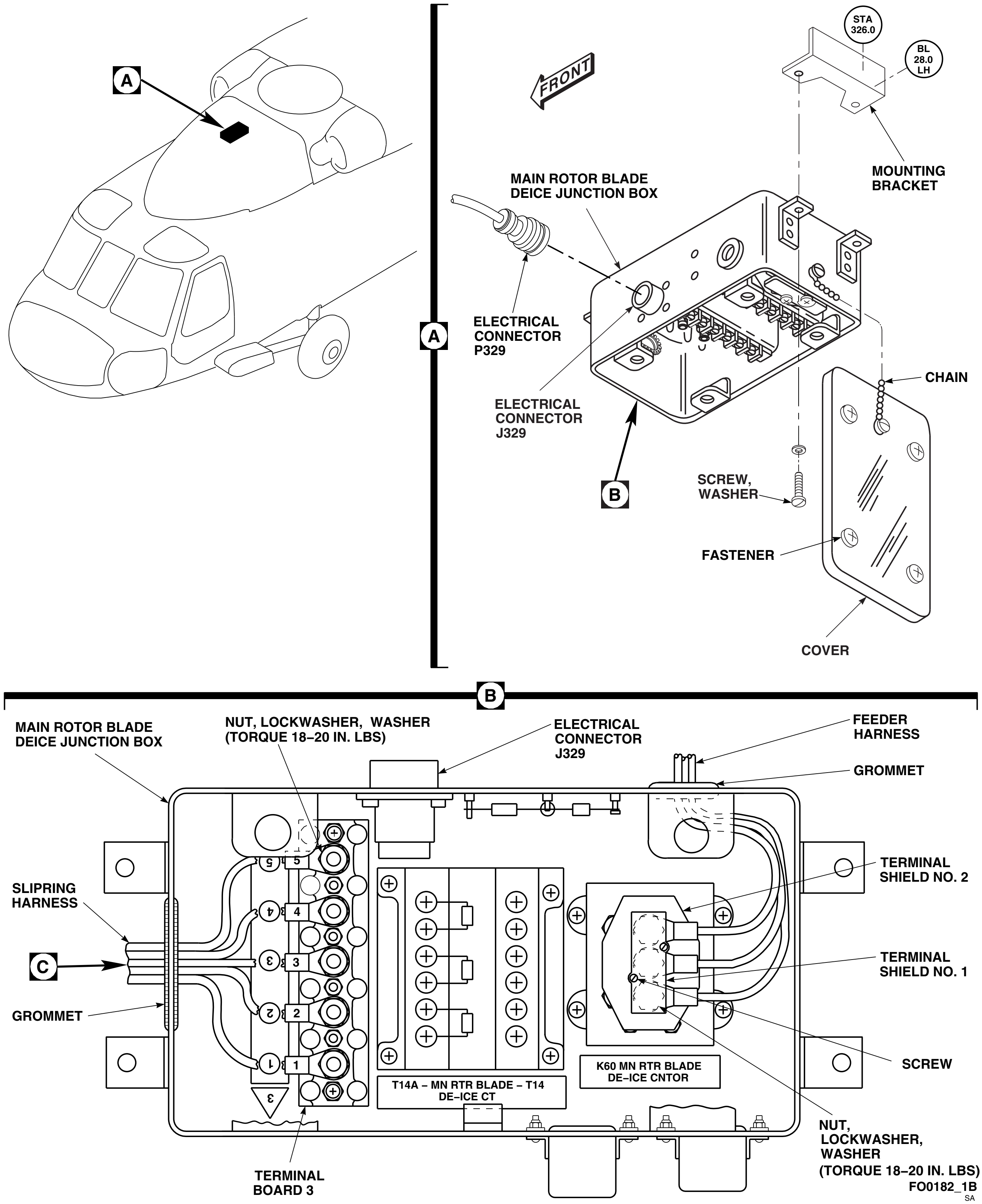
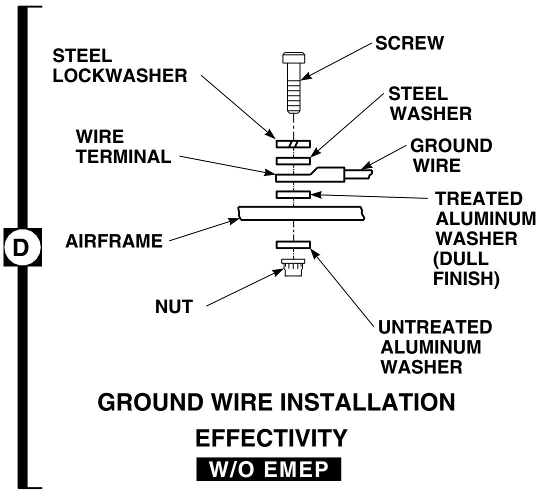
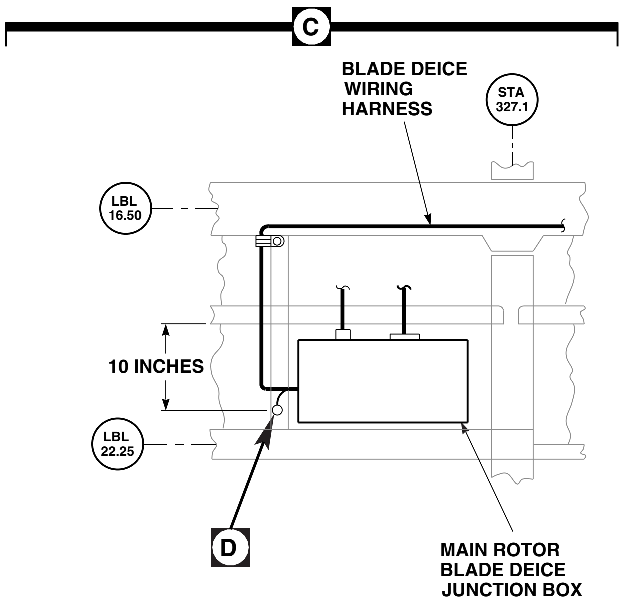
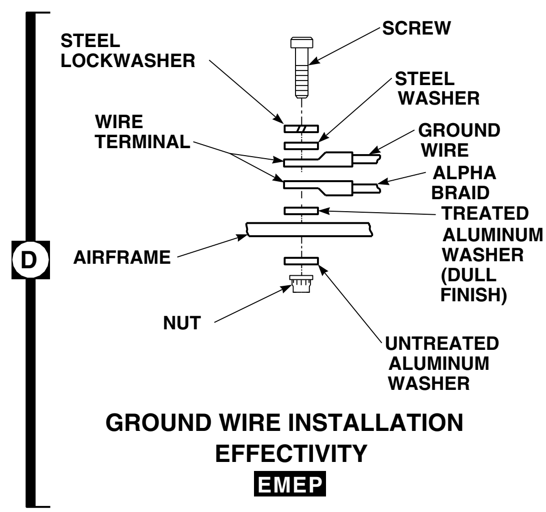


FIGURE 12-4-44-1. REPLACE MAIN ROTOR BLADE DEICE JUNCTION BOX (SHEET 1 OF 2)







FT2633_2
SA

FIGURE 12-4-44-1. REPLACE MAIN ROTOR BLADE DEICE JUNCTION BOX (SHEET 2 OF 2)

j. EMEP Tag wires and remove screw, lockwasher, washers, and nut, and disconnect ground wire and alpha braid from airframe (Figure 12-4-44-1, Sheet 2, Detail C and Detail D).

k. Remove screws and washers from main rotor blade deice junction box and lower junction box from mounting brackets (Figure 12-4-44-1, Sheet 1, Detail A).

12-4-44.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position main rotor blade deice junction box on mounting brackets and secure with screws and washers (Figure 12-4-44-1, Sheet 1, Detail A).

c. Carefully pass slipring harness wires through grommet (Figure 12-4-44-1, Sheet 1, Detail B).

d. W/O EMEP Connect slipring harness wires to terminal board 3 with nuts, lockwashers, and washers as follows, then remove tags:

WIRE CODE	TO	TERMINAL NO.
1-20-S2		1
1-10-A		2
1-20-S1		3
1-10-B		4
1-10-C		5

e. EMEP Connect slipring harness wires to terminal board 3 with nuts, lockwashers, and washers as follows, then remove tags:

WIRE CODE	TO	TERMINAL NO.
H6458E20		1
H6370C08A		2
H6373C20		3
H6371C08B		4
H6372C08C		5

f. **TORQUE NUTS TO 18 - 20 INCH-POUNDS.**

g. Carefully pass contactor, K60, wires through grommet.

h. Make sure terminal shield, No. 2, is installed on contactor, K60.

i. Connect feeder harness wires to terminals, A2, B2, and C2, of contactor, K60 using nuts, washers, and lockwashers. **TORQUE NUTS TO 18 - 20 INCH-POUNDS.**

j. Install terminal shield No. 1 on contactor, K60, using screws. Remove tags.

k. W/O EMEP Connect ground wire to airframe with screw, lockwasher, washers, and nut (Figure 12-4-44-1, Sheet 2, Detail C and Detail D). Remove tag.

l. EMEP Connect ground wire and alpha braid as follows (Figure 12-4-44-1, Sheet 2, Detail D):

(1) Inspect chemical conversion coated surfaces on stringer and mounting washers (PARA 2-6-5).

(2) Connect ground wire, H6369A20N, and alpha braid to airframe with screw, lockwasher, washers, and nut. Remove tags.

m. Remove protective cap and plug from electrical connectors.

n. Inspect and treat electrical connectors (PARA 1-7-20).

o. Connect electrical connector, P329 to J329, on main rotor blade deice junction box (Figure 12-4-44-1, Sheet 1, Detail A).

p. Install cover on main rotor blade deice junction box and secure with fasteners

q. Perform operational check of blade deicing system (PARA 12-2-5).

r. Install soundproofing panels (PARA 2-4-69).

12-4-45 BLADE DEICE DIODE, CR11, CR15, AND CR16.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Solder, Item 313, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 12-2-5

12-4-45.1 Replace Blade Deice Diode, CR11, CR15, and CR16.

12-4-45.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels (PARA 2-4-69).

c. Loosen fasteners securing cover on main rotor blade deice junction box and let cover hang by chain (Figure 12-4-45-1, Detail A).

d. Remove screws, washers, and locknuts from diode board and hang diode board by wiring.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Using soldering set, unsolder and remove diode from diode board (Figure 12-4-45-1, Detail B).

12-4-45.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Mount diode on diode board. Make sure to observe polarity as indicated on board (Figure 12-4-45-1, Detail B).

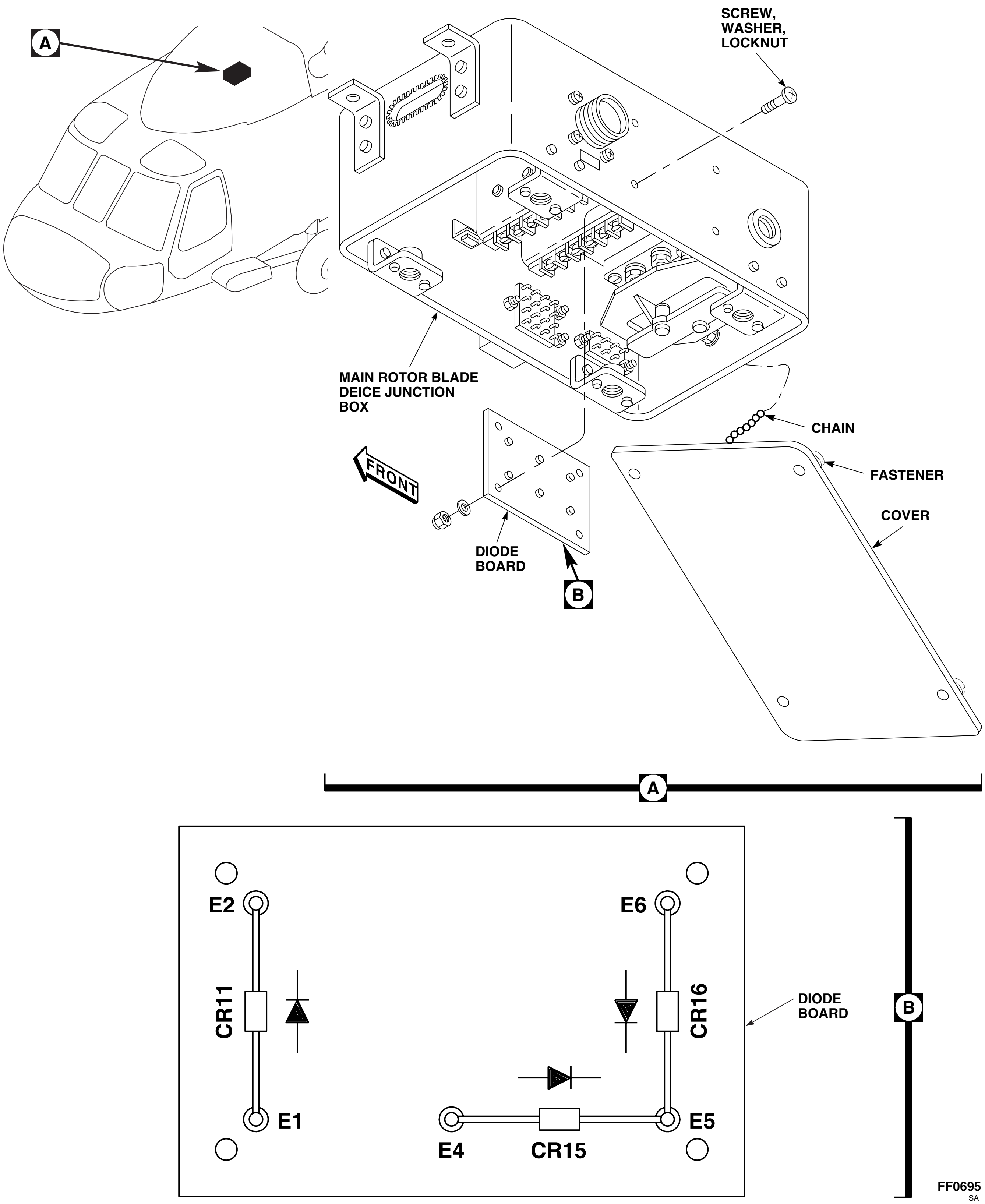


FIGURE 12-4-45-1. REPLACE BLADE DEICE DIODE, CR11, CR15, AND CR16

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Using soldering set, solder, Item 313, Appendix D, diode to diode board terminals.

d. Position diode board over mounting holes and secure with screws, washers, and locknuts (Figure 12-4-45-1, Detail A).

e. Install cover on main rotor blade deice junction box and secure with fasteners.

f. Perform operational check of blade deicing system (PARA 12-2-5).

g. Make sure area is clean and free of foreign material.

h. Install soundproofing panels (PARA 2-4-69).

12-4-46 BLADE DEICE TEST PANEL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 12-2-5

12-4-46.1 Replace Blade Deice Test Panel.

12-4-46.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen fasteners on blade deice test panel (Figure 12-4-46-1, Detail A).

c. Slowly lift blade deice test panel from instrument panel to gain access to electrical connector.

NOTE

EMEP Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapter from electrical connector, P134 (Figure 12-4-46-1, Detail B).

d. Disconnect electrical connector, P134 from J134, on blade deice test panel.

e. Install protective cap and plug on electrical connectors.

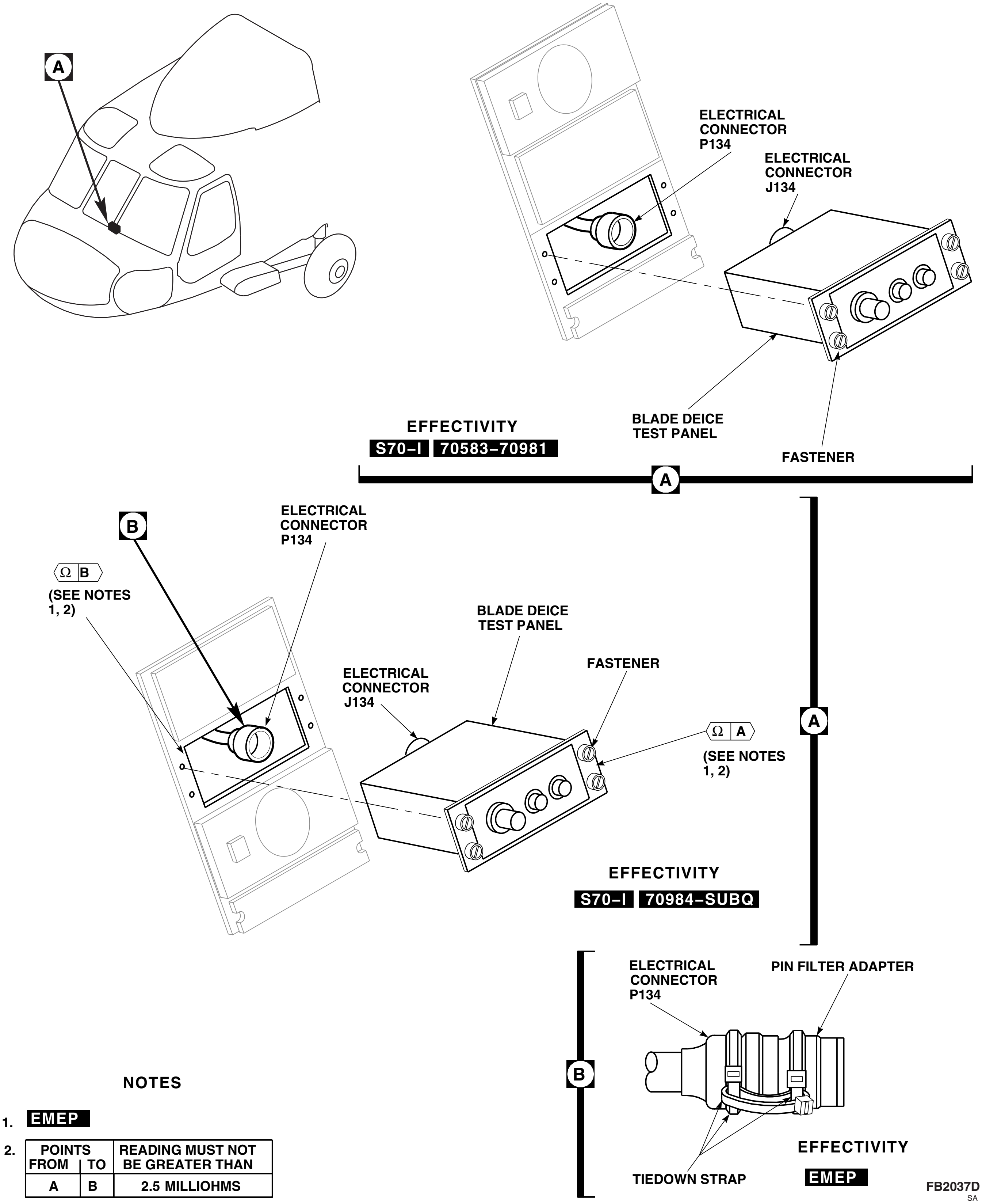
12-4-46.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. EMEP Clean surface around all mounting holes with cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.

c. EMEP Inspect chemical conversion coated mating surfaces of blade deice test panel and instrument panel for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

d. Remove protective cap and plug from electrical connectors.



NOTES

1. EMEP

2.

POINTS FROM	POINTS TO	READING MUST NOT BE GREATER THAN
A	B	2.5 MILLIOHMS

FIGURE 12-4-46-1. REPLACE BLADE DEICE TEST PANEL

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connectors, P134 to J134, on blade deice test panel (Figure 12-4-46-1, Detail A).

g. EMEP Perform EME resistance check of pin filter adapter as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, measure resistance between shell of each pin filter adapter and blade deice test panel. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, measure resistance between shell of pin filter adapter and electrical connector on blade deice test panel. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**

(3) If resistance is still beyond limits, clean surface between pin filter adapter and blade deice test panel.

h. Make sure area is clean and free of foreign material.

i. Carefully install blade deice test panel into instrument panel and secure with fasteners.

j. EMEP Perform EME resistance check of blade deice test panel as follows:

NOTE

Resistance measurements must be taken at least three times.

(1) Using digital low-resistance ohmmeter, measure resistance between information plate of blade deice test panel and instrument panel using digital low-resistance ohmmeter. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, remove blade deice test panel and apply chemical conversion coating (PARA 2-6-5).

k. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-47 BLADE DEICE TEST PANEL INDICATOR LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 12-2-5

12-4-47.1 Replace Blade Deice Test Panel Indicator Lamps.

NOTE

Replacement of both indicator lamps is the same.

12-4-47.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unscrew (counterclockwise) MAIN RTR or TAIL RTR indicator lens cap (Figure 12-4-47-1, Detail A).

c. Remove indicator lamp from lens cap.

12-4-47.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Insert indicator lamp in lens cap and screw (clockwise) lens cap onto blade deice panel (Figure 12-4-47-1, Detail A).

c. Turn on electrical power (PARA 1-6-16).

d. Press lens cap to test indicator lamp. If result is not as specified, perform operational check of blade deicing system (PARA 12-2-5).

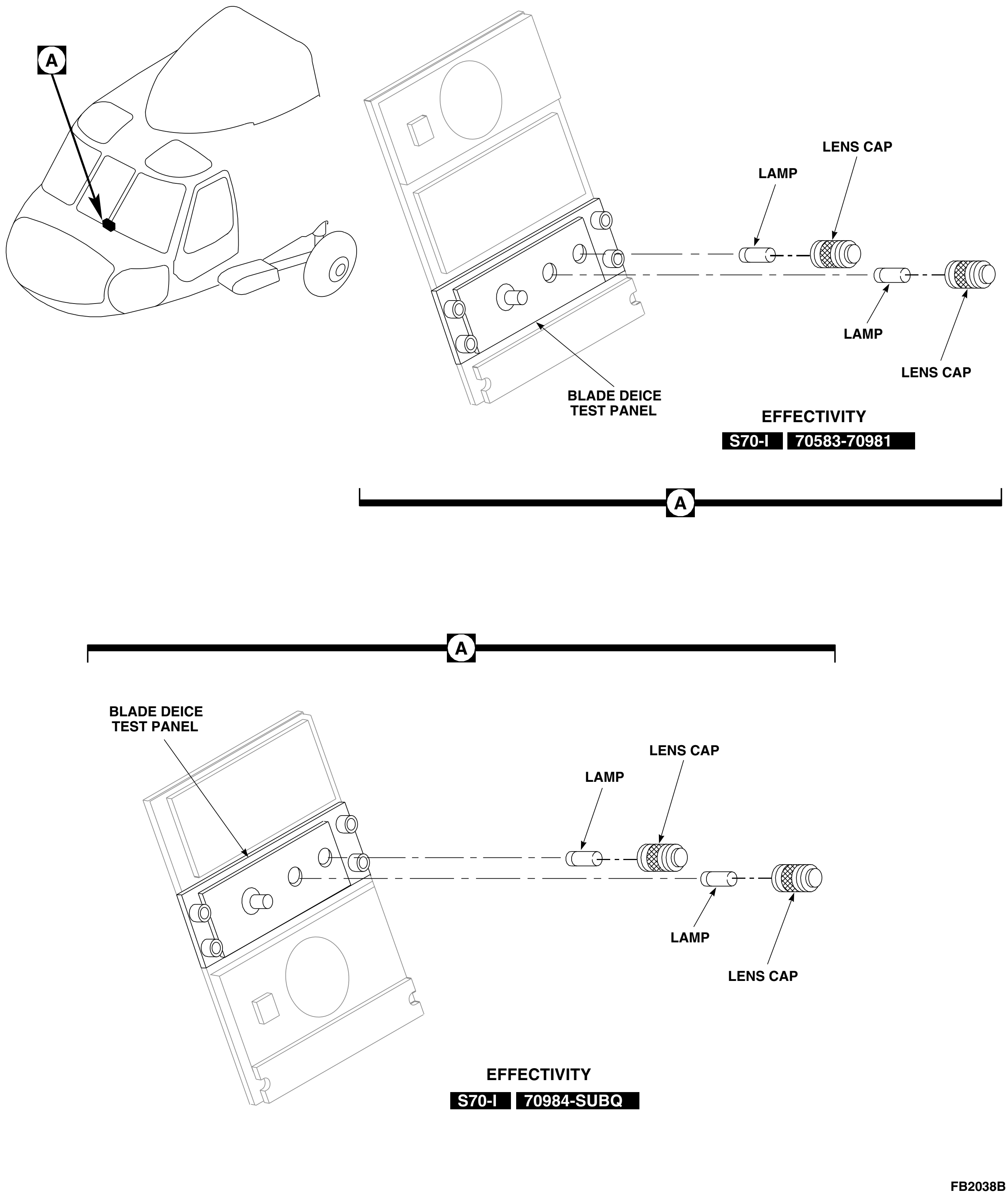


FIGURE 12-4-47-1. REPLACE BLADE DEICE TEST PANEL INDICATOR LAMPS

12-4-48 BLADE DEICE TEST PANEL INFORMATION PLATE/LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Protective Cap and Plug

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-5

12-4-48.1 Replace Blade Deice Test Panel Information Plate/Lamps.

12-4-48.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Unscrew MAIN RTR and TAIL RTR indicator lens cap, and remove both cap assemblies (Figure 12-4-48-1, Detail A).

c. Loosen setscrews and remove knob from mode select switch.

d. Remove screws from information plate.

e. Disconnect information plate electrical connector, P1, from test panel electrical connector, J1.

f. Install protective cap and plug on electrical connectors.

g. Remove moisture seal from lamp to be replaced (Figure 12-4-48-2).

h. Turn slotted lamp until slot lines up with guides in information plate.

i. Remove lamp from information plate.

12-4-48.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Place lamp in information plate receptacle (Figure 12-4-48-2).

c. Line up slot in lamp with information plate guides.

d. Turn lamp clockwise to lock in place.

e. Seal lamp receptacle by applying moisture seal.

f. Remove protective cap and plug from electrical connectors.

g. Inspect and treat electrical connector (PARA 1-7-20).

h. Carefully connect information plate electrical connector, P1, to test panel assembly electrical connector, J1 (Figure 12-4-48-1, Detail A).

i. Secure information plate with screws.

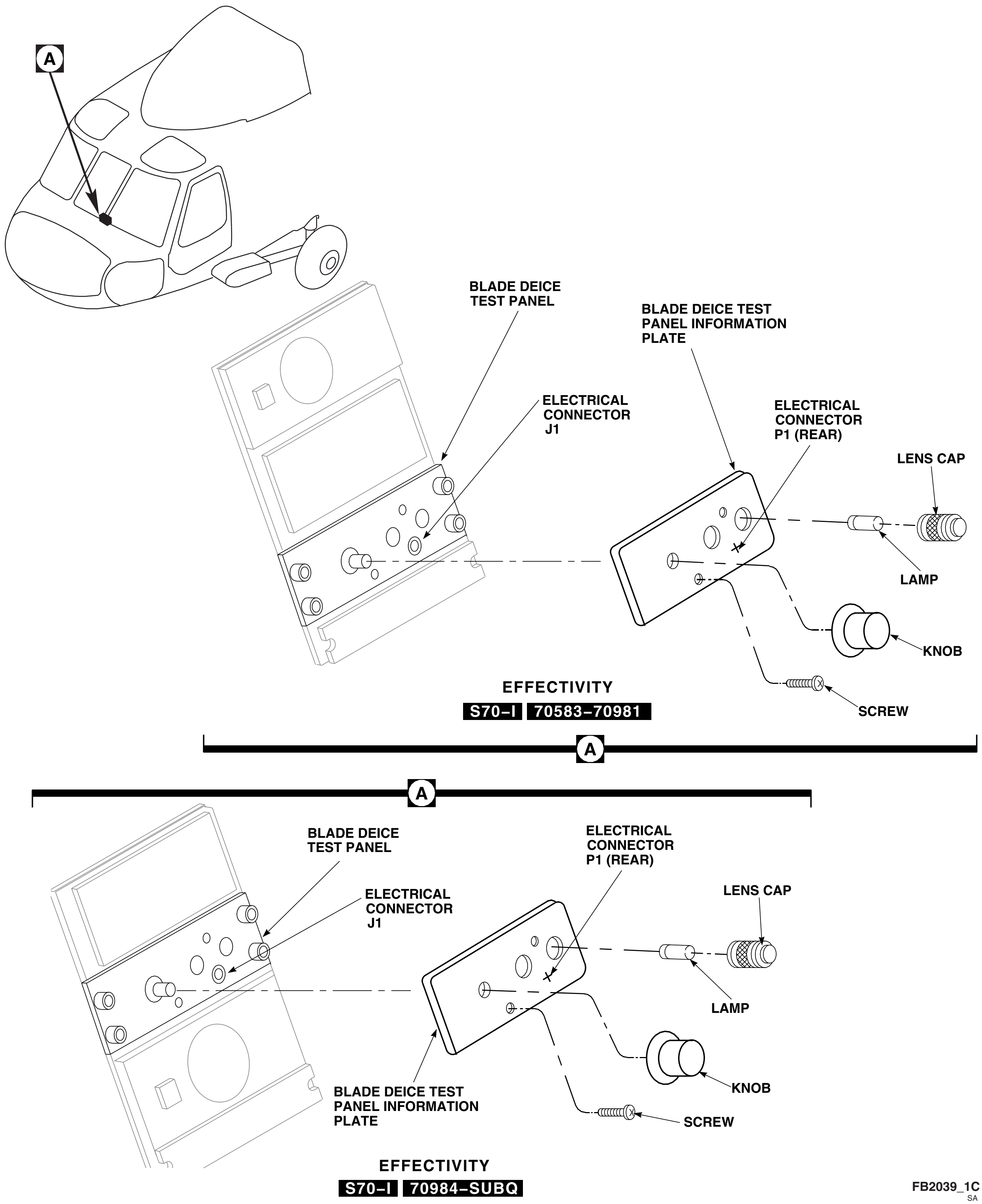


FIGURE 12-4-48-1. REPLACE BLADE DEICE TEST PANEL INFORMATION PLATE/LAMPS

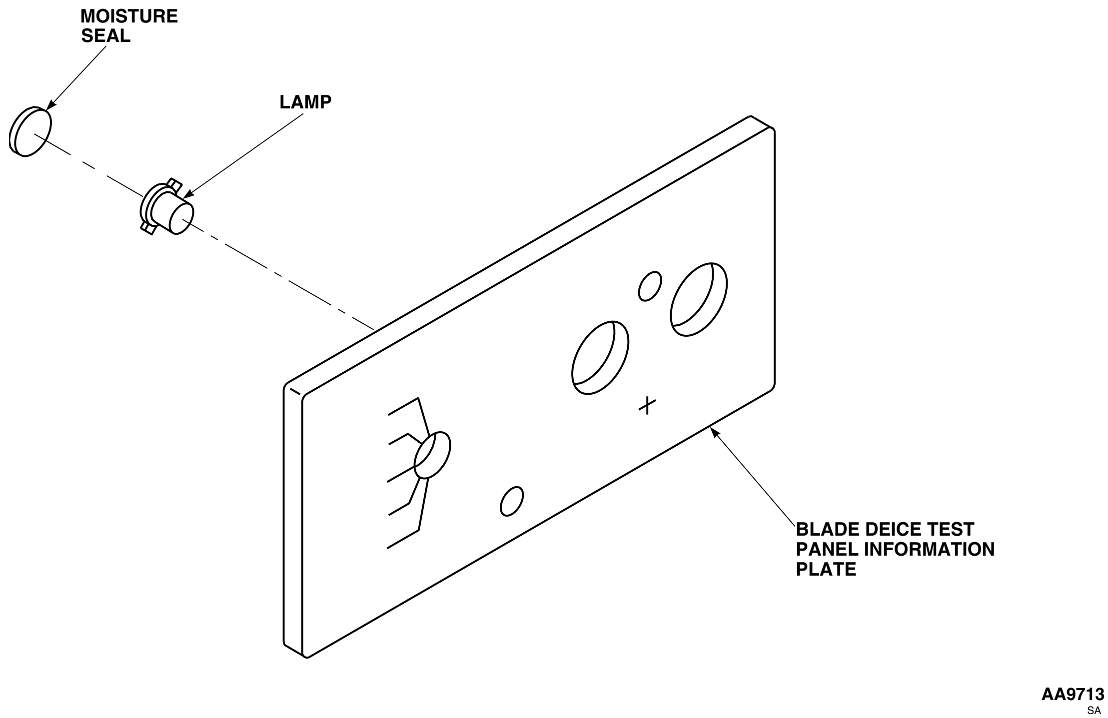


FIGURE 12-4-48-2. REPLACE BLADE DEICE TEST PANEL INFORMATION PLATE/LAMPS

j. Screw MAIN RTR and TAIL RTR lens cap assembly onto blade deice panel.

k. Line up select switch knob setscrews with flat surfaces on shaft of switch and push knob onto shaft. Make sure knob pointer is lined up with information plate stenciling. **TIGHTEN SETSCREWS.**

l. Make sure area is clean and free of foreign material.

m. Turn on electrical power (PARA 1-6-16).

n. Make sure LIGHTS NON FLT circuit breaker is in.

o. Turn LIGHTS NON FLT control to BRT, all nonflight instrument panel lights go on. If result is not as specified, perform operational check (PARA 12-2-5).

12-4-49 BLADE DEICE OAT SENSOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-5

12-4-49.1 Replace Blade Deice OAT Sensor.

12-4-49.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open nose door.

c. Disconnect electrical connector, P122, from blade deice outside air temperature (OAT) sensor (Figure 12-4-49-1, Detail A).

d. Install protective cap and plug on electrical connectors.

e. Remove sealing compound from OAT sensor shield and screws using nonmetallic scraper.

f. Remove screws and washers from OAT sensor shield.

g. Remove OAT sensor shield and OAT sensor.

12-4-49.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove remaining sealing compound from OAT sensor shield and helicopter skin using nonmetallic scraper.

c. Clean area to be sealed with a cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

d. Apply a bead of sealing compound, Item 292, Appendix D, to mounting surface of sensor to prevent water leakage into avionics compartment.

e. Install OAT sensor and OAT sensor shield using screws and washers (Figure 12-4-49-1, Detail A).

f. Shape sealing compound squeezeout to form an even, concave, continuous seal around OAT

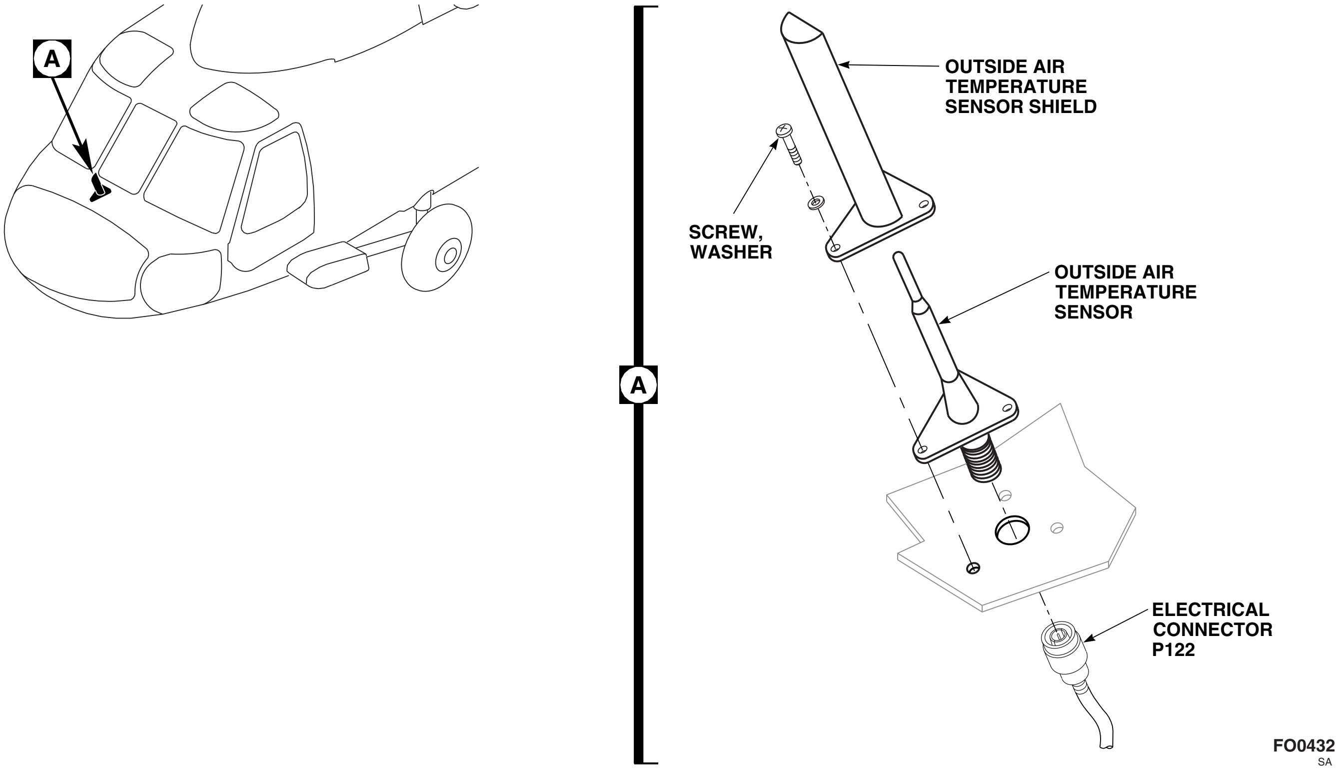


FIGURE 12-4-49-1. REPLACE BLADE DEICE OAT SENSOR

sensor mounting plate and skin using beverage stirring stick, Item 315C, Appendix D.

g. Apply sealing compound, Item 292, Appendix D, around screwheads.

h. Remove protective cap and plug from electrical connectors.

i. Inspect and treat electrical connectors (PARA 1-7-20).

j. Connect electrical connector, P122, to base of OAT sensor.

k. Close nose door.

l. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-50 BLADE DEICE ICE DETECTOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-5

12-4-50.1 Replace Blade Deice Ice Detector.

NOTE

Blade deice ice detector is installed on No. 2 engine air inlet only.

12-4-50.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and washers securing anti-icing valve cover to No. 2 engine air inlet (Figure 12-4-50-1, Detail A).

c. Slowly pull cover from engine air inlet to gain access to electrical connector and hose connection on ice detector.

d. Disconnect electrical connector, P326, and hose assembly from ice detector.

e. Install protective caps and plugs on electrical connectors and hose assembly.

f. Remove screws, washers, and ice detector.

12-4-50.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position ice detector over mounting holes in anti-icing valve cover and secure with screws and washers (Figure 12-4-50-1, Detail A).

c. Remove protective caps and plugs from electrical connectors and hose assembly.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P326, and hose assembly to ice detector.

f. Install cover to helicopter using screws and washers.

g. Perform operational check of blade deicing system (PARA 12-2-5).

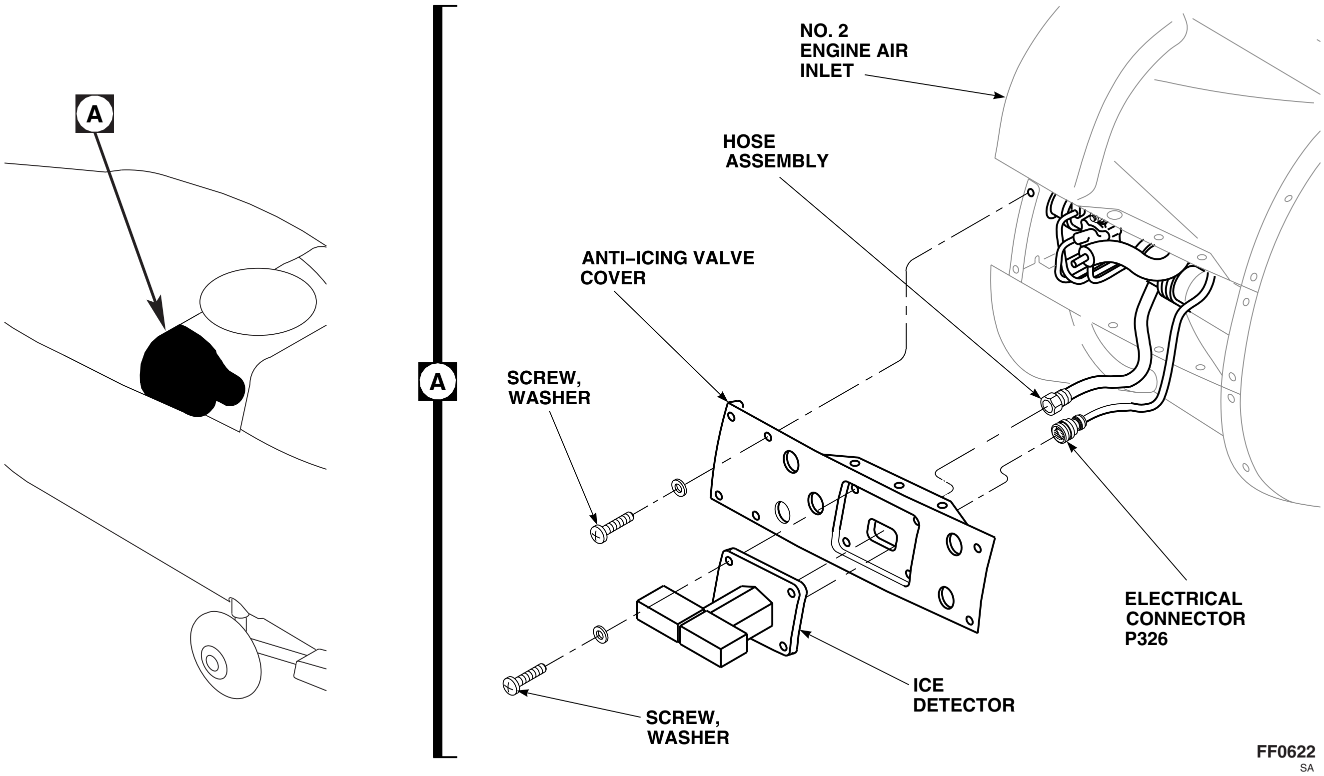


FIGURE 12-4-50-1. REPLACE BLADE DEICE ICE DETECTOR

12-4-51 BLADE DEICE ICE DETECTOR HOSE ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-5

12-4-51.1 Replace Blade Deice Ice Detector Hose Assembly.

NOTE

Blade deice ice detector hose assembly is installed on No. 2 engine air inlet only.

12-4-51.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and washers securing anti-icing valve cover to No. 2 engine air inlet (Figure 12-4-50-1, Detail A).

c. Slowly pull cover from engine air inlet to gain access to electrical connector and hose connection on ice detector.

d. Disconnect electrical connector, P326, and hose assembly from ice detector.

e. Install protective cap and plug on electrical connectors.

f. Disconnect hose assembly from tube assembly and remove hose assembly.

g. Install protective caps and plugs on hose assembly.

12-4-51.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Connect hose assembly end, with 90° bend, to tube assembly. Make sure hose assembly is positioned for connection to ice detector (Figure 12-4-51-1, Detail A).

c. Remove protective cap and plug from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P326, and hose assembly to ice detector.

f. Install cover to helicopter using screws and washers.

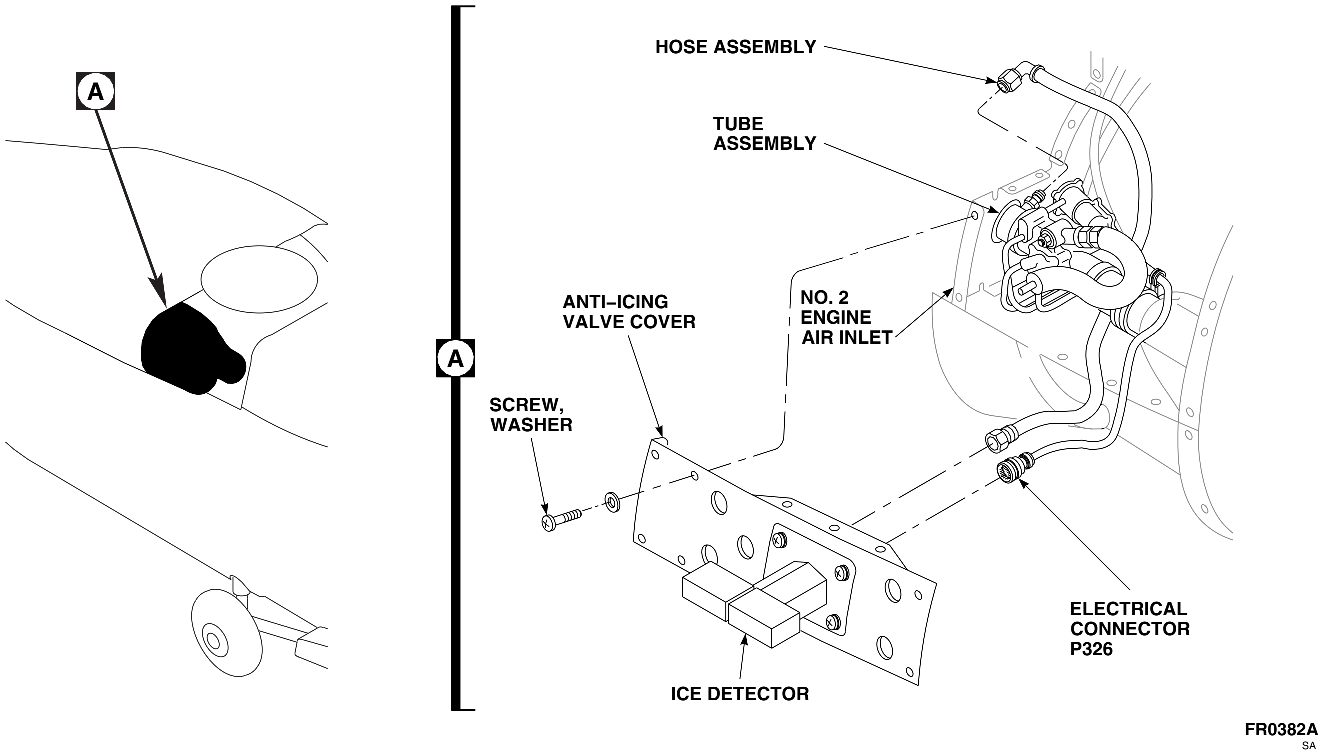


FIGURE 12-4-51-1. REPLACE BLADE DEICE ICE DETECTOR HOSE ASSEMBLY

g. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-52 TAIL ROTOR BLADE DEICE SLIPRING ROTOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 5-4-38
- PARA 6-4-54
- PARA 12-2-5
- PARA 12-3-3

12-4-52.1 Replace Tail Rotor Blade Deice Slipring Rotor.

12-4-52.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect slipring electrical connectors, P611, P612, P613, and P614, from tail rotor blade electrical connectors. Remove and discard packings (Figure 12-4-52-1, Sheet 1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove inboard retention plate (PARA 6-4-54).

e. Remove slipring rotor.

f. Remove screws, washers, and slipring cables from slipring rotor (Figure 12-4-52-1, Sheet 2, Detail C).

12-4-52.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Inspect tail rotor blade deice slipring cables (PARA 12-3-3).

c. Apply sealing compound, Item 292, Appendix D, to base of slipring cable and secure with screws and washers to slipring rotor (Figure 12-4-52-1, Sheet 2, Detail C).

d. Install inboard retention plate (PARA 6-4-54).

e. Check for proper slipring stator to slipring rotor contact as follows:

(1) Check gap between slipring stator and slipring rotor. Check at brush assembly

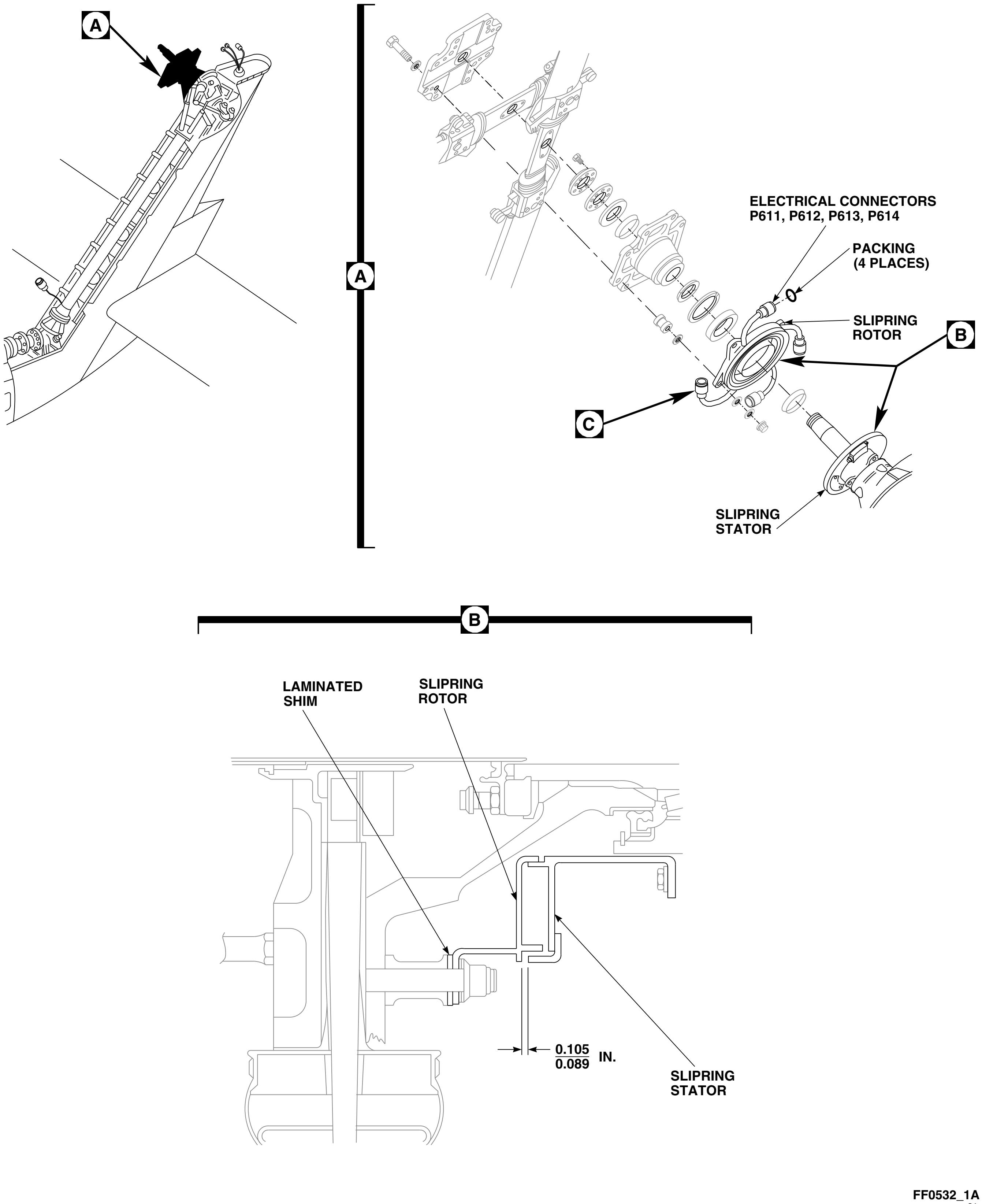


FIGURE 12-4-52-1. REPLACE TAIL ROTOR BLADE DEICE SLIPRING ROTOR (SHEET 1 OF 2)

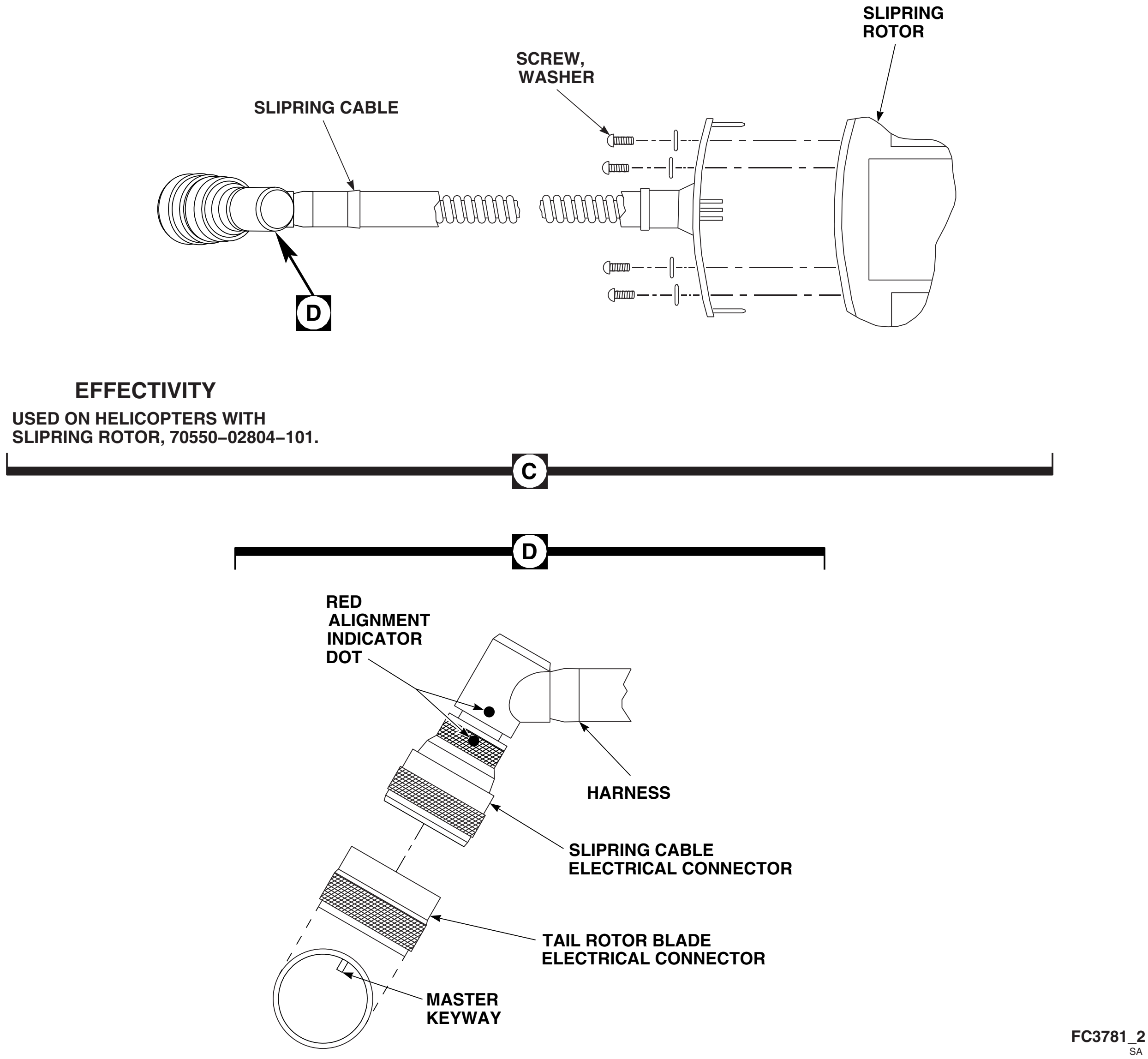


FIGURE 12-4-52-1. REPLACE TAIL ROTOR BLADE DEICE SLIPRING ROTOR (SHEET 2 OF 2)

location on slipring stator and by turning tail rotor until each of the four hard points on slipring rotor line up with brush assembly (Figure 12-4-52-1, Sheet 1, Detail B). **GAP SHALL BE 0.089 - 0.105 INCH.** Peel laminated shim washers as required to obtain gap.

(2) If shim adjustment was required, torque nuts securing retention plates, tail rotor blades, and slipring rotor (PARA 5-4-38).

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Install packings on electrical connectors (Figure 12-4-52-1, Sheet 1, Detail A).

CAUTION

To prevent damaging deice bracket assemblies, align red alignment indicator dots on each slipring cable electrical connector with master keyway on each tail rotor blade electrical connector (Figure 12-4-52-1, Sheet 2, Detail D).

i. Connect slipring cable electrical connectors, P611, P612, P613, P614, to tail rotor blade electrical connectors (Figure 12-4-52-1, Sheet 1, Detail A).

j. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-53 TAIL ROTOR BLADE DEICE SLIPRING STATOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-4-57
- PARA 12-2-5
- PARA 12-4-52

12-4-53.1 Replace Tail Rotor Blade Deice Slipring Stator.

12-4-53.1.1 Remove.

a. Remove tail rotor blade deice slipring rotor (PARA 12-4-52).

b. Disconnect electrical connector, P327 from J327, on brush assembly (Figure 12-4-53-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove bolts and washers securing slipring stator to tail gear box.

CAUTION

Damage to tail rotor pitch change shaft will result if slipring stator is not removed carefully. To prevent surface damage and subsequent corrosion of pitch change shaft, use care when removing slipring stator.

e. Remove slipring stator from tail gear box.

12-4-53.1.2 Inspect.

Inspect tail gear box housing for nicks, scratches, and corrosion. Repair as required (PARA 6-4-57).

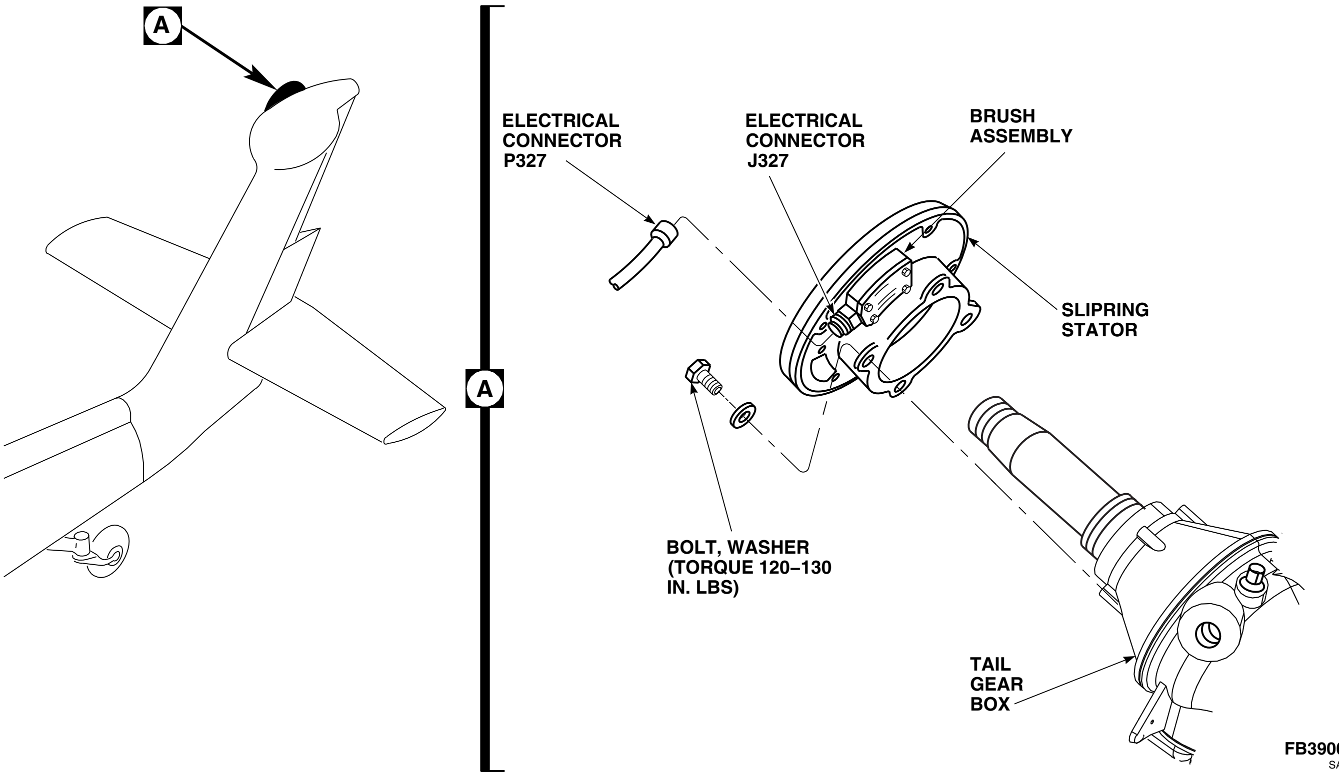


FIGURE 12-4-53-1. REPLACE TAIL ROTOR BLADE DEICE SLIPRING STATOR

12-4-53.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connectors (PARA 1-7-20).

d. Position slipring stator on tail gear box so that electrical connector, P327, reaches electrical connector, J327, on brush assembly (drain hole at bottom) (Figure 12-4-53-1, Detail A).

e. Secure slipring stator with bolts and washers. **TORQUE BOLTS TO 120 - 130 INCH-POUNDS.**

f. Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer to exposed mating area of slipring stator and tail gear box housing.

g. Remove protective caps and plugs from electrical connectors.

h. Inspect and treat electrical connectors (PARA 1-7-20).

i. Connect electrical connector, P327 to J327, on brush assembly.

j. Install tail rotor blade deice slipring rotor (PARA 12-4-52).

k. Perform operational check of blade deicing system (PARA 12-2-5).

12-4-54 TAIL ROTOR BLADE DEICE SLIPRING STATOR BRUSH ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

Materials/Parts

- Lockwire, Item 196, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 12-2-5

12-4-54.1 Replace Tail Rotor Blade Deice Slipring Stator Brush Assembly.

12-4-54.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect electrical connector, P327, from brush assembly (Figure 12-4-54-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Loosen captive bolts and remove cover, brush assembly, and gaskets from slipring rotor.

12-4-54.1.2 Inspect.

Inspect gaskets for damage. **NONE ALLOWED.** If damaged, replace gaskets.

12-4-54.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install brush assembly with gaskets, and cover and secure with captive bolts (Figure 12-4-54-1, Detail A).

c. Using lockwire, Item 196, Appendix D, lockwire captive bolts.

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P327, to brush assembly.

g. Perform operational check of blade deicing system (PARA 12-2-5).

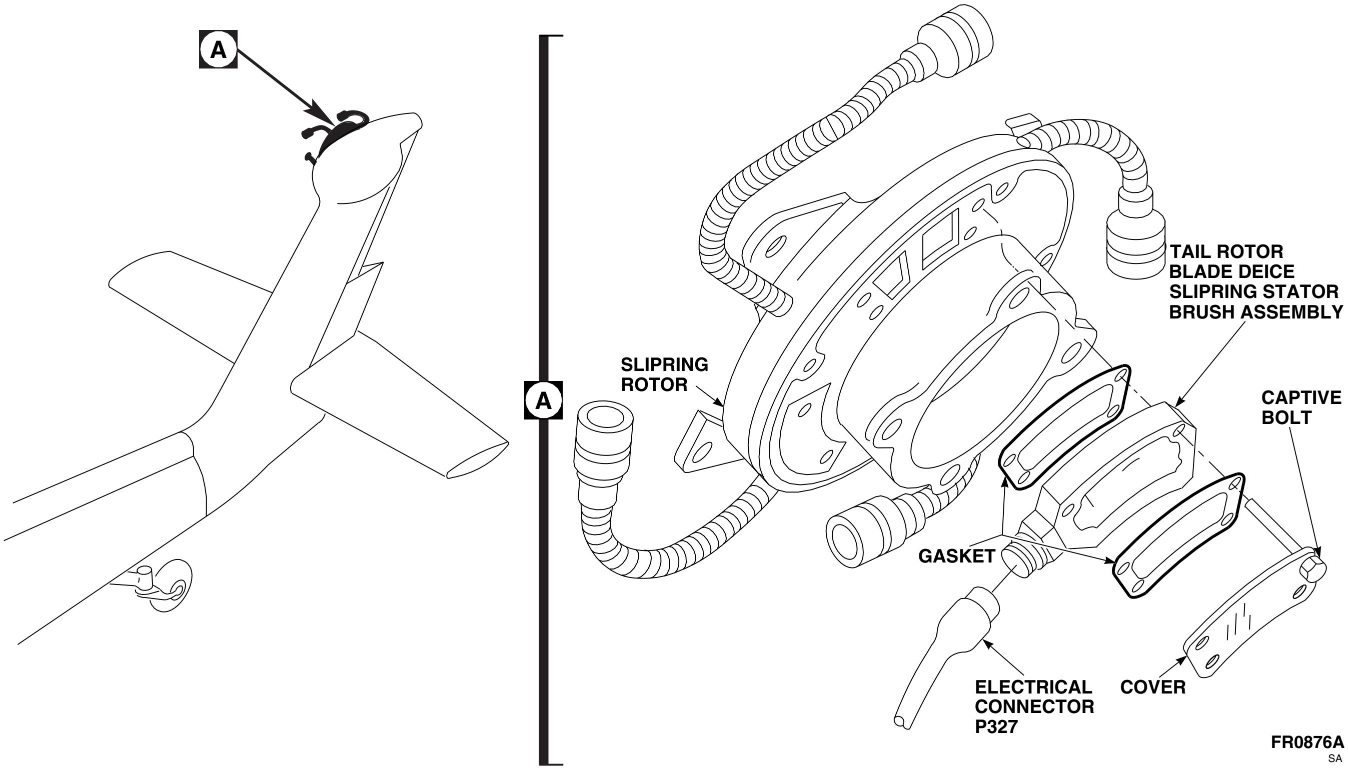


FIGURE 12-4-54-1. REPLACE TAIL ROTOR BLADE DEICE SLIPRING STATOR BRUSH ASSEMBLY

12-4-55 TAIL ROTOR BLADE DEICE SLIPRING STATOR BRUSHES AND RETAINER BOARD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Crimping Tool
- Electrical Repairer's Toolkit
- Multimeter
- Nonmetallic Scraper, Locally-Made
- Protection, Eye and Skin
- Soldering Set
- Spring Tester

Materials/Parts

- Adhesive, Item 43, Appendix D
- Adhesive, Item 64, Appendix D
- Brush, Item 82, Appendix D
- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 12-4-54

12-4-55.1 Replace Tail Rotor Blade Deice Slipring Stator Brushes and Retainer Board.

NOTE

Replace all brush pairs. Replacement of all brush pairs is the same, except as noted.

12-4-55.1.1 Remove.

a. Remove tail rotor blade deice slipring stator brush box (PARA 12-4-54).

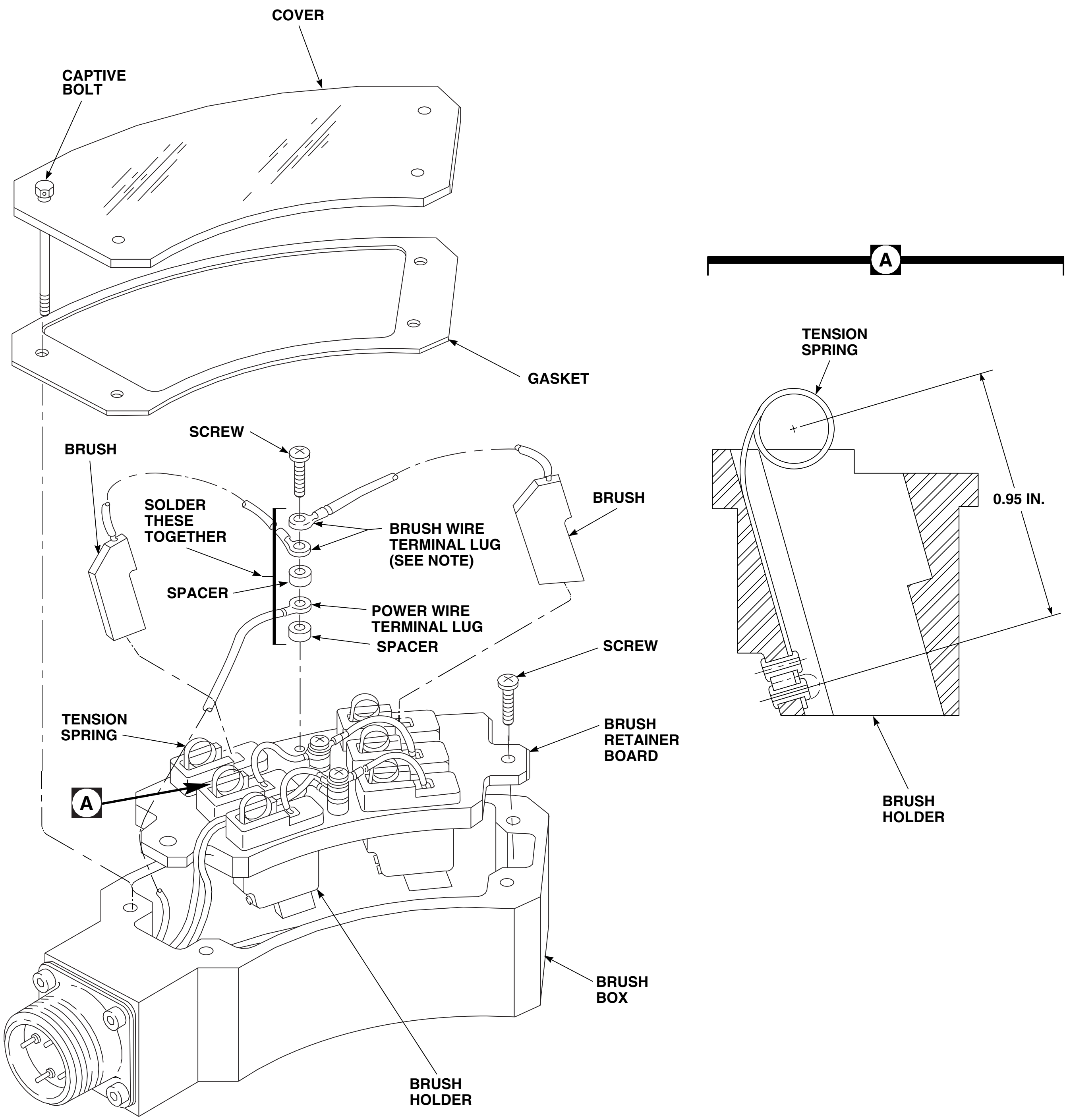
b. Loosen captive bolts and remove cover and gasket from brush assembly (Figure 12-4-55-1).

c. Remove screws holding brush retainer board to brush box.

d. Carefully pull brush retainer board up to access terminal screws.

e. Tag white power wires, noting brush pair position.

f. Remove wire terminal screws.



NOTE

IF BRUSH WIRES DO NOT HAVE TERMINAL LUGS, USE CRIMPING TOOL TO ATTACH LUGS TO WIRES.

FO0434C
SA

FIGURE 12-4-55-1. REPLACE TAIL ROTOR BLADE DEICE SLIPRING STATOR BRUSHES AND RETAINER BOARD

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

g. Using soldering set, unsolder brush wire terminal lugs and power wire terminal lug from spacers.

h. Carefully pull tension springs up and remove brush pairs.

i. Remove brush retainer board.

j. Clean brush retainer board and brush assembly with methyl ethyl ketone, Item 217, Appendix D.

12-4-55.1.2 Inspect.

a. Inspect tension springs for wear with spring tester (Figure 12-4-55-1, Detail A). **SPRING TENSION MUST BE BETWEEN 4.8 AND 5.8 OUNCES WHEN EXTENDED 0.95-INCH FROM LOWER RIVET.** If spring tension is not within limits, perform the following:

(1) Remove retaining adhesive with methyl ethyl ketone, Item 217, Appendix D, cheesecloth, Item 90, Appendix D, and nonmetallic scraper. Remove brush holder from retainer board.

(2) Install brush holder in brush retainer board with adhesive, Item 43, Appendix D, per manufacturer's instructions.

12-4-55.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Apply a coat of adhesive, Item 43, Appendix D, around any brush holders that are loose in brush retainer board.

c. If brush wire does not have terminal lug installed, perform the following:

(1) Strip insulation from end of brush wire.

(2) Using brush, Item 82, Appendix D, brush ends of brush wire.

(3) Using crimping tool, secure terminal lugs to brush wire.

CAUTION

To avoid damage to tension springs, do not apply too much pressure to springs while sliding brushes into brush holders.

d. Carefully pull tension springs up and slide brushes into brush holders. Make sure brush wire is opposite tension spring (Figure 12-4-55-1).

e. Install brush wire terminal lug, spacer, power wire terminal lug, and spacer on brush retainer board with terminal screw. Remove tags.

f. Make sure wires do not interfere with free movement of brushes. Reposition wires as required.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

g. Using soldering set, solder, Item 313, Appendix D, terminal lugs and spacers together.

h. Apply a coat of adhesive, Item 64, Appendix D, to mounting screws and terminal lugs, to prevent screws from working loose.

i. Using multimeter, check for continuity between pins of brush assembly electrical connector and each brush of brush pair.

j. Using multimeter, check for infinite resistance between brush pairs.

k. Position brush retainer board in brush assembly and fasten with screws.

l. Install gasket.

m. Position cover on brush assembly and secure with captive bolts.

n. Install tail rotor blade deice slipring stator brush assembly (PARA 12-4-54).

12-4-56 TAIL ROTOR BLADE DEICE SLIPRING STATOR BRUSH ASSEMBLY ELECTRICAL CONNECTOR, J327.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Insert/Extract Tool

Materials/Parts

- Lockwire, Item 196, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 12-4-54

12-4-56.1 Replace Tail Rotor Blade Deice Slipring Stator Brush Assembly Electrical Connector, J327.

12-4-56.1.1 Remove.

a. Remove tail rotor blade deice slipring stator brush assembly (PARA 12-4-54).

b. Remove screws securing electrical connector, J327, to brush assembly (Figure 12-4-56-1).

c. Using insert/extract tool, remove and tag wires from electrical connector, J327.

d. Remove electrical connector, J327, and gasket from brush assembly.

12-4-56.1.2 Inspect/Clean.

a. Inspect gasket for damage. **NONE ALLOWED.** If damaged, replace gasket.

b. Clean electrical connector gasket surface with methyl ethyl ketone, Item 217, Appendix D.

12-4-56.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install gasket on electrical connector, J327 (Figure 12-4-56-1).

c. Using insert/extract tool, insert tagged wires into electrical connector, J327. Remove tags.

d. Install electrical connector, J327, on brush assembly and secure with screws.

e. Using lockwire, Item 196, Appendix D, lockwire screws.

f. Install tail rotor blade deice slipring brush assembly (PARA 12-4-54).

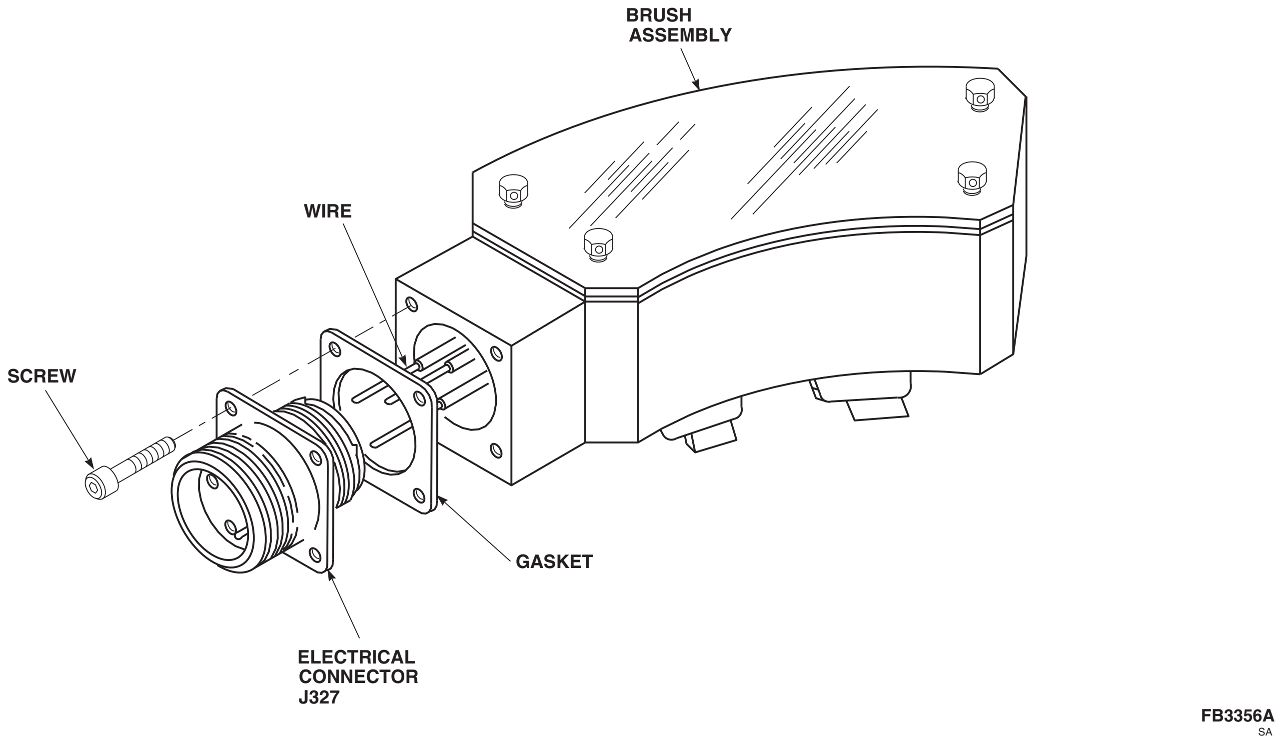


FIGURE 12-4-56-1. REPLACE TAIL ROTOR BLADE DEICE SLIPRING STATOR BRUSH ASSEMBLY ELECTRICAL CONNECTOR, J327

12-4-57 CARGO HOOK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicating Scale
- Electrical Repairer's Toolkit
- Fluorescent Penetrant Inspection Kit
- Shorting Jumper
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- ASTM E1417
- PARA 1-6-16
- PARA 1-7-20
- PARA 1-8-1
- PARA 12-2-6
- PARA 12-4-58
- PARA 12-4-59
- PARA 12-4-61

12-4-57.1 Replace Cargo Hook.

12-4-57.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open cargo hook door.

c. From under helicopter, remove screws and washers attaching access cover to helicopter. Remove access cover (Figure 12-4-57-1, Detail A).

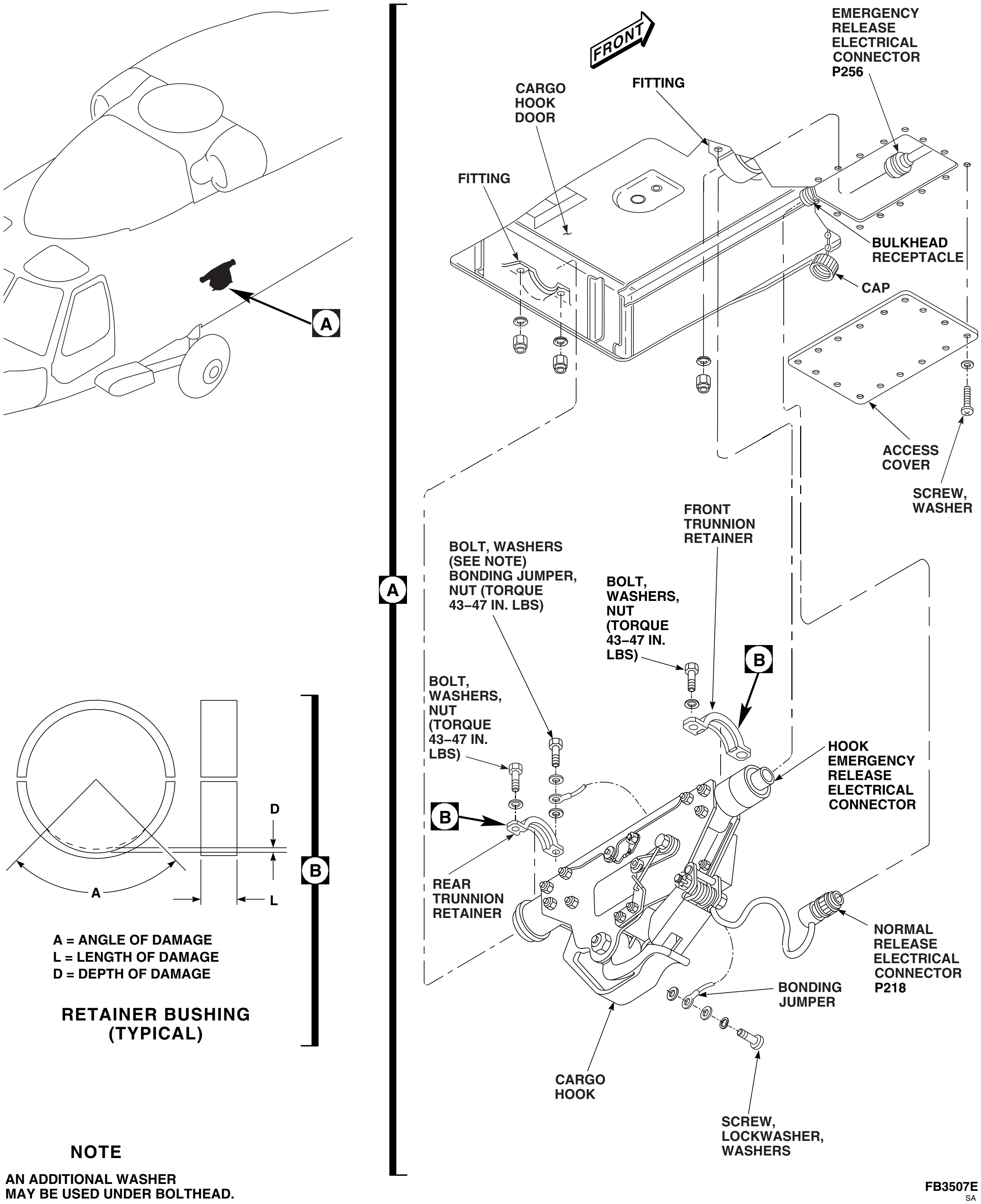


FIGURE 12-4-57-1. REPLACE CARGO HOOK

WARNING

- Injury to personnel will result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies. Short circuit explosive cartridge connector pins to prevent accidental firing.

- Injury to personnel and damage to equipment will result if explosive cartridges are inadvertently activated. All explosive cartridges shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.

d. Disconnect emergency release electrical connector, P256.

e. Install protective caps and plugs on electrical connectors.

f. Install shorting jumper on hook emergency release electrical connector (squib).

g. Disconnect normal release electrical connector, P218, from bulkhead receptacle in cargo hook well.

h. Measure axial play of cargo hook installed. Record measurement if more than 0.060-inch.

i. Measure radial play of cargo hook installed. Record measurement if more than 0.040-inch.

j. Remove bolts, washers, nuts, and trunnion retainers securing cargo hook in fittings. Disconnect bonding jumper from retainer hardware.

k. With aid of assistant, remove cargo hook from fittings.

l. If cargo hook is being replaced, remove screw, lockwasher, washers, and bonding jumper from cargo hook.

m. Close cargo hook door.

12-4-57.1.2 Inspect/Repair.

a. Inspect retainer bushing Teflon surface for wear (Figure 12-4-57-1, Detail B). **THE AMOUNT OF EXPOSED STEEL IN BUSHING INSIDE DIAMETER SHALL NOT EXCEED THE FOLLOWING LIMITS:**

(1) **ANGLE (A) UP TO 90°, LENGTH (L) 0.16-INCH AND DEPTH (D) 0.005-INCH INTO STEEL.**

(2) **ANGLE (A) UP TO 60°, LENGTH (L) 0.25-INCH AND DEPTH (D) 0.005-INCH INTO STEEL.**

(3) **ANGLE (A) UP TO 30°, LENGTH (L) 0.50-INCH AND DEPTH (D) 0.005-INCH INTO STEEL.**

(4) **IF AXIAL OR RADIAL PLAY MEASUREMENT WAS GREATER THAN THE ALLOWABLE TOLERANCE, INSPECT RETAINER BUSHING** (PARA 12-4-61).

b. Measure cargo hook trunnions (Figure 12-4-57-2). If trunnions are out-of-tolerance, replace cargo hook.

NOTE

Do not disassemble cargo hook to perform the following inspection.

c. Using fluorescent penetrant inspection kit, inspect exposed area of cargo hook, including trunnions, trunnion retainers, trunnion retaining bolts and nuts, load arm, plate edge, and bolt-hole surface areas for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace cargo hook.

d. Using dial indicating scale, check manual release tension required to open cargo hook. **TENSION SHALL NOT BE MORE THAN 10 POUNDS.** If hook has tight or rough manual release action, replace cargo hook.

e. Inspect cargo hook for internal damage as follows:

(1) Remove top shield quick-release pin screws and shield.

(2) Remove rear dirt shield attaching hardware and shield.

(3) Check cargo hook interior for metal shavings and gouged, nicked, or chipped parts.

(4) Replace cargo hook showing signs of internal damage.

f. Check keeper arm for cracks and wear. **NONE ALLOWED.** If cracks or wear are found, replace keeper arm (PARA 12-4-59).

g. Check keeper spring tension. **SPRING MUST HOLD KEEPER ARM SOLIDLY AGAINST STOP.** If tension is weak, replace keeper spring (PARA 12-4-59).

h. Check trunnion retainers for corrosion. **NONE ALLOWED.** If corrosion exists, inspect and repair as required (PARA 12-4-61).

i. Inspect cargo hook electrical parts as follows:

(1) Check wiring harnesses for cracks or cuts.

(2) Check connectors for loose, bent, or broken pins.

(3) Check connector barrel for damage.

(4) Replace cargo hook having damaged or corroded electrical parts.

j. Check service life of installed explosive cartridge (PARA 1-8-1). **INSTALLED LIFE OF CARTRIDGE MUST NOT BE OUTDATED.** If outdated, replace explosive cartridge (PARA 12-4-58).

12-4-57.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Open cargo hook door.

c. Depreserve replacement cargo hook.

WARNING

- Injury to personnel will result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short circuit explosive cartridge connector pins to prevent accidental firing.
- Injury to personnel and damage to equipment will result if explosive cartridges are inadvertently activated. All explosive cartridges shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.

d. If cargo hook has an explosive cartridge installed, install shorting jumper between connector pins of emergency release electrical connector (squib).

e. If new cargo hook is being installed, install bonding jumper as follows:

(1) Clean area where bonding jumper will be connected with dry-cleaning solvent, Item 124, Appendix D.

(2) Install bonding jumper on cargo hook with screw, lockwasher, and washers. Make sure a washer is installed on each side of bonding jumper terminal, and lockwasher is under screwhead (Figure 12-4-57-1, Detail A).

f. Clean bearing surface of cargo hook trunnions, airframe trunnions, and trunnion retainers with cleaning cloth, Item 102, Appendix D.

g. Install cargo hook in airframe trunnions.

h. Place trunnion retainers over cargo hook ends.

i. Install front trunnion retainer to fitting with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

j. Install bonding jumper on left bolt of rear retainer. Make sure washer is installed on each side of bonding jumper, with washer and additional washer, if required, under bolthead. Install rear trunnion retainer to fitting with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

k. Measure axial play of cargo hook installed. **IF AXIAL PLAY IS MORE THAN 0.060-INCH, MEASURE CARGO HOOK TRUNNIONS** (PARA 12-4-57.1.2).

l. Measure radial play at each bearing journal. **IF RADIAL PLAY IS MORE THAN 0.070-INCH, MEASURE CARGO HOOK TRUNNIONS** (PARA 12-4-57.1.2).

m. Remove protective caps and plugs from electrical connectors.

n. Inspect and treat electrical connectors (PARA 1-7-20).

o. Connect normal release electrical connector, P218, to bulkhead receptacle in cargo hook well.

p. Remove protective caps and plugs from electrical connectors.

q. Inspect and treat electrical connectors (PARA 1-7-20).

r. Remove shorting jumper on hook emergency release electrical connector (squib).

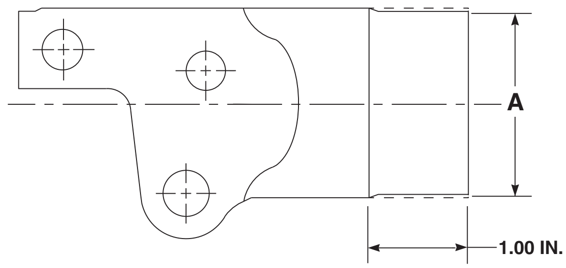
s. Connect emergency release electrical connector, P256, to hook emergency release electrical connector (squib).

t. Make sure area is clean and free of foreign material.

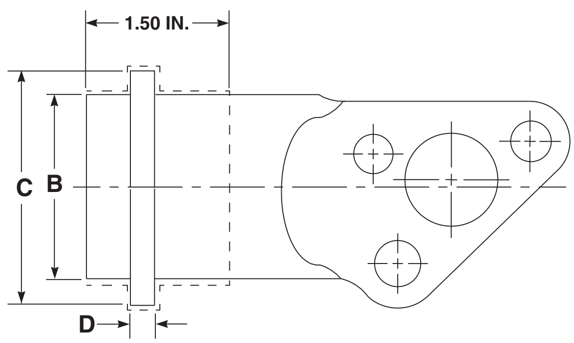
u. Install access cover on underside of helicopter fuselage with screws and washers.

v. Close cargo hook door.

w. Perform operational check of cargo hook (PARA 12-2-6).



FRONT TRUNNION



REAR TRUNNION

ZONE	INSPECTION TYPE	INSPECTION METHOD	MINIMUM DIMENSION (INCHES)
A	SERVICEABILITY	MEASURE	1.970
B	SERVICEABILITY	MEASURE	1.970
C	SERVICEABILITY	MEASURE	2.490
D	SERVICEABILITY	MEASURE	0.250

FB4594
SA

FIGURE 12-4-57-2. INSPECT TRUNNIONS

12-4-58 CARGO HOOK EXPLOSIVE CARTRIDGE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Shorting Jumper
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D
- Marking Pen, Item 211C, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 1-8-1
- PARA 12-2-6
- PARA 12-4-57
- PARA 12-4-61

12-4-58.1 Replace Cargo Hook Explosive Cartridge.

12-4-58.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open cargo hook door.

c. From under helicopter, remove screws and washers attaching access cover to helicopter. Remove cover.

WARNING

- Injury to personnel may result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short circuit explosive cartridge connector pins to prevent accidental firing.
- Injury to personnel and damage to equipment will result if explosive cartridges are inadvertently activated. All explosive cartridges shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.

d. Disconnect emergency release electrical connector, P256. Install shorting jumper on hook emergency release electrical connector (squib) (Figure 12-4-58-1).

e. Install protective caps and plugs on electrical connectors.

WARNING

- Injury to personnel may result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short circuit explosive cartridge connector pins to prevent accidental firing.

- Injury to personnel and damage to equipment will result if explosive cartridges are inadvertently activated. All explosive cartridges shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.

f. Remove explosive cartridge (squib) from cargo hook as follows:

(1) Remove quick-release pin from top shield on cargo hook. Remove top shield.

(2) Disconnect electrical connector, J218, from explosive cartridge (squib) (Figure 12-4-58-1, Detail A).

(3) Install protective cap on electrical connector, J218.

(4) Install shorting jumpers on explosive cartridge (squib) between pins 2 and 3, and between pins 1 and 4.

(5) Remove lockwire attaching explosive cartridge (squib) to thruster cartridge.

(6) Unthread explosive cartridge (squib) from thruster cartridge.

g. If explosive cartridge (squib) has been fired, remove and replace entire top shield assembly complete with thruster cartridge.

NOTE

If explosive cartridge (squib) has been fired, load beam of cargo hook will not close until after CAD (top shield) is removed from cargo hook.

h. Dispose of any unfired squib according to local directives and Calendar Retirement Schedule (PARA 1-8-1).

i. If explosive cartridge (squib) has been fired, dispose of cartridge and handle pressure cartridge assembly as expended ammunition.

12-4-58.1.2. Inspect.

Check explosive cartridge electrical connector for corrosion and cracks. Check connector wire harness for cuts, cracks, or holes and clamps for tightness, position, and security. Replace hooks having wiring or connector damage.

12-4-58.1.3. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. From under helicopter, remove screws and washers attaching access cover to helicopter. Remove cover (PARA 12-4-57).

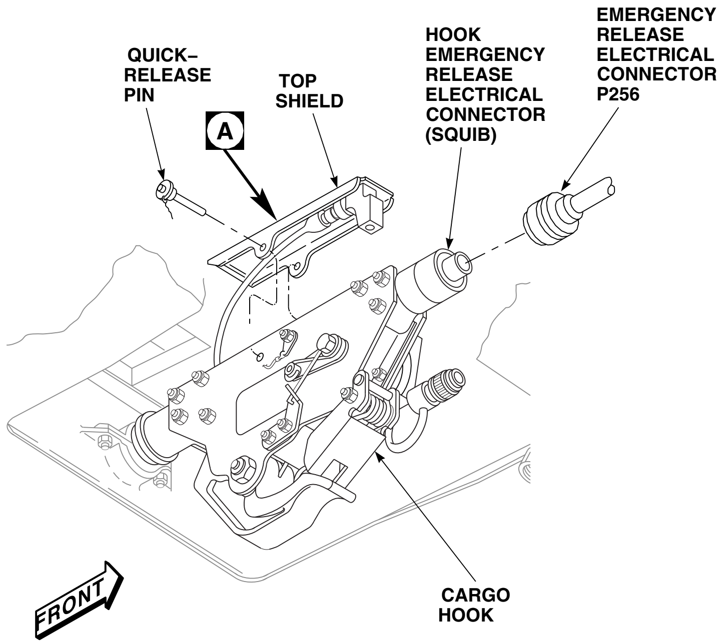
CAUTION

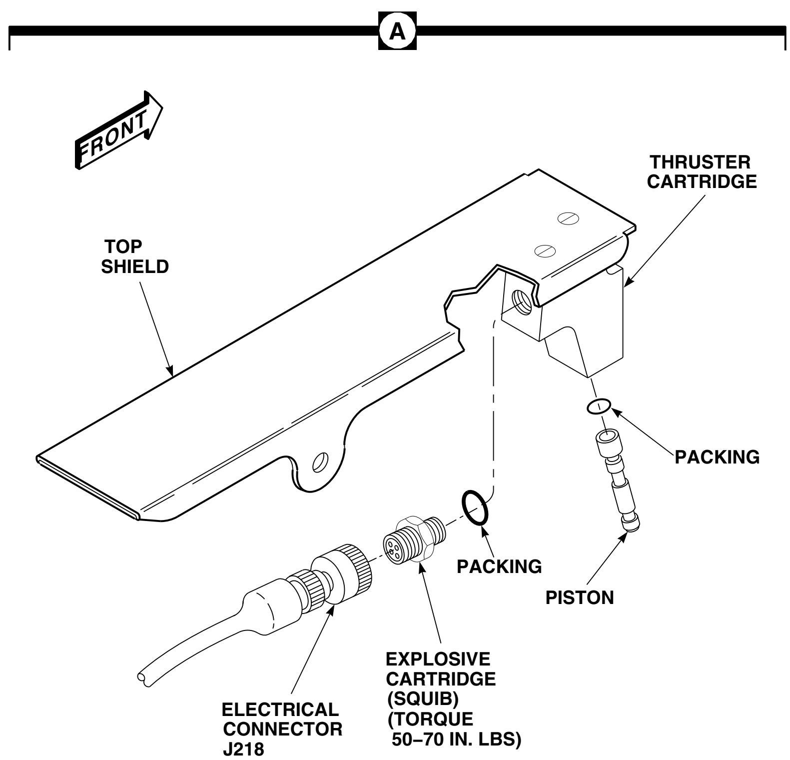
Damage to explosive cartridge will result if part is scribed, scratched, or vibroetched. Damage to corrosion resistant surface will result. Do not scribe, scratch, or vibroetch service life date on explosive cartridge.

c. Using marking pen, Item 211C, Appendix D, mark each side of explosive cartridge (four places) with the service life date (month and year).

d. Check Retirement Schedule of explosive cartridge being installed (PARA 1-8-1).

e. Remove protective caps and plugs from electrical connectors.





FB3354B
SA

FIGURE 12-4-58-1. REPLACE CARGO HOOK EXPLOSIVE CARTRIDGE

f. Inspect and treat electrical connectors (PARA 1-7-20).

WARNING

Injury to personnel may result if explosive cartridge accidentally fires. Explosive cartridge may be fired by radio or radar frequencies and static electricity. Short circuit explosive cartridge connector pins to prevent accidental firing.

g. Install shorting jumpers on explosive cartridge (squib) between pins 2 and 3 and between pins 1 and 4.

h. Install explosive cartridge in cargo hook as follows (Figure 12-4-58-1, Detail A):

(1) Calculate retirement date for explosive cartridge.

(2) Impression stamp cartridge retirement date on top shield data plate box or just below box (PARA 1-8-1).

(3) Replace packing in cartridge housing.

(4) Inspect piston in cartridge housing. **TIP OF PISTON MUST BE FLUSH TO 0.030 INCH PAST FACE OF PLUG.**

(5) Thread explosive cartridge (squib) into thruster cartridge. **TORQUE EXPLOSIVE CARTRIDGE TO 50 - 70 INCH-POUNDS.**

(6) Lockwire, Item 197, Appendix D, explosive cartridge (squib) to thruster cartridge using double twist method.

(7) Remove shorting jumpers on explosive cartridge (squib) and connect electrical connector, J218, to explosive cartridge (squib).

(8) Position top shield on cargo hook and install quick-release pin (Figure 12-4-58-1).

i. Remove shorting jumper on hook emergency release electrical connector. Connect emergency release electrical connector, P256, to hook emergency release electrical connector (squib).

j. Measure axial play of cargo hook installed. **IF AXIAL PLAY IS MORE THAN 0.060 INCH, REPLACE RETAINER BUSHINGS** (PARA 12-4-61).

k. Measure radial play at each bearing journal. **IF RADIAL PLAY IS MORE THAN 0.040 INCH REPLACE RETAINER BUSHINGS** (PARA 12-4-61).

l. Install electrical connector access cover on underside of helicopter fuselage with screws and washers.

m. Make sure area is clean and free of foreign material.

n. From under helicopter, install access cover to helicopter with screws and washers.

o. Close cargo hook door.

p. Perform operational check of cargo hook (PARA 12-2-6).

12-4-59 CARGO HOOK KEEPER AND KEEPER SPRING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 12-2-6

12-4-59.1 Replace Cargo Hook Keeper and Keeper Spring.

12-4-59.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open cargo hook door.

c. Unlock and hold load beam in open position.

CAUTION

Carefully note spring installation before removing spring ends are not the same. Incorrect installation could cause accidental loss of load.

d. While holding keeper open, remove bolt and nut securing keeper and keeper spring on cargo hook. Carefully release tension on keeper spring and remove from keeper (Figure 12-4-59-1, Detail A).

12-4-59.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Unlock and hold load beam in open position.

CAUTION

Install keeper spring carefully. Spring ends are not the same. Incorrect installation could cause accidental loss of load.

c. Place keeper spring in keeper and position on cargo hook (Figure 12-4-59-1, Detail A).

d. Install bolt through keeper, hook, and keeper spring. Install nut on bolt. **TIGHTEN NUT FINGERTIGHT.** Install cotter pin.

e. Make sure keeper snaps across hook mouth to keeper stop.

f. Perform operational check of cargo hook (PARA 12-2-6).

g. Make sure area is clean and free of foreign material.

h. Close cargo hook door.

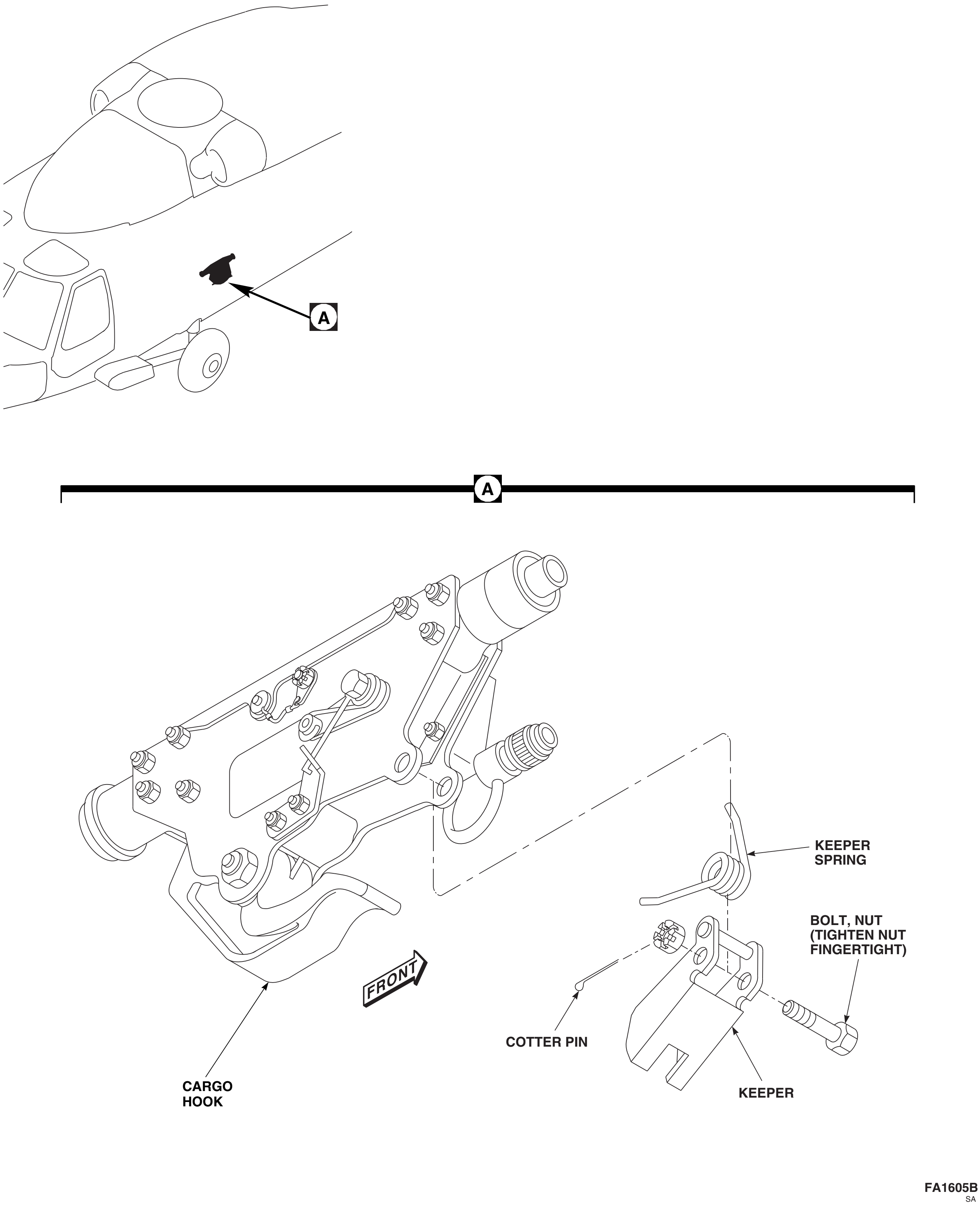


FIGURE 12-4-59-1. REPLACE CARGO HOOK KEEPER AND KEEPER SPRING

12-4-60 CARGO HOOK MANUAL RELEASE HANDLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 289, Appendix D
- Sealing Primer, Item 307, Appendix D

References

- Appendix D
- PARA 1-6-16

12-4-60.1 Replace Cargo Hook Manual Release Handle.

12-4-60.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open cargo hook door.

c. Mark lever position and remove nut from shaft (Figure 12-4-60-1, Detail A).

d. Remove spring and manual release handle.

e. Remove nuts and spring bracket from bolts on cargo hook.

12-4-60.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Clean shaft and nut threads with machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Wipe parts dry.

c. Apply sealing compound, Item 289, Appendix D, to nut and shaft threads. Allow 5 minutes drying time.

d. Install spring bracket over bolts on cargo hook and secure with nuts. **TORQUE NUTS TO 85 - 95 INCH-POUNDS** (Figure 12-4-60-1, Detail A).

e. Install manual release handle on shaft. Make sure manual release handle is in its original position.

f. Install spring as follows (Figure 12-4-60-1, Detail A and Detail B):

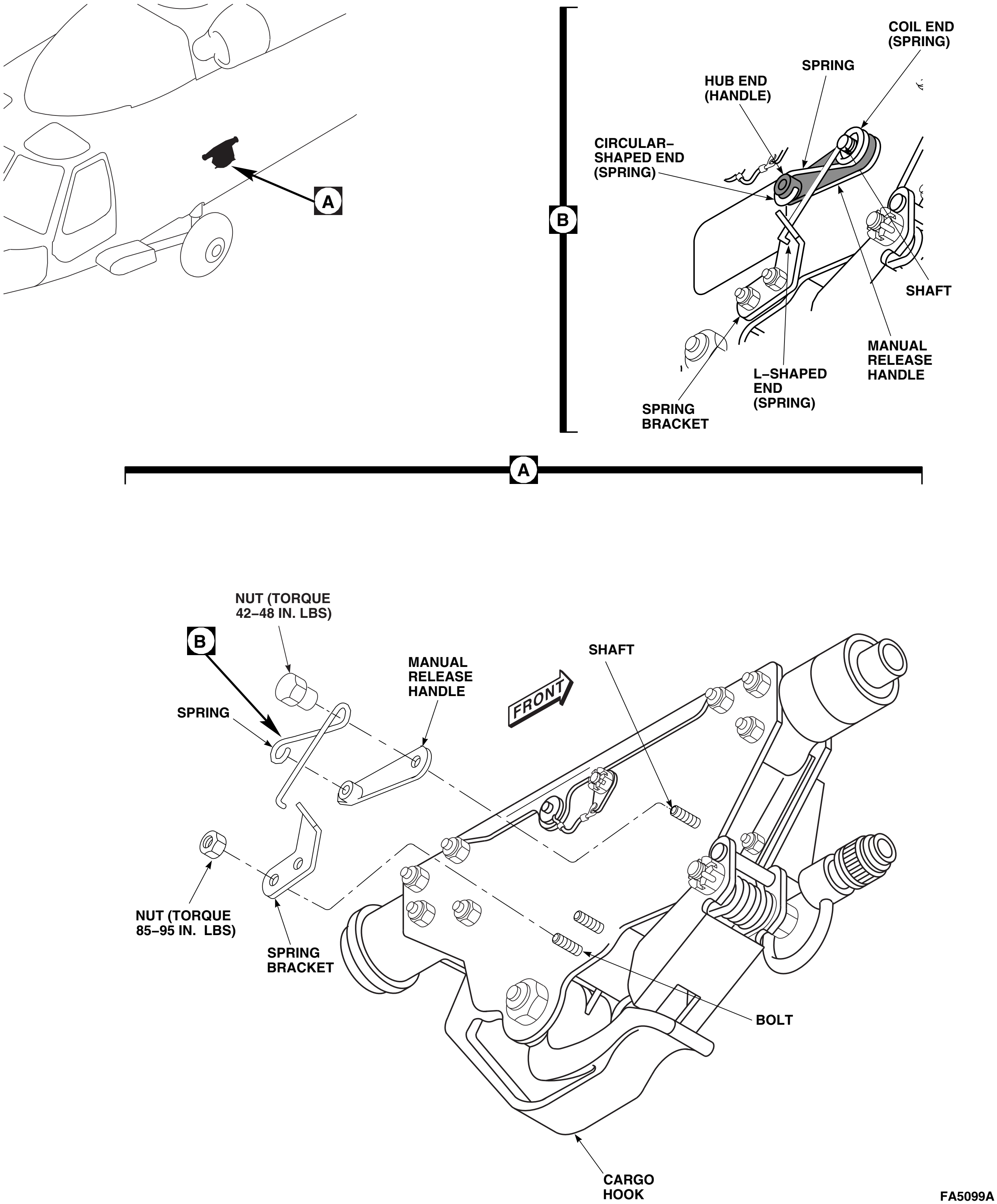


FIGURE 12-4-60-1. REPLACE CARGO HOOK MANUAL RELEASE HANDLE

WARNING

Injury to personnel and damage to equipment may result if spring is missing or installed incorrectly. Missing spring or failure to correctly install spring may result in uncommanded release of the external load.

(1) Slide L-shaped end of spring into hole in spring bracket.

(2) Move circular-shaped end of spring up and under L-shaped leg and position on hub end of manual release handle.

(3) Position coil end of spring over shaft.

g. Apply sealing primer, Item 307, Appendix D, to nut threads.

h. Install nut on manual release handle shaft. **TORQUE NUT TO 42 - 48 INCH-POUNDS** (Figure 12-4-60-1, Detail A).

i. Test hook manual release.

j. Make sure area is clean and free of foreign material.

k. Close cargo hook door.

12-4-61 CARGO HOOK SUPPORT FITTING AND RETAINER BUSHINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Cargo Hook Bushing Installation Block, Locally-Made, Figure H-194, Appendix H
- Heat Gun
- Mallet
- Nonmetallic Drift
- Platform, Plywood or Plastic (or equivalent), 20 x 20 in., Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Adhesive, Item 32, Appendix D
- Antiseize Compound, Item 76, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Removing Compound, Item 116, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry Ice, Item 125, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 231, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-16
- PARA 12-4-57

12-4-61.1 Replace Cargo Hook Support Fitting and Retainer Bushings.

12-4-61.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove cargo hook (PARA 12-4-57).

c. Cut a piece of plywood, plastic, or equivalent, 20 x 20 inches square as a platform.

d. Position platform in cargo hook well from underneath and support in place.

e. Pack support fittings in dry ice, Item 125, Appendix D, to freeze fittings for bushing removal. Retainers may be packed in a bucket of dry ice (Figure 12-4-61-1).

f. Using nonmetallic drift and mallet, tap bushings out of support fitting or retainer.

g. Remove dry ice and platform.

h. Using paint remover, Item 231, Appendix D, remove adhesive from support fitting holes. Remove paint remover residue using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

12-4-61.1.2. Inspect/Repair.

NOTE

Do all corrosion removal and blending of support fittings with abrasive cloth, Item 8, Appendix D, and abrasive cloth, Item 10, Appendix D.

a. Inspect cargo hook rear support fitting for corrosion and repair as follows (Figure 12-4-61-2):

(1) **SURFACE A. REPAIR SHALL NOT EXCEED 0.030 INCH MAXIMUM DEPTH OVER NO MORE THAN 25% OF SURFACE AREA.**

(2) **SURFACE B. CORROSION DAMAGE SHALL NOT EXCEED 40% OF SURFACE AREA. MINIMUM REMAINING GAGE THICKNESS AFTER REPAIR SHALL NOT EXCEED LESS THAN 0.250 AND 0.185 INCH.**

(3) **SURFACE C. CORROSION DAMAGE PERMITTED OVER 100% OF SURFACE AREA. MINIMUM REMAINING THICKNESS AFTER REPAIR SHALL NOT EXCEED LESS THAN 0.205 AND 0.185 INCH.**

(4) **SURFACE D. REPAIRED AREA SHALL NOT EXCEED 0.030 INCH MAXIMUM DEPTH OVER NO MORE THAN 40% OF SURFACE AREA.**

(5) **SURFACE E. REPAIRED AREA SHALL NOT EXCEED 0.030 INCH MAXIMUM DEPTH OVER NO MORE THAN 20% OF SURFACE AREA.**

b. Inspect cargo hook front support fitting for corrosion and repair as follows:

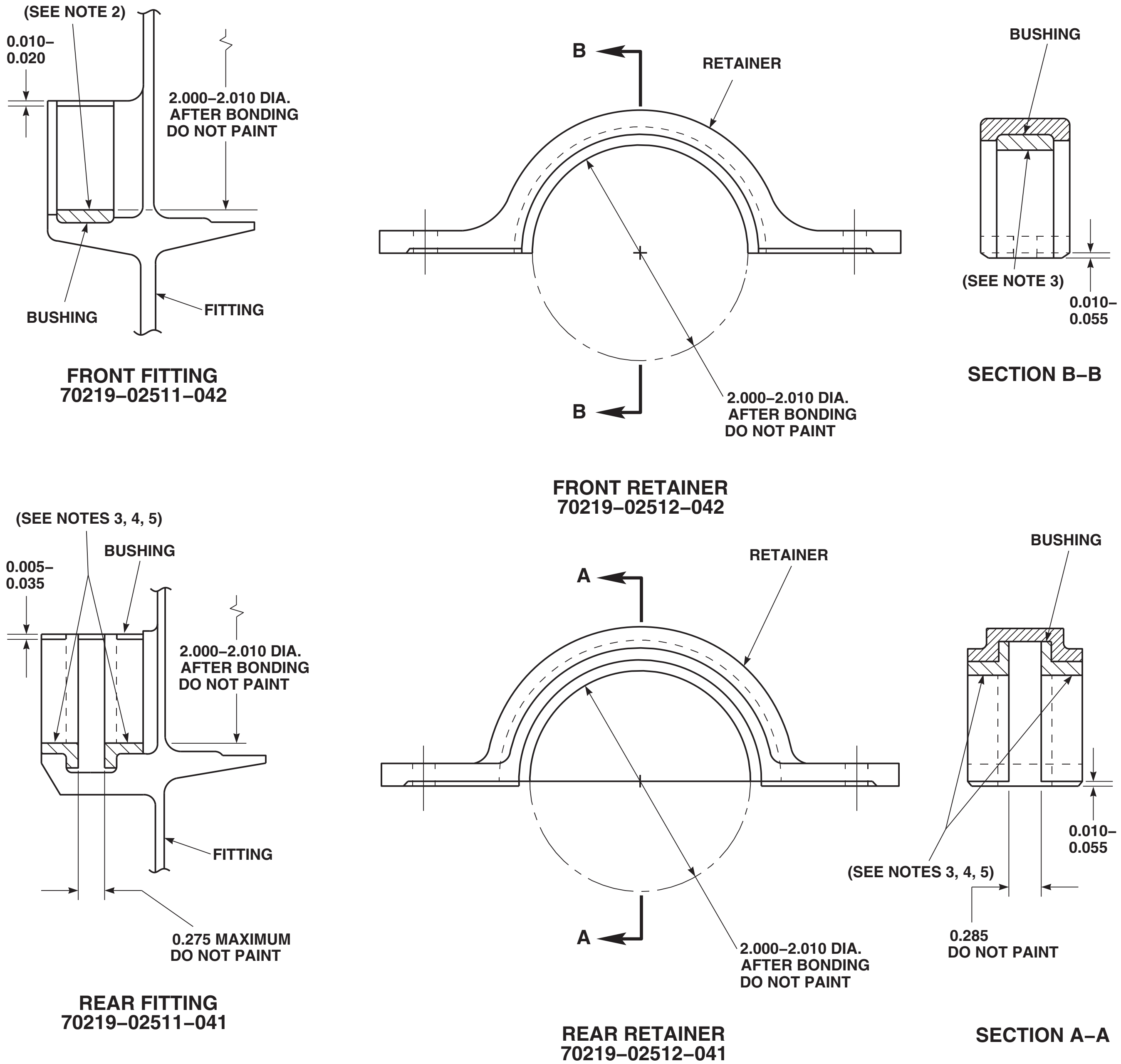
(1) **SURFACE A. MAXIMUM DEPTH OF REPAIRED AREA SHALL NOT EXCEED HEIGHT OF TANGENT POINT ON FILLET RADIUS OF SURFACE B (0.010 - 0.020 INCH ABOVE SURFACE B).**

(2) **SURFACE B. REPAIRED AREA SHALL NOT EXCEED 0.018 INCH DEPTH.**

(3) **SURFACE C. MAXIMUM DEPTH OF REPAIRED AREA SHALL NOT EXCEED HEIGHT OF TANGENT POINT ON FILLET RADIUS OF SURFACE B (0.010 - 0.020 INCH ABOVE SURFACE B).**

(4) **SURFACE D. REPAIRED AREA SHALL NOT EXCEED 0.030 INCH DEPTH AND BLEND TO MATCH FILLET RADIUS (0.010 - 0.020 INCH).**

(5) Inspect repaired areas and make sure minimum dimensions shown in Figure 12-4-61-2 are maintained.

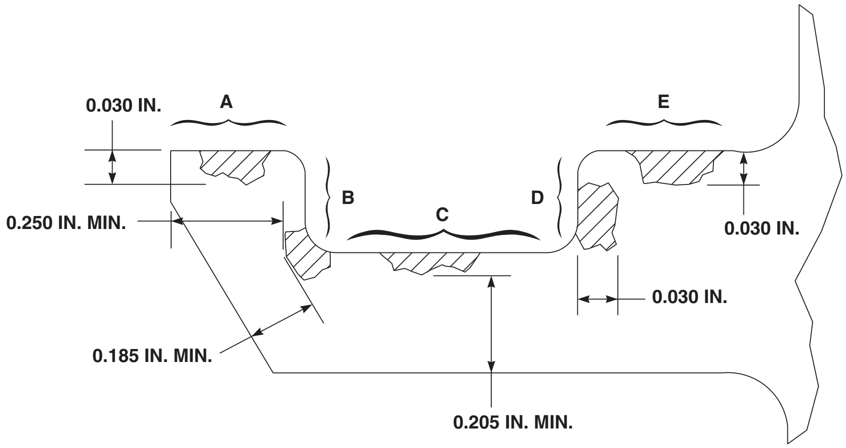


NOTES

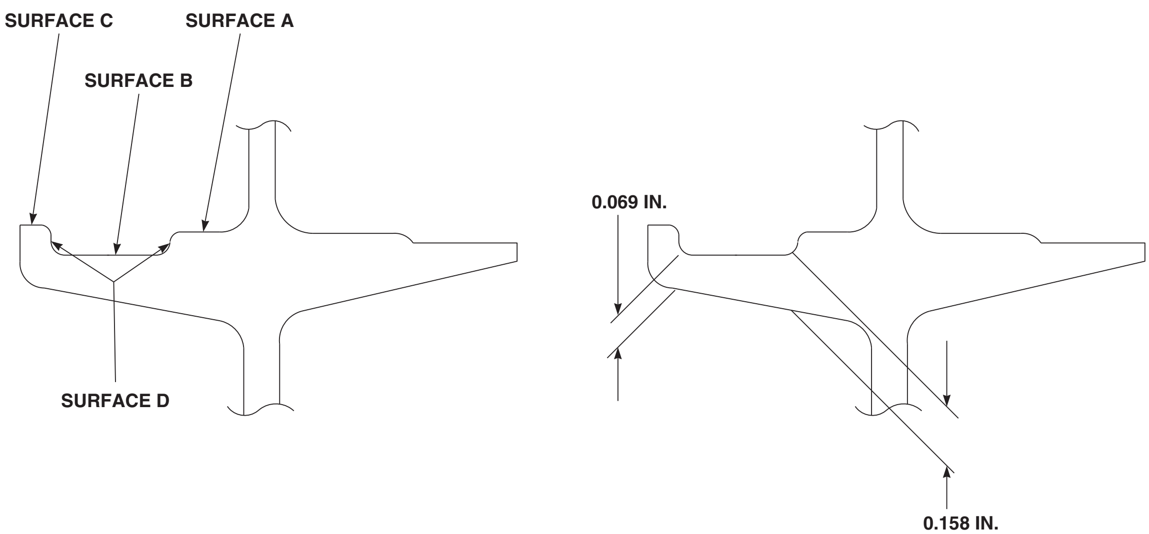
1. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.
2. MAXIMUM ALLOWABLE WEAR IS 0.008 INCH.
3. MAXIMUM ALLOWABLE WEAR IS 0.010 INCH.
4. WEAR ON RADIAL SURFACE OF CARGO HOOK THAT MATES TO FRONT AND REAR FITTING / RETAINER BUSHING IS PERMISSABLE TO MAXIMUM PLAY OF 0.070 INCH.
5. WEAR ON AXIAL SURFACE OF CARGO HOOK THAT MATES TO REAR SUPPORT FITTING / RETAINER BUSHING IS PERMISSABLE TO MAXIMUM PLAY OF 0.060 INCH.

FO0435B
SA

FIGURE 12-4-61-1. REPLACE CARGO HOOK SUPPORT FITTING AND RETAINER BUSHINGS



REAR SUPPORT FITTING



FRONT SUPPORT FITTING

FA2110
SA

FIGURE 12-4-61-2. REPAIR CARGO HOOK SUPPORT FITTINGS

c. Apply corrosion removing compound, Item 116, Appendix D, to repaired areas using acid brush, Item 84, Appendix D. Do not touch up finish in bonded area.

d. Inspect cargo hook front support fitting bushing for wear as follows (Figure 12-4-61-1):

(1) Front support fitting bushing, lower surface, **MAXIMUM ALLOWABLE WEAR IS 0.008 INCH.**

(2) Front support fitting bushing, upper surface, **MAXIMUM ALLOWABLE WEAR IS 0.010 INCH.**

e. Inspect cargo hook rear support fitting bushing for wear. **MAXIMUM ALLOWABLE WEAR IS 0.010 INCH.**

12-4-61.1.3. Install.

a. Install bushings as follows (Figure 12-4-61-1):

(1) Locally make cargo hook bushing installation blocks (Figure H-194, Appendix H).

(2) Coat outside of bushings with adhesive, Item 32, Appendix D.

(3) Install bushings in support fittings.

(4) Install bushings in retainers.

(5) Wipe off any excess adhesive.

(6) Lightly coat outside diameter of cargo hook bushing installation blocks with antiseize compound, Item 76, Appendix D.

(7) Install cargo hook bushing installation blocks between support fittings and retainers. Install bolts, washers, and nuts. **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**

(8) Wipe off any excess adhesive.

(9) Touch up remaining repaired surfaces with corrosion resistant coating, Item 118, Appendix D.

(10) Cure adhesive at 70° - 80°F (22° - 24°C) for 24 hours or at 140° - 160°F (60° - 71°C) for 1 hour using heat gun.

(11) Remove bolts, washers, nuts, and cargo hook bushing installation blocks from support fittings and retainers.

(12) Clean bushing surfaces of cargo hook trunnions, trunnion retainers, and bushing installation block(s) with cleaning cloth, Item 102, Appendix D.

b. After bonding replacement bushings, check inside diameters of bushings.

c. Fillet seal around edges of bushings with sealing compound, Item 292, Appendix D.

d. Surface treat remaining areas with sealing compound, Item 301A, Appendix D. Limit build-up to prevent contact with cargo hook when installed.

e. Install cargo hook (PARA 12-4-57).

12-4-62 CREWMAN'S CARGO HOOK PENDANT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

References

- PARA 1-7-20
- PARA 12-2-7

12-4-62.1 Replace Crewman's Cargo Hook Pendant.

12-4-62.1.1. Remove.

a. Disconnect electrical connector, P267 from J267 (Figure 12-4-62-1, Detail B).

b. Install protective cap on electrical connector, P267.

c. Install cap on electrical connector, J267.

d. Install crewman's cargo hook pendant behind copilot's seat.

12-4-62.1.2. Install.

a. Remove crewman's cargo hook pendant from behind copilot's seat.

b. Remove protective cap from electrical connector, P267 (Figure 12-4-62-1, Detail B).

c. Remove cap from electrical connector, J267, in cabin ceiling.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P267 to J267 (Figure 12-4-62-1, Detail B).

f. Perform operational check of crewman's cargo hook pendant (PARA 12-2-7).

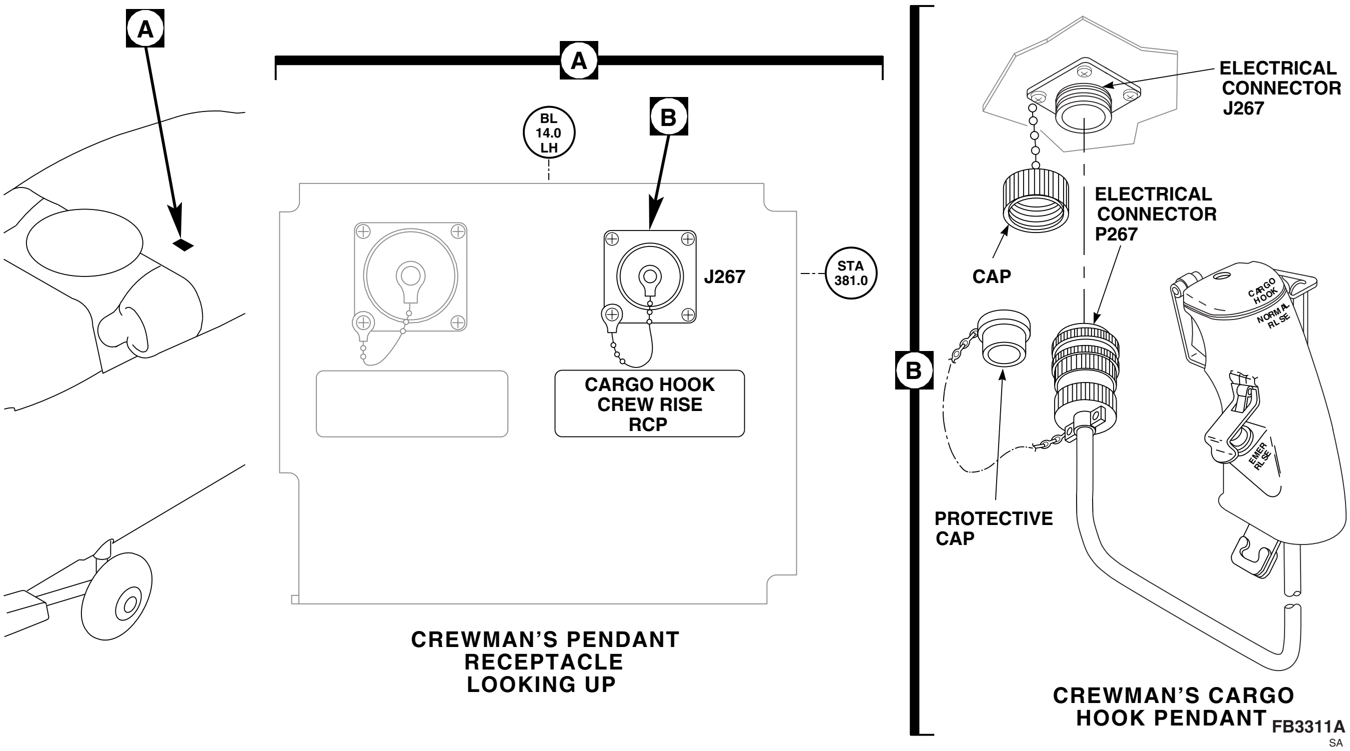


FIGURE 12-4-62-1. REPLACE CREWMAN'S CARGO HOOK PENDANT

12-4-63 CREWMAN'S CARGO HOOK PENDANT SWITCHES AND GUARDS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Conduit Pliers
- Electric Heater Gun
- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set
- Twist Drill, No. 32

Materials/Parts

- Adhesive, Item 36, Appendix D
- Insulating Compound, Item 158, Appendix D
- Sealing Compound, Item 290, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 12-2-7
- PARA 12-4-62

12-4-63.1 Replace Crewman's Cargo Hook Pendant Switches and Guards.

NOTE

Replacement of NORMAL RLSE and EMER RLSE switches is the same, except as noted.

12-4-63.1.1 Remove.

a. If installed, remove crewman's cargo hook pendant (PARA 12-4-62).

b. Using pointed tool from toolkit, remove sealing compound from screw and pin cavities.

c. If removing NORMAL RLSE switch, remove screws securing cover and switch guard and strap bracket assembly to pendant (Figure 12-4-63-1). Remove parts.

d. If removing EMER RLSE switch, use No. 32 twist drill to drive out pin securing EMER RLSE switch guard and spring to pendant. Remove switch guard and spring.

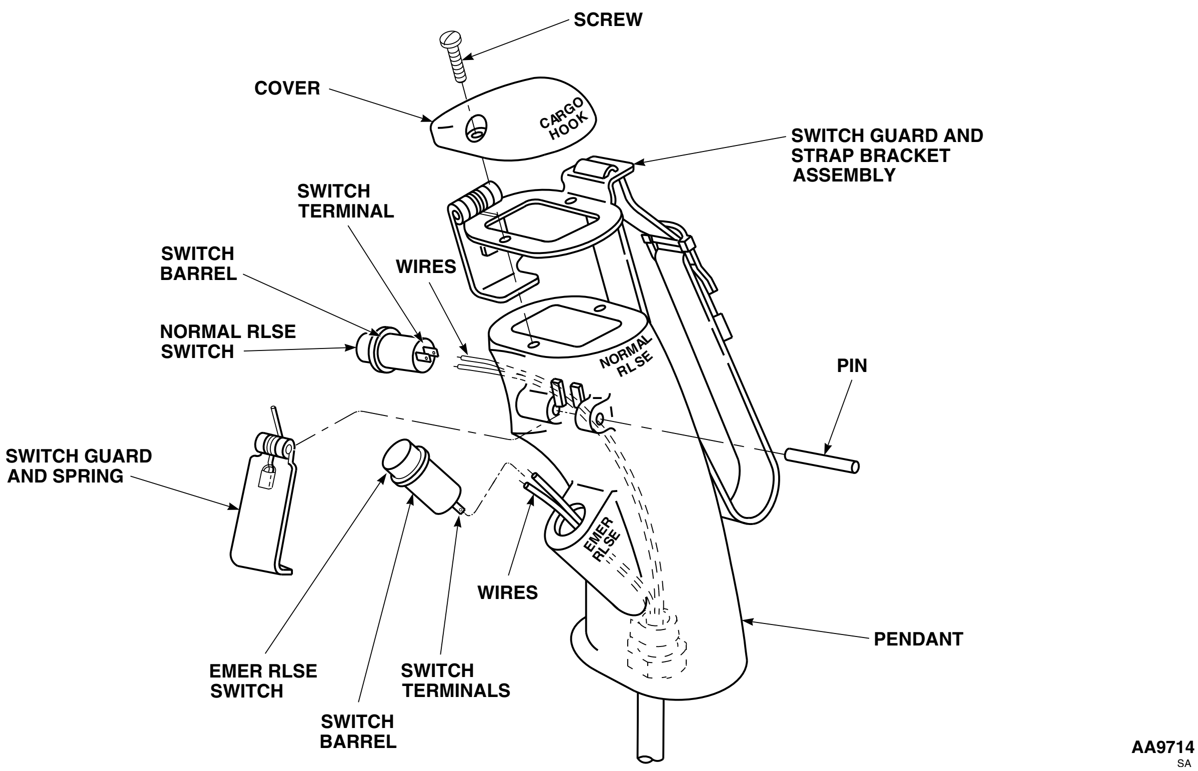


FIGURE 12-4-63-1. REPLACE CREWMAN'S CARGO HOOK PENDANT SWITCHES AND GUARDS

CAUTION

Use extreme care when heating area around switches to avoid permanent damage to components and to prevent disfigurement of molded plastic.

e. Using electric heater gun, heat area of switch until switch can be removed.

f. Using conduit pliers, carefully pull switch from pendant.

g. Remove switch only far enough to expose switch terminals at end of switch barrel.

h. Tag switch wires to identify switch connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

i. Using soldering set, unsolder wires from switch.

12-4-63.1.2 Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

a. Using soldering set and solder, Item 313, Appendix D, solder wires to switch (Figure 12-4-63-1). Untag switch wires.

b. Coat all exposed solder terminals with insulating compound, Item 158, Appendix D.

c. Coat barrel of switch with adhesive, Item 36, Appendix D.

d. Carefully press switch into pendant, routing wires to prevent interference.

e. If installing EMER RLSE switch, position switch guard and spring on pendant and secure with pin.

NOTE

If nylon screwheads are damaged, discard and replace screw.

f. If installing NORMAL RLSE switch, position switch guard and strap bracket assembly and cover on pendant and secure with screws.

g. Fill screw and pin holes with sealing compound, Item 290, Appendix D.

h. Perform operational check of crewman's cargo hook pendant (PARA 12-2-7).

i. Stow or install crewman's cargo hook pendant as required (PARA 12-4-62).

12-4-64 CREWMAN'S CARGO HOOK PENDANT CABLE ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Insert/Extract Tool

Materials/Parts

- Neoprene Sleeve, Item 219A, Appendix D
- Tags

References

- Appendix D
- PARA 12-2-7
- PARA 12-4-63

12-4-64.1 Replace Crewman's Cargo Hook Pendant Cable Assembly.

12-4-64.1.1. Remove.

a. Remove switches and switch guards (PARA 12-4-63).

b. Loosen screws on strain relief clamp (Figure 12-4-64-1).

c. Unscrew strain relief clamp from connector backshell.

d. Slide strain relief clamp along cord, to expose wire connections in connector.

e. Using insert/extract tool, tag and remove wires from connector.

f. Remove strain relief bushing and cable from bottom of pendant.

NOTE

If strain relief bushing is damaged during removal, replace it.

g. Remove strain relief bushing from cable.

12-4-64.1.2. Install.

a. Identify and tag cable assembly wires as follows:

Wire Code	Switch-Terminal	Connector Pin
C-20	EMER RLSE -1	C (SHIELDED GRD TO A)
E-20	EMER RLSE - 2	E
B-20	NORMAL RLSE - 1	B
F-20	NORMAL RLSE - 2	F

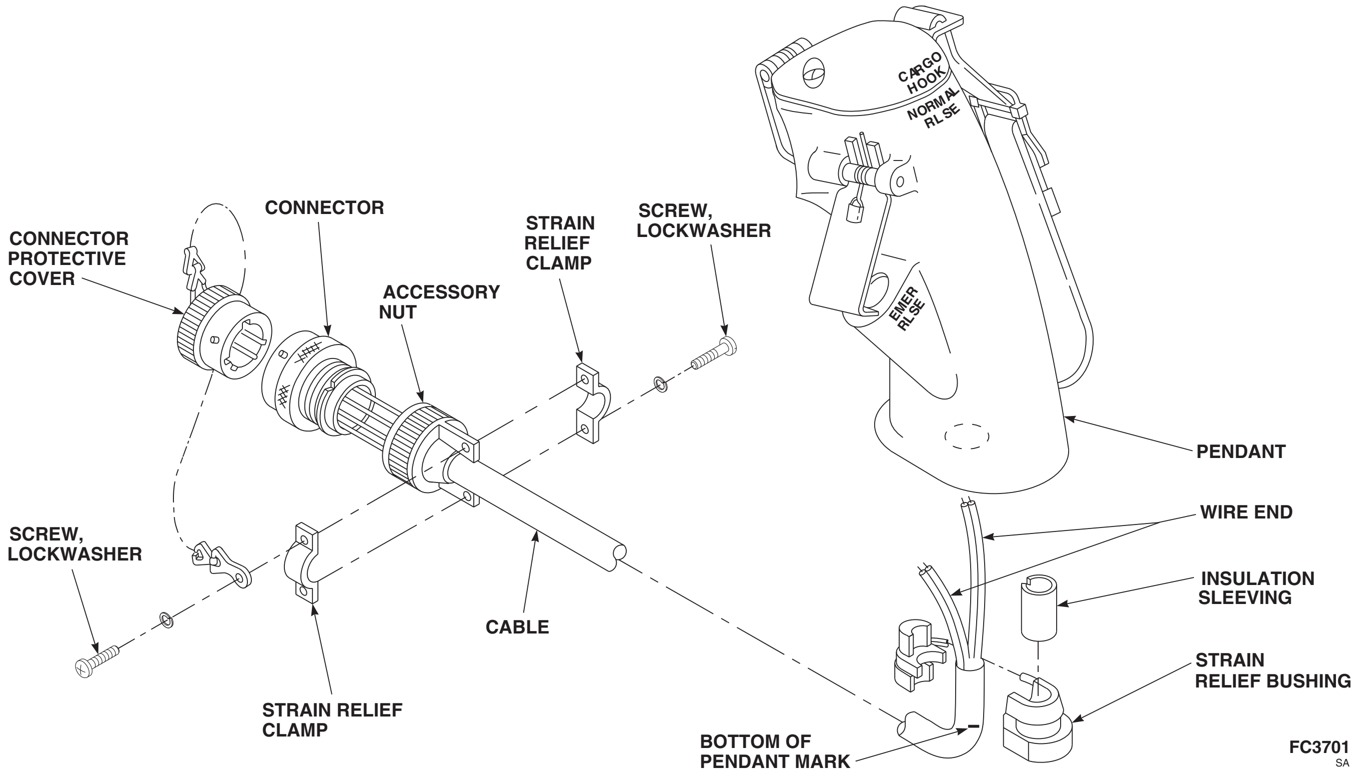


FIGURE 12-4-64-1. REPLACE CREWMAN'S CARGO HOOK PENDANT CABLE ASSEMBLY

b. Insert replacement cable through bottom of pendant (Figure 12-4-64-1).

c. Route identified wires through corresponding switch openings (NORMAL and EMER RLSE).

d. Pull wires through proper switch openings and extend them far enough past openings to allow soldering to switches.

e. Mark cable at bottom of pendant.

f. Remove cable from pendant.

g. Cut 1.5 inches of neoprene sleeving, Item 219A, Appendix D, and slit lengthwise.

h. Wrap neoprene sleeve, Item 219A, Appendix D, around cable, centered on cable mark.

i. Install strain relief bushing around cable and sleeving, leaving sleeving exposed on each end of strain relief bushing.

j. Route wires through proper switch holes while installing cable and strain relief bushing in bottom of pendant.

k. Install cable and strain relief bushing through hole in bottom of pendant, and snap into place.

l. Pull on cable to make certain that strain relief bushing and cable are properly installed.

m. Install switches and switch guards (PARA 12-4-63).

n. Slide connector strain relief clamp along cable, with threaded end towards connector backshell.

o. Using insert/extract tool, connect tagged wires to connector backshell. Untag wires.

p. Screw strain relief clamp onto connector backshell.

q. **TIGHTEN SCREWS ON STRAIN RELIEF CLAMP.**

r. Hold connector and pull cable to make certain strain relief clamp is properly attached to cable.

s. Perform operational check of crewman's cargo hook pendant assembly (PARA 12-2-7).

t. Screw connector protective cover on connector.

SECTION 12-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

12-5-1 WINDSHIELD WIPER MOTOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Insert/Extract Tool
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 149, Appendix D
- Lockwire, Item 197, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D

12-5-1.1 Repair Windshield Wiper Motor (BENCH).

12-5-1.1.1 Disassemble.

a. Remove screws, cap, and gear case housing from bearing plate (Figure 12-5-1-1).

b. Remove cam and idler gear from bearing plate.

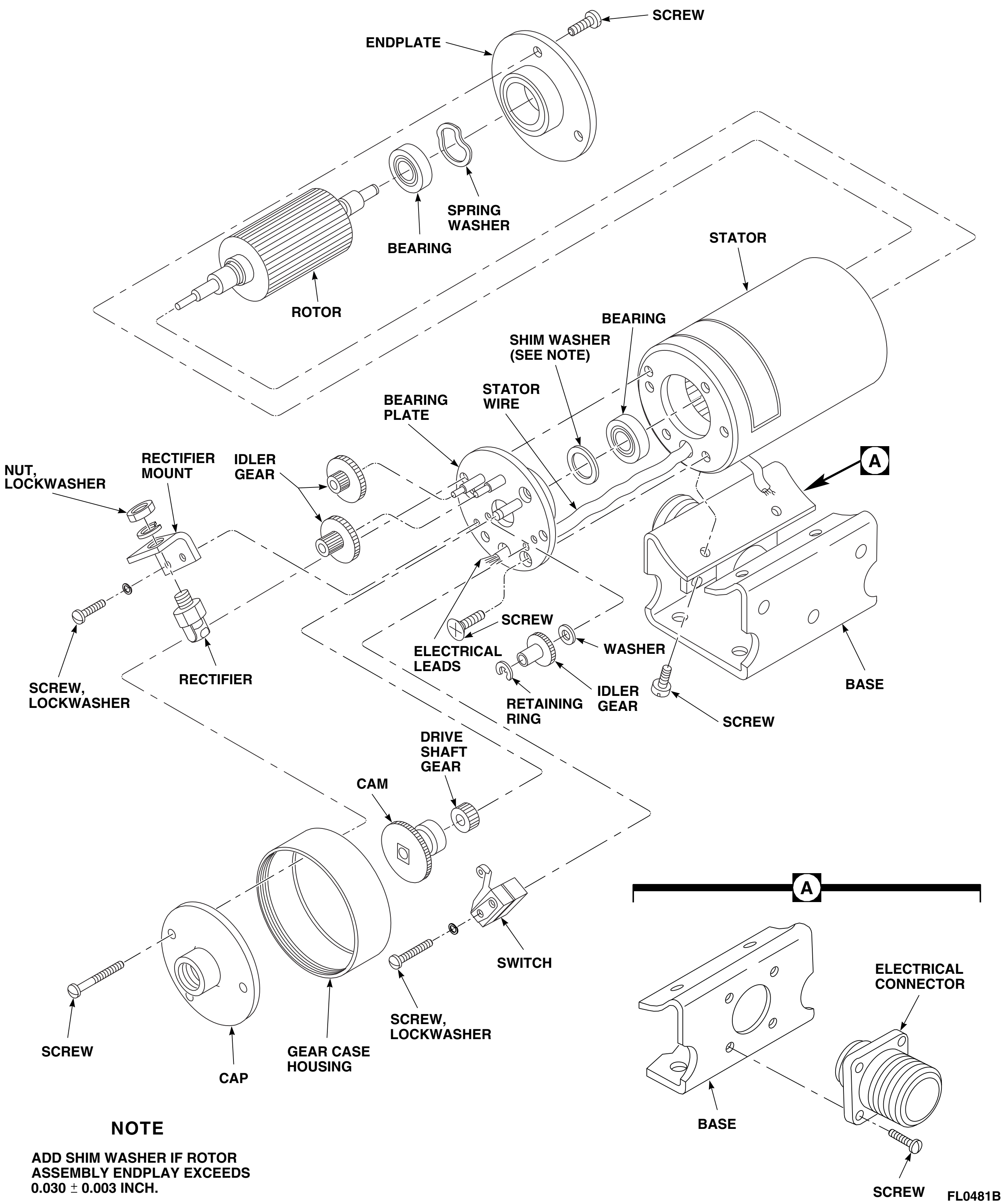
c. Remove retaining ring and remove idler gear and washer from bearing plate.

d. Tag switch and rectifier wires.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Using soldering iron, unsolder switch electrical leads and rectifier electrical leads.



NOTE

ADD SHIM WASHER IF ROTOR ASSEMBLY ENDPLAY EXCEEDS 0.030 ± 0.003 INCH.

FIGURE 12-5-1-1. REPAIR WINDSHIELD WIPER MOTOR (BENCH)

f. Remove screws and lockwashers securing switch to bearing plate.

g. Remove nut and lockwasher securing rectifier to rectifier mount.

h. Remove screws and lockwashers and remove rectifier mount from bearing plate.

i. Remove screws, endplate, and spring washer from end of stator.

j. Pull rotor and shim washer, if installed, from stator.

NOTE

Do not remove bearings or drive shaft gear from rotor unless they are worn or damaged.

k. Remove screws securing bearing plate to stator. Carefully thread stator wires through bearing plate and remove.

l. If stator is damaged, remove stator from base as follows:

(1) Remove screws securing electrical connector to base (Figure 12-5-1-1, Detail A).

(2) Using insert/extract tool, remove and tag wires from electrical connector.

(3) Remove screws securing stator to base-(Figure 12-5-1-1).

12-5-1.1.2 Clean/Inspect/Repair.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

a. Clean all metal parts, except stator, bearings, and electrical parts, with dry-cleaning solvent, Item 124, Appendix D. Blow dry using compressed air.

b. Clean stator, bearings, and electrical parts with cleaning cloth, Item 102, Appendix D.

c. Inspect bearing bore in endplate for wear. **BORE SHALL NOT EXCEED 0.8668-INCH.** If bore exceeds limit, replace endplate.

d. Inspect bearing bore in bearing plate for wear. **BORE SHALL NOT EXCEED 0.8668-INCH.** If bore exceeds limit, replace bearing plate.

e. Check bearings for wear and fit. **BEARING SHOULD FIT SNUG, BUT NOT TIGHT, AND SHOULD SPIN FREELY.** If bearing is worn or does not spin freely, replace bearing.

f. If bearing(s) are to be replaced, inspect rotor shaft for roundness. **DIAMETER SHALL NOT BE LESS THAN 0.3148-INCH.** If roundness exceeds limit, replace rotor shaft.

g. Check windings of stator for broken leads and frayed insulation. **NONE ALLOWED.** If broken leads or frayed insulation are found, replace stator.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

h. Check wiring for loose electrical connections. **NONE ALLOWED.** If connections are loose, using soldering iron and solder, Item 313, Appendix D, repair connections.

i. Check switch by pressing and releasing the switch actuating plunger. **SWITCH SHOULD HAVE SOLID ACTION AND A DISTINCT AUDIBLE CLICK SHOULD BE HEARD WHEN SWITCH IS PRESSED IN OR**

RELEASED. If action is slow, terminals are loose, or if switch plunger is worn, preventing positive action of switch, replace switch.

j. Check all parts for obvious defects, excessive wear, nicks, burrs, and dents which may interfere with proper operation of motor. Pay particular attention to gear and pinion teeth and gear posts. If practical, make minor repairs such as removing nicks and burrs. Replace all damaged, worn, or defective parts.

12-5-1.1.3 Assemble.

a. If removed, install stator as follows:

(1) Secure stator on base with screws (Figure 12-5-1-1). Using lockwire, Item 197, Appendix D, lockwire screws.

(2) Thread electrical connector wires through mount hole.

(3) Using insert/extract tool, insert tagged wires into electrical connector. Remove tags.

(4) Install electrical connector on base with screws (Figure 12-5-1-1, Detail A).

b. Carefully thread electrical leads through hole in bearing plate. Secure bearing plate to stator with screws (Figure 12-5-1-1).

c. If removed, install bearings and drive shaft gear on rotor. Install drive shaft gear on rotor with inside diameter chamfered end towards rotor.

d. Insert assembled rotor into stator. Make sure bearing on rotor is fully seated against bearing plate.

e. Install spring washer on rotor.

f. Install endplate on rotor. Make sure bearing on rotor is fully seated against endplate. Secure endplate to stator with screws. Using lockwire, Item 197, Appendix D, lockwire screws.

NOTE

Do not compress spring washer when checking rotor endplay.

g. Turn motor on its end and check rotor for endplay. **ENDPLAY SHALL NOT EXCEED 0.030 ± 0.003-INCH.** If endplay exceeds limits, perform the following:

(1) Remove screws securing bearing plate to stator. Remove bearing plate.

(2) Install shim washer between bearing and bearing plate.

(3) Install bearing plate on stator with screws.

(4) Repeat step g. until endplay is within limits.

h. Install rectifier mount on bearing plate with screws and lockwashers.

i. Secure rectifier to rectifier mount with nut and lockwasher.

j. Secure switch to bearing plate with screws and lockwashers.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

k. Using soldering iron, and solder, Item 313, Appendix D, solder electrical leads to rectifier and switch.

l. Install washer and idler gear on bearing plate. Secure with retaining ring.

m. Install two idler gears on bearing plate.

n. Install cam on idler gear.

o. Apply a light coat of grease, Item 149, Appendix D, to screws.

p. Install gear case housing and cap on stator and secure with screws.

q. Using lockwire, Item 197, Appendix D, lockwire screws.

12-5-2 BLADE DEICE TEST PANEL ROTARY SELECTOR SWITCH (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 12-2-8
- PARA 12-4-48

12-5-2.1 Replace Blade Deice Test Panel Rotary Selector Switch (BENCH).

12-5-2.1.1. Remove.

a. Remove information plate (PARA 12-4-48).

b. Remove screws and bottom cover from blade deice test panel (Figure 12-5-2-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Tag wires. Using soldering set, unsolder wiring from switch.

d. Remove nut, lockwasher, and rotary selector switch from blade deice test panel.

12-5-2.1.2. Install.

a. Place switch through opening in front of blade deice test panel and fasten with nut and lockwasher (Figure 12-5-2-1).

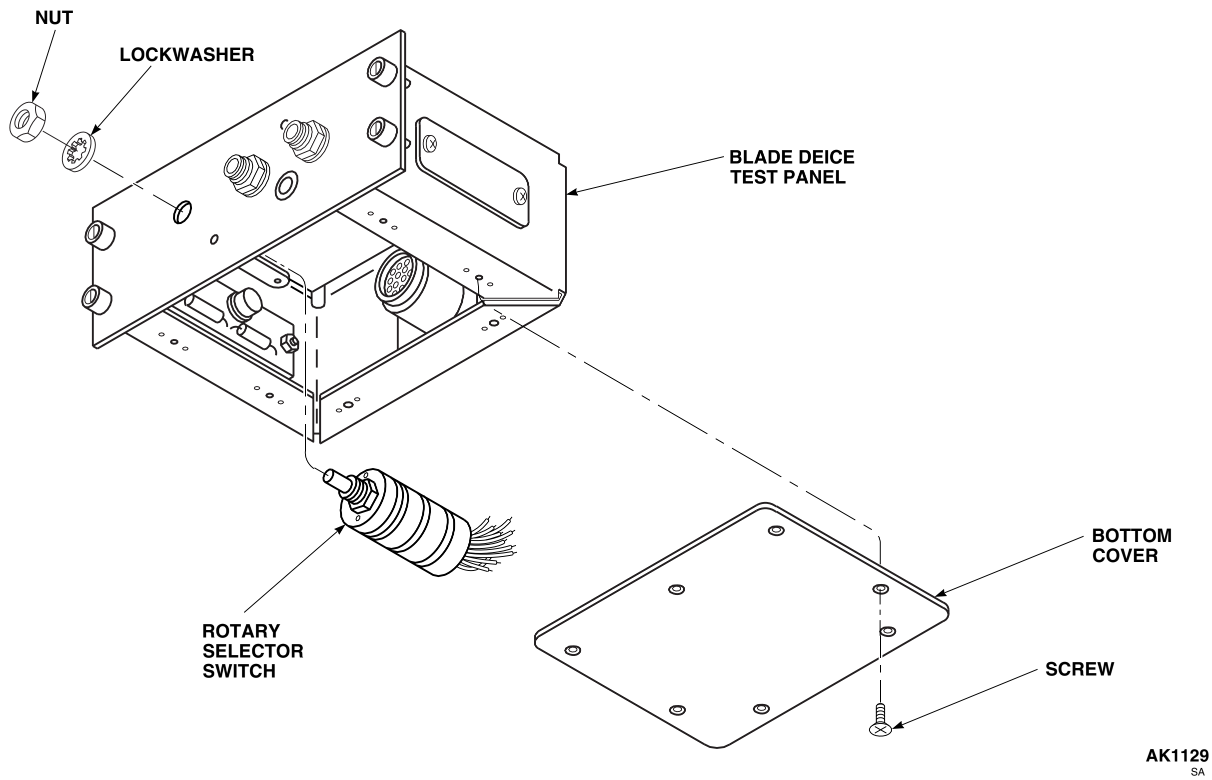


FIGURE 12-5-2-1. REPLACE BLADE DEICE TEST PANEL ROTARY SELECTOR SWITCH (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering set, solder, Item 313, Appendix D, wiring to switch. Remove tags.

c. Fasten bottom cover to blade deice test panel with screws.

d. Install information plate (PARA 12-4-48).

e. Do an operational check of blade deice test panel (PARA 12-2-8).

12-5-3 BLADE DEICE TEST PANEL INDICATOR LIGHT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 12-2-8
- PARA 12-4-48

12-5-3.1 Replace Blade Deice Test Panel Indicator Light (BENCH).

12-5-3.1.1. Remove.

a. Remove information plate (PARA 12-4-48).

b. Remove screws and bottom cover from blade deice test panel (Figure 12-5-3-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Tag wires. Using soldering set, unsolder wiring to indicator light.

d. Remove nut, lockwasher, and indicator light from panel.

12-5-3.1.2. Install.

a. Place indicator light through opening in front of blade deice test panel and fasten with nut and lockwasher (Figure 12-5-3-1).

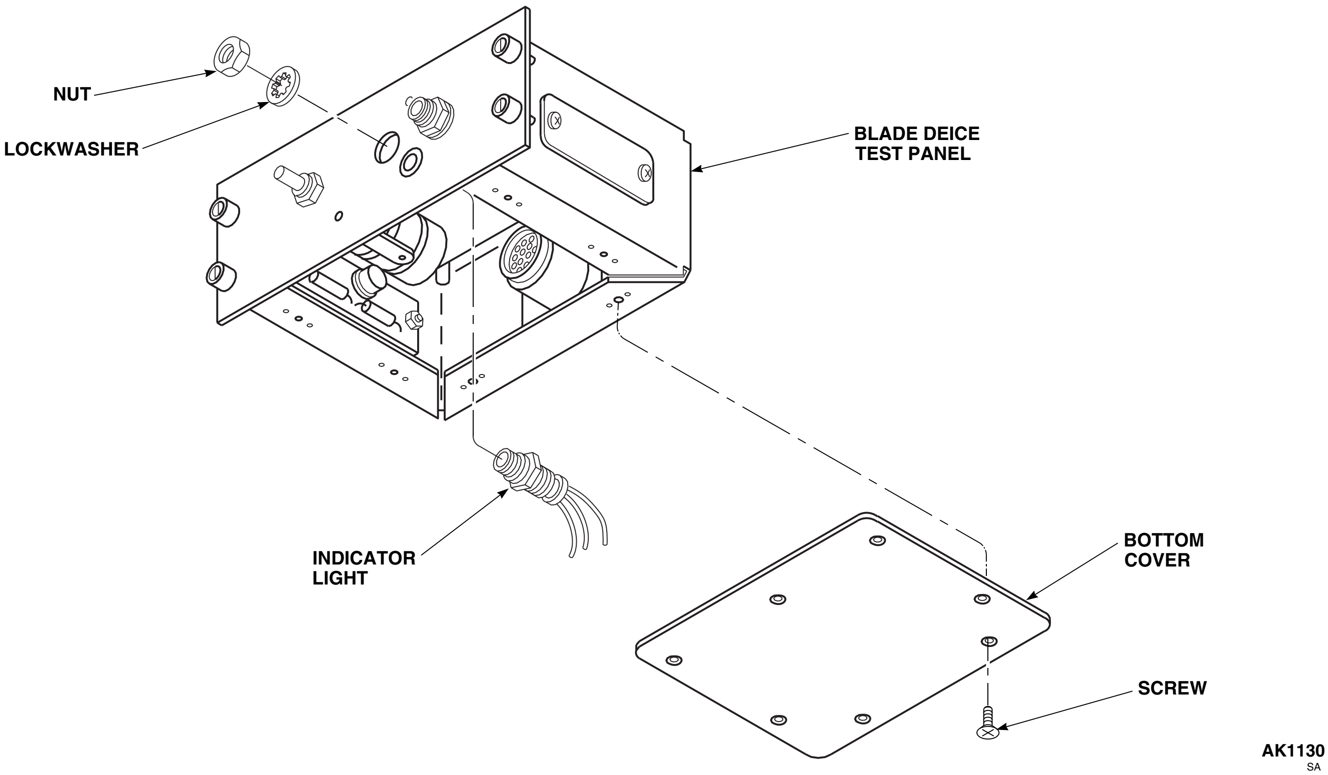


FIGURE 12-5-3-1. REPLACE BLADE DEICE TEST PANEL INDICATOR LIGHT (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering set, solder, Item 313, Appendix D, wiring to indicator light. Remove tags.

c. Fasten bottom cover to blade deice test panel with screws.

d. Install information plate (PARA 12-4-48).

e. Do an operational check of blade deice test panel (PARA 12-2-8).

12-5-4 BLADE DEICE TEST PANEL ELECTRICAL CONNECTOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Insert/Extract Tool

Materials/Parts

- Tags

References

- PARA 1-7-20
- PARA 12-2-8

12-5-4.1 Replace Blade Deice Test Panel Electrical Connector (BENCH).

12-5-4.1.1. Remove.

a. Remove screws and remove bottom cover from blade deice test panel (Figure 12-5-4-1).

b. Remove screws, washers, lockwasher, and nuts securing electrical connector, J1, and ground wire to blade deice test panel.

c. Tag wires to electrical connector, J1.

d. Using insert/extract tool, remove pins from electrical connector, J1.

12-5-4.1.2. Install.

a. Using insert/extract tool, insert tagged wires into electrical connector, J1 (Figure 12-5-4-1). Remove tags.

b. Inspect and treat electrical connector (PARA 1-7-20).

c. Install electrical connector, J1, and ground wire to blade deice test panel with screws, washers, lockwasher, and nuts.

d. Install bottom cover to blade deice test panel and fasten with screws.

e. Do an operational check of blade deice test panel (PARA 12-2-8).

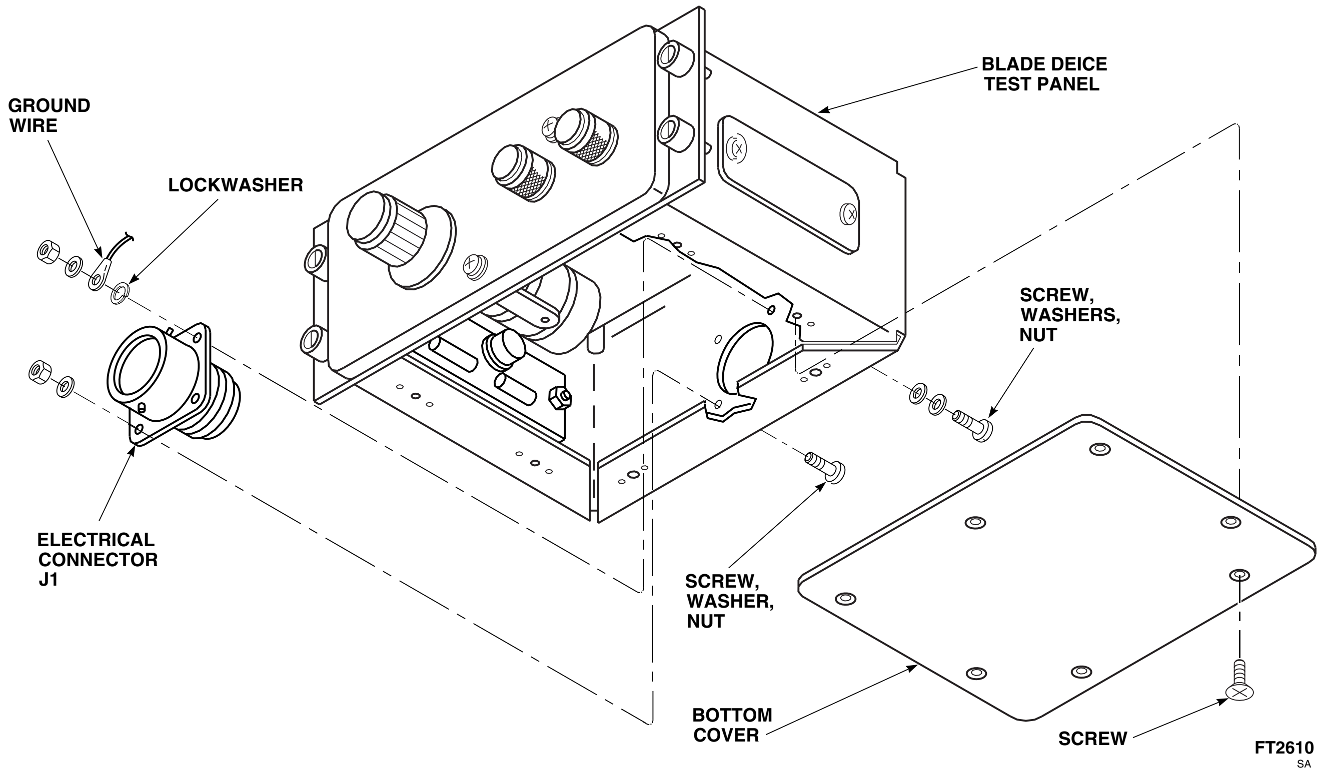


FIGURE 12-5-4-1. REPLACE BLADE DEICE TEST PANEL ELECTRICAL CONNECTOR (BENCH)

12-5-5 BLADE DEICE TEST PANEL PRINTED WIRING BOARD (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-7-20
- PARA 12-2-8

12-5-5.1 Replace Blade Deice Test Panel Printed Wiring Board (BENCH).

12-5-5.1.1. Remove.

a. Remove screws securing top and bottom covers to blade deice test panel, and remove both covers (Figure 12-5-5-1).

b. Remove screws securing electrical connector, J2, to electrical connector, P2, on printed wiring board.

c. Disconnect electrical connector, J2, from electrical connector, P2.

d. Remove screws and washers securing deck assembly to test panel, and slide deck assembly away from test panel.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Tag wires. Using soldering set, unsolder four wires connecting printed wiring board to circuit board.

f. Tag wires to electrical connector, J2.

g. Using soldering set, unsolder all wires from printed wiring board connecting to electrical connector, J2.

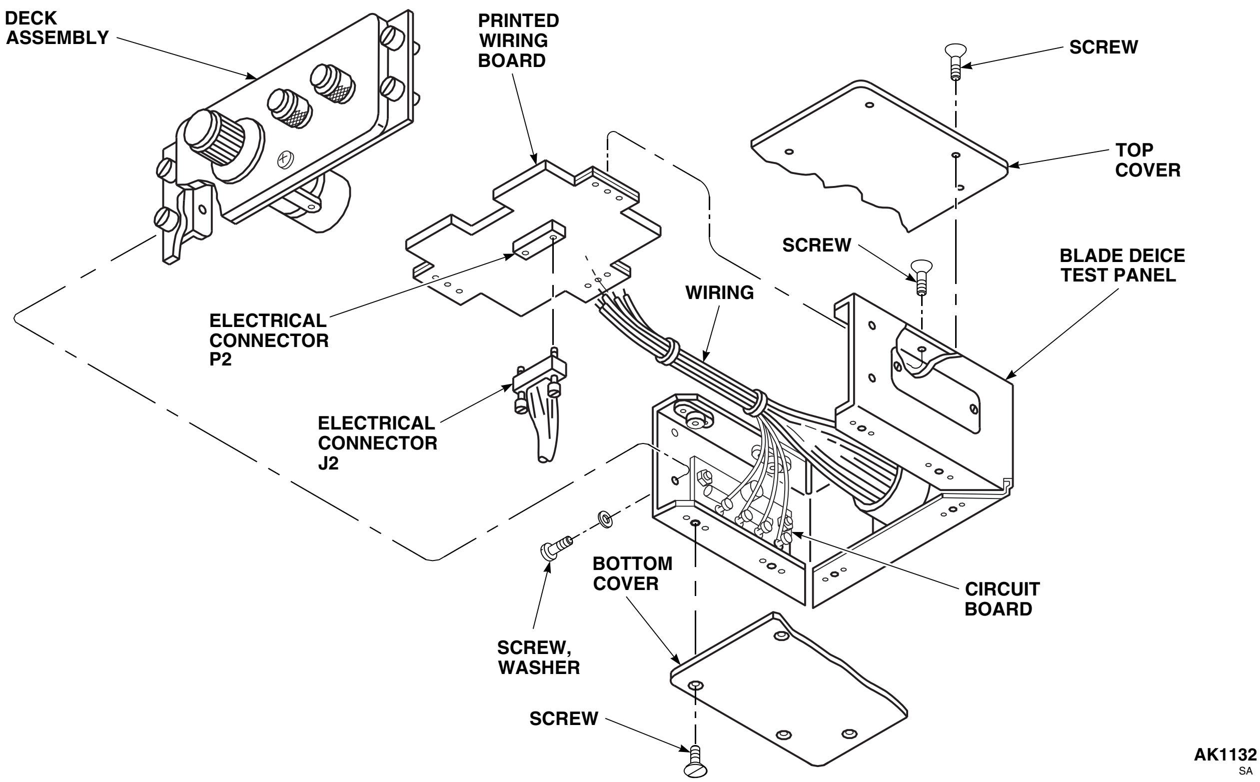


FIGURE 12-5-5-1. REPLACE BLADE DEICE TEST PANEL PRINTED WIRING BOARD (BENCH)

h. Remove screws securing printed wiring board to blade deice test panel.

i. Remove printed wiring board from blade deice test panel.

12-5-5.1.2. Install.

a. Install printed wiring board in blade deice test panel using screws (Figure 12-5-5-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering set, solder, Item 313, Appendix D, four wires connecting printed wiring board to circuit board. Remove tags.

c. Using soldering set, solder, Item 313, Appendix D, all wires connecting printed wiring board to electrical connector, J2. Remove tags.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Join electrical connector, J2, to electrical connector, P2. **TIGHTEN SCREWS.**

f. Install deck assembly to blade deice test panel with screws and washers.

g. Install top and bottom covers to blade deice test panel with screws.

h. Do an operational check of blade deice test panel (PARA 12-2-8).

12-5-6 BLADE DEICE TEST PANEL CIRCUIT BOARD (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Set

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 12-2-8

12-5-6.1 Replace Blade Deice Test Panel Circuit Board (BENCH).

12-5-6.1.1. Remove.

a. Remove screws and bottom cover from blade deice test panel (Figure 12-5-6-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Tag wires. Using soldering set, unsolder four wires from printed wiring board connected to circuit board.

c. Remove screws, washers, spacers, and nuts securing circuit board to blade deice test panel.

d. Remove circuit board.

12-5-6.1.2. Install.

a. Install screws, washers, spacers, and nuts securing circuit board to blade deice test panel (Figure 12-5-6-1).

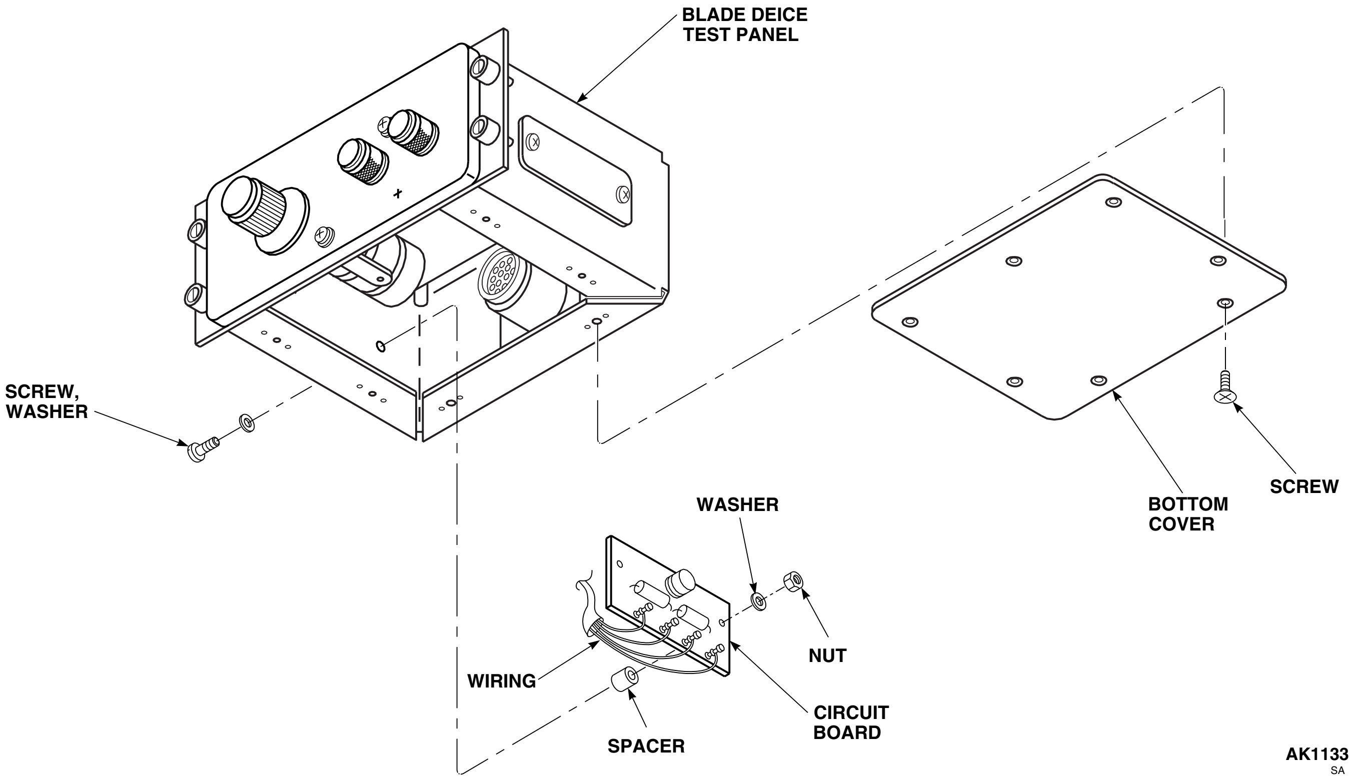


FIGURE 12-5-6-1. REPLACE BLADE DEICE TEST PANEL CIRCUIT BOARD (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering set, solder, Item 313, Appendix D, four wires to printed wiring board leading from circuit board. Remove tags.

c. Secure bottom panel to blade deice test panel with screws.

d. Do an operational check blade deice test panel (PARA 12-2-8).

12-5-7 BLADE DEICE TEST PANEL PRINTED WIRING BOARD AND CIRCUIT BOARD COMPONENTS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Solder-Removing Device
- Soldering Set

Materials/Parts

- Insulation Compound, Item 157, Appendix D
- Solder, Item 313, Appendix D

References

- Appendix D
- PARA 12-2-8
- PARA 12-5-5
- PARA 12-5-6

12-5-7.1 Replace Blade Deice Test Panel Printed Wiring Board and Circuit Board Components (BENCH).

12-5-7.1.1. Remove.

a. If replacing printed wiring board components, remove printed wiring board (PARA 12-5-5). If replacing circuit board components, remove circuit board (PARA 12-5-6).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering set, unsolder component leads by heating each lead and removing solder with solder-removing device (Figure 12-5-7-1).

CAUTION

When removing components with positive and negative termination points, note the position of the component and its leads before removal.

c. Remove component from circuit board/wiring board.

12-5-7.1.2. Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

CAUTION

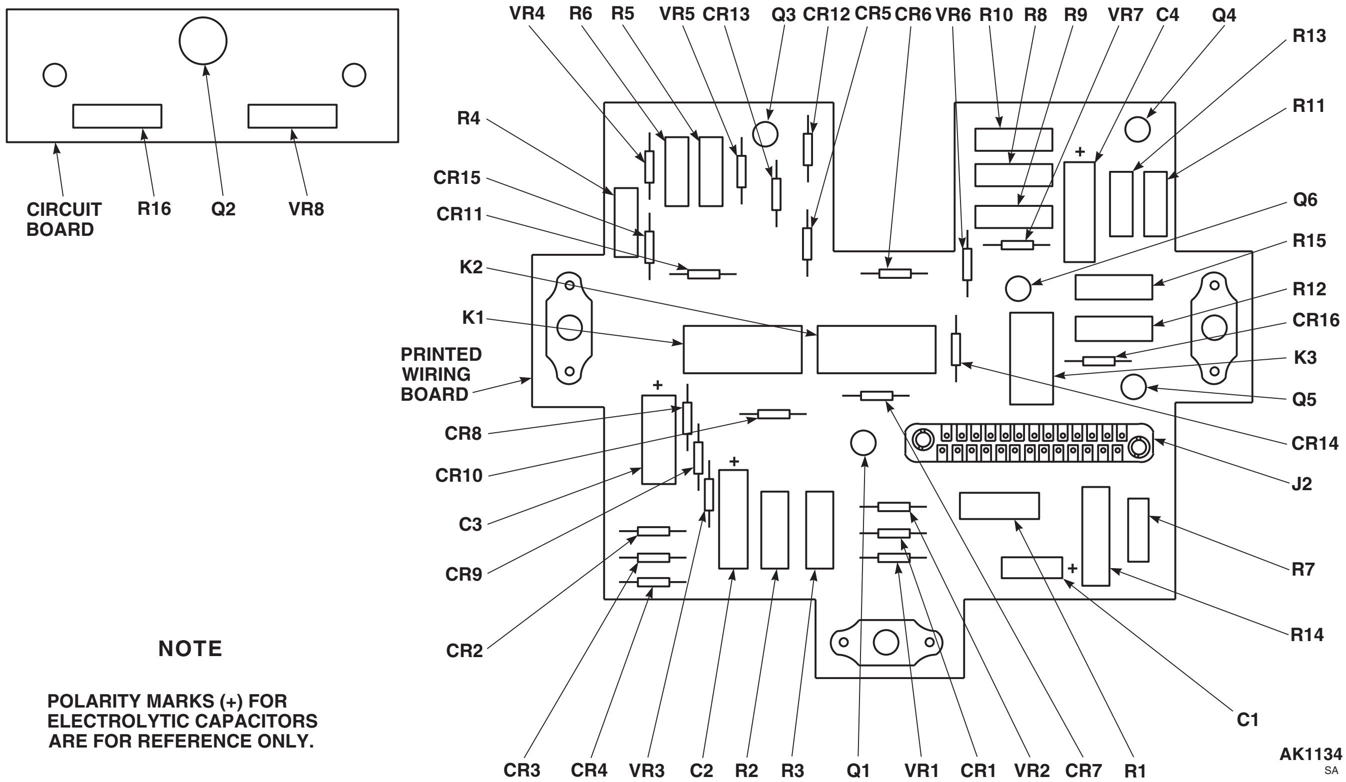
- Too much heat during soldering will cause damage to components and printed circuit.
- When installing components with positive and negative termination points, note the position of the component and its leads before installation.

a. Install replacement part on circuit board/wiring board. Using soldering set, solder, Item 313, Appendix D, leads to board (Figure 12-5-7-1).

b. Apply insulation compound, Item 157, Appendix D, to replacement part and leads.

c. Install circuit board (PARA 12-5-6) and/or printed wiring board (PARA 12-5-5) in blade deice test panel.

d. Do an operational check blade deice test panel (PARA 12-2-8).



NOTE

POLARITY MARKS (+) FOR ELECTROLYTIC CAPACITORS ARE FOR REFERENCE ONLY.

FIGURE 12-5-7-1. REPLACE BLADE DEICE TEST PANEL PRINTED WIRING BOARD AND CIRCUIT BOARD COMPONENTS (BENCH)

12-5-8 DIRECTIONAL CONTROL VALVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Adjustable Power Supply, 0 - 30 VDC
- Aircraft Mechanic's Toolkit
- Control Valves
- Electrical Repairer's Toolkit
- Flowmeter, 0 - 2 SCFM
- Flowmeter, 0 - 20 SCFM
- Fluorescent Penetrant Inspection Kit
- Manometer, 0 - 2 psig
- Nonmetallic Stiff Bristle Brush
- Plugs
- Pressure Gage, 0 - 30 psi
- Pressure Gage, 0 - 1000 psi
- Pressure Source, 1000 psi, 15 SCFM
- Regulator Valve(s), 0 - 1500 psi
- Tap and Die Set
- Torque Wrench, 30 - 150 in. lbs
- Valve Assembly Tool
- Water Tank

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 197, Appendix D
- Shipping Caps

References

- Appendix D
- ASTM E1417

12-5-8.1 Repair Directional Control Valve (BENCH).

12-5-8.1.1 Disassemble.

NOTE

Disassemble control valve only as far as necessary to remove and replace malfunctioning or damaged components.

a. Remove reducer end cap and packing from valve body. Discard packing (Figure 12-5-8-1, Sheet 1, Detail A).

b. Carefully remove pins from indicators (Figure 12-5-8-1, Sheet 1, Detail B and Sheet 2, Detail

F). Discard pins. Remove indicators and washers. Remove and discard packings.

NOTE

Solenoid plunger and spring will remain attached to arm when solenoid assembly is removed. Retain solenoid plunger and spring with solenoid assembly.

c. Using valve assembly tool, remove solenoid assembly from valve body. Remove and discard packing. If washer(s) are installed, remove and discard washers (Figure 12-5-8-1, Sheet 1, Detail A).

d. Push flapper control valve away from normally closed (N.C.) port until it stops; then disengage plunger from control arm and release flapper (Figure 12-5-8-1, Sheet 1, Detail B).

e. Remove nut. Remove and discard packing from nut (Figure 12-5-8-1, Sheet 2, Detail F).

f. Carefully press out and remove shaft from valve body. Remove flapper control valve and associated components from valve body. Remove spacer and return spring and remaining packing from valve body (Figure 12-5-8-1, Sheet 1, Detail B).

g. Remove and discard cotter pin from flapper control valve/control arm assembly (Figure 12-5-8-1, Sheet 2, Detail D). Remove washer and pin. Separate control arm from flapper control valve.

h. Remove and discard vent spring from control arm.

i. Inspect decal or identification plate for damage and readability (Figure 12-5-8-1, Sheet 1, Detail A). If damaged or illegible, remove decal or identification plate.

12-5-8.1.2 Clean.

WARNING

To prevent injury to personnel or damage to parts, all cleaning shall be done in a clean well ventilated, controlled area. Observe necessary fire precautions. All working surfaces shall be constructed of or covered with material that will not generate contaminants. Positive pressure or filtration shall be used to prevent entry of dust or other contaminants.

a. Clean metal parts using dry cleaning solvent, Item 124, Appendix D. Use nonmetallic, stiff bristle brush to remove stubborn accumulations of foreign matter.

b. Allow cleaned parts to air-dry or use cleaning cloth, Item 102, Appendix D.

12-5-8.1.3 Inspect/Repair.

a. Inspect reducer end cap as follows:

(1) Visually inspect reducer end cap for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If damage exceeds limit, replace end cap.

(2) Visually inspect reducer end cap for crossed, stripped, or flattened threads. **NONE ALLOWED.** If damage does not exceed 50%, repair end cap threads. If damage is over 50% of one thread, replace reducer end cap.

(3) Visually inspect reducer end cap for damaged or missing protective coating. **NONE ALLOWED.** If protective coating is damaged or missing, replace end cap.

(4) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect reducer end cap for cracks or fractures. **NONE ALLOWED.** If any cracks or fractures are found, replace reducer end cap.

b. Inspect nut as follows:

(1) Visually inspect nut for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If out-of-tolerance, replace nut.

(2) Visually inspect nut for crossed, stripped, or flattened threads. **NONE ALLOWED.**

If damage is less than 50% of one thread, repair nut threads. If damage is over 50% of one thread, replace nut.

(3) Visually inspect nut for damaged or missing protective coating. If protective coating is damaged or missing, replace nut.

(4) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect nut for cracks or fractures. **NONE ALLOWED.** If any cracks or fractures are found, replace nut.

c. Inspect shaft as follows:

(1) Visually inspect shaft for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If damage exceeds limit, replace shaft.

(2) Visually inspect shaft for damaged or missing protective coating. **NONE ALLOWED.** If protective coating is damaged or missing, replace shaft.

(3) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect shaft for cracks or fractures. **NONE ALLOWED.** If any cracks or fractures are found, replace shaft.

d. Inspect control arm as follows:

(1) Visually inspect control arm for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If damage exceeds limit, replace control arm.

(2) Visually inspect control arm for damaged or missing protective coating. **NONE ALLOWED.** If protective coating is damaged or missing, replace control arm.

(3) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect control arm for cracks or fractures. **NONE ALLOWED.** If any cracks or fractures are found, replace control arm.

e. Inspect flapper control valve as follows:

(1) Visually inspect flapper control valve for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If damage exceeds limit, replace flapper control valve.

(2) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect flapper control valve for cracks or fractures. **NONE ALLOWED.** If any cracks or fractures are found, replace flapper control valve.

f. Inspect solenoid assembly as follows:

(1) Visually inspect solenoid assembly for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If damage exceeds limit, replace solenoid assembly.

(2) Visually inspect solenoid assembly for damaged or missing terminals. **NONE ALLOWED.** If terminals are damaged or missing, replace solenoid assembly.

(3) Visually inspect solenoid assembly for crossed, stripped, or flattened threads. **NONE ALLOWED.** If damage is over 50% of one thread, repair solenoid assembly threads. If damage is over 50% of one thread, replace solenoid assembly.

g. Inspect valve body as follows:

(1) Visually inspect valve body for nicks, burrs, and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** If damage exceeds limit, replace valve body.

(2) Visually inspect valve body for crossed, stripped, or flattened threads. If damage is over 50% of one thread, repair valve body damaged threads. If damage is over 50% of one thread, replace valve body.

(3) Visually inspect valve body for damaged or missing protective coating. **NONE ALLOWED.** If protective coating is damaged or missing, replace valve body.

(4) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant

inspect valve body for cracks or fractures. **NONE ALLOWED.** If any cracks or fractures are found, replace valve body.

12-5-8.1.4 Repair.

a. Use proper die set to chase threads on reducer end cap. If damage is over 50% of one thread, replace reducer end cap.

b. Use proper die set to chase threads on nut. If damage is over 50% of one thread, replace nut.

c. Use proper die set to chase threads on solenoid assembly. If damage is over 50% of one thread, replace solenoid assembly.

d. Use proper tap and die set to chase threads on valve body. If damage is over 50% of one thread, replace valve body.

12-5-8.1.5 Assemble.

CAUTION

Damage to control valve will result if any type of lubricant is used when assembling control valve. Do not use lubricants when assembling control valve. Assemble in clean and well lighted area.

a. Install packing on nut (Figure 12-5-8-1, Sheet 2, Detail F).

b. Insert shaft through nut.

c. Install packing, washer, and indicator onto shaft. Insert pin through indicator and shaft.

d. Install vent spring into hole in control arm (Figure 12-5-8-1, Sheet 2, Detail D).

e. Line up and install control arm to flapper control valve and secure with pin, washer, and cotter pin.

f. Install spacer into return spring and place return spring into valve body and over control arm (Figure 12-5-8-1, Sheet 1, Detail B). Install shaft into valve body and line up spring and spacer (Figure 12-5-8-1, Sheet 2, Detail E).

g. Install flapper control valve into valve body, with tang on control arm facing out at normally open (N.O.) port position. Make sure shaft is installed with indicator facing back to normally closed (N.C.) position (Figure 12-5-8-1, Sheet 1, Detail B).

h. Engage three threads of nut into valve body.

i. Install plunger into valve body, and push flapper control valve in from N.C. position to engage plunger onto control arm tang and release flapper control valve.

j. Install spring into plunger (Figure 12-5-8-1, Sheet 1, Detail A).

k. Install packing on solenoid assembly.

NOTE

Do not install washers between solenoid assembly and valve body at this time. Washers are only installed for alignment of terminal block. Do not use more than three washers for alignment.

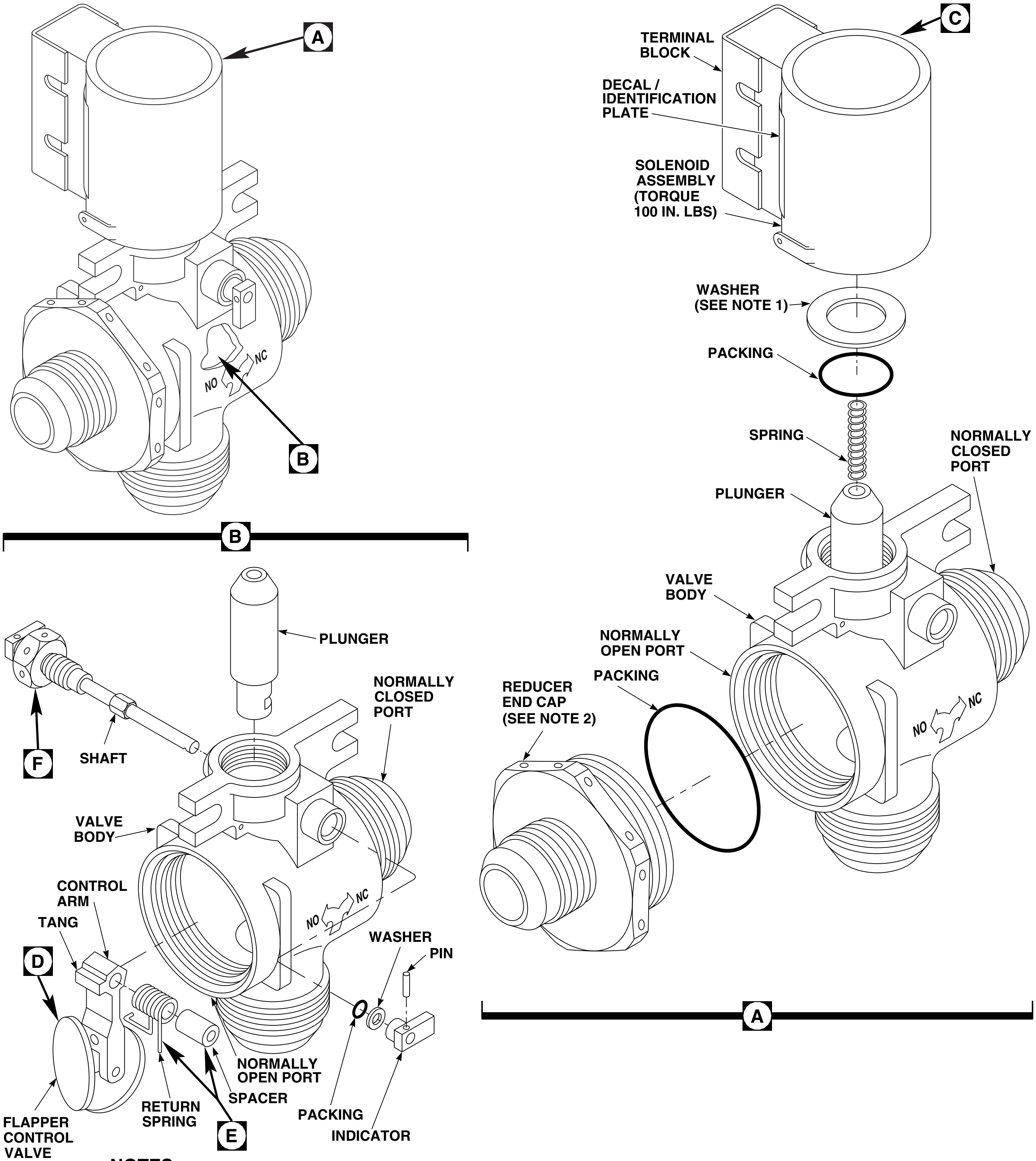
l. Using valve assembly tool, install solenoid assembly (without washers) to valve body. **TORQUE SOLENOID ASSEMBLY TO 100 INCH-POUNDS** and check alignment of solenoid terminal block as shown (Figure 12-5-8-1, Sheet 2, Detail C). If proper alignment is not achieved, remove solenoid assembly and add washers. Install solenoid assembly and retorque.

NOTE

Make sure indicator is installed facing back to normally closed position.

m. Install packing, washer, and indicator onto shaft. Insert pin through indicator and shaft (Figure 12-5-8-1, Sheet 1, Detail B).

n. Install packing on reducer end cap. Install reducer end cap into valve body (Figure 12-5-8-1, Sheet 1, Detail A). **TIGHTEN REDUCER**



NOTES

1. USE MAXIMUM OF THREE WASHERS TO MAINTAIN ALIGNMENT.
2. TIGHTEN REDUCER END CAP 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

FO0436_1E
SA

FIGURE 12-5-8-1. REPAIR DIRECTIONAL CONTROL VALVE (BENCH) (SHEET 1 OF 2)

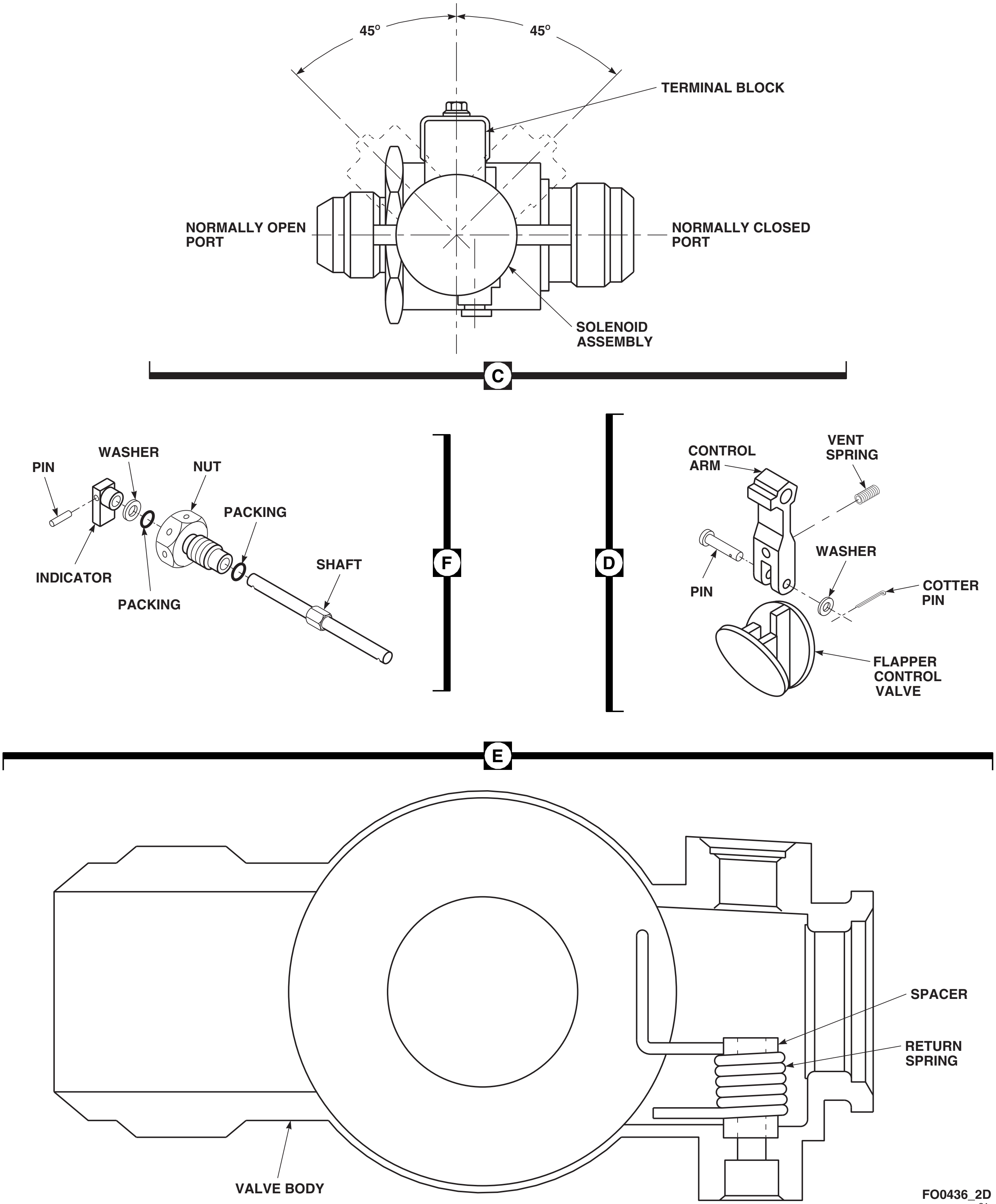


FIGURE 12-5-8-1. REPAIR DIRECTIONAL CONTROL VALVE (BENCH) (SHEET 2 OF 2)

END CAP 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

12-5-8.1.6 Test.

a. Perform external leakage and proof pressure test of directional control valve as follows:

(1) Install plugs on normally open (N.O.) port and normally closed (N.C.) port, connect pressure source to IN port and connect solenoid to adjustable power supply.

WARNING

Injury to personnel and damage to equipment will result if electrical solenoid is submerged in water tank. Make sure electrical solenoid is not submerged in water tank.

(2) Mount directional control valve into test setup as shown (Figure 12-5-8-2). Make sure solenoid is not submerged in water tank.

(3) With solenoid deenergized (Figure 12-5-8-3), apply 880 psig pressure to IN port and submerge directional control valve into water tank. Make sure solenoid is not submerged in water tank. Hold pressure for one minute; then reduce pressure to 0 psig. **THERE SHALL BE NO PERMANENT DISTORTION OR LOOSENING OF PARTS.**

(4) With solenoid deenergized, apply 440 psig pressure to IN port and submerge directional control valve into water tank. Make sure solenoid is not submerged in water tank. Hold pressure for one minute; then reduce pressure to 0 psig. **THERE SHALL BE NO EVIDENCE OF LEAKAGE THROUGH BODY OR FITTINGS. LEAKAGE PAST PACKINGS ON SHAFT SHALL NOT BE OVER FIVE BUBBLES PER MINUTE.**

(5) Repeat steps a. (3) and a. (4) with solenoid energized, using adjustable power supply at 24 VDC. Decrease voltage to 0 VDC, reduce pressure to 0 psig, and remove directional control valve from test setup.

b. Perform internal leakage test of directional control valve as follows:

(1) Install plugs on N.O. port, connect pressure source to IN port and connect solenoid to adjustable power supply and deenergize solenoid.

(2) Mount directional control valve into test setup as shown (Figure 12-5-8-4).

(3) Close V2 control valve and open V1 control valve. Close V4 regulator valve and open V3 regulator valve. Apply 5.0 psig to IN port. Connect N.C. port to flowmeter.

NOTE

If unable to reach 5.0 psig, increase pressure until flapper valve seals; then reduce pressure to 5.0 psig.

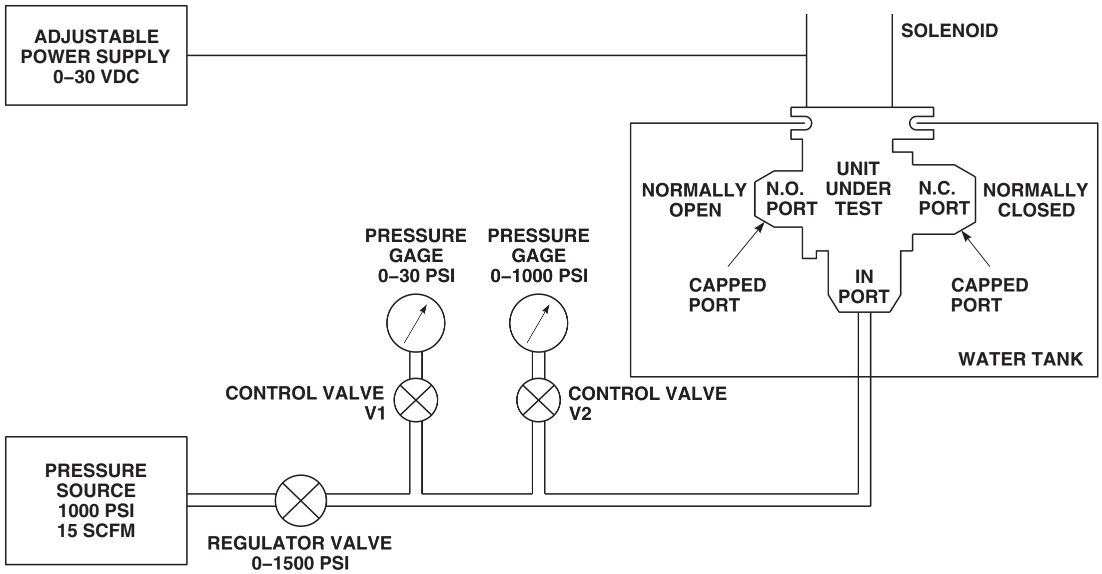
(4) Check leakage rate at 5.0 psig. Leakage rate shall not be over 0.6 standard cubic feet per minute (SCFM).

(5) Close V3 regulator valve and open V4 regulator valve. Close V1 control valve and open V2 control valve. Increase pressure at IN port to 440 psig and measure leakage flow at N.C. port. Leakage rate shall not be over 14.75 SCFM. Reduce pressure to 0 psig.

(6) Plug N.C. port. Connect N.O. port to flowmeter. Energize solenoid with 24 VDC, and repeat steps b. (3) through b. (5).

(7) Decrease voltage to 0 VDC, reduce pressure to 0 psig, and remove directional control valve from test setup.

c. Perform functional test of directional control valve as follows:

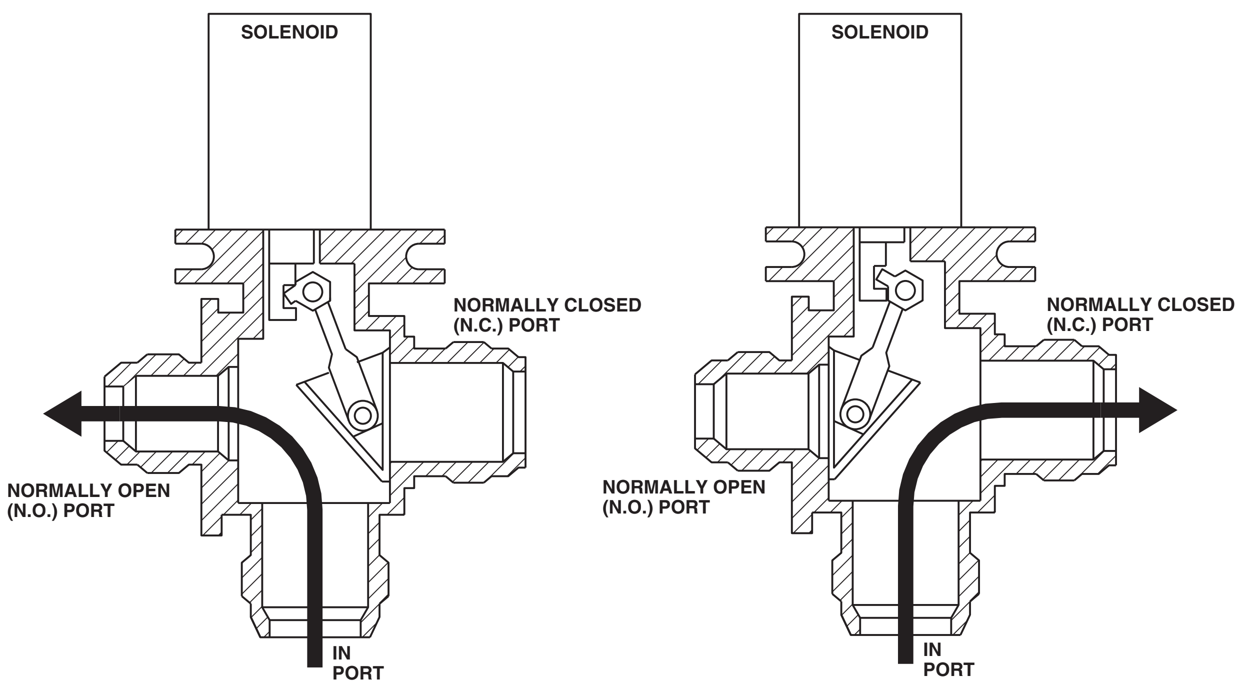


NOTE

DO NOT SUBMERGE SOLENOID ASSEMBLY IN WATER.

AB0899
SA

FIGURE 12-5-8-2. EXTERNAL LEAKAGE AND PROOF PRESSURE TEST SETUP (BENCH)



FLOW WITH SOLENOID DEENERGIZED

FLOW WITH SOLENOID ENERGIZED

AB0898
SA

FIGURE 12-5-8-3. FLOW DIAGRAM (BENCH)

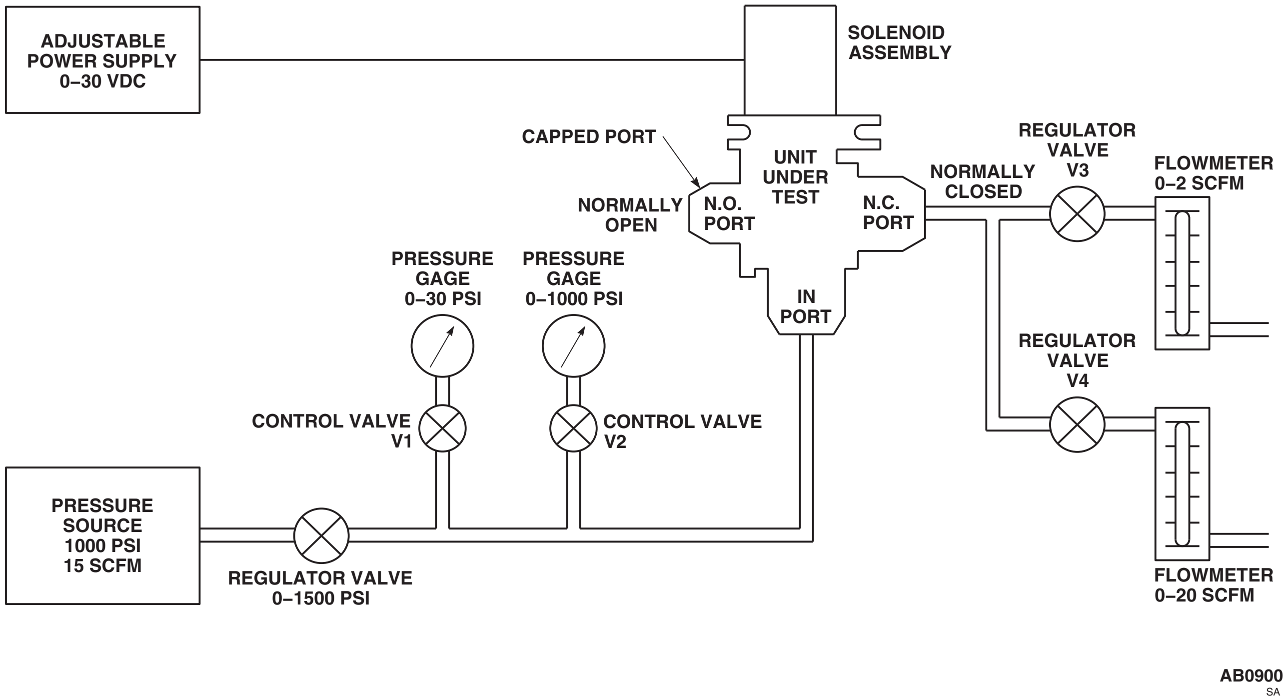


FIGURE 12-5-8-4. INTERNAL LEAKAGE TEST SETUP (BENCH)

(1) Install plug on N.O. port and open N.C. port to atmosphere. Connect pressure source to IN port through flowmeter and manometer.

(2) Mount directional control valve into test setup as shown (Figure 12-5-8-5).

(3) Apply pressure to achieve 5.0 SCFM on flowmeter. Flow reading shall be maintained with 1 psig or less.

(4) Reduce pressure to 0 psig and remove directional control valve from test setup.

d. Perform actuation test of directional control valve as follows:

(1) Open N.C. port to atmosphere.

(2) Mount directional control valve into test setup as shown (Figure 12-5-8-6).

NOTE

Make sure indicators on directional control valve indicate proper flapper valve position.

(3) Energize solenoid with 24 VDC using adjustable power supply. Close V1 control valve and open V2 control valve. Apply 100 psig maximum to IN port; then deenergize solenoid.

(4) Energize solenoid with 24 VDC using adjustable power supply. Make sure flapper valve moves to N.O. port, and flow shall be from N.C. port. Apply 100 psig maximum to IN port; then deenergize solenoid. Flapper valve shall remain at N.O. port.

(5) Reduce pressure and note pressure at point when flapper valve moves to N.C. port with solenoid deenergized. Pressure shall not be over 5.0 psig. Reduce pressure to 0 psig.

(6) Using adjustable power supply, reduce voltage to 18.0 VDC, then energize and deenergize solenoid several times. Flapper valve shall move instantly to N.O. port with solenoid energized, and return

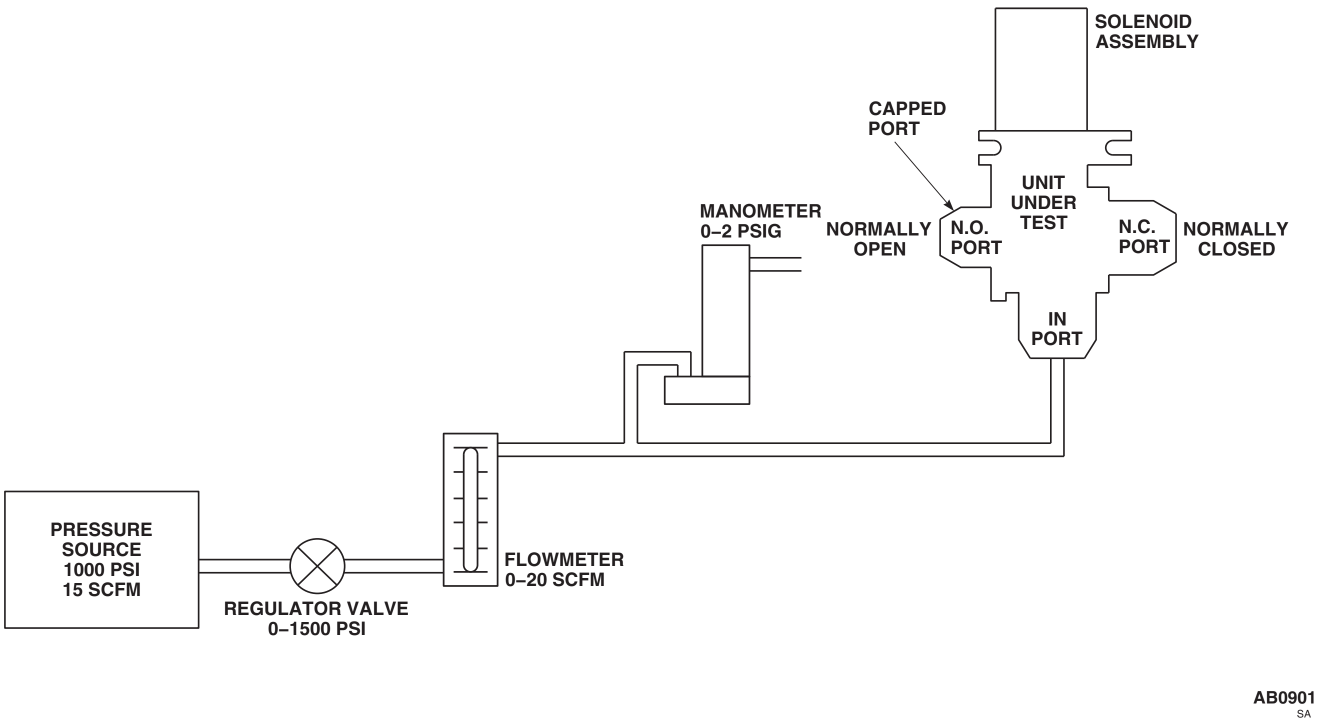


FIGURE 12-5-8-5. FUNCTIONAL TEST SETUP (BENCH)

to N.C. port when solenoid is deenergized.

(7) Using adjustable power supply, increase voltage to 30.0 VDC, and repeat step d. (6) with power at 30.0 VDC.

(8) Using adjustable power supply, increase voltage to 24.0 VDC, and repeat step d. (6) with power at 24.0 VDC.

(9) With solenoid energized to 24.0 VDC, measure current draw. Current draw at 24.0 VDC shall not be over 2.1 amperes.

(10) Reduce pressure to 0 psig and decrease voltage to 0 VDC. Remove directional control valve from test setup.

e. Perform final assembly of directional control valve as follows:

(1) Install shipping caps on all open ports.

(2) Using lockwire, Item 197, Appendix D, lockwire reducer end cap to valve body.

(3) Using lockwire, Item 197, Appendix D, lockwire nut to valve body.

(4) If removed, install decal and/or identification plate (Figure 12-5-8-1, Sheet 1, Detail A).

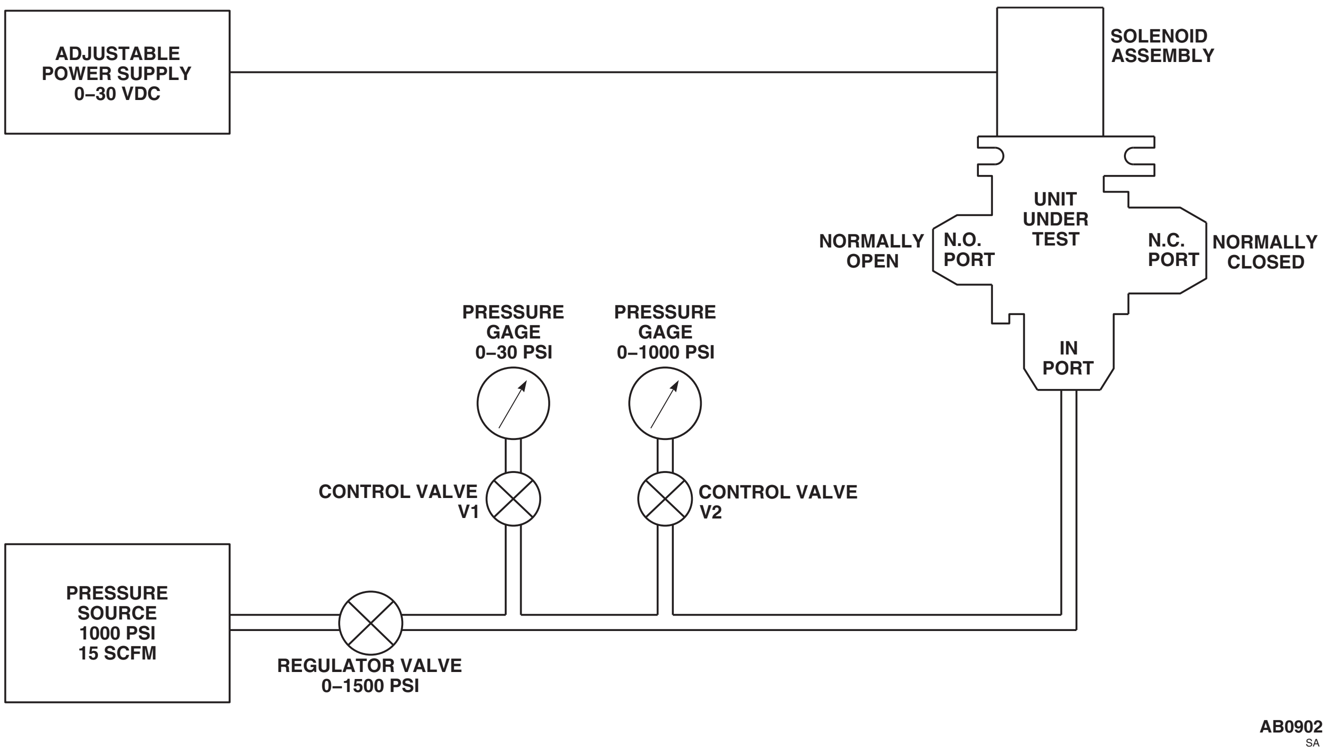


FIGURE 12-5-8-6. ACTUATION TEST SETUP (BENCH)